Proceedings

FOURTH CONFERENCE ON RESEARCH IN MATHEMATICS EDUCATION MEI 4

22nd and 23rd September 2011

St Patrick's College
Drumcondra, Dublin 9

Theme:

Mathematics Teaching Matters

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ACKNOWLEDGEMENTS

The First National Conference on Mathematics Education in Ireland – a Research Perspective (MEI1) was the brain child of Dr Seán Close with a theme of 'Opening Doors' and an organising committee of three. To date, there have been three successful conferences and the biennial MEI gathering has become an important date on the calendar for mathematics education researchers at all levels in Irish education. This year all papers have been peer reviewed.

We wish to thank the keynote speakers, presenters of research reports, session chairs and all participants in the fourth MEI conference on Research in Mathematics Education in Ireland. Our main aim – that of searching for ways to achieve discussion and consensus about what constitutes 'good' teaching for the learning of mathematics at all levels – is reflected in our conference theme - Mathematics Teaching Matters. In order to achieve this, we sought to bring together those who have an interest in mathematics education, to consider future directions for mathematics education research in Ireland in the light of recent developments and international trends, and to consider ways of improving linkages with mathematics education communities within Ireland and in other countries. We hope that the conference provides participants with an interesting and challenging programme of presentations, and with opportunities for discussion, for making contacts and renewing friendships. We are happy that Irish mathematics education is flourishing and we welcome links with the wider scientific research community. We welcome especially our presenters from outside Ireland since their participation adds richness to our national deliberations.

We also express our sincere gratitude to all those who supported the conference including: Dr. Pauric Travers, President of St. Patrick's College; Mr Seán Sherlock, TD, Minister of State, Department of Enterprise, Jobs & Innovation and Department of Education & Skills with responsibility for Research & Innovation and our sponsors - St. Patrick's College Research Committee, the Teaching Council, the Irish National Teachers' Organization, the Association of Secondary Teachers Ireland, the Irish Mathematical Society and the Centre for the Advancement of Science Teaching and Learning.

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FINNISH MATHEMATICS TEACHING: A CASE OF UNIQUELY IMPLICIT DIDACTICS

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This paper reports on a qualitative analysis of video-taped mathematics lessons taught by four case study teachers, defined locally as effective, in a provincial university city in Finland. The aim was to examine how teachers conceptualise and present mathematics to their learners and, in so doing, understand the relationship between Finnish mathematics teaching practices, as reflected in case study lessons, and Finnish success on successive PISA assessments. Analysed by means of the process of constant comparison, the data yielded two key characteristics of case study classrooms. Firstly, irrespective of their intended learning outcome, teachers exploited a series of implicit didactic strategies focused on encouraging students to infer meaning. Secondly, three culturally located activities were identified that appeared complementary to this sense of the implicit. These were the systemic encouragement of students to make notes, teachers' exploitation of the confident child and the assumed collaboration of parents.

INTRODUCTION

This paper reports on a qualitative analysis of video-taped mathematics lessons taught by four case study teachers in a provincial university city in Finland. The aim was to examine how these teachers conceptualised and presented mathematics to their learners and, in so doing, understand how they constructed those opportunities for learning contributory to that country's success on successive iterations of the Organisation for Economic Cooperation and Development's (OECD) Programme for International Student Achievement (PISA) (OECD, 2001, 2004, 2007, 2010). A key aspect of the study is that it was undertaken by an outsider looking in; a process necessitating particular methodological and ethical considerations.

Finnish Mathematical Success

The primary foci of the four cycles of the Programme for International Student Assessment (PISA) have been literacy, mathematical literacy, scientific literacy and literacy respectively. However, each assessment has included, irrespective of the main focus, substantial assessments of students' application of mathematical knowledge and skills to authentic settings, both within mathematics itself and a wider world (Adams 2003). Over these four cycles Finnish students appear not only to have performed consistently higher than those of any other European country but also comparably to those of the highest achieving nations of the Pacific Rim. This is, of course, not quite true as the figures published in the various PISA reports have not distinguished between the autonomous regions of Belgium. Taken as a whole Belgium was among the higher performing European states but, when viewed separately, the Flemish community, which is similar in size to that of Finland, performed at least as well as Finland, and often better, across all four cycles (De Meyer et al., 2002, 2005; De Meyer 2008; De Meyer and Warlop, 2010).

That said, the headline success of Finnish students has been a catalyst for such international interest in its educational processes that over “the past few years, more than 100 delegations from all over the world have beaten their way to the doors of the ministry of education in Helsinki to find out how the Finnish system works” (Crace, 2003). A number of PISA success-related explanations have been proposed by internal commentators. In general, it has been attributed to the quality of the comprehensive school system, high expectations of all participants, a well qualified and committed teaching force that enjoys high status, if not high pay, and the trust of society in general and parents in particular (Välijärvi et al., 2002; Välijärvi, 2004; Tuovinen, 2008). In particular it has been ascribed to the high quality of mathematics teacher education, competition for teacher education places and the esteem in which teaching continues to be held (Malaty, 2007) and high systemic investment in the language acquisition and competence of special educational needs students (Kivirauma and Kari, 2007).

Thus, internal commentators have indicated that Finnish PISA success may be a consequence of various cultural factors, although little evidence has been forthcoming recently with regard to the functioning of Finnish mathematics classrooms in general and the role of participants in particular (Pehkonen et al., 2007). Admittedly, research undertaken a quarter of a century ago highlighted a tradition of teacher-dominated practice that had not only changed little in fifty years but created an intelligence and emotional wasteland (Carlgren et al., 2006). More recently, but preceding Finland’s PISA successes, Norris et al. (1996, p. 29) found “rows and rows of children all doing the same thing in the same way whether it be art, mathematics or geography”, adding that they had “moved from school to school and seen almost identical lessons, you could have swapped the teachers over and the children would never have noticed the difference”. Thus, while Finnish classrooms appear to have been historically dominated by traditional approaches to teaching and learning, little is known about the ways in which participants enact their roles in the modern, PISA-driven world. Thus, it would seem timely to suggest that an examination of the practices of Finnish teachers of mathematics and the role they play in the construction of PISA success would be of benefit to policy-makers, educational researchers and curriculum developers alike.

METHODS

The data on which this paper is based derive from an EU-funded study of mathematics teaching in Belgium (Flanders), England, Finland, Hungary and Spain. The main data set comprised video recordings of four sequences of lessons taught in each country on agreed topics by teachers defined locally, in the manner of the learner’s perspective study (Clarke 2006) as effective. Each sequence comprised five lessons. Videographers were instructed to focus on the teachers whenever they were speaking. Teachers wore radio-microphones while a static microphone was placed strategically to capture as much student talk as possible. However, since attention was on how teachers structured opportunities for learning the integrity of data collection was not greatly compromised through lost student talk. After filming, each tape was compressed and transferred to CD for coding, copying and distribution. The first two lessons of each sequence were transcribed and translated into English and subtitled recordings made to facilitate analysis by all project teams. Each of the remaining lessons in each sequence was accompanied by a narrative written by the home team to inform a foreigner's viewing. The analysis presented in this paper draws predominately on the eight subtitled videos although the remaining twelve were consulted frequently.

Qualitative analysis of such data is not a straightforward process. One approach, particularly in respect of its facilitating our understanding of how "actors respond to changing conditions and the consequences of their actions" (Corbin and Strauss, 1990, p. 5), is the constant comparison process exploited by grounded theorists. Firstly, all videos, with and without subtitles, were viewed several times in order to get a *feel* for how participants constructed the playing out of their lessons. Secondly, the first video in the sequence on linear equations, for no other reason than it was the first alphabetically, was viewed again several times to identify categories of teacher activity. With each new category the video was viewed again to determine whether or not the category had been missed in the earlier sections. Once the first video had been completed the second in the sequence was subjected to the same process. However, any new category to emerge from the second video prompted a return to the first to examine whether it, too, had been missed in the earlier viewings. In this manner a set of teacher behaviours emerged on which this paper is based and which, it is suggested, represent a unique Finnish mathematics didactic tradition.

Inevitably, interpreting alien educational systems creates particular ethical issues, not least because cultural insiders, as the taxpaying funders of a system's policies, have critical rights denied the outsider who is, essentially, a guest in the classrooms of that country. Moreover, when researching contexts culturally different from their own it is important to understand that incomplete understanding of the historical, social and political agendas underpinning participants' beliefs and actions may skew interpretations of events (Liamputtong, 2010). However, a significant methodological strength of an outsider is that he or she exploits a different interpretive lens from the insider. This is because cultures so "shape the classroom processes and teaching practices within countries" (Knipping, 2003, p. 282) that teachers' practices are "deep in the background of the schooling process ... so taken-for-granted… as to be beneath mention" (Hufton and Elliott, 2000, p. 117). In this paper, therefore, we adopt the position that an outsider will see things hidden to the insider but that data interpretation should show commensurability with Tillman's (2006) conception of culturally sensitive research in its subordination of dominant theoretical perspectives to the voice of the cultural group under scrutiny.

RESULTS

The four Finnish teachers, here given pseudonyms, taught in different comprehensive schools in a provincial university city with a large teacher education department. All four, three male and one female, were involved in teacher education activities as part of the University's programme and, therefore, located in the norms and values of mathematics education within their respective community. The males, Sami, Antti and Janne were all in their early thirties, while the female, Katja, was in her mid-fifties. Sami taught his sequence of lessons on percentages to a grade 6 class, Antti taught linear equations to a grade 8 class, Janne taught polygons to a grade 6 class while Katja taught geometry to a grade 8 class.

The analysis yielded a variety of categories of teacher actions or behaviours that were, essentially, located within the teaching of the concepts of mathematics and the procedures of mathematics. Therefore, the results are presented against these two broad categorisations that tend to typify mathematics teaching internationally. Interestingly, little evidence was seen of teachers presenting non-routine or authentic problems other than as the vehicles for teacher-managed whole class introductions to new procedures or concepts.

Conceptual Knowledge

The development of students' conceptual knowledge was typically manifested in episodes of exposition, various forms of questioning and whole-class reflections on conceptually-focused tasks. In the following, examples from each of the case study teachers are presented before being discussed briefly. By way of explanation, any reference to either a teacher or a student writing should be taken as meaning that they were writing on the chalk board.

Antti and linear equations

Early in his first lesson on linear equations Antti wrote, *Definition: An equation is two expressions denoted as being of equal magnitude*. He wrote in capitals, taking almost two minutes to complete the definition while his students copied in silence. The definition was given with no comment. Next, six sentences were written on the board and each examined publicly to determine whether or not it was an equation. The third, x = 3, typically, went as follows.

Antti:	What about x equals three? Olli?
Olli:	Yes
Antti:	Yes, there's a sign of equality and expressions exist on both sides.

Following the operationalisation exercise Antti wrote, *Equation solving means finding out when the equation is true or false*". He said,

Antti:	The solution means that we think about when an equation is true... The other possibility is that it might never be true. So then it would be false... So it is settling between those two possibilities, which one it is.

Later, during another operationalisation exercise, also undertaken publicly, the following transpired.

Antti:	What should x be to make x squared equals eight true; x to the second equals eight? What would fit in x's place? Does anyone have any good suggestions? Olli, can you give us something? What would fit in x's place? When you square it, it equals eight.
Antti:	Joonas?
Joonas:	The square root of eight.
Antti:	Yes. If x is the square root of eight, this is true. So, this equation is a conditional equation. This equation is true, when x is the square root of 8.

The above exchanges seemed focused on two key concepts - the nature of an equation and a categorisation of equations as always true, conditionally true or false – introduced by means of two strategies. The first involved definitions written on the board and the second involved opportunities for students to operationalise those definitions. The definitions, which were written in capitals, were presented slowly and without comment. In this way students had time not only to copy but also to read and interpret the text. These were then operationalised through publicly discussed exercises that allowed Antti's desired categorisations to emerge.

With respect to interpretation, the written definitions, particularly in their drawing on mathematically appropriate forms of words, could be construed as explicit. However, Antti neither sought nor offered clarification, seeming to expect his students to infer meaning. Such practices may allude to something implicit to which we return below. The operationalisation exercises could also be viewed as attempts to make explicit his definitions and yet the manner in which they were managed also alluded to a form of implicit presentation. For example, when Olli correctly asserted that x=3 was an equation, Antti's reaction was to repeat the definition of an equation independently of its relation to the equation itself. He neither discussed the nature of the expressions involved nor elicited his students' understanding as to both x and 3 constituting mathematical expressions. In similar vein, when discussing Joonas' solution of $x^2 = 8$, the relationship between the square root of eight and the value of x that would make the equation true, and the role of that relationship in defining a conditional equation, was left for students to infer. Thus, despite several superficially explicit elements, Antti's presentation and operationalisation of his definitions were implicitly managed with students being left to infer meaning.

Sami and Percentages

In starting his first lesson on percentages, Sami asked, *what do we mean by the word percent?* Aku suggested that it meant one hundredth, which Sami accepted. Sami informed his students that *there are three ways to denote one hundredth* and invited them to write them in their notebooks. This they did in silence with Sami then suggesting that it should be in equation form. Salla, invited to the board, wrote without hesitation, 1/100 = 1% = 0,01. Later, Sami set an exercise whereby students constructed equations of the same form from 100 squares with various proportions shaded. Several were discussed collectively, as in this typical example:

Sami:	What about the green shading? Anna?
Anna:	Thirty hundredths.
Sami:	Yes. As a percentage, Aku?
Aku:	Thirty.
Sami:	As a decimal number, Jussi?
Jussi:	Zero point thirty.

Both the above exchanges, typical of longer sequences of interaction, appeared conceptually focused on the equivalence of different percentage-related representations. It could be argued, also, by means of Aku's assertion that one percent meant one hundredth, that Sami had provided his students with an implicit definition. However, this implicit definition was operationalised by means of instrumentally presented procedures linking the various percentage-related representations. Thus, particularly as Sami had neither sought nor offered clarification, students appeared to have been left to infer meaning, through individual interpretation of the repeated tripartite equation, as to the meaning of these linkages. Thus, as with Antti and the teaching of equations, Sami's presentation and operationalisation of his definitions were implicitly managed.

Janne and Polygons

During his first lesson on polygons, by means of an over-head projector, Janne progressively revealed rows of quadrilaterals with each row a categorical refinement of the previous. When he reached rectangles, having passed through rows of generic quadrilaterals, trapezia and parallelograms, he asked, *what properties do they have?* Katja replied, *all the angles are right angles*. Janne repeated Katja's words before asking, *how many angles have to be right angles in order to get such a figure?* Having received several different responses Janne invited his students to find out by drawing a rectangle. Eventually, after initiating a vote on the matter, he concluded that three right-angles would be sufficient. Next, having discussed other rectangular properties like equal and parallel sides, he uncovered a row of squares. A question revealed that *all the sides have the same length*, to which he commented, *yes, added to all the things we had earlier*.

In the above, Janne appeared conceptually focused on an increasingly refined categorisation of quadrilaterals. However, his exploitation of the popular vote and decision not to discuss the sufficiency of three left much for students to infer. Thus, although Janne presented a genuine problem for students to solve, a rare occurrence in case study lessons, the management of its solution alluded to an implicitly-derived concept. With respect to the categorisation of quadrilaterals, Janne's linking of the current level of quadrilateral to the previous, despite a superficial explicitness, was always implicitly managed. For example, when discussing the square, and despite making explicit the common length of all sides, Janne made no comment with respect to other properties other than to comment, *added to all the things we had earlier*.

Katja and Geometry

During her second transcribed lesson, having invited her class to construct the mid-normal to a segment of their choice, Katja engaged her class in a discussion as to why such constructions yield a perpendicular to the segment. It went:

Katja:	How do we know that this is a right-angle? What is the reason behind our knowing? (Tracing the line of the mid-normal with her finger) What did we draw here? Simo?
Simo:	A mid-normal.
Katja:	A mid-normal. What kinds of properties does a mid-normal have? Leena?
Leena:	It is perpendicular to the segment at the midpoint.
Katja:	Yes. It is perpendicular to the segment. What other property? Can you tell it us again?
Leena:	Which one? Do you mean the one that it goes through the midpoint?
Katja:	Yes. So, it goes through...?
Leena:	The midpoint.
Katja:	Through the midpoint of this segment. Yes. We could get more properties to this segment, which is... perpendicular to the base and goes through the midpoint of the base.

Unlike the previous episodes, Katja's efforts seemed focused not only on revising the properties of a mid-normal but also engaging her students in mathematical reasoning. In this respect her questioning of Leena elicited the desired properties, which Katja repeated, wrote and students copied. Interestingly, despite its comprising more explicit conceptually-focused activity than almost any other observed episode, Katja neither sought nor offered any clarification as to the meaning of any of the terms used. Thus, even in such an explicit exchange, elements were left for implicit students to infer.

Procedural Knowledge

The four Finnish case study teachers placed substantial emphasis on their students' procedural knowledge, which was also frequently developed by means of exposition, questioning or reflections on tasks.

Antti and Linear Equations

Early in his first lesson on equations Antti rehearsed answers to test questions, intended to assess students' calculator competence, posed the previous lesson. One example, written here as $\sqrt{}(7+\sqrt{}3.\sqrt{}12)$, was managed as follows:

Antti:	You can use the product rule here, 3 times 12 is 36 and the square root of 36 is 6... So, 7 plus 6 is 13 and the square root of 13 is 3.61

During his introduction to the main part of the second lesson, Antti wrote on the board x/8 + 1 = 4 and said,

Antti:	Tell me, what x should be? We have an equation. There are two expressions that are denoted as being of equal magnitude. What should x be? (He waits) Mika?
Mika:	Twenty-four.
Antti:	Twenty-four. Let's try it. Twenty-four divided by eight is three. And add one to it. Yes.

During the test answer exchange Antti sought no input from his students. Despite the reference to the product rule, nothing was said about the rule itself or its application to the particular context. In general, it appeared that Antti's intention was on alerting students to the procedures they should employ. With regard to the second example, having reminded students of the definition introduced the previous lesson, Antti accepted Mika's response, but did not ask how the solution was obtained, before presenting a checking procedure. By way of interpretation, Antti's approach to the test question, in its alerting students to the product rule, could have been construed as an explicit instruction. In similar vein, in responding to Mika's equation answer, while electing not to explore Mika's solution process, Antti presented a checking procedure. However, in both cases, procedures were introduced with no explanatory narrative; Antti neither sought nor offered any clarification of any operation, leaving students to infer both the meaning and the manner of their implementation.

Sami and Percentages

Midway through his first lesson on percentages, several text-based questions were solved individually before students were invited to share responses. Aku, having been invited to share his solution to the first, wrote, over three lines, 100% - (15% + 30%) = 100% - 45% = 55%, V: 55%. Aku's contribution went undiscussed. In similar vein, Salla wrote up the solution to the second. It took her a minute and a half while the class looked on in silence. She wrote, over several lines, 100% - (20%+12%+25%) = 100% - 57% = 43%, V: 43%. This process was repeated for all the questions in the exercise with no commentary from teacher or student.

During his second lesson on percentages Sami asked his students how they would write one sixth as a decimal. The following conversation transpired:

Sami:	Well, who can start the division? Sanni?
Sanni:	Six goes into one zero times.
Sami:	Then we put the point there (he inserts the decimal point into the calculation). Pirkka, you continue.
Pirkka:	For the sake of the point we have to add zero. Six goes into ten once (Sami calculates the remainder and writes the four into the calculation and brings down the zero to create forty)
Sami:	Then, Markus continues.
Markus:	Six goes into forty six times. Again the remainder is four (Sami adds six to the growing number on top and adds in the remainder four and so on).
Sami:	I asked for an accuracy of one hundredth. What is the answer? Aku?
Aku:	0,17

One construal of the first exchange is that the repeated pattern of correct solutions provided a model for those students who were, as yet, unsure as to the procedure they were expected to learn. The fact that Sami neither offered nor sought any additional comment supports this conjecture. It is also interesting to note that all the questions in this episode were word problems but no reference was made to such contexts during the reporting of their solutions. Throughout the second exchange Sami's emphasis appeared to be on a revision of a particular procedure. Even those students who contributed did so in ways that reinforced the mechanical nature of the task; at no point did Sami offer or invite either commentary or justification. Thus, procedures, which incorporated veneers of explicitness, offered only implicit opportunities for those who did not follow to learn.

Janne and Polygons

During Janne's five lessons on polygons, few episodes were observed in which the primary focus was on procedures. However, in the following can be seen two similar but typical examples. During his first transcribed lesson he invited his students to determine how many triangles could be found in a particular figure. After two or three minutes of independent seatwork Janne, having observed some disagreement as to the answer, asserted there were eight before asking, *how can we check if there are fewer or more than eight triangles?* He received little response and so asserted that naming them all would suffice. However, he appended his instruction with, *let's not check and just move on*. Later students spent some time on a similar problem before Janne asserted that there were seven triangles. In this case

Janne suggested that they should *not check them all. It would take too long. By naming them we could get them all*.

In both episodes Janne asserted an answer and alluded to the procedure by which students should be able to derive it. However, in both cases, he also asserted that implementing the procedure was a waste of time. Thus, students were left to infer what they could with regard to their being confident in the answer they obtained. In both cases, Janne offered procedures, derived implicitly, focused on the systematic identification of triangles that, for reasons unknown, he chose not to exploit in any observed lesson.

Katja and Geometry

As with Janne, much of Katja's teaching was focused on the development of her students' conceptual rather than procedural knowledge. However, the following highlights well how her few procedurally focused episodes were managed. During her second transcribed lesson, having spent the first lesson constructing and discussing the properties of twin circles, Katja invited her class to draw an arbitrary segment AB and then construct a mid-normal. Katja, having constructed an arbitrary segment on the board quietly constructed her arcs with no comment. Unbidden, students did the same, some seemingly independently and others taking their lead from Katja's actions. On completion of her arcs Katja told her students to mark the upper intersection of the two arcs with a letter C. The episode continued:

Katja	Connect points A and C, and B and C... What do you get? Elisa?
Elisa	A triangle.
Katja	Yes, you get a triangle... What can you say about that triangle? What can you say about the lengths of triangle ABC's sides? Ilja?
Ilja	They have the same length.
Katja	All of them? Have you drawn all the sides of equal length? Simo?
Simo	A to C and B to C are the same. A to B is not.
Katja	OK. So AC's length is equal in magnitude to BC's length. Why? (Katja writes, $\lvert AC \rvert = \lvert BC \rvert$) Leena?
Leena	The distance to the centres is the same.
Katja	So, how has it been drawn? How did you get the point C? I don't want go on about it, but I want to have clear reasons. So I am still asking why? How did we get that point? Elisa?
Elisa	It is on the mid-normal
Katja	Yes, the mid-normal. How was it drawn?
Katja	Lisa-Maria?
Lisa-Maria	By drawing twin circles.
Katja	Yes, twin circles. And this proves that they have the same length, because they have the same radius.

During the episode, which was structured by six questions eliciting responses from five students, Katja's apparent objective was the procedure for constructing a mid-normal or perpendicular bisector to a segment and why it yielded the result it did. Indeed, her comment, *I don't want go on about it, but I want to have clear reasons*, seems to confirm that objective.

However, the means by which Katja managed her students' responses alludes to something different. For example, Ilja's and Leena's contributions were received in ways that left observers uncertain as to whether they were acceptable. Even those responses that received a *yes* were followed by a question intended to refocus thinking in some way. Thus, despite an apparent explicitness, the episode comprised several implicit elements, not least of which being that no comment was offered with respect to justifying why the invocation of twin-circles *proves that they have the same length, because they have the same radius*. Furthermore, although the initial task could have been presented as an opportunity for students to engage with a non-routine problem, Katja's modelling of the procedure offered students only limited opportunity to solve the problem for themselves. Katja's behaviour appeared to reflect an implicit differentiation according to students' independence.

DISCUSSION

This study was undertaken in order to understand how Finnish teachers of mathematics facilitate their students' learning. It exploited a constant comparison analysis that precluded predetermined categorisations of mathematics didactics and facilitated an emergent understanding of culturally located patterns of behaviour. However, as the analysis proceeded, the question posed by Clarke and Xu (2008, p. 965) became increasingly pertinent; "who is responsible for the public generation of mathematical knowledge in the classroom and how is this responsibility distributed between the teacher and the students?"

The following considers this question. First, however, the analysis indicated that all four teachers emphasised the development of their students' conceptual knowledge and procedural knowledge. In this regard there seemed little evidence of culturally-informed uniqueness. This is not surprising as recent studies have shown that when broad and inclusive categories of analysis have been exploited, mathematics teachers, irrespective of their location, appear to act similarly (Andrews, 2011; LeTendre et al., 2001; Van de Grift, 2007). Indeed, with respect to teachers' development of their students' conceptual knowledge and procedural knowledge, both of which could be described as broad variables, the data showed case study teachers behaving in similar ways to those of Flanders, England, Hungary and Spain (Andrews, 2009). Similarly, there was little surprise in the ways in which these outcomes were realised, with all four teachers employing exposition, various forms of whole class discussion and whole-class reflections on generative tasks. However, in this regard, it is interesting to note that Finnish teachers appear to continue, as they have for more than fifty years, to talk for the majority of the time, with pupils giving generally short responses (Carlgren et al., 2006). Thus, at very general and possibly superficial levels, not only do Finnish teachers appear to privilege outcomes similar to those of their colleagues elsewhere but they behave in ways similar to those who have gone before them.

Of course, case study research seeks to go beyond the identification and reporting of similarities and attempts to identify those characteristic and culturally located practices typical of the system under scrutiny. In this respect, the analyses indicate that few teacher-initiated public exchanges did not involve one or more act of implicit teaching and it is this sense of the implicit that seems to characterise the unique nature of Finnish mathematics teaching. That is, irrespective of the focus of attention, whether students' conceptual knowledge or procedural knowledge, teachers' actions and utterances were consistently implicit in their facilitation of student learning. A particular manifestation is evidenced in teachers' management of publicly-posed questions. Even when an offering was accepted as correct, typically indicated by a *yes* on the part of the teacher, observers of the exchange were rarely offered insight into why the response had been accepted. Similarly, when an offering was deemed incorrect or irrelevant, teachers tended to ignore what was said. For example, during his first lesson on percentages, Sami asked students to give examples of one hundredth. For reasons known only to her, Lisa suggested *the circumference of the earth compared with the circumference of the sun*, a response that Sami ignored before turning to another student for his response, which turned out to be related to the proportion of fat in foods. In other words, the data suggest that case study teachers appeared to play no part in alerting students, whether active or passive in the exchange, to either the strengths or the weaknesses of students' offerings. In similar vein, teacher exposition was managed with few explicit concessions to student understanding. Whether it was Janne's management of the number of triangles problems, Sami's dealings with the percentages word problems or Antti's dealing with the answers to the earlier test, students were offered procedures with no explicit justification as to the warrant for either the procedure itself or the manner of its implementation. Indeed, an important component of this emergent sense of implicit didactics, as noted repeatedly above, lay in the frequently observed finding that case study teachers neither sought nor offered clarification with respect to public utterances, whether teacher or student.

However, the emergent sense of implicit didactics was complemented by three issues highlighted by the analyses. The first was that students were encouraged in both implicit and explicit ways, to make extensive notes. Implicitly, teachers wrote extensive notes on the board, as in Antti's definitions and categorisations of equations or Katja's construction of the mid-normal to a segment. In such circumstances, students were observed to spend time copying and annotating what teachers had written. Explicitly, though these were rarer, there were several occasions when teachers commented or instructed in ways that implied an instruction to make notes. For example, when asked a procedural question in Katja's second lesson on geometry, Sauli commented, *I don't know how to do it.* In response, Katja commented that, *that is why we write it down; so that it would not stay unclear to anyone*. More explicitly, Sami indicated to his students, during the first lesson on percentages, that they should *draw a picture of it in your notebook*. Interestingly, three of the four case study teachers always wrote in capitals, which not only seemed to provide more legible text for copying but also slowed the writer in ways that facilitated student note-taking.

The second complementary feature was that case study teachers regularly appeared to exploit those students perceived as likely to make appropriately meaningful contributions. For example, all four teachers invited high proportions of responses from a small number of confident and competent students, as in Sami's frequent exploitation of Aku or the fact that Janne asked Taneli, a confident boy in his class, an average of nine questions per lesson. On one occasion, lasting more than five minutes, Janne's class observed passively as Janne and Taneli conducted a conversation as though in private but clearly intended for public consumption. Significantly, such practices frequently resulted in students, particularly those like Taneli, spending long periods of their lessons waiting, having completed a task, for their peers to catch up. This latter characteristic of Finnish lessons was highlighted by Gameran, (2008), a journalist working for the Wall Street Journal, when she interviewed Fanny, a straight A student. Fanny admitted that *she sometimes doodles in her journal while waiting for others to catch up*.

Thirdly, there were several occasions during the four sequences where teachers alluded to the involvement of parents in their students' work. For example, Antti, having gone through the test results that started his first lesson, instructed his students to *return your paper tomorrow with a signature showing your parents know at home how the test went*. A more telling exchange emerged during Janne's first transcribed lesson. He had instructed his students to mark on their paper and connect three points to construct an arbitrary triangle. Joona had placed his three points in a straight line with the consequence that he was unable to identify any shape. This following ensued:

Janne:	This is a task for you. Tell me at 10 o'clock tomorrow what sort of figure you have.
Joona:	You should know because you're the teacher.
Janne:	No, I want you to find out for me. Tomorrow at 10 o'clock you tell me what kind of a figure you made. You have some kind of a figure there, a geometrical figure. Find out what it is.
Joona:	I can't make it out.
Janne:	Then you'll just have to think about it at home with mum and dad.

In this episode can be seen evidence of two distinctive elements of case study teachers' didactic tradition. Firstly, in informing Joona that he would have to solve the problem for himself, Janne further confirmed the extent to which Finnish mathematics teaching draws on implicit didactics. Secondly, in telling Joona to discuss the problem with his parents, Janne was highlighting the explicit role that Finnish parents play in the education of their children. This latter observation accords with both external and internal analyses of Finnish classrooms. In its recent review of Finnish mathematics education, the English education inspectorate highlighted the multi-faceted role of parents in the education of their children (Office for Standards in Education 2010). From the perspective of internal commentators, Välijärvi et al. (2002, p. 46) concluded that Finnish educational success draws on a

> network of interrelated factors, in which students' own areas of interest and leisure activities, the learning opportunities provided by schools, parental support and involvement as well as the social and cultural context of learning and of the entire education system combine with each other.

With regard to the particularities of mathematics, Hannula et al. (2007), writing of the impact of the home on teacher education students' learning of school mathematics, found that few students had not received help with their mathematics at home and that the family not only supported their learning but also provided positive attitudes towards the subject. Thus, the role of parents in the education of their children appears a long established characteristic of

Finnish schooling that is well understood by all participants in accordance with research from around the world showing that “parents who are involved in their children's education contribute not only to higher academic achievement, but also to positive behaviours and emotional development” (Cai 2003, p. 87).

In returning to Clarke’s and Xu’s (2008) question, one conclusion is that the responsibility for the public generation of mathematical learning lies with teachers and students, with both having clearly defined, if not publicly articulated, and distinctive roles. However, these roles offer only a partial picture with, it seems, parents playing complementary roles facilitating students’ making sense of the plethora of tasks and activities on which their teachers invite them to work. Thus, in attempting to understand the ways in which mathematics teaching plays out in Finnish classrooms, researchers need to understand the role of participants outside those classrooms in the generation of mathematical knowledge.

CONCLUSIONS

In sum, the Finnish mathematics education tradition, and its hitherto unacknowledged drawing on implicit didactics, seems to exploit a partnership between teachers, students and parents that reflective of a characteristic collective mentality of the Finnish people (Simola, 2005). A particular manifestation of this collective mentality was seen in the ways that the class, as a unit, appeared to learn together, with none of the four teachers ever being seen to differentiate the tasks they presented. Indeed, as Välijärvi (2004, p. 47) observed, an effective “school works as a community, and the results depend on its ability to employ the students' individual and special skills to benefit the common good”. In this manner, case study teachers’ reliance on the confident student and parental collaboration highlight this systemic manifestation of the collective in ways that may go further towards explaining Finnish PISA success than an analysis of classrooms on their own would ever do.

That said, the notion of the collective seems to go beyond merely describing the prosaics of mathematics teaching and learning. Simola (2005, p. 458) has commented that “there is something archaic, something authoritarian, possibly even something eastern, in the Finnish culture and mentality” and, in some ways, this accords with Välijärvi et al.'s (2002) attribution of Finnish literacy success to positive student attitudes, an engagement and interest in reading and, in particular, the tradition of reading at home during the long winter evenings. Indeed, in this latter characteristic of Finnish culture may be found a more powerful explanation of not only the Finnish mindset but also PISA success. According to Linnakylä (2002), for more than four hundred years, reading competence was a prerequisite for receiving Holy Communion and, therefore, permission to marry. Failure in the public examination, or kinkerit, meant a denial of such permission with the consequence that Finns have, for centuries, been raised in a culture of high expectations with respect to learning. Such a tradition would help explain how the mathematics didactic tradition described above came into being and continues to dominate classroom discourse. Thus, as Simola (2005, p. 465) notes,

> it is still possible to teach in the traditional way in Finland because teachers believe in their traditional role and pupils accept their traditional position. Teachers' beliefs are

> supported by social trust and their professional academic status, while pupils' approval is supported by the authoritarian culture and mentality of obedience. The Finnish 'secret' of top-ranking may therefore be seen as the curious contingency of traditional and post-traditional tendencies in the context of the modern welfare state and its comprehensive schooling.

The analysis presented in this paper confirms not only that Finnish mathematics teaching remains much as it was but also that traditional, in the Finnish sense, means something unique in the mathematics education literature. Traditional Finnish didactics seem to fall outside the descriptive frameworks used to describe mathematics teaching and learning in other countries. Indeed, the evidence of this paper cautions those looking to high achieving countries for solutions to domestic educational problems not to assume that educational achievement and didactic excellence are synonymous. As our case study teachers have shown, understanding the role of culture in the construction of student achievement is crucial. Finally, this paper reminds us that when comparing or analysing classrooms from a system other than one’s own, broad categorisations of variables, as in the study of LeTendre et al (2001), are unlikely to be helpful in identifying those practices likely to discriminate one system’s didactic traditions from another’s. Researchers need to find ways of burrowing beneath the layers of superficially similar practice to expose those characteristic behaviours typically hidden from view, especially from the insider.

REFERENCES

Adams, R. (2003). Response to ‘Cautions on OECD’s recent educational survey (PISA)’. *Oxford Review of Education*, 29 (3), 377-389.

Andrews, P. (2009). Comparative studies of mathematics teachers’ observable learning objectives: validating low inference codes. *Educational Studies in Mathematics*, 71 (2), 97-122.

Andrews, P. (2011). *The teaching of linear equations: Comparing effective teachers from three high achieving European countries.* Paper presented to Working Group 11, Comparative studies in Mathematics Education, at the Seventh Congress of the European Society for Research in Mathematics Education (CERME 7): Rzeszów, Poland.

Cai, J. (2003). Investigating parental roles in students' learning of mathematics from a cross-national perspective. *Mathematics Education Research Journal*, 15 (2), 87-106.

Carlgren, I., Klette, K., Mýrdal, S., Schnack, K. & Simola, H. (2006). Changes in Nordic teaching practices: From individualised teaching to the teaching of individuals. *Scandinavian Journal of Educational Research,* 50 (3), 301–326.

Clarke, D. (2006). The LPS research design. In D. Clarke, C. Keitel & Y. Shimizu (Eds.), *Mathematics classrooms in twelve countries: The insider's perspective*. Rotterdam: Sense Publishers.

Clarke, D. & Xu, L. H. (2008). Distinguishing between mathematics classrooms in Australia, China, Japan, Korea and the USA through the lens of the distribution of responsibility for knowledge generation: public oral interactivity and mathematical orality. *ZDM: The International Journal on Mathematics Education,* 40 (6), 963–972.

Corbin, J. & Strauss, A. (1990). Grounded theory research: Procedures, canons, and evaluative criteria. *Qualitative Sociology* 13 (1), 3-21.

Crace, J. (2003). Heaven and Helsinki. *The Guardian*, September 16.

De Meyer, I. (2008). *Science competencies for the future in Flanders: The first results from PISA 2006*. Gent: University of Gent and the Ministry of Education of the Flemish Community of Belgium.

De Meyer, I., De Vos, H. & Van de Poele, L. (2002). *Worldwide learning at age 15: First results from PISA 2000*. Gent: University of Gent and the Ministry of Education of the Flemish Community of Belgium.

De Meyer, I., Pauly, J. & Van de Poele, L. (2005). *Learning for tomorrow's problems: First results from PISA 2003*. Gent: University of Gent and the Ministry of Education of the Flemish Community of Belgium.

De Meyer, I. & Warlop, N. (2010). *Leesvaardigheid van 15-jarigen in Vlaanderen: De eerste resultaten van PISA 2009*. Gent: Universiteit Gent and Departement Onderwijs and Vorming.

Gameran, E. (2008). What makes Finnish kids so smart? *The Wall Street Journal Online*, February 29, 2008.

Hannula, M., Kaasila, R., Pehkonen, E. & Laine, A. (2007). Elementary education students' memories of mathematics in family context. In J. Woo, H. Lew, K. Park & D. Seo (Eds.) *Proceedings of the 31st Conference of the International Group for the Psychology of Mathematics Education*, Seoul.

Hufton, N. & Elliott, J. (2000). Motivation to learn: the pedagogical nexus in the Russian school: some implications for transnational research and policy borrowing. *Educational Studies*, 26 (1), 115-136.

Kivirauma, J, & Kari, R. (2007). Excellence through special education? Lessons from the Finnish school reform. *International Review of Education,* 53 (3), 283-302.

Knipping, C. (2003). Learning from comparing: A review and reflection on qualitative oriented comparisons of teaching and learning mathematics in different countries. *Zentralblatt fur Didaktik der Mathematik,* 35 (6), 282-293.

LeTendre, G, Baker, D., Akiba, M., Goesling, B. & Wiseman, A. (2001). Teachers' work: Institutional isomorphism and cultural variation in the U.S. Germany, and Japan. *Educational Researcher,* 30 (6), 3-15.

Liamputtong, P. (2010). *Performing qualitative cross-cultural research*. Cambridge: Cambridge University Press.

Linnakylä, P. (2002). Reading in Finland. In C. Papanastasiou & V. Froese (Eds.), *Reading literacy in 14 Countries*. Lefkosia: University of Cyprus Press.

Malaty, G. (2007). What are the reasons behind the success of Finland in PISA? *International Journal for Mathematics Teaching and Learning*. http://www.cimt.plymouth.ac.uk/journal/

Norris, N, Asplund, R., MacDonald, B., Schostak, J. & and Zamorski, B. (1996). *An independent evaluation of comprehensive curriculum reform in Finland*. Helsinki: National Board of Education.

Office for Standards in Education (2010). *Finnish pupils' success in mathematics: Factors that contribute to Finnish pupils' success in mathematics*. Manchester: The Office for Standards in Education, Children's Services and Skills.

Organisation for Economic Co-operation and Development (2001). *Knowledge and skills for life: first results from the OECD programme for international student assessment (PISA) 2000*. Paris: OECD.

Organisation for Economic Co-operation and Development (2004). *Learning for tomorrow's world: first results from PISA 2003*. Paris: OECD.

Organisation for Economic Co-operation and Development (2007). *PISA 2006 Science competencies for tomorrow's world*. Paris: OECD.

Organisation for Economic Co-operation and Development (2010). *PISA 2009 Results: What students know and can do – student performance in reading, mathematics and science*. Paris: OECD.

Pehkonen, E., Ahtee, M. & Lavonen, J. (Eds.) (2007). *How Finns learn mathematics and science*. Rotterdam: Sense Publishers.

Simola, H. (2005). The Finnish miracle of PISA: historical and sociological remarks on teaching and teacher education. *Comparative Education*, 41 (4), 455-470.

Tillman, L. (2006). Researching and writing from an African-American perspective: reflective notes on three research studies. *International Journal of Qualitative Studies in Education*, 19 (3), 265 - 287.

Tuovinen, J. E. (2008). Learning the craft of teaching and learning from world's best practice. The Case of Finland. In D. M. McInerney & A. D. Liem (Eds.), *Teaching and Learning: International Best Practice* (pp. 51- 77). Charlotte: Information Age Publishing.

Välijärvi, J. (2004). The system and how does it work: Some curricular and pedagogical characteristics of the Finnish comprehensive school. *Education Journal,* 32 (1), 31-55.

Välijärvi, J., Linnakylä, P., Kupari, P., Reinikainen, P. & Arffman, I. (2002). *The Finnish success in PISA - and some reasons behind it*. Jyväskylä, Institute for Educational Research, University of Jyväskylä.

Van de Grift, W. (2007). Quality of teaching in four European countries: a review of the literature and application of an assessment instrument. *Educational Research,* 49 (2), 127-152.

TELLING MATTERS IN MATHEMATICS TEACHING AND LEARNING

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Despite the current focus on the active construction of mathematical knowledge by learners, many teachers find it difficult to resist 'telling' in mathematics lessons. This telling is linked to an Initiation-Response-Evaluation (IRE) structure that is prevalent in mathematics both at primary and second-levels of education. In this paper I examine some of the reasons that teachers feel constrained to 'tell'. I also explore teacher follow-up moves other than 'evaluation' that, in the context of whole-class discussion, are commensurate with inquiry-based approaches to teaching mathematics. There is little doubt that the orchestration of such discussion makes particular demands on teachers in terms of the competing demands of 'learner' and 'subject' agency. However, the mathematical power that can thereby be devolved to students offers teachers cogent reason to problematize 'telling' practices.

INTRODUCTION

A view of the learner as an active constructor of knowledge has led to greater focus internationally on processes of problem-solving, reasoning and communication in mathematics teaching and learning. These emphases also permeate mathematics curricula in Ireland. In the primary mathematics curriculum, for example, it is stated that

> [e]xperimentation, together with discussion among peers and between the teacher and the child, may lead to general agreement or to the re-evaluation of ideas and mathematical relationships … The importance of providing the child with structured opportunities to engage in exploratory activity in the context of mathematics cannot be overemphasized. The teacher has a crucial role to play in guiding the child to construct meaning, to develop mathematical strategies for solving problems, and to develop self-motivation in mathematical activities. (DES/NCCA, 1999, p. 5)

Similar teaching and learning approaches are advocated in the revised Junior Cycle (JC) and Senior Cycle (SC) mathematics syllabi. Furthermore, mathematics is regarded as a subject that fosters the development of the key skills – information processing, critical and creative thinking, communicating, working with others and being personally effective – which are identified as central to teaching and learning at senior cycle (NCCA, 2010).

Despite these foci, the move from traditional, transmissive approaches to inquiry based approaches in the teaching of mathematics presents a considerable challenge both to pre- and in-service teachers at all educational levels. The desire to 'tell' is particularly difficult to uproot. In this paper I examine some of the reasons that teachers are resistant to 'not telling'. I also explore the ways in which teacher-led discussions can facilitate and nurture the construction of knowledge by learners with particular reference to 'talk moves'. Finally I argue that the mathematical power that is conferred on learners by the adoption of a discursive approach to the teaching of mathematics might provide teachers with the motivation to change existing 'telling' practices.

BACKGROUND

In the course of my work as lecturer in mathematics education, pre-service teachers, while appearing positively disposed towards an inquiry-based approach to their own learning of mathematics, often resist the notion of teaching in this way. Among the challenges that a not insignificant number of them pose to me are:

> *Why not just tell the pupils?*
>
> *Why not give them the formula (or procedure) and later they will figure out the 'why'?.*
>
> *If pupils learn a procedure by rote, they will be able to progress to the next topic.*

Or even,

> *This approach is all well and good for the bright pupils but not for those who are 'weak' (sic) at mathematics.*

This desire on the part of the pre-service teachers to have students well versed in procedures is not without its merits. Although the primacy of conceptual knowledge is well established, it has a deeply embedded relationship with procedural knowledge (Gray and Tall, 1994; Rittle-Johnson and Wagner Alibali, 1999; Skemp, 1978). And the suggestion that all students should have some feeling of success is doubtless one that would be advocated by any educationalist. However, underlying these challenges are more worrying attitudes, particularly about the nature of mathematics, equity and student agency. I tend to deal with such queries by asking my challengers to detail the reasons they might adopt a traditional approach to which the common reply is that 'it works'. It is difficult to know how 'it works' since there is overwhelming evidence to the contrary. In the two national assessments of achievement in mathematics that have taken place since the introduction of the PMC in 1999, there is little evidence of any improvement in the capacity of our primary pupils to engage in higher-order mathematical thinking, that is, they do well in the area of recall but their performance in 'applying and problem-solving' remains a cause of concern (Eivers, et al., 2010; Shiel, Surgenor, Close, and Millar, 2006). In the 2009 Programme for International Student Assessment (PISA), the performance of Irish 15 year-olds in mathematics was significantly below the OECD average (Shiel, Moran, Cosgrove, and Perkins, 2010). Moreover, Ireland had significantly fewer students at the higher proficiency levels (i.e. capable of solving the most difficult tasks) than the OECD average. There are problems also with uptake of higher-level Leaving Certificate (LC) mathematics. For example, in 2010, 16 per cent of the Irish students who sat the LC mathematics examination took the higher-level paper. In comparison, 64 percent of the corresponding student cohort took the higher-level paper in English and 33 percent sat the higher-level paper in Irish.[1] Performance in the 2011 LC saw little change in this situation. Furthermore, many third-level institutions are putting measures in place to redress the low levels of mathematical proficiency of students, many of whom

[1] The total number of students who sit the Leaving Certificate examination each year is approximately 55,000.

have been successful in LC mathematics at higher level (Faulkner, Hannigan, and Gill, 2009; Ní Fhloinn, 2009). Boaler and Greeno (2000, p. 196) explain this as follows:

> Students who choose mathematics as their main field of study, based upon the idea that the subject is structured, certain, and nonnegotiable, may encounter significant problems as the mathematics they learn at university becomes more advanced.

In the 2009 national assessments of mathematics in which the achievement of 2nd and 6th class pupils was assessed, it was found that textbooks are used on a daily basis by practically all primary pupils (Eivers, et al., 2010). And yet, to cite Conway and Sloane, there is a "mismatch between ambitious problem-solving oriented curriculum goals in mathematics education compared to the procedural approach and content of mathematics textbooks" (Conway and Sloan, 2005, p. 57). The situation at post-primary level is worsened because of exam pressures. In an intensive video-study of twenty mathematics lessons and six English lessons in ten different secondary schools, it was observed that

> … Mathematics was generally presented as a fixed body of knowledge, separate from other subjects. Little time or attention was devoted to the problem-solving nature of mathematics, to the application of mathematics in the world, or to alternative methods of solving mathematical problems, other than those prescribed by the text or the teacher. Learning for the examination was the central task. (Lyons, Lynch, Close, Sheerin, and Boland, 2003, p. 147)

The persistence of this approach has been a subject at this and indeed previous MEI conferences. Perhaps pre-service teachers feel that 'it works' because, as a cohort of students who are among the higher achieving students in the LC examination, it has worked for them. Yet the severe deficits in the mathematics knowledge of prospective and in-service primary and post-primary teachers in Ireland have received considerable attention (Conway, et al., 2011; Corcoran, 2009; Delaney, 2010; Hourigan and O'Donoghue, 2009; Leavy and O'Loughlin, 2006; Ní Ríordáin and Hannigan, 2007).

So, what is it that 'works'? Smith (1996) discusses this from the perspective of teachers' self-efficacy. Describing personal teaching efficacy as "an individual teacher's own sense of his or her ability to take effective action in teaching" (p. 389), he maintains that

> [t]elling, irrespective of its pedagogical strengths and weaknesses, provides a clear model for teachers of mathematics to develop a sense of efficacy. Though good telling cannot guarantee that students will learn, it narrows the scope of the content to manageable proportions, clearly defines what the central acts of teaching are and what counts as evidence of student learning, and provides structure for daily classroom life. Teachers can feel efficacious when their students accomplish the reasonable task of remembering facts and computing with the standard procedures. (ibid., p. 393)

He concludes that any attempts to reform the learning of mathematics must be based not only on understanding by teachers of new principles but also in a reconceptualization of their sense of efficacy. He proposes at least four components of teaching mathematics using inquiry based learning approach that are promising sites for building and maintaining efficacy beliefs:

1. Choosing problems
2. Predicting student reasoning
3. Generating and directing discourse
4. Judicious telling

While all of these deserve attention and indeed are not disconnected, I pay particular attention to the third of these – generating and directing discourse – in the sections that follow. I highlight the kind of 'follow-up' moves that teachers can make in the context of whole-class discussion that encourage students to express their own mathematical ideas and build on each others' contributions. Such practices can have ramifications not only for the learning of mathematics but for the student's sense of self as an individual and as a member of society.

CLASSROOM TALK

A common classroom interaction structure is the Initiation – Response – Follow-up (I-R-F) model in which the teacher initiates an exchange, the student then makes a contribution and the teacher then makes a follow-up move (Sinclair and Coulthard, 1975). When the follow-up move is 'evaluative', the pattern is described as I-R-E (Edwards and Westgate, 1994; Mehan, 1979), e.g.,

Teacher:	What is three times two?
Student:	Six.
Teacher:	Good.

It is true that many everyday conversations are characterised by this 'three –move exchange structure' (Nassaji and Wells, 2000), e.g.,

Person A:	What is the time, please?
Person B:	It is almost three-thirty.
Person A:	Thank you.

However, what distinguishes the exchange between teacher and student above from that between persons A and B is the role played by the initiator of the conversation with respect to the information sought (ibid.). In the second exchange the initiator (Person A) and the primary knower (Person B) are different, that is, Person A does not have the information required from Person B. In the classroom exchange, the teacher is both initiator and primary knower. Thus the I-R-E structure tends to reinforce the 'asymmetry of power' between teacher and pupils since the role of the pupils is to 'guess what the teacher is thinking' (Mercer and Dawes, 2008; Pimm, 1994). Variations of the I-R-E cycle have been observed in mathematics lessons. For example, Wood (1994) describes a 'funnel' pattern of interactions as one that involves repeated cycles of I-R-E where students are provided with leading questions in an attempt to guide them to a predetermined solution procedure. She gives the following example (Wood, 1994, p. 153):

T:	What is nine plus seven, Jim?
Jim:	Fourteen.

T:	Okay, seven plus seven equals fourteen (*writes on board*). Eight plus seven is just one more than fourteen which makes …?
Jim:	Fifteen.
T:	And nine is one more than eight. So fifteen plus one more is …?
Jim:	Sixteen.
T:	Good.

The teacher in this case does not actually tell the student that he is incorrect but rather guides him to the correct answer through the use of thinking strategies. However, Wood contends that it is the teacher who is engaged in the consideration of the thinking strategies while the pupil has to do little more than 'add one more' to acquire the solution. A pattern with a similar effect is the 'elicitation pattern' described by Voigt (1995, p. 179) as follows:

- The teacher proposes an ambiguous task and the students offer different answers and solutions that the teacher evaluates preliminarily.
- If the students' contributions are too divergent, the teacher guides the students towards one definite argument, solution and so on.
- The teacher and students reflect and evaluate what has transpired.

Although this interaction pattern has the appearance of being democratic, students have to align themselves with the teacher's way of thinking if they want to participate in the discourse. Their own competencies become hidden and emphasis is placed on the teacher's solution method rather than on mathematical argument (Voigt, 1995). Edwards and Westgate (1994) refer to this as 'cued elicitation' which they say is more about "an inculcation of pupils into a predetermined culture of educated knowledge and practice, than some unfolding development of individual cognition" (p. 144).

The I-R-E structure is particularly pervasive in mathematics lessons. This is so because the nature of classroom interaction is influenced by the teacher's attitude to and beliefs about the subject being taught (Solomon and Black, 2008). If mathematics is regarded as a fixed set of facts and procedures and teaching of mathematics is centred on the transmission of these facts and procedures, the I-R-E structure is apposite. However, the structure runs counter to a constructivist view of mathematics. According to Richards (1991, p. 16), it "does not produce mathematics, or mathematical discussions, but rather a type of 'number talk' that is driven by computation". It also gives little voice to students. For example, Bartolini Bussi (1998, p. 17) says that "even though it seems that a dialogue is taking place it is nothing more than a teacher monologue". O'Connor and Michaels (1996) maintain that, after an evaluation has been made, the I-R-E structure provides no opportunity for students to negotiate the meaning of their contributions.

There are follow-up moves other than 'evaluation' that can nurture greater equity in the teacher-student relationship (Nassaji and Wells, 2000). For example, in the following type of exchange, Erickson (1996) suggests that the student becomes the ‘expert’ and the teacher becomes the ‘novice’ who doesn’t yet know the new information – how the student ‘got’ it:

Teacher:	What is three times two?
Student:	Six.
Teacher:	Can you tell me how you got six?

Nystrand and Gamoran (1991) use the term ‘uptake’ to describe the process of incorporating student responses into subsequent questions. They make a distinction between ‘test questions’ which are designed to assess if the student knows what someone else thinks or has reported and ‘authentic questions’ which signal a teacher’s interest in what the student thinks. When the teacher uses authentic questions s/he opens the floor to what students have to say and this leads to substantive engagement by students. Related to the use of ‘authentic’ questions in mathematics lessons is the ‘focusing’ pattern of interaction identified by Wood (1994). In exchanges of this type, the teacher draws students’ attention to the critical aspects of the problem giving them the responsibility of resolving the situation, e.g.,[2]

T:	What is nine plus seven, Jim?
Jim:	Fourteen.
T:	Okay, why do you think nine plus seven is fourteen?
Jim:	I just guessed.
T:	Is there anything you could do to check that nine plus seven is fourteen?
Jim:	Well, ten plus seven…
Anne:	Oh, it’s sixteen!
Jim:	Yeah, seventeen and one less, sixteen.
T (*to class*):	Do you agree…?

In the focusing pattern, students are encouraged to make themselves understood and to explain their thinking. The teacher’s role is that of drawing attention to that which is commonly known (known number facts such as ‘10 + 7’ in the case above) and then focusing the students’ attention on that which is not yet understood. Since the teacher is endeavouring to get further information and to encourage student reflection, Woods suggests that this pattern of interaction reveals greater equity in the teacher-pupil relationship.

Some attention has been paid recently to the idea of ‘teacher press’. ‘Press’ is described by Staples (2007, p. 174) as a “well-balanced mixture of commitment and pressure” indicating that the teacher values the student’s contribution but expects him/her to further explicate or articulate his/her reasoning. The goal of the teacher in this instance is to make public the thoughts of the student so that all students in the class have access to them. Kazemi and Stipek (2001) make a distinction between ‘high press’ and ‘low press’. The focus in low press situations is on procedures – students are merely expected to give a summary of steps taken to

[2] Examples of exchanges are ‘invented’ unless otherwise stated.

solve a problem. In high press situations the reference point in students' explanations are mathematical concepts – students are expected to include explanation and justification in their presentation of strategies in whole-class settings and students are accountable for participation in a climate characterised by intellectual argument and debate.

A different type of follow-up is that of 'revoicing'. It is described as 'the reporting, repeating, expanding, or reformulating a student's contribution so as to articulate presupposed information, emphasise particular aspects of the explanation, disambiguate terminology, align students with positions in an argument or attribute motivational states to students' (Forman and Larreamendy-Joerns, 1998, p. 106). A straightforward example of how a teacher might revoice in a triadic exchange is as follows:

Teacher: What is three times two?

Tom: Six.

Teacher: So you think that three times two is six.

According to O'Connor and Michaels (1996, p. 71), revoicing or recasting a students contribution in group discussion serves to

- give a greater voice to this contribution;
- transform the contribution so that it can be uttered more succinctly, loudly, completely or in a different register; and
- realign students with respect to each other and to the content of the ideas at hand.

A key part of the reformulation is the use of the discourse 'so' as in 'So you think that' or 'So Tom thinks that'. Coupled with the attribution 'you think', the discourse marker 'so' opens up a new slot in the conversational space, giving the student an opportunity to comment on the correctness of the revoiced utterance. The effect of this 'layering' is to bring students' ideas in contact with each other and thus to effect involvement of all children in the conversation (O'Connor and Michaels, 1996; Rowland, 2000). Sohmer, Michaels, O'Connor, and Resnick (2009) link 'revoicing' and 'press' moves to position-driven discussions, that is, discussions in which students commit to a position and explain why. They suggest that in such a discussion students are afforded the opportunity to listen to one another and to build on one another's ideas and thus have the chance to "*make something that is highly (and rigorously) valued*, that is, the opportunity to develop and marshal evidence for a well-reasoned position about a matter of personal interest" (p. 111, italics in original).

While 'revoicing' and 'press' moves tend to be linked with democratic teaching styles, an interplay between authoritative and dialogical style on the part of the teachers who are adopting an inquiry approach to their teaching is widely reported (Brodie, 2008; Scott, Mortimer, and Aguiar, 2006). This has been attributed to the tension that teachers experience between allowing students to engage in dialogue and inducting them into the discipline of mathematics. In the context of a science, Scott et al. (2006, p. 622) describe it as follows:

> For the teacher in the lesson sequence it is not sufficient to engage students in dialogue about their everyday views of phenomena – there is the additional and central responsibility of introducing the science perspective … students need to engage in the dialogic process of exploring and working on ideas … within the context of the scientific point of view

Indeed, Chazan and Ball (1999) and Lobato, Clarke, and Ellis (2005) take issue with the common exhortation 'not to tell' on two counts. On one hand, it ignores the issue of context and, on the other, it gives little guidance on what a teacher should do. They argue for a more pragmatic approach in which teacher responses relate to the situation at hand with reference to the pupil or group of pupils and to the needs of the mathematics. Lobato et al. argue for a shift in focus from *form* to *function* in debates on 'telling'. According to them, what constitutes telling is context dependent and a teacher's actions and intentions as well as students' interpretations of the action need to be taken into account in making an assessment of the practice. Similar suggestions have been made about the practice of revoicing (Herbel-Eisenmann, Drake, and Cirillo, 2009). However, there is little doubt that the prevalence of the I-R-E structure in Irish classrooms (e.g., Lyons et al., 2003; NicMhuirí, 2011) is reflective of the 'procedures-based' mathematics to which students are exposed and that, in looking at reforms in the teaching of the subject, the quality of classroom interaction needs to be closely scrutinised.

AN EXAMPLE

The aim of my doctoral research was to find out how primary pupils might construct mathematical insight in the context of teacher-led discussion (Dooley, 2010). The research design that I used was the 'classroom design experiment' (Cobb, Gresalfi, and Hodge, 2009). My study involved three consecutive cycles, A, B and C. In Cycle A, I taught 11 mathematics lessons (11 sessions) to a 5th class in a school of middle S.E.S. In Cycle B, I taught five mathematics lessons (10 sessions) to a 5th class in a school designated as disadvantaged and in Cycle C I taught 16 lessons (23 sessions) to a 4th class in a school of middle S.E.S. Data collection and data analysis were interwoven. Retrospective analysis was conducted on micro- (between lessons) and macro- (between and after cycles of research) levels, on the basis of which changes were made to my teaching and theory was generated. While refinements were made to the tasks and tools used, my role as teacher and the agency devolved to pupils became particular foci of attention. Thus I developed an analytic framework that comprised four dimensions:

(i) Mathematical Principles – these are the constructs that pupils could be expected to develop by engagement with the task.

(ii) RBC Epistemic Actions – these actions - 'recognising' 'building-with' and 'constructing' – are the observable actions underpinning the Abstraction-in-Context (AiC) theoretical framework which has been used to describe the construction of new mathematical ideas by individuals and groups of individuals (Schwartz, Dreyfus, and Hershkowitz, 2009).

(iii) Hedges and Pronouns – these are aspects of 'vague language'. Hedges refer to words and phrases such as 'I think', 'maybe' 'possibly' which allow a speaker to make an assertion without full commitment to it. Learners often use pronouns to refer to mathematical entities which they cannot or do not name and, in particular, use the word ‘you’ as a generaliser (Rowland, 2000).

(iv) Teacher Follow-up Moves – these are categories of ‘moves’ made by the teacher in response to pupils’ contributions. They include ‘insert’ in which a teacher adds something to a learner’s contribution; ‘elicit’ which is similar to the funnelling pattern of interaction; ‘press’ in which the pupil is requested to justify and elaborate on his/her input; ‘maintain’ in which the learner’s contribution is maintained in the public realm; and ‘confirm’ in which teacher confirms that s/he has heard the learner correctly (Brodie, 2008).

At MEI 3, I argued that, in mathematics lessons “teachers have to ‘let go’ of rigid views of maths and allow ideas, even those that are incorrect to ‘stumble across each other’” (Dooley, 2009, p. 110). I elaborate on this idea further with particular reference to the fourth dimension of the framework – ‘teacher follow-up moves’ – and deal especially with the affordances of the ‘maintain’ move. The example that I use is taken from a lesson on Gauss that took place in Cycle C.

THE STORY OF GAUSS

The young Gauss astounded his teacher when, as a school pupil, he rapidly calculated the sum of the integers from 1 to 100. He did so by adding 1 to 100, 2 to 99 and so on and thus found fifty 101s. The power of this story, according to Rowland (2001), is that it provides a means of finding the sum of the first 2k integers for any integer 2k (that is, k (2k + 1)).

Mathematical Principles

The Gauss lesson that is the focus of this paper followed a lesson on the ‘Handshakes’ problem.[3] The pupils had developed an explicit formula for this problem, that is, that the number of handshakes could be determined by application of the formula, ½ $n(n-1)$, where n is the number of people. Some pupils were able to verify this formula structurally, that is, with reference to the underlying structure of the problem (Rowland, 1999). Since they had calculated the number of handshakes in the case of 100 people, that is, by multiplying 100 by 99 and halving the solution, I anticipated that they might be able solve Gauss’ problem with relative ease (that is, 4950 + 100). I also thought that they might make use of the formula although I expected that this would pose a challenge since the number of handshakes for n people is ½ $n(n-1)$, the formula for $1 + 2+ 3 + \ldots + (n - 1)$. Pupils of this age could also be expected to use ‘flexible computation methods’ (Greeno, 1991). One method of doing this is to identify ‘compatible pairs’ of numbers summing to 100, i.e.,[4]

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[3] The Handshakes problem is as follows: “If n people are at a party and each shakes hands exactly once with each other person, how many handshakes are there?”. Solution strategies include summing 1 to $(n - 1)$ or application of the formula, ½ $n(n-1)$.

[4] While the omission of ‘0’ and pairing of sums to 101 (that is, 100 + 1, 99 + 2, 98 + 3, …) might be a more obvious solution method, the pairing to 100 was the method that emerged in this lesson.

$$100 + 0, 99 + 1, 98 + 2 \ldots$$

In order to use this method to solve the sum of the first n positive integers, pupils have to observe that if n is even, (a) the number of pairs is $\frac{1}{2} n$ and (b) that $\frac{1}{2} n$ ('50' in the example above) does not have a 'compatible partner'. An interesting pattern emerges when sums of decades are examined. Using the notation $s(m, n) = m + (m + 1) + \ldots + n$,

s(1, 10) = 55; s(11, 20) = 155 (that is, s(1, 10) + 100); s(21, 30) = 255 (that is, s(1, 10) + 200) etc.

Similarly,

s(1, 100) = 5050; s(101, 200) = 15050 (= s(1, 100) + 10000) etc.

One of the cognitive conflicts (Piaget, 1985) that pupils have to overcome in order to use either of these patterns effectively is the 'illusion of linearity' (De Bock, Van Dooren, Janssens, and Verschaffel, 2002), that suggests that $s(1, kn) = ks(1, n)$. For example, pupils might assume that

s(1,100) = 10s(1,10), that is, s(1,100) = 10 (55).

On a more global level, another principle that pupils might be expected to attain is the extension of this example (the sum of 1 – 100) to other classes, such as the sum of the first *n*-hundred integers.

Analysis of Anne's Construction of Insight

Early in the lesson, Anne proposed that the sum of the first 100 integers might be found by doubling the sum of one to five (that is, fifteen) and then multiplying 30 by ten, that is [5],

56	Anne:	If you add one to five, that's fifteen …
57	TD:	Hm, hm.
58	Anne:	… and then fifteen and fifteen is thirty, so then if you multiply that by ten.
59	TD:	Ok, possibly that would get it for you. Fiona?

Fiona then proposed a similar idea, that is, that the answer might be found by adding the numbers from 1 – 50 and doubling that sum. Both assumed that a linear relationship between partial sums applied and I did not correct this assumption. Indeed I gave the impression that the idea might produce a correct solution, (although the word 'possibly' is a linguistic hedge that, according to Rowland (2000), is used as a means of distancing oneself from full commitment to a proposition). Anne's insight about the inefficacy of this strategy occurred a

[5] The following transcript conventions are used: TD: the researcher/teacher (myself); Ch: a child whose name I was unable to identify in recordings; Chn: A group of children speaking in unison; //: encloses utterances overlapping that of next or previous speaker; ... : a hesitation or short pause; [...]: a pause longer than three seconds; ( ): inaudible speech; [ ]: lines omitted from transcript because they are extraneous to the substantive content of the lesson.

short while later and was prompted by an observation made by Alan. The talk turns[6] chosen for analysis are turns 65 – 86 and 91 – 96 since they provide the trace of her constructions.

	Pupil Action	Teacher Move
65 TD: Alan?		Direct
66 Alan: Em, well, I don't think Anne's one is right.	Alan evaluates idea proposed by Anne.	
67 TD: Why not?		Press
68 Alan: Cos ninety-nine plus ninety-eight plus ninety-seven plus ninety-six to ninety would be around over five hundred and when …	Estimation of sum of '90's'	
69 //Ch: Oh!	Exclamation	
70 TD: Ok, so you are thinking that, you think, you disagree with Anne because you are thinking, what Alan is doing now … Alan is thinking ninety - I haven't forgotten you now, Enda, alright, I will be with you in a moment - you are thinking ninety plus ninety-one plus ninety-two plus ninety-three would give you approximately how much?		Maintain Elicit
71 Alan: Em, I don't know.		
72 TD: But it's…		Elicit
73 Alan: But it would probably be over five hundred.	Estimation of sum of 90s	
74 TD: It would be over five hundred, so in that section, if you are thinking about all those numbers there that would give you about, even just adding ninety to a hundred so you are thinking that would give you about five hundred - I will be with you in about one minute alright. Barry?		Maintain Direct
75 Barry: Eh, well, I disagree with Anne as well because eh I counted, I counted up all the numbers up to ten and I got fifty-five.	Further evaluation of Anne's idea	
76 TD: You counted, ok so actually in your head, so you went, so ok so you added, Barry, you went one plus two plus three and so on up to ten (*writes on blackboard*) and you just counted that section there and you said that's about fifty-five. And then Alan was saying that ninety plus ninety-one (*writes on blackboard*) up as far as a hundred would give you. About how much what would that give you if you just added those numbers there, ninety, about how much, you don't have to give the exact answer, about how much would that be if you went ninety, ninety-one, ninety-two, ninety-three added all those numbers there, what would that be about? Barry?		Maintain (Barry) Maintain (Alan) Elicit
77 Barry: A hundred and eighty one.		
78 TD: It would be about a hundred and eighty one, that's if you went ninety		Maintain

plus ninety-one plus ninety-two plus ninety-three …		
79 Barry: No eh…		
80 TD: … plus ninety-four plus ninety-five plus ninety-six plus ninety-seven plus ninety-eight plus ninety-nine plus a hundred? About how much would that be?		Elicit
81 Barry: Eh … a thousand.	Estimation of sum of 90s	
82 TD: It could be yeah, it could probably be a thousand.		Maintain
83 Ch: Ah!		
84 TD: Ok, so that would give you about … Do you agree with that, that that would give you about a thousand?		Maintain
85 Barry: No around nine hundred because there's not really a hundred in the nineties.	Estimation of sum of 90s	
86 TD: In the nineties, so you thinking that that would be around nine hundred. Ok. So this is now what we are thinking now, alright that eh this number here is about nine hundred and Enda what you might do for me if you just take out ... is just write down your idea in your diary for me please. Alright?		Maintain Maintain
[ ][6]		
90 TD: Yes, so it's not just, and as Barry said that if you were going ninety, or Alan made this point, that ninety, ninety-one all the way up to a hundred gives you about nine hundred, who else said about, oh yes and Barry was saying that one up as far as ten gives you fifty-five. And we are thinking all the numbers one plus two plus three plus four plus five plus six plus seven plus eight plus nine plus ten plus eleven plus twelve (speaking quickly) ... counting up, adding up all the numbers from one up to a hundred. Anne?		Maintain (Alan) Maintain (Barry)
91 Anne: I don't … my answer wouldn't work.	Anne re-evaluates her early input.	
92 TD: What were you thinking your answer was?		Elicit
93 Anne: I thought it would be thirty multiplied by a hundred but it wouldn't work.	Anne repeats earlier idea.	
94 TD: Why would it not work?		Press
95 Anne: Em, because you would have to em … cos I did eh one plus two plus three plus four plus five and then em I got fifteen and then I added fifteen and fifteen equals thirty but then it would be more em … because you would have to add six,	Anne rationalises why her earlier idea would not work.	

[6] Turns 87 - 90 involved an observation by Enda that addends of 100 (e.g., 50 +50) would not yield the solution to the 'story'.

seven and that …		
96 TD: Ok, so you are … Eh Colin?		

Teacher Moves

The main uptake move I used in the excerpt above was the 'maintain' move. In turn 70, I revoiced Alan's input in two different ways. First I repeated his idea, "So you are thinking that …" and then I broadcasted his idea to the class in " … what Alan is doing now … Alan is thinking ninety". This is similar to a move I made in other lessons when I wanted attention to be paid by the class to a key idea. In this case I probably judged that Alan's reasoning would serve as a source of cognitive conflict relating to the illusion of linearity. In a further endeavour to keep this idea in the public realm, I asked him on two occasions to give an estimate of the sum of the 90s decade. I did not correct his estimate of "about five hundred" in turn 74 but maintained it, probably because I deemed this estimate to be sufficient to provide the challenge required. I maintained Barry's idea (made in turn 75) in a similar manner and tried to link his reasoning with that of Alan by writing their ideas on the blackboard (see Figure 1). The effect of maintaining both ideas did not open up the discussion to other members of the class as Barry made inputs in turns 77, 79 and 81. The 'elicit' move that I made after 'maintaining' (i.e., "About how much what would that give you if you just added those numbers there?") may have been one of the reasons. Barry's estimated sum of "one-hundred and eighty" was possibly the sum of 90 and 91. However, I misinterpreted it and made an 'elicit' move in order to encourage him to make a more accurate estimate of the sum of the numbers in the 90s decade. Both of these 'elicit' moves represent a return to a more authoritative style on my part. My recollection is that I did so in an effort to provoke in the pupils of this class the cognitive conflict that was required. As was the case with Alan, my decision to maintain his input of 'a thousand' (turn 81) was probably influenced by the fact that this estimate was sufficient to do so. The fact that I did not evaluate this but asked for comment on it (turn 84) may have been the reason that Barry then changed his mind in turn 85. In turn 90, I reviewed the ideas proposed by Barry and Alan and repeated the problem to the class. When Anne changed her mind, I pressed her for a reason and this led to an articulation by her of a new construction (turn 95).

95 Anne: Em, because you would have to em … cos I did eh one plus two plus three plus four plus five and then em I got fifteen and then I added fifteen and fifteen equals thirty but then it would be more em … because you would have to add six, seven and that …

The 'it' in the latter part of this sentence is most likely a referent to the sum of six to ten. Her "it would be more" indicates that she has constructed the idea of the 'non-linearity' by observing that $s(1, 5) \neq s(6, 10)$, thus correcting her earlier misapprehension (turn 58).

Later in the lesson, Anne engaged in similar reasoning to compare $s(1, 100)$ with $s(101, 200)$.

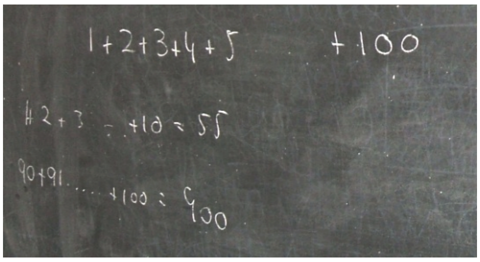

Figure 1: Sample of blackboard work

DISCUSSION

This excerpt is not presented as exemplary teaching – it is open to critique on several grounds, most notably the extent to which I maintained control of the discussion. However, it provides an example of a position driven discourse where Alan and Barry took positions on Anne's idea. Their motive for doing so may be that I left her erroneous idea in the public domain when she first proposed it. On foot of the contributions of Alan and Barry, Anne adopted a different position on her original argument although the justification she tendered was different from that suggested by either. It is not that I relinquished control of the discussion. As mentioned above, my 'elicit' moves may have served to inhibit wider engagement in the discussion. On the other hand I judged it important to engender cognitive conflict on the linearity issue. As argued by Boaler (2003, p. 12) "teaching is a practice that takes place at the intersection of hundreds of variables that play out differently in every moment". Moreover, I equivocated on Enda's endeavours to interject. The main reason for this is that Enda made frequent contributions to class discussions but his thinking was not always accessible to the other pupils. I wanted to bring other pupils' 'storylines' to the fore. He did, however, make an input in turns 87 and 89 regarding the unsuitability of a method proposed by another student earlier in the lesson and I returned to his line of reasoning later in the lesson.

An alternative move that I might have made in response to Anne's conjecture in turns 56 - 58 might have been to evaluate or funnel her response. This may have had the effect of speeding up the lesson. After all, thirty-three[7] talk turns elapsed before Anna re-evaluated her position. I argue that what happened in the interim is more typical of 'real' mathematical activity. According to Battista (1999, p. 447):

> In an appropriate classroom culture of inquiry, students, like scientists, construct, revise, and refine theories to solve and make sense of perceived problems... In problem-centred inquiry-based learning ... the development of viable theories and concepts is often confused, irregular and untidy. Students like scientists make mistakes and struggle to make sense of ideas. Consequently, teachers attempting to implement problem-centred inquiry-based instruction must understand that often, in the meaningful learning promoted by such instruction, grappling with ideas and even becoming confused (within reason) are natural components of students' sense-making.

[7] The thirty-three talk turns are those between turn 59 and 91.

This ‘confused, irregular and untidy’ state runs counter to the neat structure that is inherent in the recitation and evaluation of ‘correct’ procedures. It is risky for teachers and students alike. In her description of her embarkation on an innovative approach to teaching mathematics to her class, Mau (2007) speaks of her apprehension concerning how her students might view her when she adopted a different role, that is, one in which she was more learner than authority in the classroom. She talks in particular about the notion of “getting comfortable with being uncomfortable” (ibid., p. 349). However, while the teacher has to be more ‘learner’ than authority, s/he also has to induct the students into the discipline of mathematics. Boaler (2003) describes the interplay between human and disciplinary agency as a ‘dance of agency’ – a dance in which students and teacher are engaged. Lampert (1990) suggests that the teacher has to work on two teaching agendas simultaneously – one related to the knowledge of mathematics and the other concerning mathematical practices. S/he also has to remain cognisant of the personal and social risks posed to students who either propose or refute an idea in the public domain of the classroom (Asterhan and Schwartz, 2009; Edwards and Westgate, 1994; Lampert, Rittenhouse, and Crumbaugh, 1996). The establishment of classroom norms where ideas are discussed and debated is a gradual process. Pupils in the class where the lesson on Gauss took place were accustomed to mathematical debate. The fact that Anne changed her mind in a public fashion and made further conjectures later in the lesson suggests that she was not personally affronted by the challenges posed by Alan and Barry. Indeed the tone of the discussion was harmonious and non-confrontational. In particular, both Alan and Barry prefaced their criticism of Anne’s idea with ‘Well’, a maxim hedge that can be used in conversation to lessen the threat to one’s ‘positive face’ (that is, desire for approval) while critiquing his/her idea (Rowland, 2000). Asterhan and Schwartz (2009, p. 170) contend that engagement in such discussion not only helps students to construct knowledge collaboratively but also to become “civilized, rational and empathic discussants”. In the excerpt above I suggest that the pupils concerned – and other tacit participants – constructed not only an understanding of the inefficacy of ‘linearity’ for solving the Gauss problem, but also how to reason and argue in a civilized and harmonious manner.

I do not propose that the ‘maintain’ move is a necessary condition of inquiry-based learning. As mentioned above, its effect has to be examined from the perspectives of form and function. However, it seems reasonable to suggest that, in this instance, the act of *telling* Anne that she was incorrect rather than upholding her ‘incorrect’ conjecture would teach her – and the other class members – a different (and, arguably, not so desirable) lesson about the nature of mathematics and social interaction.

On a further note, these pupils engaged in sophisticated levels of algebraic reasoning – described by Franke, Carpenter and Battey (2008, p. 337) as the ability to “generate, use, represent, and justify generalizations about fundamental properties of arithmetic”. In the context of the current foci on improvements in mathematics learning, it is imperative that the bridging framework (NCCA, 2011) which aims to facilitate improved continuity between mathematics in primary and post-primary schools, addresses the capacity of young learners to reason in this manner.

CONCLUDING REMARKS

The crisis that exists in mathematics education has been met by a number of responses, most notably Mathematics Recovery, an initiative for primary schools in DEIS areas and Project Mathematics, a project which currently being rolled out to second-level schools. The recent DES strategy on literacy and numeracy (DES, 2011) seeks to address the issue in a wide-ranging manner. In particular, it acknowledges that children's experiences of mathematics across all levels of education (including Early Childhood Care and Education) are significant and that there are important roles to be played by family and community. All of these moves are welcome but, as argued by Waldron (2011), the idea of *purpose* is a core dynamic in transforming practice. The rich possibilities inherent in harnessing learner agency provide, I suggest, such purpose and offer cogent reason to resist the practice of telling.

REFERENCES

Asterhan, C., and Schwartz, B. (2009). Transformation of robust misconceptions through peer argumentation. In B. Schwartz, T. Dreyfus and R. Hershkowitz (Eds.), *Transformation of knowledge through classroom interaction* (pp. 159 - 172). London and New York: Routledge.

Bartolini Bussi, M. G. (1998). Joint activity in mathematics lessons: A Vygotskian analysis. In F. Seeger, J. Voigt and U. Waschescio (Eds.), *The culture of the mathematics classroom* (pp. 13 - 49). Cambridge: Cambridge University Press.

Battista, M. T. (1999). Fifth graders' enumeration of cubes in 3D arrays: Conceptual progress in an inquiry-based classroom. *Journal for Research in Mathematics Education, 30*(4), 417 - 448.

Boaler, J. (2003). Studying and capturing the complexity of practice - the case of the 'dance of agency'. In N. A. Pateman, B. J. Dougherty and J. T. Zilliox (Eds.), *Proceedings of the 27th Conference of the International Group for the Psychology of Mathematics Education* (Vol. 1, pp. 3 - 16). Honolulu: PME.

Brodie, K. (2008). Towards a language of description for changing pedagogy. In O. Figueras, J. L. Cortina, S. Alatorre, T. Rojano and A. Sepulveda (Eds.), *Proceedings of the Joint Meeting of PME 32 and PME-NA XXX* (Vol. 2, pp. 209 - 216). Mexico: Cinvestav-UMSNH.

Chazan, D., and Ball, D. (1999). Beyond being told not to tell. *For the Learning of Mathematics, 19*(2), 2 -10.

Cobb, P., Gresalfi, M., and Hodge, L. L. (2009). A design research perspective on the identity that students are developing in mathematics classrooms. In B. Schwartz, T. Dreyfus and R. Hershkowitz (Eds.), *Transformation of knowledge through classroom interaction* (pp. 223 - 243). London and New York: Routledge.

Conway, P., Murphy, R., Delargey, M., Hall, K., Kitching, K., Long, F., et al. (2011). *Learning to Teach Study (LETS): developing curricular and cross-curricular competencies in becoming a 'good' secondary teacher*. Cork: School of Education, UCC.

Conway, P., and Sloan, F. (2005). *International trends in post-primary mathematics education: Perspectives on learning, teaching and assessment*. Dublin: NCCA.

Corcoran, D. (2009). Mathematics teaching as a shared enterprise: Learning by 'doing'. In D. Corcoran, T. Dooley, S. Close and R. Ward (Eds.), *Proceedings of the Third National Conference on Research in Mathematics Education* (pp. 34 - 52). Dublin: St. Patrick's College.

De Bock, D., Van Dooren, W., Janssens, D., and Verschaffel, L. (2002). Improper use of linear reasoning: An indepth study of the nature and the irresistibility of secondary school students' errors. *Educational Studies in Mathematics, 50*, 311 - 334.

Delaney, S. (2010). *Knowing what counts: Irish primary teachers' mathematical knowledge for teaching*. Dublin: Colaiste Mhuire, Marino Institute of Education.

DES/NCCA (1999). *Primary curriculum: Mathematics*. Dublin: The Stationery Office.

DES (2011). Literacy and numeracy for learning and life: The National Strategy to improve literacy and numeracy among children and young people 2011-2020. Retrieved from www.education.ie.

Dooley, T. (2009). Insight in primary mathematics: Teacher 'moves' that facilitate the 'stumbling of ideas across each other'. In D. Corcoran, T. Dooley, S. Close and R. Ward (Eds.), *Proceedings of the Third National Conference on Research in Mathematics Education* (pp. 102 - 111). Dublin: St. Patrick's College.

Dooley, T. (2010). *The construction of mathematical insight by pupils in whole-class conversation (Unpublished doctoral dissertation).* University of Cambridge, Cambridge.

Edwards, A. D., and Westgate, D. P. G. (1994). *Investigating classroom talk* (2nd ed.). London and Washington, DC: Falmer Press.

Eivers, E., Close, S., Shiel, G., Millar, D., Clerkin, A., Gilleece, L., et al. (2010). *The 2009 national assessments of mathematics and English reading*. Dublin: Stationery office.

Erickson, F. (1996). Going for the zone: The social and cognitive ecology of the teacher-student interaction in classroom conversations. In D. Hicks (Ed.), *Discourse, learning and schooling* (pp. 29 - 62). Cambridge: Cambridge University Press.

Forman, E. A., and Larreamendy-Joerns, J. (1998). Making explicit the implicit: Classroom explanations and conversational implicatures. *Mind, Culture and Activity, 5*(2), 105 - 113.

Franke, M. L., Carpenter, T. P., and Battey, D. (2008). Content matters: Algebraic reasoning in teacher professional development. In J. J. Kaput, D. W. Carraher and M. L. Blanton (Eds.), *Algebra in the early years* (pp. 333 - 359). New York and London: Lawrence Erlbaum Associates.

Gray, E. M., and Tall, D. O. (1994). Duality, ambiguity and flexibility: A proceptual view of simple arithmetic. *Journal for Research in Mathematics Education, 26*(2), 115 - 141.

Greeno, J. G. (1991). Number sense as situated knowing in a conceptual domain. *Journal for Research in Mathematics Education, 22*(3), 170 - 218.

Herbel-Eisenmann, B., Drake, C., and Cirillo, M. (2009). "Muddying the clear waters": Teachers' take-up of the linguistic idea of revoicing. *Teaching and Teacher Education, 25*(2), 268 - 277.

Hourigan, M., and O'Donoghue, J. (2009). Working towards addressing the mathematics subject matter knowledge needs of prospective teachers. In D. Corcoran, T. Dooley, S. Close and R. Ward (Eds.), *Proceedings of the Third National Conference on Research in Mathematics Education* (pp. 133 - 143). Dublin: St. Patrick's College.

Kazemi, E., and Stipek, D. (2001). Promoting conceptual thinking in four upper-elementary mathematics classrooms. *The Elementary School Journal, 102*(1), 59 - 80.

Lampert, M. (1990). When the problem is not the question and the solution is not the answer. *American Educational Research Journal, 27*, 29 - 63.

Lampert, M., Rittenhouse, P., and Crumbaugh, C. (1996). Agreeing to disagree: Developing sociable mathematical discourse. In D. R. Olson and N. Torrance (Eds.), *Handbook of psychology and education: New models of learning, teaching and school* (pp. 731 - 764). Oxford: Basil Blackwell.

Leavy, A., and O'Loughlin, N. (2006). Pre-service teachers' understanding of the mean: Moving beyond the arithmetic average. *Journal of Mathematics Teacher Education, 9*, 53-90.

Lobato, J., Clarke, D., and Ellis, A. (2005). Initiation and eliciting in teaching: A reformulation of telling. *Journal for Research in Mathematics Education, 36*(2), 101 - 136.

Lyons, M., Lynch, K., Close, S., Sheerin, E., and Boland, P. (2003). *Inside classrooms: The teaching and learning of mathematics in a social context*. Dublin: Institute of Public Administration.

Mau, S. T. (2007). Under the microscope: Looking closely at one's own teaching and learning. In W. G. Martin, M. E. Strutchens and P. C. Elliott (Eds.), *The learning of mathematics: Sixty-ninth yearbook* (pp. 345 - 355). Reston, VA: NCTM.

Mehan, H. (1979). *Learning lessons: Social organization in the classroom*. Cambridge, MA: Harvard University Press.

Mercer, N., and Dawes, L. (2008). The value of exploratory talk. In N. Mercer and S. Hodgkinson (Eds.), *Exploring talk in school* (pp. 55 - 71). London and Los Angeles: Sage.

Nassaji, H., and Wells, G. (2000). What's the use of 'triadic dialogue': An Investigation of teacher-student interaction. *Applied Linguistics, 21*(3), 376 - 406.

NCCA (2010). *Leaving Certificate Mathematics Syllabus: Foundation, Ordinary and Higher Level (for examination in 2012 only).* Dublin: Stationery Office.

NCCA (2011). *Mathematics bridging framework.* Retrieved from http://action.ncca.ie/en/mathematics-bridging-framework

NicMhuirí, S. (2011, September). *Look who's talking now - The role of talk in relation to mathematical thinking and problem solving*. Paper prepared for the MEI 4 Conference, St. Patrick's College, Dublin.

Ní Ríordáin, M., and Hannigan, A. (2007). Assessing the level of suitably qualified teachers teaching mathematics at post-primary education in Ireland. . In S. Close, D. Corcoran and T. Dooley (Eds.), *Proceedings of the Second National Conference on Research in Mathematics Education* (pp. 274 - 295). Dublin: St. Patrick's College.

Nystrand, M., and Gamoran, A. (1991). Student engagement: When recitation becomes conversation. In H. C. Waxman and H. J. Walberg (Eds.), *Effective teaching: Current research* (pp. 257 - 276). Berkeley, CA: McCutchan.

O'Connor, M. C., and Michaels, S. (1996). Shifting participant frameworks: Orchestrating thinking practices in group discussion. In D. Hicks (Ed.), *Discourse, Learning and Schooling* (pp. 63 - 103). Cambridge: Cambridge University Press.

Piaget, J. (1985). *The equilibration of cognitive structures*. Chicago, IL: University of Chicago Press.

Pimm, D. (1994). Spoken mathematical classroom culture: Artifice and artificiality. In S. Lerman (Ed.), *Cultural perspectives on the mathematics classroom* (pp. 133 - 147). Dordrecht and London: Kluwer.

Richards, J. (1991). Mathematical discussions. In E. Von Glasersfeld (Ed.), *Radical constructivism in mathematics education* (pp. 13 - 51). Dordrecht and London: Kluwer.

Rittle-Johnson, B., and Wagner Alibali, M. (1999). Conceptual and procedural knowledge of mathematics: Does one lead to the other? *Journal of Educational Psychology, 91*(1), 175 - 189.

Rowland, T. (1999). 'i' is for induction. *Mathematics Teaching, 167*, 23 - 27

Rowland, T. (2000). *The pragmatics of mathematics education: Vagueness in mathematical discourse*. London: Falmer Press.

Rowland, T. (2001). Generic proofs: Setting a good example. *Mathematics Teaching, 177*, 40 - 43.

Schwartz, B., Dreyfus, T., and Hershkowitz, R. (2009). The nested epistemic actions model for abstraction in context. In B. Schwartz, T. Dreyfus and R. Hershkowitz (Eds.), *Transformation of knowledge through classroom interaction* (pp. 11 - 41). London and New York: Routledge.

Scott, P., Mortimer, E., and Aguiar, O. (2006). The tension between authoritative and dialogic discourse: A fundamental characteristic of meaning making interactions in high school science lessons. *Science Education, 90*(4), 605 - 631.

Shiel, G., Moran, G., Cosgrove, J., and Perkins, R. (2010). *A summary of the performance of students in Ireland on the PISA 2009 test of mathematical literacy and a comparison with performance in 2003*. Dublin: Educational Research Centre.

Shiel, G., Surgenor, P., Close, S., and Millar, D. (2006). *The 2004 national assessment of mathematics achievement*. Dublin: Educational Research Centre.

Sinclair, J. M., and Coulthard, M. (1975). *Towards an analysis of discourse: The English used by teachers and pupils*. London: Oxford University Press.

Skemp, R. (1978). Relational understanding and instrumental understanding. *Arithmetic Teacher, 26*(3), 9-15.

Smith III, J. P. (1996). Efficacy and teaching mathematics by telling: A challenge for reform. *Journal for Research in Mathematics Education, 27*(4), 387 - 402.

Sohmer, R., Michaels, S., O'Connor, M. C., and Resnick, L. (2009). Guided construction of knowledge in the classroom: The troika of talk, tasks and tools. In B. Schwartz, T. Dreyfus and R. Hershkowitz (Eds.), *Transformation of knowledge through classroom interaction* (pp. 105 - 129). London and New York: Routledge.

Solomon, Y., and Black, L. (2008). Talking to learn and learning to talk in the mathematics classroom. In N. Mercer and S. Hodgkinson (Eds.), *Exploring talk in school* (pp. 73 - 90). London and Los Angeles: Sage.

Staples, M. (2007). Supporting whole-class collaborative inquiry in a secondary mathematics classroom. *Cognition and Instruction, 25*(2), 161 - 217.

Voigt, J. (1995). Thematic patterns of interaction and sociomathematical norms. In P. Cobb and H. Bauersfeld (Eds.), *The emergence of mathematical meaning* (pp. 163 - 201). Hillsdale, NJ: Lawrence Erlbaum Associates.

Waldron, F. (2011, July). *Re-imagining initial teacher education: Purpose, passion and possibility*. Keynote address delivered at the ITE Conference, St. Patrick's College, Dublin.

Wood, T. (1994). Patterns of interaction and the culture of mathematics classrooms. In S. Lerman (Ed.), *Cultural Perspectives on the Mathematics Classrooms* (pp. 149 - 168). Dordrecht: Kluwer.

TEACHER AND STUDENT LEARNING THROUGH STANDARDS: INTERNATIONAL COMPARISON

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As global mathematics performance exams now make it clear how students compare to their international peers, each nation is attempting to guide their students' learning of mathematics by communicating expectations and providing structures through content standards and curricula. This paper compares four different international mathematics content standards, describes the instructional roadmaps they present, and discusses how they consider and frame teaching and student learning differently. While standards are designed to be intellectually rigorous, specific, and coherent, they also need to allow flexibility to accommodate students' diverse ideas and learning trajectories. Using the analogy of trip planning and driving, I argue that teachers need additional support for understanding the aims of the standards they teach within to guide their students to the desired learning outcomes/destination.

The purpose of this paper is to compare and analyse different mathematics content standards (curricula) internationally, outline instructional paths they present, and consider teaching and student learning of mathematics through the varied paths. Standards provide roadmaps of instruction by considering current research on student learning and teaching, weighing the local needs (which may come from outside the educational community), and prioritizing and ordering how content topics should be organized, connected, and developed over time. Textbooks are written according to the standards (National Council of Teachers of Mathematics, 1989, 2000; Resnick & Resnick, 1985), and teachers then teach in classrooms using the textbooks to frame student learning in particular ways (Clements, D. H., Sarama, J., & DiBase, A-M., 2004; Fuson, K. C., Carroll, W. M., & Drueck, J. V., 2000). Teachers also come to better understand student learning of mathematics through teaching (Franke, Carpenter, Levi, & Fennema, 2001; Sherin, 2000). In all these ways, standards work to frame the way mathematics is experienced in schools.

We are becoming increasingly aware of where our students position themselves among international peers with results of international mathematics performance studies (e.g., TIMSS, PISA). What our students learn depends largely on what they are taught, and by contrasting and examining learning paths defined by different standards internationally, the paper aims to highlight ways in which students experience mathematics. I first compare and analyse California Content Standards (CCS; California Board of Education, 1997), Common Core State Standards (CCSS; Common Core State Standards Initiative, 2010), and Japanese National Standards (Japanese Course of Study – JCS; Japan's Ministry of Education, 2009) in the subject area of mathematics to illustrate different ways student learning is experienced in classrooms. Currently, in the United States, many states are adapting CCSS, and the analysis of these three standards will illuminate the transition process and the aims of the new CCSS. From there, I will extend the discussion to the Irish Curriculum (IC; National Council of

Curriculum and Assessment). I consider each set of standards as an instructional roadmap, and discuss how to support teachers as they navigate student learning on each path.

METHODS

I analysed and contrasted four mathematics content standards (CCS, CCSS, JCS, and IC). First, I created separate tables listing different learning objectives for each. In order to focus our analysis and discussion, I then extracted the objectives that pertain to whole number addition and subtraction and created a new table to organize and show how similar learning objectives are structured across the four standards. I noted different descriptions they used to discuss student learning of mathematics and how these differences may reflect different values and priorities in the different standards. I also examined instructional suggestions (e.g., activity ideas) presented in the standards as well as textbook curricula to understand various implementations of these standards.

RESULTS AND DISCUSSION

Treatment of Whole-Number Addition and Subtraction in Three Standards

Each set of standards is organized differently with different content strands. The focus topic of this analysis -- whole number addition and subtraction -- was found in the strands in the three standards listed in Table 1.

Table 1: Content strands relevant to whole-number addition and subtraction

California Content Standards	Common Core State Standards	Japanese Course of Study
Grades K – 4	Grades K – 4	Grades 1 - 3
• Number Sense • Algebra and Functions	• Operations and Algebraic thinking • Number and Operations in Base ten • Measurement and Data	• Numbers and Operations • Quantitative Reasoning

CCSS places the whole number addition and subtraction in three different strands, while CCS and JCS include it in two strands. Although the difference is small, CCSS considers the learning of whole number addition and subtraction to apply to more areas of mathematics.

Different sets of standards place different levels of emphasis (compared to other topics) on whole number addition and subtraction in each grade level. Table 2 shows the findings.

At first glance, the way in which the number of CCS learning objectives increases dramatically with grade level stands out. This may reflect curriculum that is often referred to as a “mile wide and an inch deep” (Schmidt, McKnight, & Raizens, 1997). CCSS and JCS have numbers of learning objectives that span more consistently across grade levels.

Table 2: Total numbers of learning objectives and the numbers of whole-number addition and subtraction learning objectives in each set of standards

	California Content Standards		Common Core State Standards		Japanese Course of Study	
	Add/Sub	*Total*	Add/Sub	*Total*	Add/Sub	*Total*
K	1	*18*	5	*22*	--	--
1st	10	*29*	11	*20*	5	*17*
2nd	6	*35*	9	*26*	4	*21*
3rd	1	*49*	1	*25*	3	*32*
4th	1	*54*	1	*28*	0	*31*

In Grade 1, whole-number addition and subtraction occupies approximately 1/3 of all CCS standards (10/29), while it occupies more than 1/2 of CCSS standards (11/20). For JCS, whole number addition and subtraction occupies a little less than 1/3 of the standards (5/17). In Grade 2, approximately 1/6 for CCS (6/35), 1/3 for CCSS (9/26), and 1/5 of JCS (4/21) address the topic. Thus, in each set of standards, whole number addition and subtraction is gradually de-emphasized as students come to learn other topics. CCSS maintains the largest ratio placed on the topic in these grade levels, possibly suggesting their focus on solid development of whole number addition and subtraction at the first and second grade levels.

Another interesting difference is that in Japan, the standards do not include the kindergarten level, and formal mathematics instruction starts at Grade 1. Japanese students are then given instruction for whole-number addition and subtraction up to Grade 3, while US students continue to receive instruction through Grade 4. Japanese students begin later and are expected to move more quickly than the students in the US.

For all three standards, the last objective that addresses whole-number addition and subtraction discusses some level of proficiency with the vertical algorithm. Thus, all standards appear to have the same ultimate learning goal with respect to whole number addition and subtraction.

Descriptions Used in Different Standards

While all standards address similar student learning goals, the descriptions they use differ greatly from one another. For 'learning to add and subtract multi-digit numbers' in Grades 1 and 2, the standards describe similar learning goals shown in Table 3.

CCS uses words such as "facts" and "memory" to suggest their emphasis on fluency development achieved from procedure-focused practices. CCSS mentions a list of possible student strategies along with the learning goals, to give teachers ideas of likely student thinking and possible approaches for teaching the topic. For JCS, they suggest students "think" and "invent" their own methods to solve problems, as well as use what they have learned previously to apply to the new problems (single-digit method as a foundation for two-digit operations, for example). The emphasis is on students' own ideas and connections among ideas.

Table 3: Descriptions of learning goals (Grade 1 and Grade 2)

California Content Standards	Common Core State Standards	Japanese Course of Study
Grade 1; Number Sense 2.1	*Grade 1; Operations and Algebraic Thinking 6*	*Grade 1; Number and Operations 2a*
Know the addition facts (sums to 20) and the corresponding subtraction facts and commit them to memory.	Add and subtract within 20, demonstrating fluency for addition and subtraction within 10, use strategies such as counting on, making ten, decomposing a number leading to a ten; using the relationship between addition and subtraction; and creating equivalent but easier known sums.	Consider/invent the methods for addition and subtraction using two single-digit numbers and perform the operations successfully.
Grade 2: Number Sense 2.1	*Grade 2: Number and Operations in Base Ten 5*	*Grade 2: Number and Operations 2a*
Understand and use the inverse relationship between addition and subtraction (e.g., an opposite number sentence for 8+6=14 is 14-6=8) to solve problems and check solutions.	Fluently add and subtract within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	Think/invent methods for addition and subtraction as the reverse process of addition. They will understand that they could use the single-digit addition and subtraction methods as a foundation for two-digit operations ...

Themes for Instructional Trajectories

I started this paper with the analogy of a roadmap. Based on the analyses so far, I describe the different roadmap approaches (instructional trajectories) for the three standards as follows (Table 4). CCS focuses on the end product and proficiency with the product. It gives a destination without detailed description of the roads, thus the general knowledge of the driver (teacher) is crucial in bringing the passengers (students) to the destination. Without knowing alternate routes, the teacher may not want to deviate from what he/she is familiar with. CCS also gives many more destinations than other standards, which may suggest additional stops along the way.

CCSS emphasizes processes and strategies by providing multiple alternate route options. Like an online map program, CCSS gives a variety of possible routes, and the driver (teacher) is informed in advance of the various options and is thus able to navigate the trip in his/her own way.

Table 4: Themes for instructional trajectories for three standards

California Content Standards	Common Core State Standards	Japanese Course of Study
End Product and Proficiency	Process and Strategies	Invention, Consideration, and Extension
Knowing the Destinations	Multiple Alternate Road Options	Flexibility in Rerouting the Paths
Destination Address General knowledge of the driver is crucial	Online map program Multiple possibilities are given, and the driver is informed in advance	GPS As the trip unfolds, the driver decides and navigates the road, depending on the road/traffic challenges

JCS puts priority on students' own invention, strategies, and extension of ideas. It presents flexibility in rerouting the paths on the journey and remains open-ended. Like a GPS (Global Positioning System) for drivers, it provides information to the teacher to navigate as the teaching journey unfolds. It allows for flexibility to respond to possible situations such as changes in current traffic and students' diverse ideas.

How Standards are Incorporated into US and Japanese Textbooks

In illustrating how these different standards are reflected in textbook materials, I looked at one US textbook curriculum (*Everyday Mathematics*; Pearson Education, 2008) and one Japanese textbook curriculum (*Study with Your Friends Mathematics*; Gakkotosho, 2009). I chose these textbooks because they are each one of the most-commonly used textbooks in their respective countries. I examined Grade 2 textbooks for lessons that addressed the objective of seeing inverse relationship between addition and subtraction (please see the standards objectives listed in Table 3). The US lesson reflects the ideas from CCS and not CCSS, as CCSS is not yet incorporated in the textbook materials. Figure 1 shows the examples.

Figure 1: Examples of US and Japanese lessons for inverse relationships between addition and subtraction

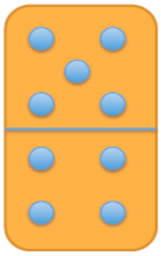
US Lesson

Everyday Mathematics

Second Grade, Volume 2

Lesson 2.6

Using dominoes to generate related addition and subtraction facts.



$$\begin{array}{r}5\\+\ 4\\\hline 9\end{array}\qquad\begin{array}{r}4\\+\ 5\\\hline 9\end{array}\qquad\begin{array}{r}9\\-\ 5\\\hline 4\end{array}\qquad\begin{array}{r}9\\-\ 4\\\hline 5\end{array}$$

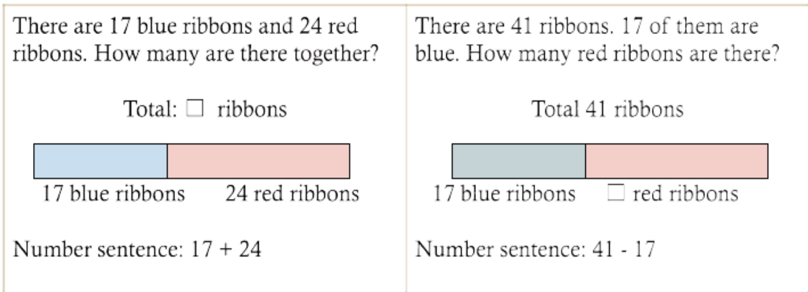
Japanese Lesson

Study with Your Friends Mathematics

Grade 2, volume 2

Unit "Addition and Subtraction"

"There are 17 blue ribbons and 24 red ribbons, whose total is 41. Make a math problem using the situation."



The US lesson focuses on the practices based on games to develop "fact power" which is explained in the teachers' instructional manual as the ability to recall number facts instantly. Students explore the number relationships through collaborative games, and notice the inverse relationship through observation of domino patterns. For the Japanese lesson, students share the problems they create and their solution methods, discuss similarities and differences among their problems, the operations used, and the relationships between the problems, operations, and solutions. They also focus on representations of problem situations using tape diagrams. These two lessons present very different flavours for teaching and learning the same content, that closely reflect the themes of instructional trajectories discussed above for CCS and JCS (Table 4). The US lesson guides students to fluency based on numerical pattern recognition, while the Japanese lesson focuses on analysis of the problem context. The learning objective is the same, while these different instructional approaches not only frame students academic development differently but also teach students what mathematics learning is about and who they are as mathematics learners.

IRISH CURRICULUM

In seeing the Irish Primary School Curriculum (IC), we first notice the very large numbers of objectives included in each grade level. There are even more objectives than CCS discussed above. There are 55 objectives for Grade 1, 60 for Grade 2, 70 for Grade 3, and 87 for Grade 4. Looking further, the objectives are also separated into more-specific strands, and further divided into sub strands. For example, for the content topic of whole number addition and subtraction (analysis focus of this paper), IC separates the learning objectives in terms of addition and subtraction independently for the first two years of primary school before combining them together to make it into one sub strand called addition and subtraction. This may reflect the emphasis and belief that content topics are better considered and taught-learned separately and specifically.

This specificity is also reflected for younger children's learning. For IC, the discussion of content learning starts in infant classes, which designate two years prior to entering primary schools. For junior and senior infant classes, discussed separately, they identify different content learning objectives specifically. One strength of this specificity can be that it makes it easier for teachers and others to make connections among different learning objectives across grade levels.

Table 5: Whole-number addition learning objectives across grade levels in the Irish Curriculum

Jr Infant	**Combine sets of objects, totals to 5** *add one more to a given set, combine two sets, state total, record pictorially*	
Sr Infant	**Combine sets of objects, totals to 10** *use appropriate strategies: counting all, counting on, counting on the number strip, starts at 5, count on 3, where am I? oral counting without the number strip, combine two or more sets, state total, record*	
First	**Develop an understanding of addition by combining or partitioning sets, use concrete materials 0-20** *find all the addition combinations to make up a given number:* *11 + 1 = 12, 2 + 6 + 4 = 12* *record addition: orally, pictorially, in number sentences, in jumps on the number line*	**Add numbers without and with renaming within 99** *estimate sum by adding the tens, check estimates using manipulatives add numbers using concrete materials, notation boards, number lines and hundred squares,* *use mental calculations,* *record using number lines, number sentences and algorithm*
Second	**Develop an understanding of addition by combining or partitioning sets**	**Add numbers without and with renaming within 99** *estimate simple sums within 99* *use mental calculations* *record using notation boards, number lines, number sentences and algorithm* *emphasise addition of 10 to multiples*

		of 10, to other numbers (36 + 10) *add multiples of 10 to numbers (45 + 20)*
Third		**Add and subtract, without and with renaming, within 999** *estimate sums and differences (rounding where necessary),* *check estimates* *record using horizontal and vertical presentation*
Fourth		**Add and subtract, without and with renaming, within 9999** *estimate sums and differences* *check estimates without and with a calculator*

The common structures and descriptions used in each objective help teachers make connections across student learning goals from one grade level to the next. The specificity may also help teachers to understand different aspects of student learning of the larger content topic, to support students more effectively. There is also a risk that it may encourage teachers to view and understand student learning of mathematics more discretely without connection. The Irish teachers may need extra support in bridging their understanding of how these small learning experiences coherently come together in classrooms for their students learning. In terms of the themes for instructional trajectory, I found IC to be similar to CCSS, an online map program, with more detailed and specific information for each small step.

On the National Council of Assessment and Curriculum web site, they list a few activity examples and suggestions for teachers. One example is shown in Figure 2.

Figure 2: Activity examples for Irish Curriculum

Special numbers strategy (with estimation)

This strategy looks for numbers that make patterns, for example tens or hundreds;

(a) 3, 5, 7, 4, 6

3 and 7 are ten, 6 and 4 are ten, that's 20; add the 5, this totals 25

(b) 37, 54, 71, 42, 69

Older children could group the tens using a mixture of rounding and compatibility, for example 37 and 42 is about 80 …

While listed as an estimation activity, this activity incorporates student understanding of ten-ness and hundred-ness based on number combinations, one of the important concepts for early mathematics learning. For younger students, teachers may give only two or three numbers at a time (to make ten), and for older students, they may use more numbers or larger numbers in a problem set. The focus of the lesson should be, as always, to help students discuss their thinking and problem solving process, and teachers may focus the attention on how to help students see connections among their differing ideas and the common mathematics concept behind them. Thus this activity reflects the emphasis of IC, which value different student strategies and alternate teaching routes.

CONCLUSION

In comparing different sets of standards, I found that they each have different approaches for teaching and student learning of mathematics. The structures of the standards, as well as the descriptions they use, illuminate different ideas and beliefs about how students best learn and how teachers best teach mathematics. These differences are reflected in textbook curricula and lessons and thus classroom teaching and learning. In the United States, as schools are currently in the process of transitioning to using CCSS, instruction is moving away from fact mastery and memorization to using students' own strategies and thinking. IC appears to have a similar focus. While CCSS and IC give strategy examples to be used, JCS discusses students' own invention and use of the strategies. This is not to say Japanese teachers are left to figure out student strategies on their own: the strategies are often outlined and explained in teachers' instructional manuals to help teachers understand potential student thinking and various instructional approaches for teaching the content topics. In CCSS, too, there are supplemental materials available to guide teacher understanding of student learning of mathematics, by outlining the progressions of and connections among learning objectives across grade levels.

In terms of Ireland Curriculum, the NCCA (National Council of Curriculum and Assessment) discusses, on their web site, the importance of young children being able to think and communicate mathematically, solve problems, recognize situations to apply mathematics, and use appropriate technologies. For the curriculum that support students' active engagement in mathematics, they present similar image of student learning of the subject as CCSS; such as the focus on language/communication, problem solving, guided discovery, and attention to individual differences. With the increasing pressure to support our children to become mathematically proficient in international arena, we seem to be all facing similar challenges to balance the academic rigor and student diversity.

Education is a complex system, and standards alone won't determine how students learn in schools. Other parts of the system, such as curriculum, assessment (testing), and teacher education, all need to work together to frame and support student learning. When we change standards, we also need to change other parts of the system accordingly, or the aims of the new standards will be lost in the old system. In order for the new standards to work, for example, the textbook curricula need to be revised and testing systems needs to be altered to assess student learning accordingly.

When thinking about standards as an instructional roadmap, the crucial question may be how to make learning more flexible and interactive to value student diversity in our schools. JCS seems closest to this ideal. Japanese textbooks and teachers' instructional manuals provide detailed guidelines for teacher action research, assessment, and instruction. These materials focus on student thinking, valuing coherence from one year to the next, while maintaining flexibility. Teachers also teach multiple grades throughout their career to understand how students learn over time, to see how learning objectives connect across years (Shimahara & Sakai, 1995). Japanese teachers are also supported by lesson study that helps them hone their craft (Lewis & Tsuchida, 1997; Murata, 2011), and they take a more integral part in educational decision making. In order for CCSS and IC to be used successfully to make an impact on student learning, the critical step may be to support teacher learning through supplemental materials and professional development opportunities so they will understand the intentions of CCSS and IC, and be able to guide their students to successfully learn in the instructional trajectories.

REFERENCES

California Department of Education. (1997). Mathematics Content Standards. Retrieved June 2011, from: http://www.cde.ca.gov/be/st/ss/

Clements, D. H., Sarama, J., & DiBase, A-M. (Eds.). (2004). *Engaging young children in mathematics: Standards for early childhood mathematics education.* Mahwah, NJ: Erlbaum.

Council of Chief State School Officers & National Governor's Association. (2010, June). Common core state standards for mathematics. Retrieved June 2011, from: http://www.corestandards.org/assets/CCSSI_Math%20Standards.pdf

Franke, M., Carpenter, T., Levi, L. & Fennema, E. (2001). Capturing teachers' generative change: A follow-up study of professional development in mathematics. *American Educational Research Journal, 38*(3). 653-689.

Fuson, K. C., Carroll, W. M., & Dreck, J. V. (2000). Achievement results for secon and third graders using the Standards-based curriculum, Everyday Mathematics. *Journal for Research in Mathematics Education, 31*(3). 277-295.

Gakkkotosho Publication. (2009). *Study with your friends: Mathematics for elementary school.* Tokyo: Author.

Lewis, C. and Tsuchida, I. (1998). A lesson is like a swiftly flowing river: Research lessons and the improvement of Japanese education. *American Educator*, Winter, 14-17 & 50-52.

Murata, A. (2011a). Conceptual overview of lesson study: Introduction. In L. Hart, A. Alston, & A. Murata (Eds.). *Lesson study research and practice in mathematics education: Learning together*. NY: Springer. 1-12.

Ministry of Education, Culture, Sports, Science, and Technology. (2009). *New Course of Study for Mathematics for Elementary Schools.* Retrieved June 2011, From: http://www.mext.go.jp/a_menu/shotou/new-cs/youryou/syo/san.htm

National Council of Curriculum and Assessment. (2011). Primary School Curriculum Online, Mathematics. Retrieved June 2011. From:

http://www.curriculumonline.ie/en/Primary_School_Curriculum/Mathematics/Mathematics.html

National Council of Teachers of Mathematics (1989). *Curriculum and evaluation standards for school mathematics.* Reston, VA: the Author.

-- (2000). *Principles & standards for school mathematics*. Reston, VA: the Author.

Pearson Education. (2008). *Everyday Mathematics*. San Francisco: the Author.

Resnick, D. P. & Resnick, L. B. (1985). Standards, curriculum, and performance: A historical and comparative perspective. *Educational Researcher, 14*. 5-20.

Schmidt, W. H., McKnight, C. C., & Raizens, S. A. (1997). *A splintered vision: An investigation of U.S. science and mathematics education.* Boston: Kluwer.

Shimahara, N. & Sakai, A. (1995). *Learning to teach in two cultures: Japan and the United States*. Gerland Science.

WANT TEACHING TO MATTER? THEORIZE IT WITH LEARNING…

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"There is nothing as practical as a good theory." (Kurt Lewin, cited in Shaffer, 1993, p. 42)

My purpose in this paper is to provide a compelling argument for using a theory of teaching that capitalizes on an articulation of learning, by presenting such a theory as a case. I begin with three examples of lessons, followed by a few questions, to substantiate the need for articulating the learning process a mathematics lesson may engender. Then, I present perspectives on mathematics learning and knowing identified to underlie mathematics teachers' practices. I elaborate one perspective as an example of an account of teaching (a 7-step, cyclic process) informed by mathematics learning. Finally, I use this account to discuss the rationale for and role of using a theory of learning in conceptualizing mathematics teaching, including implications for mathematics teacher education and links to brain studies.

This paper is about mathematics teaching as a purposeful human endeavour. It draws on a common, twofold observation: (a) people can and often do learn mathematics without teaching or through interactions with others who did not intend to promote that learning and (b) teaching often fails to engender learning for which it was planned and implemented. Ever since Erlwanger's (1973) seminal study of 'Benny', research in our field repeatedly demonstrated these assertions. At issue is how to explain and deliberately practise teaching that actually engenders others' learning of sought mathematics. This conference theme stresses the dire need to figure out ways to think about and carry out mathematics teaching that does promote students' substantial understanding and use of mathematics set out as goals for their learning (Mason, 2008b). I thank the organizers for the opportunity to address this issue, while advancing a central thesis: *Mathematics teaching matters* if it engenders students' substantial learning. This entails a need to *articulate and explicitly integrate* into teaching a lucid understanding of both how mathematics learning may progress and how teaching can *methodically* foster it in others (Runesson, 2008; Scherer & Steinbring, 2007; Steffe, 1990; von Glasersfeld, 1995a; Wood, 2005). As a constructivist, I focus on learning as a cognitive change process, and argue for being theoretical about mathematical conceptualization so that teaching matters. The premise is that, just as we do not accept a student's answer to a mathematical problem without reasoning, we should require of ourselves to solve pedagogical problems by reasoning why teaching activities may bring forth (or not) students' learning.

The approach to the effectiveness of mathematics teaching that I present here does not focus on measuring the impact of particular pedagogical interventions on student outcomes. As much as such measures are important, I focus on explaining how and why mathematics teaching may engender intended learning in students—providing a framework for considering how achievements can be fostered. I believe that a theoretical approach is indispensable when we try to (a) understand why particular teaching attempts do or do not lead to the sought learning and (b) modify unsuccessful attempts so they do. No matter how thoughtful mathematics teaching is—it cannot determine students' learning (Pirie & Kieren, 1992). Yet, theorizing teaching with learning can empower mathematics teachers' and teacher educators' design and adjustment of learning situations for better promoting the desired progression.

ILLUSTRATING THE ISSUE

To illustrate the central thesis of this paper, I present three examples of mathematics lessons. When reading these examples, I hope the reader asks the following questions about each: Assume you agree with the goal that was set for students' learning; (1) Do you think the lessons described would have brought forth such learning and *why*? (2) Whether your answer was 'yes' or 'no', would you modify the lesson and if so – how and why? (3) Do you think that students were actively engaged in the learning process and if so – in what sense? By examining each of the examples through these three questions, it seems that mainly judging *how* a lesson unfolds (tasks, problems, student activities, teacher intervention) is insufficient for deciding its effectiveness (Clarke, 2008; Jaworski, 2007). Rather, an explanation of how someone might shift from not knowing to knowing a particular piece of mathematics seems needed. The actual outcome of each lesson is provided below the figures. The reader is encouraged to form her or his own view about each example, then compare to that outcome, and reconsider whether or not modifications were needed, which, and why.

The first example took place in a grade-5 classroom (Tzur, Simon, Heinz, & Kinzel, 2001). The teacher's goal was for students to make sense of and use the long-division algorithm. First, he engaged *small groups in hands-on activities* of decomposing 2- and 3-digit numbers into equal-size groups by manipulating base ten blocks. For example, to solve 203÷7 they exchanged two 'mats' of 100 cubes each into 20 'longs' of 10 each, and placed two such 'longs' into each group. Realizing the remaining 6 'longs' could not be divided equally into groups, they exchanged them for 60 single cubes, added the initial 3 cubes, and divided 9 singletons into each group ('seeing' 29 as answer). In the next lesson, he wrote a long-division problem on the board and led students through the algorithmic steps while, at each step, pointing to the corresponding step with the blocks. He then asked students to work on a few more problems without the blocks, which many seemed unable to do. Thus, he asked students to bring back the blocks and re-engaged them in manipulating blocks to solve division problems.

The second example took place in a grade-7 classroom (see Jin & Tzur, 2011b; based on Jin's dissertation); the teacher's goal was for students to make sense and use of the algorithm of simplifying algebraic fractions. He first engaged students in solving a few problems that pertained to the previous day's lesson—asking them to decide which of eight expressions was an algebraic fraction ($5x-7$; $(b-3)/(2a+1)$; $3x^2-1$; $4/(5b+c)$; $2/3\pi$; $(x^2-xy+y^2)/2x-1)$; m/n; and $m(n+p)/7$). Then, he engaged them in simplifying *numerical fractions* ($4/8$; $-16/42$; $-6/9$; $3/(-12)$), explicitly reminding they learned this *method* at the primary school. He followed by whole-class sharing and monitoring of individual students' solutions, which led to work on simplifying gradually more challenging algebraic fractions (e.g., $a/2a$; n^2/mn; $-8ab^2c/-12a^2b$; $(a^2+4a+4)/(a^2-4)$; $4a^2b/-6ab^2$; $-4m^3n^2/2m^2n$; $(3x^2+x)/-x+x^2)$; $(y^2-9)/(-2y^2+6y)$). Key to his expected 'smooth transition' to the new mathematics was the application of a known method to the novel 'objects'.

The third example took place in a Grade-4 classroom; the teacher's goal was for students to transform their reasoning about numerical operations—from additive to multiplicative. First in this transition was to understand multiplication not merely as 'repeated addition' but as a distribution of items of one composite unit across the items of another composite unit to generate a third type of unit (Steffe, 1992b). Prior to this lesson, she assessed and found that students conceived of number as an abstract composite unit. Then, she taught and engaged students in a pair-game called *Please Go and Bring for Me* (PGBM). In PGBM, one player repeatedly sends the other to produce and bring same-size towers made of plastic, connecting cubes (e.g., please bring a tower of 4 cubes; asked 3 times). Then, the 'sender' asks a set of four questions: How many towers did you bring? How many cubes are in each tower? How many cubes did you bring in all? How did you figure out this total? Once work with tangible materials seemed to run well, the teacher began using constraints (e.g., cover towers, see Figure 1a). This impeded counting singletons and helped students develop a method to keep track of how many composite units (towers) they have accounted for while accruing the total number of 1's (e.g., raising fingers per hidden towers). Then, to promote transition from operations on figural (substitute) units to abstract units, she changed the game so the 'sender' asked the 'bringer' to *pretend* but not actually bring a given number of towers. Finally, two ways of modelling (Lesh & Doerr, 2003) the situation were encouraged. One focused on using diagrams that shifted from depictions of towers with cubes, to empty towers with numerals, to lines with numerals above them, to numerals alone (see Figure 1b). The other focused on using an equation into which given numbers were inserted and the answer was then calculated (see Figure 1c). The multiplication symbol ('x') was introduced via linking 'times' with students' activity (see Tzur, 2000)—how many times they brought a tower.

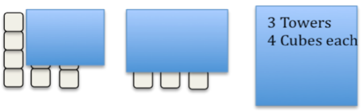

Figure 1a: Covered towers

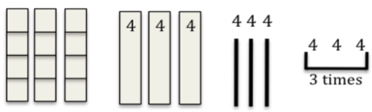

Figure 1b: Tower modelling

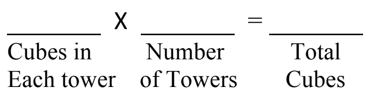

Figure 1c: Equation modelling

Outcomes of these lessons: In the first, almost all students seemed unable to understand the long-division algorithm. In the second, all students seemed to meaningfully solve the first three examples, but (based on errors) the more challenging problems seemed difficult for low-achieving students. In the third, all students seemed to begin constructing a multiplicative operation on whole numbers, including figuring out and justifying their answers.

These three examples, and potential alternatives for teaching the goal concepts (e.g., traditional show-and-tell), bring to mind a host of issues. Addressing these issues can guide mathematics teachers in their daily dealings with two fundamental questions: *What* to teach next and *how* to teach it? Not claiming the list is by any means conclusive, the issues include:

1. When we advocate for and use specific teaching methods, such as working in small groups, using manipulative and technological tools, shifting between representations, even giving a lecture—how do we explain why these methods may bring about the intended learning in students? And if they do not—how do we explain why and adjust them?

2. In all three examples, students were differently engaged in actively solving problems; when we say that learning is active and teaching should build on this premise—what do we mean by ‘activity’, how do we select one that matters, and how do we link this choice and use to the teacher’s work? How do we choose one activity over another?
3. When advocating balance of activities (e.g., small group and whole class; manipulating tangible objects and abstract symbols), how do we decide what is the proper balance?
4. When we advocate for teaching via tasks (problem situations), how do we ‘tailor’ them to fit with both what students already know and the target understandings?
5. More generally, when contemplating among theories of teaching (e.g., ‘spiral’, models and modelling, problem-based learning)—how do we determine if they are useful? How do we link these to grand theories of learning, such as constructivist or social-cultural?
6. Simply put—when we refer to particular ways of teaching as ‘effective’ (Cai, Perry, Wong, & Wang, 2009; Wang & Cai, 2007a, 2007b), or ‘they matter’, we mean … what?
7. Finally, when we attempt to promote teachers’ professional development of sought teaching—on what basis do we choose goals for their learning and guide the design and implementation of experiences that can promote the needed change (Simon, 2006, 2007)? For instance, the second example, found in the work of Chinese teachers and termed ‘Bridging’ (Jin & Tzur, 2011a; based on Jin's dissertation), seemed highly effective—but how do we explain why it works and how do we teach it to teachers?

To address these issues, the next section elaborates on perspectives that seem to underlie teaching practices.

PERSPECTIVES ON KNOWING AND LEARNING

My argument for theorizing teaching with learning draws on a growing body of research on the considerable impact that teachers’ ways of thinking about mathematics knowing and learning have on their pedagogical practices (Bishop, 1991; de Ponte & Chapman, 2006; Thompson, 1992). At the core of this impact seems to be the need for teachers to ‘translate’ their academic knowledge into classroom activities they anticipate would foster student learning (Pajares, 1992; Phillip, 2007). Correa et al. (2008) asserted that, in spite of different meanings assigned to terms like ‘teacher beliefs’ (or ‘orientations’, or ‘perspectives’), most researchers seem to agree that how teachers think greatly impacts how they may design and implement learning situations. Wilson and Cooney (2002) succinctly summarized this premise: “what teachers believe is a significant determiner of what gets taught, how it gets taught, and what gets learned in the classroom” (p. 128). Said differently, how a teacher teaches seems to be rooted in her or his views about mathematics, learning, and teaching (Cooney, 1985; Kember, 1998; Kember & Kwan, 2000; Richardson, 1996; Torff & Warburton, 2005), and differences found in teachers’ practices reflect, to a large extent, their different views (Bryan, Wang, Perry, Wong, & Cai, 2007; Xie & Carspecken, 2008). Of particular importance seem to be teachers’ epistemological stances about mathematical knowing and how one comes to such knowing (Ernest, 1989a, 1989c; Steinbring, 1998;

Thompson & Saldanha, 2000; Tzur, 2008), because these stances provide a basis for one's pedagogical content knowledge (Ball, 2000; Shulman, 1986, 1987).

Teaching Perspectives.

The framework I present here draws on Ernest's (1989c) contention that if reform in teaching is to take place, it must address not only what teachers do but also their deeply held beliefs about the nature of mathematics, mathematics learning, and its teaching. To link these, he described key components of teachers' knowledge systems (Ernest, 1989a, 1989b). About the *nature of mathematics*, Ernest distinguished three typical views: (a) *instrumentalist*—who considers mathematics as a set of tools, facts, rules, and skills; (b) *Platonist*—who considers it as a static, cohesive body of knowledge that exists outside the knower and can be discovered; and (c) *Problem solver*—who considers it as a dynamic field of human invention that exists within and among people through interactive processes of inquiry. About *pedagogy*, Ernest suggested three views, respectively: (i) *Instructor*—who focuses on transmitting skills and facts to students in a clear and organized fashion, so they master and perform those correctly; (ii) *Explainer*—who focuses on students' first-hand grasp of pieces from the unified body of knowledge; and (iii) *Facilitator*—who focuses on students' construction of understandings that foster their competence and confidence in solving and posing mathematical problems. Accordingly, Ernest identified two views that constitute a teacher's stance on learning, *passive reception* of knowledge by submissive, compliant students and *active construction* of concepts via problem solving, inquiry, and development of student autonomy.

My colleagues and I have extended Ernest's work while substantiating it in empirical studies on how teachers may modify their work in the context of principles and standards of reform in the US (National Council of Teachers of Mathematics, 2000). Using the *Account of Practice* (Simon & Tzur, 1999) strategy of inquiry (Denzin & Lincoln, 1994), we identified three perspectives about knowing and learning that seemed to underlie mathematics teachers' practices (Heinz, Kinzel, Simon, & Tzur, 2000; Simon, Tzur, Heinz, Kinzel, & Smith, 2000; Tzur, et al., 2001). Initially, teachers seemed to adhere to a *traditional perspective*, which consists of a view of mathematics as existing outside and independent of the knower, of a behaviour-oriented view of learning, and a passive view of learners. Thus, their teaching included lectures, show-and-tell methods, reading and following examples from books—essentially efforts to transmit ideas to their students. Via reform-oriented undertaking, they seemed to develop practices rooted in a *perception-based perspective* (PBP). A PBP is marked by a view of mathematics that is like the traditional—a part of the external world, independent of human activity, and accessible through perception and exercise; a Platonic (discovery) view of learning—attaining meaningful, interconnected understandings through first-hand experiences; and a view of learners that is starkly different than the traditional—students learn mathematics when being allowed to actively perceive mathematical objects, operations, and relationships. The third, *conception-based perspective* (CBP) may then evolve through a paradigm shift (Tzur, 2008) in a teacher's view of mathematics as being constructed through mental activity and reflection—inseparable from a knower. Thus, a teacher focuses on the key role that student available (assimilatory) schemes play in what they can perceive, and on learning as a process of building-up of new mathematical schemes via mental transformation of assimilatory ones (Piaget, 1970, 1985; Steffe, 1992a, 2010; von Glasersfeld, 1991, 1995b). Not surprisingly, these two lines of work—our three perspectives and Ernest's typology—seem quite similar.

Building on both lines of work, a doctoral student from Monash University, Xianyan Jin, with whom I worked along with Prof. Peter Sullivan, had set out to study perspectives that may underlie practices of mathematics teachers in China. Based on her dissertation, we (Jin & Tzur, 2011b) elaborated a novel perspective, termed *Progressive Incorporation* (PIP). The scope of this paper precludes detailing it. Suffice is to say that it seems to underlie teachers' attention to assessing and deliberately reactivating students' existing knowledge ('old'). In those Chinese teachers' views, such reactivation serves as a means for students' meaningful grasp of new mathematical ideas, because it lets them apply known methods to solve new problems and, in turn, incorporate the 'new' into the 'old'. In particular, the PIP was inferred from teachers' use of and rationale for a distinct lesson component, called Bridging (Jin & Tzur, 2011a). Problems students solved during that lesson component required methods they learned long before the lesson (e.g., simplifying numerical fractions as a lead into simplifying algebraic fractions, illustrated in Example 2 above). Those teachers explicated an expectation that students' use of the 'old' methods would foster smooth transition to and incorporation of the 'new' knowledge. The focus on reactivating students' existing knowledge seemed to go beyond PBP's focus on learning as an active process. Yet, the teachers' views of both 'old' and 'new' knowledge seemed consistent with a view of mathematics as existing outside the knower (Xie & Carspecken, 2008, explained that this view of the knower and the known should be considered dialectically). We suggested (Jin & Tzur, 2011b) that a PIP fits between PBP and CBP, as summarized in Table 1. We argued that a PIP-rooted practice may engender students' learning envisioned by CBP, while not requiring the teacher's explicit awareness of such a vision and the epistemological paradigm shift it entails (Tzur, 2008).

Perspective	View of Knowing	View of Learning	View of Teaching
Traditional (TP)	Independent of the knower, out there	Passive reception	Transmit knowledge (lecturing); *Instructor*
Perception-Based (PBP)	Independent of the knower, out there	Discovery via active, first-hand perception	Point out via first-hand experiences; *Explainer*
Progressive-Incorporation (PIP)	Dialectically independent and dependent on knower	Active (mentally); known is required to start; new incorporated to 'old'	'Engineer' situations conducive to learning; purposeful *Guide*
Conception-Based (CBP)	Dynamic; depends on knower's assimilatory schemes	Active construction of 'new' as transformation in the known (reflection)	Engage in problem solving and reflection; proactive *Facilitator*

Table 1: Placing PIP within Ernest's and Simon & Tzur et al.'s lines of work

Looking back at the three examples presented in the previous section, the first (long-division), second (algebraic fractions), and third (multiplicative reasoning) were found in the work of teachers who seemed to adhere to PBP, PIP, and CBP perspectives, respectively. It should be noted that to infer into a teacher's perspective, a lesson plan and its unfolding is insufficient. Rather, interviewing a teacher about her or his rationale for highly specific aspects of the lesson, before (plan) and after (recall of events), is required. Teacher explanations for how and why each of the example lessons was taught indicated their underlying perspective. About the long-division lesson, the teacher stated an expectation that the hands-on experience with base ten blocks would guide students to also 'see' (perceive) the meaning of steps in the algorithm. The idea that this intended mathematics could be discovered, if properly pointed to, indicated a teacher's view of its existence apart from the knower (students like teacher) and a view of learning as an active process of discovery. About the fraction lesson, the teacher stated that starting with numerical fractions was used to allow *all* students, particularly low-achieving, to successfully reactivate their known ('old') method of simplifying fractions as a bridge to similarly applying it to the 'new' objects (algebraic fractions). That is, the teacher focused on assessing where students are and on reactivating knowledge he supposed was abstracted by all of them. Then, he explained, the lesson moved from these concrete, simple examples to gradually more demanding problems, which would allow them to link (incorporate) the 'new' to the 'old'. His view indicated that the mathematics still existed apart from students and learning was thought of as appending—not as transformation. About the multiplicative reasoning lesson, the entire plan and implementation drew on a CBP. The teacher assumed that reactivating students' assimilatory conceptions was needed to initiate their use and promote their transformation into the intended mathematics, while gradually decreasing prompting and increasing abstractness of objects on which students operated. The following sub-section presents key features of a CBP—*reflection on activity-effect relationship*, which I propose as a sound candidate for theorizing teaching with learning.

Preliminaries to a Learning-Based Theory of Teaching.

As a constructivist, I adhere to the core assumption common to Dewey's (1933), Piaget's (1970, 1971, 1985), Skemp (1979), and Vygotsky's (1978, 1986) grand theories, that thinking cannot be understood apart from the 'historical process' in which one's knowing evolved. Building on these seminal works, my colleagues and I have proposed a framework for articulating a mechanism of cognitive change when learning a new mathematical conception, and a corresponding account of mathematics teaching (Simon & Tzur, 2004; Simon, Tzur, Heinz, & Kinzel, 2004; Tzur, 2007; Tzur & Lambert, in press; Tzur & Simon, 2004). The *mechanism of reflection on activity-effect relationship* (Ref**AER*) is the core of this framework. To illustrate Ref*AER, I have included examples (in parentheses) pertaining to the PGBM game (Example 3 above). *Ref*AER* commences with a learner's assimilation of problem situations into her *extant (assimilatory) conceptions* (e.g., she can operate *additively* on number as a composite unit). The learner's assimilatory conceptions set her situation and goal—a desired, anticipated state that regulates her activity (e.g., find how many cubes are in all towers after distinguishing them as units apart from 1's that comprise them). The *learner's* situation/goal then triggers, *and regulates from within the mental system* (Piaget, 1985), execution of a mental activity (e.g., accruing the total of cubes while keeping track of each tower). While 'running' the activity, she may notice (a) discrepancies between the actual effects and the expected result and (b) novel effects (e.g., the total is not equal to her anticipated effect (error) of *adding* the number of towers and the number of cubes in each tower). Through reflection on and reasoning about solutions to similar problems, she abstracts a new, invariant relationship between an activity and its anticipated effects (e.g., the total—product—of 1's in a set of equal-size groups corresponds to the accrual of all 1's distributed across each composite unit). This invariant *AER* brings forth a reorganization of the original situation, that is, the learners' previous assimilatory conceptions. The crucial implication for instruction is the clear distinction between a teacher's goal for students' learning (here, multiplicative reasoning as distribution of units) and the student's goal in the activity (bring towers, ask proper questions, account for the number of cubes).

Accordingly, the Ref**AER* framework defines *conception* as the abstract relationship between an activity and its effects, implying that an activity is a constituent of a conception (e.g., coordinating number of composite units and the given unit rate). This view differs from a view of activity as a way to engage learners, to which von Glasersfeld (1995b) referred as 'trivial constructivism'.

This view of a conception and the learning mechanism is consistent with and draws on recent studies on brain and learning (Bransford, Brown, & Cocking, 1999; Medina, 2008). Recently, I have linked it to a model of the brain (Tzur, 2011, accepted for publication) by separating and distributing the three parts of a scheme (von Glasersfeld, 1995b)—situation/goal, activity, and effects—across three major neuronal systems in which they seem to be activated (Baars, 2007): a 'Recognition System', which includes the sensory input/buffer and various long-term memories; a 'Strategic System', which includes the Central Executive; and an 'Engagement-Emotive System'. More details of how brain and mind processes that constitute *Ref*AER* are coordinated when a learner solves known problems and/or constructs a novel mathematical conception can be found in (Tzur, 2011). Here, a key point needs attention. The coordinated, brain-mind model articulated learning as transformation in one's anticipation via two basic types of reflection. Reflection *Type-I* consists of *continual, automatic* comparisons executed by the mental system—*between the goal* it sets for the activity and *subsequent effects* produced and noticed. The effects either match the anticipation or not. By default, the mental system runs an activity-sequence to its completion as determined by the goal. Yet, the execution may stop earlier if (a) the mental system detects unanticipated sub-effects or (b) a different goal became the regulator. Reflection *Type-II* consists of comparisons *across (mental) re-presentations of experiences*, each containing a record of a 'run' of the activity and its effect (*AER*), and sorted/recorded as match or no-match. Critically, the brain may or may not execute Type-II reflection (i.e., it does not happen automatically). The recurring, invariant *AER* across those compared records are linked with the situation(s) in which they were found anticipatory of the proper goal and registered as a new mathematical concept.

Accordingly, *Ref*AER* postulates that a new conception is constructed in two stages. The first, *participatory*, is made possible by reflection Type-I and is marked by an evolving anticipation a learner can access *only when and if somehow prompted* for the novel, provisional *AER* (Tzur & Lambert, in press, linked this stage with the Zone of Proximal Development—ZPD). The second, *anticipatory* stage requires reflection Type-II and is marked by spontaneous, independent, reactivation, running, and justifying the novel anticipation. Although *Ref*AER* was developed independently, it is consistent with Skemp's (1979) theory; the reflection types and stage distinctions seem to extend his work.

When used as a core of a model of teaching, the Ref**AER* framework implies engaging students in solving tasks that serve *three principal functions*: (a) promoting assimilation into and reactivation of existing schemes, (b) orienting learners' attention to novel, intended effects, and (c) guiding reflection on and distinction/construction of the intended conceptions.

That is, a teacher needs to create, evaluate, and adjust problems to fit within the ever-changing student understandings. This threefold function of tasks implies a *cyclic process of seven principal teaching activities*, which Tzur (2008) proposed as an extension to Simon's (1995) hypothetical learning trajectory. The starting point of this cycle is not arbitrary. Rather, to capitalize on assimilation, teaching proceeds from and builds on student 'old' knowledge. Based on our recent studies of Chinese teachers' practices and the PIP (Jin & Tzur, 2011b), the 7-step cycle below integrates aspects of their lesson structure (examples from Chinese teachers' work and the PGBM game are used, although PGBM was *not* part of their teaching).

1) <u>Specifying students' assimilatory conceptions</u> by inferring into mental processes that might underlie their work on previous tasks: goals *they* seemed to set, mental activities *they* seemed to execute, and effects *they* seemed to notice and link to the activities (e.g., to find the total of cubes a student may add the number of composite units and the number of cubes in each, or simply count all cubes one-by-one without keeping track of the number of towers). In the Chinese teachers' work, assessing students' existing ('old') knowledge seemed to be at the core of the decisions about both what and how to teach next. The focus is on making sure that *all* students have a solution method (a goal-directed activity) known to them that they could initiate and use prior to and as lead into the new learning.

2) <u>Specifying a goal for students' learning</u> by decomposing the conceptions students are expected to learn into their four constituents—goals, activities, effects, and invariant relationships (e.g., anticipate the need to keep track and coordinate composite units with 1's in each—the unit rate—and symbolize this coordination abstractly). These constituents seemed to underlie Chinese teachers' focus on including, in every lesson, not only a central mathematical goal for students' learning but also three levels of attaining it (basic, focal, and difficult/demanding). Such levels were instantiated, for example, via the different 'objects' on which students were to carry out their activities.

3) <u>Identifying a mental activity sequence</u> that can be reactivated in students as a means to generate intended effects and relationship (e.g., separating the production of a composite unit by iterating the unit of 1 from the production of a set of such composite units, and then coordinating both to figure out the total of 1's). Chinese teachers used the term 'method' (Fang Fa) in reference to this aspect. They distinguished it as the underlying commonality between 'old' and 'new' knowledge, and focused on reactivating it early in a lesson.

4) <u>Selecting and sequencing tasks</u> by designing an initial task and follow-up prompts and questions. This is based on a teacher's reasoned hypotheses about (a) how students may assimilate the tasks and begin solving them via their current conceptions, (b) notice intended effects, and (c) perform *both types of reflection* on new activity-effect relationship (e.g., *PGBM* as a game in which children produce one tower at a time, create a set of towers, ask and answer questions about the components of the set, figure out and justify the total of cubes—starting with counting by ones and noticing the possibility to coordinate accrual of cubes by keeping track of towers). In the Chinese teachers' work, step *a* seemed supported by the *Bridging* (Jin & Tzur, 2011a) lesson component (e.g., simplifying numerical fractions in Example 2), in which students could reactivate the old method when operating on known objects. Steps *b* and *c* seemed supported by their teaching with Variation (Gu, Huang, &

Marton, 2006), as a focus on particular changes of problems and solutions would promote noticing of what stays the same across situations. Those teachers' selection of tasks to fit within difficulty levels seemed consistent with the work of Sullivan et al. (Sullivan, Mousley, & Zevenbergen, 2004) on the creation and use of enabling (low-level) and challenging (high-level) prompts.

5) Engaging students in the task sequence while making sure they solve it by using *their* goal in regulating *their* activity. This involves a delicate challenge of negotiating students' interpretations of the task so their goal is compatible with the teacher-intended goal, while not attempting to govern students' necessarily internal goals. When students are working on tasks, a teacher interjects follow-up prompts that enable/challenge their thinking based on where they are conceptually (e.g., "Pretend you bring 6 towers of 3 cubes each" as a follow-up to the child's actual bringing of 5 such towers). The Chinese teachers drew on a view of the critical role that errors serve in learning and continually asked students who solved problems incorrectly to share their solutions. This process seemed to promote both types of reflection for all students, as they repeatedly compared one's own solution with others; a similar outcome seems to be encouraged via letting students solve problems in small groups and then discuss different solutions as a whole class.

6) Examining students' progress by repeatedly assessing their actual work: which goals seem to regulate their activities, which activities they seem to use for realizing their goals, which effects they seem to notice anew, and which comparisons/linkages (reflections) they seem to perform. When students make mistakes or solve problems correctly, and when teachers ask them to share, monitor, and critique one another's solutions (not just the correct ones), ongoing assessment of students' progress provides teachers with ample opportunities to prompt reflection type I (compare actual, different effects of activities to what one anticipated) and reflection type II (across different 'runs' of the same activity/method). Combined, both types seem to be greatly supported and lead to students sorting—with justification—*what is* from *what is not* (Xie & Carspecken, 2008).

7) Introducing follow-up prompts by interjecting follow-up questions and comments (e.g., "You found that there are 18 cubes in 6 towers of 3; can you use this to figure out how many cubes would you have if you brought 7 towers of 3?"). Such follow-ups provide the teacher with a pedagogical means to *indirectly* orient and re-structure students' attention and awareness (Mason, 1998, 2008a), so they reflect on intended effects of their goal-regulated activity and relate those effects with the activity in anticipation.

I believe that this 7-step cycle contributes to a theory of mathematics teaching that is grounded in a framework of learning as a cognitive change—transformation in one's anticipation of AER through two types of reflective processes. It extends the previous 7-step cycle (Tzur, 2008) by including particular lesson components, found in Chinese teachers' practices, which seem helpful in structuring learning opportunities for students. It is situated in the larger framework of categories about teachers' perspectives on mathematics knowing and learning (TP, PBP, PIP, and CBP), and illustrates a conception-based perspective and practice. The following section elaborates on implications of such a framework.

Discussion.

In this section, using as a backbone the above framework of teachers' perspectives, learning as transformation in anticipation, and 7-step teaching cycle, I discuss five main reasons for my contention that teaching matters when it is theorized with learning. My first reason is that a theory of learning makes possible the creation and use of inferences into the mental realm. Behaviourists (Gagné, 1965; Guthrie, 1942; Skinner, 1953) considered this realm as a 'black box' and advocated for ignoring it altogether. Downgrading the mental realm seems to also characterize social-cultural theories of learning (Goos, 2008; Leach & Scott, 2003; Lerman, 1996, 2000, 2006; Wertsch & Toma, 1995; Wickman & Östman, 2002), such as situated cognition (Lave, Smith, & Butler, 1988; Lave & Wenger, 1991; Wenger, 1998). In contrast, a framework that articulates what knowing mathematics means, and how learning—changes in mental operations and structures—may occur allows teachers to focus on and promote mathematical experiences as a cognitive phenomenon. The framework above provides teachers with a tool for interpreting what constitutes a mathematical conception (anticipation of AER), how it may evolve (two stages), and what processes underlie this evolution (two types of Ref*AER). This tool served in analyzing students' multiplicative reasoning that was detailed in Example 3.

My second reason is that such a theory enables teachers to *specify* changes needed in students' thinking in terms of *both* what they bring to a learning situation (assimilatory conceptions) and what a teacher may intend for their learning. Moreover, it explicates the critical role that students' existing mathematical knowing plays in *commencing* their learning and thus not only the need but also criteria for how this knowing could be reactivated. By inferring into students' goals (cognitive *and* social-emotional), activities, and effects noticed/anticipated, a teacher can make an informed assessment of students' current understandings and use it as a basis for engendering the sought transformation. And by using Bridging tasks teachers can develop a practice and an underlying perspective for *strategically* reactivating students' assimilatory conceptions in service of the desired cognitive change.

My third reason is that a theory seems needed if one is to successfully translate a hypothetical learning trajectory (Simon, 1995) into a set of teaching moves that effectively promote intended conceptualization in students. Various practices such as small group discussions, manipulation of tangible, figural, and/or abstract objects, variation of solutions and problems, and continual translations among representations, can be explained via explicit articulation of both types of reflection (comparison) to which teachers should orient students' attention. Indeed, a teacher cannot directly 'give' students the intended anticipation. But she or he can indirectly orient students' reflections—noticing of mismatches between anticipated and actual effects (Type-I) and most crucially similarities and differences across situations (Type-II, non automatic in the brain!) to foster abstraction of an anticipatory stage of a new concept. Understanding these two reflection types is particularly helpful in designing not just each problem/task but also sequencing them, so that against the background of variation students construct an intended, invariant AER. Furthermore, the theory informs such sequencing in terms of specifically selecting and decreasing prompts so that students advance beyond the participatory stage into the robust, transfer-enabling anticipatory stage. The theory can also guide increases in levels of difficulty, symbolization, and abstraction according to changes in students' levels of anticipation. The sequence of tasks and prompts in the PGBM game illustrates how the theory was used to this end; the sequences of Bridging and Variation problems used by Chinese teachers could be analyzed via this theory in spite of the different perspective (PIP) that seemed to underlie their practice. On the other hand, the ineffective use of shifts between concrete and abstract manipulation of objects in Example 2 indicated lack of gradual orientation of students' reflective processes. In short, by explicating the cognitive change mechanism as constituted by two types of reflection on AER, the theory seems to enable effectively planning and implementing problem-based learning-teaching processes.

My fourth reason is that an integrated theory of learning, which includes analysis of teaching perspectives, can help to (a) make explicit teachers' implicit theories and (b) inform setting goals and choosing activities for their professional development with student learning as a driving force (Moyer & Milewicz, 2002; Simon, 2006, 2007; Thompson, Carlson, & Silverman, 2007; Watson, 2008; Zaslavsky, 2008). Used by teacher educators, the theory enables inferring into perspectives that may underlie teachers' practices—the frameworks into which they are likely to assimilate pedagogical and/or mathematical tasks in which they are engaged as part of professional development efforts. That is, a theory of learning applied to teachers' pedagogical approaches corresponds to inferring into students' assimilatory mathematical conceptions. Because at the heart of each perspective is a view of knowing and learning, the theory can help assessing and guiding teachers' progress toward practices that are more conducive to students' learning. Jin & Tzur (2011b) proposed that for the development of teachers who adhere to a perception-based perspective (PBP; illustrated in Example 2), a progressive incorporation perspective (PIP) is a more feasible goal than a conception-based perspective (CBP), because a transition to PIP does not require the paradigm shift in a stance on mathematical knowing as dependent on the knower, which the CBP entails (Tzur, 2008).

My fifth reason is that a detailed theory like the one above endows mathematics educators with a model of the mind that can be linked to the vast, growing body of research about how the brain functions when people do mathematics. In recent years, articulating and studying brain-mind links has become the focus of interest for both communities—neuroscientists and educators (see for example Dehaene, 1997; Dehaene, Molko, Cohen, & Wilson, 2004; Dehaene, Piazza, Pinel, & Cohen, 2003; Kelly, Jensen, & Baek, accepted for publication; Medina, 2008; Zull, 2002). I made a serious effort to contribute to such articulation of an integrated model. Currently, in collaboration with a cognitive neuroscientist and fMRI expert, we are in the midst of testing aspects of that integrated model by studying how the brain processes a transition from ordering whole numbers to ordering unit fractions, which requires inverting one's anticipation (e.g., $8>5$ but $1/8<1/5$). Key here is that such an integrated, brain-mind model informs generation and testing of hypotheses regarding what, when, and where to look for in the brain (Varma, McCandliss, & Schwartz, in press). Examining these hypotheses can shed light on how teaching may be conducive to learning consistent with the functioning of the organ—brain—that endows us, humans, with mathematics.

Taken together, the aforementioned reasons help to answer the seven questions presented earlier in this paper. The first question asked about how one determines if/why a specific teaching method is effective (or not). As explained above, any method can be designed and modified via analyzing how it (a) reactivates students' assimilatory conceptions, (b) orients their reflection (both types), and/or (c) judiciously provides and eliminates prompts. These three criteria, when applied to Example 2, suggest that students' assimilatory conceptions of dividing the blocks were not reactivated for reflection as a means to abstract and symbolize an algorithm for division. Rather, students could at best use these conceptions to follow the teacher's rather separate, non-gradual, and abstract solution to a long division problem. Because students were yet to reflect on effects of distribution activities, the algorithm was inaccessible to them and the return to using heavy-prompted manipulation of the blocks did not foster the needed transformation to an anticipatory stage. The second question asked about the meaning of activity. In this framework, the answer is straightforward—activity refers to the mental processes a knower uses while solving/posing problems (Lompscher, 2002; Simon, et al., 2004; Tzur & Simon, 2004). A mental activity may or may not be accompanied by observable behaviours (bodily actions, utterances), which serve as indicator for inference into the mental, goal-directed activity. And selecting an activity that matters is based on explicitly linking between goal-directed activities available to students via their assimilatory conceptions and the intended mathematics. This is also the answer to the third question about how to tailor tasks (problem situations) to fit with existing and sought understandings. Answering the fourth question (comparing theories of learning) goes beyond the scope of this paper. Here, I only emphasize a central point—if a theory of mathematics teaching is to be useful it should include the mental realm and be consistent with what is known about the brain (at the least, it should articulate why it does not contradict such knowledge). The sixth question asked what do we mean by 'effective teaching', my answer is teaching that engenders intended transformations in students' *thinking* and hence, necessarily, capitalizes on their assimilatory conceptions to foster spontaneous, independent (prompt-less), anticipatory stage of mathematical ideas. The seventh question, about the role a theory of teaching can play in teacher development, has been addressed above. All together, these questions and answers amount to a fundamental stance I wished to advance in this paper: When mathematics educators (teachers included) propose particular pedagogical interventions we, as a field and 'consumers' of their suggestions, should expect and look for a rationale that includes articulation of the pertinent learning process.

Concluding Remarks

This paper focused on teaching as a human endeavour designed and implemented to engender transformation in students' mathematical thinking. It depicted thinking as anticipation and awareness (justification) of relationships between goals, mental activities, and effects. An extended, 7-step cycle was proposed for fostering such transformation via students' assimilation of tasks into their existing knowledge and two reflection (comparison) types that enable students to: (a) competently and confidently know where/how to begin and (b) develop an anticipatory (prompt-less) stage of understanding a new, intended idea. My work with children with learning disabilities in mathematics (e.g., using the PGBM to teach 4th graders whose mathematics was barely that of 1st graders) suggests that this learning-rooted account of teaching enables restoration of their spark for learning and progress toward important mathematics they could understand (e.g., reason multiplicatively). In this sense, this paper illustrates that teaching matters when theorized with learning.

REFERENCES

Baars, B. J. (2007). A framework. In B. J. Baars & N. M. Gage (Eds.), *Cognition, brain, and consciousness: Introduction to cognitive neuroscience* (pp. 31-57). Amsterdam, The Netherlands: Elsevier.

Ball, D. L. (2000). Bridging practices: Interweaving content and pedagogy in teaching and learning to teach. *Journal of Teacher Education, 51*(3), 241-247.

Bishop, A. J. (1991). Mathematical values in the teaching process. In A. J. Bishop, S. Mellin-Olsen & v. J. Dormolen (Eds.), *Mathematical knowledge: Its growth through teaching* (pp. 195-214). Dordrecht, The Netherlands: Kluwer.

Bransford, J. D., Brown, A. L., & Cocking, R. R. (Eds.). (1999). *How people learn: Brain, mind, experience, and school*. Washington, DC: National Research Council - National Academy Press.

Bryan, C., Wang, T., Perry, B., Wong, N.-Y., & Cai, J. (2007). Comparison and contrast: similarities and differences of teachers' views of effective mathematics teaching and learning from four regions. *ZDM: The International Journal on Mathematics Education, 39*(4), 329-340.

Cai, J., Perry, B., Wong, N.-Y., & Wang, T. (2009). What is effective mathematics teaching: A study of experienced mathematics teachers from Australia, the Mainland China, Hong Kong-China, and the United States. In J. Cai, G. Kaiser, B. Perry & N.-Y. Wong (Eds.), *Effective mathematics teaching from teachers' perspectives: National and cross-national studies* (pp. 1-36). Rotterdam, The Netherlands: Sense.

Clarke, D. (2008). The mathematics teacher as curriculum maker: Developing knowledge for enacting curriculum. In P. Sullivan & T. Wood (Eds.), *The International Handbook of Mathematics Teacher Education: Knowledge and Beliefs in Mathematics Teaching and Teaching Development* (Vol. 1, pp. 133-151). Rotterdam, the Netherlands: Sense.

Cooney, T. J. (1985). A beginning teacher's view of problem solving. *Journal for Research in Mathematics Education, 16*, 324-336.

Correa, C. A., Perry, M., Sims, L. M., Miller, K. F., & Fang, G. (2008). Connected and culturally embedded beliefs: Chinese and US teachers talk about how their students best learn mathematics. *Teaching and Teacher Education, 24*, 140-153.

de Ponte, J. P., & Chapman, O. (2006). Mathematics teachers' knowledge and practices. In A. Gutiérrez & P. Boero (Eds.), *Handbook of research on the psychology of mathematics education: Past, present, and future* (pp. 461-494). Rotterdam, The Netherlands: Sense.

Dehaene, S. (1997). *The number sense: How the mind creates mathematics*. NY: Oxford University.

Dehaene, S., Molko, N., Cohen, L., & Wilson, A. (2004). Arithmetic and the brain. *Current Opinion in Neurobiology, 14*, 218-224.

Dehaene, S., Piazza, M., Pinel, P., & Cohen, L. (2003). Three parietal circuits for number processing. *Cognitive Neuropsychology, 20*, 487-506.

Denzin, N. K., & Lincoln, Y. S. (1994). Introduction: Entering the field of qualitative research. In N. K. Denzin & Y. S. Lincoln (Eds.), *Handbook of qualitative research* (pp. 1-17). Thousands Oaks, CA: Sage.

Dewey, J. (1933). *How we think: A restatement of the relation of reflective thinking to the educative process*. Lexington, MA: D.C. Heath.

Erlwanger, S. H. (1973). Benny's Conception of Rules and Answers in IPI Mathematics. *Journal of Children's Mathematical Behavior, 1*(2), 7-26.

Ernest, P. (1989a). The impact of beliefs on the teaching of mathematics. In P. Ernest (Ed.), *Mathematics teaching: The state of the art* (pp. 249-254). New York: Falmer.

Ernest, P. (1989b). The knowledge, beliefs and attitudes of the mathematics: A model. *Journal of Education for Teaching, 15*, 13-33.

Ernest, P. (1989c). *Mathematics teaching: The state of the art*. Basingstoke: Falmer.

Gagné, R. M. (1965). *The conditions of learning*. New York: Holt, Rinehart, and Winston.

Goos, M. (2008). Sociocultural perspectives on learning to teach mathematics. In B. Jaworski & T. Wood (Eds.), *The international handbook of mathematics teacher education: The mathematics teacher educator as a developing professional* (Vol. 4, pp. 75-91). Rotterdam, the Netherlands: Sense.

Gu, L., Huang, R., & Marton, F. (2006). Teaching with variation: A Chinese way of promoting effective mathematics learning. In L. Fan, N.-Y. Wong, J. Cai & L. Shiqi (Eds.), *How Chinese learn mathematics: Perspectives from insiders* (pp. 309-347). Singapore: World Scientific Publishing.

Guthrie, E. R. (1942). Conditioning: A theory of learning in terms of stimulus, response, and association. In N. B. Henry (Ed.), *The Forty-First Yearbook of the National Society for the Study of Education (Part II - The Psychology of Learning)* (pp. 17-60). Chicago: The National Society for the Study of Education.

Heinz, K., Kinzel, M., Simon, M. A., & Tzur, R. (2000). Moving students through steps of mathematical knowing: An account of the practice of an elementary mathematics teacher in transition. *Journal of Mathematical Behavior, 19*, 83-107.

Jaworski, B. (2007). Theory and practice in mathematics teaching development: Critical inquiry as a mode of learning in teaching. *Journal of Mathematics Teacher Education, 9*(2), 187-211.

Jin, X., & Tzur, R. (2011a). *'Bridging': An assimilation- and ZPD-enhancing practice in Chinese pedagogy*. Paper presented at the The 91st Annual Meeting of the National Council of Teachers of Mathematics (Research Pre-Session).

Jin, X., & Tzur, R. (2011b). *Progressive incorporation of new into known: A perspective on and practice of mathematics learning and teaching in China*. Paper presented at the Annual Conference of the Association of Mathematics Teacher Educators.

Kelly, E., Jensen, S., & Baek, J. (Eds.). (accepted for publication). *NeuroMathEducation*. Cambridge, MA: Cambridge University Press.

Kember, D. (1998). Teaching beliefs and their impact on students' approach to learning. In B. Dart & G. Boulton-Lewis (Eds.), *Teaching and Learning in Higher Education* (pp. 1-25). Melbourne: Australian Council for Educational Research.

Kember, D., & Kwan, K. P. (2000). Lecturers' approaches to teaching and their relationship to conceptions of good teaching. *Instructional Science, 18*(5), 469-490.

Lave, J., Smith, S., & Butler, M. (1988). Problem solving as everyday practice. In R. I. Charles & E. A. Silver (Eds.), *The Teaching and Assessing of Mathematical Problem Solving* (1 ed., Vol. 3, pp. 61-81). Reston, VA: Lawrence Erlbaum

National Council of Teachers of Mathematics.

Lave, J., & Wenger, E. (1991). *Situated Learning: Legitimate Peripheral Participation* (1 ed. Vol. 3). Cambridge, UK: Cambridge University Press.

Leach, J., & Scott, P. (2003). Individual and sociocultural views of learning in science education. *Science and Education, 12*(1), 91-113.

Lerman, S. (1996). Intersubjectivity in mathematics learning: A challenge to the radical constructivist paradigm? *Journal for Research in Mathematics Education, 27*(2), 133-150.

Lerman, S. (2000). A case of interpretations of social: A response to Steffe and Thompson. *Journal for Research in Mathematics Education, 31*(2), 210-227.

Lerman, S. (2006). Socio-cultural research in PME. In A. Gutiérrez & P. Boero (Eds.), *Handbook of research on the psychology of mathematics education: Past, present, and future* (pp. 347-366). Rotterdam, The Netherlands: Sense.

Lesh, R., & Doerr, H. M. (2003). *Beyond consructivism: Models and modeling perspectives on mathematics problem solving, learning, and teaching*. Mahwah, NJ: Lawrence Erlbaum.

Lompscher, J. (2002). The category of activity as a principal constituent of cultural-historical psychology. In D. Robbins & A. Stetsenko (Eds.), *Voices within Vygotsky's Non-Classical Psychology: Past, Present, Future* (pp. 79-99). New York: Nova Science.

Mason, J. (1998). Enabling teachers to be real teachers: Necessary levels of awareness and structure of attention. *Journal of Mathematics Teacher Education, 1*(3), 243-267.

Mason, J. (2008a). Being mathematical with and in front of learners: Attention, Awareness, and Attitude as sources of differences between teacher educators, teachers, and learners. In B. Jaworski & T. Wood (Eds.), *The international handbook of mathematics teacher education: The mathematics teacher educator as a developing professional* (Vol. 4, pp. 31-55). Rotterdam, the Netherlands: Sense.

Mason, J. (2008b). PCK and beyond. In P. Sullivan & T. Wood (Eds.), *The International Handbook of Mathematics Teacher Education: Knowledge and Beliefs in Mathematics Teaching and Teaching Development* (Vol. 1, pp. 301-322). Rotterdam, the Netherlands: Sense.

Medina, J. (2008). *Brin rules: 12 principles for surviving and thriving at work, home, and school*. Seattle, WA: Pear.

Moyer, P. S., & Milewicz, E. (2002). Learning to question: Categories of questioning used by preservice teachers during diagnostic mathematics interviews. *Journal of Mathematics Teacher Education, 5*(4), 293-315.

National Council of Teachers of Mathematics. (2000). *Principles and Standards for School Mathematics*. Reston, VA: Author.

Pajares, M. F. (1992). Teachers' beliefs and educational research: Cleaning up a messy construct., 62(3), 307-332. *Review of Educational Research, 62*(3), 307-332.

Phillip, R. E. (2007). Mathematics teachers' beliefs and affect. In F. K. Lester (Ed.), *Handbook of research on mathematics teaching and learning* (pp. 257-315). Charlotte, NC: Information Age Publishing.

Piaget, J. (1970). *Genetic epistemology* (E. Duckworth, Trans.). New York: Columbia University.

Piaget, J. (1971). *Biology and knowledge* (B. Walsh, Trans.). Chicago: The University of Chicago.

Piaget, J. (1985). *The equilibration of cognitive structures: The central problem of intellectual development* (T. Brown & K. J. Thampy, Trans.). Chicago: The University of Chicago.

Pirie, S. E. B., & Kieren, T. E. (1992). Creating constructivist environments and constructing creative mathematics. *Educational Studies in Mathematics, 23*(5), 505-528.

Richardson, V. (1996). The role of attitude and beliefs in learning to teach. In J. Sikula (Ed.), *Handbook of research on teacher education* (pp. 102-119). New York: Simon and Schuster.

Runesson, U. (2008). Learning to design for learning: The potential of learning study to enhance teachers' and students' learning. In P. Sullivan & T. Wood (Eds.), *The International Handbook of Mathematics Teacher Education: Knowledge and Beliefs in Mathematics Teaching and Teaching Development* (Vol. 1, pp. 153-172). Rotterdam, the Netherlands: Sense.

Scherer, P., & Steinbring, H. (2007). Noticing children's learning processes – teachers jointlyr reflect on their own classroom interaction for improving mathematics teaching. *Journal of Mathematics Teacher Education, 9*(2), 157-185.

Shaffer, D. R. (1993). *Developmental Psychology*. Pacific grove, CA: Brooks/Cole.

Shulman, L. S. (1986). Paradigms and research programs in the study of teaching: A contemporary perspective. In M. C. Wittrock (Ed.), *Handbook of Research on Teaching* (pp. 3-36). New York: Macmillan.

Shulman, L. S. (1987). Knowledge and teaching: Foudnations of the new reform. *Harvard Educational Review, 57*, 1-22.

Simon, M. A. (1995). Reconstructing mathematics pedagogy from a constructivist perspective. *Journal for Research in Mathematics Education, 26*(2), 114-145.

Simon, M. A. (2006). Pedagogical concepts as goals for teacher education: Towards an agenda for research in teacher development. In S. Alatorre, J. L. Cortina, S. Mariana & A. Méndez (Eds.), *Proceedings of the Twenty Eighth Annual Meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (Vol. 2, pp. 730-735). Merida, Yucatán, Mexico: Universidad Pedagógica Nacional.

Simon, M. A. (2007). *Constraints on what teachers can learn from their practice: Teachers' assimilatory schemes.* Paper presented at the Thirty-First Conference of the International Group for the Psychology of Mathematics Education, Seoul, Korea.

Simon, M. A., & Tzur, R. (1999). Explicating the teacher's perspective from the researchers' perspectives: Generating accounts of mathematics teachers' practice. *Journal for Research in Mathematics Education, 30*(3), 252-264.

Simon, M. A., & Tzur, R. (2004). Explicating the role of mathematical tasks in conceptual learning: An elaboration of the hypothetical learning trajectory. *Mathematical Thinking and Learning, 6*(2), 91-104.

Simon, M. A., Tzur, R., Heinz, K., & Kinzel, M. (2004). Explicating a mechanism for conceptual learning: Elaborating the construct of reflective abstraction. *Journal for Research in Mathematics Education, 35*(3), 305-329.

Simon, M. A., Tzur, R., Heinz, K., Kinzel, M., & Smith, M. S. (2000). Characterizing a perspective underlying the practice of mathematics teachers in transition. *Journal for Research in Mathematics Education, 31*(5), 579-601.

Skemp, R. R. (1979). *Intelligence, Learning, and Action*. New York: Wiley.

Skinner, B. F. (1953). *Science and human behavior*. New York: Macmillan.

Steffe, L. P. (1990). Adaptive mathematics teaching. In T. J. Cooney & C. R. Hirsch (Eds.), *Teaching and learning mathematics in the 1990s* (pp. 41-51). Reston, VA: National Council of Teachers of Mathematics.

Steffe, L. P. (1992a). *On the nature of learning theory.* Paper presented at the International Congress of Mathematics Education (ICME-7), Quebec, Canada.

Steffe, L. P. (1992b). Schemes of action and operation involving composite units. *Learning and Individual Differences, 4*(3), 259-309.

Steffe, L. P. (2010). Articulation of the Reorganization Hypothesis. In L. P. Steffe & J. Olive (Eds.), *Children's fractional knowledge* (pp. 49-74). New York: Springer.

Steinbring, H. (1998). Elements of epistemological knowledge for mathematics teachers. *Journal of Mathematics Teacher Education, 1*(2), 157-189.

Sullivan, P., Mousley, J., & Zevenbergen, R. (2004). Describing elements of mathematics lessons that accommodate diversity in student background. In M. J. Hoines & A. B. Fuglestad (Eds.), *Proceedings of the 28th Conference of the International Group for the Psychology of Mathematics Education* (Vol. 4, pp. 257-264). Bergen, Norway: Bergen University College.

Thompson, A. G. (1992). Teachers' beliefs and conceptions: A synthesis of research. In D. A. Grouws (Ed.), *Handbook of research on mathematics teaching and learning* (pp. 127-146). New York: Macmillan.

Thompson, P. W., Carlson, M., & Silverman, J. (2007). The design of tasks in support of teachers' development of coherent mathematical meanings. *Journal of Mathematics Teacher Education, 10*(4), 415-432.

Thompson, P. W., & Saldanha, L. A. (2000). Epistemological analysis of mathematical ideas: A research methodology. In M. L. Fernández (Ed.), *Proceedings of the 22nd Annual Meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (Vol. 2, pp. 403-408). Tucson, AZ.

Torff, B., & Warburton, E. C. (2005). Assessment of teachers' beliefs about classroom use for critical thinking activities. *Educational and Psychological Measurement, 65*, 155-179.

Tzur, R. (2000). An integrated research on children's construction of meaningful, symbolic, partitioning-related conceptions, and the teacher's role in fostering that learning. *Journal of Mathematical Behavior, 18*(2), 123-147.

Tzur, R. (2007). Fine grain assessment of students' mathematical understanding: Participatory and anticipatory stages in learning a new mathematical conception. *Educational Studies in Mathematics, 66*(3), 273-291.

Tzur, R. (2008). Profound awareness of the learning paradox (PALP): A journey towards epistemologically regulated pedagogy in mathematics teaching and teacher education. In B. Jaworski & T. Wood (Eds.), *The international handbook of mathematics teacher education: The mathematics teacher educator as a developing professional* (Vol. 4, pp. 137-156). Rotterdam, the Netherlands: Sense.

Tzur, R. (2011). Can dual processing theories of thinking inform conceptual learning in mathematics? . *The Mathematics Enthusiast, 8*(3), 597-636.

Tzur, R. (accepted for publication). Brain-mind theory-driven task design. In E. Kelly, S. Jensen & J. Baek (Eds.), *NeuroMathEducation*. Cambridge, MA: Cambridge University Press.

Tzur, R., & Lambert, M. A. (in press). Intermediate participatory stages as ZPD correlate in constructing counting-on: A plausible conceptual source for children's transitory 'regress' to counting-all. *Journal for Research in Mathematics Education*.

Tzur, R., Simon, M., Heinz, K., & Kinzel, M. (2001). An Account of a teacher's perspective on learning and teaching mathematics: Implications for teacher development. *Journal of Mathematics Teacher Education, 4*(3), 227-254.

Tzur, R., & Simon, M. A. (2004). Distinguishing two stages of mathematics conceptual learning. *International Journal of Science and Mathematics Education, 2*, 287-304.

Varma, S., McCandliss, B. D., & Schwartz, D. L. (in press). Scientific and pragmatic challenges for bridging education and neuroscience. *Educational Researcher*.

von Glasersfeld, E. (1991). Abstraction, re-presentation, and reflection: An interpretation of experience and Piaget's approach. In L. P. Steffe (Ed.), *Epistemological foundations of mathematical experience* (pp. 45-67). New York: Springer-Verlag.

von Glasersfeld, E. (1995a). A constructivist approach to teaching. In L. P. Steffe & J. Gale (Eds.), *Constructivism in education* (pp. 3-15). Hillsdale, NJ: Lawrence Erlbaum.

von Glasersfeld, E. (1995b). *Radical constructivism: A way of knowing and learning*. Washington, D.C.: Falmer.

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes*. Cambridge, MA: Harvard University Press.

Vygotsky, L. S. (1986). *Thought and language*. Cambridge, Massachusetts: MIT.

Wang, T., & Cai, J. (2007a). Chinese (Mainland) teachers' views of effective mathematics teaching and learning. *Identities, Cultures, and Learning Spaces, 39*(4), 287-300.

Wang, T., & Cai, J. (2007b). United States teachers' views of effective mathematics teaching. *ADM Mathematics Education, 39*, 315-327.

Watson, A. (2008). Task transformation is the teacher's responsibility. In O. Figueras, J. L. Cortina, S. Alatorre, T. Rojano & A. Sepulveda (Eds.), *Proceedings of the 32nd Annual Conference of the International Group of Psychology of Mathematics Education* (Vol. 1, pp. 146-153). Morelia, Michoacán, Mexico: Michoacán University of Saint Nicholas of Hidalgo.

Wenger, E. (1998). *Communities of Practice: Learning, Meaning, and Identity*. NY: Cambridge University.

Wertsch, J. V., & Toma, C. (1995). Discourse and learning in the classroom: A sociocultural approach. In L. P. Steffe & J. Gale (Eds.), *Constructivism in education* (1 ed., Vol. 1, pp. 159-174). Hillsdale, NJ: Lawrence Erlbaum.

Wickman, P.-O., & Östman, L. (2002). Learning as discourse change: A sociocultural mechanism. *Science Education, 86*(5), 601-623.

Wilson, M. S., & Cooney, T. J. (2002). Mathematics teacher change and development. In G. C. Leder, E. Pehkonen & G. Torner (Eds.), *Beliefs: A hidden variable in mathematics education* (pp. 127-147). Dordrecht, the Netherlands: Kluwer.

Wood, T. (2005). Understanding Mathematics Teaching: Where We Began and Where We are Going. *Journal of Mathematics Teacher Education, 8*(3), 193-195.

Xie, X., & Carspecken, P. F. (2008). *Philosophy, learning and the mathematics curriculum: Dialectical materialism and pragmatism related to Chinese and U.S. mathematics curriculums*. Rotterdam, The Netherlands: Sense.

Zaslavsky, O. (2008). Meeting the challenges of mathematics teacher education through design and use of tasks that facilitate teacher learning. In B. Jaworski & T. Wood (Eds.), *The international handbook of mathematics teacher education: The mathematics teacher educator as a developing professional* (Vol. 4, pp. 93-114). Rotterdam, the Netherlands: Sense.

Zull, J. E. (2002). *The art of changing the brain: Enriching the practice of teaching by exploring the iology of learning*. Sterling, VA: Stylus.

GOOD MATHEMATICS TEACHING AND CLASSROOM NORMS – VIEWS OF SECONDARY SCHOOL STUDENTS

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This article describes research on students' views of what constitutes good mathematics teaching. A questionnaire with open ended questions was given to 106 students in six different mathematics classes with five different teachers, in four different upper secondary schools in Iceland. The questions were designed to get at students concerns and opinions of what constitutes good mathematics teaching and where the strengths of their particular teacher lie.

The students of all the teachers considered 'clear explanations of the content' as the teachers' main advantages as well as thorough mathematical knowledge and incisive teaching, while they also appreciated highly personal qualities such as patience, cheerfulness, and care for their students' progress. Remarks on the characteristics of good teaching witnessed different societal norms in the groups as well as different beliefs about students' own roles, the roles of others, teachers included, and the general nature of mathematical activity in school.

INTRODUCTION

The goal of this study is to contribute to an increased knowledge of the Icelandic upper secondary school, which has been subject to radical development during the last 40 years. The first part of that period was characterized by great expansion, while the second part has been characterized by coordination of curricula, adjustment to a different clientele and subsequent decentralization.

This article describes a study of students' view of what constitutes good mathematics teaching. Some teachers have the reputation that they are 'good', while others are, as most people, favoured by some, with others ignoring them or even blaming them for their own low achievement.

It is not easy to define good mathematics teaching, as the working habits of each teacher depend on the social norms and traditions dominant in the country, the specific school and in the interplay between the teacher and the individuals in each group. The assumed qualities of the teaching appear to be flavoured by how the teacher succeeds in forming traditions and social norms in interaction with the group.

THE ICELANDIC UPPER SECONDARY SCHOOL

The Icelandic upper secondary school has transformed during the last 40 years from being an elite school to becoming a school for all. Historically, the school rests on a Danish Latin School tradition and it was adjusted to Danish curriculum guides, at least in mathematics, until the 1940s or even the 1960s (Bjarnadóttir, 2010, pp. 100–102, 133–139, 163–168). Due to domestic circumstances the academic year in Iceland was only eight months. To account for the same syllabus as the Danes, with the addition of one more foreign language, Danish, the Icelandic students graduated one year later than Danish students, at the age of 20. In spite of gradually shorter summer breaks, presently three months, graduation with a matriculation examination is still at the age of 20. Certainly, a few students graduate earlier, with a considerable number graduating a year or two later.

Until 1953, the number of non-vocational upper secondary schools in Iceland was only three, all of them classical grammar schools. Their purpose was generally considered to produce competent candidates for professional university studies (Bjarnadóttir, 2010, pp. 132, 193, 318–330). Between 1966–1985 a number of new schools were established, many of them comprehensive schools, including a variety of vocational and technical education schools in addition to the traditional grammar school syllabus. Each operated based on their separate legislative act until 1988 (Law no. 107/1988; no. 72, 1989).

In 1989 and 1999 national curriculum guides were published (Ministry of Education, 1989; 1999), comparable to a leaving certificate syllabus, although no standardized examinations were established. The leaving certificate mathematics syllabus for the natural science stream is historically chosen so as to be a suitable preparation for engineering studies. It contains enumerative combinatorics; introduction to vector geometry; trigonometry, polynomial, trigonometric, exponential and logarithmic functions, sequences and series, and introduction to calculus. Schools may offer elective courses in more advanced courses of these and other topics, such as number theory. Following the new legislative act no. 92/2008, a new curriculum guide was published (Ministry of Education, 2011), which will provide the schools with considerably more freedom to organize their courses and study streams than previous guides.

Only a little research has been condducte on teaching in Icelandic upper secondary schools. Macdonald et al. (2007) researched the views of five lower secondary school teachers on teaching and learning science. It concerns a different school level, a different school subject, and the views of teachers, not students. Studies on mathematics teaching are missing. It is therefore of interest to learn about students' views on good mathematics teaching in Icelandic upper secondary schools. If the new legislative act of 2008 and the national curriculum guide of 2011 are to lead to reforms, researching its function and degree of success would be valuable.

THEORETICAL BACKGROUND

It is a task of general didactics, and mathematics didactics in particular, to research the environment that creates favourable conditions for mathematics learning, both for individuals and in study groups, such as a classroom. Cobb and Yackel (1996) suggest that conditions for learning depend both on individual psychological factors and social norms. The theories of Cobb and Yackel will be further explored below.

The question of good mathematics teaching has been widely considered, not least in connection with the international comparative studies, TIMSS and PISA. Many articles in international journals discuss comparisons of teaching to South-East and East Asia, where performances in the comparative studies have been higher than in Europe and North America. Groves (2009) has provided an overview of research papers of that kind. Groves' conclusion is that what is considered to be good mathematics teaching is deeply rooted in cultural values and beliefs. Furthermore, many factors influence teaching habits and possibilities to alter them, such as teachers' social status, in addition to stability in curricular matters and pedagogy.

Shimizu (2009) has studied what characterizes exemplary mathematics instruction in Japanese classrooms from the learner's perspective. Shimizu analyzed learners' views on what constitutes a 'good' mathematics lesson in post-lesson video-stimulated interviews with 60 students in three 'well taught' eight-grade mathematics classrooms. The most frequent response (45%) referred to students' understanding and thinking in the classroom. Other frequent responses referred to whole class discussion (26.7%), students presenting their ideas in the classroom (16.7%), other students' presentations and explanations (6.7%) and to teacher's explanation (16.7%).

COBB AND YACKEL'S FRAMEWORK

Cobb and Yackel (1996) have outlined an interpretive framework, where psychological factors, such as students' beliefs about their own roles and the role of others in the classroom, are coordinated with general social norms in the group. They analyzed classroom processes by using a framework of three layers in which psychological constructivist analyses of individual activity are coordinated with interactionist analyses of classroom interactions and discourse. The coordination of interactionism and psychological constructivism is the primary defining characteristic of the version of social constructivism that Cobb and Yackel refer to as the *emergent approach* or the *emergent perspective*. The framework is illustrated in Table 1.

SOCIAL PERSPECTIVE	PSYCHOLOGICAL PERSPECTIVE
1. Classroom social norms	2. Beliefs about own role, others' roles, and the general nature of a mathematical activity in school
3. Sociomathematical norms	4. Mathematical beliefs and values
5. Classroom mathematical practices	6. Mathematical conceptions and activity

Table 1. An interpretive framework for analyzing individual and collective mathematical activity at the classroom level (Cobb & Yackel, 1996, p. 177).

Each layer in Table 1 embodies a conjectured relationship between an aspect of the classroom microculture and the activity of the individuals who participate and contribute to it. The three layers concern the norms and beliefs which matter in a group learning mathematics, the norms, beliefs and values in direct connection with mathematics as a school subject, and the specific manner of doing mathematics and the conceptions and beliefs which the student develops.

For example, Cobb and Yackel take individual students' beliefs about their own role, others' role and the general nature of mathematics to be the psychological correlates of general classroom social norms. The conjectured relationship between classroom social norms and individual beliefs then implies that a teacher, who initiates and guides the recognition of classroom social norms, is simultaneously supporting individual students' reorganization of the corresponding beliefs. Cobb and Yackel consider that this explanatory power makes the framework particularly relevant to engaging in classroom-based research.

An example of the uppermost layer could be the ways of communication between the students and between themselves and the teacher (Skott, Jess & Hansen, 2007). For example a norm could be that the students work independently and ask the teacher to assist them when they need help, that they work together on a task in groups or that they observe what happens at the blackboard and are prepared to be questioned there.

Similarly, students' beliefs and values concerning mathematics and mathematical activity are mutually related to socio-mathematical norms which shape the mathematics teaching to the group. The socio-mathematical norms could for example be what may be counted as a sufficient explanation and what may be considered as a complete solution to a mathematical task. Such norms emerge among students and their teachers in their interactive communication.

The third layer concerns the subject content in a more narrow sense, while also there, students' beliefs on mathematics concepts are related to mathematical norms which have been shaped in the group on what are accepted truths in the topic concerned.

THE SEARCH FOR GOOD MATHEMATICS TEACHING

In order to seek answers to the question on the characteristics of good mathematics teaching, a survey was prepared. Five teachers (A1, B1, C1, D1 and D2) in four upper secondary schools, A, B, C and D respectively, were chosen as a convenience sample. The four schools are different and they have different roles.

- ✓ School A is a classical grammar school with a centuries-long history and it can select its students from a number of applicants.
- ✓ School B is called 'college bridge'; it offers a bridge for those who have not completed a matriculation examination into a university college at the normal age.
- ✓ Schools C and D are modern modular based comprehensive schools offering a variety of courses and streams, a number of them leading to a matriculation examination.

Two teachers, A1 and B1, were chosen for their reputation of being 'good' mathematics teachers, and C1 for a reputation for being a progressive teacher applying information technology. Two teachers, D1 and D2, were added to the research group upon a request from the headmaster of school D. All the teachers were teaching a typical syllabus of a natural science stream; polynomial, trigonometric and logarithmic functions. References to the teachers are written in masculine form, although the teachers were both men and women.

The teachers were visited in 2–5 lessons each. The researcher observed the lessons and a short questionnaire with open-ended questions was given to the students. This article concerns the results of two questions from the questionnaire, making some use of observations of teaching methods by the researcher and using replies to three other questions as background information. The two questions were:

- ✓ Which do you consider your teacher's main strength as a mathematics teacher?

- ✓ Anything you would like to add.

The students were thus not required to list the teacher's perceived negative qualities, while they had an opportunity to express themselves freely to the second question.

In three more questions the students were asked to which stream they belonged and how they had been doing in mathematics until and in the present course. The students in the groups in concern were either attending the natural science stream, the economics stream or preparing for higher technical education. Students in Schools A, C and D had generally been doing above average in earlier studies in mathematics.

All the five teachers participated willingly in the research as did the students, who had reached the age of 18 so their parents did not have to be contacted for permission. The students were informed that they could skip the questions if they chose to do so. All the students in the five groups replied willingly to the questionnaire, except two foreign exchange students, one by Teacher A1 and one by Teacher C1. A total of 106 students replied to the questionnaire (A1, 23 replies; B1, 34 replies; C1, 12 replies; D1, 17 replies; D2, 20 replies).

School A was visited in autumn term 2008. This classical grammar school has fixed classes where all the students in each class have the same programme in all subjects. All students attending this particular class were taking the natural science stream. They had generally been doing well in mathematics according to answers to the questionnaire. The students were 'questioned at the board', where they wrote their solutions to exercises prepared at home. The teacher advised them and commented as needed. The other students were expected to follow and correct their own papers if necessary. They were seldom assisted in their seats during the observation and the rooms were too small for the teacher to approach individuals easily.

School B was visited in spring term 2009. It offers a general preparatory programme for university college studies. The students were older than the normal 18–19 year age and some were former drop-outs. They had returned to school to improve and add to their knowledge in mathematics to prepare for higher technical or economics studies. Many felt that they had not been doing well in mathematics earlier, although presently it was going better. Although the students were divided into two groups, they studied the same syllabus with the same teacher. They were therefore recorded as one group. The students were attended to as a group, not as individuals in their seats.

School C, a modern modular-based comprehensive school, mainly accepting students from a particular urban area, was visited in spring term 2010. Most students were attending the natural science or economics stream and their performance in the lower-secondary compulsory school and first years in the present school had been above average. The teacher, C1, made extensive use of information technology with confidence. The teacher conducted the class from an electrical board, turning towards the students and asking them as a group, where some students responded more actively than others. Afterwards, the teacher walked around the room and assisted the students in their seats.

School D, a modern modular-based comprehensive school, accepting students from a certain semi-rural area, was visited in spring term 2010. There also, most students were attending the natural science or economics stream and their performance in the lower-secondary compulsory school and first years in the present school had been above average. Teachers D1 and D2 taught the same syllabus in parallel. D1's teaching was characterized by a presentation at the board, trying to assist individuals in their seats as time allowed afterwards. Teacher D2 rearranged the syllabus in different order from the textbook and from what D1 did. The syllabus was to be covered swiftly to allow time for recap. The teaching was characterized by very short presentations at the board while most of the time was devoted to walking around the room, allowing plenty of time to each student who reached for assistance.

RESULTS

As the questions were open-ended, the students could give as long replies as they pleased. The replies contained varying numbers of items, differently phrased. In reviewing, the same or similar items emerged in many replies in all the schools. After consideration the researcher chose ten categories and counted how often they occurred in some form or another concerning each teacher. The total results of the most frequent categories of remarks were gathered into Table 2 on the relative frequency of the remarks in percentages.

	Characteristics	**A1**	**B1**	**C1**	**D1**	**D2**
S1	Explains well, communicates well, reaches out to students, knows how to deliver the content.	78	59	25	63	47
S2	Is knowledgeable, interested in the content and mathematics in general.	43	18	42	37	12
S3	Incisive teaching, is well organized, returns homework promptly.	35	15	42	11	0
S4	Patient, reacts positively to queries and requests for assistance.	35	59	8	11	12
S5	Keen on everyone understanding, shows goodwill to all, sympathetic, fair.	22	53	25	37	12
S6	Funny, cheerful, jolly, in a good mood, relaxed.	13	26	25	0	6
S7	Keeps discipline, strict, firm.	30	0	25	0	6
S8	Talks fast, is in a hurry.	0	0	17	11	0
S9	Few, no advantages.	0	0	0	0	24
S10	Disorganized, confusing, complicates matters.	0	0	0	11	12

Table 2. A summary of the most frequent remarks by 106 students on their teacher's strengths, percentages.

Table 2 indicates that the 23 students of teacher A1 valued primarily the teacher's clear explanations, extensive knowledge, incisive teaching, patience and good discipline, but also keenness on everyone understanding. Other remarks on A1's teaching:

> "Possesses deep understanding of the content" (3 remarks), "... tries to widen students' understanding and insight into mathematics in general", "... questions students also at the board, which encourages one to do the homework" (2 remarks), "... wants everyone to be questioned at the board and solve a problem, but is slightly flexible, if you could not do the problem you are to solve, you can take another that you were able to do", "... I like also how well the teacher is prepared to help us with the problems we were to solve as homework if someone is very confident at the board meanwhile", "... solutions of tests and homework are good, clear", "... runs tests suitably often", "... tries to connect the syllabus to reality and make it practical", "... adds extra knowledge, which I appreciate", "... sometimes the teacher starts talking outside the syllabus which is always rewarding!", "... excellent teacher", "very competent teacher", "quality teacher" "... simply teaches well."

Table 2 reveals that the 34 students of Teacher B1 appreciated most the teacher's clear explanations, patience and goodwill to all. Other remarks on Teacher B1' teaching:

> "... one is encouraged to ask if one does not quite understand", "... the main point is that the teacher never assumes students' knowledge so that one never has inferiority feeling for the sake of lack of knowledge in math and dares to pose silly questions and make mistakes", "... talks to you as an equal, speaks in an understandable language, uses good metaphors", "... is a top-quality copy of a beautiful person in every understanding", "... good, excellent, the very best teacher (18 remarks) I have ever had during my whole schooling."

The 12 students of Teacher C1 valued most the teacher's knowledge and incisive teaching, in addition to clear explanations, cheerfulness, fairness and good discipline. Other remarks on Teacher C1's teaching are:

> "Treats the material <u>sometimes</u> thoroughly and is <u>most often</u> ready to treat it more closely if one does not understand what the matter is", "... keeps the plan well, could however treat the content better and more thoroughly ...", "Very good and cheerful teacher."

Table 2 indicates that the 17 students of Teacher D1 appreciated their teacher's clear explanations, knowledge and keenness on everyone understanding, while there were some remarks on hurriedness. Other remarks on Teacher D1' teaching are:

> "... tries to let you understand what you are doing", "... tries hard to explain so that everyone should understand", "... takes care to assist everyone", "... does not have enough time to assist everyone, is far too stressed", "... spends too much time at the board instead of assisting everyone for a while", "... does not respect a raised hand, "... goes often away without finishing helping you."

Teacher D2 earned some praise from his 20 students for clear explanations, while a considerable group did not give the teacher any credit. Other remarks on Teacher D2's teaching:

> "I understand what the teacher is ... explaining", " ... gets to the heart of the matter", "... allows us to do the tasks ourselves", "... explains the material in his own way, not just directly from the book", "... tries to plunge deeper into the material and wants the students

to understand the core of the matter and why all this works", "Does not explain well and twists one's questions, does not use good teaching methods. Treats all methods in several lessons and everything becomes mixed in your head", "… I find the teacher too disorganized, ... does not allow us to compute ourselves, switches topics, etc.", "... I find [Teacher D2] and Teacher D1 the best teachers."

DISCUSSION

Table 2 indicates several general points. Students appreciate clear explanations from their teacher, even if their content cannot be determined from the survey. They are also impressed by their teacher's wisdom, knowledge and understanding of the syllabus and organizing skills. Besides that they appreciate the teacher being patient, cheerful, sympathetic and fair, reacting positively to questions and requests about assistance, showing goodwill and being keen on everyone understanding. Keeping good discipline or even being strict is also counted as an advantage.

The research by Shimizu (2009) concerns students' experience of one particular lesson, while the current research concerns students' overall experience of the teacher and his qualities. The two surveys are therefore not fully comparable. They have, however, several traits in common. Japanese students appreciate being able to present their ideas in classroom (16.7%), which is also mentioned in Iceland but only in School A. Students' understanding has the highest score (45.0%) in the Japanese classroom. For the Icelandic teacher, the highest score is on clear explanations, also frequently mentioned in Japan (16.7%), and could be interpreted as leading to students' perceived better understanding. Furthermore, Icelandic students value highly that their teacher is keen on everyone understanding. One may therefore conclude that students' main expectations from a mathematics teacher and a mathematics classroom is their own 'understanding' by their teacher's 'clear explanations', a result that concerns the universal purpose of students attending schools. But how?

Surveying comments on individual teachers reveals different expectations and indications on different social norms in the classrooms. This may be discussed with reference to the Cobb and Yackel 1996 framework. We shall exclusively discuss social norms and students' beliefs presented in the uppermost layer in the framework, as the questionnaire did not offer opportunities to regard the two lower layers.

In School A students frequently mention their teacher returning homework and tests promptly and the good support from being questioned at the board. This encourages students to do their homework. They appreciate that the teacher is somewhat flexible and that a student who could not solve the problem assigned could take another that the student was able to solve. This may be considered as a social norm in the uppermost layer in Cobb and Yackel's framework, which furthermore is related to students' beliefs on the roles of themselves and others in the classroom and general mathematical activity on schools, here to prepare for the lessons and be called up to the board. Teacher A1 is credited for being prepared to assist students with their homework, while someone is working confidently at the board. The norm is that the student at the board has priority, while the teacher assists other students as possible.

The remarks from School B indicate that the role of the teacher is to explain and the role of the students to ask questions when they need. The students are encouraged to ask if they do not understand, and a norm has been established that they dare to ask silly questions and make mistakes, presumably related to Teacher B1's good mood and goodwill towards the students, which the teacher is highly credited for.

Teacher C1 is mainly credited with firm mathematical knowledge and organization, and secondly for being cheerful, fair and explaining well. The remarks do not cover teaching methods, although in visits to the classroom the researcher noticed the teacher's extensive use of information technology which made it easier to face the students. The students did not mention this; they seemed to regard it as a norm of the classroom. Complaints concerned the teacher's hurriedness when covering the syllabus and explaining, but not lacking assistance in seats. Teacher C1 was observed helping students in their seats after presenting at the board. The norm seemed to be that the presentation had priority, while assistance in seats was secondary and that students were comfortable with that order.

The majority of Teacher D1's students mentioned clear explanations but there was some mention of hurriedness. Teacher D1 did not seem to reach all those who requested assistance in their seats, if a notice is taken of a remark that the teacher spends too much time at the board instead of helping individuals in their seats; that the teacher does not react to a raised hand and often leaves without finishing helping the student. These remarks suggests that the students' beliefs about social norms in the classroom and their own and the teacher's roles, which have developed in the classroom, are that the students arrange themselves in a queue by their raised hands and that the teacher is to take care of individuals rather than explain at the board, while the teacher is possibly trying to avoid that role. A total of 17 students sat in Teacher D1's classroom. Even if the number is not large, one may assume that it is too much for the teacher to assist each student in one lesson. If such a norm has been developed, it may create a tension between the teacher and the students, and contribute to perceived hurriedness.

Similar norms were reflected in students' remarks about Teacher D2, while they did not mention hurriedness. The teacher allowed plenty of time to explain when asked about particular problems. The remarks were mixed, nearly half the group found the explanations clear, while remarks on disorganization were observed.

In general, social norms and mathematical activity, as defined in the uppermost layer of Cobb and Yackel's framework, were that students receive teacher explanations. The teacher's role was to explain, while students' role was to ask if they did not understand the teacher or the task they were working on. The students of School A, a classical grammar school found it normal to deliver their knowledge at the board and receive individual assistance there if necessary. That implies that the teacher chooses who is called up. Students may ask for individual assistance if the teacher does not have to assist the student at the board. In School B, the bridge to college for mature students, a norm seemed to be established that students were able to take the initiative to pose even silly questions. In School D, a comprehensive school with a modular programme, students considered the role of the teacher to assist students individually in their seats. The students may then have to wait until the teacher reaches them and they may then sit passively while waiting.

None of the comments mention as a role of the teacher to provide hints but leave it to the students to elaborate on them. This may though be included in assistance at the board and in the seats. Neither was there any remark about working together with other students on a mathematical activity and consulting with other students before asking the teacher.

Students' remarks regarding Teachers A1 and B1, with whom the students were most content, lead to the conclusion that their classroom social norms were different. Both groups appreciated clear explanations highly. In addition, students of Teacher A1 appreciated the teacher's knowledge and interest in mathematics, while the students of Teacher B1 appreciated most their teacher's patience and care to reach everyone. The student groups were also different, as the students of School A were selected into the school, while students in School B were returning back to school and were generally older than average. Both teachers seem however to have succeeded in creating such norms in cooperation with their students that both they themselves and their students are comfortable in their roles, and they are able to avoid tension, for example on covering the content too fast, or not being able to react to queries from individuals.

In relation to Groves' (2009) results that good teaching is deeply rooted in cultural values and beliefs, our research may indicate that this applies not only to different cultural geographical areas, such as South-East Asian and Western countries, but that different cultures may be created in different schools in the same country, not the least if the students' personal stories and the history and traditions of the schools are different.

Thus the values and beliefs of students in Schools A and B on own roles and the roles of their teacher differ considerably, as do their own personal stories and the stories and traditions of their schools. The personal stories of students in School B are different from the stories of others, as the students are returning back to school after a period of employment. They value patience and positive reactions to queries and requests for assistance considerably higher than students in other schools. School A differs from the other schools, being a selective classical grammar school with a centuries-long history and traditions, while Schools B, C and D are offspring of the expansion of the upper secondary level that began in 1966. Students in School A value most clear explanations and their teacher's knowledge. Their beliefs about their own roles are to prepare for the lessons and a social norm in their class is to be called up to the board to deliver the homework, which was not mentioned in remarks from students in other schools.

FINAL WORDS

The survey described above indicates that students value teacher's personal qualities and conduct, such as cheerfulness, patience and care for students, while what they appreciate most is that the teachers explain the syllabus in a way such that the students perceive that they understand it. The survey does not reveal fully what is needed to achieve that, but clear presentation, discipline in the classroom, absence of hurriedness and firm knowledge and interest in mathematics contribute to that perception.

The survey also indicates that good mathematics teaching is characterized by classroom social norms where the teacher is comfortable in his role and the students accept and know their own role in creating those norms. Highly positive norms have been created by Teachers A1 and B1, even if they and their schools are different, and in various respects this applies also to the other teachers participating in the survey.

REFERENCES

Bjarnadóttir, K. (2010). *Mathematical education in Iceland in historical context – Socio-economic demands and influences*. Saarbrücken: Lap Lambert Academic Publishing. Available at http://rudar.ruc.dk/bitstream/1800/2914/1/Chapter0_IMFUFA.pdf.

Cobb, P. og Yackel, E. (1996). Constructivist, emergent, and sociocultural perspectives in the context of developmental research. *Educational Psychologist*, 31, 175 –190.

Groves, J. (2009). Exemplary mathematics lessons: a view from the West. *ZDM The International Journal of Mathematics Education, 41*(3), 385–391.

Law no. 107/1988, no. 72/1989, no. 92/2008. [Legislation on the Icelandic upper secondary school].

Ministry of education (1990). *Námskrá handa framhaldsskólum*: *námsbrautir og áfangalýsingar*. Reykjavík.

Ministry of education (1999). *Aðalnámskrá framhaldsskóla – Stærðfræði*. Reykjavík.

Ministry of education (2011). *Aðalnámskrá framhaldsskóla*. Reykjavík. Retrieved August 10, 2011 http://www.menntamalaraduneyti.is/utgefid-efni/namskrar/nr/3954,

Shimizu, Y, (2009). Characterizing exemplary mathematics instruction in Japanese classrooms from the learner's perspective. *ZDM The International Journal of Mathematics Education*, *41*(3), 311–318.

Skott, J., Jess, K. and Hansen, H.C. (2007). *Matematik for lærerstuderende: Delta, fagdidaktik*. Frederiksberg: Forlaget Samfundslitteratur.

Thórólfsson, M., Macdonald, A. and Lárusson, E. (2007). Sýn fimm grunnskólakennara á nám og kennslu í náttúruvísindum. *Tímarit um menntarannsóknir, 4*, 83–99. Avaiable at http://www.fum.is/FUM_4_arg.pdf.

DESIGNING RICH SETS OF TASKS FOR UNDERGRADUATE CALCULUS COURSES

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Research has shown that the types of tasks assigned to students affect their learning. Studies have found that many mathematical tasks created for second and third level students promote instrumental rather than relational understanding, and imitative rather than creative reasoning. In this paper, we describe some frameworks that aim to guide teachers when writing tasks. From these frameworks a set of task types that are deemed appropriate for undergraduate students, and that foster mathematical habits of mind, have been selected: these are evaluating mathematical statements; example generation; analysing reasoning; conjecturing; generalising; visualisation; and using definitions. We report on an effort to design a set of tasks for an introductory calculus course using these principles and give examples of exercises on the topic of continuity. The teacher's role in the implementation of such tasks is also discussed.

INTRODUCTION

Skemp (1976) identified two types of understanding: relational and instrumental. Relational understanding refers to conceptual understanding and involves knowing both what to do and why to do it. Skemp explained instrumental understanding as 'rules without reasons' (p. 21) and conjectured that for many students and even teachers 'the possession of such a rule, and ability to use it, was what they meant by understanding' (p. 21). The main goal of a mathematics lecturer is to foster mathematical understanding in their students; however, many authors have expressed the view that mathematics at third level suffers from an over-emphasis on procedures and memorisation. Dreyfus (1991) asserted that many students learn a large number of standardised procedures in their university mathematics courses but fail to gain insight into how or why these procedures were developed. Cuoco, Goldenberg and Mark (1996) remarked that curricula are usually given in terms of the content of mathematics courses, but that the thought processes that created the mathematical results studied in the courses are rarely mentioned. They advocated inclusion of 'mathematical habits of mind' in order to 'give students the tools they will need in order to use, understand and even make the mathematics that does not yet exist' (p 376).

Closely related to mathematical understanding is the type of thinking or reasoning that students are expected to employ. Lithner (2008) has characterised reasoning into two main categories: imitative reasoning, and creative mathematically-founded reasoning. The term imitative reasoning refers to the use of memorisation or well-rehearsed procedures, while creative mathematically founded reasoning involves new (to the student at least) arguments that are mathematically well-founded. Boesen, Lithner and Palm (2010) used Lithner's framework to analyse the relation between the types of tasks assigned to high-school students and the types of reasoning employed by the students in solving these tasks. They found that when faced with familiar tasks students employed imitative reasoning and, in contrast, used creative mathematically-founded reasoning to tackle unfamiliar tasks. Boesen et al. (2010) claim that the solutions to familiar tasks required little or no conceptual understanding and they conjecture that exposure to these types of tasks alone limits the students' ability to reason and gain conceptual understanding. In other words, the use of familiar tasks fosters instrumental understanding but not relational understanding.

Studies carried out in universities in the UK (Pointon and Sangwin, 2003) and in Sweden (Bergqvist 2007) have found that, at least in introductory mathematics courses, students are mostly required to carry out procedural calculations and rarely need to use any higher order thinking skills. The consequences of this phenomenon were explored by Selden, Selden, Hauk and Mason (2000) in a study of second year calculus students. The students were given two tests. The first asked them to demonstrate familiar techniques; the second required them to use these techniques to solve non-routine problems. The study found that more than half of the students could not solve any non-routine problems although the first test had shown that they were technically competent. It seems that these students had instrumental but not relational understanding of the material. Selden et al. (2000) recommend that lecturers should regularly assign non-routine problems to students in order to develop their thinking skills.

The Irish Context

We are not aware of any research that has studied the types of tasks assigned to students in introductory calculus courses in third level institutions in Ireland. However much has been written recently on the teaching of mathematics at second level, and in particular on the examination system. Since the majority of our first year undergraduates proceed directly from second level education to third level, we outline here some studies that we feel are relevant to understanding the mathematical background of our freshman classes.

Students undertake a state examination in Ireland at the end of their time in post-primary school, called the Leaving Certificate examination, the results of which determine entry into third level education. The Chief Examiner's Report (2005) on the Leaving Certificate Mathematics Examination reported that students' strengths were 'most evident in procedural questions where a definite sequence of familiar steps was required' (p. 48) and that weaknesses related to 'inadequate understanding of the mathematical concepts and a consequent inability to apply familiar techniques in anything but the most familiar of contexts' (p. 49). Unwarranted levels of difficulty were caused by questions requiring students to display an understanding of underlying concepts. Problems were also encountered due to students' under-developed problem-solving and decision-making skills.

Elwood and Carlisle (2003) noted that the way in which questions were structured on the Leaving Certificate examination papers year-on-year seemed to reward the learning of rules and their application in familiar mathematical contexts, and that the traditional style of questions appearing on the papers possibly closely reflected the items pupils were exposed to from their textbooks. They suggested that this predictability of the examination questions was likely to influence not only how the students prepared for these examinations but also how they were taught. Conway and Sloane (2005) speak of this as the 'backwash effect' examinations have on curriculum, 'shaping both what is taught and how it is taught' (p. 32) by determining what is deemed to be valuable knowledge. They describe how ‘teaching to the test’ can enable students to perform well on examinations without engaging in higher levels of cognition.

Lyons, Lynch, Close, Sheerin and Boland (2003) carried out a video study of twenty post-primary mathematics lessons in ten schools in Ireland. They found that teaching was strongly didactic, and that a procedural rather than a conceptual approach prevailed. Hourigan and O’Donoghue (2007) also observed two second-level senior-cycle Irish classrooms over 10 weeks and found them to be focused on examinations and on the mastery of algorithmic procedures as isolated skills, with ready-made mechanisms provided for the pupils to aid memorisation. Teachers tended to do the pupils’ thinking for them and a ‘learned helplessness’ in the pupils was repeatedly reinforced. Hourigan and O’Donoghue (2007) contended that the inability of such students to successfully make the transition to tertiary level mathematics education lies in the substantial mismatch between the nature of their pre-tertiary mathematics experiences and their subsequent mathematics-intensive tertiary level courses. They described mathematics-intensive courses at third level as requiring students to be independent learners, displaying conceptual skills and the ability to solve unfamiliar problems. However, the fostering of such skills was not observed in the second level classrooms studied.

Mason (2002) explains how students who can perfectly complete routine problems are not necessarily prepared to ‘tackle non-routine problems which require juxtaposing ideas, adapting familiar techniques, or are simply more complex than the ones with which they are familiar’ (p. 33). In order to enable our students to overcome their possible weaknesses in applying techniques in unfamiliar contexts and to develop ‘mathematical habits of mind’, it is necessary to focus primarily on tasks which foster relational understanding. Skemp (1976) describes the development of instrumental understanding as requiring the learning of an increasing number of fixed ‘plans’ whereby students can find their way to the answers of a fixed number of questions. Whereas relational understanding can be explained as relying on the learners’ building of a conceptual structure from which he/she can answer an unlimited number of questions. It is such relational understanding that will combat ‘learned helplessness’ and enable students to think independently. To this end, in this paper we will consider some frameworks which aim to guide the creation of tasks that will foster good mathematical habits of mind and aid the development of relational understanding. As many of these frameworks were designed to be used at second level we will report on the task types that we feel are most useful for third level courses. We will illustrate these task types by giving examples of tasks on the theme of continuity.

TASK FRAMEWORKS

Before we consider task frameworks we will explain what is meant by ‘mathematical habits of mind’. Cuoco et al. (1996) proposed that students need to conjecture, experiment, visualise, describe, invent and generalise. They also advocate the precise use of mathematical language. Bass (2005) claimed that these habits of mind are precisely those used by research mathematicians, and that these practices can, and should, be fostered at all educational levels.

Mason's Framework

How then can we design tasks which will allow our students to develop these habits of mind? Mason and Johnston-Wilder (2004a) suggest that questions posed to students should involve the practices employed by mathematicians when conducting research. To this end they propose that the following words be used when designing tasks: 'exemplifying, specialising, completing, deleting, correcting, comparing, sorting, organising, changing, varying, reversing, altering, generalising, conjecturing, explaining, justifying, verifying, convincing, refuting', (p 109). Watson and Mason (1998, p. 7) grouped the words in this list and gave a table of questions and prompts that could be used to generate the thinking processes in each section. For example, the prompts associated with the processes of exemplifying and specialising, and generalising and conjecturing are shown in Table 1.

Exemplifying Specialising	Give me one or more examples of … Describe (show, choose, find, …) an example of … Is … an example of …? What makes … an example? Find a counter example of… Are there any special examples of …?
Generalising Conjecturing	What happens in general? Of what is this a special case? Is it always, sometimes, never …? Describe all possible … as succinctly as you can. What can change and what has to stay the same so that … is still true?

Table 1: Watson and Mason's (1998) questions and prompts

Swan's Framework

Swan (2008) described his framework of five tasks types which he selected in order to promote conceptual understanding amongst secondary school students. These task types encourage the development of skills such as classifying, interpreting, comparing, evaluating and creating. The five task types are: classifying mathematical objects; interpreting multiple representations; evaluating mathematical statements; creating problems; analysing reasoning and solutions. In the first of these, students devise or apply classifications. In doing this, they gain an appreciation for the properties of objects and for the importance of definitions and mathematical language. In the second task type, interpreting multiple representations, students must link different representations of the same object. Swan claims that this process leads to the development of new mental images. The third type of task involves students deciding whether mathematical statements are always, sometimes, or never true and justifying their reasoning. These tasks encourage the use of examples and counter examples. The fourth task type asks students to create problems for their classmates to work on. This naturally involves the notions of doing and undoing. Students are asked to compare solutions or to diagnose errors in the last task type. This allows them to appreciate the possibility of more than one solution to a problem and enhances their reasoning skills.

Swan (2007) trialled the use of these task types with teachers on a professional development course. It is interesting to note that, during the study, many of the teachers changed their beliefs and practices and moved from a teacher-centered to a student-centered approach to teaching.

Pointon and Sangwin's Framework

Sangwin (2003) approached the chore of categorising mathematical tasks used in university courses from a different perspective. Taking into account the backwash effect of examinations, he took the view that 'any attempt to elaborate on what is meant by mathematical skills must be based on an analyis of what *in reality* we ask students to do' (p. 814) through the assessment criteria for a particular course. Together with Pointon (Pointon & Sangwin, 2003), he developed a taxonomy with eight classes of mathematical questions, distributed over two levels as shown in Table 2.

1. Factual recall	5. Prove, show, justify (general argument)
2. Carry out a routine calculation or algorithm	6. Extend a concept
3. Classify some mathematical object	7. Construct an instance
4. Interpret situation or answer	8. Criticize a fallacy

Table 2: Pointon and Sangwin's (2003) mathematical question taxonomy

Tasks falling into categories 1-4 are said to be those of 'adoptive learning' which involves students applying well-understood knowledge in bounded situations. On the other hand, tasks which typically require higher cognitive processes, such as those described by categories 5-8 (and sometimes 4), are deemed to require students to behave in a more sophisticated way mathematically and are characteristic of 'adaptive learners'.

Comparison of Frameworks

There is some overlap evident in the frameworks outlined above, for instance, Mason and Johnston-Wilder (2004a) speak of 'exemplifying' and 'specialising', while Pointon and Sangwin (2003) use the term 'construct an instance' to describe the same mathematical skill. However, due to the difference in the motivation or perspective of the authors involved, some of the frameworks focus on different aspects of mathematical thinking than others. Recall that Pointon and Sangwin were interested in categorising the types of mathematical skills actually assessed in first year undergraduate mathematics courses, while Mason and Johnston-Wilder aspired to describe the types of questions students *should* be asked in order to facilitate the development of mathematical habits of mind, and Swan (2008) focussed on task types teachers could use to foster conceptual understanding (in particular) in their students. Nonetheless, one thing many of those discussed have in common is the broad categorisation of the task types they comprise into two levels. This is also true of the theories used to describe types of mathematical understanding and reasoning. Skemp (1976) wrote of ‘instrumental’ versus ‘relational understanding’, Pointon and Sangwin (2003) use the terms ‘adoptive’ and ‘adaptive learning’, while Lithner (2008) speaks of ‘imitative’ and ‘creative reasoning’. Similarly, learners are described as being in ‘accepting’ or ‘asserting mode’ by Mason and Johnston-Wilder (2004a). Each of these descriptions can be facilely understood as distinguishing between a student behaving mathematically as a competent practitioner or as an expert. As a competent practitioner, a student can succeed on familiar, routine tasks, but in order to construct and reconstruct mathematics for himself, a student must have developed the skills of an expert. The latter typically require higher level cognitive processes such as reflection, creativity and criticism and stem from the mathematical habits of mind advocated by Cuoco et al. (1996) and Bass (2005).

TASKS DESIGNED

During the academic year 2010/2011, both authors taught courses on differential calculus to groups of first year students in their respective third level institutions. The courses were delivered to large mixed-ability groups of students and were calculus rather than analysis courses. The emphasis in this type of introductory calculus module is often on procedures and calculations.

In redesigning a set of homework tasks for our first year calculus courses, we realised that some of the task types discussed in the previous section seemed more appropriate for our groups of students than others, as most of the frameworks discussed above concern the creation of tasks for secondary school pupils. In drawing on these frameworks, we wanted to help our students make the transition from thinking about mathematics in a procedural or instrumental manner to using the habits of mind of a research mathematician. In particular, we wanted our students to gain experience of behaving like an expert. Thus, for example, instead of asking students to prove that a mathematical statement is true, we sometimes asked them to decide if it was true or not and to justify their answer. The act of making a decision about a mathematical statement was novel and challenging to our students (Breen & O’Shea 2011).

From the variety of types of tasks and prompts discussed earlier, the seven task types that we felt were most useful in first year undergraduate calculus courses in an Irish context were: **evaluating mathematical statements**; **example generation** (constructing an instance); **analysing reasoning**; **conjecturing**; **generalising**; **visualisation**; **using definitions** (to classify a mathematical object, for instance). We do not include tasks that ask students to prove a result here, not because we do not think these tasks are useful, but because we feel that they are present in most mathematics courses.

In order to illustrate these task types we will concentrate on the topic of continuity and give some tasks under the headings above.

Examples

Evaluating Mathematical Statements

Suppose f is continuous and never zero on [a,b].

Is it possible that $f(z)<0$ for some z in [a,b] and $f(w)>0$ for some w in [a,b]? Explain.

Example Generation

Give an example of the following:

1. A function f which is continuous at $x = 5$.
2. A function f which is not continuous at $x=5$ because $f(5)$ is not defined.
3. A function f which is not continuous at $x = 5$ because $Lim_{x \to 5} f(x)$ does not exist.
4. A function f which is not continuous at $x = 5$ because $Lim_{x \to 5} f(x) \neq f(5)$.

Analysing reasoning

Consider the following argument. Decide whether the reasoning used is satisfactory, justifying your answer.

Statement: Let f and g be functions that are continuous everywhere, then f/g is continuous everywhere.

Proof: Let c be a real number. Since f and g are continuous at c we know that $Lim_{x \to c} f(x) = f(c)$ and that $Lim_{x \to c} g(x) = g(c)$. Thus $Lim_{x \to c} f(x)/g(x) = f(c)/g(c)$ and so f/g is continuous at $x=c$. Since c is arbitrary, f/g is continuous everywhere.

Conjecturing

Suppose f is a continuous function on [a,b] and suppose that $f(a)<0$, and that $f(b)>0$.

What can you say about the number of times that the graph of f crosses the x-axis?

What about the number of times the graph of f touches the x-axis?

Generalising

The function $f(x) = (x^2 - 1)/(x - 1)$ has a removable discontinuity at $x = 1$.

(a) Describe a family of functions which have a removable discontinuity at $x = 1$.

(What can change and what must stay the same?)

(b) Describe two families of functions which have a removable discontinuity at $x = c$.

Visualisation

Draw a rough sketch of the graphs of the following functions:

1. A function *f* which is continuous everywhere except at $x=-3$ and $x=4$.
2. A function *g* which is continuous everywhere except for removable discontinuities at $x=-3$ and $x=4$.
3. A function *h* which is continuous everywhere except for a removable discontinuity at $x=-3$ and a non-removable discontinuity at $x=4$.

Use of definitions

Let $\left[x\right]$ denote the integer part of x. Is the function $f(x)=\left[x\right]$ continuous on the interval (0,1)? What about on the interval [0,1]? How about on the interval [0,1)?

(Further tasks and students' reactions to them can be found in Breen & O'Shea (2011).)

In designing calculus tasks we were mindful of the need to create a rich variety of tasks as it was imperative that the coursework should not become predictable nor a particular type of task become over-familiar. Mason and Johnston-Wilder (2004a) advocate what they term a 'mixed economy' (p. 6) of tasks as no single strategy or task type has proved to be universally successful in developing mathematical thinking. Moreover, Schoenfeld (1992) explains how any one task could not be expected to foster all types of mathematical thinking and thus, he emphasizes the need for balance in designing a set of tasks, so that as many different 'dimensions' of mathematical thinking as possible can be addressed. To ensure that students encounter a range of different possibilities, Bell (1993) also advises that the structure and format of tasks presented to students should be varied.

IMPLEMENTATION OF TASKS

Conditions for effective learning are complex: designing a rich set of tasks is not sufficient in itself and there is no guarantee that the learner will successfully develop the desired mathematical skills on the completion of such tasks. Researchers have drawn a distinction between the intended, implemented and attained curricula (Mason & Johnston-Wilder, 2004a). It is particularly important to be aware of the 'didactic tension' present in a classroom or teaching situation by which the more clearly the behaviour expected from learners in undertaking a task is expressed in the instructions accompanying it, the more the teacher deprives the learners of the conditions needed for learning and understanding (Mason & Johnston-Wilder, 2004b). Henningsen and Stein (1997) warn that it is not enough for a teacher to select and appropriately set-up worthwhile mathematical tasks: he/she must also provide support for students' cognitive activity in order to ensure that the complexity and cognitive demands of the tasks are not eroded during implementation. Furthermore, Stein, Grover and Henningsen (1996) found that those mathematical tasks which were designed to be the most cognitively demanding (e.g. involving conjecturing, justifying, interpreting) were those most likely to decline into somewhat less-demanding activities in implementation. Their research also found that tasks which built on students' prior knowledge were most likely to maintain high-level cognitive engagement.

Bell (1993) recommends that tasks should be attempted by students initially and only when their responses have been given should the teacher intervene to offer hints or help towards a solution. He also asserts that learning takes place more effectively when the learning situation is 'managed' by the teacher by, for instance, adjusting the challenge of a task presented to keep it at an appropriate level for all learners. Lampert (1990) advocated a public demonstration of mathematical thinking in the classroom, through engaging in mathematical arguments with her students, encouraging the students to make conjectures and to muster appropriate evidence to support or challenge each others' assertions in a discursive manner. She believed that students would not learn a different way of thinking about what it means to do mathematics simply by being told what to do or having mathematical problems explained to them. Mason (2002) also suggests that possibly what 'students need most is to be in the presence of someone who is 'being mathematical''(p. 4). In short, the practices of the teacher matter.

CONCLUDING REMARKS

In this paper, we have drawn on the work of Mason and Johnston-Wilder (2004a), Swan (2008), and Pointon and Sangwin (2003) to gather together a set of task types that are appropriate for use in an undergraduate course. The task types have been chosen to promote the mathematical habits of mind described by Cuoco et al. (1996).

Mason (2002) contends that 'in a sense, all teaching comes down to constructing tasks for students…This puts a considerable burden on the lecturer to construct tasks from which students actually learn' (p. 105). In doing so, a balance must be struck between sufficient and excessive challenge in order to avoid disillusioning students and destroying their confidence. Keeping this in mind, tasks similar to those outlined above were interspersed with more familiar, sometimes procedural, tasks on the homework sets distributed to students. Some preliminary feedback from students describing their reactions to a small number of routine and non-routine tasks and their understanding of the purposes of the tasks is described elsewhere (Breen & O'Shea, 2011). Responses gathered from the students indicated that they had a mature understanding of the purposes of different tasks. We hope to undertake further analysis of this data and to gather further data to examine aspects of these tasks more closely.

REFERENCES

Bass, H. (2005). Mathematics, mathematicians and mathematics education, *Bulletin of the American Mathematical Society*, 42, 417-430.

Bell, A. (1993). Principles for the design of teaching, *Educational Studies in Mathematics*, 24(1), 5-34.

Bergqvist, E. (2007). Types of reasoning required in university exams in mathematics, *Journal of Mathematical Behavior*, 26, 238-370.

Boesen, J., Lithner, J, & Palm, T. (2010). The relation between types of assessment tasks and the mathematical reasoning students use. *Educational Studies in Mathematics*, 75, 89-105.

Breen, S. & O'Shea, A. (2011). The use of mathematical tasks to develop mathematical thinking skills in undergraduate calculus courses – a pilot study. In Smith, C. (Ed) *Proceedings of the British Society for Research into Learning Mathematics*, 31(1), 43-48.

Conway, P.F. & Sloane, F.C. (2005). *International trends in post-primary mathematics education: perspectives on learning, teaching and assessment*. Dublin: NCCA.

Cuoco, A., Goldenberg, E.P. & Mark, J. (1996). Habits of mind: an organising principle for mathematics curricula. *Journal of Mathematical Behaviour*, 15, 375-402.

Dreyfus, T. (1991). Advanced mathematical thinking processes. In D. Tall (Ed). *Advanced Mathematical Thinking.* Dordrecht: Kluwer Academic Publishers.

Elwood, J. & Carlisle, K. (2003). *Examining gender: gender and achievement in the Junior and Leaving Certificate Examinations 2000/2001*. Dublin: National Council for Curriculum and Assessment.

Henningsen, M. & Stein, M.K. (1997). Mathematical tasks and student cognition: classroom-based factors that support and inhibit high-level mathematical thinking and reasoning, *Journal for Research in Mathematics Education,* 25(5), 524-549.

Hourigan, M., & O'Donoghue, J. (2007). Mathematical under-preparedness: the influence of the pre-tertiary mathematics experience on students' ability to make a successful transition to tertiary level mathematics courses in Ireland. *International Journal of Mathematical Education in Science and Technology*, 38 (4), 461-476.

Lithner, J. (2008). A research framework for creative and imitative reasoning. *Educational Studies in Mathematics*. 67 (3), 255-276.

Lyons, M., Lynch, K., Close, S., Boland, S. & Sheerin, E. (2003). *Inside classrooms: a study of teaching and learning in Irish second-level schools.* IPA: Dublin.

Mason, J. (2002). *Mathematics teaching practice: a guide for university and college lecturers.* Chichester: Horwood Publishing Ltd.

Mason, J., & Johnston-Wilder, S. (2004a). D*esigning and using mathematical tasks*, St Alban's: Tarquin Press.

Mason, J. & Johnston-Wilder, S. (2004b). *Fundamental constructs in mathematics education,* Oxon: RoutledgeFarmer.

Pointon, A., & Sangwin, C. (2003). An analysis of undergraduate core material in the light of hand-held computer algebra systems, *International Journal of Mathematical Education in Science and Technology*, 34(5), 671-686.

Sangwin, C. (2003). New opportunities for encouraging higher level mathematical learning by creative use of emerging computer assessment, *International Journal of Mathematical Education in Science and Technology,* 34(6), 813-829.

Schoenfeld, A. H. (1992). A *framework for balance: Balanced Assessment Project*, Unpublished document, prepared for NSF-sponsored Calculus assessment workshop.

SEC (State Examinations Commission), *Chief Examiner's Report on Leaving Certificate Mathematics* 2005. Available at http://www.examinations.ie/archive/examiners_reports/cer_2005/LCMathematics

Selden, A, Selden, J., Hauk, S. & Mason, A. (2000). Why can't calculus students access their knowledge to solve non-routine problems? In *Research in collegiate mathematics educations IV* ed. E. Dubinsky, A. H. Schoenfeld and J. Kaput, 128-153. Providence, RI: American Mathematical Society.

Skemp, R. (1976). Relational understanding and instrumental understanding. *Mathematics Teaching*, 77, 20-26.

Stein, M.K., Grover, B.W. & Henningsen, M. (1996). Building student capacity for mathematical thinking and reasoning: an analysis of mathematical tasks used in reform classrooms, *American Educational Research Journal,* 33(2), 455-488.

Swan, M. (2007). The impact of task-based professional development on teachers' practices and beliefs: a design research study, *Journal of Mathematics Teacher Education*, 10, 217-237.

Swan, M. (2008). Designing a multiple representation learning experience in secondary algebra, *Educational Designer*, 1(1), 1-17.

Watson, A. and Mason, J. (1998). *Questions and prompts for mathematical thinking*. Derby: Association of Teachers of Mathematics.

A STUDY OF THE BENEFITS OF USING THE HISTORY OF MATHEMATICS IN TRANSITION YEAR

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In this study we designed a module of twenty lesson plans on the history of mathematics for implementation in transition year in a second level school. The module was implemented by the first author in a transition year class in an all girls school. Students provided feedback before, during and after the module and we analysed their feedback. The two main results of our qualitative analysis are that there was an instant change of all students' perception of mathematics for the better after the first lesson plan and also that there was a sustained change of all students' perception of mathematics for the better after all twenty lesson plans. This shows that the history of mathematics has a powerful immediate (and sustained) positive impact on students' perception of mathematics. We also discuss some positive changes in students' attitudes in relation to mathematics and humanity, creativity, beauty, practical power and history. We feel this paper helps to fill a gap in the current research on using the history of mathematics in second level schools in Ireland.

INTRODUCTION

The discussion about whether or not the history of mathematics should be incorporated into the teaching and learning of mathematics has been going on for many years. Some examples of suggestions on why the history of mathematics should form part of the teaching and learning of mathematics are that it may increase students' understanding and could enable them to see mathematics as a human subject (Fried, 2007; Gulikers & Blom, 2001, p. 229) and it may positively change students' and teachers' attitudes (Barbin, 2000; Lui, 2003). According to Jankvist (2009a) there are two main approaches, which are "history as a goal" and "history as a tool". He argues that there are three ways of integrating the history of mathematics into mathematics education; the illumination approach, the module approach and the history-based approach. Our research takes the form of the module approach, whereby we have designed twenty lesson plans on topics from the history of mathematics that students do not generally experience in the Irish curriculum.

According to Puig and Rojano (2004) there is "no need to further discuss the necessity or usefulness of studying the history of mathematics for mathematics education" and instead current research papers are making recommendations for "empirical investigations on the effectiveness of using history" (Jankvist, 2009b). We feel that our research helps fill the gap in this area because we analysed student feedback to our module on the history of mathematics. At second level Lit, Siu and Wong (2001) implemented an experiment over a period of three weeks in Hong Kong. The effects of teaching Pythagoras' Theorem using the history of mathematics on students in the experimental group was compared to a control group who were taught using current practise. Using statistical analysis they assessed if students' enjoyment and attitude changed after completion of the module. The results showed that the enjoyment level of the students in the experimental group rose compared to a drop in the control group. More recently a study was carried out by Jankvist (2009b) in the Danish

Upper Secondary school with 23 students. He designed two modules on “history as a goal”. The first module was on the history of error correcting codes and the second was on the history of RSA Cryptography. The module contained both the teaching material and the classroom activities for the teacher to use. The modules had a positive effect on the students. Our current research adds to the two empirical studies just mentioned above because firstly, our work took place in an Irish setting and secondly the person who implemented the module in our case was both a fully qualified second level mathematics teacher and a co-designer of the module.

The practical issues, in using the history of mathematics, for the teacher have been outlined in numerous papers (Tzanakis & Arcavi, 2000, p. 203) with the main difficulties associated with its use being the lack of teaching resources, preparation time and pedagogical knowledge on behalf of the teacher. Our project also helps to resolve these difficulties by creating many new lesson plans, along with background information on various historical topics and student activities for the teacher. There are some resources on using the history of mathematics for teaching at second level (Ó Cairbre, Watson & McKeon, 2006; Ó Cairbre, 2009).

IRISH CONTEXT

Mathematics is one of three compulsory subjects at the second level in the Irish education System. Does the curriculum include the history of mathematics? Yes, it is stated in the Junior and Leaving Certificate Syllabi that the history of mathematics is an objective of the course.

> “They should be aware of the history of mathematics and hence its past, present and future role as part of our culture.” (NCCA 2000 Junior Certificate Mathematics Syllabus, Objective J, p. 4)

Currently in Ireland the NCCA, who design and publish the mathematics curriculum for all levels of the Irish Education System, are revising the mathematics syllabi. In 2008 they began implementing a trial in 24 pilot schools of a new programme known as Project Maths. The new syllabus has moved out of its pilot stage and in September 2010 began to be phased in across all secondary schools in Ireland over a 4 year period and will be completed by June 2015. The main aim of the curriculum in Project Maths is teaching for understanding. Also other objectives are to present mathematics to the student from the concrete to the abstract and to develop their problem solving skills. This new programme also refers to the history of mathematics.

> “ Develop positive attitudes to mathematics as a result of being able to use mathematical methods successfully; acknowledge the beauty of form, structure and pattern in mathematics; and appreciate its history and its role in our lives.” (NCCA 2010 Project Maths Junior Certificate Syllabus, Objective F, Section 1.3,)

So why then is it not taught by many teachers? One may argue that it is never examined by the State Examinations Commission (SEC) who design the terminal examinations. However, some reports, in recent years, have highlighted the lack of pedagogical knowledge of some qualified mathematics teachers of all levels (Conway & Sloane, 2005). Ní Ríordáin and Hannigan (2009) argue that 48% of mathematics teachers surveyed in secondary schools in

Ireland were not qualified mathematics teachers according to the Teaching Council Standards. The U.S. Department of Education (2008) concluded that there is a strong correlation between students' achievement level and teachers' content knowledge.

In the Irish System there is a fourth year between the Junior and Leaving Certificate Courses. It is not a compulsory year set by the Department of Education and is often optional in many schools. This year is called Transition Year. Approximately 75% of schools provide this one year course (see the website NCCA.ie). This year is not examinable by the SEC and allows students to develop new skills and ideas about mathematics. It gives the teacher the freedom to show different aspects of mathematics and gives the perfect opportunity for students to engage with the history of mathematics. The majority of time is supposed to be spent on non syllabus material.

PROJECT OUTLINE AND METHODOLOGY

Our project had three parts:

(a) We designed a module of twenty lesson plans on a wide variety of topics from the history of mathematics for use by teachers in transition year. Each lesson plan lasted forty minutes. We also designed questionnaires and reflective journals for student feedback.

(b) The first author, who is a qualified second level mathematics teacher, implemented the twenty lesson plans to twenty-one transition year students in an all girls school. The lesson plans were implemented over a 3 month period in Spring 2010. The first author also gathered the questionnaires and the reflective journals that were filled out by the students. Each student was given a student number at the beginning of the module to protect their identity and allow them the freedom to express their thoughts and opinions without prejudice.

(c) We analysed the student feedback using Grounded Theory (Strauss & Corbin, 1990). We felt this qualitative approach, rather than a quantitative approach, was an appropriate one to obtain students' thoughts and feelings about mathematics and related issues. We had no preconceived ideas on what the feedback would produce. A motto for Grounded theory is "All is data", which means that the results in Grounded Theory should be fully justified and backed up by the data. We allowed concepts and categories to emerge. The first step in the so called coding process in Grounded Theory involves breaking down the data and then labeling the individual concepts. Then the individual categories are formed by grouping concepts that pertain to the same phenomena. Finally, the Core Category (if it exists) is the central phenomenon around which all the other categories are integrated.

In relation to (b) above, the first author used a multitude of teaching styles including discussion, chalk and talk, DVD and videos, group work, individual study and active learning methodologies. The classes were structured to bring students on a mathematical journey from a long time ago to current day applications. Most importantly the topics were chosen to show students the humanity in mathematics that seems devoid in the teaching and learning of mathematics (Rickey, 1996). The lesson plans had four main objectives:

1. Demonstrate mathematical topics as they developed through time.

2. Enable students to see the connections within mathematics and with other school subjects (music, art, science etc.) and also with the practical world.
3. Link the history of mathematics to the junior and leaving certificate curriculum.
4. Discuss the effect mathematics has on the cultural and social development of society (and vice versa).

Table 1 below gives a list of the lesson plans implemented after students completed the pre-questionnaire. Although most of the topics were chosen by the authors, it is interesting to note that some were chosen as a result of students asking to learn more on a particular area. In this way students had some ownership of the lesson plan topics and this made a very positive impression on the students.

Lesson Plan Number	Titles of Lessons
1	What does mathematics mean to you?
2	Practical power of mathematics
3	Beauty in mathematics
4	The history of number writing
5	Using old number systems and Egyptian doubling
6	Symbolism in algebra
7	Archimedes in his bath
8	Death in mathematics
9	The fly on the wall
10 & 11	Napier's bones and Donald duck in mathemagicland
12	The golden ratio
13	Fibonacci numbers
14	Pythagoras' Theorem
15	A lucky dip into mathematical history
16	The story of Pi
17	Women in mathematics
18	Two main pillars of mathematics
19 & 20	Secret gathering of mathematicians including special guest, William Rowan Hamilton

Table 1: Lesson plans implemented after students completed the pre-questionnaire

We will now give a very brief summary of two of the lesson plans. Lesson plan 14 was on Pythagoras' theorem. The class began with a discussion of Pythagoras' life, his secret society of followers and society at the time. The class then went on to look at various proofs of the theorem and the practical uses of Pythagoras' theorem. The students were then introduced to Pythagorean Triples which then led on to a discussion of Fermat's last theorem and Andrew Wiles who proved Fermat's last theorem in the 1990's.

Women in mathematics was the focus of lesson plan 17. It was a historical overview of some of the greatest female minds in the mathematical world, like Hypatia, Sophie Germain and Sonya Kovalevskaya. We discussed who they were as people, why they chose to study mathematics, the contributions they made to the field of mathematics and the impact that had on society.

Data Sources on Student Feedback

There were three main stages of data collection:

i. Before implementation of the module.
ii. During the module (on-going basis).
iii. After completion of the module.

In relation to (i) a pre-questionnaire was distributed to the students without them having any prior knowledge regarding the module. Table 2 below is the list of questions each student answered when completing the pre-questionnaire. In relation to (ii) the students engaged in a continuous recording of their thoughts and opinions on the material being taught, with a reflective journal. Students recorded in their journal at the end of each lesson. The journal was used to analyse students' engagement with each individual topic and their enjoyment of the class. It also allowed students an opportunity to express their opinion on the teaching and learning of mathematics. Table 3 illustrates the questions in the student journal. In relation to (iii) the post-questionnaire (which is the exact same as the pre-questionnaire in table 2) was distributed to the students after the completion of the module.

1. What does mathematics mean to you?
2. Do you enjoy mathematics? Why?
3. Do you consider mathematics interesting? Why?
4. Do you think mathematics plays an important role in today's society? Why?
5. Would you like to study mathematics at third level?
6. Do you consider mathematics to be creative? Why?
7. Any further comments regarding mathematics?

Table 2: List of questions in pre-questionnaire

1. What did you think of this class?
2. What did you enjoy about today's class?
3. What did you learn in today's class?
4. What would you like to learn more about from today's class?
5. Were there any thought provoking moments in today's class?
6. Is there anything else you would like to write about today's class?

Table 3: Questions in the student journal

MAIN RESULTS

In our Grounded Theory approach, the Core Category that emerged was that all students' perception of mathematics changed for the better. In fact, an instant change of all students' perception of mathematics for the better, was evident from our analysis of the feedback from lesson plan 1. Also, a sustained change of all students' perception of mathematics for the better, was clear from our analysis of the responses in the post-questionnaires (compared to the pre-questionnaires).

Lesson plan 1 began with a discussion of the students' responses to the pre-questionnaires and then a discussion of what mathematics means to both authors. The students were then taken on a journey covering the evolution of number from when it was not thought of as an idea to the modern day where it is an incredibly powerful idea. The class also discussed how number can slot into infinitely many situations because it is an idea.

Instant Change

The instant change of perception mentioned above illustrates the powerful immediate impact that the history of mathematics can have on students' perception of mathematics, because after doing mathematics for about 12 years in school all the students changed their perception of mathematics for the better after just 40 minutes of lesson plan 1. Some students commented on the immediate impact after lesson plan 1 mentioned above:

> Student 5: I thought the class was interesting. I learned some interesting facts that I never even thought about before even though I have been studying maths since I was 5. I never thought of the numbers like that.

Some changed their perception of mathematics so much after lesson plan 1 (LP1) that they looked forward to the next class:

> Student 4: Today's class was really really interesting. It gave me a new outlook on maths as a whole and I found that I really appreciate it. History is one of my favourite subjects, and to be able to incorporate it into maths is fascinating. It makes maths more enjoyable and I'm looking forward to class next week.

Many students had previously thought numbers were just there and were not created by people:

Student 13: I thought it was interesting. I never really thought about maths in that way before. I changed how I think about maths being created. I never thought about how numbers started, I thought they were just there.

Some got quite emotional after LP 1:

Student 12: Wow! I really liked this class. It was very enlightening and I learned basic things I ever knew about maths. I loved this class.

Sustained Change

The sustained change mentioned in the Core Category above shows that all students' change of perception of mathematics for the better was sustained until after the 20 lesson plans. Quite a few students made a massive leap from thinking mathematics had no history to now realising it has a rich history:

Student 10: Mathematics means so much more to me now. I used to think that it was mainly numbers and equations and solving but it's not. There is a whole background to maths. Maths wasn't just made up on some random day. Maths was made up by amazing mathematicians over the years. (Question 1 - post-questionnaire)

Some students became more confident in their ability to do mathematics after their perception of mathematics changed for the better:

Student 17: I think that maths is a good subject and it's good to be able to work out things but I don't really understand it completely. I don't know why I'm answering the questions that I am and when I'll ever need to answer these again. Teachers don't really explain what the questions are, they just tell you how to do it and then we're expected to do this with everything in maths. I think maths is about working out difficult equations and understanding how and why you do it. (Question 1 - pre-questionnaire)

Maths means a lot more to me now than it did at the start of this course. I have a more open mind about maths. I have now realised how much we actually use maths everyday and how different our world would be without it. I'm now more aware of where maths came from and what it's all about. Maths means a lot more to me now and I'm more confident in doing honors maths next year. (Question 1 - post-questionnaire)

The following student not only changed her perception of mathematics from being a stressful subject to an interesting one but also (like some other students) greatly admired the role played by female mathematicians in history:

Student 21: STRESS!!!A compulsory subject for Leaving Cert. It's pretty boring, we've it everyday. It's a lot of hard work, it's very hard. Honors for the leaving cert. is supposed to be impossible even if you did well in your Junior Cert. I'll probably attempt honors for the start of fifth year and then drop to pass for the actual Leaving Cert. There's a lot of work involved. Teachers don't explain things properly (they usually only tell you once or twice), then if you ask them again because you don't fully understand it, they tell you that don't need to understand it - just learn it! But that's hard to do when you don't know what you're learning! (Question 1 - pre-questionnaire)

> It's a subject we do in school. It can be really interesting. It's just not taught properly. There's beauty in maths and a history behind. There have been so many female mathematicians. They've fought so women can have the same education. They loved maths and thought it was fun. They did it by choice - because they had a passion for it not because they had to. (Question 1 - post-questionnaire)

Some Other Positive Impacts

The five main categories (that were all related to the Core Category above) were students' changes of attitude for the better in relation to mathematics and Humanity, History, Creativity, Beauty and Practical Power (meaning the powerful applications of mathematics to science, engineering, film making and many other areas). Here is a selection of responses related to the five categories mentioned above:

Humanity and History (quotes from the post-questionnaire):

A number of students made the big leap to now realising it is alright if they don't understand something immediately. This completely changed their approach to doing mathematics.

Student 17: I do enjoy maths now because I realised that it's okay if you don't understand something the first time it's explained to you because it took mathematicians thousands of years to understand some things that we know today.

Some students mentioned specific mathematicians and how related short stories would have enhanced their interest in mathematics in the Junior Certificate:

Student 14: I think it's good to learn about the background of maths before you learn a new topic. Take Pythagoras Theorem for example. If we learned about his life and his school and the secret society, then maybe students would be more interested in learning it. I know that it made me more interested.

A number of students now realise that mathematics is not a boring textbook anymore but it's a subject created by humans and new ideas and theories are still being created. This had a huge impact and resulted in a greater appreciation for mathematics by the students.

Student 21 : It's just an idea someone had but they turned it into a proof so now we study it. There's this amazing history behind maths that I never knew about. There's so many people behind maths. It's not just a boring textbook anymore.

Creativity (quotes from the post-questionnaire):

Student 21: It's creative because it's an abundance of ideas. People came up with maths in their heads. They then created it just like an art or music piece. It's an idea therefore it's creative.

Student 15: I do now consider maths to be creative as I now know that everything in maths was once formed by someone. It did not just appear from nowhere. It was thought up by someone.

Student 23: I consider maths extremely creative after the past few months because some of it came from nothing, it's just peoples ideas. Maths solves many problems and is created by peoples imagination.

The revelation that there can be beauty in mathematics was a very positive experience for many students. Here are some responses on LP 3 in that regard.

Beauty

Student 1: Maths can be as beautiful as a song or painting.

Student 11: I enjoyed today's class but at first I found it hard to understand the beauty maths has but as we got further into the class I began to understand it more.

Student 14: The thought provoking moment in today's class was how years ago people solved problems in maths not to find answers but purely for the beauty of maths.

Practical Power (all quotes after LP 2)

The history of mathematics gives plenty of nice examples of the powerful applications of mathematics in science and the wider society. As a result of this, many students became more aware of the important role that mathematics has had (and still has) in our society. Here are some responses by students in relation to the practical power of mathematics:

Student 17: I enjoyed finding out how cartoons need maths to be made. Without maths cartoons would still be in 2D instead of 3D. I never thought that maths would be involved in that.

Student 14 : I found this class very interesting as I learned how maths is used in different ways in everyday life. It was fun and not boring. I also found myself thinking about different things in maths and also found myself realising that areas in maths that I found boring before weren't so boring as I understood how they were used in everyday life.

Student 22: Maths is a hell of a lot more practical than it seems.

CONCLUSIONS

For all students there was an instant change of perception of mathematics for the better as a result of the first lesson plan. Also, there was a sustained change of all students' perception of mathematics for the better after completion of the twenty lesson plans on the history of mathematics. Other major positive impacts of the module were in relation to changes in students' attitude corresponding to mathematics and humanity, creativity, history, beauty and practical power. All these changes help to greatly enhance the students' awareness and appreciation of mathematics. Furthermore, teachers will now have some new resources and Teaching and Learning plans at hand for use in Project Maths. Some previous empirical studies on the use of the history of mathematics in other situations in mathematics education have also shown that it can have a positive impact on some students.

RECOMMENDATIONS

We feel that the history of mathematics would have a positive impact on students' attitudes towards mathematics in any second level year. Our resources are versatile in the sense that they can be used as a full module or as individual lesson plans in transition year and also shorter pieces of appropriate lesson plans could be implemented separately, at the discretion of the teacher, in the Junior and Leaving Certificate Cycle. Future work could involve an analysis of student feedback to appropriate parts of our resources outside of transition year.

REFERENCES

Barbin, E. (2000). Integrating history: research perspectives. In J. Fauvel and J. van Maanen (eds.), *History in Mathematics Education, The ICMI Study*. (pp. 63–90). Dordrecht: Kluwer Academic Publishers.

Conway, P. & Sloane, F. (2005). *International Trends in Post-Primary Mathematics Education: Perspectives on Learning, Teaching and Assessment.* NCCA, Dublin.

Fried, M. N. (2007). Didactics and history of mathematics: Knowledge and self-knowledge. *Educational Studies in Mathematics*, 66, 203–223.

Gulikers, I. & Blom, K. (2001). 'A historical angle', a survey of recent literature on the use and value of the history in geometrical education. *Educational Studies in Mathematics*, 47, 223 – 258.

Jankvist, U. (2009a). A categorization of the "whys" and "hows" of using history in mathematics education. *Educational Studies in Mathematics*, 71, 235 – 261.

Jankvist, U. (2009b). *On Empirical Research in the Field of Using History in Mathematics Education.* Retrieved April 15, 2011 http://tsg.icme11.org/document/get/569.

Lui, P. (2003). Do teachers need to incorporate the history of mathematics in their teaching? *The Mathematics Teacher* 96(6), 416–421.

Lit, C.-K., Siu, M.-K., & Wong, N.-Y. (2001). The use of history in the teaching of mathematics: Theory, practice, and evaluation of effectiveness. *Educational Journal*, 29(1), 17–31.

NCCA (2000). *Mathematics Junior Certificate Syllabus*. Dublin: The Stationery Office.

NCCA (2010). *Project Maths Junior Certificate Syllabus*. Dublin: The Stationery Office.

Ní Ríordáin, M. & Hannigan, A. (2009). Assessing the level of suitably qualified teachers teaching mathematics at post-primary education in Ireland. *The Proceeding of Third National Conference on Research in Mathematics Education, MEI 3, St. Patrick's College Drumcondra, Dublin.*

Ó Cairbre, F., Watson, R. & McKeon, J. (2006). *A Resource for Transition Year Mathematics Teachers*. Department of Education and Science, Dublin.

Ó Cairbre, F. (2009). The power of the short story and the big picture in mathematics education in schools, universities and for the general public. *The Proceeding of Third National Conference on Research in Mathematics Education, MEI 3, St. Patrick's College, Drumcondra, Dublin.*

Puig, L. & Rojano, T. (2004). The history of algebra in mathematics education. In K. Stacey, H. Chick and M. Kendal (Eds.), *The teaching and learning of algebra: The 12th ICMI study* (pp. 189-224). London: Kluwer Academic Publishers.

Rickey, V. (1996). The necessity of history in teaching mathematics. In R. Calinger (Ed.), *Vita mathematica: Historical research and integration with teaching (*pp. 251-256). Washington, DC: Mathematical Association of America.

Strauss, A. & Corbin, J. (1990). *Basics of Qualitative Research - Grounded Theory Procedures and Techniques.* Sage Publications.

Tzanakis, C. & Arcavi, A. (2000). Integrating history of mathematics in the classroom: An analytic survey. In J. Fauvel, & J. van Maanen (Eds.), *History in mathematics education, The ICMI Study* (pp. 201– 240). Dordrecht: Kluwer Academic Publishers.

U.S. Department of Education (2008). *Final Report of the Mathematics Advisory Council*. Retrieved April 09, 2011 http://www.ed.gov/MathPanel.

ASSESSING REASONING AND PROBLEM-SOLVING IN JUNIOR CYCLE MATHEMATICS: LESSONS FROM A PILOT STUDY

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This paper describes a pilot study conducted in the course of developing a new standardised test of mathematics achievement for students in Second year in post-primary schools. The analysis of a pool of 95 items, many of which were embedded in realistic contexts, illustrates the challenges inherent in developing a test prior to the introduction of Project Maths in all Junior Cycle classrooms. The pilot data showed overall difficulty levels for the three test booklets ranging from 37% to 44% correct. Differences in performance on the items in favour of female students were not statistically significant. However, substantive differences were observed between students attending or not attending schools in the Schools Support Programme under DEIS. Item discrimination was a problem for fewer than 10% of items whereas over 25% were identified as being too difficult. In discussing the outcomes, it is suggested that multiple forms of the test at different levels of difficulty should be developed to accommodate the wide range of achievement across students.

BACKGROUND AND CONTEXT

Introduction

The past decade has seen considerable debate on and change in Junior Cycle Mathematics. This has been influenced by a number of factors including: (i) revision of the primary school mathematics curriculum; (ii) developments in the use of ICTs in mathematics classrooms; (iii) dissatisfaction with the mathematical knowledge and competence of recent school and college graduates expressed by employers; and (iv) poor results of Irish 15 year-olds on the OECD PISA mathematics surveys.

Revision of the Primary School Mathematics Curriculum (PSMC).

In 1999 a revised primary school mathematics curriculum was introduced in which more emphasis was to be placed on problem-solving, discussion, and use of technology (calculators and computers) in teaching methods, along with minor changes in content (DES/NCCA, 1999). One of the content changes involved the introduction of a new Data strand, part of which included informal work on Probability and Chance. Probability was not part of the Junior Cycle mathematics curriculum at that time but was part of the Senior Cycle curriculum. The revised PSMC further highlighted the discontinuity between the primary and post-primary mathematics curricula as the JC curriculum at the time did not permit calculators either in classwork or in the JC examination, whereas the PSMC encouraged their use from

Fourth class onwards. (In response to the subsequent introduction of calculators into the JC curriculum, a study was carried to examine the effects of access to calculators on the JC curriculum (Close et al., 2004, 2008)).

With the introduction of the revised PSMC, national assessments in mathematics were carried out at the Fourth class level in 1999 and 2004 and in Second and Sixth classes in 2009 (Shiel & Kelly, 2001; Shiel, Surgenor, Close & Millar, 2006; Eivers et al., 2010). Overall performance in Fourth class in 1999 and in 2004 was not significantly different, indicating no change in overall achievement from just before the revised curriculum was introduced to immediately after). However, there were significant improvements on two mathematics content areas (Data and Shape & Space) and one skill process (Reasoning). In both assessments, relative weaknesses were identified in the content areas of Measures and Number, and the process skills of Applying & Problem-solving. In Second and Sixth Class in 2009, performance on the process skills of Applying and Problem-Solving and on the content area of Measures was poor, relative to other process skills and content areas, especially at Sixth class. All three studies indicated little use of new technologies in primary classrooms apart from the electronic whiteboard.

Developments in the use of ICTs in mathematics classrooms.

ICTs are now pervasive in everyday life and the workplace. Despite this, there has been limited use of the technology in teaching of mathematics in schools in many countries. This is particularly true of Irish primary and post-primary schools as evidenced by recent studies (DES 2008a, 2008b; Eivers et al., 2010; OECD, 2011). This is less likely to be due to costs and lack of suitable hardware and software than it was 10 to 20 years ago. Many post-primary schools are now well equipped with laptop or desktop computers and digital projectors/electronic whiteboards as well high speed wireless broadband and networking. Any software that has been used in schools has been mainly aimed at providing explanations and practice of specific curriculum related concepts and skills, a task arguably more readily and cheaply done with textbooks. There has been little use of software and computer-based learning environments aimed at developing higher order thinking and problem solving skills in mathematics, including collaborative problem-solving. This lack could be partly ascribed to the focus of the JC mathematics curriculum on teaching isolated concepts and skills and the assessment of these in timed pen-and-paper group examinations. This focus lends itself to use of the new digital technologies to support and supplement existing traditional teaching methods rather than encouraging the development of more interactive enquiry-based and self-directed learning which would be more in keeping with the way these technologies are used in the wider society.

Dissatisfaction with the mathematical knowledge and competence of recent school and college graduates expressed by employers.

Until recently one of the reasons attributed to Ireland's success in attracting multi-national industries to the country was the availability of a young highly educated workforce. However, despite this view, there has been criticism from the business community about the quality of Mathematical and Scientific education in schools, particularly with regard to the excessive emphasis on rote learning and teaching towards state examinations, while there have been calls for curricula and assessments that foster the development of critical thinking, self-directed learning, communication skills, and team work. A 2008 report by the Expert Group on Future Skills Needs on raising mathematical achievement made these criticisms and recommended that students be engaged in:

> discussing real-world/ business problems and how the mathematics involved might be applied to solve them. Problem solving, critical thinking and logical reasoning should be encouraged. There is a need to make connections within mathematics across subjects such as science, engineering and technology, business and finance, and social sciences (EGFSN, 2008).

In 2010 Engineers' Ireland produced a *Report of the Taskforce on Education of Mathematics and Science at Second Level* that expressed concern with the approach to the Mathematics education in Second Level schools including "the teaching of rote learning' rather than 'learning by understanding' and the rather bland teaching of the subject with little contextual links with everyday life. The report also expressed concern about the implementation of Project Maths and called for it to be well resourced as there were no appropriate text books available for the new programme.

There have also been numerous reports about the concerns of industry in national newspapers over the past decade. As recently as January of this year, a spokesperson for US companies speaking at a Dublin Chamber of Commerce meeting stated that they wanted problem-solving capabilities rewarded in examinations to encourage the innovation and creativity needed in the economy (Irish Times, 2011). Clearly there has been, and is, ongoing dissatisfaction among the business and industry community with mathematics curricula and teaching, particularly in the post-primary schools.

Poor results of Irish 15 year-olds on the OECD PISA mathematics surveys.

Since 2000, Ireland has participated in the OECD Programme for International Student Assessment (PISA), which includes a test of mathematical literacy. PISA defines mathematical literacy as: 'an individual's capacity to identify and understand the role that mathematics plays in the world, to make well-founded judgements and to use and engage with mathematics in ways that meet the needs of that individual's life as a constructive, concerned and reflective citizen' (OECD, 2009, p. 84). The PISA mathematics assessment focuses on real-world problems, moving beyond the kinds of situations and routine problems typically encountered in classrooms. It emphasises the notion of 'mathematising' which involves starting with a problem in a real-world context, identifying the relevant mathematics, and reorganising the problem according to the mathematical concepts identified. The next step is to gradually trim away the reality so that the mathematics problem can be solved. The final step is to make sense of the mathematical solution in terms of the real situation. PISA mathematics assesses four content domains (Space & Shape, Change & Relationships, Quantity and Uncertainty) and three competency clusters (Reproduction, Connections and Reflection).

In the first three cycles (2000, 2003, 2006), the mean mathematics score of 15-year olds in Ireland was not significantly different from the average for OECD countries. In 2009, students in Ireland achieved a mean score that was significantly below the OECD average, (by one sixth of a standard deviation), indicating a significant drop in mathematics achievement – the second largest decline across OECD countries since 2003 (when mathematics was a major assessment domain in PISA), though most of the decline occurred between 2006 and 2009 (OECD, 2010). Ireland's ranking in mathematics in 2009 was 26th out of 34 OECD countries and 32nd out of 65 participating countries. The percentage of students at/below Level 1 in Ireland on the PISA mathematics proficiency scales (20.8%) is just below the OECD average (22%). Ireland had significantly fewer students at the higher proficiency levels (at/above Level 5) than the OECD average (6.7% compared to 12.7%), indicating poor performance among high achievers. In Ireland, males achieved a higher mean score than females but the difference was not statistically significant (Perkins et al., 2010). In PISA 2003, performance was also reported by content area. Students in Ireland achieved mean scores above the OECD average on two content areas (Uncertainty and Change & Relationships), at the OECD average in one area (Quantity), and below the OECD average in one (Space & Shape). Subsequent analyses of the PISA items revealed that they were quite different from typical (at the time) Junior Certificate mathematics items in that they were strongly contextualized (OECD 2004; Cosgrove et al., 2005). Furthermore, a majority of the Space & Shape items were found to be drawn from the Junior Certificate mathematics content area of Applied Arithmetic & Measure, and hence assessed problem solving. Close (2006) reported no overlap between PISA mathematics and Junior Certificate Geometry, and minimal overlap with Algebra.

A number of explanations for the decline in performance in PISA mathematics in Ireland have been put forward (see Shiel et al., 2010). These include an increase in the proportion of students who speak a language other than English at home, a sharp decline in the performance of students in Transition year, a small decline in early school leaving (meaning that more weaker students sat the PISA tests) and a minor change in sampling methodology. However, none of these changes, in and of themselves, account for the large drop in performance, and other possibilities are being examined, including an increase in student disengagement from testing, as evidenced by increasing levels of skipped items and incomplete tests.

Project Maths

The combined influence of the ongoing issues and developments described above led to the setting up of Project Maths in 2008 by the NCCA , following a period of consultation with stake-holders. Project Maths is a revised curriculum that is being introduced into post-primary schools with an increased focus on problem solving and applications. The Project Math curriculum is being phased in for both Junior Cycle and Senior Cycle over an eight-year period. It commenced with 24 pilot schools in September 2008 and it is planned that it will be fully implemented in all post-primary schools, across all classes, by June 2015. According to its website:

> Project Maths aims to provide for an enhanced student learning experience and greater levels of achievement for all. Much greater emphasis will be placed on student understanding of mathematical concepts, with increased use of contexts and applications that will enable students to relate mathematics to everyday experience. The initiative will also focus on developing students' problem-solving skill.
> (www.projectmaths.ie/overview/)

These aims suggest a move towards Realistic Mathematics Education (Freudenthal, 1981) and enquiry-based or problem-based teaching and learning (Capraro and Slough, 2009). Such a radical move needs to be underpinned by considerable resources and materials in the form of guidelines, textbooks, software, and tests as well ongoing professional development support for the teachers involved.

In 2010, the DES-appointed Project Maths Implementation Support Group published its first report (DES, 2010). The Group made a number of recommendations designed to support the rollout of PM including: more time for maths; bonus points for higher maths; improved qualifications for maths teachers; provision of authentic maths examples by business and industry; synergies with ICT in schools initiatives; volunteer programme; and mentoring of more able maths students. To date, apart from examination grades of students in the 24 pilot schools, no data have been published on the effects of Project Maths on teaching and learning in mathematics classes, or on student performance.

Aim of the Study

The aim of the present paper is to describe efforts to develop standardised mathematics tests for Junior Cycle mathematics based on the Project Maths curriculum. The ongoing introduction of the Project Maths curriculum and its partial implementation in schools presents unique issues and problems for the development of standardised mathematics tests to enable teachers to adequately assess their students' performance on the new programme. These issues and problems are described and discussed in the remainder of this paper.

DEVELOPING MATHEMATICS TESTS FOR JUNIOR CYCLE

Need for Standardised Tests at Junior Cycle

The decision to develop a new standardised test of mathematics at Junior Cycle Level was motivated by several factors:

1. The absence of standardised tests based on the current Irish Junior Cycle curriculum and normed on students in Ireland that could be used by schools or for research purposes
2. The outcomes of a study commissioned by the NCCA, that established a need for standardised tests of mathematics (and reading) at lower secondary level, and pointed to the potential diagnostic value of such tests for parents, teachers and students (Shiel, Kellaghan & Moran, 2010). The study also highlighted the value of involving subject teachers in implementing and scoring tests and interpreting results.
3. The proposal in the Department of Education and Skills' *Literacy and Numeracy for learning and Life: The National Strategy to Improve Literacy and Numeracy in*

Schools (DES, 2011) that post-primary schools would administer standardised tests in mathematics (and reading) to students at the end of Second year and use outcomes for school self-evaluation and improvement.

4. The proposal in the *DES National Strategy* to extend national assessments at primary level to Second year at post-primary (this would require a large item bank, particularly if adaptive, computer-based testing were to be administered).

In embarking on the development of a pool of mathematics items that would be suitable for Second-year students in 2014 and later, it was recognised that the pace of implementation of Project Maths in schools would need to be taken into account. For example, Second year students in the pilot study described in this report would not be exposed to Project Maths until they reached Fifth year (though, of course, their teachers might introduce aspects of the Project gleaned from working at other class levels). Similarly, the gradual introduction of Project Maths (an incremental approach, with new curriculum strands being added each year) might mean students in our development studies would have encountered some topics within Project Maths but not others.

Test Framework and Item Development

Since implementation of the Project Maths syllabus had only begun in September 2010 for all schools and would not be fully implemented until June 2015, the assessment framework and specifications for the proposed JC standardised mathematics tests were based on both the existing JC maths syllabus and the incoming Project Maths syllabus, for which there is a common syllabus for First year, though not for Second or Third years. The intention was that the majority of items designed for the test would reflect the Ordinary Level syllabus with some more challenging items included for Higher level students and some easier items for Foundation level students. The initial focus of test development would be on producing tests for administration towards the end of Second year, beginning in 2014.

The test framework has three dimensions in the style of the PISA frameworks as follows:

- *Mathematics content* (Statistics & Probability; Geometry & Trigonometry; Number & Measure; Algebra & Functions);
- *Cognitive process skills* (Recall relevant facts & terms; Implement standard procedures; Reason & connect; Apply, analyse, & solve problems); and
- *Task contexts* (Practical contexts - personal, social/occupational, scientific; Pure mathematical contexts - mainly tasks which assess individual concepts or skills in isolation).

Using the framework a block of 95 items were written by a team of experienced mathematics teachers involved in the implementation of Project Maths. The items assessed mainly the process skills of reasoning and problem-solving in practical contexts (c. 70% of the items), and were organised into three booklets. These were then subjected to a pilot study in April 2011. The classification of the items by framework dimensions and item format is given in Tables 1 to 4 below.

Table 1: Item frequency by content strands

Content Strand	No. of Items
Statistics & Probability	27
Geometry & Trigonometry	15
Number & Measure	30
Algebra & Functions	23
Total	95

Table 2: Item frequency by process skills

Skill Category	No. of Items
Recall facts & terms; Implement procedures	28
Reason & connect; Apply, analyse & solve problems	67
Total	95

Table 3: Item frequency by context category

Context Category	No. of Items
Practical contexts	67
Mathematical contexts	28
Total	95

Table 4: Item frequency by item format

Item Format	No. of Items
Multiple choice	19
Open response	76
Total	95

METHOD

Sample

The following principles underpinned the selection of a sample for the pilot study:

(i) Allow for an achieved sample of about 900 students in Second year (at least 300 responses per mathematics booklet item); (ii) Confine the sample of schools to the greater Dublin area; (iii) Exclude schools that participated in PISA 2009 or were participating in PISA 2011 field trial; (iv) Exclude all-Irish schools and Project Math pilot schools.

Schools were implicitly stratified by DEIS status, school type, gender mix and size (number of students in Second year). Twelve of 108 eligible schools in the sampling frame were selected. There were 1315 students in Second year in the 12 schools in 2009/10. Nine of the 12 originally selected schools participated in the study. Three replacement schools were invited to take part but all declined. However, a volunteer school was permitted to take part, bringing the achieved sample to ten schools (5 DEIS and 5 Non-DEIS) with a total of 1026 students.

Weighting

Due to non-participation the achieved sample differed from the population (the three schools that declined to take part were non-DEIS secondary schools, while the volunteer school was a DEIS vocational school) (Table 5). In order to reduce any bias in the item statistics due to non-response, the data were weighted prior to analysis. Because of the relatively small numbers involved, weights were based on six cells: 3 school types (community & comprehensive, secondary, vocational) by 2 disadvantaged categories (DEIS, non-DEIS). Weighting had the effect of making the sample characteristics similar to the population characteristics in terms of weighting variables.

Procedures

The three booklets containing 96 items in total (33, 32 and 31 items respectively) were administered to the 1026 students in their schools during the last week of March and the first two weeks of April. The booklets were distributed to the students on a rota basis so that no student was sitting beside another student with the same booklet and each booklet was taken by approx. one third of the students. They were allowed 45 minutes to complete their test booklets. Students had access to calculators. The booklets were then scored by a team of trained markers and the scores entered in a database for analysis.

Analysis

Analysis involved computing the weighted percentage of students responding correctly to each item and these were averaged across booklets. Item percent correct scores were also averaged for groups of students (males, females; those attending/not attending schools in the Schools Support Programme under DEIS). The statistical significance of differences between groups was evaluated using t-tests, with adjusted t-values to take multiple comparisons into account.

Point biserial coefficients were also computed for test items. Point biserials are correlations between dichotomous items (that is, items scored as right or wrong) and a continuous scale (overall performance on the test). A value below 0.30 may mean that the item does not discriminate well between higher- and lower-performing students.

Finally, each item below criterion level (i.e., below a difficulty of .25 and/or a point-biserial of .30) was tagged and examined separately to ascertain why there might be a problem.

RESULTS

1026 students in ten schools participated in the pilot study. The mean weighted item percentage scores on the three test booklets were: Booklet 1 - 37.02% (N = 352); Booklet 2 - 43.76% (N = 340); and Booklet 3 - 40.86% (N = 334). These are an improvement on earlier pre- pilot study means but are still relatively low for a potential standardised test.

Gender and DEIS Comparisons

Average item percent correct scores were compared for male and female students for each booklet and across all booklets. Table 5 (top half) shows that though there were differences in favour of female students they were not large enough to reach statistical significance. Similarly, average item percent correct scores were compared for students in DEIS and non-DEIS schools. Here, differences were all large and significant, ranging from 12% (in favour of non-DEIS students) on Booklet 1, to 20% on Booklet 2, and to 17% on Booklet 3.

Table 5: Average weighted item percent correct scores, by gender and school DEIS status

	% Correct		Difference between Means			
Booklet	Male	Female	Diff	t	df	p
1	35.3	38.3	-2.97	-0.739	64	> .05
2	41.2	45.4	-4.29	-0.789	62	>.05
3	39.4	41.9	-2.52	-0.663	60	>.05
	DEIS	Non-DEIS	Diff	t	df	p
1	27.7	39.4	**-11.68**	-2.982	64	< .05
2	28.1	47.7	**-19.60**	-3.800	62	< .01
3	27.0	44.4	**-17.32**	-4.886	60	<.01

Statistically significant differences in bold. t-values (adjusted for multiple comparisons) for booklets 1, 2 and 3 were 2.459, 2.461 and 2.462 respectively for p. = 0.5/3, and 3.049, 3.053 and 3.057 respectively for p. = .01/3. ..

Table 6 shows that, although differences in percent correct scores in favour of females were not statistically significant, females were ahead of males, by at least 5%, on about half of the items in each booklet. The findings for gender and DEIS status are consistent with findings for these two factors in the recent national survey of primary school mathematics (Eivers et al., 2010).

Table 6: Summary of item differences, by gender

	Number of Items	Percent of Items
Booklet 1		
5% + Diff for Males	3	9.1
5% + Diff for Females	14	42.4
Diff < 5%	16	48.5
Booklet 2		
5% + Diff for Males	1	3.1
5% + Diff for Females	16	50.0
Diff < 5%	15	46.9
Booklet 3		
5% + Diff for Males	4	12.9
5% + Diff for Females	11	35.5
Diff < 5%	16	51.6
All Booklets		
5% + Diff for Males	8	8.3
5% + Diff for Females	41	42.7
Diff < 5%	47	49.0

Item Characteristics

The item characteristics of individual items were also examined in terms of their ability to discriminate among higher and lower achieving students. Table 7 shows the numbers of items in each booklet with difficulty levels less than 0.25 and discrimination levels (Point Biserial) less than 0.30. Whereas just 7 items did not meet the discrimination criterion of 0.30, 26 items had difficulty levels of less than 25%.

Table 7: Numbers of items with serious problems by booklet*:

		Number of Items with Potential Problems		
	Total Number of Items	Difficulty < 0.25 Only	Point Biserial <0.30 Only	Below criterion for both
Booklet 1	33	8	1	3
Booklet 2	32	8	1	0
Booklet 3	31	7	2	0
All Booklets	96	23	4	3

*Criteria: Difficulty Level < .25; Point Biserial Coefficient < .30.

Inspection of the 26 very difficult items showed that they all involved higher order cognitive skills of reasoning, applying, analysing and solving multi-step problems. Eight of the items related to the Probability & Statistics strand, 5 of which were on probability, a topic the students in the pilot study sample would not have been taught as it is on the Project Maths syllabus but not on the current Junior Cycle syllabus. A further 8 items involved applied algebra including interpreting, substituting or transposing verbal or symbolic formulae relating to practical situations or geometric patterns. Here again these latter skills are a major focus of PM but are given limited attention in the current syllabus. The other 7 items required the solution of more complex multi-step arithmetic problems. Nineteen of the 23 items were set in a practical context and just three of them were multiple-choice.

Cronbach's alpha coefficients were computed to establish the overall reliability of students' test scores. These coefficients ranged in value from .88 (booklet A) to .90 (booklets B and C), indicating that individual scores on the test had reasonably good reliability.

CONCLUSIONS

The pilot study described above highlights a number of issues which need to be considered further in the development of standardised mathematic tests for post-primary schools including:

- The difficulty of developing standardised mathematics tests when a new PM syllabus is being implemented on a phased basis. An added complication in this regard is that there is no prescribed programme for Second year students in post-primary schools though there is one for First year students (DES/NCCA, 2010). A consequence of this is that some topics may be covered in Second year in some classes, and in Third year in others.
- The ongoing difficulty for Irish post-primary students in responding to test items involving higher-level reasoning and problem-solving in mathematics, a difficulty borne out by recent PISA surveys (Shiel et al., 2010). The inclusion of a substantial number of these kinds of items on a test may mean keeping the mean score relatively low. The full implementation of Project Maths should lead to an improvement in this situation, given its greater focus on use of practical contexts and applications to develop mathematical understanding and problem-solving skills.

- The difficulty in developing a single standardised mathematics test for students in both DEIS and non-DEIS schools, given the typically large differences in performance between the two types of school. A preliminary screening test (and, in the longer term, computer adaptive testing) leading to easier and more diffiuclt test forms may be a way to offset this problem.
- The somewhat poorer performance of males compared with females. Although not significantly different in this study, the performance of males and females on PM-type tasks is worth monitoring closely.

The next stage in the process involves developing a larger pool of items for a second pilot study, to take place in autumn 2011. The new pool will include some additional reasoning and problem-solving items, but will also include items designed to ascertain students' conceptual understanding and procedural skill (including Recall and Implement) in each of the four curriculum content strands identified in the framework. This may also reduce the overall difficulty of the test.

REFERENCES

Capraro, R.M., & Slough, S.W. (2009). *Project-based learning an integrated science, technology, engineering, and mathematics (STEM) approach.* Rotterdam, Sense Publishers.

Close, S. (2006). The junior certificate curriculum and the PISA mathematics framework. *Irish Journal of Education, 37,* 53-78.

Close, S., Oldham, E., Shiel, G., Dooley, T., Hackett, D., & O'Leary, M. (2004). *The effects of calculator use on mathematics in schools and in certificate examinations; Final report on phase 1*. Dublin: St. Patrick's College.

Close, S., Oldham, E., Surgenor, P., Shiel, G., Dooley, T., & O'Leary, M. (2008). *The effects of calculator use on mathematics in schools and in certificate examinations: Final report on phase II*. Dublin: St. Patrick's College.

Conway, P., & Sloane, F. (2005). *International trends in post-primary mathematics education.* Dublin: NCCA.

Cosgrove, J., Oldham, E. & Close, S. (2005) Assessment of mathematics in PISA 2003: achievements of Irish 15-year olds in an international context. In S. Close, T. Dooley & D. Corcoran (Eds.) ,*Proceedings of the First National Conference on Research in Mathematics Education* (pp. 205-219. Dublin: St Patrick's College.

DES/NCCA. (1999). *Primary school curriculum. Mathematics.* Dublin: Stationery Office.

DES (2008a). *ICT in Schools: Inspectorate Evaluation Studies.* Dublin: Stationery Office.

DES (2008b) *Investing effectively in information and communications technology in schools, 2008-2013: The Report of the Minister's Strategy Group.* Dublin: Stationery Office

DES. (2011). *Literacy and numeracy for learning and life: A draft plan to improve literacy and numeracy in schools*. Dublin: Department of Education and Skills.

DES (2010). *Report of the Project Maths Implementation Support Group.* Dublin: Author.

Department of Education and Skills. (2010b). *Better literacy and numeracy for children and young people: A draft plan to improve literacy and numeracy in schools*. Dublin: Department of Education and Skills.

DES/NCCA. (2010). *Junior Certificate Mathematics Syllabus;Foundation, Ordinary and Higher Level.* Dublin: Stationery Office.

Eivers, E., Close, S., Shiel, G., Millar, D., Clerkin, A., Gilleece., L. & Kiniry, J. (2010). *The 2009 National Assessments of Mathematics and English Reading*. Dublin: Stationery Office.

Engineers Ireland (2010) *Report of the Task Force on Education of Mathematics and Science at Second Level.* Dublin. Engineers Ireland. Accessed at: http://www.engineersireland.ie/public/20100211-Mathematics_and_Science_at_Second_Level.pdf

Expert Group on Future Skills Needs (EGFSN) (2008). *Statement on raising national mathematical achievement*. Report to the Tánaiste & Minister for Enterprise, Trade and Employment and the Minister for Education and Science; November 2008. Dublin: The National Policy and Advisory Board for Enterprise, Trade, Science, Technology and Innovation.

Freudenthal, H. (1981). Major problems in mathematics education. *Educational Studies in Mathematics*, 12, 133-150.

National Council for Curriculum and Assessment. (no date). *Project Maths – Bridging framework*. Accessed at http://www.ncca.ie/en/Curriculum_and_Assessment/Post-Primary_Education/Project_Maths/Syllabuses_and_Assessment/PM_bridging_framework.html

OECD (2004). *Learning for tomorrow's world: First results from PISA 2003*. Paris: Author.

OECD (Organisation for Economic Co-operation and Development). (2009). *PISA 2009 Assessment framework: Key competencies in reading, mathematics and science.* Paris: Author.

OECD. (2010). *PISA 2009 results: Learning trends – Changes in student performance since 2000 (Volume V)*. Paris: Author.

OECD. (2011). *PISA 2009 results: Electronic reading (Vol. 6).* Paris: Author.

Perkins, R., Moran, G., Cosgrove, J., & Shiel, G. (2010). *PISA 2009: The performance and progress of 15-year-olds in Ireland –summary report*. Dublin: Educational Research Centre.

Shiel, G., Kellaghan, T., & Moran, G. (2010). *Standardised testing in lower secondary education* (Research report no. 12). Dublin: National Council for Curriculum and Assessment.Accessed at http://www.ncca.ie/en/Publications/Reports/Standardised_Testing_In_LowerSecondary_Education.pdf

Shiel, G., Moran, G., Cosgrove, J., & Perkins, R. (2010). *A summary of the performance of students in Ireland on the PISA 2009 test of mathematical literacy and a comparison with performance in 2003.* Dublin: Educational Research Centre.

Shiel, G., & Kelly, D. (2001). *The 1999 National Assessment of Mathematics Achievement*. Dublin: Educational Research Centre.

Shiel, G., Surgenor, P., Close, S., & Millar, D. (2006). *The 2004 National Assessment of Mathematics Achievement*. Dublin: Educational Research Centre.

LEARNING TO TEACH IN THE CONTEXT OF REFORM-ORIENTED MATHEMATICS

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This paper draws on the Learning to Teach Study (LETS), the first of its kind on the Postgraduate Diploma in Education (PGDE) in Ireland, funded by the Department of Education and Skills (DES), which developed and implemented a study of initial teacher education in the PGDE in post-primary education. Adopting an interpretive approach, LETS involved the collaborative development of three interview protocols and a survey by the research team. Seventeen (n=17) students were interviewed three times over the course of PGDE programme, and one hundred and thirty three students completed a detailed survey on their learning to teach experience (n=133, i.e. response rate of 63% of the 212 students in the PGDE 2008/09 cohort, of whom thirty were learning to teach mathematics. We highlight two key findings in this paper: (i) the low to moderate levels of support for student teachers on teaching practice allied with (ii) the challenges experienced by student teachers in enacting the meaning of the 'real world' a theme central to current reform-oriented mathematics in Irish post-primary schools. We locate these claims in terms of the issues they raise if we consider assisted performance central to learning to teach and crucially important in efforts to reform teaching.

INTRODUCTION

Experiencing student teaching and becoming a teacher of post-primary mathematics in 2011 and beyond presents a fundamental challenge for student teachers and schools - as well as for university-based teacher educators. The challenge, which poses many dilemmas for ITE, revolves around the difference between student teachers' experience of learning maths during their own schooling (what Lortie in 1975 termed the 'apprenticeship of observation') and what is being proposed as the future direction of mathematics in Ireland, with the current roll out of Project Maths[1]. At one level, this phenomenon is only about mathematics teaching but at another, it is one of the fundamental dilemmas of teacher education (Ball, 1996; Boaler, 2002): how should the next generation of teachers be educated and assessed in a reform-oriented era? Central issues to the teaching of any subject are conceptions of knowledge, assumptions about learning, teaching and assessment in the domain, and the role of curriculum resources. Project Maths has drawn significantly, although not exclusively, on both Realistic Mathematics Education (RME) and situated cognition (see Conway and Sloane, 2006; Lubienski, 2011). This raises yet another issue, that is, how student teachers and teacher educators understand and enact the 'real-world', as an important dimension of problem solving, in the context of maths teaching. In discussing mathematics teaching traditions, we refer to existing approaches, such as those described in the Lyons, Lynch, Close, Sheerin and Boland (2003) video study, as conventional and to Project Maths as reform-oriented

[1] Full details of the Project Maths initiative may be found at http://www.projectmaths.ie

mathematics. This paper reports on some of the findings from a DES-funded study on learning to teach at post-primary level (Conway et al., 2010), focusing on the dynamics of preparing post-primary mathematics teachers in the reform-oriented context of Project Maths.

MATHEMATICS: POLICY AND PRACTICE

The teaching of mathematics has become a policy priority around the world for a number of reasons. In Ireland, the concern about mathematics is no less urgent than elsewhere. We now turn to the recent policy context, focusing on the review of mathematics at post-primary level led by the National Council for Curriculum and Assessment (NCCA, 2005; Conway & Sloane, 2006) and ongoing reforms under the banner of Project Maths (NCCA, 2007). The focus on mathematics is of particular interest in terms of teacher education as it brings to the fore the question: how do student teachers in initial teacher education learn to teach in ways that are different to those which they experienced in their own school experience of mathematics? A related question is how do teacher education programmes design a curriculum of mathematics education to foster reform-oriented mathematics teaching? Current government thinking stresses the need to develop a knowledge economy, and it is envisaged that mathematics is a crucial subject that is needed by citizens to fully participate in both the knowledge economy and society. Indeed policy papers and debate following publication of international comparative studies of second level mathematics achievement over the last decade have highlighted the importance of mathematics and the need to reform both the subject and its teaching. Specifically in relation to teacher education, the National Competitive Council (2009) asserted that if Ireland is to become one of the top OECD countries in regard to mathematical performance then it is important that the mathematical knowledge and competence of teachers is enhanced in both their pre-service and in-service formation. Project Maths planning and piloting began in 2007. This reform-oriented maths curriculum is intended to strengthen the emphasis on real life applications, promote greater maths literacy across the school population, focus strongly on context and applications and problem solving, encourage greater take up of mathematics at higher level and provide a solid foundation which prepares students for a wide range of careers in the knowledge economy such as science, engineering, technology and business. All teachers, including those engaged in initial teacher education, bring Lortie's (1975) 'apprenticeship of observation' with them into their classrooms. These 2000 hours have not only instilled particular images of teaching and learning but have also shaped their understanding of mathematics. Student teachers may not have the subject knowledge needed to enact the reform-based images of teaching when they come to teacher education. Drawing on the well-developed literature on teacher beliefs, Morine-Dershimer and Corrigan (1996) summarise a number of findings concerning student mathematics teachers:

- Students had a narrow diet of mathematical experiences themselves as pupils
- Students brought with them influential role models of 'good' mathematics teachers
- Students very often rejected ideas from teachers whose views were different from those of their model teacher

- Students chose to observe and talk to those teachers who were most like their model teacher. (p. 119)

In addition to beliefs about teaching, student teachers also brought with them beliefs concerning the nature of mathematics. Their exposure to different styles of teaching in the classroom did not necessarily alter their views about mathematics. Echoing Veenman's (1984) comprehensive review of fifty years of research on student teachers' concerns, Haggarty (1995) also reminds us how student teachers' preoccupation with classroom management concerns came to bear in relation to thinking about the relevance of different teaching styles. Student teachers simply wished to survive teaching practice. Thus practical work might be rejected because of likely classroom control problems, rather than used because of its potential in improving conceptual understanding.

Lyons et al. (2003) provide a contemporary picture of mathematics teaching in Irish post-primary schools. Their findings suggest a high level of uniformity in terms of how mathematics lessons are organised and presented in second level schools. The dominant teaching styles employed by teachers are demonstration followed by practice. Demonstration is based on exercises or examples from the text books and includes teacher-led question and answer sessions and reviews. The Lyons et al. video study demonstrates powerfully that this conventional approach is pervasive in Irish second level schools. Such a view of teaching has also been highlighted by the Inspectorate Reports[2] concerning the teaching of mathematics in post-primary schools:

> The predominant methodology used in lessons observed was traditional - teacher presentation of work at the board followed by the setting of work for individual student practice. While teachers reported the use of other methodologies in lessons not observed it is, nonetheless, recommended that an expanded range of teaching methodologies be used, along with supporting materials, in recognition of students' different preferred learning styles.

Oldham (2001), commenting on the Third International Mathematics and Science Study (TIMSS) results for Ireland, argues convincingly that the predominant teaching style in Ireland was whole class exposition with low calculator usage. The TIMSS study, involving a survey of teachers, focused on the 1987 syllabus which did not advocate any particular style of mathematics teaching and which did not permit the use of the calculator in examinations. This appears to be in marked contrast to the experiences of other English speaking countries. The TIMSS study found that the majority of mathematics teachers internationally expressed the view that it was very important for students to be able to think creatively, to understand how mathematics is used in the real world and to be able to provide reasons to support their solutions. Based on teacher self-report data, teachers in Ireland did not rate these skills highly (Oldham, 2001). Similarly, Lyons et al. (2003) found in their study of mathematics teaching that teachers do not attach a high level of importance to these skills. Taken together, the TIMSS study and the Lyons et al. study seem to suggest that Irish teachers regard

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[2] Inspectorate Reports may be accessed at http://www.education.ie

mathematics in traditional ways, that is, a given body of knowledge to be transmitted to learners. Most believed that the way to improve learning was to use varied teaching methodologies and to provide lots of practice at the subject. Learning itself was most often equated with the memorisation of formulae and procedures rather than in terms of problem solving and thinking creatively. Being able to justify solutions or displaying real life understanding of mathematics use was considered not very important by teachers. The 'drill and practice' approach conforms with teachers' perceptions of mathematics as also evidenced by TIMSS findings. Ireland was second highest out of thirty-seven countries with regard to remembering formulae and procedures. The country was lowest of all however with respect to understanding how mathematics is used in the real world, in the bottom quarter as regards students being able to provide reasons to support their conclusions and thirty fifth in ability to think creatively (Oldham 2001). In conclusion, an instrumental understanding approach fits in well with being able to memorise formulae and their associated 'rules without reasons'. However a high degree of relational understanding is needed so as to exhibit competency in the other three tasks mentioned above. It appears that Irish students are being taught using an instrumental understanding approach.

REFORM-ORIENTED MATHEMATICS: THE PROJECT MATHS VISION

However, there are many challenges in bringing teachers around to the view of mathematics underlying Project Maths and in particular its emphasis on more active learning methods and contexts in the classroom. Project Maths is the government's response to previous policy calls, as noted above, for mathematics reform which emerged from a range of studies and the NCCA-led review of post-primary mathematics education. In calling for reform, the government is conscious of Irish students' level of attainment in mathematical literacy. The OECD Programme for International Student Assessment (PISA) defines mathematical literacy as "an individual's capacity to identify and understand the role that mathematics plays in the world, to make well-founded judgements and to engage with mathematics in ways that meet the needs of that individual's life as a constructive, concerned and reflective citizen" (OECD, 2006, p. 72). This conceptualisation of mathematical literacy is based on the Realistic Mathematics Education (RME) movement, which emphasises the notion of "mathematising". RME grew out of the work of Hans Freudenthal[3] a Netherlands based mathematician turned mathematics educator. It stresses the importance of problem solving, be it real world-focused or not but with an emphasis on the former, as both a source of learning and site in which mathematical ideas can be applied. The use of real-life materials fits very well with the philosophy of the Realistic Mathematics Movement (RME) which has become increasingly influential in mathematics curriculum development over the last few decades (Conway and Sloane, 2006; OECD, 2003). Goffree (1993, p. 89) writes that in RME "…reality does not only serve as an application area but also as the source of learning." The real world serves as a representation of a mathematical concept or technique. Schroeder and Lester (1989, p. 33) write of this representation being a movement from the concrete, 'every

[3] A classic text on RME is Freudenthal. (1991).

day of things, problems and applications of mathematics" to the abstract world "mathematics symbols, operations and techniques", and/or vice versa. The PISA framework is based on the assumption that mathematics is a human activity, and reflects the strong influence of both RME and situated cognition.

METHOD

LETS is an empirical research project into how students on the Post-Graduate Diploma in Education (PGDE) in University College Cork (UCC) develop teaching competence as post-primary teachers. The LETS study was undertaken over three years (2007-10) and involved the participation of an experienced research team from the School of Education, UCC. The team brought a variety of experiences and insights to the research task, and within the overall socio-cultural understanding that framed this study, were able to contribute their specialised knowledge in areas such as inclusion, equality and diversity, second language teaching, the teaching of mathematics and of science etc. The research team itself can be seen in this context as a community of learners, participating together in the task of achieving an understanding of the process of learning to teach. The principles of the interpretive research genre (Mertens, 2010) informed the LETS research project.

Table 1: Overview of semi-structured interviews with student teachers

Part	**Interview domains: January 2009**	**Interview domains: March 2009**	**Interview domains: May 2009**
A	Background, previous experience, motivation to learn to teach	Update on progress learning to teach	Opportunity to learn to teach
B	Opportunity to learn to teach	Opportunity to learn to teach	Critical incidents in learning to teach and in school
C	Critical incidents in learning to teach and in school	Critical incidents in learning to teach and in school	Understanding of subject teaching, inclusion and reading
D	Understanding of subject teaching, inclusion and reading literacy	Understanding of subject teaching, inclusion and reading literacy	Future plans
E	Summary: SWOT 1	Summary: SWOT 2	Summary: SWOT 3

The methods used included semi-structured interviews, analysis of documents and a survey questionnaire (see Table 1 for overview of interview foci). Using a multiple-case study research design, seventeen student teachers were interviewed on three occasions over the course of an academic year. A survey focused on the prior experiences and beliefs (e.g. about learning) that student teachers bring to the PGDE, including a measure of their teaching efficacy and knowledge about reading literacy and inclusion, was completed in March 2009 by 133 of the 212 students of the 2008/2009 PGDE cohort (a response rate of 63%). For the purposes of this paper, we draw on items from the first section of the survey, which focused on students' experiences in their teaching practice school, and on excerpts from interviews with student teachers of mathematics.

FINDINGS

In this section, we discuss two of the findings from the mathematics component of the *Learning to Teach Study* (Conway et al., 2010):

- The dynamics and consequences of support in schools on learning to teach mathematics
- The appeal and associated challenges of problem solving-oriented mathematics be it real-world focused or not

Claim 1: The Dynamics And Consequences Of Support In Schools On Learning To Teach Mathematics

A fundamental assumption guiding LETS is that learning to teach is best undertaken in a context in which student teachers experience gradual and supported entry into full classroom responsibility. This assumption, growing out on Vygotsky's *zone of proximal development* (ZPD), is based on research on learning as assisted performance (Tharp & Gallimore, 1988) and more recent teacher education studies (Grossman et al., 2009). These suggest that the 'sink or swim' model of learning to teach ultimately undermines teaching as it provides far less opportunity to develop a wide repertoire of skills since the pressure to survive consigns student and beginning teachers to reliance on their apprenticeship of observation. Framed in terms of a continuum from 'graduated and supported' to 'sink or swim' models of learning to teach, how can we characterise the experiences of the PGDE students in our study? In order to address this question vis-à-vis mathematics, we draw on both survey data. A five-item scale (Cronbach Alpha = 0.70) was constructed to assess the extent of support in schools perceived by the PGDE students. The items used to construct the scale we detail below were Likert items (Likert, 1932) with the following response categories: strongly agree, agree, undecided, disagree, strongly disagree.

The five items in the *Support in School* scale were:

- Got lot of help planning lessons from school staff
- Had chance to talk daily about lesson progress with teachers
- Felt supported by staff in school
- Had access to resources in school, textbooks etc
- Felt supported in my main subject

A one-way ANOVA, with two-way Tukey test posthoc comparisons, to assess the differences in means between support for students in different subjects, when teaching as their main PGDE subject, was significant overall (df 9/123, F=2.64, p=0.008). A number of observations are noteworthy (see Table 2).

Table 2: 'Support in Schools' for main and second PGDE subject

PDE Subject: Main & Second	Mean (Main)	n	Mean (Second)	N
Maths	14.2	10	9.9	31
French	8.5	9	7.7	5
Other language	12.6	3	11.0	5
Science (incl Bio)	11.2	25	12.3	9

Overall, the mean scores (from a low of 8.5 in French to a moderate score of 14.2 in maths; Score scale: min = 0 and max = 20) suggest a low to moderate degree of support was available to student teachers in both their main and second subjects. However, there was a statistically significant difference (Tukey posthoc, p=0.05) between the degree of support in schools for student teachers of Maths (mean=14.2, sd 2.8) compared to French (mean=8.5, sd 3.9). No other two-way comparison of mean scores on the 'Support in School' scale, between Maths, Science, French or Other Languages, were significant. In relation to support for teaching Maths, Science, French and Other Modern Languages when this was a second PDE subject, overall there was somewhat less support overall (not statistically significant). There were no between-subject differences in the degree of *Support in School* for the second subject in two-way comparisons. However, the ranking was different with Science rather than Maths having a higher mean. Consistent with *Support in School* for the main PDE subject, student teachers teaching French again perceived themselves as receiving less support than student teachers in other subject areas. Furthermore, student teachers for whom maths was a second subject reported a much lower level of support in school (mean 9.9, sd 3.8) than did those for whom maths was their main subject (mean=14.2, sd 2.8). In fact the support in school for students for whom maths was a second subject was almost as low as that for those for whom French was their main subject. As such, the difference between support levels was almost statistically significant comparing those for whom maths was their primary versus second subject.

The type of support experienced by student teachers is explored in greater detail in the full LETS report from which this paper is drawn (Conway et al., 2011). Two of the main aspects were the availability of mentoring and of opportunities to observe other teachers at work. Aoife, one of the participants in the LETS study, was in many ways typical, in that she reported a moderate degree of support in school from one of the mathematics teachers, who also afforded her the opportunity to observe in the classroom on three occasions. The main benefit that Aoife reported from these observations was learning how to structure lessons time-wise, that is, into ten-minute blocks of time around a focal idea in order to maintain greater student engagement. These are valuable insights. Nevertheless, she also felt that in the final analysis learning to teach was very much a solo endeavour despite the support she was given:

Interviewer: So in terms of teaching your subject, to what extent is what you have learned in your school placement helped you in terms of how you teach?

Aoife: Well I think what I have learned in my school placement, <u>you almost learn from yourself, from your own mistakes. Like I said I don't have a lot of interaction with the other teachers</u> there, so I am only learning from my own mistakes and in that way <u>I have learned an awful lot</u>. When I think back to how I did things at the start of the year, I couldn't imagine doing it now but at the start of the year...

Claim 2: The Appeal And Associated Challenges Of Problem Solving-Oriented Mathematics

The reform-based ideals in mathematics education espoused by Project Maths advocate the use of contexts and real life examples as critical. In addition, the PISA definition of mathematical literacy expects mathematics students to be able to identify and understand the role that mathematics plays in the real world. To what extent do the pre-service teachers in this study understand and use the concrete, ‘every day of things, problems and applications of mathematics” and move to the abstract world of “mathematics symbols, operations and techniques” (Schroeder & Lester, 1989, p. 33) in their classroom teaching? One informative revelation is that some of the pre-service teachers have discovered the limitations of traditional teaching. Excerpts from the interviews illustrate this. Julie (interview 1, p. 16) explains:

> There have been lots of times in maths anyway where I’ve gone in and my lesson plan is, so I will tell them this and we will do these examples and then I will tell them this and we will do more examples. And the telling them just doesn’t get through.

How did Julie grapple with this? She recounts in a subsequent interview:

> <u>I decided to abandon the textbook</u>…and I just said ‘OK, let’s just do the pie charts about your interests and <u>lets do a bio graph of how you came to school</u>. And I made questions about them…<u>it captured their interest</u>.

She discovered as Goffree (1993, p. 89) explained that, “…reality does not only serve as an application area but also as the source of learning”. Julie related the mathematics being studied in class to the real life of her pupils. They described a part of their day-to-day life using mathematics. Of course this approach to teaching mathematics also satisfies the appreciation objective of the current junior certificate syllabus. Growing in confidence, Julie transferred this real life approach to another topic – sets. She described how she formed various sets, sub-sets, etc. based on data provided by the class themselves. This data was relevant and meaningful to the pupils. Using this data, she explained the set operations so that these concepts such as union, intersection etc. meant something to the pupils. Crucially, for

Julie, “they were happy enough with it”. She had experienced success. She had used the pupils’ own experiences to promote mathematical understanding.

Padraig (Interview 2, p. 3) also reports positively on students’ engagement with real life materials. He recounted bringing in authentic utility bills to help his students learn about “percentages, interest rates, etc. Practical daily examples which illustrate the relevance of mathematics to students”. He enjoyed it when the “students tend to pick up on it and then they get involved in those classes…so the students will arrive with examples as well”. Padraig was also able to give other examples of real life examples to help with his teaching of integers - these included thermometers, lifts in a building and identifying sub-terrain levels.

An interesting interpretation of ‘real’ was one offered by a trainee teacher of mathematics and French, Emma. She recounts struggling to teach a fundamental property of integer arithmetic, namely the convention that two like signs multiplied or divided together will always produce a positive sign and two unlike signs multiplied or divided together will produce a negative sign. Pupils struggle with this and whereas an instrumental understanding approach is often taken by teachers i.e. learn the rule off by rote and just use it, sometimes pupils like to know the reasons behind the rule. Emma explains:

> When I was teaching them before Christmas, the operations, you know, positive and negative numbers, and I teach maths grinds as well at the weekends and I have done for about 6 years so I know that people up to Leaving Cert still get confused as to whether 2 negatives makes a positive, so there is still always a bit of doubt in people's minds. So I found this thing on the internet where the positive would be represented by love and the negative would be represented by hate. So if you loved to love, you love, so that would be 2 positives. And if you hate to hate then you love, so that would be 2 negatives making a positive. If you hate to love, then you still hate, so it is negative. And if you love to hate, then you hate so it is still negative. I am trying to get it straight in my head. So it was really really successful and even when we would be doing the operation by saying, so we have -4 x -3 what do we get? Or what is the sign? And they would say 'love' and then, 'oh no, a positive!' <u>And no student in my class had difficulties with that, even though there are some very weak students, but because they had a nice little story and I had done up a poster with big hearts and a cross face, and that is perfect for a 13 year old girl, so it was brilliant.</u> That probably was one of the biggest things this year as a best resource in maths anyway. (Emma, Interview 2, pg. 3)

This is a potentially revealing episode. The analogy used, successfully, by Emma is mathematics free. As such, it appears that no mathematics was used to clarify a mathematical idea. However, such an approach cannot be condoned. The love/hate analogy does not equip students with a mathematical reason as to why the rule holds. Relational understanding of the issue would require that a student could give a sound, mathematical reason as to why this rule works and crucially, they would be able to explain by reference to a real life example the convention in operation[4]. Crucially for Emma, it worked because no student had difficulties with applying the convention, therefore she must have taught it successfully.

Caron also stressed the importance of the ‘real world’ (Interview 2, p. 7). Consistent with emerging overlap in the role of the ‘real world’ in both science and mathematics reforms, she expressed a view that:

> A good mathematics teacher is no different to a science teacher…in that maths again has to be firmly anchored in what is around you…that would be key for me teaching maths…actually bringing the maths out of the class if you like.

The reference to the real world can be challenging for student teachers. Aoife, a student teacher of science and mathematics explains that:

> I know we have been told all year that if you make the situation a bit more real to the kids that they will be more inclined to learn it. <u>In maths there aren't always too many opportunities to make the situation realistic</u>, geometry for example, I mean what if the kids hear about the angles of things around them, you know, it is very hard to get them to be excited over angles. But with the statistics, fortunately... Maybe someone else has a good angles class to tell me about and I will tell them about my statistics experience.(Aoife, Interview 2, p. 8)

The above interview excerpt reveals that although the student teachers are being told of the importance of real world examples in their teaching, it is hard at times to adopt such an approach in their teaching of certain topics. However Thomas, another science and mathematics student teacher, offers a perceptive view when he states that such an approach would be a lot more difficult in mathematics than in science because “science is a lot more close to real life” (Interview 2, p. 8). This seems to suggest that enabling student teachers to become adapt at incorporating real life approaches in mathematics will require some time and effort in the methods course. The mathematisation of RME is a level above the simple application of real life examples in conventional teaching. The challenge for mathematics teacher educators involved in teacher education is to show student teachers, that like science, mathematics *is just as close* to real life. However, not all problem solving in mathematics is or needs to be ‘real world’ focused, nevertheless there is considerable potential in rooting much school-based mathematics in so-called ‘real world’ contexts. The brief account above all suggest that student teachers are engaging with ITE coursework and are grappling with importance problem solving, particularly so called real world-oriented problems, in their mathematics teaching. Crucially, the student teachers are working toward what Chazan, Callis and Lehman (2007, p. 78-79) describe as a present-oriented approach to mathematics teaching. Such an approach helps students make connections between their lives and the mathematics they are learning in school. They are simply “finding mathematics in the world around us” (*ibid*., p. 79).

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[4] See for example Steward (2010, p. 37)

CONCLUSION

This brief report, based on the *Learning to Teach Study*[5], highlights two aspects of learning to become a post-primary mathematics teacher which, we think, are central to current reforms: the nature of support experienced by student teachers in their teaching practice placements and the manner in which they understand and begin to enact a more problem solving-oriented approach to mathematics in their teaching. The findings on the nature of support suggest that mathematics student teachers experience at best moderate levels when it is their first subject, with low levels of support when it is their second teaching subject. This is of concern given the power of the apprenticeship of observation and the likelihood that it will become even more powerful when those learning to teach feel especially challenged by expectations of reform-oriented teaching. As such, the need for structured support (e.g. West & Staub, 2003) in student teaching becomes, we argue, even more important in a reform-oriented era (for discussion of teaching reform-oriented mathematics see Boaler, 2002; Darling-Hammond & Bransford, 2005; and for an initial study of Project Maths see Lubienski, 2011). Both structured support and a deeper engagement with pedagogy during and after initial teacher education (Conway, Murphy, Hall & Rath, 2011) are needed to help student teachers of mathematics to make sense of and enact new understandings of problem solving, be it real world-focused or not, and its role in mathematics. Otherwise, efforts to promote new ways of thinking about and teaching mathematics in Irish post-primary schools are likely to be short circuited.

REFERENCES

Ball, D. L. (1996). Teacher learning and the mathematics reforms: what we think we know and what we need to learn, *Phi Delta Kappan,* 77.

Boaler, J. (2002). *Experiencing school mathematics: traditional and reform approaches to teaching and their impact on student learning*. London: Lawrence Erlbaum.

Chazan, D., Callis, S. & Lehman, M. (2007). *Embracing Reason: Egalitarian Ideals and the Teaching of High School Mathematics.* New York: Routledge.

Conway, P. F. & Sloane, F. C. (2006). *International trends in post-primary mathematics education: Perspectives on learning, teaching and assessment*. Dublin: NCCA. Available online at: http://www.projectmaths.ie/learningResources/reference.asp

Conway, P.F., Murphy, R., Delargey, M. Hall, K., Kitching, K. Long, F., McKeon,J. Murphy, B., O'Brien, S. & O'Sullivan, D. (2011). *Learning to Teach Study (LETS): Curricular and Cross-curricular competences in Initial Teacher Education: Executive Summary.* School of Education, University College Cork.

Conway, P. F., Murphy, R., Hall, K. and Rath, A. (2011). Leadership and teacher education: Deepening engagement with pedagogy. In *Leading and Managing Schools*, ed. H. O'Sullivan, H. and J. Burnham-West, 89-110. London: Sage.

[5] LETS Executive Summary is available at: http://www.ucc.ie/en/education/research/

Conway, P.F., Murphy, R., Delargey, M., Hall, K., Kitching, K., Long, F., McKeon, J., Murphy, B., O'Brien, S., and O'Sullivan, D. 2011. *Learning to Teach (LETS): Developing curricular and cross-curricular competences in becoming a 'good' secondary teacher - Executive Summary*. (A study funded by the Department of Education and Skills.). http://www.ucc.ie/en/education/research/

Darling-Hammond, L., & Bransford, J. (Eds.) (2005). *Preparing Teachers for a Changing World: What Teachers Should Learn and Be Able to Do.* Washington: US National Academy of Education.

Freudenthal, H. (1991). Revisiting mathematics education. Dordrecht: Kluwer A.P

Goffree, F. (1993). Hans Freudenthal: Working on mathematics education, *Educational Studies in Mathematics*, 25 (1-2), 21-58

Grossman, P., Compton, C., Igra, D., Ronfeldt, M., Shahan, E., & Williamson, P. (2009). Teaching practice: A cross-professional perspective. *Teachers College Record*, 111,(9), 2055-2100.

Haggarty, L. (1995). *New Ideas for Teacher Education: A Mathematics Framework*. London: Cassell.

Likert, R. (1932). A technique for the measurement of attitudes. *Archives of Psychology*, 140, 1–55.

Lortie, D.C. (1975). *Schoolteacher*. Chicago: University of Chicago Press.

Lubienski, S. (2011). Mathematics education and reform in Ireland: An outsider's analysis of Project Maths. *Bulletin of the Irish Mathematical Society*, 67, 27-55

Lyons, M., Lynch, K., Close, S., Sheerin, E. and Boland, P. (2003). *Inside Classrooms: the teaching and learning of mathematics in social context*. Dublin: IPA.

Mertens, D. M. (2010). *Research and Evaluation in Education and Psychology: Integrating Diversity with Quantitative, Qualitative, and Mixed Methods (3rd edition)*. London: Sage.

Morine-Dershimer, G., Corrigan, S.(1996).Teacher beliefs. In H.J. Walberg & G.D. Haertel, (Eds.). *Psychology and educational practice*. San Pablo. CA: McCutchan Publishing.

National Competitiveness Council (2009). *Statement on Education and Training*. Dublin: Forfás.

National Council for Curriculum and Assessment (2005). *Proposals for the Future Development of Senior Cycle Education in Ireland,* Dublin: NCCA. Available online at: www.ncca.ie

NCCA (2009). *Senior cycle towards learning: listening to schools*. Dublin: NCCA.

OECD (2003). *The PISA 2003 assessment framework: Mathematics, reading, science and problem solving knowledge and skills*. Paris: OECD.

OECD (2006). *Assessing scientific, reading and mathematical literacy: A framework for PISA 2006*. Paris: OECD.

Oldham, E. (2001).'The culture of mathematics education in the Republic of Ireland: Keeping the faith?' *Irish Educational Studies* 20, 266-277.

Schroeder, T.L., & Lester, F.K. (1989). Developing understanding in mathematics via problem solving. In P.R. Trafton (Ed.), *New directions for elementary school mathematics* (pp. 31-56). Reston: NCTM.

Stewart, I. (2010). *Professor Stewart's Cabinet of Mathematical Curiosities*. London: Profile Books.

Tharp, R.G., & Gallimore, R. (1988). *Rousing Minds to Life: Teaching, Learning, and Schooling in Social Context*. Cambridge: Cambridge University Press.

Veenman, S. (1984). Perceived problems of beginning teachers. *Review of Educational Research,* 54, 2, 143-178.

Vygotsky, L. (1978). *Mind in Society" The Development of Higher Psychological Processes*. Harvard University Press.

West, L. & Staub, F.C. (2003). *Content-focused Coaching: Transforming Mathematics Lessons*, Portsmouth, NH: Heinemann.

LESSON STUDY IN PRACTICE: STAFF OWNERSHIP OF MATHEMATICS TEACHER PROFESSIONAL DEVELOPMENT

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In-post teacher professional development is a complex process where established practices constitute a 'black box' to the outside researcher (Goldsmith, Doerr & Lewis, 2009). Over three different teacher development studies these researchers detected four "shifts in attentional foci" as evidence of teacher learning. This study documents some further shifts in focus observed among members of a school staff, over a two year engagement, in small class-related groups, with lesson study aimed at improving the teaching of mathematics in an educationally disadvantaged school setting. The research is framed by two questions: Can engagement in lesson study enhance the teaching of mathematics in a particularly challenging school setting? How might such enhancement be characterised and measured? It builds on a research project which characterised the learning of student teachers engaged in lesson study using a community of practice heuristic, where notions of engagement, imagination and alignment were crucial to participants' learning and the development of a mathematics teacher identity (Corcoran, 2008). The disparate shifts in focus found in this study point to substantial and sustained learning at staff or 'community of practice' level (Wenger, 1998).

INTRODUCTION

The continuing professional development (CPD) of mathematics teachers in Ireland is an issue of growing importance to the international mathematics education community and in Ireland, as the title of this conference Mathematics Teaching Matters attests. A recently published National Strategy to Improve Literacy and Numeracy (Department of Education and Skills, 2011) posits that:

> We continue to be fortunate in Ireland in attracting some of the most able and talented people to the teaching profession. Teachers play a key role in providing a quality education. ... A quality teacher workforce must be cultivated and sustained within a framework of strong school leadership. This is essential in order to achieve improved student learning outcomes in literacy and numeracy (ibid. p. 27).

These sentiments move the debate about students' mathematical attainment at national level (Eivers, Close, Shiel, Millar, Clerkin, Gilleece and Kiniry, 2010) and on the international stage (Shiel, Moran, Cosgrave and Perkins, 2010) beyond 'single' issues such as mathematics teacher knowledge (Delaney, Ball, Hill, Schilling and Zopf (2008), out-of-field teaching (Ní Ríordáin and Hannigan, 2009), or curriculum appropriateness (Oldham and Close, 2009) to such 'hard' questions as 'what exactly is a "quality education" in terms of mathematics? How might "a quality teacher workforce be cultivated and sustained?" There is general consensus that teaching is a complex process the outcomes of which are of concern to the society at large, but defining and agreeing 'good' teaching is just as complex. In Ireland, until recently, teaching was considered to be "invisible and silenced, the silent discourse of the reform process" (Sugrue, 2004). Meanwhile, in an attempt to penetrate the 'black box' of classroom teaching on an international stage, Krainer (2005) identified three positions which interested parties can take on what constitutes 'good' teaching:

- Refusing norms: It is not possible to state clear criteria and indicators for good teaching which can be applied in a general way.
- Establishing norms: Teachers need a clear orientation for good teaching; educational policy needs a sound basis for monitoring and steering the educational system.
- Negotiating norms: All stake holders are expected to articulate their views on good teaching and thus become co-constructors of norms for what constitutes good mathematics teaching.

He continues that "research is about increasing our understanding of teaching, and about making normative assumptions about good teaching explicit, and also about further developing teaching". In this paper I report on a study where the participants in the study unveiled and made public their teaching and in doing so adopted a "negotiating norms" stance in pursuit of improved mathematical attainment goals.

THEORETICAL FRAMEWORK AND RELATED LITERATURE

This study draws on two strands of literature, that relating to mathematics teacher CPD and that concerning lesson study. First, I refer to a framework for teacher learning that recognises four domains where it might occur. These are related to the pattern of CPD for mathematics teachers in Ireland and will be related to international research findings on teacher learning from CPD. Then, I raise some salient points concerning the practice of lesson study as an opportunity for teacher learning.

Support for Primary Mathematics Teaching

Concomitant with the introduction of the revised primary mathematics curriculum (Government of Ireland, 1999) a dedicated body was set up to promote its implementation. The Primary Curriculum Support Programme (PCSP) under the auspices of the statutory National Council for Curriculum and Assessment (NCCA) consisted of teachers who had been seconded from primary schools to give in-service support to teachers on various curricular subjects. Often the same presenters covered a different curricular subject each year. PCSP personnel gave in-service courses in mathematics education at regional level to groups of teachers released from teaching for two to three days annually. This service continued for the first two years of implementation. It was then adulterated to the provision of regional 'cuiditheoirí', who were available to clusters of schools all over the country in an advisory capacity, for a number of subjects. A new Primary Professional Development Service (PPDS) was formed in 2008 and a team of advisors continues to provide support primarily to schools in Urban Bands 1 and 2 DEIS schools in the areas of literacy and numeracy. With regard to mathematics they promote two 'ready-made' programmes: Maths Recovery and Ready Set Go-Maths. This initiative means of 'establishing norms' is considered beneficial by the teachers who participate, although studies of mathematical attainment in many socially disadvantaged schools continue to cause concern (Eivers, et al., 2010).

Support for Project Maths

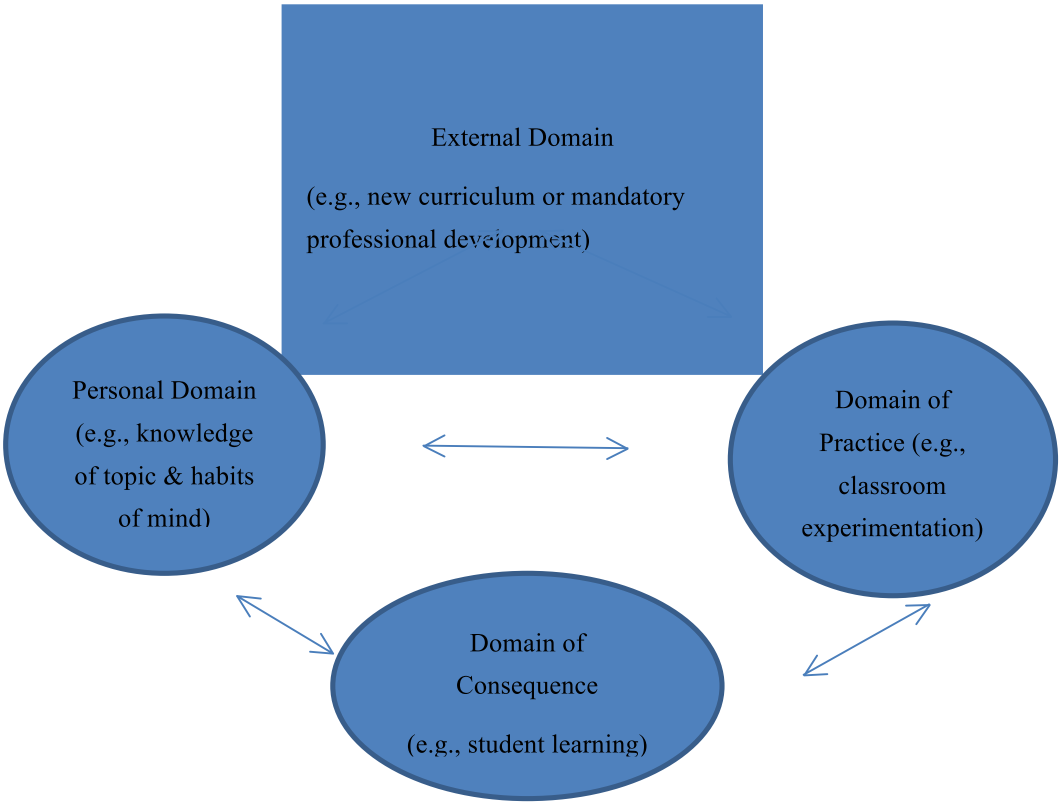
In second level schools an augmented version of this model of CPD is being used to implement the new mathematics syllabuses for Junior and Senior cycle- Project Maths. Having been piloted in 24 schools, project maths is now being rolled out nationwide with three rounds of workshops per year to initiate practising teachers into providing for "an enhanced student learning experience and greater levels of achievement for all" (Project Maths Development Team, 2011). An extensive website provides considerable support materials for curriculum implementation in the form of teaching and learning plans, learning and teaching resources, material created by teachers, examination papers and video records of teachers and children doing Project Maths. The Project Maths CPD pilot initiative was deemed to be successful and its implementation nationally is still in the early stages and growing organically. While the Project Maths approach is innovative in Irish terms, it still takes the form of work-shop style training, which has tended to be the typical professional development for mathematics teachers. It may be that further shifts are needed to other alternative forms of CPD, which are even more school-based, teacher-centred and focused on the vagaries of student learning. In other words, a genuine shift from 'establishing norms' to 'negotiating norms' may be needed to effect "an enhanced student learning experience and greater levels of achievement for all students" (ibid).

A Framework for Domains of Teacher Learning

The ultimate goal of teacher professional development is the improvement of students' learning through the improvement of teaching, which has a direct effect on student learning. Based on the Clarke & Hollingsworth (2002) framework, (see figure 1) one can think of teacher learning as arising out of the External Domain of mandatory CPD, as has been the expectation for the primary mathematics curriculum and Project Maths. However, the framework is not linear and movement (i.e., teacher learning) from the External Domain to the Domain of Consequence is far from straightforward. Maass (2011) in a small study of six teachers, found that teachers' beliefs about effective teaching (Personal Domain) had a major impact on whether or not they intended to change their classroom practice (Domain of Practice) and identified two types of teachers - the *static* and the *process type* - who demonstrated very different reactions to a professional development intervention.

Clark & Hollingsworth suggested two processes – *enaction* and *reflection*- which were indicative of teacher growth. Building on their work, Goldsmith, Doerr and Lewis (2009) added a third process that of noticing (Ainley & Luntley, 2007; Jacobs et al., 2007; Mason, 2002). They found that "shifts in *noticing* (or attention to) elements of practice may help drive teachers' *enactions* and *reflections* and, in turn, be further sharpened by them" (Goldsmith et al., 2009, p. 99). Their study referred to three different forms of mathematics teacher professional development of which one was lesson study. In the case of lesson study, teachers were found to shift from an 'evaluative mindset' which looked at a lesson globally to a 'knowledge-building mindset', "focusing more heavily on both knowledge sources within the school (e.g., student thinking) and external knowledge sources (e.g., research, outside expertise and programs)." They were able to trace shifts in teacher development through analysis of individual lesson study cycles across all four domains of teacher growth.

Figure 1: A model for professional growth (adapted from Clarke & Hollingsworth, 2002)



Implications for CPD

Research findings further indicate that efforts to achieve growth and improvement in teaching mathematics can be achieved in four major areas (Doerr, Goldsmith & Lewis, 2010). First, teacher learning which impacts on pupils' achievement can be achieved through building teachers' mathematical knowledge and their capacity to use it in practice (Hill, Rowan & Ball, 2005). The focus here is on the practice of teaching (Fernandez, 2005; Lewis, Perry and Hurd, 2009) and on mathematical proficiency (Kilpatrick, Swafford, & Findell, 2001) that extends beyond a mere understanding of procedures and concepts. Second, improved teaching results from building teachers' capacity to notice, analyse and respond to students' thinking. This finding results from teacher participation in Cognitively Guided Instruction (CGI) professional development (Fennema, Carpenter, Franke, Levi, Jacobs and Empson (1996) and was found to be sustained long after the professional development had ceased (Jacobs, Franke, Carpenter, Levi, & Battey, 2007). Third, successful professional development programmes "build teachers' productive habits of mind since learning to improve one's teaching practice is challenging, effortful work" (Doerr, Goldsmith & Lewis, 2010, p. 2). If teachers are to promote "a mathematical disposition" in their pupils, they must themselves develop in an "integrated way" the aptitudes characteristic of a "mathematical disposition" (De Corte, 2004). Fourth, build collegial relationships and structures that support collegial work (Jaworski, 2006). Teachers working together on mathematical problems and sharing solutions can help teacher learning, as can working together to analyse pupils' mathematical thinking (Borko, Jacobs, Eiteljorg, & Pittman (2008). Participation in a community of practice devoted to lesson study can result in productive power distribution among the members and considerable personal agency in that communal space (Corcoran, 2011).

Lesson Study

Lesson study is a form of continuing professional development which appears particularly suitable to mathematics teaching (Corcoran & Pepperell, 2011). The grass roots practice of lesson study is widespread though out Japan and growing in popularity in the US, the UK and elsewhere. A typical lesson study cycle consists of four-six parts. 1. The collaborative study of resource materials and planning of the research lesson. 2. The teaching/observing of the research lesson. 3. Discussing the research lesson. 4. Revising the research lesson. 5. Teaching the new version of the lesson (optional). 6. Sharing reflections on the new version of the lesson (Krainer, 2011). Successful participation in lesson study by US teachers was first theorised by Frernadez, Cannon and Chokshi (2003) as requiring three essential lenses, the researcher lens, the curriculum developer lens and the student lens, which they found characterised Japanese teachers. Participation in lesson study has been found to help build collective teacher efficacy (McLoone, 2010). For an in-depth exposé on lesson study see Hart, Alston & Murata (2011).

METHODOLOGY

This study began in autumn 2008 when the researcher was invited to 'help' a local inner city school to enhance its mathematics teaching as part of the school staff's commitment to achieving improved student performance goals in mathematics, in accordance with their DEIS goals. It builds on a previous research project which characterised the learning of student teachers engaged in lesson study using a community of practice heuristic, where notions of *engagement*, *imagination* and *alignment* were crucial to participants' learning and the development of a mathematics teacher identity (Corcoran, 2008). Following that study, which had included investigation of three research lessons based in an Urban Band 1 DEIS school, I contended that lesson study appeared to be a potent recipe available to teachers to address the challenges associated with teaching mathematics for equity in designated disadvantaged schools.

The research project is framed by two questions:

- Can engagement in lesson study enhance the teaching of mathematics in a challenging school setting?
- How might such enhancement be characterised and measured?

In this study, small groups of five/six teachers participated in five lesson study cycles, one per term. Data for this paper include DVDs of the five research lessons, teachers' lesson plans and 'original' teacher-designed teaching materials, some samples of students' work, my field notes of the collaborative planning and review meetings that I attended and a research lesson data collection sheet arising out of participation in lesson study cycle 5.

The Lesson Study Setting

Better Lessons National School is an inner city Urban Band 1 vertical school with approximately 200 pupils on rolls and a staff of twenty one teachers. There are twelve mainstream class teachers and eight teachers with either Learning Support or English as an Additional Language posts. In autumn 2008, the school staff directed a teacher with responsibility for mathematics to approach St Patrick's college for 'help' in raising mathematics achievement scores. The teacher, Ms Atwell had already undertaken personal professional development in mathematics education and had successfully completed an in-service diploma course. Her special interest in mathematics education was evident in the school where she had researched children's attitudes to mathematics (McLoone, 2007). I adopted a "reflective rationality" approach as researcher (Krainer, 2011). Because of my interest in supporting teacher development I welcomed the opportunity to work with and learn from teachers as a staff *in situ.*

Introductory Sessions

I attended the school for one hour on a weekly basis for four sessions to meet groups of up to eight teachers who were released by others to attend for informal discussion on how to teach mathematics differently. They expressed a desire to increase the use of problem-solving and the use of manipulatives. For example, I presented some problem scenarios relating to fractions that they might try in class during the following week and we discussed examples of children's work which the teacher brought. Participants might be said to be collaborating by discussing the teaching of mathematics and they might be said to be "reflective practitioners" (Schön,1983) but I felt the time could be more usefully spent and proposed helping to facilitate a cycle of lesson study. I was subsequently invited to address a staff meeting about my research, and to outline the practicalities for teachers engaged in lesson study. I stressed that the lesson study research project would be a valuable learning opportunity for me and sought ethical approval to engage with the school for an extended period. The project was also supported by a research grant from the Research Committee of the college.

THE LESSON STUDY RESEARCH

In this study, small groups of four/five teachers participated in five lesson study cycles, one per term extended over three school years, so that all teachers had an opportunity to participate in at least one cycle of lesson study. I observed all research lessons and acted as Knowledgeable Other to the staff in their engagement with lesson study (Watanabe and Wang-Iverson, 2005).

Table 2: Research lessons taught according to class level

	Autumn term	Spring term	Summer term
2008-2009	Introductory sessions	Third class: Measures	Senior infants: Number
2009-2010	XXXXXX	Sixth class: Chance	Second class: Number
2010-2011	Fifth class: Fractions	YYYYYY	XXXXXX

XXX Lesson study did not take place due to my unavailability. YYY Lesson study did not take place due to changes in school personnel.

The project generated a huge amount of data of different grain sizes and analysis was possible on many levels, a) analysis of the five research lessons as mathematics teaching events, b) analysis of teachers' learning from participating in lesson study (both individual and collective), c) analysis of my own learning from participation in the process. For the purposes of this paper I will draw from various data sources to attempt to answer the two questions. While it recognised that the research lesson is not by any means the only site of teacher growth during a lesson study cycle, it is however, the pivot of teacher development activity, rooted as it is in the Practice Domain. Local details of who would teach what research lesson were all arranged by the teachers themselves. Because I was present at each research lesson and made video-recordings of each one, they have become a primary source of data for me.

In Better Lessons NS a DVD of each research lesson and a copy of the lesson plan is stored in the staff room as a resource for use by teachers. This retrospective use of the research lessons as "actionable artefacts" was cited as a possible benefit of lessons study (Lewis, Perry & Murata, 2006). With wider mathematics education community involvement, such artefacts have potential to become part of a "negotiating norms" approach to good mathematics teaching and contribute to "shared instructional products" proposed by Morris & Hiebert (2011) as an effective means of improving teaching at national level.

Brief Synopses of Research Lessons

The first research lesson, taught by Ms Atwell, was a problem-based lesson for third class which enquired if the daily milk consumption of the class could be met by the yield from a cow (Corcoran, 2010). The participant teachers appeared to enjoy the experience but as a once off event, notwithstanding the multiple hours of collaborative work involved in the lesson study cycle, it would be difficult to place lesson study cycle 1 on any of the teacher learning domains (Clarke & Hollingsworth, 2002). In an attempt to relate the professional learning of the teacher of the research lesson to the framework, she and I engaged in video-stimulated recall of the lesson eight months later and the teachers comments on the research lesson were audio-recorded and analysed, indicating a wealth of data still to be mined in relation to teacher learning over time (Corcoran & McLoone, in preparation).

Research lesson 2 was with Senior Infants and took place towards the end of the school year. Again, a problem-based approach to the number strand was used. Children calculated quantities of lollipops that Mammy might distribute over a week to her two/three children, where they each got different amounts. Lollipop sticks were used as children counted and observing teachers were struck by how valuable the DVD of the lesson would be to the teacher of that class in the following year, since it was a record among other things of the children's counting strategies. This lesson was also interesting to the teacher of the research lesson and others observing, because when she summarised the lesson with a series of flashcards that individual children placed in correct order – number bonds totalling 6 - on the board, she asked if anyone recognised a pattern. To her surprise and delight, one child spotted that 4 + 2=6 was the same as 2+4 =6. Others joined in seeing the other 'turn arounds'. Another child drew attention to the 0, 1 … 6 sequence in descending and ascending order. Teachers in Better Lessons NS would all have tended to describe their classes as weak at mathematics and in need of much drill and practice. By research lesson 2, lesson study participants were beginning *to notice* and pay closer attention to *children's ways of thinking and being mathematical*.

Research lesson 3 was with a fifth class and focused on the curricular strand of Chance. The opening activity chosen by the participating teachers was a mathematical game which children had played as part of a home/school programme. Some teachers were unsure of the underlying mathematics and this lesson study cycle appeared to contribute to building mathematical knowledge as teachers come to terms with teaching a topic which was relatively new to them. There appeared to be a growing openness and preparedness to take risks among the group in the third cycle. In each of the first three cycles of lesson study teachers had cooperated to create resources – tally sheets, charts, flash cards etc.,- all useful for the pedagogy of the mathematics involved in the lesson.

Research lesson cycle 4 was notable because a teacher-designed resource was the basis of the mathematics lesson itself. During the preparation meetings we talked about a context for addition and subtraction sums. The pros and cons of using money as a context were discussed and it was decided to base the lesson on choosing items from a 'real' menu for Mario's Café. Food items, realistic prices and possible solution strategies were discussed in detail. Questions about the orthodoxy of writing HTU as a heading for columns of figures to be added were debated, where children were calculating amounts in euro and cents to a total of ten euro. This lesson was very active as second class children worked in pairs to choose their meal items and totted and counted what they owed. Besides the video-recorder I used to record the lesson, teachers around me observing the lesson were busy taking notes and making short video clips on their own flick cameras. The lesson went over the allotted time because children and teachers were so engaged there appeared to be reluctance to bring the lesson to a close. I had to leave early and it was not possible for me to be present at the post-lesson discussion on that occasion. However, one of the teachers undertook to 'fill me in' by forwarding video clips she had made of children finishing their tasks. I was happy to believe that the teachers in Better Lessons NS felt ownership of the lesson study process at that stage and were sufficiently initiated to 'go it alone'. In trying to answer research question two, I find it is difficult to enumerate instances of individual teacher learning in small group work such as this. However, I offer one artefact from lesson study cycle 5 as evidence of collective teacher growth in all four domains outlined by Clarke & Hollingsworth (2002).

DISCUSSION OF FINDINGS

Lesson study cycle 5 was with a sixth class. The teacher of the research lesson had been a participant observer of lesson study cycle 3, from which I infer that the lesson study process was becoming 'embedded' in the culture of the school. The chosen topic was fractions and during the first planning meeting, I was asked to suggest a challenging lesson to develop children's thinking about fractions. I suggested the Marilyn Burns "Fractions with Cookies" lesson (1987) and dropped in a copy of the chapter to the school. Teachers worked away to prepare the research lesson and I attended another hour long meeting nearer the date of the research lesson where they discussed how children would be asked to record their findings on sharing different amounts of cookies among four children. The mathematical depth of this lesson and its value as a teacher professional development task has been documented by Fernandez (2005) and I was keen to see what the Better Lessons lesson study team would make of it. On the research lesson day, observing teachers were handed a sheet with seven key data collection points listed. In my opinion, the creation of this sheet was an impressive step forward in teacher growth. The attentional focus of each data collection point was on the Domain of Consequence, a process of attending to children's thinking, often through the Domain of Practice and occasionally through the Personal Domain of teacher beliefs. The teachers in this study began their studying children's work at a very practical and 'practice' oriented level.

a) We want to observe how the children will tackle the challenge, e.g., will they opt to use scissors etc.

Their data collection then moved to evidence of the children's development of the concept of fractions.

b) What does equal mean to the children? Do they realise the shares will be the same size each?

c) Can they represent numbers as fractions? How will they record this?

d) Do they understand the concept of numerator and denominator?

There are resonances of teachers who have experienced disappointment in their pupils' previous performance in the next data collection point.

e) Have they retained the symbolism of fractions which they have covered in previous years?

Evidence of either positive or negative answers to this question, and any of the others, could be a very useful starting point for another lesson study cycle. The personal beliefs of teachers about the 'situatedness' of student learning of mathematics are in evidence in the next question.

f) We think children from bigger families might be better at sharing. We want to see if this has any relevance.

Finally, again in the Domain of Consequence, the teachers ask themselves:

g) Will any of them [pupils] grasp the concept of equivalent fractions?

CONCLUSION

The intricate processes by which the teachers in this study, made disparate shifts in "attentional foci" from a broad aspiration of wanting to teach mathematics using problem-solving and manipulatives, to seeking detailed evidence of children's manner of learning specific important mathematical ideas arose through school-wide engagement in lesson study. They appear to indicate substantial and sustained learning at staff or 'community of practice' level, where teachers examine their own practices from the perspective of Consequence and student outcomes, while supporting each other in the on-going task of improving their teaching of mathematics. Research lessons were being taught and observed using the three essential, successful lesson study teacher lenses (Fernandez et al., 2003). *Enactment* of critical inquiry (Jaworski, 2006) into 'reform –oriented' mathematics lessons and *reflection* on mathematics teaching has become public and is shared. *Noticing children's responses* is explicit and has become integral to teaching. In the Domain of Practice the focus of this staff is now directed towards responsibility for their own growth in "good teaching" by considered and informed studying of teaching, using lesson study protocols. The substantial time invested and the creative, flexible system support available to them has resulted in many opportunities for active learning at Better Lessons National School.

REFERENCES

Ainley, J., & Luntly, M. (2007). Toward an articulation of expert teaching practice. *Teaching and Teacher Education*, 23, 1127-1138.

Borko, H., Jacobs, J., Eiteljorg, E., & Pittman, M. (2008). Video as a tool for fostering productive discussions in mathematics professional development. *Teaching and Teacher Education,* 24, 417–436.

Burns, M. (1987). Fractions with "Cookies," in *A collection of math lessons: from grades 3 through 6*. Sousalito, CA: Math Solutions.

Clarke, D. & Hollingsworth, H. (2002). Elaborating a model of teacher professional growth. *Teaching and Teacher Education*, 18(8), 947- 967.

Corcoran, D. (2011). Learning from Lesson Study: Power Distribution in a Community of Practice. In Hart, L., Alston, A. & Murata, A. (Eds.). *Lesson Study Research and Practice in Mathematics Education: Learning Together*. USA: Springer.

Corcoran, D., & Pepperell, S. (2011). Learning to Teach Mathematics Using Lesson Study, in *Mathematical Knowledge in Teaching*. In T. Rowland & K. Ruthven (Eds.) *Mathematical Knowledge in Teaching*. London: Springer.

Corcoran, D. (2010). Potentials and Pitfalls of Facilitating an Authentic Problem-based Mathematics Lesson. Paper presented at SMEC 2010, Dublin City University, September.

Corcoran, D. (2008). *Developing Mathematical Knowledge for Teaching: A Three-Tiered Study of Irish Pre-service Primary Teachers*. Unpublished doctoral thesis. University of Cambridge.

Corcoran, D, & McLoone, M. (in preparation). Revisiting a research lesson through video-stimulated recall : multiple opportunities to notice and reflect on teaching.

De Corte, E. (2004). Mainstreams and Perspectives in Research on Learning (Mathematics) from Instruction. *Applied Psychology: An International Review*, 53 (2) 279-310.

Delaney, S., Ball, D.L., Hill, H.C., Schilling S.G., & Zopf, D. (2008). "Mathematical knowledge for teaching": Adapting U.S. measures for use in Ireland. *Journal of Mathematics Teacher Education* 11(3).

Department of Education and Skills (2011). Literacy and Numeracy for Learning and Life: The National Strategy to Improve Literacy and Numeracy among Children and Young People 2011-2020. Available online at: www.education.ie.

Doerr, H. Goldsmith, L. & Lewis, C. (2010). Mathematics Professional Development: Professional Development Research Brief. National Council of Teachers of Mathematics. Available online at:www.nctm.org/uploadedFiles/Research_News_and_Advocacy.

Eivers, E., Close, S., Shiel, G., Millar, D., Clerkin, A., Gilleece, L., & Kiniry, J. (2010). The 2009 National Assessments of Mathematics and English Reading. Dublin: Educational Research Centre.

Fennema, E., Carpenter, T. P., Franke, M. L., Levi, L., Jacobs, V. R., & Empson, S. B. (1996). A longitudinal study of learning to use children's thinking in mathematics instruction. Journal for Research in Mathematics Education, 27(4), 403–434.

Fernandez, C. (2005). Lesson study: A means for elementary teachers to develop the knowledge of mathematics needed for reform-minded teaching? *Mathematical Thinking and Learning*, 7 (4), 265–289.

Fernandez, C., Cannon, J. & Chokshi, S. (2003). A US—Japan lesson study collaboration reveals critical lenses for examining practice, *Teaching and Teacher Education,* 19, 171-185.

Goldsmith, L., Doerr, H. & Lewis, C. (2009). Opening the Black Box of Teacher Learning: Shifts in Attention. In Tzekaki, M., Kaldrimidou, M. & Sakonidis, H. (Eds.) *Proceedings of the 33rd Conference of the International Group for the Psychology of Mathematics Education*, Vol. 3, pp. 97-104. Thessaloniki, Greece: PME.

Government of Ireland (1999). *Primary School Mathematics Curriculum*. Dublin: Stationery Office.

Hart, L., Alston, A., & Murata, A. (Eds.) (2011). *Lesson-study research and practice in mathematics education: Learning together*. Dordrecht, The Netherlands: Springer.

Hill, H., Rowan, B., & Ball, D. (2005). Effects of teachers' mathematical knowledge for teaching on student achievement. *American Educational Research Journal, 42* (2), 371–406.

Jacobs, V., Franke, M., Carpenter, T., Levi, L., & Battey, D. (2007). Professional development focused on children's algebraic reasoning in elementary school. *Journal for Research in Mathematics Education*, 38 (3), 258–288.

Jaworski B. (2006) Theory and Practice in Mathematics Teaching Development: critical inquiry as a mode of learning in teaching. *Journal of Mathematics Teacher Education*. Special Issue, 9, 2, 187-211.

Kilpatrick, J., Swafford, J., & Findell, B. (Eds.) (2001). *Adding it up: Helping children learn mathematics*. Washington, DC: The National Academies Press.

Krainer, K. (2011). Teachers as stakeholders in mathematics education research. In Ubuz, B. (Ed.) *Proceedings of the 35th Conference of the International Group for the Psychology of Mathematics Education.* Vol. 1, pp. 47-62. Ankara, Turkey: PME.

Krainer, K. (2005). What is "good" mathematics teaching, and how can research inform practice and policy? Editorial. *Journal of Mathematics Teacher Education*, 8, 75-81.

Lewis, C., Perry, R., & Hurd, J. (2009). Improving mathematics instruction through lesson study: A theoretical model and North American case. *Journal of Mathematics Teacher Education, 12* (4), 285–304.

Lewis, C. Perry, R. and Murata, A. (2006). How Should Research Contribute to Instructional Improvement? The Case of Lesson Study, *Educational Researcher*, 356 (3), 3-14.

Maass, K. (2011). How can teachers' beliefs affect their professional development? ZDM Mathematics Education. Published online at: www.springer.com

Mason, J. (2002). *Researching your own practice: The discipline of noticing*. Routledge-Falmer: London.

McLoone, M. (2007) Examining the relationship between children's attitudes and achievement in mathematics: A study of twenty children in a mixed fifth and sixth class in one of Dublin's inner-city schools. Unpublished diploma thesis, St Patrick's College.

McLoone, M. (2010).The Impact of Participating in Lesson Study on Teachers' Beliefs and Attitudes in Relation to Teaching Mathematics. Unpublished master's thesis, St Patrick's College.

Morris, A., & Hiebert, J. (2011). Creating Shared Instructional Products: an alternative approach to improving teaching. *Educational Researcher*, Vol. 40, 1, pp.5-14.

Ní Ríordáin, M., & Hannigan, A. (2009). Out-of-Field Teaching in Post-Primary Mathematics Education: an analysis of the Irish context. Limerick: NCE-MSTL

Oldham, E. and Close, S. (2009). Solving problems in mathematics education: challenges for Project Maths. In D. Corcoran, T. Dooley, S. Close & R. Ward (Eds.) *Proceedings of the Third National Conference on Research in Mathematics Education* (MEI3), pp.295 – 308. Dublin: St Patrick's College.

Project Maths Development Team (2011). Learning and Teaching for the 21st Century. Available online at: www.projectmaths.ie

Schön, D. (1983). *The Reflective Practitioner*. New York: Basic Books.

Shiel, G., Moran, G., Cosgrove, J., and Perkins, R. (2010). *A Summary of the Performance of Students in Ireland on the PISA 2009 Test of Mathematical Literacy and a Comparison with Performance in 2003.*Dublin: Education Research Centre.

Sugrue, C. (2004). Whose Curriculum is it Anyway? Power, Politics and Possibilities in the Construction of the Revised Primary Curriculum, in C. Sugrue (Ed.) *Curriculum and Ideology: Irish Experiences, International Perspectives.* Dublin: Liffey Press.

Watanabe, T. and Wang-Iverson, P. (2005).The Role of Knowledgeable Others. In (Eds.) P. Wang-Iverson and M. Yoshida) *Building Our Understanding of Lesson Study,* pp. 85-92. Philadelphia, PA: Research for Better Schools.

Wenger, E. (1998). *Communities of Practice*. UK: Cambridge University Press.

RETHINKING APPROACHES TO THE LEARNING AND TEACHING OF MATHEMATICS FOR 3-6 YEAR-OLD CHILDREN IN EARLY EDUCATION SETTINGS IN IRELAND.

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In Ireland, there is currently a great deal of attention by policy makers on standards in Mathematics. This is evidenced by the recent publication of the discussion paper Draft National Plan to Improve Literacy and Numeracy in Schools (Department of Education & Skills, 2010), quickly followed by Literacy and Numeracy for Learning and for Life: The National Strategy to Improve Literacy and Numeracy Among Children and Young People (DES, 2011). While these documents pay some attention to the need to consider issues related to learning and teaching mathematics in early childhood, there is need for much more explication of its importance for children's subsequent mathematics learning. This paper discusses current provision for mathematics learning for children aged 3-6 years in early education settings in Ireland. In doing so it identifies and discusses a number of issues that should be addressed to ensure optimal support for mathematics learning for all young children and across the range of early education settings. In particular the need for specific types of guidance and support for educators is highlighted.

INTRODUCTION

The Draft National Plan to Improve Literacy and Numeracy in Schools (Department of Education & Skills, 2010) i.e. the consultation document which preceded the Literacy and Numeracy for Life (2011), asserted that the teaching and learning of mathematics in Ireland requires even more attention than literacy and that there are systemic issues in mathematics education that require attention. It recognised 'the education system needs to refocus on the development of the concepts and skills needed to establish the sound foundations for … numeracy development' (p. 11). Literacy and Numeracy for Life (DES, 2011) states that early childhood education should 'provide appropriate developmental experiences that foster a firm foundation for later literacy and numeracy skills' (p. 47). This view is entirely consistent with research indicating that the foundation of children's mathematical development is established in the earliest years (e.g. Saracho & Spodek, 2008; Sarama & Clements, 2009) and which identifies the early childhood years as especially important for mathematics development. It seems clear then that one obvious way to address problems in relation to mathematics attainment in Ireland is to focus on early childhood mathematics learning and teaching. This involves a consideration of current provision for 3-to-6-year old children in early education settings.

In the Irish context then, the issues of pedagogy and curriculum for early childhood mathematics need attention. Internationally, recent years have seen some key developments in relation to both. For instance, in the United States, in recognition of the need for pedagogical guidance for early childhood mathematics teaching, a range of experts including mathematicians, mathematics educators, researchers, curriculum developers, teachers and policy makers came together to agree the Standards for Early Childhood Mathematics Education (Clements, Sarama & DiBiase, 2004). Seventeen research-based recommendations were developed, several of which pertain directly to pedagogy. The standards also address the issue of curriculum and there are a number of recommendations regarding ways of ensuring a high-quality curriculum for very young children. This paper discusses current provision for mathematics learning for children aged 3-6 years in early education settings in Ireland. In doing so it identifies a number of issues that should be addressed to ensure optimal support for early mathematics learning for all children.

BACKGROUND

Traditionally young children in Ireland can be enrolled in primary schools from the age of four, and currently half of all four-year-olds and almost all five-year-olds experience their early education in infant classes in primary schools. Also, there are approximately 1,600 three-year-old children, deemed to be at risk of educational disadvantage, enrolled in half-day pre-school sessions in Early Start units in primary schools. From 2010, state provision of early childhood education has been extended to all children (aged 3-4 years) in the year prior to starting primary school (Free access to 2 hours 15 minutes a day, 5 days a week, 50 weeks of the year). This development is designed to give children access to appropriate programme-based activities in the year before they start primary school. Participation…provides children with their first formal experience of early learning…(Office of the Minister for Children and Youth Affairs, 2009). According to reports in the media in September 2011 70,000 three and four year-old children availed of the offer of a free preschool place.

A Curriculum Framework

Aistear: The early childhood curriculum framework presents early learning (for children aged birth to six) using four themes. These comprise of well-being; identity and belonging; communicating and exploring and thinking (NCCA, 2009).

The Aistear framework supports learning by identifying aims and learning goals in relation to each of the four themes. Since Aistear is a curriculum framework and not a curriculum, it doesn't specify content for specific areas of learning such as mathematics or literacy. Rather it provides the framework into which curricula fit and with which pedagogical strategies can be aligned. In relation to mathematics one learning goal is specified in the Aistear Framework: that children 'will develop counting skills, and a growing sense of the meaning and use of numbers and mathematical language in an enjoyable and meaningful way' (NCCA, 2009, p. 35). So how do early childhood educators in Ireland decide what mathematics to teach, how to teach it and when to teach it?

A Range of Curricula

For children attending preschool in the year prior to starting school, there is currently no common structured systematic mathematics curriculum in place. Providers are free to use any curriculum guidelines they wish to use. They are also free to use none. There are a range of curricula in use including High-Scope, Montessori and Early Start. The mathematical content of each of these curricula and any specific pedagogical strategies that they employ in supporting and developing children's mathematical learning depend to a large extent on their underlying theoretical perspectives, but also on when and for what population they were designed.

Here in Ireland the Primary School Curriculum (Government of Ireland, 1999a) provides a systematic mathematics curriculum for those children aged 4-6 attending infant classes in primary schools. The curriculum is based on a number of key principles and provides a detailed statement of content in the form of skills and concepts to be acquired, and learning objectives to be achieved. The Primary School Curriculum: Mathematics: Teacher Guidelines (Government of Ireland, 1999b) accompanies the curriculum statement and these are described as '... an aid and resource for teachers and schools as they encounter the curriculum and begin to implement its recommendations.' (p. 66). The guidelines seek to explore a wide range of approaches and methodologies that develop new emphases and give expression to new thinking on teaching and learning. While the curriculum emphasises some key pedagogical considerations, for example, the importance of children's experiences as key reference points in learning mathematics, the guidance for early infant classes does not specify in any detail current emphases in relation to mathematics learning and teaching in early childhood. There is no discussion of the big ideas of early mathematics, nor of the ways in which children can interact with these at different levels of thinking. In terms of pedagogy, for instance, scant attention is paid to how play can be used to support early mathematics learning. Also mathematical experiences building on children's play are not well articulated in the guidance for infant teachers.

The Delivering Equality of Opportunity in Disadvantaged Schools (DEIS) scheme focuses on addressing the educational needs of children and young people from disadvantaged communities. As part of the DEIS Action Plan, professional development programmes were provided to enable teachers to implement internationally-recognised numeracy programmes in designated schools (for further details of these programmes see http://www.education.ie/servlet/blobservlet/si_deis_numeracy.pdf). One significant criticism of those selected for use in with infant-aged children in DEIS schools is that they focus only on number, though they are presumably supplementing the more comprehensive Primary School Curriculum which introduces children to other aspects of mathematics besides number.

The different traditional philosophical orientations associated with the provision of education (for 4-6 year old children) and care (for 3 year old children) should be regarded as a significant factor. In short, educators of 3 year-old children have in the main tended to implement a largely play-based curriculum wherein mathematics learning is informal and incidental. Mathematical ideas may be introduced or expanded on as, and if, they arise in everyday play and routines. This lack of planning may result in very little mathematics learning. On the other hand, educators working in primary schools with children in infant classes are required to work with a structured curriculum and with pre-specified objectives. In practice research has shown (e.g. Dunphy, 2009) that this has often resulted in teachers utilising over-formalised approaches (especially in relation to teaching number-related concepts), wherein the curriculum content is prioritized and children's involvement in the processes of mathematics (e.g., problem-solving, reasoning, communicating) is relatively neglected. Also worth noting is the proposition that the development of children's dispositions for mathematics learning (e.g., persistence, curiosity, willingness to experiment) will not be optimally addressed in such circumstances. Indeed children's dispositions may in fact be damaged (Katz, 3003).

RETHINKING APPROACHES TO EARLY MATHEMATICS LEARNING AND TEACHING IN IRELAND

The last decade in particular has seen a vast increase in research related to a number of aspects of early mathematics including curriculum and pedagogy (e.g. NCTM/NAEYC 2002; Clements, Sarama & DeBiase, 2004; Saracho & Spodek, 2008). Clements & Sarama (2009), as a result of examining the research, conclude that early childhood classrooms in the United States underestimate children's ability to learn mathematics and are generally ill-suited to helping them learn in this area. Systematic and structured learning experiences are important and necessary elements in building sound foundations for mathematical understandings and later achievement of children in the age range 3-6 years (e.g. Clements & Sarama, 2009). While some of the elements which contribute to high-quality provision as indicated by the research are in place in Ireland, there is still a great deal of room for improvement.

A key issue for policy makers in Ireland is the diversity in levels of qualifications amongst early childhood educators working with 3-6 year old children. Critical to the quality of provision are the educators. One issue for many of these is in relation to what is considered appropriate mathematics for the youngest children. Research indicates a general acceptance amongst many early educators that specific mathematical experiences are not necessary (e.g Starkey, Klein & Wakeley, 2004); that a laissez faire approach will result in mathematics learning (e.g. Ginsburg et al., 2005) or that play in itself is sufficient to promote mathematics learning (e.g. Ginsburg, Cannon, Eisenband & Pappas, 2006). There is much work to be done in ensuring that all educators of young children are aware of the research on mathematics learning in early childhood: what is possible, necessary and desirable. Educators must be knowledgeable about how young children learn, but they must also understand mathematics and have a reasonable standard of mathematics education themselves. They must be able to recognise and develop the mathematics in children's activities and behaviours (e.g. Ginsburg & Ertle, 2008).

Planned mathematics activity for young children has always been a feature of The Primary School Curriculum but the weight of the evidence suggests that a planned approach with specified core content needs to be extended to three-and four-year-old children who avail of the preschool year, since it is now clear that planned mathematics education should start at this point. This may be resisted by some pre-school educators, many of whom may consider that introducing mathematics in an organised way to young children is inappropriate, unnecessary or even detrimental. Such a viewpoint is rooted to a large extent in historical considerations and social theories concerning what young children are capable of, but also in consideration of the nature of mathematics and mathematics learning (e.g. Balfanz, 1999). However, the weight of recent research evidence is such that there is now a widespread acceptance amongst early childhood mathematics education specialists (e.g. Clements, Sarama & DiBiase, 2004) that it is imperative that we move beyond this perspective to providing worthwhile and challenging mathematics education for the youngest children. In reviewing the use of mathematics curricula with very young children in the United States, Clements & Sarama (2009) show that high-quality mathematics education can avoid many of the learning difficulties that may otherwise arise. Here in Ireland we need to think carefully about the nature of the mathematics education offered across the range of early education settings in order to avoid some of the pitfalls identified in research.

FOCUSING ON MATHEMATICS IN EARLY CHILDHOOD

The Draft National Plan to Improve Literacy and Numeracy in Schools (Department of Education & Skills, 2010) anticipates that good quality learning activities of the sort recommended in Aistear can make what it terms 'a very significant contribution to improving children's acquisition of …numeracy'(p. 27). This view concurs with the recommendation that

> mathematical experiences for very young children should build largely upon their play and the natural relationships between learning and life in their daily activities, interests and questions (Clements, Sarama & DiBiase 2004, p.4).

In the statement of principles of early learning and development Aistear identifies six pedagogical concerns regarding how children learn and develop (NCCA, 2009)

These are

- holistic learning and development
- active learning
- play and hands on experience
- relevant and meaningful experiences
- communication and language
- the learning environment

Adopting these principles and the general goals of Aistear across the range of early education settings is a critical first step.

Starting then from the premise that we agree the 'how' of mathematics learning and teaching in early childhood there are a number of other central issues that need to be addressed with educators of 3-6 year old children. These relate to both the 'what' and the 'when' of early mathematics education. Specifically they include the following:

- Focusing on the teaching and learning of mathematics in the early education settings
- Identifying and building on the mathematics that children bring to the early education setting
- Specifying both general and specific goals for early mathematics education
- Using a systematic structured approach to developing mathematical ideas.

FOCUSING ON THE TEACHING AND LEARNING OF MATHEMATICS IN THE EARLY EDUCATION SETTING

The creation of a mathematics rich environment is important in developing mathematics in early childhood education. As part of that provision the availability of a range of mathematical materials and resources for children's use is key. However, early childhood educators need to understand clearly that playing with such artefacts is not in itself likely to lead to mathematics learning. Even though children might be focused on mathematically-related objects in their play (for example magnetic numerals), they will need to be drawn into mathematically related talk (e.g. Tudge & & Doucet, 2004). Such talk should include, as appropriate, opportunities for children to articulate mathematical ideas and understandings from their interactions with objects and to explain, reason, justify and hypothesise. Otherwise the potential for mathematical development and learning will not be realised. The explicit use of mathematical language in the course of daily routines in early education settings further develops children's understandings. Good quality provision will also include sequenced and structured mathematics learning since we know clearly that general enrichment is not enough and that children need specific mathematical experiences (e.g. Starkey, Klein & Wakeley, 2004). Mathematical experiences must be planned and supported by the educator utilizing a range of strategies, including play, to engage children and develop their learning.

In the US, a key position statement by two influential groups asserted that mathematics education for 3-6 year-old children should take at least two forms: that which arises as teachers seize emerging opportunities (in play and in daily routines) to both introduce and expand mathematical ideas; and carefully planned experiences that focus children's attention on a particular mathematical idea or set of related ideas (NAEYC/NCTM, 2002). In short what is recommended is a balance between child-initiated and teacher-initiated activity.

IDENTIFYING AND BUILDING ON THE MATHEMATICS THAT YOUNG CHILDREN BRING TO THE EARLY EDUCATION SETTING

We know that young children, through their active participation in everyday activity in their homes and their communities, can learn a great deal about mathematics. Learning arising from these kinds of opportunities is considered to be informal i.e. it 'is gleaned from everyday activities in what are not normally considered instructional settings …such knowledge is generally represented nonverbally or verbally and often learned incidentally' (Baroody, Lai & Mix, 2006, p. 188). As children participate in informal learning opportunities they develop fundamental ideas in relation to number, shape and space, patterns (algebra), measurement (e.g. Ginsburg et al., 2006). We know that informal learning is crucial for later success and is generally seen as the foundation for mathematics learning in school (e.g. Baroody, Lai & Mix, 2006). Informal learning, for instance, provides opportunities for children to learn about the role and importance of numbers and of processes such as counting. When parents (and educators) pay more attention to ways in which they can expand on what children are already doing i.e. the naturally occurring mathematical experiences that children encounter, then there is great potential for learning (e.g. Tudge & Douchet, 2004).

Some experiences are more helpful in this regard, and some learning experiences are more enabling in relation to providing children with opportunities for informal learning in mathematics. There is however, a paucity of information regarding young children's involvement in everyday experiences with mathematics. In the United States one study (Tudge & Doucet, 2004) revealed that over 60% of three-year-old children studied were never involved in explicit mathematical activity either in the course of play with mathematical objects, or lessons in either the home or the early childcare setting. We have no reason to believe that the situation is any different in relation to the informal mathematics learning of young children in Ireland.

The quantity and quality of children's informal experiences and knowledge are key factors for mathematical success at school and these cannot be taken for granted; nor should they be disregarded. Some children will have more/better opportunities to develop an informal knowledge of mathematics. Productive informal experiences are less likely for children from low-income families and so-called middle–class parents tend to provide their children with a broader range of mathematically rich activities and more complex tasks and do so more often than working-class parents (Baroody, Lai & Mix, 2006).Gaps in children's informal learning arise mainly from lack of relevant experiences, though for some children with special educational needs that gap may arise from other sources. These gaps need to be identified and addressed at the earliest possible opportunity and this has implications for teachers' knowledge and skills in assessing and intervening to fill these gaps. Teachers need to actively elicit children's informal knowledge and experiences and work with these to help children to build mathematical understanding. Thus the research suggests a need to provide greater mathematical enrichment for all children, but this is especially important for children living in poverty.

For three-and four-year-old children in Ireland, a real opportunity to engage in extensive, worthwhile and challenging informal mathematics arises at preschool and in the first year at school.

SPECIFYING BOTH GENERAL AND SPECIFIC GOALS FOR EARLY MATHEMATICS EDUCATION

Sarama & Clements (2004) argue that the goal for the early years should be mathematical literacy, not any narrow concept of numeracy. The latter usually is (mis)understood as dealing mainly with numbers, whilst their research suggests that geometry and patterning are foundational to maths learning. There is now general agreement amongst both mathematics educators and the early education community (e.g. Sarama, Clements & DiBiase, 2004) about what aspects of mathematics it is appropriate for young children to encounter in early education settings.

During the preschool years children develop a broad knowledge, a metacognitive framework (Munn, 1994, p. 7) related to numbers with which to develop further ideas about this form of communication. This developing understanding of the purpose of numbers is just one aspect of young children's developing number sense or their feel for numbers (Dunphy, 2007).

Number sense is key to children's developing mathematical understanding (e.g. Griffin, 2004; Dunphy, 2006).

Early mathematics is broad in scope and must deal with more than the topic of number and simple shapes (e.g. Ginsburg et al., 2006). Early maths is also deep, as is evidenced when considering, for example, all that is involved in learning to count meaningfully. Mathematics encompasses ideas related to shape and space, measure, pattern (algebra) and data. Mathematical proficiency in each of these strands of mathematics entails four aspects as follows: an understanding of key mathematical ideas; an ability to use basic skills such as counting or addition appropriately, flexibly and efficiently; an ability to solve mathematical problems, demonstrating creativity as appropriate; and an interest and a confidence in mathematics, i.e. a disposition towards mathematics (Kilpatrick, Swafford & Findell, 2001, as cited in Baroody, Lai & Mix, 2006).

USING A SYSTEMATIC STRUCTURED APPROACH TO DEVELOPING MATHEMATICS WITH YOUNG CHILDREN (AGED 3-6 YEARS)

Many mathematics educators now argue that early childhood educators need to be provided with quite specific goals for early childhood mathematics and information on how to achieve them (e.g. Ginsburg & Seo 1999; Clements, Sarama & DiBiase, 2004; Baroody et al., 2006). Aistear provides general guidance on pedagogy but no specifics in relation to content. The Primary School Curriculum provides content objectives for teachers of infant classes but falls short on specifying a comprehensive range of appropriate pedagogies for early childhood mathematics teaching.

The issues then for early childhood mathematics are as follows:

- Specifying clear goals for mathematics learning for all children in the age range 3-6 years
- Specifying how the aims and goals of Aistear can be achieved, in an integrated way, in relation to mathematics development
- Specifying pedagogical strategies that can support children towards achieving the mathematical goals.

General cognitive prerequisites for mathematics are language development, perceptual development and spatial development (e.g. Goswami & Bryant, 2010). It follows then that a child with poor spatial skills is likely to have difficulty in acquiring number. Since language development is fundamental to all learning, a curriculum rich in opportunities for young children to develop language, to talk and to take part in discussion and to engage in collaborative thinking with the educator and other children is important for mathematical development as well as for literacy development. But equally, early mathematics learning can contribute to later learning of other subjects especially literacy (e.g. Sarama & Clements, 2004; Goswami & Bryant, 2010). In particular, there is reason to believe that geometry and a spatial sense are key thinking tools for young children across all areas of learning (Sarama & Clements, 2004). A systematic focus on engaging children in describing their thinking, sharing opinions, justifying their ideas and discussing mathematically-related stories are all practices that have a major effect on children's language and their thinking (e.g. Ginsburg et al., 2006). This interrelatedness in learning gives early learning its holistic character.

THE IMPORTANCE OF UNDERSTANDING MATHEMATICS LEARNING TRAJECTORIES

In the United States, there are currently moves to focus educators on learning trajectories i.e. developmental progressions. These have been developed for all of the big ideas related to the different aspects of mathematics (Sarama & Clements 2004, 2009). Research demonstrates that the learning trajectories approach is potentially one of the most powerful ways of helping educators understand and promote young children's mathematical development (Sarama & Clements, 2009). The authors claim that it focuses educators on teaching children rather than on merely implementing the curriculum (Baroody et al., 2006). According to the research, educators who understood the learning trajectory for specific topics such as addition '…were most effective in teaching small groups and encouraging informal, incidental mathematics at an appropriate and deep level.' (Sarama & Clements 2004, p.188). The learning trajectories help to sequence the teaching and learning but also help educators to understand the mathematics involved from children's perspectives. Educators are encouraged to use the learning trajectories to decide where children 'are' and to sequence activities appropriately. Learning trajectories have three parts:

> a specific mathematical goal, a developmental path along which children develop to reach that goal, and a set of instructional activities that help children move along that path (Clements & Sarama, 2009, p.ix).

Furthermore, it is claimed that teachers who have mastered the learning trajectories approach 'understand the math, the way children think and learn about math and how to help children learn it better'(p.ix).

THE CHALLENGE OF PROFESSIONAL DEVELOPMENT

Ginsburg et al. (2005, p. 190) are clear that early childhood mathematics education involves a strong but sensitive adult role, the introduction of challenge with the development of young children's natural interest in mathematics. We know that many early childhood educators are uncomfortable about teaching mathematics (e.g. Ginsburg et al., 2005). Also the fact that mathematics is a highly sequential, hierarchical domain in which certain ideas and skills must be learned before others (e.g. Clements & Sarama, 2009) is an issue that requires serious attention in pre-service and in-service education. The nature of mathematics presents many early educators in Ireland (and in other countries also) with a challenge in terms of pedagogy. Guidance on specific mathematical goals and on how these may be achieved in an integrated way in early education settings is essential to support these educators.

CONCLUSION

The Primary School Curriculum (DES, 1999) does not provide infant-class teachers with sufficient guidance as to how to enable children to achieve the curriculum objectives. However it is strong in relation to offering a broad and deep mathematics education to young children. Curricula in use in preschool settings in Ireland may not promote access to all the strands of mathematics. Programmes used in DEIS schools must be seen as complementary to the Primary School Mathematics Curriculum, not as a replacement for it. Aistear, while strong on general pedagogical strategies, is not designed to be explicit about specific mathematics goals, nor is it designed to give explicit support in how these goals may be realised in integrated ways in early education settings.

What is obvious then is that in order to achieve optimal learning of mathematics in early education settings in Ireland a common comprehensive set of specific goals need to be articulated. They need to be explicated by providing educators with strong guidance on how these goals are pursued in an integrated way in early education settings. Both the Primary School Curriculum and Aistear have important contributions to make to the articulation of a mathematics plan for early childhood education but neither alone can provide the level of support that educators need to ensure high quality mathematics learning across the age range 3-6 years. Rethinking approaches to mathematics teaching and learning in early education settings must be informed by current research in relation to what mathematics should be taught, how it should be taught and when it should be taught.

REFERENCES

Balfanz, R. (1999) Why do we teach young children so little mathematics? Some historical considerations. In J. Copley (Ed.) *Mathematics in the early years* (pp. 3-10). Reston, VA: The National Council of Teachers of Mathematics/ National Association for the Education of Young Children.

Baroody, A., Lai, M. & Mix, K, (2006). The development of young children's early number and operation sense and its implications for early childhood education. In B. Spodek, & O. Saracho, (Eds.) *Handbook of research on the education of young children* (2nd ed.) (pp.187-221). New Jersey: Erlbaum Associates.

Bowman, B., Donovan S., & Burns, S. (Eds.) 2001. Eager to learn: Educating our preschoolers. *Report of Committee on Early Childhood Pedagogy, Commission on Behavioral and Social Sciences and Education National Research Council*. Washington DC: National Academy Press.

Clements, D., Sarama, J. & DiBiase, A. (2004) (Eds.) *Engaging young children in Mathematics: Standards for early childhood mathematics education*. Mahwah, NJ.: Lawrence Erlbaum & Associates.

Clements, D., Sarama, J. (2009) *Learning and teaching early math: The learning trajectories approach*.UK: Routledge.

Department of Education and Skills (DES) (2010). *Better literacy and numeracy for children and young people: A draft National Plan to improve literacy and numeracy in schools*. Dublin: DES.

Department of Education and Skills (DES) (2011). *Literacy and numeracy for life and learning: The National Strategy to improve reading and writing among children and young people 2011-2020.* Dublin: DES.

Dunphy, E. (2006). *An exploration of young children's number sense on entry to school in Ireland.* Ed D., Open University.

Dunphy, E. (2007) The primary mathematics curriculum: enhancing its potential for developing young children's number sense in the early years at school. *Irish Educational Studies*, 26, (1), 5-25.

Dunphy, E. (2009) Early childhood mathematics teaching: Challenges, difficulties and priorities of teachers of young children in primary schools in Ireland. *International Journal of Early Years Education*, 17, (1), 3-16.

Ginsburg, H. & Ertle, B. (2008) Knowing the mathematics in early childhood mathematics, In O. Saracho, & B. Spodek, (2008) *Contemporay perspectives on mathematics in early childhood education* (pp.45-66). Information Age Publishing.

Ginsburg, H. and Seo, K. (1999) The mathematics in children's thinking. *Mathematical Thinking and Learning*, 1, (2),113-129.

Ginsburg, H. Cannon, J., Eisenband, J. & Pappas, S. (2006) Mathematical thinking and learning. In K. McCartney, & D. Phillips (Eds.), *Blackwell handbook of early child development* (pp. 208-228). Blackwell, Malden, MA.

Ginsburg, H., R. Kaplan, J. Cannon, M. Cordero, J. Eisenband, M. Galanter & Morganlander, M. (2005). Helping early childhood educators to teach mathematics. In M. Zaslen and I. Marting-Bak (Eds.), *Critical issues-early childhood professional development* (pp.171-202). Baltimore: Brooks Publishing.

Goswami, U. & Bryant, P. (2010) Children's cognitive development and learning. In R. Alexander (Ed.), *The Cambridge Primary Review Research Surveys* (pp.141-169). London: Routledge.

Government of Ireland (1999a) *Primary School Curriculum: Mathematics*. Dublin, Ireland: Stationary Office.

Government of Ireland (1999b) *Primary School Curriculum: Mathematics: Teacher Guidelines.* Dublin, Ireland: Stationary Office.

Griffin, S. (2004) Building number sense with Number Worlds: A mathematics programme for young children. *Early Childhood Research Quarterly*, 19, 173-180.

Katz, L. (2003) *The right of the child to develop and learn in quality environments. International Journal of Early Childhood,* 35, 1&2, 13-22.

Munn, P. (1994) The early development of literacy and numeracy skills. *European Early Childhood Education Research Journal*, 2, (1), 5-18.

National Association for the Education of Young Children & National Council of Teachers of Mathematics (2002) *Position statement: Early childhood mathematics: Promoting good beginnings*. Accessed on www.naeyc.org/about/positions/pdf/psmath.pdf , March 3 2009.

National Council for Curriculum and Assessment. (2009) *Aistear: The early childhood curriculum framework*. Dublin: NCCA.

Office of the Minister for Children and Youth Affairs, 2009 *Guide for parents to the free pre-school year in early childhood care and education (ECCE).* Dublin, Ireland: OMCYA.

Saracho, O. & Spodek, B. (2008) *Contemporary perspectives on mathematics in early childhood education.* United States: Information Age Publishing.

Sarama, J. & Clements , D. (2004) Building blocks for early childhood mathematics. *Early Childhood Research Quarterly*, 19, 181-189.

Sarama, J. & Clements, D. (2009) *Early childhood education mathematics research: Learning trajectories for young children*. UK: Routledge.

Starkey, P., Klein, A & Wakeley, A. (2004) Enhancing young children's mathematical knowledge through a pre-kindergarten mathematics intervention. *Early Childhood Research Quarterly*, 19, 99-120.

Tudge, R. & Doucet, F. (2004) Early mathematical experiences: Observing young Black and White children's everyday activities. *Early Childhood Research Quarterly*, 19, 21-39.

LOOKING BACK TO LOOK FORWARD: THE MATHEMATICAL IDENTITY OF PROSPECTIVE MATHEMATICS TEACHERS

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The study reported in this paper investigates the mathematical identity of a group of 'prospective teachers': undergraduate students of mathematics who undertook elective modules on mathematics education and classroom experience. They were required to write mathematical autobiographies, the protocol for which was drawn from another study, in this case of the mathematical identity of (primary) student teachers (MIST). The autobiographies were analysed to see if the themes emerging from MIST were present also in the prospective teachers' narratives, and to compare the responses of the two cohorts. While there were many similarities, differences reflected the differing experience and focus of the two groups. The work is being continued in a new study with other cohorts of students.

INTRODUCTION

Following the publication of Wenger's (1998) social theory of learning, a research focus has emerged on teachers' and prospective teachers' *mathematical identity*: the relationship an individual has with mathematics, including knowledge and experiences as well as perceptions of oneself and others. In Ireland, a project on 'The Mathematical Identity of Student Teachers' (MIST) has been conducted with student teachers who were taking a Bachelor's degree programme in order to qualify as primary teachers, and who were specialising in mathematics. It was carried out at two colleges of education, one in Belfast and the other in Dublin. Qualitative data were collected, using questionnaires completed by the participating students followed by focus group discussions; several themes with regard to mathematical identity were identified (Eaton & OReilly, 2009b). The study reported in this paper extends the work to prospective second-level teachers. It is hoped that a forthcoming project, 'Mathematical Identity using Narrative as a Tool' (MINT), will be able to build on the methodology and findings of both studies.

The present study was carried out with undergraduates who were specialising in mathematics at a university in Dublin, and who as part of their programme were taking elective modules on mathematics education. Using a protocol drawing on that employed in MIST, the students were asked to write mathematical autobiographies; these were used as tools to reveal significant aspects of and influences on their mathematical identities. For this paper, the autobiographies were analysed to address two research questions:

1. To what extent are the themes that emerged in MIST relevant and exhaustive for the data collected from the new cohort of students?

2. If the themes are sufficiently relevant for comparisons to be made, what within each theme are the similarities and differences between typical responses of the two cohorts?

The paper begins with a brief outline of the research that informed the MIST study, followed by a summary of the findings of that study; an outline is given of other relevant work done in Ireland. Then, the methodology of the present study is described. The findings are reported and discussed, comparing and contrasting the identities of the mathematics undergraduates with those of the student teachers in the MIST study. In the concluding section, some implications for MINT are noted.

LITERATURE REVIEW

The links between affect in mathematical thinking (including beliefs, attitudes and self-efficacy) and classroom practice has been an area of active research for over two decades (Ernest, 1988; Fang, 1996; Hannula, 2004). The experiences that students have during their own formative years in the classroom as pupils have been shown to have a major impact on their behaviour as teachers (Ball, 1988; Ernest, 1988; Hill, 2000). Boaler & Greeno (2000, p173) point out that "identities develop in and through social practice." This is the setting for the MIST study outlined in the introduction. It combined two notions: *mathematical identity* as defined above (Grootenboer, Smith & Lowrie, 2006; see also Wenger, 1998), and *narrative* as a tool to help access identity (Clandinin & Connelly, 2000; Kaasila, 2007). Funding for the project was provided by SCoTENS, the Standing Committee on Teacher Education North and South.

Using a grounded theory approach, seven main themes were found in the MIST data; for details, see Eaton & OReilly (2009b). We identify the main features of each of these in turn:

1. **Student teachers' self-reflection on learning and teaching** (a clarification of the original title) relates to how students' exploration of their mathematical identity leads them to deepen their insight into learning and teaching mathematics.
2. **The role played by key figures in the formation of mathematical identity** focuses not only on teachers and family members, but also on peers and society at large.
3. **Ways of working in mathematics** explores what students find effective in learning mathematics and why, either through individual endeavour or through collaboration.
4. **How learning mathematics compares with learning in other subjects** considers the particular characteristics of learning mathematics, usually at school, which distinguish the process from learning in other subject areas.
5. **The nature of mathematics** draws from a broad range of students' perceptions touching on the philosophy of mathematics and on what doing mathematics is about.
6. **'Right' and 'wrong' in mathematics** concerns students' perception that what is important in mathematics is to find the correct answer, and also a more general notion around the unambiguous nature of mathematical truth.

7. **Mathematics as a rewarding subject** examines the extent to which students enjoy the subject, often relating to how they persist with it or to significant moments of insight.

The themes themselves are not always sharply distinguished one from another. For example, **'right' and 'wrong' in mathematics** often appeared as a sub-theme of **the nature of mathematics**; this latter theme was often intertwined with the theme **how learning mathematics compares with learning in other subjects** or **ways of working with mathematics**. Also, students' recollection of key figures (especially teachers) often gave rise to self-reflection on how mathematics might best be learnt or taught.

As well as the MIST study, the current work uses mathematical autobiography. This approach was used at another teacher education college in the Republic of Ireland; in that case the subjects of study were graduates – from many subject areas, not necessarily mathematics – who were reading for a Higher Diploma in Education in order to qualify as primary teachers (Oldham, 1997, 2005). Other studies in Ireland relating to mathematical identity include those by Corcoran (2008), Liston & O'Donoghue (2009) and Meehan & Paolucci (2009).

METHODOLOGY

The subjects in the present study were third or fourth year undergraduate students in an honors mathematics programme at a university in Dublin who elected to take two modules on mathematics education (one in each semester) in the academic year 2010-2011. The first module addressed theoretical and practical aspects of the subject. For the second module, the students were placed in second-level schools and acted as classroom assistants in three classes per week. This provided an immediate service to the community – introducing people with a good background in mathematics into classrooms where they could be of assistance to both teachers and students – and also gave the undergraduates a chance to find out if they would like to pursue a career in teaching. The students who took the two modules are designated in this paper as 'prospective teachers.'

As part of the assessment of the first module, the students were asked to write mathematical autobiographies. The specifications are shown in Fig. 1 (which replicates the handout given to the class); they are taken directly from the MIST questionnaire protocol.

Mathematical autobiography

Think about your total experience of mathematics. Write about the dominant features that come to mind. [Take about 10 – 15 minutes]

Now think carefully about all stages of your mathematical journey from primary school (or earlier) to university mathematics. Consider

- Why you chose to study mathematics at third level
- Influential people
- Critical incidents or events
- Your feelings or attitudes to mathematics

- How mathematics compares to other subjects
- Mathematical content / topics

With these and other thoughts in mind, describe some further features of your relationship with mathematics over time.

[Writing time for this section might be about an hour; aim to write 1000-1500 words (typed double-spaced)]

[Taken from Eaton & OReilly (2008-2010) with a view to facilitating research comparisons with intending primary teachers specialising in mathematics]

Fig 1: Specifications for the mathematical autobiography assignment

The resulting essays provide data for the present study, in lieu of the focus groups used in the MIST study. Analysis is based on the work of 29 students who took both modules. It focused on identifying occurrences of the themes that had emerged from MIST and judging whether or not these themes were relevant and exhaustive for the present cohort of students. Building on the grounded theory approach of MIST, data coding was carried out as follows. As the essays were read, a key phrase from each distinguishable item (typically a few sentences), together with its MIST theme codes and information to identify both the essay and the location within the essay, were entered into an Excel spreadsheet. Initially six essays were coded by two of the authors, to check for consistency and allow for preliminary discussion of the adequacy of the MIST themes; four more essays were coded by both authors as a further check on consistency. The filtering facility of the spreadsheet was used in order to focus on different themes. A fuller account of the methodology is provided by Eaton, Oldham and OReilly (2011).

FINDINGS

In the 39 readings (of the 29 essays, ten read by two of the authors), 647 items were identified and coded according to the MIST themes. The maximum number of themes assigned to any item was four; in all, 941 theme assignments were made to the items. The two coders reconciled any differences in coding through discussion rather than resorting to recoding the essays. In addressing the first research question, they noted that 93.5% of the items were assigned to at least one of the seven themes. This evidence substantiated their judgement that the seven themes identified above for MIST were relevant and broadly sufficient if not quite exhaustive for the new data. To address the second research question, quotations relevant to instances of each theme in turn are presented below, drawing comparisons with the MIST data as appropriate; a more general overview of the comparisons and contrasts is presented at the end of this section.

1. Student teachers' (here, prospective teachers') self-reflection on learning and teaching

In summarising his reflections on learning mathematics, one student states that his own engagement was strongest when an underlying understanding of a topic was emphasised:

> But I do think the moments that I most engaged with the subject [were] those where I felt there really was some kind of understandable method and reasoning behind it all, which is perhaps something which is worth keeping in mind and trying to sustain in discussions of overall mathematics education. {Student 13}

Moreover, he recommends that this insight should underpin good mathematics education. Another student recalls how his experience of poor teaching motivated him to explain mathematics to his peers:

> Her approach to teaching the subject was to learn off the formulae and learn how to solve each one separately. I was frequently at odds with her, because the way she was teaching it excluded any understanding of the subject, and that only the people who inherently understood the subject had a chance of doing so. There was a beneficial side effect to this, in that I had my first (definitely positive) experiences in teaching as a direct result of this; my friends often came to me if they needed something explained rather than asking the teacher, and I derived a great amount of satisfaction (and still do!) from helping people to understand something they are having trouble with. {Student 38[1]}

Yet another student testifies to the gradual decline in her engagement with mathematics in the first three years of secondary school:

> I feel that, for me to learn, I need to be quite interactive. Being passive in class led to me starting to be bored and not asking questions … This was not so much the case in first year, but as the years went on I slowly began to not quite grasp everything. I began just learning how to do things as opposed to why to do things. This is not the way my brain likes to learn and so I started forgetting things I needed to know for the next part and so on. By the time I reached Junior Cert, I had gone from top of my class to a B student. {Student 22}

The preference for being taught for understanding and in an engaging manner echoes findings from the MIST data. Other aspects of this theme common to both datasets are the importance of effort and perseverance in studying and of high expectations by teachers. However, some pedagogical aspects, such as supporting both individual learning and cooperative learning, are articulated more explicitly in the MIST data than here (Eaton & OReilly, 2009a).

2. The role played by key figures in the formation of mathematical identity

A detailed study of the key figures influencing this cohort of students and their engagement with mathematics is made elsewhere (Eaton, Oldham & OReilly, 2011). There we reported that the main emphasis relates to teachers at primary and secondary school (see also theme 3, **ways of working with mathematics**, below), with significant references also to family members. Another example of a student's experience of positive influence from his family is:

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[1] The numbers refer to the full list of students submitting autobiographies, not just to those in the present study.

> Of course, the encouragement from my family has been instrumental also. My brother, an air traffic controller, and my father, a pilot instructor, taught me early in life that mathematics played a pivotal role in everyday life; and it only encouraged me to work harder at something that was already beginning to appeal to me. {Student 33}

All of this is rather similar to what was observed in the MIST data. However, the variety of interaction of the students in this study with mathematics at an early stage appears both broader and deeper. For example, students mentioned exposure to Mathematical Olympiad problems, meeting with mathematicians, and reading 'popular' mathematical books (such as Doxiadis' *Uncle Petros and Goldbach's Conjecture* or Du Sautoy's *The Music of the Primes*) before entering university. Moreover, there is evidence that they had considerable satisfaction in socialising in the context of science and mathematics, a feature not noted in the MIST data:

> At the Young Scientist [Exhibition], I got to learn that I wasn't actually quite the fiend I was starting to think I was, thanks to many smooth and companionable social interactions, even friendships, while there. This was perhaps where the first real urge to go to College came to me. I knew that I wanted to get to know more people, and to be myself and relax in the company of others, (something I was having difficulty with in my current School, to a certain extent), and in my mind, College became the doorway to all of that. {Student 21}

> The Mathsoc was an important part of my college maths experience. I became quite friendly with the people in the society when I started college, and became the First Year Rep. I then became auditor in second year, which I'm quite proud of; there have been very few second year auditors, and very few female auditors, I think I might have been the first person to be both. … The older members of the society were definitely influential in my switch to maths.… {Student 30}

Finally on this theme, a student draws attention to the crucial role of encouragement:

> The journey that has led me to study mathematics at third level has been an eventful one: I have had some very positive and negative experiences, and I've [encountered] people who may not have understood at the time just how influential of a role they played. And, still, I couldn't feel [more strongly] that my mathematical journey would have come to end long before now if it were not for one thing: encouragement. {Student 33}

3. Ways of working in mathematics

Sub-themes in this area related to educational level (primary, second level and third level). In primary school, ways of working were directed almost exclusively by the class teacher:

> I suppose that the 'times tables' will never leave my memory until I die. In fact I am very grateful to my teacher Mrs D for drilling these into us. I do remember that every morning we had to recite our spellings and whatever the chosen multiplication tables were for the week. I enjoyed learning them off and I certainly never understood back then how useful they would be to me in the future. {Student 7}

> Every couple of weeks we would have what were called “Maths Fun Days” in which the class was separated into teams and given problems to solve. Each team had a designated runner who would, once the problem was solved would run and swap the answer for the next question. The team who finished all the questions first won (i.e. it was a race) and that team would get a reward. By relating the actual classroom to somewhere where you have fun, the students began to relate the subject to enjoyment and therefore were inspired to work harder when it came to the more arduous aspects of mathematics. {Student 18}

By contrast, for second level, the focus was less on the teacher’s role and more on student-initiated aspects:

> [José {pseudonym}, a classmate] was a competitor like no other. I never beat him in all the tests we did that year, and we did plenty of them. I remember the first test I did in Mr T’s class. I got 85%, a poor performance. I don’t think I scored that low again until 5th year. I was very hard on myself that year because I was used to being in my comfort zone where nobody challenged me. It was tough at the time but I now believe I learnt more from defeat than victory. I really thought I got the better of José in the summer test but 95% was only good enough for second place. {Student 28}

> During my year out, I was able to learn and perfect methods of self regulated study. I had been able to get a glimpse of this while studying Applied Maths, as the only force driving the study was my own ambition. {Student 26}

With regard to third level, some students reflected on their learning strategies, noting the success or otherwise of different approaches:

> But I hardly ever took time to cement the knowledge by solving problems. … It took me years to realise both understanding and [practice are] needed to internalise the information. Now that I have, I am both learning better and getting more joy out of my studies. {Student 12}

> In a few of my subjects throughout the years my ability to pass the exam has come down to my ability to learn off large theorems. Hardly good for maths! {Student 10}

At third level also, collaborative approaches manifested themselves:

> The key to college is interaction with fellow students and the lecturers as well as a great deal of personal study. Fellow students and lecturers can recommend better material to study than if I walked into the library and picked out a book filled with different notation on the same subject. {Student 25}

> [However] in College the lecture theatres were packed with students [who] didn’t know what was really going on, but were convinced they were the only one, so no one asked any questions. Then in a flurry of panic, several of us would run to the library to try get the recommended reading, only to find to our horror that [the books] had already been taken out by [a] college-savvy student. From there, it would escalate, until in desperation, we would turn to each other, and then the learning would truly begin. {Student 21}

This evidence is rather similar to that reported in the MIST study: collaboration was not part of the "social practices" (Boaler & Greeno, 2000, p172) at second level, but developed naturally, or by necessity, at third level.

4. How learning in mathematics compares with learning in other subjects

Comparisons between learning mathematics and learning other subjects tended to involve aspects of the previous and/or the next theme (themes 3 and 5). Examples for the present theme are therefore restricted to just two that illustrate this point:

> In secondary school, particularly in exam years, mathematics for me was a break away from my other subjects, I was confident in mathematics and I loved solving questions and figuring out what was going on and what 'had to be done'. It was a relief to be able to answer questions without trawling through books or looking through plays or novels for quotes. {Student 39}

> [While] I am very fond of the idea of studying the likes of Philosophy or English, I feel that the incessant essays and the inevitable lack of certainty would most likely drive me mad long before I graduated! The logical certainty of maths and the satisfaction that I derive from solving problems far outweighs the intermittent longing for an essay or two to write, or something to debate which doesn't have one "correct" answer. {Student 38}

Overall, the comparisons made by the students in this study between learning mathematics and learning other subjects are remarkably similar to those in MIST.

5. The nature of mathematics

This theme is considered in more depth by Eaton et al. (2011), so is addressed only briefly here despite its prominence in the data. Many (though not all) of the students appreciate the abstract and exact nature of the subject, as highlighted rather poetically by one student:

> [Whenever] one has an idea, rendering it mathematically is akin to exposing an object to a high autumn wind: it is stripped of all needless embellishment…. Mathematics is spare and rigorous and it has a cold, gaunt beauty…. [The] one feature common to all mathematics is precise ordered thought. {Student 16}

Another student makes similar points and considers how the nature of mathematics compares to that of other disciplines, perhaps echoing theme 4 (**learning other subjects**):

> I've never been one for the endless pinhead-dancing of philosophy, or the futile what-ifs [of] historians. To me that is all a waste of time. But maths relates to what is. It possesses exactitude, balance, perhaps truth. And truth is worth seeking. {Student 10}

Common aspects of this theme across the two studies (the present study and MIST) include the abstract nature of mathematics, the vastness, power and applicability of the subject and what it is like to work with it.

6. 'Right' and 'wrong' in mathematics

This was the theme least in evidence. Some students did address the issue in terms of what they perceived as the unambiguous nature of mathematical truths or the precise forms of argument, rather than with regard to arriving at unique correct answers:

> We all love that feeling of spending hours on a question and finally producing the correct answer. But, likewise, we despise the hours we agonise over a question and don't manage to come up with any solution. One of our biggest flaws as math students is that we can't let go of an unsolved problem – we continue to persist until some conclusion is drawn. In honesty, I think that's what retains our interest; we love to search for the solutions to the almost-insolvable questions. {Student 33}

A minority view was expressed as follows:

> Mathematics suited my personality, I like straight answers. The way I see it, an answer is either right or wrong. Maths was for me. I loved sums and would scratch away with my pencil into the wee small hours. I would promise myself a treat, maybe a bar of chocolate or a packet of biscuits, if I got some exercises right and then I would frantically go checking the back of the text book to see if my answers tallied. {Student 17}

These views – and the rare occurrence of this theme – correspond closely with the MIST data. However, in the latter there was also some concern expressed about the experience of student teachers in giving private lessons ('grinds') where they tried to resist pressure with regard to their students seeking the 'right' answer simplistically.

7. Mathematics as a rewarding subject

Students wrote about the 'joy and frustration' of their mathematical experience, with one student expressing his experience (again rather poetically) as follows:

> Reflecting on my total experience of Mathematics, I think first of the highs and lows: the joy when I am enabled to gain deeper insight into a particular topic and see how all the different elements cohere, rather like stepping back from a stained glass window; then the despair when nothing works, when no pattern emerges from the masses of coloured squares. {Student 16}

Strong expressions emerged of satisfaction, both in a general sense and in relation to particular topics:

> When I was asked to think about my total experience of mathematics the first memory that I recalled was that of the functional analysis class I had straight before hand. … To be honest I was feeling a little overwhelmed … However when I relaxed and thought about the influence maths has had over my life and the enjoyment I have gotten out of it, I realised that the only place I could feel challenged and pursue my love of maths was probably in a maths class in University. {Student 6}

> [Statistics] has been a long time passion. … It borrows some difficult theoretical mathematical ideas which I enjoy. It takes these and applies them to real life in almost every aspect of life, is something I appreciate. {Student 31}

Similar expressions were in evidence in the MIST data. However the pre-service primary teachers did not articulate, as these students did, their appreciation of how mathematics affected their thinking:

> I am in my third year now and every day I have learnt something new and exciting. I am really fascinated with the power that mathematics has over my mind. It is a special world that you can escape to when you want some personal time and a breather from the sometimes ordinariness of life. {Student 7}

> Maths has taught me to think, but it has also taught me to follow what I like. And, if you ask me, that is a good lesson to learn. {Student 1}

Comparison of the Two Cohorts

In illustrating the seven themes above, we have noted more similarities than differences between the two cohorts of students, pre-service (primary) teachers and prospective (second-level) teachers. Of course the manner of collecting the data, a questionnaire followed by small focus groups for one cohort, and mathematical autobiographies – written prior to undertaking the classroom experience – for the other, can be expected to have a significant bearing on the narratives expressed. We can expect more homogeneity in the former (as participants respond to and elaborate on each other's stories in the focus groups); the latter is likely to give rise to more individual accounts. This is indeed what happened, but probably even to a larger extent than the modes of collecting the data can explain. Although individuality amongst the pre-service teachers was expressed in their experience of home and school, their common experience in teacher education gave rise, it seems, to a more homogeneous collective lens through which to perceive mathematical identity. Moreover the diversity and individuality of the prospective teachers was accentuated by the fact that four of them reported having experience as students in education systems outside Ireland.

Overall, it is evident that the registers adopted by the two cohorts were different. Essentially these correspond to different "figured worlds" arising from diverse social practices of learning mathematics (Boaler & Greeno, 2000, p173). The prospective teachers adopted a stance of mathematicians-in-the-making, drawing more broadly and deeply from the detail of mathematics. On the other hand, the pre-service teachers adopted more the stance of teachers-in-the-making. This observation can be quantified tentatively by noting that about 28% of items in the MIST study were instances of theme 2 (**the role played by key figures**); the corresponding figure for the mathematics students was only about 20%. Conversely, for theme 7 (**mathematics as a rewarding subject**) only about 13% of the items in the MIST data provided instances, while the comparable figure for the mathematics students was around a quarter. In other words, plausibly enough, the two cohorts may be differentiated somewhat in terms of a focus on *people* (by the student teachers) versus a focus on *the subject* (by the mathematics students). Differences between instances of each of the other themes were no more than about two percentage points.

At the beginning of this section, we pointed out that 93.5% of items were assigned to at least one of the seven themes. Omitting themes 4 (**other subjects**) and 6 (**'right' and 'wrong'**), this figure is reduced only slightly to 91.8%. This reduction is of a similar order to that for the MIST coding.

Summarising the discussion on research question 2, we can say that the distinctive features of the two cohorts may be said to derive from the primary focus of each; for one it is education, while for the other it is mathematics. However, each has engaged significantly in the 'other' focus. Collectively it appears their mathematical identities can be adequately analysed using five of the seven MIST themes.

CONCLUSION

This paper addressed the mathematical identity of a group of undergraduate students who were reading for degrees in mathematics, and who were sufficiently interested in mathematics education and mathematics teaching to take a one-semester module providing 'classroom experience' – hence being designated for the purposes of the study as 'prospective teachers'. Participants in the module were assigned to secondary schools and attended three classes per week in the role of teaching assistants. The work reported here is based on an analysis of mathematical autobiographies written by the students as part of the assessment of a module on mathematics education, completion of which was a prerequisite for undertaking the classroom experience module.

For this study, the main aim was to consider the mathematical identities of the students as revealed in their autobiographies. In particular, the research sought to find if the themes that emerged from an earlier study (MIST), undertaken with student teachers who were in primary teacher education courses and were specialising in mathematics, were applicable also for the group of prospective teachers defined as above. The foregoing report and discussion has indicated that the themes do apply to the prospective teacher group, to an extent broadly comparable to that found in MIST, but with variations of emphasis between and within the individual themes. It is therefore natural to ask if the themes are robust enough to be used in larger-scale studies. A promising approach is to reduce the number of themes by two and to clarify the detail in the definitions of the remaining five. It is expected that the new SCoTENS-funded project, MINT (Mathematical Identity using Narrative as a Tool), will make significant progress in identifying a framework for exploring mathematical identity that is likely to be relatively stable across different groups and different coders.

REFERENCES

Ball, D. (1988). Unlearning to teach mathematics. *For the Learning of Mathematics*, 10 (2), 10-16.

Boaler, J. & Greeno, J. (2000). Identity, agency and knowing in mathematics worlds. In J. Boaler (Ed.), *Multiple perspectives on mathematics teaching and learning*, Stamford, CT: Ablex, 171-200.

Clandinin, D. & Connolly, M. (2000). *Narrative enquiry: experience and story in qualitative research*. San Francisco: Wiley.

Corcoran, D. (2008). *Developing mathematical knowledge for teaching: a three-tiered study of Irish pre-service primary teachers*. PhD Thesis (University of Cambridge).

Eaton, P., Oldham, E. & OReilly, M. (2011). Nature and nurture: An analysis of mathematical identity of distinct cohorts of prospective teachers. Paper accepted for presentation at the 36th Annual ATEE Conference, Riga, 24-28 August 2011.

Eaton, P. & OReilly, M. (2009a). Exploring mathematical identity as a tool for self-reflection amongst pre-service primary school teachers: “I think you have to be able to explain something in about 100 different ways”. Presented for *Proceedings of 10th Mathematics Education into the 21st Century Project* conference, Dresden.

Eaton, P. & OReilly, M. (2009b). Who am I, and how did I get here? Exploring the Mathematical Identity of Student Teachers. In D. Corcoran et al. (Eds.), *Proceedings of 3rd National Conference on Research in Mathematics Education MEI 3* (pp. 228-237). Dublin: St. Patrick’s College.

Eaton, P. & OReilly, M. (2010a). What other people think and why it matters? An investigation of key influences on mathematical identity. In M. Montané & J. Salazar (Eds.), *ATEE 2009 Annual Conference Proceedings* (pp. 290-298). Brussels: ATEE.

Eaton, P. & OReilly, M. (2010b). What is mathematics and why do we study it? The views of student teachers. Paper presented at the 35th Annual ATEE Conference, Budapest, 26-30 August 2010.

Ernest, P. (1988). The attitudes and practices of student teachers of primary school mathematics. In A. Borbas (Ed.), *Proceedings of 12th International Conference on the Psychology of Mathematics Education, Hungary,* Vol. 1 (pp. 288-295). Veszprém: OOK.

Fang, Z. (1996). A review of research on teacher beliefs and practices. *Educational Research,* 38, 47-65.

Grootenboer, P. J., Smith, T., & Lowrie, T. (2006). Researching identity in mathematics education: The lay of the land. In P. Grootenboer, R. Zevenbergen, & M. Chinnappan (Eds.) *Identities, cultures and learning spaces* (Proceedings of the 29th annual conference of MERGA, Vol. 2, pp. 612-615). Canberra: MERGA.

Hannula, M. (2004). *Affect in Mathematical Thinking and Learning*. PhD Thesis (University of Turku).

Hill, L. (2000). Theory, practice and reflection: A pre-service mathematics education programme. *Teachers and Teaching: Theory and Practice,* 6 (1), 23-39.

Kaasila, R. (2007). Using narrative inquiry for investigating the becoming of a mathematics teacher. *ZDM-International Journal of Mathematics Education*, 39 (3), 205-213.

Lerman, S., Amato, S. et al. (2009). Studying student teachers’ voices and their beliefs and attitudes. In R. Even & D. Ball (Eds.), *The Professional Education and Development of Teachers of Mathematics.* 15th ICMI Study (pp. 73-82). New York: Springer.

Liston, M. & O'Donoghue, J. (2009). The importance of pre-service teachers' conceptions of mathematics and approaches to learning for the future of mathematics education in Ireland. In D. Corcoran et al. (Eds.), *Proceedings of 3rd National Conference on Research in Mathematics Education: MEI 3* (pp. 228-237). Dublin: St. Patrick's College.

Meehan, M. & Paolucci, C. (2009). Why it's different with Mathematics: Prospective teachers' reflections on what makes teaching post-primary Mathematics unique. In D. Corcoran et al. (Eds.), *Proceedings of 3rd National Conference on Research in Mathematics Education: MEI 3* (pp. 250-261). Dublin: St. Patrick's College.

Oldham, E. (1997). Encountering mathematical thinking: Reactions and reflections of a group of student-teachers. In J. McCall & R. Mackay (Eds.), *Partnership and cooperation: 21st Annual Conference of the Association for Teacher Education in Europe (ATEE)*, *Glasgow*, *September 1996* (pp. 88-102). Glasgow: University of Strathclyde Faculty of Education.

Oldham, E. (2005). Solving problems with student-teachers: reactions and reflections 1995-2005. In S. Close, T. Dooley & D. Corcoran (Eds.), *First National Conference on Research in Mathematics Education (MEI 1)*, *St. Patrick's College, Drumcondra*, *16-17 September 2005* (pp. 263-277). Dublin: St. Patrick's College.

Wenger, E. (1998). *Communities of Practice: Learning, meaning and identity*. Cambridge: Cambridge University Press.

EXPLORING THE WHOLE NUMBER KNOWLEDGE OF CHILDREN IN GRADE 1 TO GRADE 4: INSIGHTS AND IMPLICATIONS

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This paper provides a snapshot of the whole number knowledge of nearly 2000 Australian primary school children that was gained through use of a one-to-one assessment interview. The assessment data was related to a corresponding growth point framework that describes learning trajectories in counting, place value, addition and subtraction strategies, and multiplication and division strategies. The findings highlight the broad distribution of growth points in each domain for each grade level, and the wide distance between the lowest and highest growth points in each grade level and each domain. This demonstrates the complexity of classroom teaching and highlights the challenge of meeting each student's learning needs. Particular issues related to children's learning emerged from the data, and these suggest important themes for teacher professional learning and for refining mathematics curriculum. These include: (1) interpreting 2-digit and 3-digit numbers; (2) using reasoning strategies as opposed to counting-based strategies in addition and subtraction; (3) and strategies for solving partially modelled and abstract problems in multiplication and division.

INTRODUCTION

Education is well established as a significant factor in breaking the cycle of poverty for marginalised people throughout the world (Zappalà, 2003). Education provides knowledge that ultimately empowers people to access further education, employment and active citizenship. Sadly, educational outcomes for Australian children living in low Socio-Economic Status (SES) communities and Aboriginal and Torres Strait Islander communities are lower than for children not living in these communities (Commonwealth of Australia, 2008; Zevenbergen & Nieske, 2008). Thus, the current Australian Federal Government has both a continued and renewed emphasis on closing the education gap between these groups of Australians. One initiative launched by the Federal Government is a series of Pilot Projects that seek insight about how to close the literacy and numeracy gap for Australian children. This paper reports on one pilot, *Bridging the Numeracy Gap* (Gervasoni, Parish, Hadden, Turkenburg, Bevan, Livesey, & Croswell, 2011) that is a collaboration between 44 school communities, the Catholic Education Offices (CEOs) in three dioceses in south-eastern Australia (Ballarat, Sandhurst and Sale), the CEO Western Australia, and Australian Catholic University.

Key approaches for improving mathematics learning in this Pilot were: one-to-one interview-based mathematics assessment using the *Early Numeracy Interview* and associated framework of growth points (Clarke, Cheeseman, Gervasoni, Gronn, Horne, McDonough, Montgomery, Roche, Sullivan, Clarke, & Rowley, 2002; Gervasoni, Turkenburg, & Hadden, 2007) using this data to guide instruction and curriculum development at individual, class and whole school levels (Gervasoni & Sullivan, 2007); and using the *Extending Mathematical Understanding* Program (Gervasoni, 2004) in the second year of schooling (six-year-olds) to provide intensive specialised instruction for children who were mathematically vulnerable.

This paper reports on one aspect of this project; using a one-to-one interview-based assessment and an associated framework of growth points to gain insight about primary school children's whole number knowledge. Based on the insights gained about this knowledge, the implications for classroom instruction and teacher professional learning will be discussed with the view to enhancing mathematics learning for all.

USING FRAMEWORKS AND INTERVIEWS TO IDENTIFY CHILDREN'S WHOLE NUMBER KNOWLEDGE

Clinical assessment interviews are now widely used by teachers in Australia and New Zealand as a means of assessing children's mathematical knowledge. This is due to the experience of three large scale projects that informed assessment and curriculum policy formation in Victoria, NSW and New Zealand: *Count Me In Too* (Gould, 2000) in NSW, the Victorian *Early Numeracy Research Project* (Clarke, Cheeseman, Gervasoni, et al., 2002) and the *Numeracy Development Project* (Higgins, Parsons, & Hyland, 2003) in New Zealand.

A common feature of each of these projects was the use of a one-to-one assessment interview and an associated research-based framework to describe progressions in mathematics learning (Bobis, Clarke, Clarke, Thomas, Wright, Young-Loveridge & Gould, 2005). Teachers participating in each project indicated that the benefits of the assessment interview, though time-consuming and expensive, were considerable in terms of creating an understanding of what children know and can do, and for subsequently planning instruction. Indeed, an important feature of clinical interviews is that they enable the teacher to observe children as they solve problems to determine the strategies they used and any misconceptions (Gervasoni & Sullivan, 2007). They also enable teachers to probe children's mathematical understanding through thoughtful questioning (Wright, Martland, & Stafford, 2000) and observational listening (Mitchell & Horne, 2011).

The insights gained through this type of assessment inform teachers about the particular instructional needs of each student more powerfully than scores from traditional pencil and paper tests, the disadvantages of which are well established (e.g., Clements and Ellerton, 1995). Bobis et al. (2005) concluded that one-to-one assessment interviews and associated frameworks assisted to move the focus of professional learning in mathematics from the notion of children carefully reproducing taught procedures to an emphasis on children's thinking. This is an important outcome at a time when it is broadly accepted that the traditional focus on taught procedures for calculating can negatively impact on children's number sense (Clarke, Clarke, & Horne, 2006) and may impede children's development of powerful mental reasoning strategies for calculating (Narode, Board, & Davenport, 1993). Thus, because of the deep insight about children's mathematical knowledge gained through the use of one-to-one assessment interviews, the *Early Numeracy Interview* was chosen as the assessment tool for the *Bridging the Numeracy Gap* (BTNG) project. It was anticipated that the data obtained would be insightful for teachers and that the cohort data would provide a rich snapshot of children's whole number knowledge for principals and system leaders.

THE EARLY NUMERACY INTERVIEW AND GROWTH POINTS

One feature of the BTNG project was the assessment of nearly 2000 Preparatory to Grade 4 (5-9-year-old) children's whole number knowledge at the beginning of each year using the *Early Numeracy Interview* (Department of Education Employment and Training, 2001). *The interview* was refined during the project and renamed the *Mathematics Assessment Interview (MAI)* in 2010 (Gervasoni et al., 2011). The data examined in this paper was drawn from the interviews and the associated framework of growth points, so it is important that both are clearly understood. An outline is provided below.

The *Early Numeracy Interview* (Department of Education Employment and Training, 2001), developed as part of the *Early Numeracy Research Project* (ENRP, Clarke et al., 2002), is a clinical interview with an associated research-based framework of growth points that describe key stages in the learning of nine mathematics domains. The principles underlying the construction of the growth points were to: describe the development of mathematical knowledge and understanding in the first three years of school in a form and language that was useful for teachers; reflect the findings of relevant international and local research in mathematics (e.g., Steffe, von Glasersfeld, Richards, & Cobb, 1983; Fuson, 1992; Mulligan, 1998; Wright, Martland, & Stafford, 2000; Gould, 2000); reflect, where possible, the structure of mathematics; allow the mathematical knowledge of individuals and groups to be described; and enable a consideration of children who may be mathematically vulnerable.

The growth points form a framework for describing development in nine domains, including four whole number domains that are the focus of this research: Counting, Place Value, Addition and Subtraction, and Multiplication and Division. The processes for validating the growth points, the interview items and the comparative achievement of children in project and reference schools are described in full in Clarke et al. (2002).

To illustrate the nature of the growth points, the following are the growth points for Addition and Subtraction. These emphasise the strategies children use to solve problems.

1. Counts all to find the total of two collections.
2. Counts on from one number to find the total of two collections.
3. Given subtraction situations, chooses appropriately from strategies including count back, count-down to & count up from.
4. Uses basic strategies for solving addition and subtraction problems (doubles, commutativity, adding 10, tens facts, other known facts).
5. Uses derived strategies for solving addition and subtraction problems (near doubles, adding 9, build to next ten, fact families, intuitive strategies).
6. Extending and applying. Given a range of tasks (including multi-digit numbers), can use basic, derived and intuitive strategies as appropriate.

Each growth point represents substantial expansion in knowledge along paths to mathematical understanding (Clarke, 2001). They enable teachers to: identify any children who may be vulnerable in a given domain; identify the zone of proximal development for each child in each domain so instruction may be customised and precise; and identify the diversity of mathematical knowledge in a class. The whole number tasks in the interview take between 15-25 minutes for each student and are administered by the classroom teacher. There are about 40 tasks in total, and given success with a task, the teacher continues with the next tasks in a domain (e.g., Place Value) for as long as the child is successful. Teachers reported that the *Early Numeracy Interview* (ENI) provided them with insights about children's mathematical knowledge that might otherwise remain hidden (Clarke, 2001). This was an important reason for using this assessment instrument during the *Bridging the Numeracy Gap* project.

GAINING INSIGHT ABOUT CHILDREN'S WHOLE NUMBER KNOWLEDGE

The data examined in this paper were collected in 44 school communities associated with the Catholic Education Offices of Ballarat, Sandhurst and Sale in Victoria (south-eastern Australia), and the Catholic Education Office of Western Australia. Participants included nearly 2000 Grade 1 to Grade 4 (6-9-year-old) children who were assessed at the beginning of 2011 by their classroom teachers using the *Mathematics Assessment Interview* (MAI).

The detailed interview record sheets were independently coded by Australian Catholic University (ACU) research staff to determine the growth points children reached in the domains of Counting, Place Value, Addition and Subtraction Strategies, and Multiplication and Division Strategies (see Appendix A). Independent coders were used to increase the validity and reliability of the data. The growth points for each student were entered into a database and analysed to determine the percentage of children on each growth point in each whole number domain and grade level. This enabled a rich snapshot of children's whole number knowledge to be obtained, and also enabled the researchers to: (1) measure children's growth in mathematical understanding over the 3-year period of the project; (2) inform the refinement of the interview; and (3) measure the impact of the EMU intervention program (Gervasoni, 2004) for participating children in 2009 and 2010. Creating a snapshot of children's whole number knowledge at the beginning of 2011 is the focus of this paper.

INSIGHTS ABOUT CHILDREN'S WHOLE NUMBER KNOWLEDGE

Examination of the 2011 MAI growth point distributions for nearly 2000 children provides a rich picture of the current whole number knowledge of children across the first five years of schooling, and enables associated insights and issues to be identified. The following section explores the findings for the domains of Counting, Place Value, Addition and Subtraction Strategies, and Multiplication and Division Strategies.

Counting Knowledge

The 2011 Counting growth point distributions for children in Grade 1 to Grade 4 are shown in Figure 1. The data highlights that there is a wide distribution of growth points in each grade. For example, the growth points for Grade 1 (6-year-old) children range from GP0 to GP5, and from GP1 to GP6 for Grade 4 (9-year-old) children. This demonstrates the complexity and challenge of classroom teaching, and the need for activities and instruction to be customised for individuals. Apparent also is the growth in knowledge that occurs from one grade to the next; the median Counting growth point increased by one growth point in each grade level.

Examination of the Counting growth point distribution for the Grade 4 children provides some insight about the growth of children's knowledge across the first four years of primary school. By the beginning of Grade 4 (fifth year at primary school), about 50% of children were working towards Growth Point 6 – extending and applying their counting knowledge. Thus, assisting children to make this growth point transition needs to be a key focus for Grade 4 teachers. In contrast, about 10% of Grade 4 children could not yet count forwards and backwards by ones beyond 110, and another 20% could not yet skip count by 2s, 5s and 10s from zero. It is anticipated that these children (30%) would struggle with accessing some aspects of the Grade 4 curriculum and may benefit from customised instruction.

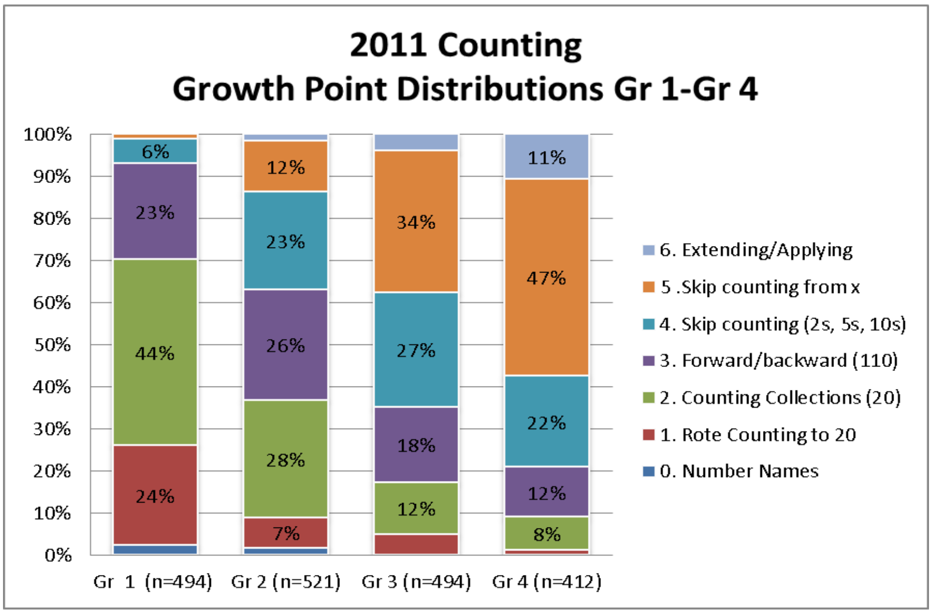

Figure 1. Counting growth point distributions for children in Grades 1– 4.

Place Value

The Place Value knowledge assessed by the MAI includes children's abilities to read, write, order and interpret numbers. The growth point distributions are shown in Figure 2.

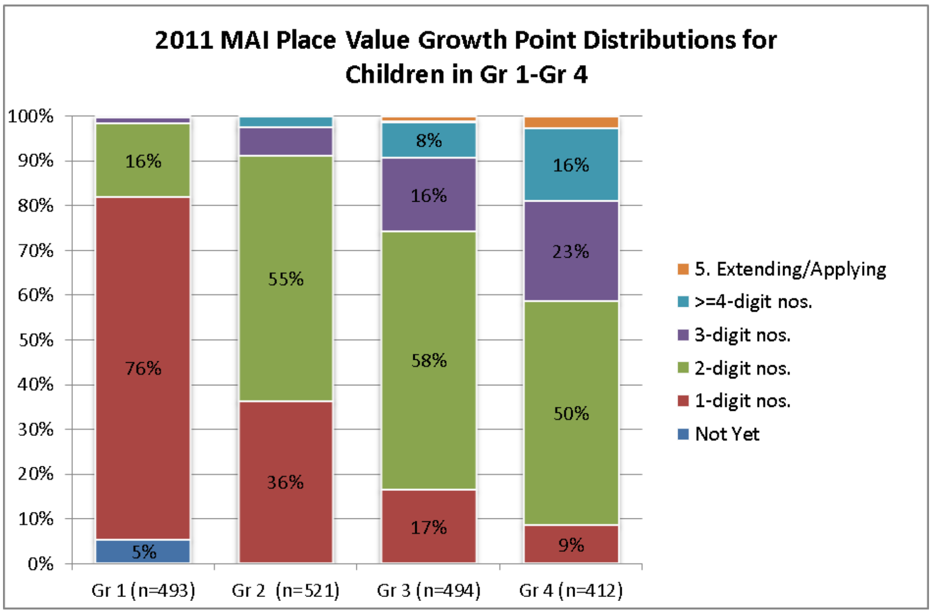

Figure 2. Place Value growth point distributions for children in Grades 1– 4.

Figure 2 also indicates a wide distribution of growth points in each grade, but less growth is apparent from grade to grade compared with Counting. Indeed the median growth point remains GP2 (2-digit numbers) from Grade 2 to Grade 4. Reaching GP2 and then GP3 (3-digit numbers) are significant issues for children in these grade levels.

By Grade 4, 50% of children remained on GP2 in Place Value, and almost 10% were still on GP1. This means they could read, write, order and interpret 2-digit and 1-digit numbers respectively. Thus the focus for 60% of Gr 4 children is learning to interpret 2-digit and 3-digit numbers. This implied struggle with understanding 3-digit numbers well into Gr 4 is important for teachers to recognise, as some may assume that children who can read and write these numbers can also appreciate them as quantities. This is not so. Indeed, it was the interpretation of quantities rather than the ability to read, write and order numerals that posed difficulty for children in this cohort (Gervasoni et al., 2011). Highlighting the complexity of planning instruction for Grade 4 children is the fact that 40% can already understand 3-digit numbers, while 15% have reached GP4. This wide variation in the range of numbers that children understand has implications for their approaches to calculating and solving number problems, and for the instruction planned by teachers.

Addition and Subtraction Strategies

A key focus for learning and teaching in addition and subtraction is the development and use of reasoning strategies for calculating as opposed to counting-based strategies. However, the 2011 growth point distributions for Addition and Subtraction Strategies (Figure 3) indicate that many children in all grades relied on less efficient counting-based strategies for calculating.

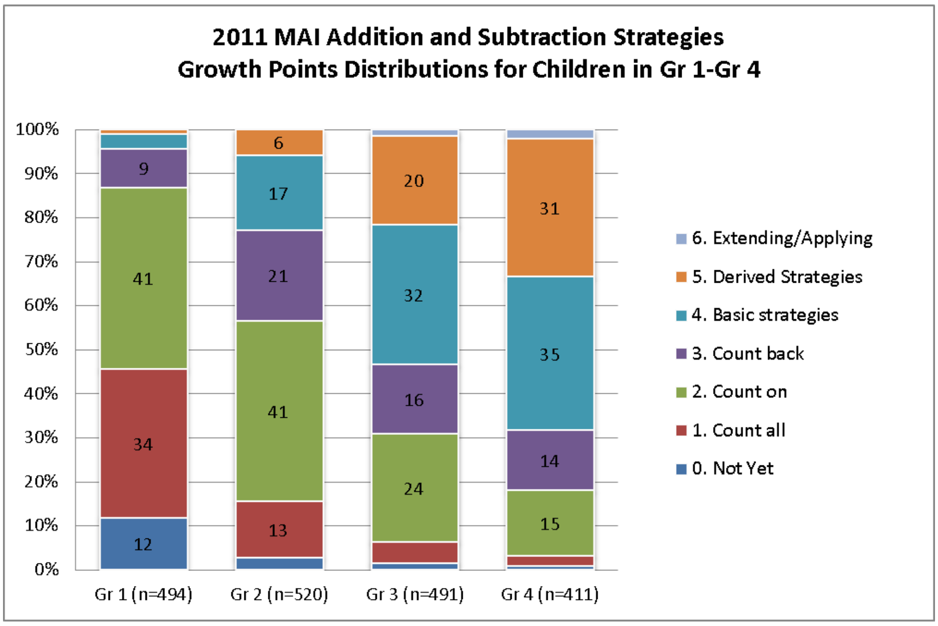

Figure 3. Addition & Subtraction growth point distributions for children in Grades 1– 4.

The data in Figure 3 indicate that 96% of Grade 1 children, 75 % of Grade 2 children, 46% of Grade 3 children and 30% of Grade 4 children used counting-based strategies for calculations involving 1-digit numbers, such as 4+4 and 10-3. The fact that so many Grade 4 children remain reliant on counting-based strategies for calculating, and that almost no Grade 4 students could solve mental calculations involving 2-digit and 3-digit numbers (GP6), is at odds with the tasks typically found in Grade 4 text books that involve calculations with much larger numbers. The data in Figure 3 also highlights the wide distribution of growth points in each grade, and also the growth in knowledge from one grade to the next.

Multiplication and Division Strategies

In the Multiplication and Division domain, the key issue for learning and teaching is children's ability to perform calculations without models being present to assist the calculating, first in partial modelling situations and also when models are completely removed. Figure 4 shows the 2011 growth point distributions for Multiplication and Division Strategies.

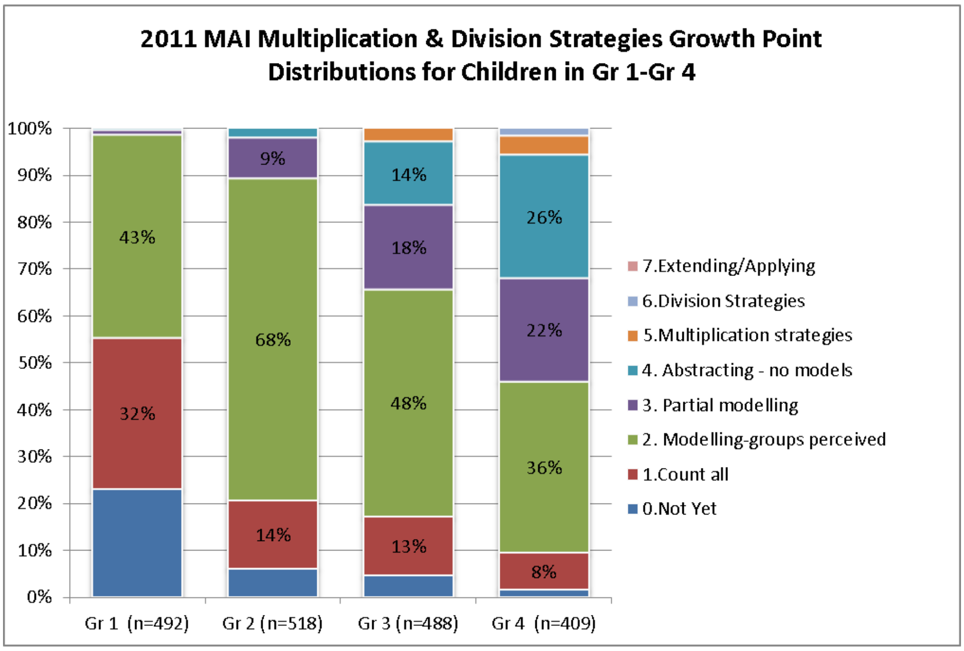

Figure 4. Multiplication and Division growth point distributions for Grade1– 4 children.

An outstanding feature of the growth point distributions shown in Figure 4 is the large number of children in each grade on GP2 (models all objects to solve multiplicative and sharing situations). In Grade 3 only 35% of children were able to solve multiplicative problems in partial modelling or abstract situations. This increased to 50% of Grade 4 children. Overall, it is likely that children need more experience with an expanded range of multiplicative problems presented in partial modelling situations in order for their multiplicative reasoning to develop. Other features of the data are the wide distribution of growth points in each grade, and the similarity in growth point distributions for the lower 20% of children in Grade 2 and Grade 3. This suggests that instruction for these children may not be sufficiently meeting their needs. Also, two-thirds of Grade 4 children were not able to solve simple multiplication problems without partial models being provided. This finding is contrary to the expectations inherent in many mathematics programs and text books.

DISCUSSION

The findings presented in the previous section highlight several important issues related to the instructional needs of children. First is the broad distribution of growth points in each domain and each grade level, and the wide distance between the lowest and highest growth points in each grade level and each domain. This highlights the complexity of classroom teaching and of meeting each student's learning needs. The data indicate that children in each grade have diverse learning needs, and this calls for customised instructional responses from teachers. It is likely that teachers need to make individual decisions about the instructional approach for each student, and that there is no 'formula' that will meet all children's instructional needs. Thus it is essential that classroom teachers are (1) able to identify children's current knowledge in order to plan suitable learning opportunities for them, and (2) have the ability to refine the curriculum and customise learning opportunities for individuals according to the range of knowledge represented in a teaching group. Principals and teachers participating in the BTNG Pilot (Gervasoni et al., 2011) noted that classroom teachers who used MAI data to refine the curriculum and design instruction for individual children, increased the capacity of their school communities to provide effective instruction for all.

The second point to note is that progress through the growth points is clearly obvious from one grade level to the next in each domain. However, it is also clear that progress from one growth point to the next is more challenging for some growth points and in some domains, than for others. For example, progress from GP2 to GP3 in Place Value took considerable time for many children. Knowledge about these challenging points in children's learning and about how to assist children to reach them is necessary to enable teachers to be most effective, and may be a useful focus for professional learning programs.

A third issue highlighted by the growth point distributions is the significant number of children on very low and very high growth points in most grade levels and domains. These children may be particularly vulnerable if teachers are not able to cater for their particular instructional needs learning in the classroom. Of particular interest are the children on the lowest growth points who are not thriving in the context of the regular classroom. These children are easily identified by the growth point framework and may benefit from a specialised program.

A fourth point emerging from the data is that few children reached the highest growth points in each domain, even in Grade 4. This provides confidence that the MAI may be useful throughout the primary school, and certainly beyond Grade 4. Indeed, some schools have used the MAI with children in Grades 5-7 and few children have reached the ceiling in each domain. However, these data also emphasise the importance of creating high quality learning environments that enable all to thrive and reach these higher growth points.

Finally, the analyses suggest that several growth points represent challenging aspects of whole number learning. These include: (1) interpreting 2-digit and 3-digit numbers; (2) using reasoning strategies as opposed to counting-based strategies in addition and subtraction; and (3) solving partially modelled and abstract problems in multiplication and division. It is recommended that professional learning opportunities for teachers focus on these challenging growth points and the associated powerful pedagogical actions and tools that will assist students to learn successfully.

CONCLUSION

The findings presented in this paper suggest that there is no single ‘formula’ for describing children’s whole number knowledge or the instructional needs of children in a particular grade. Meeting the diverse learning needs of children is a challenge, and requires teachers to be knowledgeable about how to identify each child’s current mathematical knowledge and customise instruction accordingly. This calls for rich assessment tools capable of revealing the extent of children’s knowledge in a range of domains, and an associated framework of growth points capable of guiding teachers’ curriculum and instructional decision-making. Growth points help teachers to identify children’s zone of proximal development in mathematics so as to create appropriate learning opportunities, and in order to adjust activities to increase engagement and remove features that are creating barriers to learning. Thus, reference to a framework of growth points helps to ensure that instruction for children is closely aligned to their initial and on-going assessment, and is at the ‘cutting edge’ of each child’s knowledge (Wright et al, 2000).

Assisting children to learn mathematics is complex, but teachers who are equipped with the pedagogical knowledge and actions necessary for responding to the diverse needs of individuals are able to provide children with the type of learning opportunities and experiences that enable them to thrive mathematically. This must be the goal of all school communities.

ACKNOWLEDGEMENTS

The research reported in this paper was funded by the Australian Government as part of the *Bridging the Numeracy Gap in Low SES and Indigenous Communities Project*. The author acknowledges gratefully the contribution of teachers, parents and students in the 44 school communities involved in the research, and of her inspirational colleagues in the research team, Kate Bevan, Melissa Croswell, Teresa Hadden, Carole Livesey, Linda Parish, and Kathie Turkenburg.

REFERENCES

Bobis, J., Clarke, B., Clarke, D., Thomas, G., Wright, R., Young-Loveridge, J. & Gould, P. (2005). Supporting Teachers in the Development of Young Children’s Mathematical Thinking: Three Large Scale Cases. *Mathematics Education Research Journal*. 16(3), 27–57.

Clarke, D. (2001). Understanding, assessing and developing young children's mathematical thinking: Research as powerful tool for professional growth. In J. Bobis, B. Perry & M. Mitchelmore (Eds.), *Numeracy and beyond: Proceedings of the 24th Annual Conference of the Mathematics Education Research Group of Australasia* (Vol. 1, pp. 9-26). Sydney: MERGA.

Clarke, D., Cheeseman, J., Gervasoni, A., Gronn, D., Horne, M., McDonough, A., Montgomery, P., Roche, A., Sullivan, P., Clarke, B., & Rowley, G. (2002). *Early Numeracy Research Project Final Report*. Melbourne: Australian Catholic University.

Clarke, B., Clarke, D., & Horne, M. (2006). A longitudinal study of mental computation strategies. In Novotná, J., Moraová, H., Krátká, M, & Stehlíková, N. (Eds). *Proceedings of the 30th Conference of the International Group for Psychology of Mathematics Education, Vol. 2*, pp. 329-336. Prague: PME.

Clements, M. A., & Ellerton, N. (1995). Assessing the effectiveness of pencil-and-paper tests for school mathematics. In B. Atweh & S. Flavel (Eds.), *Galtha: MERGA 18: (Proceedings of the 18th Annual Conference of the Mathematics Education Research Group of Australasia,* pp. 184–188). Darwin, UNT.

Commonwealth of Australia (2008). *National Numeracy Review Report*. Retrieved 18 March 2010, from COAG Web site: http://www.coag.gov.au/reports/docs/national_numeracy_review.pdf.

Department of Education Employment and Training. (2001). *Early numeracy interview booklet*. Melbourne: Department of Education, Employment and Training.

Fuson, K. (1992). Research on whole number addition and subtraction. In D. A. Grouws (Ed.), *Handbook of research on mathematics teaching and learning* (pp. 243-275). New York: Macmillan.

Gervasoni, A., Parish, L., Hadden,T., Turkenburg, K., Bevan, K., Livesey, C., & Croswell, M. (2011). Insights about children's understanding of 2-digit and 3-digit numbers. In J. Clark, B. Kissane, J. Mousley, T. Spencer & S. Thornton (Eds.), *Mathematics: Traditions and [New] Practices (Proceedings of the 23rd biennial conference of The Australian Association of Mathematics Teachers and the 34th annual conference of the Mathematics Education Research Group of Australasia, Vol 1*, pp. 315-323). Alice Springs: MERGA/AAMT.

Gervasoni, A., & Sullivan, P. (2007). Assessing and teaching children who have difficulty learning arithmetic. *Educational & Child Psychology*, 24(2), 40-53.

Gervasoni, A., Hadden, T., & Turkenburg, K. (2007). Exploring the number knowledge of children to inform the development of a professional learning plan for teachers in the Ballarat diocese as a means of building community capacity. In J. Watson & K. Beswick (Eds), *Mathematics: Essential Research, Essential Practice Hobart: MERGA (Proceedings of the 30th annual conference of the Mathematics Education Research Group of Australasia,* 305-314. Hobart: MERGA.

Gervasoni, A. (2004). *Exploring an intervention strategy for six ans seven year old children who are vulnerable in learning school mathematics.* Unpublished PhD thesis, La Trobe University, Bundoora.

Gould, P. (2000). Count Me In Too: Creating a choir in the swamp. In Improving numeracy learning: *What does the research tell us? (Proceedings of the ACER Research Conference* 2000, pp. 23–26). Melbourne: Australian Council for Educational Research.

Higgins, J., Parsons, R., & Hyland, M. (2003). The Numeracy Development Project: Policy to practice. In J. Livingstone (Ed.), *New Zealand annual review of education*. (pp 157–174). Wellington: Victoria University of Wellington.

Mitchell, A. & Horne, M. (2011). Listening to children's explanations of fraction pair tasks: when more than an answer and an initial explanation are needed. In J. Clark, B. Kissane, J. Mousley, T. Spencer & S. Thornton (Eds.), *Mathematics: Traditions and [New] Practices (Proceedings of the 23rd biennial conference of The Australian Association of Mathematics Teachers and the 34th annual conference of the Mathematics Education Research Group of Australasia, Vol 1*, pp.515-522). Alice Springs: MERGA/AAMT.

Mulligan, J. (1998). A research-based framework for assessing early multiplication and division. In C. Kanes, M. Goos & E. Warren (Eds.), *Teaching mathematics in new times: Proceedings of the 21st annual conference of the Mathematics Education Research Group of Australasia (Vol. 2, pp. 404-411)*. Brisbane: MERGA.

Narode, R., Board, J., & Davenport, L. (1993). Algorithms supplant understanding: Case studies of primary students' strategies for double-digit addition and subtraction. In J. R. Becker & B. J. Preece (Eds.), *Proceedings of the 15th Annual Meeting of the North American chapter of the International Group for the Psychology of Mathematics Education* (Vol. 1, pp. 254–260). San Jose: Center for Mathematics and Computer Science Education, San Jose State University.

Steffe, L., von Glasersfeld, E., Richards, J., & Cobb, P. (1983). *Children's counting types: Philosophy, theory, and application*. New York: Praeger.

Wright, R., Martland, J., & Stafford, A. (2000). *Early Numeracy: Assessment for teaching and intervention*. London: Paul Chapman Publishing.

Zappalà, G. (2003). *Barriers to Participation Financial, Educational and Technological. A report into the barriers to societal participation among low-income Australians.* Camperdown, NSW: The Smith Family. http://www.thesmithfamily.com.au/webdata/resources/files/TSF_barrieries_2003.pdf Retrieved 18 March 2010.

Zevenbergen, R., & Nieske, R. (2008). *Reforming mathematics classrooms: a case of remote Indigenous education. Working Paper 2*, Brisbane: Griffith Institute for Educational Research, Griffith University.

APPENDIX A: ENRP 2010 REFINED ASSESSMENT FRAMEWORK

Growth Points for the Number Domains

Notes:

- *Growth points are not necessarily hierarchical, but involve increasingly complex reasoning and understanding.*
- *It must be emphasised that conclusions drawn in relation to placing students at levels within this framework are based on a 30-minute (approx.) interview only. On-going assessment by the teacher during class will provide important further information for this purpose.*
- *Student understanding may be reported as a "0". This should not be taken as an indication of "no knowledge" or "no understanding", but rather as an indication of a lack of evidence of "1".*
- *The Growth Points below were revised in 2010 as part of the Bridging the Numeracy Gap Project.*

A. COUNTING

0. Not apparent.
   Not yet able to say sequence of number names to 20.
1. Rote counting
   Rote counts the number sequence to at least 20, but is not yet able to reliably count a collection of that size.
2. Counting collections
   Confidently counts a collection of around 20 objects.
3. Counting by 1s (forward/backward, including variable starting points; before/after)
   A count forward and backwards from various starting points between 1 and 100; knows numbers before and after a given number.
4. Counting from 0 by 2s, 5s, and 10s
   Can count from 0 by 2s, 5s, and 10s to a given target.
5. Counting from *x* (where *x* >0) by 2s, 5s, and 10s
   Given non-zero starting points, counts by 2s, 5s,10s to given target.
6. Extending and applying counting skills
   Counts from non-zero starting points by any 1-digit number, and applies counting skills in practical tasks.

C. STRATEGIES FOR ADDITION & SUBTRACTION

0. Not apparent
   Not yet able to combine & count 2 collections of objects.
1. Count all (two collections)
   Counts all to find the total of two collections.
2. Count on
   Counts on from one number to find the total of two collections.
3. Count back/count down to/count up from
   In subtraction contexts, chooses suitably from count-back, count-down-to & count-up-from strategies.
4. Basic strategies (doubles, commutativity, adding 10, tens facts, other known facts)
   Given an addition or subtraction problem, strategies such as doubles, commutativity, adding 10, tens facts, and other known facts are evident.
5. Derived strategies (near doubles, adding 9, build to next ten, fact families, intuitive strategies)
   Given an addition or subtraction problem, strategies such as near doubles, adding 9, build to next ten, fact families and intuitive strategies are evident.
6. Extending and applying addition and subtraction using basic, derived and intuitive strategies
   Given a range of tasks (including multi-digit numbers), can solve them mentally, using the appropriate strategies and a clear understanding of key concepts.

B. PLACE VALUE

0. Not apparent
   Not yet able to read, write, interpret and order single digit numbers.
1. Reading, writing, interpreting, and ordering single digit numbers
   Can read, write, interpret and order single digit numbers.
2. Reading, writing, interpreting, and ordering two-digit numbers
   Can read, write, interpret and order two-digit numbers.
3. Reading, writing, interpreting, and ordering three-digit numbers
   Can read, write, interpret and order 3-digit numbers.
4. Reading, writing, interpreting, and ordering numbers beyond 1000
   Can read, write, interpret & order numbers beyond 1000.
5. Extending and applying place value knowledge
   Can extend and apply knowledge of place value in solving problems.

D. STRATEGIES FOR MULTIPLICATION & DIVISION

0. Not apparent
   Not yet able to create and count the total of several small groups.
1. Counting group items as ones
   To find the total in a multiple group situation, refers to individual items only.
2. Modelling multiplication & division
   Models all objects to solve multiplicative and sharing situations.
3. Partial modelling multiplication & division (some objects perceived)
   Solves multiplication and division problems where objects are partially modelled or perceived.
4. Abstracting multiplication and division
   Solves multiplication and division problems where objects are not all modelled or perceived.
5. Basic, derived and intuitive multiplication strategies
   Can solve a range of multiplication problems using strategies such as commutativity, skip counting and building up from known facts.
6. Basic, derived and intuitive strategies for division
   Can solve a range of division problems using strategies such as fact families and building up from known facts.
7. Extending and applying multiplication and division
   Can solve a range of multiplication and division problems (including multi-digit numbers) in practical context.

THE EFFECTS OF SOCIAL INTERACTIONS ON ENGAGEMENT WITH MATHEMATICS

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In spring 2010 nine undergraduate students were interviewed about their experiences of first year mathematics. All nine students had passed their first year examinations and had engaged with mathematics to some extent. The aim of this research was to provide insight into the reasons why these students had engaged. It is clear from the interviews that these students experienced mathematical difficulties in their first year; however they all had a plan or coping mechanism to help them succeed. They regularly mentioned social interactions with students and staff members. Consequently the main focus of this paper is to examine these social interactions and how they formed part of the students' coping mechanisms. These interactions can be broken into two categories: interactions with teaching staff; and interactions with friends and peers. When negative experiences are related by the students we note that they always find a way to engage with other supports available.

INTRODUCTION

The Department of Mathematics and Statistics at the National University of Ireland Maynooth (NUIM) has many supports in place to help students if they experience difficulties. In addition to lectures, each student is assigned to a weekly small-group tutorial where attendance is taken and students also have homework graded every week. Online courses and additional workshops have been designed to help students with weak mathematical backgrounds, and the department runs a very successful Mathematics Support Centre (MSC). For the remainder of this paper when we mention mathematics supports we mean all the services mentioned here. The main service of the MSC is the drop in centre where students can seek help with any mathematics related problem. The MSC operates in an informal and friendly atmosphere, encouraging the students not to be fearful or embarrassed to attend and seek help. The engagement level of students with the available supports varies and consequently we decided to investigate the reasons and motivations why certain students engage with mathematics supports and others do not.

In September 2009 seven students, who were repeating their first year of mathematics, were interviewed and it subsequently emerged that they had not generally engaged with mathematics supports. The dominant theme to emerge from these interviews was the embarrassment and fear suffered by these students (Grehan, Mac an Bhaird, & O'Shea, 2010, 2011). Four different facets of fear were identified: fear of failure; fear of showing a lack of knowledge or ability; fear of being singled out; and fear of the unknown. Other factors identified in our study as important in students' non-engagement with mathematics support included: the demoralising effect of failing first semester examinations; the anonymity of large classes; and to a lesser extent the lack of awareness of support services. Very few of these students mentioned working on mathematics with their friends or peers and very rarely discussed interactions with lecturers or tutors. Analysis of these interviews showed that the students who did not avail of supports were often not aware that they had a problem, or were unwilling to admit to it (to themselves or others) until it was too late.

To obtain further insight into the reasons students engage or do not engage with mathematics supports we decided to conduct interviews with students with similar mathematical backgrounds to the first group, but who had generally engaged with mathematics supports. These interviews took place in spring 2010. Ten students were chosen at random and nine were interviewed. Preliminary results (Grehan, Mac an Bhaird, & O'Shea, to appear) have shown that these students encounter many of the same difficulties as the first group. However, the majority have some plan or strategy to overcome these difficulties. Other factors that emerged as significant in promoting engagement were: personal motivation; the willingness to admit to having problems; the appreciation of mathematics as a cumulative subject; and social interactions between students and peers and staff. Analysis of all of these factors is ongoing but these social interactions and relationships form the main focus of this paper.

In contrast to the first group, these students frequently mentioned interactions with peers and staff. Many of them had discussed mathematics with their friends and classmates. Most mentioned working in groups to solve problems or tackle assignments. Some negative experiences are also related but we will see how these students found some alternative way to engage with mathematical support in these situations.

LITERATURE REVIEW

Several studies have shown the positive impacts of students engaging with mathematics, and specifically with mathematics support. In a paper by colleagues at NUIM (Mac an Bhaird, Morgan, & O'Shea, 2009) it was shown that the MSC has a positive effect on the grades of students who attend. It is of interest that mathematics support is particularly beneficial to those with a weak mathematical background. However, many at-risk students do not avail of mathematics support. Similar studies by Dowling and Nolan (2006) and Pell and Croft (2008) discuss the benefits of students engaging with mathematics support but also reinforce the findings that it is not always the students most in need of support who attend the MSC.

Similar results were found during an investigation by Symonds (2008) at Loughborough University in the UK. She interviewed a small number of students and found that there was a very strong correlation between engagement with mathematics and a student's mathematics module grade.

A separate paper by Mac an Bhaird and O'Shea (2009) suggests that students deemed at risk in first year are more likely to engage than those not deemed at risk. This study took place in NUIM and is especially relevant to our ongoing research. The authors admit that the statistics do not shed light on the reasons behind the lack of engagement of some students.

The study carried out by Symonds (2008) suggests that a lack of awareness of the location of services and a lack of awareness of the facilities and supports available to them form part of the reason students do not engage with mathematics support. However, she openly questions whether these are the real reasons for non-engagement and suggests that students who regularly engage overcome these issues with a lack of awareness of location and supports available:

> Such students were able to overcome these barriers in order to avail themselves of the support facilities. This poses the question: would simply implementing the above suggestions (advertising, actively seeking out students who need help, staff changes) be enough to improve the uptake of support amongst failing students? (P.140)

Other authors have found that the fear of showing a lack of knowledge or ability negatively impacts on students' willingness to ask questions (Ryan, Pintrich, & Midgley, 2001). They explore why students do not ask for help:

> Students who feel insecure about their abilities-academically or relating socially to other students-are more likely to avoid help seeking. Students who are focused on their reputation-academic or social-are more likely to avoid help seeking. (P.110-111)

This suggests that students' goals can play a role in students' choice to seek help. Early work in this area includes papers by Dweck (1986) and Pintrich (1994). Goal theory, until quite recently, was limited to the idea of performance and mastery goals. Students with mastery goals were understood to be motivated by a desire to understand a topic in depth and increase their knowledge and understanding, whereas students with performance goals were understood to be motivated by the desire to do well in examinations and other extrinsic factors. However, psychologists have recently suggested that these definitions are an over-simplification. Research from Dowson and McInerney (2003) and Levy, Kaplan and Patrick (2004) highlighted their independent discovery of several social goal orientations. These orientations cover varying ideas such as sense of belonging to a group; desire to receive praise from peers or teachers, and wanting to assist others with their academic and personal development.

Hannula (2006) has studied in detail the social aspect of student motivation. He believes that one of the primary facets of how students are motivated is the psychological need for social belonging. He discusses the possibility that students are motivated by goals such as wanting to contribute to group project work and affected by the social norms of the classroom environment. He showed how some students' choices were being dominated by social goals.

In a recent paper from researchers at Loughborough (Inglis, Palipana, Trenholm, & Ward, to appear) it was found that students who attend lectures and the MSC have higher levels of attainment than those who do not. One of the reasons suggested in this paper is the face-to-face interaction in lectures and the MSC as compared to online learning supports. The social perspective on learning was highlighted by Lave and Wenger (1991) when they introduced the concept of 'communities of practice'. Wenger (2004) defines this as follows: 'communities of practices are groups of people who share a concern or passion for something they do and learn how to do it better by interacting regularly', (p. 1). In their view, learning is a social activity which takes place within a community and is facilitated by the sharing of ideas and understanding both between teachers and students and between peers. They see learning as a form of apprenticeship where students learn both from the practices of experts and by taking part in meaningful activities in the field. These studies show that social goals and interactions can have an effect on engagement. It is this effect that we will explore in the remainder of this paper.

METHODOLOGY

In September 2009 we decided to investigate the reasons behind non-engagement with mathematics supports. We began by conducting an analysis of the engagement of students who had failed first year mathematics. Subsequently we randomly selected students who had similar mathematical backgrounds to students in the first group but who had passed first year examinations and had engaged with mathematics supports. Similar mathematical background is determined by two categories: Leaving Certificate mathematics result; and the result of a diagnostic test at the beginning of the first semester. We interviewed nine of these students.

All the interviews were conducted by the first author. Each interview lasted for approximately forty minutes. The questions were open-ended and concerned the students' mathematical education prior to enrolling in NUIM, their impression of mathematics at university, and their views on the mathematics support available to them and their reasons for engaging or not engaging with mathematics supports.

The interviews were transcribed by the first author. The transcripts had all identifying characteristics of the interviewees removed prior to coding. The transcriptions were coded separately by the three authors using Grounded Theory. The methods for analysing qualitative data using Grounded Theory are laid down by Strauss and Corbin (1998). They emphasise how to utilise one's personal and professional experience to allow you insight ("sensitivity") into the data but also caution against the bias that these experiences naturally lend to the process. Both cohorts of students frequently commented on the specific details of their mathematical experiences in NUIM. On these occasions, the three authors utilised their knowledge of the structure of mathematics support at NUIM to interpret the students' comments.

The codes were compared and contrasted, and from this process concepts and categories emerged from the data. This process and the categories that emerged from both sets of interviews are discussed in Grehan et al., (2010, 2011, to appear).

RESULTS

The main focus of this paper is to examine the social interactions and relationships mentioned by the second group of students during the course of their interviews. These interactions can be broken into two categories: interactions with teaching staff; and interactions with friends and peers. Although these students frequently mentioned difficulties, unlike our first group they were rarely deterred by these negative experiences; they almost always found some other way to engage with the supports available. We argue that these social interactions and relationships are part of the coping mechanisms that they use to succeed.

Data from our first group (Grehan et al., 2010, 2011, to appear) showed that those students rarely mentioned social interactions in relation to mathematics. This second group mention such interactions regularly. They also refer to social interactions with teachers, peers and parents at second level but we will focus purely on their experiences at third level in this paper.

Relationships/Interactions with Teaching Staff

In contrast to the first group, it is clear from our interviews that these students place high value on their interactions with lecturers, tutors and MSC support staff. While most interactions with staff are positive we will also see that negative experiences do occur. The first group of students had difficulties with all of the mathematical supports on offer including lectures, tutorials and assignments. The second group generally compensated for a difficulty with one of these supports by attending or seeking help at another.

Students were asked to comment on the importance of lectures and commented frequently about the interactions they had with lecturing staff. Several students discussed how approachable they perceived their lecturers to be and how at ease this put them. Adrian mentioned several times how he viewed the teacher-student relationship as more like a friendship than a formal relationship:

> The attitude that they have, not just with the mathematical content, but teachers in general in college, it's a much more mature relationship. I mean, it's more like a conversation between friends, that's how I perceive it like.

Other students commented on how they found some lecturers intimidating and unapproachable. Class size was mentioned by some as a contributing factor to the view that one could not ask a question in a lecture. Students admitted to being too embarrassed to ask questions in large classes and suggested that this may be part of the reason lecturers appear to be unapproachable. Janice talks about her view of mathematics lecturers as being intimidating:

> Well I got help, the only people I didn't go to were the lecturers. They were the only ones, in 1st year I would have found slightly intimidating, now this year I have no problems going to my lecturers. I would have been slightly intimidated going to them last year but I went to everyone else and I had no problem sticking up my hand and saying I don't get that!

It should be noted that Janice is no longer studying maths and the lecturers she now feels comfortable in approaching are from other departments. She clearly says that although she found her lecturers intimidating that she went to "everyone else" and had no issue asking them for help. As we will see later, Janice regularly attended the MSC and sought help from her friends. A negative experience did not dissuade her from availing of other aspects of mathematics support.

For many students, a perception that the lecturer or tutor wanted to help them seemed to influence their level of engagement. Adrian said that he was nervous about studying mathematics at university but that the attitudes of the staff he encountered encouraged him:

> You feel like it's somebody who really wants you to learn rather than somebody who is kind of spitting at you. …. Em, what was different? (The lecturer's) attitude really and the fact I really felt they wanted you to understand things.

Aoife expresses a similar sentiment when talking about the atmosphere in tutorials and her tutor:

> You knew that you were being watched, not even that you were watched, that you were being helped. And there was someone you could go and talk to about certain problems…. My tutor…he was, I found him really good, he wasn't real pushy or annoying, he was kinda like "ah you'll be grand" kind of thing but you could tell he would actually help you. I know he sound a bit blasé but he'd always come over and help you know that kind of way? I just felt more at ease.

Students also commented on the MSC tutors and their interactions with them. The majority of students were highly positive about these interactions and several commented on how the tutors were like friends who would help with mathematical problems. David says:

> I think the biggest thing about the MSC is the psychological effect. To know that somebody will help you, for me that was the biggest thing. It also, it tied in with the atmosphere that I had of the teacher wanting you to understand things and more like your friend explaining something to you rather than somebody just spouting the information.

A common facet of students' unwillingness to interact or seek help from lecturers was their perception that office hours were not for "normal" students, or that lecturers do not really want students to use them. Here Shane explains:

> They always seemed really busy and like, I know they have office hours and stuff but I was always like, that's for proper problems, not for a problem with your assignment…like (you would be) putting them out by going to them.

Commenting again on office hours Shane adds:

> I know they just have to put it up there, but I wouldn't see it as something they want you doing.

Shane eventually stopped attending mathematics lectures due to feeling he was not getting anything out of the classes. However, he would take his borrowed lecture notes and his assignments to the MSC and seek help there. This is another example of a negative experience not preventing a student from engaging with another form of mathematical support.

Students commented in depth about their interactions and relationships with their tutors. Many of the students spoke about tutors' attitudes to questions. Here Janice speaks about the one to one support she received in the MSC:

> You felt you could ask and the fact that they never laughed at you. Cause that's the big fear asking a question that they're gonna turn round and laugh at you and none of them ever looked at you to say you're stupid.

Aoife comments that the attitudes of the tutors in the MSC made it easier for her to go and to ask questions:

> And you were talked to like you were an adult, you were never talked down to. No matter how stupid the question was or how small or how large, you were always helped, you were never kinda put down.

Students also reported that they felt more at ease asking tutors for help rather than lecturers because they feel closer in age and experience to them. Janice says:

> You'd be afraid to go to a lecturer, they're doing this for ages and they've got their PhDs. … (tutors) have just gone through the college thing and I'm sure they didn't find it a doss the whole way through. So you think they might at least think to themselves, 'OK, well I kind of understand where she's coming from'.

Some students commented on the negative aspects of engaging with tutors. Siobhán found interacting with her tutor difficult and was too intimidated or embarrassed to engage in the class:

> Everyone was too afraid to speak when he asked a question. And it, awww, it was just an hour of silence!

Siobhán may have been too embarrassed to engage in tutorials but she overcame this difficulty by regularly attending the MSC where she had no issue with asking for help from the tutors.

Again negative experiences were not a major setback to these students. When confronted by an MSC staff member with whom he did not interact well, Shane developed "favourites", tutors he knew were a good fit for him. These students were all aware of their own learning style and that links into knowing what tutors worked best for them. Shane states:

> I suppose you get to know the tutors in there and you'd know who to go to, you know who'd be better for you.

It is clear from the evidence above that these students value their interactions with staff members. Students detail the personal interactions and relationships they have with their lecturers and tutors and it is apparent that their engagement with mathematics supports is affected by those relationships. They use these relationships to help themselves cope with their difficulties with mathematics. It is clear from the majority of comments mentioned by students that a positive interaction with lecturers or tutors encouraged students to ask for help, made them more comfortable and less likely to be embarrassed or afraid. We note that even if these students had negative experiences with a staff member they still engaged somehow. They may have altered their engagement, stopped attending lectures or tutorials perhaps, but they compensated elsewhere, studying with friends, in the MSC, or studying independently.

Influence of Friends/Peers on Engagement

Whilst the second group of students were selected because they engaged with mathematics support it emerged that the influence and experiences of peers and classmates were vital in determining the shape and form of that engagement. One of the main differences between this cohort of students and the first group was the formers' apparent willingness to work on mathematics with their friends and classmates. The first group of students rarely mentioned interactions with friends or peers in relation to mathematics. They were also reluctant to admit to themselves or others that they were struggling with mathematics and it is possible that this was a contributing factor to the lack of social interactions related by them during their interviews. As we shall see the second group rarely had such issues.

Each student had some negative experiences with regards to their interactions with friends and peers but we will outline how they altered their engagement so as to get help another way. We will again attempt to portray how these interactions and cooperation with their peers formed part of a general strategy aimed at helping themselves cope with their difficulties with mathematics.

Working with others on mathematics generally took two forms: working in groups in the MSC and meeting with friends or peers outside of class to work together. When questioned about the positives and negatives of group work it was apparent that most of these students were quite aware of their own learning style, as touched upon above. It was clear they had thought about how working in groups helps or hinders them in respect to completing their mathematics assignments.

Several students mentioned deciding to attend the MSC based on recommendations from friends. Aoife specifically mentions how she had a negative reaction when she heard about the MSC in a lecture but how a recommendation from peers swayed her into attending for the first time:

> But one of the girls who lived downstairs, you know we'd go to lectures together. And she was living with two other girls who (did) 3rd or 4th year maths and they were telling us how helpful the MSC was. You know when you hear about something and you're like 'ugh'. But then the second week we went and it was such a lifeline like.

Again we see how negative experiences or perceptions are not critical setbacks for these students. Shane, who later became a regular attendee, expressed similar reservations before attending based on a friend's recommendation:

> Well I wasn't sure about it now, I didn't think, maybe how popular it'd be. So I thought, you know, if I go in maybe I'll be the only one in there.

When questioned about group work in the MSC, students mentioned how beneficial it was to be able to discuss ideas with peers. They also noted how working in groups left them more independent of the MSC tutors, and remarked on the sense of team work that built up from working on problems together. Jason says:

> I found group work the best. I dunno, you could just ask somebody, bounce an idea off somebody or somebody would ask you, "what do you think?" and you're less reliant on the tutors as well then.

They also spoke about the benefits of learning from peers rather than teachers. Sorcha says:

> It was just, like one of you knew the first step, the other knew the other step, it was easier to explain to each other rather than having a lecturer or someone trying to explain it to you where it's like impossible!

Several of these students stated the unease and embarrassment they had with asking questions in the MSC when working by themselves, and how working in a group helped negate that fear. Lisa says:

> I'd feel a bit more shy cause you feel like you're the only one yourself saying "I don't know that". Where there's obviously a few of you (in a group) saying we all don't know.

Not all students were comfortable participating in group work in the MSC however. A common worry about working with peers in the MSC was that of embarrassment. Janice says:

> Well just the whole you don't want to stick up your hand cause you'd feel embarrassed if you were keeping everyone back.

As we will see below Janice was not deterred by her fear of working in a group in the MSC and found help from her friends.

Group work was not confined to the MSC. Adrian discusses how he and some friends would get together and do some mathematics:

> Group work, if I'm working on my own project I like it if I'm working with somebody who I feel is working back with me. Sometimes after one of the maths classes the schedule is convenient for Callan Hall (lecture theatre), myself (and two others) we work together and we solve things on our own and it is great.

Janice also worked a lot on mathematics outside the MSC. Her hours as a first science student meant it was difficult for her to attend the MSC. A lot of first science students experience this difficulty but this did not prevent Janice from seeking help elsewhere. She managed to find a group of friends to help her with her assignments. She mentions this help several times through her interview and it is clear she feels it had a big impact on her engagement and final grade:

> Every week about five of us got round the table, one was an expert at maths, the other was good at maths, between the two of them they'd work it out and then they'd help me and a couple of the others who were really struggling…To be honest I don't think I'd have passed if I hadn't (gotten help from friends).

We have detailed both the positive and negative influences of interactions with peers and classmates. We have seen that students were not afraid to admit to themselves or others that they were struggling with mathematics and it is apparent that working with peers or classmates on mathematics was an important aspect of this characteristic behaviour. Again it is clear that the majority of these students feel comfortable doing mathematics with their peers. Some took advice on what type of support works and others mention the reassurance of working in groups where they feel everyone has the same difficulties. They often feel more comfortable asking questions in a group rather than as an individual. Although negative experiences occur and negative perceptions exist, these students alter their type of engagement to make sure they get help with their difficulties. It is also apparent that immersing themselves in group work is part of a coping mechanism to allow them to deal with their difficulties with mathematics.

CONCLUSION

In this paper we focussed on the influence of social interactions on students' engagement with mathematics. It emerged from our analysis that students who engage encounter many of the same difficulties as non-engaging students (Grehan et al., 2010, 2011, to appear), in particular they had to grapple with fear and embarrassment. Other authors (Hannula, 2006, Ryan et al., 2001) have shown how embarrassment can affect a student's willingness to ask questions and affect their engagement with mathematics. However, few of our engaging students were deeply affected by the difficulties they encountered and further investigation of this issue is required. For example, we could consider how "resilient" these students are to setbacks (Williams, 2003). We have detailed how the two types of interactions, with teaching staff or peers, were used as part of a general strategy to overcome the variety of difficulties with mathematics.

Most of the students who participated in this study had positive experiences with tutors and to a lesser extent with lecturers. It is clear that students highly value the interactions and relationships they have with teaching staff and that they form part of the coping mechanisms discussed above. This is consistent with Inglis et al. (to appear) who highlighted the importance of face-to-face interactions in lectures and the MSC and found a link between those who attended and their attainment. Our first group of interviewees rarely had the benefit of such interactions whereas the second group listed numerous examples of the face-to-face interactions that Inglis et al. (to appear) discuss. In particular many students in this group speak of the importance of feeling that a lecturer or tutor wants to help, and of the kind of relationship they had with their lecturer or tutor. They seem to suggest that a friendly atmosphere in classes and a mature relationship with their teachers encourages them to want to understand and to seek help when needed.

All interviewees in the second group were involved in working with and interacting with peers in relation to mathematics. Many visited the MSC with friends or on the recommendation of a friend. All nine students used group work at some point in the year. For some it was a way of dealing with the fear of asking a question, a 'safety in numbers' strategy, while for others it was a means of getting help outside the MSC. It was obviously also a way of learning from peers and we can see shades of Lave and Wenger's (1991) communities of practice when students comment on working in groups. However, we did not explicitly see many of the social goal orientations explored by Dowson and McInerney (2003) but a further study as detailed below may shed more light on this issue.

The issue of why these students engage is clearly a complex one. It is apparent that the positive experiences of the second group detailed above contributed to a positive learning experience. The first group rarely speak about their experience of mathematics in a positive light. Furthermore we have only considered the social aspect of their engagement for this paper. Other features such as their motivations, their goals and their willingness to admit to having difficulties are equally as important. We have completed interviews with twenty five additional students and a full analysis of all three cohorts will be required to gain a deeper understanding of these issues. We have also amended the interview structure to include additional questions on student motivations and goal orientations.

REFERENCES

Dowling, D. & Nolan, B. (2006). Measuring the effectiveness of a maths learning centre – the Dublin City University experience. *Proceedings of the CETL MSOR Conference 2006,* 51-54.

Dowson, M. & McInerney, D.M. (2003). What do students say about their motivational goals?: Towards a more complex and dynamic perspective on student motivation. *Contemporary Educational Psychology,* 28 (1), 91-113.

Dweck, C.S. (1986). Motivational processes affecting learning. *Journal of Educational Psychology,* 93, 35-42.

Glaser B.G. & Strauss A. (1967). *Discovery of Grounded Theory. Strategies for Qualitative Research.* Sociology Press.

Grehan, M., Mac an Bhaird, C., & O'Shea, A. (2010). Why do students not avail of mathematics support? A case study of first year students at the National University of Ireland Maynooth. *Proceedings of the British Congress for Mathematics Education 2010,* 254-258.

Grehan, M., Mac an Bhaird, C., & O'Shea, A. (2011). Why do students not avail themselves of mathematics support? *Research in Mathematics Education,* 13 (1), 79-80.

Grehan, M., Mac an Bhaird, C., & O'Shea, A. (accepted for publication). How do students deal with difficulties in mathematics? *Proceedings of the CETL MSOR Conference 2010.*

Hannula, M.S. (2006). Motivation in Mathematics: Goals reflected in emotions. *Educational Studies in Mathematics*, 63, 165-178.

Inglis, M., Palipana, A., Trenholm, S., & Ward J. (accepted for publication). Individual Differences in Students' Use of Optional Learning Resources. *Journal of Computer Assisted Learning.*

Lave, J. & Wenger, E. (1991). *Situated Learning: Legitimate Peripheral Participation*. Cambridge: Cambridge University Press.

Levy, I., Kaplan, A., & Patrick, H. (2004). Early Adolescents' Achievement Goals, Social Status, and Attitudes Towards Cooperation with Peers. *Social Psychology of Education,* 7 (2), 127-159.

Mac an Bhaird, C., Morgan, T., & O'Shea, A. (2009). The impact of the mathematics support centre on the grades of first year students at the National University of Ireland Maynooth. *Teaching Mathematics and its Applications,* 28 (3), 117-122.

Mac an Bhaird, C. & O'Shea, A. (2009). What type of student avails of mathematics support and why? *Proceedings of the CETL MSOR Conference 2009,* 48-51.

Pell, G. & Croft, T. (2008). Mathematics support – support for all? *Teaching Mathematics and its Applications,* 27, 167-173.

Pintrich, P. R. (1994). Continuities and discontinuities: Future directions for research in educational psychology. *Educational Psychologist*, 29, 137–148.

Ryan, A.M., Pintrich, P.R., & Midgley, C. (2001). Avoiding seeking help in the classroom: Who and why? *Educational Psychological Review*, 13 (2), 93-114.

Strauss, A. & Corbin, J. (1998). *Basics of Qualitative Research: Techniques and Procedures for Developing Grounded Theory*. London: Sage Publications.

Symonds, R. (2008). Evaluating Students' Engagement with Mathematics Support. PhD Thesis. Loughborough University.

Wenger, E. (2004). *Communities of practice: a brief introduction.* Retrieved May 18 2011 from http://www.ewenger.com/theory/communities_of_practice_intro_WRD.doc

Williams, G. (2003). *Student inclination to work with unfamiliar challenging problems: the role of resilience.* In Barbara Clarke, Alan Bishop, Rod Cameron, Helen Forgasz, Wee Tiong Seah. Making Mathematicians. Brunswick, Victoria: Mathematical Association of Victoria.

STUDENTS' MISCONCEPTIONS CONCERNING INFINITY

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We investigate some misconceptions concerning the cardinality of infinite sets. Students from three different groups were asked to complete a short questionnaire which contained questions designed to give insight into students' concept images of infinity, cardinality and the comparison of infinite sets. The study highlighted some of the conceptions of students associated with finite, countable and uncountable infinite sets. These different conceptions and misconceptions might arise from: the meaning of words in everyday language; the incorrect use of properties which are true for some situations but not for all; the misuse of the bijection criterion; the different presentations of sets; or the idea of infinity as a number.

INTRODUCTION

Many students find the transition from computational mathematics to abstract mathematics very difficult. Therefore for mathematics lecturers at university, it is worth knowing how students learn about advanced mathematical concepts such as infinity. Teacher content knowledge has been identified as being extremely significant for successful teaching and learning; Shulman (1986) referred to this as *pedagogical content knowledge.* According to him, teachers need to understand the content they teach deeply, they need to know how the key elements of a concept might be misunderstood by students, and need to have useful ways of representing these key ideas in order to help students to overcome their misconceptions. In our view, pedagogical content knowledge is also vital for lecturers at third level, and when we speak of teachers in the remainder of this paper we include teachers at all levels. In this article we report on a preliminary case study which aims to explore students' concept images of infinity, especially their concept images of finite sets, countable infinite sets, and uncountable infinite sets.

It is known that students meet a wide range of information while they are learning mathematics, and the development of mathematical concepts will naturally depend on their previous beliefs and experience. Therefore, much work has been carried out by the mathematics education community in an effort to describe how students understand mathematical concepts. One of the most important contributions to this area is the work of Tall and Vinner (1981), which describes the distinction between the terms concept definition and concept image associated with any mathematical concept. The term *concept definition* is used to refer to a mathematical definition of the given concept; Tall and Vinner state this as "a form of words used to specify that concept" (p. 152), and the term *concept image* is used to mean all the mental pictures, the visual representations, the impressions and the experiences of the individual that are associated with the concept. Przeniolo (2004) described concept image as "the cognitive structure containing all kinds of associations and conceptions related to the concept" (p. 104). Using the ideas of Tall and Vinner (1981), many studies have been carried out to explore students understanding of a mathematical concept. Alcock and Simpson

(2009) summarised the results of many studies of this kind and also reported on students' ignorance of the status of mathematical definitions in advanced mathematics.

In most real life contexts, people acquire an understanding of a concept without the need for a proper dictionary definition, and for that reason, concept image plays an essentially important role in real life. While, on the contrary in mathematical contexts, definitions have a crucial role in the acquisition of concepts. Most mathematics teachers would expect students to base their answers on definitions when working on a problem; however students often consult only their concept image. Edwards and Ward (2004) showed that, students' inability to understand the distinction between everyday life definitions and mathematical definitions has an influence on their understanding of mathematical concepts. Przeniolo (2004) studied students' conceptions of the limits of functions, and found that students regarded their intuitions as a definition of the concept.

There are many reasons for the difficulties students face when trying to understand a mathematical concept. One of these difficulties arises when a concept has terms that are used in real life language, or when the terms have a meaning that is the opposite of the mathematical meaning. As students hold these real life conceptions for a long time, they are slow to disappear after students have been introduced to mathematical concepts that use these same everyday terms. Cornu (1991) named these conceptions of ideas that are formed from the colloquial meanings of words as *spontaneous conceptions*. He observes that many mathematical terms, such as 'limit' for example, have "a significance for students before any lessons begin, and that students continue to rely on these meanings even after they have been given a formal definition" (p. 154).

Several studies have documented students' conceptions of infinity. Since infinite iterative processes are essential to many undergraduate concepts, Dubinsky, Weller, Stinger and Vidakovic (2008) investigated students' conceptions of these processes. They applied APOS Theory (Dubinsky and McDonald 2001) to give explanations of how students might think about the infinity concept. APOS Theory aims to describe how understanding of a mathematical concept might take place. That is, conceptions are described as passing through mental stages of **A**ctions, **P**rocesses, and **O**bjects, and then are organised in **S**chema to make sense of the problem situation the individual deals with. According to APOS Theory, a mental action occurs when an individual carries out an iterative process (i.e., step-by-step). When this action is repeated, he/she can reflect upon it and imagine repeating it over time and can describe the steps without actually doing them; in this case the mental action has been interiorised to become a process. When an individual reflects on the process and can move from seeing it as carried out over time to seeing it as being carried out at a moment in time, he/she becomes aware of the process as a totality then he/she thinks of it as an object. And when the individual organises a series of objects, he/she has reached the schema level. Dubinsky et al. (2008) showed that students' difficulties with the infinity concept lay in their conception of the state at infinity as being an incomplete process; they see infinity not as an entity in its own right but as a repeated action like counting. The students had not seen the unending process as a totality; that is they had not constructed the mental object required for

APOS theory and they considered infinity as a process and not an object, and that led them to misconceptions.

We will be concerned with the attempts of students to compare infinite sets. Other authors, (Tsamir 2003) have noted that the criteria that students use to carry out this type of task include one to one correspondence, inclusion, and the notion that all infinite sets have the same cardinality. Furthermore, Tirosh (1999) reported that students intuitively resort to processes involving the elements of the set (e.g. keep adding a number, or keep subdividing a line) to determine if a given set is infinite. She also found that students sometimes determined a set to be finite by comparing it with another set which was considered to be infinite; for example they might argue that the given set is proper subset of an infinite set, so it has fewer elements than the infinite set, and hence it is finite. Tirosh and Tsamir (1996) have indicated that different representations of the same infinite sets often give rise to different answers when students are asked to compare them. They found that, representing the sets $\{1,2,3,\dots\}$ and $\{1,4,9,\dots\}$horizontally (i.e. placing them side by side), encouraged part-whole consideration of the difference of the number of elements, that is the notion that the whole set has more elements than its subset. While arranging the sets vertically (i.e. one set over the other), triggered the use of the 1-1 correspondence criterion, and writing the sets numerically-explicitly (e.g.1^2 appears below 1, 2^2 below 2, 3^2 below 3, etc.) encouraged justifications of 1-1 correspondence more than the vertical representation did and more often than the geometric representation which is drawing pictures of the elements of sets.

In line with the aforementioned works, we present our study to examine conceptions related to finite and infinite sets. The study showed that, students have different concept images associated with finite, countable and uncountable infinite sets and they used these in their reasoning to answer the questions we asked. We will investigate some misconceptions displayed by the students in this study.

METHODOLOGY

In this study a questionnaire was designed to investigate students' understanding concerning the infinity concept and the comparison of infinite sets. It was administered to three separate groups of students at NUI Maynooth. These three groups took three different analysis modules which were taught by the same lecturer, they all learned about infinity, cardinality and countability. The questionnaire was anonymous, participation in the study was voluntary and 35 students took part. Twelve of these students were first year students studying for a degree in Mathematics and Theoretical Physics, 13 students were out-of-field mathematics teachers who were enrolled on a postgraduate course and ten students came from a second year science class which was aimed at students who wished to study pure mathematics in their third year. For all three groups of students, the analysis module was their first exposure to rigorous mathematics. The questionnaire was given to the students during their classes and completed by them in 20 minutes. It consisted of 7 questions which explored topics concerning finite, countable and uncountable sets. The first two questions were open-ended questions to elicit students' intuitive opinions about infinity, questions 4 and 7 involved geometric sets, question 5 involved mathematical sets of numbers, and questions 3 and 6 consisted of sets from real life. Most of the questions on the questionnaire were employed in previous studies. In some cases we modified the questions, and in other cases we changed the situation that was presented to a situation that was applicable in our work. A copy of the questionnaire can be found in the appendix.

The method we used in our analysis was qualitative because we aimed to gather a deep understanding of students thinking, so the analysis we will present is based on students' responses to the questions we asked and in particular the reasons they gave for their responses. We started by analysing each student in each group separately and in order to obtain more information on the concept images students hold for finite and infinite sets the analysis was first carried out individually. We kept note of the misconceptions the students have and we met several times to discuss the results of each student and then each group. We discussed what we found many times to reveal the similarity, difference and dominance of students' conceptions in all groups. Then we classified the conceptions of each group into categories and then we gathered the categories of all groups until we reached our final results concerning the different kinds of conceptions students have regarding the infinity concept.

RESULTS

The primary analysis which we carry out here is based on the types of arguments written by the students to justify their answers to the problems on the questionnaire. Our main concern in this article was to see how students regard finite, countable and uncountable infinite sets. We tried to explore the dominant concept images which the students used most frequently. Our results showed that, there are different types of misconceptions related to finite sets and the countability of infinite sets. We break these misconceptions down into five types: misconceptions based on daily experience; misconceptions based on the misuse of properties; misconceptions concerning the use of the bijection criterion; misconceptions based on the understanding of sets; and misconceptions concerning the idea of infinity. We will explain each type as follows.

Misconceptions Based on Daily Experience

Daily experience can affect students' conceptions, especially when the terms being used in a mathematical concept have a different meaning in everyday life. Cornu (1991) called those conceptions "spontaneous conceptions". Our study showed that, many students hold those kinds of conceptions. The term 'countable' has a significant meaning in English that is 'can be physically counted'. Some students use the terms finite and countable interchangeably. The notion that the elements of a finite set can be counted and so the set is countable can be seen in the responses to Question 3:

> M is finite as all the melodies that have been composed are a set number, they are countable. (1.8)

This spontaneous conception could also lead students to think that an infinite set is a set such that its elements are uncountable or can not be counted, and we can see this clearly in this argument:

> H is infinite as since there is no way of knowing how long the earth will go on for, and how many melodies will be composed. We must assume that this number is uncountable, and therefore H is infinite. (1.5)

The practice of using the terms 'countable' and 'uncountable' for the elements of a set and not for the set itself might be a reason for other students to assume that all infinite sets are uncountable; rather than all uncountable sets are infinite. A student responded to Question 5:

> Yes, A is equivalent to N, because there is an infinite number of elements in both sets and both are uncountable. (T.11)

Students seem to associate the word 'cardinality' with the number of elements of a set and this seems to foster the belief that the cardinality of uncountable sets is unknown (since we cannot count the elements of the set). For example, in response to Question 7 one student said:

> $\overline{AB}$ and $CDEF$ are uncountable sets, so we don't know their cardinality. Therefore we cannot say which is bigger. (2.1)

We also found that some students used the phrase 'can be counted' in relation to both finite sets and countable infinite sets. Those students who hold these conceptions seem to contradict themselves, for instance one student commented according to Question 3(a) that:

> M is finite. As there is a number of tunes, we can count them. (2.1)

And according to Question 6(a) the same student commented that:

> Both sets G and P are countable as we can count them: $G := \{1,2,3,....\} = N$ and $P := \{1,2,3,.....\} = N$ (2.1)

Counting seems to be used in two different ways here.

Misconceptions Based on the Misuse of Properties

Tirosh (1991) found that many students assumed incorrectly that "all methods suitable for comparing finite sets are adequate for infinite sets as well" (p. 204). Our results showed that many students seem to have misconceptions that arise from using theorems or properties that have been shown to hold true for countable infinite sets when considering uncountable sets. We sort these misconceptions into two types. Firstly, it is true that, every subset of a countable set is countable. Some students incorrectly assume that every subset of an uncountable set is uncountable. Students' justifications in Questions 5 and 7 were evidence of these conceptions. The main argument used to justify their answers was: A is not equivalent to N because A is a subset of R so it is an uncountable infinite set. For instance, regarding Question 5 this student claimed that:

> No, N countable, the set A is uncountable as it's a subset of uncountable set R (1.7).

Another student answering Question 7 argued that:

> Yes, both are subsets of R and equivalent to R so therefore equal (T.5).

We can see that the reason used to justify the answers above is incorrect. N is also a subset of R, but those students in this situation did not think of that, and they seem only to remember that N is a subset of R if we ask them the question directly. They seem to have a conception that R contains only intervals or line segments.

Secondly, it is true that all infinite countable sets are numerically equivalent to each other, but in fact we found that many students assume also that all uncountable infinite sets are equivalent. We notice these conceptions most obviously in the arguments with respect to Question 7. The most used argument here was that all the sets are uncountable; therefore they have the same cardinality. For example:

> Yes, The line $\overline{AB}$ contains an uncountably infinite number of points and the square's cardinality is uncountably infinite, and as result, numerically equivalent (1.4).

Another student commented that:

> Yes, Unsure, on how to explain. Need to look (to) my notes!
>
> Idea $\rightarrow$ both AB and CDEF $\rightarrow$ Uncountable (T.4).

Misconceptions Based on the Misuse of the Bijection Criterion

The students have learned in the course that the bijection criterion is the main criterion that should be used to compare infinite sets. But our findings showed that many students did not use this criterion to determine their responses in the four problems that dealt with comparison of infinite sets. Of the students who used this criterion, none of them used it in all comparisons of infinite sets. The most use was found in their answers to Question 6, maybe because the sets in that question are countable and the elements can be seen clearly and therefore the bijection between these sets is more obvious. We also found that students who used the bijection criterion used it correctly in some problems and incorrectly in other problems. To illustrate an incorrect use of bijection, a student when answering Question 5 thought that since both sets are infinite then there must be a bijection between them; hence he or she claimed that:

> Yes, they are both infinite sets. A bijection will map $N \rightarrow A$. (1.1)

Even when students invoked the bijection criterion they rarely wrote down a specific map. Some students made diagrams for their answers to Question 6 to illustrate the bijection, and others used other informal terms rather than bijection (i.e. you can match/pair up the elements of the comparison sets), like:

> Yes, for each glass, there is a plate. They can be paired up. (1.8)

Similarly, when trying to use the bijection criterion to show sets were not equivalent, students did not put forward an argument as to why a bijection could not exist but just stated the fact. For example in answer to Question 7 one student claimed:

> No, each point in the square cannot map directly 1-1 onto the line (T.8).

Misconceptions Based on the Understanding of Sets

We found that some students were unable to write the sets F and K which are given in Question 6 (b) and Question 6 (c) in a mathematical way and that led them to use incorrect justifications for their claims. The elements of sets F and K are difficult to write down and require the use of some notation. However, many students tried to write them as subsets of N. For instance (note E below denotes the even numbers):

> No, As F = {2,4,6,8,…} = Even no.'s, G = N, N is (supposedly!) bigger than E (1.2).

A different student expressed F incorrectly in Question 6(b), however he or she concluded that a bijection could exist between G and F and argued that:

> Yes, G = {1,2,3,…}, F = {2,4,6,…}. # G = # N. There is a bijection from G to F (1.10).

These students seem to grasp the essential ideas concerning countable sets but their inability to translate the description of the sets into mathematical notation hinders their efforts.

In line with this, many students interpreted the set A in Question 5 incorrectly; they thought of (1.25, 3.79) as {1.25, 3.79} and therefore for them A seems to contain only 2 numbers and thus is not equivalent to N. Some of them argued that:

> No, the set A only has 2 numbers; therefore all the natural numbers cannot biject onto it. (T.8)

Perceptions of the Concept of Infinity

For a long time the concept of infinity has been a cause of debate for many philosophers and mathematicians, so it is no wonder that students also have difficulties with understanding the notion of infinity. Many of the difficulties with infinity seem to arise when students think of infinity as a number, albeit one that is not reachable. In the questionnaire, students were asked to explain the idea of infinity and many of them thought of it as an unreachable number, for example:

> Representation of a number that [can] never be reached as there will always be more numbers bigger (1.7).

Another student thought of infinity as the largest number, but one that does not exist:

> Infinity is the highest possible number, but by the same token, it does not exist. It means there is no such largest number (1.2).
>
> It is endlessness, not number itself but the largest natural number can tend towards (1.9).

Another student mentioned that:

> I would say it as the property of numbers that they do not have an end or largest value, and we call infinity the theoretical largest number (2.2).

Another one commented that:

> I would say it is a concept. Infinity may not really exist in concrete terms. It is the idea that if you counted forever, the "last number" would be infinity. Or more realistically, you would say the last number is infinity as you would never be able to actually reach it (2.9).

The view that infinity is a number might be responsible for students thinking that all infinite sets are equivalent as all have an infinite number of elements. To show that, a student wrote regarding Question 4 (b):

> No, Infinity + Infinity = Infinity and Infinity = Infinity (2.9).

Another student commented that:

> No, Since set AB has an infinite amount of points, CD can not contain more elements than an infinite amount (1.9).

A different student supposed that there is only one kind of infinity and argued that:

> Infinity cannot be greater that infinity, they are of equal cardinality. You could also probably get a bijection from one set to the other (1.5).

CONCLUSIONS

In this study, we aimed to examine students' conceptions about the infinity concept, and their methods of comparing finite sets, countable sets and uncountable infinite sets. From our findings we see that students hold many different types of conceptions related to infinite sets. We found that everyday language is one reason for students' misunderstandings. Many of our students supposed that a countable set is a set for which its elements could be (physically) counted (from the real life use of the word), and by this meaning a countable set is a finite set. Similarly for these students an infinite set means an uncountable set. Moreover in accordance with Tirosh's (1991) study in which many students used methods that are applicable for finite sets for infinite sets also, in our study we found that many students supposed properties that are true for countable sets to be true for uncountable sets also. That is, they think that any subset of an uncountable set is also uncountable, and all uncountable sets are equivalent. Furthermore the study indicated that although some students understand that the bijection criterion is the main method to determine the comparison of infinite sets, they still thought that all infinite sets are equivalent and were unable to apply the bijection criterion correctly. In addition, we found that some students were not able to express real life sets mathematically in a correct way; for example they tried to express F the set of forks as $F = \{2,4,6,\ldots\}$. We can see that they tried to use 2 here to indicate 2 elements (forks), but writing F in this way caused some of them to think that F has fewer elements than N. The problems our students had with representations of sets echo the findings of Tirosh and Tsamir (1996). What is more, students understanding of infinity as a number that is unreachable might be a reason to think that all infinities are the same and all sets with infinite cardinalities are equivalent.

Most of the difficulties that the students in this study encountered with the concept of infinity have been reported in previous studies (see Tirosh (1991) for an overview). However, we have not been able to find other studies that mention student's difficulties with the spontaneous conception related to the word 'countable'.

In conclusion, we feel that a mathematics teacher or lecturer must be aware of the misconceptions related to any specific concept they teach, and it is beneficial for them to know most, if not all, difficulties students might encounter with it. We hope our results contribute to give some view of these difficulties with the concept of infinite sets.

REFERENCES

Alcock L. and Simpson A. (2009): *Ideas From Mathematics Education, an Introduction for Mathematicians*, The Higher Education Academy: MSOR network, Birmingham.

Cornu B. (1991): Limits, In D. Tall (Ed.) *Advanced Mathematical Thinking*, Dordreht, Kluwer, pp. 133-166.

Dubinsky E., Weller K., Stinger C. and Vidakovic D. (2008): Infinite iterative processes: The Tennis Ball Problem, *European Journal of Pure and Mathematics*, vol.1, No.1, pp. 99-121.

Dubinsky E. and McDonald M. (2001): 'APOS: A constructive theory of learning in undergraduate mathematics education research' in Derek Holton, et al. (eds.), The teaching and Learning of Mathematics at University Level: An ICMI Study, Kluwer Academic Publishers, Dordrecht, pp. 273-280.

Edward B. S. and Ward M. B. (2004): Surprises from Mathematics Education Research: Students (Mis)use of Mathematical Definitions, *The American Mathematical Monthly*, Vol. 111, no. 5, pp. 411-424.

Przenioslo M. (2004): Images of the limit of function formed in the course of mathematical studies at the university, *Educational Studies in Mathematics* 55, pp. 103- 132.

Shulman L. S. (1986): Those Who Understand: Knowledge Growth in Teaching, *Educational Researcher*, Vol. 15, No. 2, pp 4-14

Tall D. and Vinner S. (1981): Concept Image and Concept Definition in Mathematics with particular reference to Limits and Continuity, *Educational Studies in Mathematics*, 12, pp. 151-169.

Tirosh D. (1991): The role of students' intuitions of infinity in teaching the cantorian theory, In D. Tall (Ed.) *Advanced Mathematical Thinking*, Dordreht, Kluwer, pp. 199- 214

Tirosh D. (1999): Finite and infinite sets: definitions and intuitions, *International Journal of Mathematical Education in Science and Technology*, 30, 3: pp. 341-349.

Tirosh D. and Tsamir P. (1996): The role of representations in students' intuitive thinking about infinity, *International Journal of Mathematical Education in Science and Technology* 27, pp. 33-40.

Tsamir P. (2003): Primary Intuitions of Instruction: The Case of Actual Infinity, in Selden A. et al. (Eds.) *Research in Collegiate Mathematics Education. V*, CBMS Issues in Mathematics Education, vol. 12, pp. 79-96.

Vinner S. (1991): The Role of Definitions in The Teaching and Learning of Mathematics, in D. Tall (ed.) *Advanced Mathematical Thinking*, Dordreht, Kluwer, pp. 65-81.

APPENDIX

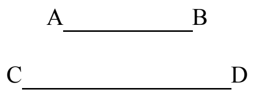
The Questionnaire problems:

1. How would you explain the idea of infinity to a friend of yours?
2. Why is N infinite?
3. Let M be the set of all melodies (tunes) that have been composed until now. Let H be the set of all melodies that could be composed.

   (a) Is M finite or infinite? Explain!

(b) Is H finite or infinite? Explain!

4. Look at the line segments AB and CD below:



Answer the following by putting a circle around the correct reply:

(a) Is the set of points on AB finite or infinite?

Finite Infinite Don't know

Explain your answer!

(b) Is the cardinality of the set CD greater than the cardinality of the set AB?

Yes No Don't know

Explain your answer!

5. Compare the set A = (1.25, 3.79) with N the set of natural numbers. Are they numerically equivalent?

   Yes No Don't know

   Explain your answer!

6. An infinite dinner table is set in a restaurant. Each person is served a glass, a plate, three knives and two forks. Let F be the set of forks on the table, K be the set of knives on the table, G be the set of glasses on the table, and P the set of plates on the table.

   (a) Is the cardinality of G equal to the cardinality of P?

   Yes No Don't know

   Explain your answer!

   (b) Is the cardinality of F equal to the cardinality of G?

   Yes No Don't know

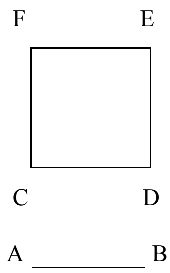
   Explain your answer!

   (c) Is the cardinality of F equal to the cardinality of K?

Yes No Don't know

Explain your answer!

7. Look at the diagram below.



Is the line segment AB numerically equivalent to the square CDEF?

Yes No Don't know

How did you come to this conclusion?

PERSPECTIVES ON MATHEMATICS CONTENT COURSES FOR ELEMENTARY TEACHERS

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This cross case analysis synthesizes results from two qualitative studies on mathematics content courses for prospective elementary teachers: one from the U.S. interviewed students, one from Canada interviewed instructors. Results were examined for convergent themes. Salient commonalities were found. Two discussed here will be presented from both the student and instructor perspective: the role of affect in student learning and the role of connections to the elementary classroom.

BACKGROUND

Pervasive concerns about the adequacy of the mathematical preparation of elementary teachers (Ma, 1999; Ball, Hill, & Bass, 2005; Rowland, Huckstep, & Thwaites, 2005) have prompted many institutions of higher education to require specialized mathematics content courses for prospective elementary teachers. These courses, referred to here as *Math for Teachers* (MFT) courses, aim to provide understanding of elementary mathematics concepts in order to develop prospective teachers' confidence and flexibility in teaching mathematics (Kilpatrick, Swafford, & Findell, 2001; Williams, 2008). Such courses were endorsed in the U.S. report of the National Mathematics Advisory Panel (Greenberg & Walsh, 2008), which argued for increasing the number of courses typically required and reiterating that these courses should be taught by mathematicians. As a result, MFT courses are often taught in mathematics departments by mathematics faculty. The majority of these MFT courses have a similar core content including number systems and algorithms, patterns and introductory number theory, measurement and geometry, probability and data analysis. There is, however, a growing concern that many MFT courses are not meeting the need of elementary teachers to develop a deep fundamental understanding of elementary mathematics. Ball, Hill, and Bass (2005) propose that more is needed and that there is a *specialized content knowledge* (SCK) needed to teach mathematics. Examples of this SCK include teachers' abilities to: (a) analyze and interpret students' mathematical thinking and ideas, (b) use multiple representations of mathematical concepts, and (c) define terms in mathematically correct and accessible ways (Ball, Thames, & Phelps, 2008; Hill, 2010; Thames & Ball, 2010).

In an effort to begin to understand issues surrounding these courses, we conducted a cross-case analysis of two existing qualitative studies. One study highlighted the instructor perspectives on an MFT course; the other study illuminated student perspectives. The *instructor-focused* study examined instructors' perceptions on a MFT course at several institutions of higher education in south-western Canada (see Oesterle & Liljedahl, 2009). The purpose of the study was to provide insights into the instructors' instructional approaches in the course and how their beliefs impacted pedagogical decisions. Data from the semi-structured, individual interviews (~1 hour in duration) were analyzed for emergent themes using constant comparative analysis. Several themes were evident, including: instructor identity, tensions, resources, classroom methods, relationships, student knowledge, student affect, orientation to mathematics, and orientation to teaching. The *student-focused* study explored the perspectives of 12 elementary education majors (i.e., prospective elementary teachers) who had completed MFT courses at a university in the south eastern United States (see Hart & Swars, 2009).The impetus for the study was grounded in concerns about the poor success rates of elementary education majors enrolled in these courses. Data collection involved semi-structured, individual interviews (~1 hour in duration). Constant comparative analysis was applied to the data, which revealed three major themes: (a) domains of mismatch (b) affective reactions, and (c) classroom practices. The domains of mismatch theme had three sub-themes: mismatch with elementary classroom, mismatch in programmatic emphasis, and mismatch in mathematics content.

The research design for both the individual studies was a within-case analysis that examined the experiences of each group of participants: instructors or students in MFT courses. A full description of data collection and analyses is provided in the original studies (Hart & Swars, 2009, Oesterle & Liljedahl, 2009). The present study reports the findings of a cross-case analysis of the results from the two studies.

OVERVIEW OF RELEVANT RESEARCH

These two extant studies were framed by prior research in three domains: prospective elementary teachers' affect and beliefs; prospective elementary teachers' experiences in post-secondary mathematics content courses; and perspectives on and characteristics of effective post-secondary mathematics instruction.

Research on teacher affect has a long history closely linked to teacher beliefs (McLeod & McLeod, 2002). According to Goldin (2000), the affective system is just one of several internal systems of representation that has several distinguishing sub-domains, including emotions, attitudes, beliefs, values, ethics, and morals. Ball (1988) connects affect specifically to teacher learning when she stated, "The experiences they [teachers] have had with mathematics shape their *feelings* about the subject and about themselves in relation to it" (Ball, 1988, p. 42). Much less is known about university instructors' beliefs about mathematics (e.g., Ernest, 1989).

There is also limited research on prospective elementary teachers' experiences in mathematics content courses, including MFT courses. Most studies were largely conducted in the context of mathematics pedagogy courses. A few studies were found. Royster, Harris, and Schoeps (1999) found that elementary education majors showed the greatest positive changes in dispositions toward mathematics in comparison to other majors in a reform-based mathematics content course. Lubinski & Otto (2004) and Philipp et al. (2007) focused specifically on mathematics content courses for *prospective elementary teachers* using a reform perspective. Both studies found that the prospective teachers developed more reform-oriented beliefs.

Finally, a few studies examined the student perspective on effective teaching of mathematics at the post-secondary level (Powell-Mikle, 2003; Schulze & Tomal, 2006; Weinstein, 2004). Characteristics of effective teachers included less time lecturing and the development of a caring learning environment. Speer and Wagner (2009) and Wagner, Speer, and Rossa (2007) examined the knowledge needed by mathematics instructors to implement inquiry-oriented curricula in post-secondary mathematics courses. Both studies asserted that there are forms of knowledge beyond content knowledge that are needed in reform-oriented teaching such as awareness of typical ways students think (both correctly and incorrectly) about mathematics.

THEORETICAL PERSPECTIVE AND RESEARCH QUESTION

This cross-case analysis, as well as the two original studies, is grounded in phenomenological interpretation (Burch, 1990). Students and instructors individually reported their perspectives on the MFT courses. Through the interviews, participants reflected on and retrospectively identified the significant or memorable events from their MFT experiences. Through this process, they recovered and verbally reenacted the meaningful components of their lived experiences.

For this cross-case study, we were interested in determining what intersections might exist between the perspectives of instructors of a MFT course and the perspectives of students in the MFT courses. More specifically, we asked this research question: *What convergent themes exist in instructor perspectives and student perspectives on mathematics content courses for prospective elementary teachers?*

METHODS

Participants and Setting (Source Studies)

The student-focused study involved twelve students (11 females and 1 male) from one urban university in the southeastern U.S. The students had completed three or four MFT courses. They were randomly selected from 4 cohorts of students in the elementary teacher preparation program with a combined size of 99 students, thus representing approximately 12% of the total population. Collectively they had taken 42 sections of MFT courses. At the time of the original study, all of the students were in the last semester of the program and completing student teaching.

Although the instructor-focused study gathered data from ten instructors, for the purpose of this paper examples will be drawn from the transcripts of only two: Harriet and Bob (pseudonyms). The divergent perspectives they offer on teaching the MFT course provide a sufficient basis for illustrating two common themes identified in this analysis. Harriet and Bob are both experienced instructors, having taught in mathematics departments for 22 and

13 years respectively. Harriet is relatively new to teaching the MFT course but had taught the course six times over three years, while Bob taught it nine times over nine years. Both have Master's degrees in mathematics but neither took mathematics education courses nor had formal teacher training. Harriet was initiated into teaching the MFT course by a colleague with a Master's degree in mathematics education who taught MFT courses for many years. Bob's first forays into teaching the course were guided by his institution's curriculum, the text, and informal discussions with colleagues.

Data Collection and Analysis

Themes from the two source cases were developed independently by researchers in the original locations prior to conducting the cross-case analysis. The results from the two original studies were independently reviewed and analyzed by the three researchers on this paper. They subsequently engaged in discussion to develop shared meanings related to each of the themes and then examined them independently for convergence. Agreed upon themes were identified and they then created a matrix with the themes from the two studies. For this paper, we will discuss two commonalities across the themes: *connections to the elementary classroom* and *student affect*, as shown in Figure 1.

Student-Focused Study Themes	Instructor-Focused Study Themes
affective reactions	*classroom environment*
classroom practices	*instructor identity*
mismatch with elementary classroom	***student affect***
mismatch of mathematics content	*orientation to mathematics*
mismatch of programmatic emphasis	***orientation to teaching***
	resources
	student knowledge

Figure 1. Themes and commonalities of the two extant studies

CROSS-CASE ANALYSIS

The Students' Perspectives on Connections to the Elementary Classroom

After experiencing other courses in their teacher preparation program, the students were aware of disconnects between their experiences in the mathematics content courses and other experiences in the program. The students frequently described an inability to position the mathematics content coursework within their growth as educators, which led to perceptions of lack of usefulness or relevance of the courses as evidenced by this statement: "I mean a lot of us were always questioning, you know, why we have to take these math courses . . . It's not even necessary." Similarly, another student stated, "The reason why we were taking those courses was never brought to our attention. We had no clue why we were taking those classes, no clue. . . it seemed very unnecessary." A third student commented:

> It [mathematics courses] had no connection to elementary schools. Anybody could take those courses. I don't think we ever talked about kids. I seriously don't think we ever talked about teaching or students or anything like that. I just don't remember ever that connection.

The following statements further support this sentiment. "They're [mathematics instructors] blind to what we are actually doing with our lives;" "[Elementary] Students were never even brought up… I mean, students or when you get your own classroom were never brought up;" and "We're thinking we're learning something about how to be teachers. But, in reality we're learning how to get through their math courses."

Students pointed out that a few instructors attempted to model pedagogical methods that might be used in an elementary classroom or they used materials that elementary students might use. For example:

> [The Number & Operation course instructor] was big on you know . . . we would do that in class, it wasn't like, take this home and do this but we actually [did activities in class] and we did a lot of group work and that was really good too.

Another student provided a specific example:

> We had to do a project in geometry where we had to go to like a specific county and get a book and find a geometry lesson, but that was just a project and we presented it to the class . . . so just different things like that, just to make it real to us.

Ideas for changing the courses were proffered, "Get some teachers who were actually qualified in elementary . . . They actually know what children are going through and that would help;" or "They [mathematics instructors] could talk to elementary teachers;" and also, "Maybe have us students say, hey, this is what is going on in my [elementary] classroom."

The Instructors' Perspectives on Connections to the Elementary Classroom

Harriet's descriptions of her goals and strategies for teaching the MFT course are permeated with comments related to mathematics-for-teaching knowledge (Ball & Bass, 2003) and how her students' learning relates to their future as teachers. When asked if there is anything that she teaches MFT students about fractions that she would not teach other students, she states:

> The fact that there are different models, there are different ways of picturing what's going on, and that they are appropriate for . . . what may work well for some situation, or for some [elementary school] student, may not work for some other one.

She also emphasizes connections between mathematical ideas both within and across grade levels. She explains:

> At all times I connect it [the course content], as far as I can, to what goes on at different levels. What you might do with a grade 1 class, how that connects to what they're going to see in, you know grade 4 or 5 or something like that, how that connects to what they might do in high school and how that connects to what I'm doing in Calculus. Because they've got to see how it's connected, and how we build bigger and bigger . . . understandings of sets of numbers, or calculations.

Harriet does not just pay lip-service to these ideas. She describes assignments that allow her students to build their mathematics-for-teaching knowledge, such as analysis of pupil errors and discussion of alternative solutions.

In contrast, Bob makes very little reference to mathematics-for-teaching knowledge. His emphasis is instead on developing a strong understanding of fundamental mathematics and communication skills. Varieties of algorithms and models form part of his course content, but he does not specifically address how they can be applied differently at various grade levels.

Bob needed to be pressed by the interviewer to consider what aspects of the course content might be particularly relevant to prospective teachers as opposed to general learners of mathematics. Initially his comments revolve around his teaching methods, such as the use of group work and manipulatives, but he makes no reference to any special mathematics knowledge for teaching. Eventually he describes challenging his students to think about the kinds of questions that they will encounter as teachers:

> . . . what kinds of questions will you encounter? And why is it important that you be able to communicate your ideas effectively . . . why should you understand this material to the most . . . fundamental and basic level, and understand all of the structure?

He adds:

> When you get some of these obtuse questions, that are seemingly . . .obtuse, you have to be able to appreciate it and be able to differentiate whether that's something that can lead you into a teachable moment.

His response appears to be a justification for his goals of developing strong mathematics content knowledge and communication skills. For Bob, mastery of the subject content along with general pedagogical skills, seem to be sufficient for the teaching of mathematics—a traditional and prevalent point of view (Hill et al., 2007).

The Students' Perspectives on Student Affect

A second theme across the interviews was students' affective reactions to the coursework experiences. Many statements described negative emotions, for example, they used words such as "emotional wreck", "so stressed", "very belittling", "discouraged", "terrified", "struggling", and "frustrating". One student asserted, "I felt like I was just hanging on. Just trying to dig myself out of a hole, and I kept falling down."

The students also portrayed the courses as having deleterious influences on their mathematics teaching efficacy beliefs and self-efficacy beliefs. Most often, feelings of negative affect were described along side of courses using *traditional* approaches to instruction. The students frequently mentioned “lecture,” “note-taking,”, and “power point presentations,” and asserted the “classes were not hands-on.” In describing how the courses impacted teaching efficacy beliefs, one student stated:

> I felt less confident [about teaching mathematics] when I walked out of those classes because it’s just so much and it just seemed so unnecessary . . . It was just very discouraging.

In response to a question on how the courses prepared her to teach elementary mathematics, a student stated the courses made her, “Feel less prepared. Feeling more scared, definitely.” Another student attributed this negative impact on her teaching efficacy to the attitude of the instructor of the course, “The attitude was if you don’t get this [math content], you won’t be able to teach it, basically.”

Students also commented on how the experiences in the courses influenced their self-efficacy beliefs in mathematics, as seen in this student’s statement:

> [I felt] terrified, struggling, especially in geometry. It was just, it was very frustrating because I didn’t get it. I didn’t understand why we’re doing what we were doing, how we were coming out with the answer, and especially if I didn’t get the answer right.

Another student stated:

> Like geometry… I came out of there in tears. I felt very disappointed. I felt stupid. I felt alone. And, I know that I am an intelligent person, or I have the potential to learn something. If I don’t know it, I’m willing to give up my time and my efforts. But, I felt like my efforts didn’t matter.

Similarly, another student said:

> It (mathematics courses) made me feel so low in math. Even though I knew those math courses, I would never be teaching that stuff . . . It totally lowered my self-esteem in mathematics.

The students also described instructors’ dispositions (i.e., caring, helpfulness, availability), as affecting their experiences. On the negative side, one student commented,

> They [mathematics instructors] don’t care. It just seemed like it, like, they’re there. They do what they do, and then they leave. That’s just the way it was […] you show up and then leave and not investing.

Several students described some instructors as uncaring. One said: “It was very hard. I didn’t feel like the teacher cared whether we did well or not … It was very hard and trying to tell her to slow down, she wouldn’t have it.” Still another student stated, “I would go ask him questions and stuff like that, and he just would not help.” The perception that the instructor did not care clearly affected the students' experience in the course, as expressed in the following comment.

> I've never made a C, so I was very upset, and the teacher had no compassion and was willing to help but in a very belittling way. 'I don't see how you don't get this.' Those kind of things. And the attitude that I felt was that, he could care less if you pass or fail, you either sink or you swim. That didn't help . . .

In a few cases students spoke about instructors who exhibited positive dispositions, including availability, helpfulness, and caring. One student stated, "If I had any questions, I could go to the instructor … she was available for me to go to them to talk to." Another student described an instructor as, "Very compassionate and very willing to help. [The instructor] was very willing to change her style of teaching to reach your level."

As a whole, however, this group of prospective elementary teachers walked away from much of the MFT experience with negative feelings. Although there is substantial evidence that this population is prone to high degrees of math avoidance and math anxiety, it appears their experience in the MFT courses did little to alter that.

The Instructors' Perspectives on Student Affect

Both Bob and Harriet describe their students as suffering from mathematics anxiety and lacking confidence in their ability to do mathematics. However, there are considerable differences in their perspectives and pedagogical approaches to these negative affective states.

Harriet observes that her students "are very anxious around problem solving. They are just terrified, most of them, of a problem they haven't seen before." Her efforts to address this seem to be centered on changing their ideas of what the enterprise of mathematics is all about. She tries to convince them to enjoy mathematics and she shifts the focus away from getting the right answer toward less threatening goals:

> We're supposed to have fun with this . . . you may never have seen it; you might not get all the way through it. But what I'm looking for is how far did you get, and how well can you explain what it is that you got.

By the end of the course she hopes her students have grown in confidence and also

> . . . they have more of a sense of play . . . I think they're more flexible. They think they're more flexible. They're not as scared if . . . that someone will ask them a question that they can't answer.

Bob describes his students as believing that mathematics is arbitrary and incomprehensible: "So many things seem magical to them". He affirms that "it's not your standard sort of math group, it's one that has encountered some challenges along the way, and it hasn't always left them with a positive impression of mathematics." In his view, their confusion and anxiety is closely linked to their skills:

> In many cases, some of the very elementary arithmetic operations are in fact, confused in their minds and so when they hit upon things, in particular when you hit rational numbers, as an example, that's one place where students have a great deal of anxiety and they would demonstrate poor understanding of ideas.

More than once he describes the MFT course as a second start for these students. He attempts to reshape their beliefs and attitudes by providing them with opportunities to see the logical structure of mathematics and deepen their understanding. For Bob, the course "focuses on a very sound fundamental ability to appreciate it [mathematics], in a theoretical way, why things work, as opposed to technical aspects of how do you do mathematics." However, although he believes that improved skills will lead to increased appreciation and confidence, he confesses that the realities of the course conspire against this occurring. Early in the interview he expresses a wish that his MFT students develop a love of math, but when asked about whether this goal is accomplished, he admits "in terms of the other goal, for love of math? Unfortunately, the course is so packed, that in some ways, I think they do get a little bit beaten by the end, and they're just tired." This statement illustrates Bob's realization that the volume of content covered in a limited time is at odds with his affective goals.

DISCUSSION

Although the two groups of participants in these studies were in different settings, they provide two distinct viewpoints on a similar experience: MFT courses. When juxtaposed the data reveal salient commonalities. They provide important insights into issues and concerns around creating experiences in MFT courses that best support elementary prospective teachers' learning of mathematics; they also enrich our understandings of the realities of MFT classrooms, revealing both the affordances and the constraints.

The student voices emphatically call for the need for connecting the mathematics to the elementary classroom. Without this connection, the students were not able to find relevance in their learning. This need is also recognised in the literature (Philip, 2007). Ball and Bass (2003) specifically advocate for this link:

Practice in solving the mathematical problems they will face in their work would help teachers learn to use mathematics in the ways they will do so in practice, and is likely also to strengthen and deepen their understanding of the ideas (p. 13).

The instructor-focused study reveals how differently instructors may perceive the need for incorporating these connections. Harriet is very aware that these links help to motivate her students, helping them to see why a deeper understanding of mathematics is required of them in an MFT course as compared to their previous mathematics courses. Bob, however, was unable to identify aspects of the course content that might be particularly relevant to prospective teachers as opposed to general learners of mathematics and he does not see making connections as an explicit part of his course. His orientation is towards the mathematics. One reason for this may be that as a mathematician his lack of experience in elementary classrooms limits his ability to do so. However, Harriet also lacks such experience. Another possibility is that Bob takes such connections for granted, assuming students will pick up strategies for *how to teach* in a methods course. His inability to identify content in his course that would be particularly relevant to future teachers of mathematics as opposed to general mathematics students reflects a lack of awareness of specialized mathematics knowledge-for-teaching. For Bob, subject content knowledge and pedagogical knowledge are distinct. He sees his role as supporting the development of the former.

With regard to the theme of affect, the student-focused study reports an alarming number of negative comments, indicating the coursework may have actually increased students' anxiety and decreased self-efficacy. Both instructors were acutely aware of the impact of affect and described their students as coming into the course with high mathematics anxiety and lack of confidence. However, their perceptions about the cause of the anxiety and strategies for addressing it were quite different. For Bob, the source is students' lack of fundamental skills. As a result, his solution is to help them see the logical structures of mathematics and develop these skills, though he acknowledges that the sheer volume of the material he must cover, in fact, adds to his students' stress. For Harriet the source is negative past experiences and a perception of mathematics as rigid. Her efforts focus on moving students away from the 'one right answer' view of mathematics, helping them develop more flexibility in approaching mathematical problems and to just have 'fun.'

From the students we hear that traditional instructional methods of lecture, power point presentations, and drill and practice tended to elevate anxiety and decrease efficacy, while reform approaches such as small group work, hands-on learning, and opportunities to share and discuss were less stressful and increased efficacy. They also shared that the instructor having a caring manner, an approachable demeanour, and a perceived willingness to help supported their learning, which is consistent with previous studies (Powell-Mikle, 2003; Royster, Harris, & Schoeps, 1999). Regardless of their perceptions of the source of their students' anxieties, knowledge of research on student affect and beliefs could help inform instructor choices with respect to how to address student concerns.

The voices of the students in the student-focused study lend support to concerns that mathematicians in mathematics departments may be unprepared to take on the task of preparing elementary teachers. The lack of connections of content with the elementary classroom and traditional teaching approaches seem to contribute to frustration and anxiety as well as decreased self-efficacy. However, the instructor-focused study shows that though lack of explicit connections to elementary learning may occur, this need not be so. The differences between Harriet and Bob in this regard may have been the result of the mentorship Harriet received, suggesting a potential means for supporting the mathematicians who teach these courses.

Another side of this issue is that mathematicians, at their best, have much to offer future teachers, even at the elementary school level (Hodgson, 2001; Williams, 2008). Jonker (in review) describes mathematicians in mathematics departments as "stewards of their discipline", "passionate about mathematics", and "eager to share their excitement with students and concerned about the place of mathematics in the world." The challenge is to create opportunities for conversations between mathematics educators and mathematicians so that students in MFT courses are better prepared to teach mathematics to elementary children.

REFERENCES

Ball, D. L. (1988). Unlearning to teach mathematics. *For the Learning of Mathematics*, *8*(1), 40-48.

Ball, D. L., & Bass, H. (2003). 'Toward a practice-based theory of mathematical knowledge for teaching.' *Proceedings of the Annual Meeting of the Canadian Mathematics Education Study Group,* 2-14. Edmonton, Canada.

Ball, D. L., Hill, H. C., & Bass, H. (2005). 'Knowing mathematics for teaching.' *American Educator* (Fall): 14-46.

Ball, D. L., Thames, M. H., & Phelps, G. (2008). Content Knowledge for Teaching: What Makes It Special? *Journal of Teacher Education, 59*(5), 389-407.

Burch, R. (1990). Phenomenology, Lived Experience: Taking a Measure of the Topic. *Phenomology + Pedagogy (8),* 130-160.

Ernest, P. (1989). The knowledge, beliefs and attitudes of the mathematics teacher: A model. *Journal of Education for Teaching, 15*(1), 13-33.Hart, L. C. & S. Swars (2009). 'The lived experiences of elementary prospective teachers in mathematics content coursework.' *Journal of Teacher Development* 13(2): 159-172.

Greenberg, J., & Walsh, K. (2008). *No common denominator: The preparation of elementary teachers in mathematics by America's education schools*. Washington, DC: National Council on Teacher Quality

Hart, L. C. & Swars, S. L. (2009). The lived experience of preservice elementary teachers in mathematics content courses. *International Journal of Teacher Development, 13*(2), 159-172.

Hill, H. C. (2010). The Nature and Predictors of Elementary Teachers' Mathematical Knowledge for Teaching. *Journal for Research in Mathematics Education*, *41*(5), 513-545.

Hill, H. C., Sleep, L., Lewis, J.M., & Ball, D. L. (2007). 'Assessing teachers' mathematical knowledge: What knowledge matters and what evidence counts?' In F. K. Lester (ed.), *Second handbook of research on mathematics teaching and learning*, 111-155. Charlotte, NC: National Council of Teachers of Mathematics.

Hodgson, B. (2001). 'The mathematical education of school teachers: Role and responsibilities of university mathematicians.' In D. Holton (ed.), *The teaching and learning of mathematics at university level: An ICMI study,* 501-518. New York: Kluwer Academic Publishers.

Jonker, L. (in review). ‘Prospective elementary school teachers in a mathematics department.’ *Journal of Mathematics Teacher Education*.

Kilpatrick, J., Swafford, J. & Findell, B. (2001). *Adding + it up: Helping children learn mathematics.* Washington, DC: National Academy Press.

Lubinski, C., & Otto, A. (2004). Preparing K-8 preservice teachers in a content course for standards-based mathematics pedagogy. *School Science and Mathematics*, *104*, 336.

Ma, L. (1999). *Knowing and teaching elementary mathematics: Teachers’ understanding of fundamental mathematics in china and the United States*. Mahwah, NJ: Lawrence Erlbaum.

Oesterle, S. & Liljedahl, P. (2009). Who teaches math for teachers? *Proceedings of the 31st International Conference for Psychology of Mathematics Education – North American Chapter*. Atlanta, USA.

Philipp, R. A. (2007). ‘Mathematics teachers’ beliefs and affect.’ In F. K. Lester (ed.), *Second handbook of research on mathematics teaching and learning,* 257-315. United States: Information Age Publishing.

Rowland, T., Huckstep, P., & Thwaites, A. (2005). ‘Elementary teachers’ mathematics subject knowledge: The knowledge quartet and the case of Naomi.’ *Journal of Mathematics Teacher Education* 8: 255-281.

Royster, D. C., Harris, M. K., & Schoeps, N. (1999). ‘Dispositions of college mathematics students.’ *International Journal of Mathematical Education in Science and Technology* 30(3): 317-333.

Schulze, E. & Tomal, A. (2006). ‘The chilly classroom: Beyond gender.’ *College Teaching* 54(3): 263-269.

Swars, S., Hart, L., Smith, S., Smith, M., & Tolar, T. (2007). A Longitudinal Study of Elementary Pre-Service Teachers' Mathematics Beliefs and Content Knowledge. *School Science and Mathematics*, *107*(8), 325-335.

Thames, M. H., & Ball, D. L. (2010) What Math Knowledge Does Teaching Require? *Teaching Children Mathematics*, *17*(4), 220-229.

Weinstein, G. L. (2004). ‘Their side of the story: Remedial college algebra students.’ *Mathematics and Computer Education* 38(2): 230-240.

Williams, P. (2008). *Review of Mathematics Teaching in Early Years Settings and Primary Schools.* Retrieved August 25, 2008, from http://www.teachernet.gov.uk/docbank/index.cfm?id=12685

MATHEMATICAL TRANSFER: STUDIES OF STUDENTS' ABILITIES IN TRANSFERRING MATHEMATICAL KNOWLEDGE TO CHEMISTRY

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It is recognised that there is a mathematics problem in chemistry; for example, many undergraduate students appear to be unable to utilize basic calculus knowledge in a chemistry context—calculus knowledge which would have been taught to these students in a mathematics context. However, there also appears to be a dearth of literature addressing the causation of this problem. Consequently, this has spurred the following two questions: 1) Can students transfer mathematical knowledge to chemistry? and 2) What are the possible factors associated with students being able to successfully transfer mathematical knowledge to a chemistry context? These questions were investigated, in terms of basic mathematical knowledge which chemistry students need for chemical kinetics and thermodynamics, using the traditional view of the transfer of learning. Two studies (Study 1 and Study 2) have been carried out amongst two samples of undergraduate students attending Dublin City University. Findings suggest that the mathematical difficulties which students encounter in a chemistry context may not be because of an inability to transfer the knowledge, but may instead be due to insufficient mathematical understanding and/or knowledge of mathematical concepts relevant to chemical kinetics and thermodynamics.

INTRODUCTION

Anecdotal evidence alludes to a mathematical problem in chemistry, while certain publications (Beressen, 2005; Steiner, 2008; Monk, 2006 & Yates, 2007 for example) explicitly highlight the problem. Bangash and Mustafa (2002) attribute the mathematical difficulties, which students encounter while learning chemistry, to insufficient mathematical understanding; also, Potgieter, Harding. and Engelbrecht (2008) claim that areas in chemistry with a particularly strong mathematical element are problematic for students. Bearing this small sample of opinions, in addition to others in mind, we decided to investigate undergraduate students' ability to transfer fundamental mathematical knowledge, pertinent to chemical kinetics and thermodynamics, from a mathematics context to a chemistry context. More specifically, we framed our questions as follows:

(1) Can students transfer (apply) mathematical knowledge (labelled mathematical items), relevant to chemical kinetics and thermodynamics, from a mathematics context to a chemistry context? (hereafter referred to as the *Transfer Question*)

(2) What effect does evidencing an ability to explain mathematical knowledge in a mathematics context have on students' ability to transfer it to a chemistry context? (hereafter referred to as the *Explaining and Transfer Question*)

RESEARCH METHODOLOGY

Study 1 and Study 2 were undertaken during the academic years 08/09 and 09/10 respectively. Each sample, in both studies, comprised of 2nd-year undergraduate students who were attending Dublin City University (DCU). During Study 1, 30 students participated while during Study 2, 24 students participated; all of the participants were volunteers. Furthermore, all of the participants were completing a non-optional chemical kinetics and thermodynamics module as part of their science-based degree, which could have been in the field of Chemical and Pharmaceutical Science, Analytical Science or Biotechnology for example. Lastly, the students in each sample would have completed and passed a calculus module ($\geq$ 40% pass rate) during the 1st year of their studies; this module should have equipped the students with basic calculus concepts applicable to chemical kinetics and thermodynamics. In order to answer our questions, *Diagnostic Tools* were developed.

Development of Diagnostic Tools

There were four Diagnostic Tools in total. Diagnostic Tool 1 assessed students' understanding of mathematical items, in a mathematics context, in the form of: calculating slope (Item 1); sketching a line with positive slope (Items 2 and 3); sketching a line with negative slope (Item 4); generating an expression for slope (Item 5); generating an expression for derivative (Item 6); interpreting derivative (Item 7); usage of exponentials (Item 8); usage of natural logarithms (Item 9); proportionality (Item 10); graphing an exponential function (Item 11) and graphing a natural logarithmic expression (Item 12). Diagnostic Tool 2 assessed all of these mathematical items in a chemistry context.

Diagnostic Tool 3 assessed students' understanding of mathematical items, in a mathematics context, in the form of: graphing a function (Item 13); evaluation of an integral (Item 14); and graphing an integral (Item 15). Diagnostic Tool 4 assessed these same items in a chemistry context.

Each of the items contained a Part (A) and Part (B) in both the mathematics context and the corresponding chemistry context. The Part (A) determined whether students answered correctly in both contexts; it allowed the investigation of the *Transfer Question.* The Part (B) required students to explain their reasoning; it thus allowed the investigation of the *Explaining and Transfer Question*. Items 2-5 did not contain a Part (B) during Study 1 because: at the time, it was considered to be replication of the Part (B) in Item 1, where Item 1 was deemed to be similar to Items 2-5.

An example of one the mathematical items, namely: *Item 12; Graphing a Natural Logarithmic Expression*, is shown in Figure 1.

Item 12: Graphing a Natural Logarithmic Expression

Mathematics Context

Given the relationship:

$$Lny = c - mx$$

(A) Draw a graph that represents the relationship in Figure 1. Label the axis accordingly.

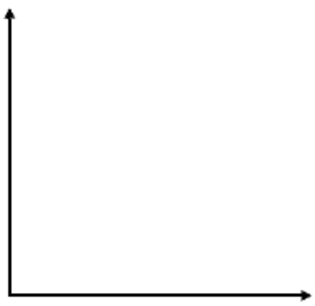

Figure 1

(B) Explain your reasoning

Chemistry Context

A student is studying the chemical reaction:

$$A + B \rightarrow P,$$

where A and B are reactants and P is the product. After graphing the Ln of the concentration of A, obtained at different times: (i.e. the graph of $Ln[A]_t$ against time (t)), the student finds that the graph corresponds to the relationship:

$$Ln[A]_t = Ln[A]_0 - kt$$

showing that the rate of the reaction is 1st order with respect to A.

(A) Sketch the relationship in Figure 1. Label the axis appropriately.

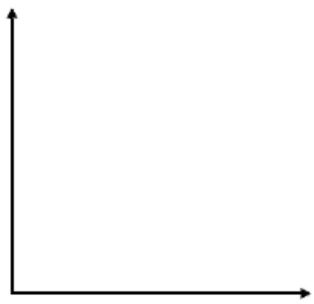

Figure 1

(B) Explain your reasoning.

Figure 1: Item 12; Graphing a Natural Logarithmic Expression

Students were administered all of the Tools separately and were allowed approximately half an hour to complete each one. Each tool was administered a week apart so as to avoid, for argument sake, the possibility of Diagnostic Tool 1 helping students to answer the same items in Diagnostic Tool 2, due to a ‘training effect’/ ‘memory effect’ (Britton, New, Sharma and

Yardley, 2005); by so doing this with all of the Diagnostic Tools, it was envisaged that this would provide a more accurate investigation of students' ability to transfer (the *Transfer Question*).

The validity of the items, in terms of the mathematics in both contexts being similar, was confirmed by collaboration with the School of Mathematical Sciences and the School of Chemical Sciences. The internal consistency of the items, in terms of each respective matching item in both contexts being answered similarly by the sample of students, was checked for, in both Diagnostic Tool 1 and 2, and Diagnostic Tool 3 and 4.

DATA ANALYSIS

Both the *Transfer Question* and the *Explaining and Transfer Question* were investigated using the traditional view of the transfer of learning (Lobato, 2003). Essentially, the traditional view of transfer investigates the transfer of learning as an all-or-nothing affair—students either transfer knowledge (or cognition in respect of a particular subject) either entirely or not at all. The traditional view also considers how and why students transfer.

What this means is that in the investigation of the *Transfer Question*, we considered the students to have transferred a mathematical item if they answered that mathematical item absolutely correctly in the mathematics context (learning context) and in the corresponding chemistry context (transfer context). This predominant cognitive focus of the traditional view is in contrast to the actor-oriented view of transfer (Lobato, 2003). The actor-oriented perspective investigates not just how and why the transfer of knowledge (cognition) by students occurs, but also the effect (if any), the situations/contexts that students transfer from and to, has on students' ability to transfer. While the traditional view of transfer does not consider the effect of the situation/context on students' ability to transfer, it still, nonetheless, like the actor-oriented approach, considers how and why knowledge can be transferred by students. Thus, the *Explaining and Transfer Question* operated within the traditional view of transfer.

For each item, categorical-statistical tests were used to determine if there was statistically-significant transfer (that is, an association between answering correctly in a mathematics context and correctly in a chemistry context). Depending on the data set, either the Chi-Squared Test or Fisher's Exact Test was used. For example, considering Item 4 (Sketching a Line with Negative Slope; shown in Figure 2) the manner in which students responded determined which cell they were positioned within, as shown in Table 1.

Looking at Table 1, of the 23 students who answered Item 4 correctly in a mathematics context, 16 of these answered correctly in the corresponding chemistry context. Of the 16 students who answered correctly in the chemistry context, all of these students answered correctly in the corresponding mathematics context. The basic laws of probability dictate that the number of students that would be expected to answer correctly in both the mathematics context and chemistry context due to chance alone is determined from multiplying the probability of being correct in a mathematics context by the probability of being correct in a chemistry context and then multiplying this value by the sample size. The value obtained (shown in parentheses in Table 1) was 12.3—the expected number of students answering correctly in both contexts due to chance alone. The actual value observed was 16 which raised the question of whether this value was significant. Using the relevant categorical-statistical test—in this case, Fisher's Exact Test— for the two-by-two contingency table, it was found that 16 was statistically significant at a confidence level of 95%. Thus, the conclusion was: if a student answers Item 4 correctly in a mathematics context, the likelihood of them answering the same item in a chemistry context correctly not due to chance alone is strong. This type of analysis was performed for all of the items in both studies.

Chemistry Context	Mathematics Context		
	Correct	***Incorrect***	***Total***
Correct	16(12.3)	0(3.7)	16
Incorrect	7(10.7)	7(3.3)	14
Total	23	7	30
			p = 0.02

Table 1: Testing for an association between being correct in a mathematics context and being correct in a chemistry context for Item 4 in Study 1.

The *Explaining and Transfer Question* involved the asking of the following question amongst ourselves: what constitutes students evidencing an ability to explain their mathematical reasoning in a mathematics context? This was investigated using the principles of qualitative data analysis as described by Cohen (2000). This type of analysis is used to distil key categories of explanation from students' Part B responses.

The approach comprised of a number of stages: 1) reading students' qualitative responses; 2) identifying the themes/meanings running through these responses; and 3) organising the themes/meanings into categories. An inter-rater reliability approach (Cohen, 2000) was used to decide what the categories were. An inter-rater reliability approach involves a number of researchers analysing qualitative data for 'categories of meanings' that appear to emerge from the data. The researchers compare what they consider to be categories of meaning with other researchers in order to reach a consensus. Working in conjunction with a researcher from the School of the Chemical Sciences and a researcher from the School of Mathematical Sciences a consensus was reached as to what categories emerged. Once the categories were decided upon, an inter-rater reliability approach was again used to decide whether each category was reflective of either an ability to explain or an inability to explain.

For each Item, testing for the presence of an association between evidencing an ability to explain in a mathematics context and being able to transfer was carried out using categorical statistical tests. The manner in which the categorical statistical tests were used was similar to the manner in which they were used to investigate the significance of the transfer observed, during the investigation of the *Transfer Question*. In addition to investigating whether students who evidenced an ability to explain in a mathematics context associated with transfer, the degree to which students explained in a mathematics context was investigated. It wanted to be seen if students who evidenced a certain degree of explanation for any particular item associated with the transfer of that item more so than other students. The work of Tall et. al (2001) was used to ascertain the degree to which students explained in a mathematics context. Categorical statistical tests were again used to investigate the appearance of an association between evidencing a certain degree of explanation for a particular item and being able to transfer that item.

Item 4: Sketching a Line with Negative Slope

Mathematics Context

L_1 as shown in Figure 1 passes through the Point 'P' and has a slope = 2.

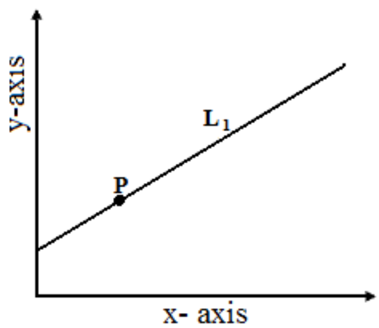

Figure 1

(A) Sketch in Figure 1: a line (L_4) that passes through the point P and has slope = -1.

(B) Explain your reasoning

Chemistry Context

The line in Figure 1 shows the graph of concentration of reactant with respect to time over a certain interval (Δt) . Its rate of decrease over this interval is equal to 2.

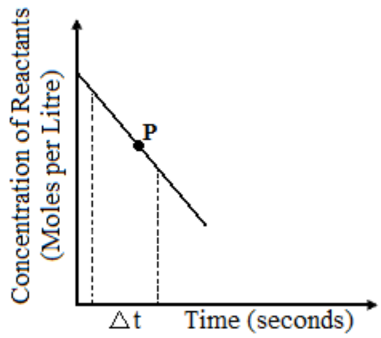

Figure 1

(A) Sketch in Figure 1: a line (L_4) that passes through the point P and has a value for the rate of decrease of the reactant with respect to time = 1.

(B) Explain your reasoning.

Figure 2: Item 4; Sketching a Line with Negative Slope

RESULTS

The Transfer Question

Tables 2 and 3 show both the number of students who were able to transfer each item; the significance of the transfer observed for each item is also shown in these tables.

Mathematical Item	Observed Frequencies and Expected Frequencies (in Parentheses) in the Contingency Tables.				Transfer p-values
	Correct in MC* and CC**	Correct in MC and Incorrect in CC	Incorrect in MC and Correct in CC	Incorrect in MC and CC	
1. Calculating Slope.	25(25.2)	2(1.8)	3(2.8)	0(0.2)	1.00
2. Sketching a Line with Positive Slope.	20(17.5)	5(7.5)	1(3.5)	4(1.5)	0.02
3. Sketching a Line with Positive Slope.	20(16.8)	4(7.2)	1(4.2)	5(1.8)	0.01
4. Sketching a Line with Negative Slope.	16(12.3)	7(10.7)	0(3.7)	7(3.3)	0.02
5. Generating an Expression for Slope.	13(10.7)	7(9.3)	3(5.3)	7(4.7)	0.12
6. Generating an Expression for Derivative.	4(1.7)	2(4.3)	5(7.3)	20(17.7)	0.04
7. Interpreting Derivative.	19(18.3)	3(3.7)	6(6.7)	2(1.3)	0.60
8. Usage of Exponentials.	10(4.3)	3(8.7)	0(5.7)	17(11.3)	0.00
9. Usage of Natural Logarithms.	9(5.2)	4(7.8)	3(6.8)	14(10.2)	0.01
10. Proportionality.	11(7.4)	2(5.6)	6(9.6)	11(7.4)	0.01
11. Graphing an Exponential Function.	1(0.4)	2(2.6)	3(3.6)	24(23.4)	0.36
12. Graphing a Natural Log Expression.	1(0.4)	2(2.6)	3(3.6)	24(23.4)	0.36
13. Graphing a Function.	4(1.7)	6(8.3)	1(3.3)	19(16.7)	0.03
14. Evaluation of an Integral.	2(0.5)	3(4.5)	1(2.5)	24(22.5)	0.06
15. Graphing an Integral.	2(0.4)	2(3.6)	1(2.6)	25(23.4)	0.04

MC* *- Mathematics Context;* ***CC***** *- Chemistry Context.*

Table 2: Results from the Contingency Tables in Study 1 that Were Used to Investigate the Significance of Observed Transfer for Each Item.

Mathematical Item	Observed Frequencies and Expected Frequencies (in Parentheses) in the Contingency Tables.				Transfer p-values
	Correct in MC* and CC**	Correct in MC and Incorrect in CC	Incorrect in MC and Correct in CC	Incorrect in MC and CC	
1. Calculating Slope.	19(18.2)	4(4.8)	0(0.8)	1(0.2)	0.21
2. Sketching a Line with Positive Slope.	13(13.1)	8(7.9)	2(1.9)	1(1.1)	1.00
3. Sketching a Line with Positive Slope.	13(11.7)	7(8.3)	1(2.3)	3(1.7)	0.27
4. Sketching a Line with Negative Slope.	15(13.8)	7(8.3)	0(1.2)	2(0.8)	0.10
5. Generating an Expression for Slope.	10(6.7)	6(9.3)	0(3.3)	8(4.7)	0.01
6. Generating an Expression for Derivative.	4(1.7)	0(2.3)	6(8.3)	14(11.7)	0.02
7. Interpreting Derivative.	15(15.3)	1(0.7)	8(7.7)	0(0.3)	1.00
8. Usage of Exponentials.	2(1.2)	12(12.8)	0(0.8)	10(9.2)	0.49
9. Usage of Natural Logarithms.	10(7.8)	1(3.2)	7(9.2)	6(3.8)	0.08
10. Proportionality.	11(10.3)	2(2.7)	8(8.7)	3(2.3)	0.63
11. Graphing an Exponential Function.	0(0.0)	3(3.0)	0(0.0)	21(21.0)	1.00
12. Graphing a Natural Log Expression.	6(4.6)	6(7.5)	5(6.5)	12(10.6)	0.44
13. Graphing a Function.	4(2.6)	5(6.4)	3(4.4)	12(10.6)	0.36
14. Evaluation of an Integral.	4(1.7)	3(5.3)	2(4.3)	15(12.8)	0.04
15. Graphing an Integral.	3(1.8)	4(5.3)	3(4.3)	14(12.8)	0.31

MC* *- Mathematics Context;* ***CC**** *- Chemistry Context.*

Table 3: Results from the Contingency Tables in Study 2 that Were Used to Investigate the Significance of Observed Transfer for Each Item.

It can be seen, from a statistically-significant perspective, that transfer was observed across both studies. We can also see that less statistically-significant transfer was observed during Study 2 compared with Study 1. One of the reasons for this may be the smaller sample size used in Study 2 (24 instead of 30). Statistically-significant transfer aside, in terms of whether students could transfer, transfer was evidenced by some students in both studies, for all of the items (not necessarily the same students evidencing transfer for each of the items), except for Item 11.

The Explaining and Transfer Question

From a statistically-significant perspective, Table 4 shows that for 8/11 items (the majority of the items) requiring an explanation during Study 1, students who evidenced any type of correct explanation for these items in a mathematics context associated with transferring them. Likewise, during Study 2, for the majority of the items (10/15) requiring an explanation, students who evidenced an ability to correctly explain the reasoning for these items in a mathematics context associated with transferring them, as shown in Table 5.

Mathematical Item	Observed Frequencies and Expected Frequencies (in Parentheses) in the Contingency Tables.				Transfer p-values
	Ability to Explain and Transfer	Ability to Explain but not Transfer	Inability to Explain and Transfer	Inability to Explain and not Transfer	
1. Calculating Slope.	13(11.7)	1(2.3)	12(13.3)	4(2.7)	0.34
6. Generating an Expression for Derivative.	4(0.8)	2(5.2)	0(3.2)	24(20.8)	0.00
7. Interpreting Derivative.	17(12.0)	1(6.0)	3(8.0)	9(4.0)	0.00
8. Usage of Exponentials.	10(5.7)	7(11.3)	0(4.3)	13(8.7)	0.00
9. Usage of Natural Logarithms.	9(3.9)	4(9.1)	0(5.1)	17(11.9)	0.00
10. Proportionality.	8(3.3)	1(5.7)	3(7.7)	18(13.3)	0.00
11. Graphing an Exponential Function.	0(0.0)	0(0.0)	0(0.0)	30(30.0)	1.00
12. Graphing a Natural Log Expression.	1(0.4)	2(2.6)	3(3.6)	24(23.4)	0.36
13. Graphing a Function.	4(1.7)	6(8.3)	1(3.3)	19(16.7)	0.03
14. Evaluation of an Integral.	2(0.3)	3(4.7)	0(1.7)	25(23.3)	0.02
15. Graphing an Integral.	2(0.3)	2(3.7)	0(1.7)	26(24.3)	0.01

Table 4: Results from the Contingency Tables in Study 1 that Were Used to Investigate the Significance of Evidencing an Ability to Explain and Transfer.

Mathematical Item	Observed Frequencies and Expected Frequencies (in Parentheses) in the Contingency Tables.				Transfer p-values
	Ability to Explain and Transfer	Ability to Explain but not Transfer	Inability to Explain and Transfer	Inability to Explain and not Transfer	
1. Calculating Slope.	19(17.4)	3(4.6)	0(1.6)	2(0.4)	0.04
2. Sketching a Line with Positive Slope.	13(11.4)	8(9.6)	0(1.6)	3(1.4)	0.08
3. Sketching a Line with Positive Slope.	12(9.5)	7(9.5)	0(2.5)	5(2.5)	0.04
4. Sketching a Line with Negative Slope.	7(6.2)	3(3.7)	8(8.7)	6(5.3)	0.68
5. Generating an Expression for Slope.	10(6.3)	5(8.8)	0(3.8)	9(5.3)	0.00
6. Generating an Expression for Derivative.	4(0.7)	0(3.3)	0(3.3)	20(16.7)	0.00
7. Interpreting Derivative.	11(6.9)	0(4.1)	4(8.1)	9(4.9)	0.00
8. Usage of Exponentials.	1(0.8)	9(9.2)	1(1.2)	13(12.8)	1.00
9. Usage of Natural Logarithms	7(3.3)	1(4.7)	3(6. 7)	13(9.3)	0.00
10. Proportionality.	11(6.0)	2(7.0)	0(5.0)	11(6.0)	0.00
11. Graphing an Exponential Function.	0(0.0)	1(1.0)	0(0.0)	23(23.0)	1.00
12. Graphing a Natural Log Expression.	3(1.3)	2(3.7)	3(4.7)	16(14.2)	0.08
13. Graphing a Function.	4(1.0)	1(4.0)	1(4.0)	18(15.0)	0.00
14. Evaluation of an Integral.	3(0.8)	2(4.2)	1(3.2)	18(15.8)	0.02
15. Graphing an Integral.	3(0.7)	3(5.3)	0(2.2)	18(15.8)	0.01

Table 5: Results from the Contingency Tables in Study 2 that Were Used to Investigate the Significance of Evidencing an Ability to Explain and Transfer.

CONCLUSION

In terms of the *Transfer Question*, the problem students appear to have with certain mathematical items in a chemistry context may not be due to difficulties in being able to transfer but instead be as a result of students simply not possessing the necessary mathematical knowledge in a mathematics context. If we look at columns 3 and 4 in Tables 2 and 3, we can see that across both studies, large numbers of students were unable to answer certain mathematical items; thus if students are unable to answer the mathematical items in a mathematics context, they cannot be expected to transfer mathematical knowledge which they do not possess.

Such a finding is similar to the findings of Potgieter et al. (2008) in their investigation of students' ability to transfer mathematical knowledge necessary for the Nernst equation which is used in chemistry. They found that the problem students appear to have with the mathematics required for this equation in a chemistry context, are most likely mathematical, and not due to an inability to transfer. Furthermore, our findings support the views of Bangash and Mustafa (2002) who claim that the problem which chemistry students have with mathematics is mathematical. The question of why certain students could answer some of the items correctly in a chemistry context but not in the corresponding mathematics context, is perhaps due to a 'training effect' in the chemistry context, with students not being able to transfer the mathematical representations they have somewhat learned in the chemistry context to a mathematics context.

Our observance of transfer, statistically-significant wise, for the items which were transferred across both Study 1, and Study 2, needs external validation. In fact, one could argue that the very observance of transfer for any particular item, irrespective of whether it was statistically-significant transfer, needs external validation.

There is good evidence in terms of the *Explaining and Transfer Question* that students who evidence an ability to explain their answer with respect to a particular mathematical item in a mathematics context associate with answering the equivalent item in a chemistry context—in other words transfer. Bishop (1985) proposes that students will only be able to transfer mathematics if they have developed an appreciation of mathematical meaning, which is dependent on the mathematical environment of which the student is a part. Boaler (1993) adds further impetus to the findings from the *Explaining and Transfer Question*, whereby she claims that understanding which allows for the development of links between different contexts will most likely develop if students are encouraged to communicate and challenge mathematics.

The findings from the *Explaining and Transfer Question* also need to be externally validated. Moreover, it is important to be aware that an ability to explain one's reasoning in mathematics context may not necessarily be a factor in allowing students to transfer; there may be a third variable at play due to the inherent fallacy of categorical statistical tests. Such variables were not investigated during this study.

REFERENCES

Bangash, F.K. and Mustafa, S. (2002). Essentials of Mathematics in Teaching Chemistry. *Proceedings of 10th IOSTE Symposium*, Vol.2, Foz do Iquaso, Parana, Brazil, July 2002.

Berressem, J. (2005) *Time to face up to the maths problem,* Education in Chemistry, 45, 1

Bishop, A. (1985). 'The Social Construction of Meaning – A Significant Development for Mathematics Education', For the Learning of Mathematics. 5 (I), pp. 24-29.

Boaler, J. (1993). Encouraging the Transfer of 'School' Mathematics to the 'Real World' Through the Integration of Process and Content, Context and Culture. *Educational Studies in Mathematics*. 25, pp. 341-373.

Britton, S., New, P.B., Sharma, M.D., and Yardley, D. (2005). A Case Study of the Transfer of Mathematics Skills by University Students. *International Journal of Mathematical Education in Science and Technology.* 36 (1), pp. 1-13.

Cohen, L., Manion, L. And Morrison, K. (2000). *Research Methods in Education*.5th ed. London: Routledge.

Lobato, J. (2003). How Design Experiments Can Inform a Rethinking of Transfer and Vice Versa. *Educational Researcher* 32 (1), pp. 17-20.

Monk, P. (2006). *Maths for Chemistry*. Oxford: Oxford University Press.

Potgieter, M., Harding, A. and Engelbrecht, J. (2007). Transfer of Algebraic and Graphical Thinking between Mathematics and Chemistry. *Journal of Research in Science Teaching.* 45 (2), pp. 197-218.

Steiner, E.(2008). *The Chemistry Maths Book*. 2nd ed. Oxford: Oxford University Press.

Tall, D., Gray, E., Bin Ali, M., Crowley, L., DeMarois, P., McGowen, M., Pitta, D., Pinto, M., Thomas, M. and Yusof, Y., (2001). Symbols and the Bifurcation between Procedural and Conceptual Thinking. *Canadian Journal of Science, Mathematics and Technology Education*, 1, pp. 81-104.

Yates, P.(2007). Chemical Calculations. 2nd ed. Florida: Taylor & Francis Group

PROBLEM POSING AND PRE-SERVICE PRIMARY TEACHERS: AN INITIAL STUDY

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Problem solving is an important component of primary mathematics. Central to the activity of problem solving is the task of problem posing. This study examines the skills and predispositions of pre-service teachers when posing problems for use in primary classrooms. The preliminary study was carried out with 36 pre-service teachers at the end of their first semester of a teacher education programme. Participants were asked to construct a problem for use in a primary classroom. Analysis of the problems revealed that problems were very similar across participants and tended to be single step problems with one possible solution. Problems were predominantly computational and focused on procedures and operations relating to quantities. The mathematical content of problems was situated within the number and measurement strands of the primary curriculum. We identify experiences, models and structures that have been shown to be effective in bringing improvements in problem posing in international contexts. Recommendations are made for practice in teacher education contexts to support pre-service teachers in developing the pedagogical skills required to teach problem solving.

INTRODUCTION

Problem solving is a critical domain in mathematics education. Francisco and Maher (2005), in a longitudinal study investigating the conditions for promoting reasoning in problem solving, state that 'providing students with the opportunity to work on complex tasks as opposed to simple tasks is crucial for stimulating their mathematical reasoning' (p. 731). Central to the activity of problem solving is the task of problem posing. Problem posing is a fundamental component of teaching and learning mathematics. Problem posing involves formulating problems that require children to draw on mathematical knowledge and skills in order to answer those problems; or involves the creation of new problems from old ones. These problems usually take the form of word problems.

Traditional efforts at presenting word problems in school mathematics have been criticized in terms of students' lack of sense making (Schoenfeld, 1991; Silver, Shapiro and Deutsch, 1993) and for their lack of effectiveness in making mathematics seem connected to real world contexts and thus seem meaningful and real for learners (Lave, 1992). Ng (2002) found that textbooks provide a high portion of routine, closed-ended problems and problems with exactly sufficient information. Heuristics suggested by curricula are not all covered by the traditional textbook, and textbooks should include more open-ended problems, non-routine problems, authentic problems and problems with insufficient or extraneous information as well as other traditional problems (Fan and Zhu, 2007). Part of the reason accounting for the shortfall of traditional approaches to teaching problem solving is the ways in which text book writers, particularly in the Irish context, conceptualized the role of problems solving in mathematics instruction. Problem solving was seen as an activity that occurred *after* children studied mathematical skills and concepts (hence we saw the placement of word problems at the ‘end of the chapter’) as opposed to being a means to acquiring new mathematical knowledge and understandings.

Preparing Prospective Teachers to Teach Problem Solving

Recent reform views of mathematics posit problem solving as ‘a vehicle for learning new mathematical ideas and skills’ (NCTM 2000, p. 182). This involves reconceptualising mathematics teaching and learning as teaching through problem solving. Carpenter, Fennema, Peterson, Chiang and Loef (1989) emphasise the importance of posing problems to students and listening to how students describe the way they have solved such problems. However, Fosnot (1989) explains that teachers are unfamiliar with such teaching because they are products of systems that emphasised drill and procedure. Any program designed to equip pre-service teachers with the requisite mathematical and pedagogical understandings related to problem solving must take into consideration the mathematical experiences of these same pre-service teachers prior to entry to teacher education. The following studies reveal insights into the influence of school mathematical experiences on prospective teachers views of problem solving and reveal the degree of disconnect between learners experiences of problem context, problem structure and the strategies employed to solve problems.

A study of prospective secondary mathematics teachers’ knowledge of problems and problem solving (Chapman, 2005) revealed that participants made sense of problems in terms of the traditional, routine problems they had experienced from their school mathematical experiences. Their understanding of the problem solving process also aligned with classroom approaches they had experienced as learners. A study of 139 pre-service teachers carried out by Contreras & Martinez-Cruz (2003) found that many of them provided incorrect solutions to additive word problems due to incorrect interpretation of the solution in relation to the problem context. Similarly, Simon (1990) identified difficulties that pre-service primary teachers demonstrated in connecting the semantic features of the word problems and their understanding of division to the procedures that they employed to divide. There is also evidence of the disconnect between pre-service teachers real-world knowledge and the approaches they used to solve school word problems (Verschaffel and De Corte, 1996). Furthermore the same study found that pre-service teachers did not use their real-world knowledge of the problem contexts when considering the children’s responses to the same problems.

Selection of tasks and models

The selection of a problem solving task will depend on the classroom situation of the teacher. Francisco and Maher (2005) explain that students must be provided with the opportunity to work on complex tasks as opposed to simple tasks as such tasks are crucial for the development of mathematical reasoning. Wilburne (2006) has explained that tasks, the best mathematical problems one can employ in the classroom, are non-routine mathematical problems that encourage rich and meaningful mathematical discussions, those that don’t exhibit any obvious solutions, and those that require the student to use various different strategies to solve them.

The selection of a problem-solving model to use as a structure within which to situate classroom-based problem solving activity is also an important consideration. Current empirical research specifically reveals the importance of having students justify solutions and fully exploring extension activities that may emanate from the problem solving exercise with them (Elmore, 1996; Francisco and Maher, 2005; Hoffman and Spatariu, 2007). O'Shea (2009) investigated the use of Polya's problem solving model and found the four stage procedure is helpful to provide structure to the problem solving lesson and ensure students engage with the problem, develop a unique method of solution and reflect on the challenge of the situation. O'Shea (2009) concluded that students are successful in their mathematical problem-solving interactions when they are presented with challenging mathematical problems that reflect their current level of understanding and when they are comfortable with a problem solving model. When students engage in reflection it can surprise the teacher and reveal what children are able to do which can often be far more than what the teacher might have given them credit for.

The provision of structured experiences focusing on problem posing has been shown to bring about improvements in how pre-service teachers think about and use word problems. While pre-service teachers often see word problems as opportunities to practice skills that children have already been taught, Chapman (2004) reported improvements in how pre-service teachers view word problems. They shifted from viewing word problems as consisting of a 'cover story' and practice with operations they already know to viewing word problems as consisting of a cover story and both mathematical and semantic structures. Their interpretation of multi-structure word problems improved as did their ability to represent the structure of word problems using a variety of representations (verbally, symbolically, concretely and pictorially) to shown the meaning of operations. A number of studies also examined instructional practices that help prospective teachers to develop their knowledge of problem solving for teaching. A study carried out by Arbaugh & Brown (2004) focused on helping prospective teachers select worthwhile problems by developing their understanding of the relationship between a task and the kind of thinking that task requires of learners. Prospective teachers engaged in task sorting activities that used level of cognitive demand as a determinant. The researchers found that participants enhanced their skills in analyzing problems in terms of the types of thinking required by students to solve them (as opposed to the tendency of novices to focus on surface features of problems). Other studies have focused on developing prospective teachers' understanding of problem solving through the provision of experiences engaging in the problem solving process themselves as learners (Ebby, 2000; Szydlik, Szydlik, & Benson, 2003) and the provision of academic courses designed to target the problem-solving process and pedagogical processes related to problem solving (Chapman, 2005; Roddick, Becker & Pence, 2000).

Approaching the Teaching of Mathematical Problem Solving

There has been very little research examining Irish pre-service teachers understandings of and dispositions towards problem solving. Studies of in-service teachers approaches to teaching mathematical problem solving highlight the emphasis placed on exploring lower level processes with less emphasis placed on teaching higher level mathematical processes such as reasoning and problem solving (O'Shea, 2003; O'Shea, 2009). Furthermore, in a review by the NCCA (2008) of the implementation of the Primary School Curriculum it appears teachers struggle with engaging classrooms of students in group collaborations, which are central to successful problem solving situations. Entry level pre-service teachers carry a multitude of attitudes, beliefs and experiences relating to mathematics and problem solving emulating from their experiences as mathematics learners in the Irish secondary school system. While the secondary mathematics curriculum is under extensive review and redesign, it may take many years for the efforts of Project Mathematics to come to fruition. Consequently we draw on some data arising from review of the leaving certificate programme prior to the implementation and roll out of Project Mathematics.

In a discussion paper reviewing mathematics in post-primary education (NCCA, 2005), the National Council for Curriculum and Assessment comment on the ways that mathematics is taught at second level, with particular reference to the ways in which the teaching of mathematics reflects current trends internationally in mathematics education. With reference to the teaching of *problem solving (*or investigations) at the senior cycle, the report states

> 'The exploratory, open-ended style associated with investigations does not seem to fit Irish teachers' and students' views of mathematics. Possible reasons for this may lie in the culture of mathematics teaching in this country .. , in the demands that this approach would make on teacher knowledge, skills and attitudes, and in the fact that such work is not currently subject to assessment in the examination.' (p. 5)

Further insights into the problem solving capacities of Irish secondary school students, a proportion of whom continue on to become teachers of mathematics, can be gleaned from performance on international mathematics assessments. The Programme for International Student Assessment (PISA), coordinated by the Organisation for Economic Cooperation and Development (OECD), tests and compares school children's performance on 'real-life' tasks considered relevant for effective participation in adult society and for life-long learning. Student achievement in PISA is categorised along a 6-level continuum. Irish students are proficient at tasks associated with levels 1-3, yet do not compare as well when dealing with mathematical reasoning and developing approaches to analysing, evaluating and working with complex mathematical problems (levels 5-6). According to the PISA 2006 results (Eivers, Shiels & Cunningham, 2007) at levels 5 and 6 Ireland fares slightly less well than the OECD average and considerably poorer than countries such as Korea and Hong Kong where over 27 per cent of students reached level 5 or higher (as compared to 10 percent of Irish children).

Several studies have been carried out of the mathematical understanding of entry-level pre-service teachers. The outcomes of this research provide insights into the problem-solving abilities of pre-service teachers. Corcoran (2005) carried out a study of 71 pre-service teachers to whom she administered eleven items released from the PISA 2000 assessment (OECD, 2002). Performance on tasks was mapped onto the PISA proficiency levels and results compared against Irish 15 year olds. In examining the two tasks categorized at highest proficiency level, level 6, analysis of responses revealed that less than 10% of student teachers got both items at level six fully correct. More than 20% got both items totally incorrect. The pattern of low performance at task difficulty levels five and six was found to be similar to Irish 15-year-olds performance in PISA 2003, where Irish students performed less well on items requiring higher order mathematical skills. Overall, the author identified a ceiling at proficiency level 4 for up to 80% of students. Analysis of the mathematical tasks with the lowest performance highlighted that more than 50% of student teachers show a concerning low level of the process skills identified in the primary curriculum of applying and problem solving skills together with communicating and expressing skills linked with formal reasoning. Other studies concur that post-primary graduate perform best at lower order mathematical skills such as memorisation of formulae and procedures as opposed to thinking creatively, providing reasons for solutions, or engaging in mathematical problem solving (Hourigan and O'Donoghue 2007; Leavy and Sloane, 2010; NCCA 2005).

METHOD

This study is a preliminary investigation of the understandings that pre-service teachers have of problem solving, with particular attention on the activity of problem posing. As such, the study represents an initial point of departure in assessing the understandings and beliefs about problem solving the Irish pre-service teachers have on entry to teacher education programs. The information arising from this study will be used to inform the design of instructional approaches to facilitate their learning of problem solving.

Participants

Participants were 38 pre-service primary teachers enrolled in the postgraduate diploma in primary education. All participants had primary degrees prior to enrolment in the teacher education diploma. Participants had completed their first semester of their three semester program. They had completed a semester-long course in the pedagogy of mathematics (10 hours) and had completed a 2 week teaching placement in a middle grades primary classroom. During the pedagogy of mathematics courses there was no explicit instruction on problem-solving or on problem posing. On the last class session of the semester, participants were asked to reflect on their experience of teaching practice and on their observations of children engaging in doing mathematics. They were then asked to discuss in groups their experiences of writing problems for children while on teaching practice. They were provided with two discussion questions to consider during the course of their discussion: what makes a good problem? What different types of stimuli can we provide when designing problems? Participants were then asked to write a problem that they might pose to a group of middle grades children.

Following discussion of these problems, participants were informed of the purpose of this study. Participants were invited to provide a copy of their task for the instructor at the end of the session. Participation was voluntary and anonymous. These written problems serve as data upon which the following analysis is based.

ANALYSIS OF DATA

The sources of data used for the study were the problems participants constructed at the end of the semester. They were analysed according to three categories. These categories focused primarily on problem type, problem feature and mathematics content (Crespo, 2003). Within these categories, a number of criteria were identified upon which to classify problems, examples are: single step versus multistep, single solution versus multiple solution, Focus on procedures versus focus on concepts, low cognitive demand versus high cognitive demand, computational versus exploratory/puzzle-type, arithmetic versus beyond arithmetic. Problems were also categorized in terms of which of the strands of the primary curriculum the problem was situated in: number, shape and space, algebra, measures or data. The criteria above are not necessarily orthogonal nor discrete and we expected some overlap between criteria. For example, a problem that was procedural in nature might also fall into the criteria of being computational. However there is also the potential that a problem may not follow a general pattern and be categorized on opposite ends of similar dimensions. Hence, qualitative notes were also taken for each of the problems to provide context and narrative for the coding and categorization.

RESULTS

Before examining the results we need to be cognizant that study participants are early stage pre-service teachers with only one semester of coursework completed. Also, these pre-service teachers have had no instruction in problem solving or problem posing. Gathering data at this stage of their careers is, however, important as it provides information regarding the point of departure for instruction and provides signposts for the direction that teaching trajectories should take in order to address the needs of pre-service teachers for teaching problem solving.

Analysis of the problems posed by pre-service teachers reinforces the findings of similar international studies. Two thirds of the problems posed by participants were *single step* problems all of which had *one possible solution*. For the most part, these problems focused on *procedures related to operations* on numbers and correspondingly fell into the number or measurement strands of the primary curriculum. A typical problem was similar to the following (response 16):

> If each Justin Bieber ticket costs €25 and 678 people want to attend. How much money would the concert organizers raise?

While one-third of problems were not single step problems, these problems almost always involved two steps, both of which involved the use of an operation, for example addition followed by subtraction. Therefore, while they may be viewed as multi-step problems, both steps were procedural in nature. The following is an example of a problem (response 4) that was coded as multistep and fell within the strand of measures.

> John was swimming in his pool last week. Paul and Stephen jumped in and dispersed ¼ of the pools contents. The amount of water that remained in the pool was 3,000 litres. How much was in the pool to begin with?

In terms of cognitive demand, the majority of problems posed (89%) were considered to be of low cognitive demand. When considering cognitive demand, we tried to consider the depth of mathematical knowledge required to complete the posed problem. Problems were categorized as consisting of low cognitive demand if they primarily involved the recall of a fact, term or procedure. While most of the one step problems were low in cognitive demand, so too were the multistep step problems which consisted of the implementation of procedures. Problems were categorized as high cognitive demand if they involved mathematical reasoning, higher order mathematical thinking, making logical arguments or developing conclusions. In the following task (response 8), which is an example of one of the four high cognitive demand tasks constructed, the problem is multistep, may have more than one solution (depending on how different children 'fix' the variables), and involves thinking extending beyond the implementation of mathematical procedures.

> How many points do Wolves need to be certain of staying in the premiership? [Present supplementary data such as the league tables, remaining fixtures etc.]

The majority of the problems were computational and focused on operational procedures. The selection of problems of this nature may have reflected participants' own school experiences of mathematics or have reflected the nature of the mathematics they taught while on teaching practice. Only four of the 38 problems were not computational; two of these problems were problems focusing on volume. Both problems (responses 29 and response 33- Figure 1) are presented below:

> Sara and John are making a cake for their mother. They need 300ml of milk. However, they only have a 500ml container and a 200ml container. These containers do not have measures on them. How would you solve this problem?

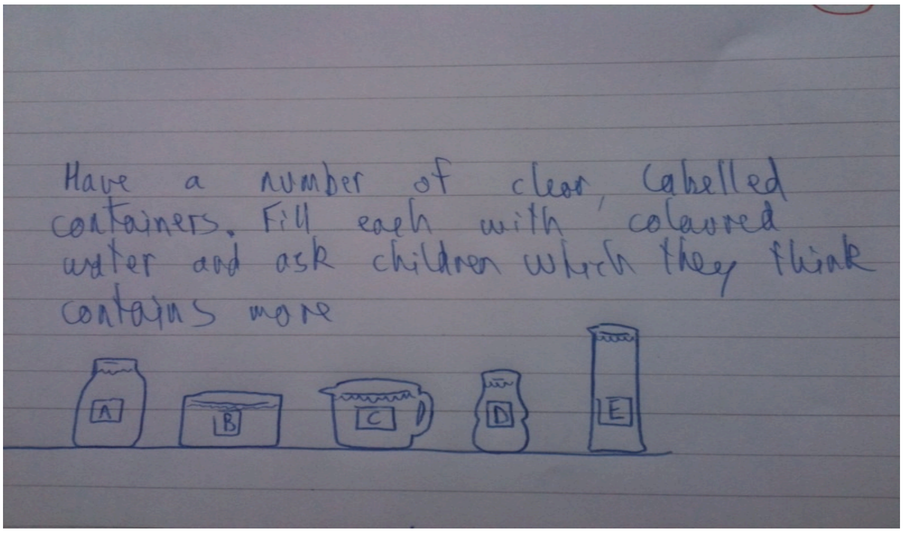

Figure 1: Response 33

As can be see, both problems involve mathematical reasoning and some element of problem solving. Neither problem involves operations on quantities.

In terms of the mathematical content of the problems, problems were located in a number of the strands of the primary school mathematics curriculum. The table below provides an indication of how the problems were dispersed across the curriculum.

Strand	Number of problems (%)
Number	12 (33%)
Shape and Space	1 (3%)
Measurement	18 (47%)
Algebra	0 (0%)
Data	5 (14%)

Table 1: Dispersion of problems across strand units

Examination of the table reveals the dominance of measurement and number problems. The presence of number is somewhat masked in that many of the measurement problems were in fact problems that involved operations on quantities but used measurement as a context within which to situate the problem. In the problem that follows (response 29), we can see that it is a multi step problem involving the addition of three quantities and subsequent subtraction of the quantity from a starting quantity. It is primarily an operations problem firmly located within the realm of number work but using length of string as the context. Consequently many of the measurement problems were, in fact, number problems.

> If a piece of string measuring 1m 30cm was cut into sections of 8cm, 42cm and 68cm, and used to tie some balloons. What length of string would remain from the original piece of string?

There are several limitations posed by this study. Very little guidance was provided for study participants in terms of the target group for the problem. No specific grade level was specified and there was no guidance provided in terms of the ability of the children involved. This lack of clarity may account for some of the generality in terms of the problems constructed.

DISCUSSION

The goal of teaching mathematics through problem solving is that students will develop a deep understanding of mathematical concepts, skills and procedures. This relationship between understanding and problem solving is symbiotic. In order to solve problems, the learner must have conceptual understanding of the mathematics. Understanding, then, enhances problem solving. At the same time, a problem causes disequilibrium and requires that the learner make connections between the mathematics they already know and apply it to the problem situation as presented. Consequently, engaging in problem solving develops understanding. Not all teachers are adept at asking the appropriate questions in a way that enhances learning because they do not have the sufficient classroom experience or practice in associated discourse (Knott, 2010) and therefore in creating active classrooms, posing and solving cognitively challenging problems, promoting reflection, metacognition and facilitating broad ranging discussions. If we are to move towards teaching mathematics through problem solving, there are many considerations that need to be addressed in pedagogy courses to prepare prospective teachers to teach problem solving. Firstly, designing and/or choosing problematic tasks is not a trivial endeavour. The mathematics that we wish children to learn must be embedded within the tasks. The tasks must also be accessible to learners in that they build on knowledge that learners already have, while at the same time be engaging and draw on contexts and situations that are both interesting and relevant to learners. Pre-service teachers need also to be aware of the landscape of problem solving and of the ways in which children engage in problem solving. This involves attention to common approaches children use, pre-conceptions and misconceptions, in addition to possible learning trajectories that we might expect children to follow when engaging in problem solving activities. It is also useful to highlight attention to the role that textbooks play in guiding the nature of problem solving activities that Irish primary children are exposed to; in addition to engaging pre-service teachers in critiquing the quality and nature of textbook representations of problems.

REFERENCES

Arbaugh, F. & Brown, C.A. (2004). What makes a mathematical task worthwhile? Designing a learning tool for high school mathematics teachers. In R. R. Rubenstein & G. W. Bright (Eds.), *Perspectives on the teaching of mathematics* (pp. 27-41). Reston, VA: National Council of Teachers of Mathematics.

Ball, D.L. (1996). Teacher Learning and the Mathematics Reforms: What We Think We Know and What We Need to Learn. *Phi Delta Kappan*, 77(7), 500-50

Carpenter, T.P., E. Fennema, P.L. Peterson, C. Chiang and M.Loef, (1989). Using Knowledge of Children's Mathematics Thinking in Classroom Teaching: An Experimental Study. *Review of Educational Research*, 20(4), 499-531.

Chapman, O. (2004). *Helping pre-service elementary teachers develop flexibility in using word problems in their teaching.* Paper presented at the annual meeting of the North American Chapter of the International group for the Psychology of Mathematics Education, Toronto, Canada.

Chapman, O. (2005). Constructing pedagogical knowledge of problems solving: Preservice mathematics teachers. In H. L. Chick & J. L. Vincent (Eds.), *Proceedings of the 29th PME International Conference*, *2*, 225-232.

Contreras, J.N. & Martinez-Cruz, A.M. (2003). Preservice elementary teachers' solution processes to problematic addition and subtraction word problems involving ordinal numbers and their interpretations of solutions. In N. A. Pateman, B. J. Dougherty & J. T. Zillox (Eds.), *Proceedings of The 27th Conference of the International Group for the Psychology of Mathematics Education*. Honolulu: University of Hawaii, 2, 197-204.

Corcoran, D. (2005). An Exploration of the Mathematical Literacy of Irish Students Preparing to be Primary School Teachers. *Proceedings of the First National Conference on Research on Mathematics Education (MEI 1)*. Editors: Sean Close, Therese Dooley and Dolores Corcoran St. Patrick's College of Education.

Crespo, S. (2003) Learning to pose mathematical problems: Exploring changes in pre-service teachers' practices. *Educational Studies in Mathematics*, 52(3), 243-270

Elmore, R., (1996). Getting to scale with good educational practice. *Harvard Educational Review, 66* (1), 1-26.

Fan, L. & Zhu, Y. (2007). Representation of problem-solving procedures: A comparative look at China, Singapore, and US mathematics textbooks. *Educational Studies in Mathematics,* 66(1), 61-75.

Fosnot, C.T., (1989). *Enquiring Teachers Enquiring Learners: A Constructivist Approach to Teaching*. New York: Teacher's College Press.

Francisco, J.M. & Maher, C.A. (2005). Conditions for promoting reasoning in problem solving: Insights from a longitudinal study. *Journal of Mathematical Behaviour* 24, 361-372.

Hoffman, B., & Spatariu, A. (2007). *The effect of self-efficacy and metacognitive prompting on math problem-solving efficiency*. Paper presented at the annual meeting of the American Psychological Association, San Francisco, August 19, 2007.

Hourigan, M. and J. O'Donoghue 2007. Mathematical underpreparedness: The influence of pre-tertiary mathematics experiences on students' ability to make a successful transition to tertiary level mathematics courses in Ireland. *International Journal of Mathematics Education in Science and Technology, 38*(4), 461-477.

Knott, L. (2010). Problem posing from the foundations of mathematics, *The Montana Mathematics Enthusiast*, 7(3), 413-432.

Lave, J. (1992). Word problems: A microcosm of theories of learning. In P. Light and G. Butterworth (Eds.), *Context and cognition: Ways of learning and knowing* (pp. 74-92). New York: Harvester Wheatsheaf.

Leavy, A.M. and F. Sloane, 2010. *Developing Statistical Literacy in Primary Level Mathematics*. Dublin: Stationary Office.

National Council of Teachers of Mathematics Commission on Standards for School Mathematics, (2000). *Principles and Standards for School Mathematics.* Reston, VA: NCTM.

NCCA, (2005). *Review of mathematics in post-primary. A discussion paper*. Dublin: NCCA.

NCCA, (2008). Primary Curriculum Review: Phase 2. www.ncca.ie, accessed 10-01-2009.

Ng, L.E. (2002). *Representation of problem solving in Singaporean primary mathematics textbooks*. Unpublished M.Ed dissertation. National Technological University, Singapore.

OECD, (2003). *Assessing Scientific, Reading and Mathematical Literacy*. OECD: Paris.

O'Shea, J. (2003). *Mathematical Problem Solving: The Beliefs and Experiences of Class Teachers*. Unpublished Masters Dissertation, University of Limerick.

O'Shea, J. (2009). *Endeavouring to teach mathematical problem solving from a constructivist perspective: The experiences of primary teachers*. Unpublished doctoral dissertation, University of Limerick.

Roddick, C., Becker, J. R., & Pence, B. J. (2000). Capstone courses in problem solving for prospective secondary teachers: Effects of beliefs and teaching practices, *Proceedings of the 24th PME Conference,* Vol. 4, 97-104.

Schoenfeld, A, H. (1991). On mathematics sense making: An informal attack on the unfortunate divorce of formal and informal mathematics. I J.F. Voss, D.N. Perkins and J.W. Segal (Eds.), *Informal reasoning and education* (pp. 311-343). Hillsdale, NJ: LEA.

Silver, E. A., Shapiro, L. J., & Deutsch, A. (1993). Sense making and the solution of division problems involving remainders: An examination of middle school students' solution processes and their interpretations of solutions. *Journal for Research in Mathematics Education, 24*(2), 117-135.

Simon, M.A. (1990). Prospective elementary teachers' knowledge of division. In Booker, G.; Cobb, P.; Mendicuti, T. (Eds.), *Proceedings of the 14th PME International Conference*, *3*, 313-320.

Szydlik, J.E., Szydlik, S.D., & Benson, S. R. (2003). Exploring changes in preservice elementary teachers'mathematics beliefs. *Journal of Mathematics Teacher Education, 6*(3), 253-279.

Verschaffel, L. and De Corte, E. (1996). Number and arithmetic. In A. J. Bishop, K. Clements, C. Keitel. J. Kilpatrick, & C. Laborde (eds.) International handbook of mathematics education [pp. 112-115]. Dorcrecht: Kluwer Academic Publishers.

Wilburne, J.M. (2006). Preparing preservice elementary school teachers to teach problem solving. *Teaching Children Mathematics*, 12(9), 454–463.

MATHEMATICS TEACHING MATTERS: MAKING COMPLEX MATHEMATICAL IDEAS ACCESSIBLE TO PRIMARY LEVEL CHILDREN

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Reform of mathematics curricula has led to the expansion of topics such as algebra, data and probability being taught to primary level children. This new subject matter can be challenging for primary teachers to teach as some teachers have not previously engaged with these topics as students themselves and also some of these areas have traditionally been considered secondary school topics. Furthermore, while there is a wealth of concrete manipulatives available to teach topics such as number and shape and space there is less so for these 'newer' areas of mathematics curricula. This paper reports on the combined efforts of teacher educators and pre-service teachers to conceptually engage 4th class primary children with algebraic concepts, function and variable, through the use of appropriate models. Lesson Study was the primary method used to support a focus on examining teaching and the effectiveness of models through the design and implementation of 'study lessons'. Insights into children's learning of algebraic concepts are presented in addition to an examination of how good teaching using manipulatives can lead to powerful mathematical reasoning by young children. Video evidence of the findings will be presented and discussed.

INTRODUCTION

The field of mathematics education is dynamic. There has been considerable debate internationally regarding what constitutes important and powerful mathematics in primary education. For the most part of the last century, computation skills were considered as a critical foundation for mathematical development into the later years. In reaction to this focus on computational skills there has been considerable emphasis placed on problem solving, on the balance between instrumental and relational understanding (Skemp, 1976) and on procedural and conceptual understanding (Hiebert & Lefevre, 1986). Furthermore, there has been debate regarding the nature of the mathematics content considered suitable for primary children. Examination of the national curricula of Ireland, Australia, and the United States (AEC, 1990; Government of Ireland, 1999; NCTM, 1989) highlights the broadening of mathematical goals through the inclusion of Data Handling, Probability, and Algebra as new foci of study in the past two decades. These developments which emphasize relational and conceptual understanding, combined with the broadening of the mathematical content of primary level curricula, have implications for teaching. Traditional modes of instruction, which draw on didactic approaches and value instructional methods such as drill and practice, do not provide sufficient support for the development of the types of mathematical understandings valued in the new millennium.

PROVIDING ACCESS TO POWERFUL MATHEMATICAL IDEAS

Mathematics education research has been referred to as a 'design science' (Wittman, 1998) wherein a central activity is to concentrate on constructing 'artificial objects' (p. 94) and investigating the differential effects of these objects on the development of mathematical understanding. Such research involves the design, testing, and the process of reiteration and modification of objects and the subsequent measurement of the impact of these objects on learning. There are a number of ways children can be supported in accessing powerful mathematical ideas with an emphasis on both objects and instructional approaches; all with a view to developing good teaching practices in classrooms. While there are many tools (emerging technologies, for example) and approaches used in primary classrooms to develop understanding of complex mathematical concepts, we briefly outline two approaches we drew on in the design of lessons that supported primary children in gaining access to algebraic concepts: models and representations.

The use of *models and manipulatives* in primary mathematics education is well established, particularly in relation to place value ideas and number work. There is evidence, however, that the manipulatives are not automatically helpful in the development of mathematical understanding. Young children need support in making connections from the analogies embodied by the tools/manipulatives to the mathematical ideas. The following quote exemplifies the thinking around the use of models in Realistic Mathematics Education (RME) wherein "the starting point is in the contextual situation of the problem that has to be solved . The premise here is that students who work with these models will be encouraged to (re)invent the more formal mathematics" (Gravemeijer, 1999, p. 159). Research examining children's understanding of mathematics has also identified useable conceptual *representations* that are accessible to children. Valuable representations are being generated that support learning in the emerging areas of primary mathematics, in particular with regard to algebraic thinking (Bellisio & Maher, 1998), data exploration (Curcio, 1987; Jones, Thornton, Langrall, Mooney, Perry & Putt, 2000) and probability (Watson & Moritz, 1998).

SUPPORTING TEACHERS IN DEVELOPING CHILDREN'S MATHEMATICAL UNDERSTANDING

Reform of mathematics curricula has broadened the scope of the mathematics content being taught at primary level and provided a spotlight on a variety of mathematical processes deemed critical to the development of mathematical understanding. The introduction of new mathematics content, such as data and probability, means that some of our teachers are teaching mathematics content that they themselves were not exposed to as learners. Teachers' lack of familiarity with new mathematics can pose challenges for the fidelity of the implementation of revised national curricula. Research in the United States found that, following reform of the mathematics curriculum, the actual mathematical experiences of elementary children reflected little of the experiences of the intended curriculum (Cooney, Shealy, & Arvold, 1998; Steele, 2001; Vacc & Bright, 1999). Discrepancies between desired and implemented curricula, which frequently occur during the implementation of reform curricula, are not a new problem in mathematics education and neither is it limited to the Irish education system. It is not surprising then, that reviews of the implementation of the Primary

School Mathematics Curriculum (NCCA, 2005) identified that teachers feel underprepared to teach certain areas of mathematics. Teaching is a complex activity. Teaching mathematics at the primary level is particularly challenging as teachers are not mathematics content experts i.e. the majority of primary school teachers do not have mathematics to degree level. Primary teachers need considerable support navigating the multiple terrains that contribute to the mathematics education of children. This terrain incorporates attention to learning theory, cognitive development, assessing understanding, and the use of models and tools which help children gain access to powerful mathematical ideas.

This paper focuses on one aspect of good mathematics teaching - the design and selection of tools that mediate learning. We examine the mathematics content area of algebra; a domain traditionally considered to 'belong' to the domain of secondary level mathematics. This focus was in part motivated by the results of the 2004 National Assessment of Mathematics Achievement which reported that 38% of inspectors were dissatisfied with the availability of resources for teaching Algebra in fourth class. Furthermore, a sizeable 60% of inspectors were dissatisfied with the use of resources for teaching this same strand (Shiel, Surgenor, Close, & Millar, 2006).

METHOD

Participants

This study was carried out with 21 final year pre-service primary teachers during the concluding semester of their teacher education program. Participants had completed their mathematics education courses (three semesters) and all teaching practice requirements (at junior, middle and senior grades) and self selected into mathematics education as a cognate area of study. Four participants were male; the remainder were female. Two participants were international Erasmus students.

Lesson Study

All pre-service teachers, and three mathematics educators, engaged in *Japanese Lesson Study* (Fernandez & Yoshida, 2004; Lewis, 2002; Lewis & Tsuchida, 1998). Lesson Study is an approach for studying teaching that utilizes detailed analyses of classroom lessons. Lesson study was used in this study to facilitate the examination of both the planning of lessons and the implementation of those lessons in classrooms and thus provided an avenue to design tools and sequences of instruction to support the development of algebraic reasoning with primary children.

The research was conducted over a 12-week semester. The central activity in lesson study was for participants to work collaboratively on the design and implementation of a study lesson. Participants were organized into groups of 5-6 to engage in the phases of Lesson Study. The first phase involved the *research and preparation* of a study lesson involving researching algebraic topics and the construction of a detailed lesson plan. The *implementation* stage involved one pre-service teacher teaching the lesson in a 4th class primary classroom while the other group members, and the researchers, observed and evaluated classroom activity and student learning. Group members then *reflected on and improved* the original lesson design through discussing their classroom observations and modifying the lesson design in line with their observations. The *second implementation* stage involved re teaching the lesson with a second class of primary children and *reflecting* upon observations. The second implementation was videotaped. The cycle concluded with each of the four lesson study groups making a presentation of the outcomes of their work to their peers and lecturers at the end of the semester.

EXPLORING ALGEBRAIC CONCEPTS IN THE PRIMARY CLASSROOMS

This paper reports on the work of two lesson study groups. Both groups were focusing on algebraic concepts more often associated with secondary school mathematics: function and variable. Research has shown that both concepts, however, can be introduced in primary classrooms provided that adequate supports are provided to children when reasoning about these concepts. We provide a description of the teaching of both concepts, along with a focus on the tools used to facilitate the development of algebraic reasoning, as an argument that 'Good Teaching Matters'.

1. Exploring the Concept of 'Function'

Function and the Primary School Mathematics Curriculum (PSMC)

While a formal treatment of function does not occur until secondary school in the Irish education system, the concept of function can be introduced informally in the primary classroom. Indeed, function can be introduced in tandem with work on patterns and relations. Work examining relationships with numerical patterns and analysis of how patterns grow and change can be used as a precursor to informal work with functions. However, for primary level children to develop a conceptual understanding of function and successfully analyze the change that comes about as a result of a function, they must be provided the opportunity to model problem situations with objects and use representations to draw conclusions. Within the context of functions, we use the 'function machine' to model the problem situation and provide the use of simple tables to represent the change that occurs as a result of the function.

The function machine

Function machines have been used in mathematics education to support children's understanding of functions. We used a physical model of a function machine, as opposed to traditional pictures or diagrams, to support primary level children in identifying important characteristics of functions. For younger learners, it was critical that these characteristics were represented concretely and explicitly identified. As can be seen in the images below, the function machine itself was a relatively unsophisticated object with a clearly identified input (image 2), output (image 3) and buttons which represent the function (image 1).

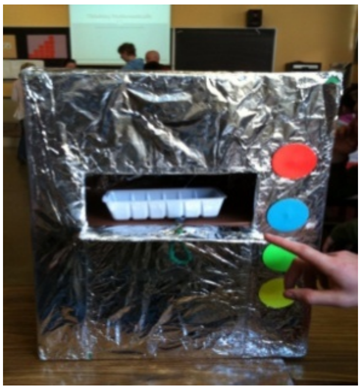


Image 1: Front of function machine with input drawer

Image 2: Input section of the function machine (front)

Image 3: Output section of the function machine (back)

Introducing the concept of function to children

The concept of function was introduced using the analogy of baking in a 'faulty oven'. The following scenario is presented by the pre-service teacher: *'Yesterday something very strange happened to me! Well… I love baking and yesterday I made 2 cakes* (picture/photo displayed on the white board using projector) *and put them into the oven. When I took the cakes out something strange had happened to them, when I opened the oven there seemed to be more than 2 cakes. Look what I found! There were 4.'* The pre-service teacher presented a number of such scenarios from the faulty oven and encouraged pupils to identify the rule/pattern which determined the output from the oven. Links were then made between the 'faulty oven' and the idea of a function/rule machine, the pre-service teacher stated *'I soon realised that my oven isn't an oven at all…It's a magic machine called the 'Maths Rule Generator'. Do you want to see it?'*

Developing the concept of function

The 'Function machine' was introduced to children and the language of function was developed through guided exploration of the function machine. The pre-service teacher then led the class through a series of activities the objective of which was to help children explore the relationship between input, output and functions/rules. Prepared materials were placed in the input slot by a child, the function button was then pressed, and the pre-service teacher removed the 'baked' materials from an output slot at the back of the box. She leads the class in trying to figure out the rule for each of the buttons. In each case the children were encouraged to predict the rule by looking at the relationship between the input and the output. As a new pair of values (input and output) was presented, such predictions were tested (confirmed or rejected). Throughout the lesson, the input, output, and rule (both guess and actual) were represented on a recording sheet. During the demonstration the rules for the various buttons varied. As this was the children's first introduction to functions, we slowly increased the complexity and challenge as the lesson progressed. Initial 'rules' focused on addition and multiplication (e.g. orange (x2 or double), blue (+15), green (x9)). In order to facilitate challenge, the final button (yellow in this case) was a two-step rule or function (x2+3).

Primary children writing functions

Children were given the opportunity to generate and identify rules/functions while working in collaborative groups. Groups were provided with record sheets, these sheets reflected many of the features of the original function machine – they used the same language (input/output) and contained the same perceptual cues to identify rules/functions (coloured buttons similar to that found on the function machine). One child in each group acted as the ‘rule generator’. Others in the group providing inputs (between 0 and 10) and the ‘rule generator’ worked out and shared the output (using a secret rule e.g. +5). The information was recorded on the recording sheet thereby facilitating prediction and checking of the rule.

As previously outlined only one example of a two-step rule or function was taught during the lesson. However when creating functions pupils were encouraged to devise a two-step rule if they wished. The following is a transcript of the class trying to solve a two-step problem created by one of the children following the work on functions. This example highlights that teaching mathematics using effective models and representations helps develop sophisticated mathematical reasoning and understanding among children.

Teacher	Robert has made up this two-step rule that only works for even numbers. Would someone like to give an input?
Cian	2
Robert	Output is 7
	Robert records 2 as the input and 7 as the output on the white board
Calum	10
Robert	Output is 11
	Robert records 10 as the input and 11 as the output on the white board
David	4
Robert	8
	Robert records 4 as the input and 8 as the output on the white board
Teacher	Can anyone guess Roberts’ rule?
	Children nod their heads to indicate they cannot find the rule
Teacher	We’ll give you a hint. The first step is addition and the second step is division.
	Children nod their heads to indicate they cannot find the rule
Teacher	The first step is plus 12.
	Children raise hands
Laura	Plus 12 divided by 2.
Robert	You are right.

2. Exploring the Concept of 'Variable'

Variables are an important concept in algebra. Variables are symbols (e.g. letters/frames) that take the place of numbers or ranges of numbers (Van De Walle, 2004). The National Council of Teachers of Mathematics (2008) outlines three key ideas of variables namely: variables as unknowns e.g. 5 + y = 30; variables as quantities that vary in joint variation e.g. A+B =24; and variables in generalizations of patterns e.g. a + b = b + a. For the purposes of this study the first idea of variables, i.e. variables as unknowns, was the focus of study. This aligns with the objectives for this area of Algebra in the Primary School Mathematics Curriculum (1999). To develop primary children's conceptual understanding of variables it was necessary, like when introducing functions, to present the concept in a concrete and accessible manner. The variable wheel was chosen as a useful model for primary level children. This was chosen as it is a concrete manipulative which introduces the concept of a variable in a fun, interesting and age-appropriate manner, while simultaneously modelling the concept of a variable as an unknown.

The variable wheel

The variable wheel is essentially two strips of paper, one with letters on it and the other with numbers on it. The letter strip is wrapped around the number strip so that the letters align to a number. These strips can be rotated so that any one variable (letter) can have a variety of different values depending on the rotation (images 4 and 5). We commenced activities by having A aligned with 1, B aligned with 2, etc. The variable wheel allowed children to explore the idea of a variable as a letter that can stand for any member of a set of numbers and also enabled children to substitute numbers for variables.

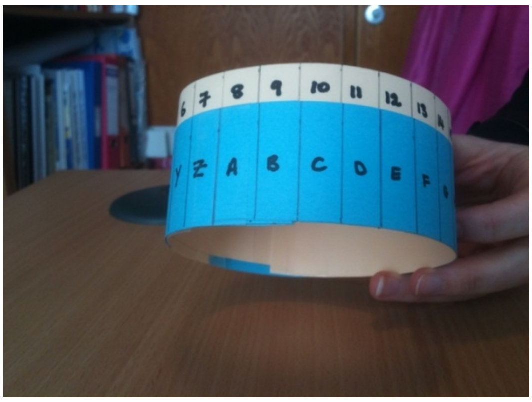


Image 4 : Variable Wheel **Image 5: Rotated Variable Wheel**

Introducing the concept of a variable to children

The idea of a variable was introduced initially by eliciting from children what they thought the word 'variable' meant. To help children the pre-service teacher asked questions such as '*Does anybody know what it means to vary? For example, the weather varies, or the speed of the car varies on your way to school.'* The first activity carried out with the variable wheel was to have children find the value of the letters in their name. The children found the value of their names, using the variable wheel, e.g. Amy was 39 because A=1, M=13 and Y=25. Answers were compared to their predictions and outcomes discussed.

To introduce the concept of a variable, children were then asked if they thought the results would be the same if they **varied** the values on their variable wheel and worked backwards letting a = 26 and z =1? This was an important step in the lesson as we wanted children to realise that a variable is a letter that can represent **any** number depending on the problem (i.e. a=1 in one problem but a=17 in another problem). Children were allowed to twist the variable wheel and come up with a new value for their name. From this, it was hoped that the children would realise that any letter can represent any number and vice versa and that letters (i.e. variables) don't have to be set values.

In the initial lesson, on completing the above task, pupils demonstrated a misconception that a letter always held the same value e.g. a = 1, e = 5. On reflection it was noted that there was insufficient emphasis placed on the idea of variable and values of letters varying. As a result during the second teaching of the lesson the pre-service teachers placed a greater emphasis on the concept of variables when using the variable wheel. It was found that children grasped the concept easily as a result.

Development of the concept of a variable

The development of the lesson consisted of using relevant story problems that required a variable to solve them. Contexts suitable for primary school children include sport, games, clothes, food, etc. In each problem there was an unknown value to be found. Children were allowed to pick any letter to represent the unknown. A simple table of what was known was drawn up. A number sentence was then written and solved (linkage with equations).

In the first problem the context of sport was chosen. The following problem was presented: '*There were 76 points scored altogether in a rugby match. Ireland scored 40 points. How many points did France score?'* This information was transferred onto a simple table by the pre-service teacher and children (see figure 1). Children were told '*As we do not know how many points France scored we can use a letter/variable to represent how many points France scored.'* Children were asked *'What letter might we use as the variable in this problem for France?'* A suggestion was F. This was written in the table (see figure 2). The next step was to write the story problem as a number sentence using the chosen variable i.e. 40 + F = 76. Children solved F = 36. A number of more challenging but similar problems were then presented to provide practice and consolidate the learning related to variable.

Ireland	France	Total
40		76

Figure 1: Table for a variable problem

Ireland	France	Total
40	F	76

Figure 2: Table (with variable) for a variable problem

Primary children writing variable problems

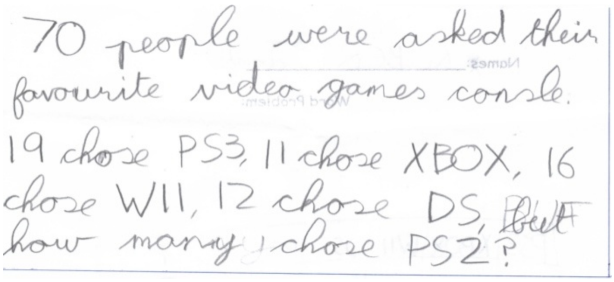
Following whole class work solving story problems using variables the children were provided with an opportunity, in pairs, to create their own story problems that required a variable to solve. To motivate the children we told them that their story problems would be placed in envelopes and given to another class or another group in their class to solve. However, it was important to model to the class how to write story problems and clear directions were given. One way of modelling how to create a story problem was to use information from the class itself and fill in a prepared chart.

There are ______ in a class.
_______ like soccer.
_________ like swimming.
How many children like rugby?

A class survey was conducted and the prepared chart was completed. This made the problem context meaningful for the children. In addition it linked to the data strand of the curriculum.

Figure 3: Model for a story problem

The following is one example of a story problems created by fourth class pupils in a Limerick City primary school.


70 people were asked their favourite video games consle. 19 chose PS3, 11 chose XBOX, 16 chose WII, 12 chose DS, But how many chose PS2?

70 people were asked their favourite video games console. 19 chose PS3, 11 chose Xbox, 16 chose WII, 12 chose DS. How many chose PS2?

Figure 4: Sample story problem created by group of children

Pushing the boundaries: Extension activity

Once children solved and created story problems involving variables it was decided to challenge the children by presenting a number sentence that had a variable and asking them to write a story problem that represented (the context for) the number sentence. This involved flexible reasoning on the part of children and supported differentiation for more able children. The following are examples of two story problems written by the same group in response to two different number sentences. These examples illustrate the outcomes that good teaching can have on the development of mathematical reasoning.

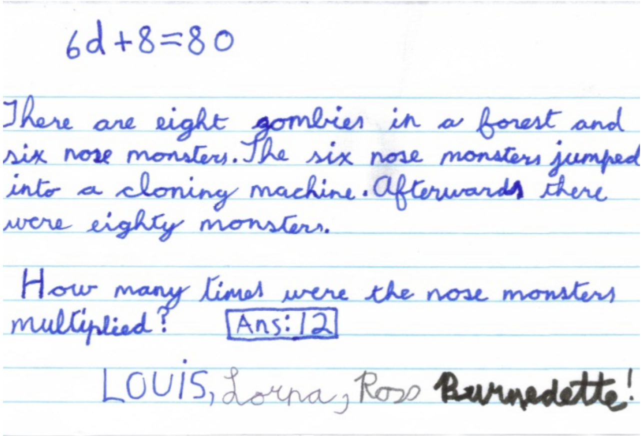


Figure 5: Story problem to represent 6d + 8 = 80

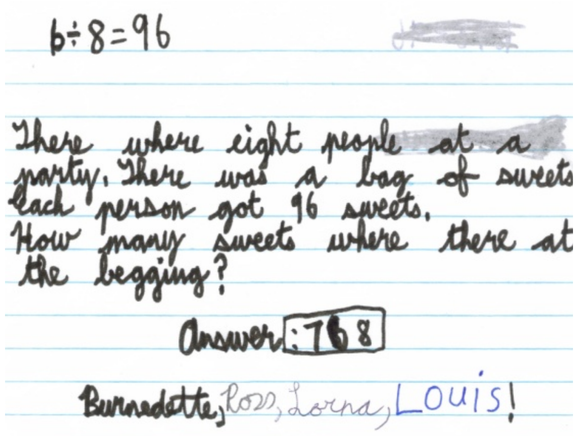


Figure 6: Story problem to represent b÷8 = 96

A COMPARISON

The mathematical understandings described in the sections above are not trivial. In particular, the skills required for the group of children to write the story problems (see figure 5 and 6 above), that represented a presented number sentence, are complex. These problems were written following just one hour of instruction on variables and using the variable wheel. As a contrast we present the results of pre-service primary teachers (n=356) when asked to write a number story for the number sentence b ÷ 3 = 11 (similar to problem in figure 6). While a relatively large proportion correctly constructed a word problem (74.4%), a significant proportion of them (25.6%) demonstrated difficulty in constructing a story problem. Examples of incorrect responses from pre-service primary teachers are shown in figures 7 and 8 below.

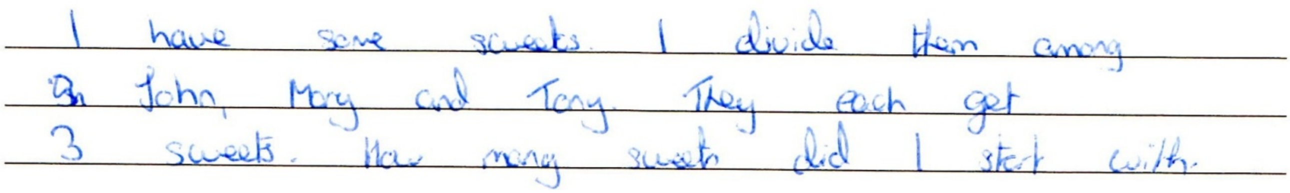


Figure 7: Sample 1 of pre-service teacher's error

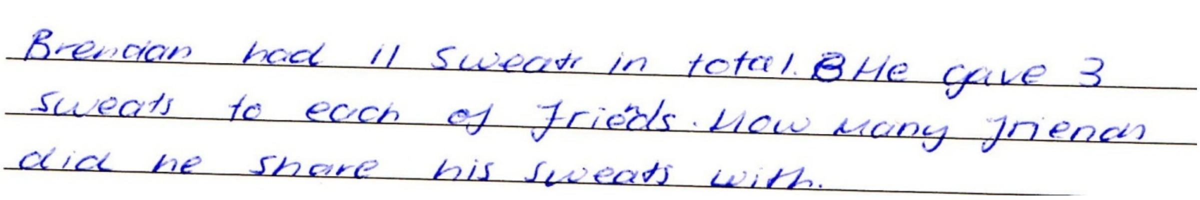


Figure 8: Sample 2 of pre-service teacher's error

Contrast this performance with those of the 4th class children in figures 5 and 6 and we can get some sense of the extent of conceptual understanding of variable held by the 4th class children.

IN CONCLUSION

Learning occurred not only for the children involved in the study, but also for the pre-service primary teachers. Pre-service teachers commented on the development of their own mathematical understanding of algebra *'As a result of trial and error that I undertook in the lesson plan my own understanding of algebra and how to write equations improved'* (Variables group: Rosemary). The development of more positive attitudes towards algebra was also mentioned by most participants: *'Algebra is not as abstract as we originally thought, once it is presented in context'* (Variable group end-of-semester presentation).

The development and use of the tools and representations described in this study were critical in supporting relatively young children in reasoning about functions and variables. Our research indicates that some 4th class children exceeded the expectations generally set for algebraic reasoning at the primary level. Yet, these were ordinary Irish children in ordinary Irish classrooms. They were taught by relatively inexperienced teachers (in that they were pre-service teachers). What made the difference to the relatively sophisticated mathematical understandings that were recorded was the attention that we paid, using the Lesson Study framework, to *teaching.* This attention to teaching encompassed concentration on and consideration of pedagogical approaches, mathematical language, the scope and sequence of instruction, and most importantly within the context of algebra – the design of tools and representations that made algebraic concepts accessible to young children. Yes, good teaching matters.

REFERENCES

Australian Education Council. (1990). *A national statement on mathematics for Australian schools.* Carlton, Australia: Author.

Bellisio, C.W. & Maher, C. (1998). What kind of notation do children use to express algebraic thinking? In S. Berenson, K. Dawkins, M. Blanton, W. Coulombe, J. Kolbe, K. Norwood, & L. Stiff (Eds.), *Proceedings of the twentieth Annual Meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education* (Vol 1, pp. 161-165). Columbus, OH: The ERIC Clearinghouse for Science, Mathematics and Environmental Education.

Cooney, T.J., Shealy, B.E. & Arvold, B. (1998). Conceptualizing belief structuresof preservice secondary mathematics teachers. *Journal for the Research in Mathematics Education, 29,* 306-333.

Curcio, F. (1987). Comprehension of Mathematical Relationships Expressed in Graphs. *Journal for Research in Mathematics Education, 18,* 382-93.

Fernandez, C. & Yoshida, M. (2004). *Lesson Study. A Japanese Approach to Improving Mathematics Teaching and Learning.* Erlbaum: Mahwah, New Jersey.

Government of Ireland (1999). *Primary School Mathematics Curriculum*. Dublin: Stationary Office.

Gravemeijer, K. (1999). How emergent models may foster a constitution of formal mathematics. *Mathematical Thinking and Learning, 1*(2), 155-177.

Hiebert, J., & Lefevre, P. (1986). Conceptual and procedural knowledge in mathematics: An introductory analysis. In J. Hiebert (Ed.), *Conceptual and procedural knowledge: The case of mathematics* (pp. 1-27). Hillsdale, NJ: Lawrence Erlbaum Associates.

Jones, G.A., Thornton, C.A., Langrall, C.W., Mooney, E.S., Perry, B., & Putt, I.J. (2000). A framework for characterizing children's statistical thinking. *Mathematical Thinking and Learning, 2, 2*69-307.

Lewis, C. (2002). Lesson *Study: A handbook of teacher-led instructional improvement.* Philadelphia: Research for Better Schools.

Lewis, C. and Tsuchida, I. (1998). A lesson is like a swiftly flowing river: How research lessons improve Japanese education. *American Educator, Winter,* 12-17, 50-52.

National Council of Teachers of Mathematics Commission on Standards for School Mathematics (1989). *Curriculum and Evaluation Standards for School Mathematics.* Reston, VA: Author.

National Council of Teachers of Mathematics (2008). *Navigating through Algebra in Grades 3-5*.

NCCA (2005). Primary Curriculum Review Phase 1. Final Report. NCCA: Merrion Square, Dublin.

Shiel, G., Surgenor, P., Close, S. & Millar, D. (2006). *The 2004 National Assessment of Mathematics Achievement.* Dublin: Educational Research Centre.

Skemp, R.R. (1976). Relational and instrumental understanding. *Mathematics Teaching*, *77*, 20-26.

Steele, Diana F. (2001). The interfacing of preservice and inservice experiences of reform-based teaching: A longitudinal study. *Journal of Mathematics Teacher Education*, 4, 139-172.

Vacc, N.N., & Bright, G.W. (1999). Elementary preservice teachers' changing beliefs and instructional use of children's mathematical thinking. Journal for Research in Mathematics Education, 30(1), 89-110.

Van DeWalle, J. (2004). *Elementary and Middle School Mathematics Teaching Developmentally.* London: Pearson

Watson, J. M., & Moritz, J. B. (1998). Longitudinal development of chance measurement. *Mathematics Education Research Journal*, 10(2), 103-127.

Wittman, E.C. (1998). Mathematics education as a "design science." In A. Sierpinska & J. Kilpatrick (Eds.), *Mathematics Education as a research domain: A search for identity* (Part 1, pp. 87-104). Dordrecht, The Netherlands: Kluwer Academic.

TOWARDS AN INSTRUCTIONAL MODEL TO SUPPORT TEACHING AND LEARNING IN MATHEMATICS

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The revision of mathematics syllabi and the associated teacher professional development under Project Maths place greater emphasis on student understanding of mathematical concepts, on the development of mathematical skills, and on the application of knowledge and skills to solving problems in both familiar and unfamiliar contexts.

This paper proposes an instructional model or framework that could be adopted by teachers. It focuses attention on the essential elements of good teaching and learning practice. Its development has grown out of the experiences of Project Maths to date and it draws on a range of insights from current research into teaching and learning. The framework is structured around a key idea: thinking about the learning and learning about the thinking, by means of which teachers can gain insights into students' thinking that will provide them with evidence of learning and enable them to adapt classroom experiences appropriately.

INTRODUCTION

Project Maths, a curriculum and assessment initiative in Irish post-primary mathematics education, sees the phased implementation of syllabus change in mathematics at second level. The revision of the syllabi and the associated teacher professional development place greater emphasis on student understanding of mathematical concepts, on the development of mathematical skills, and on the application of knowledge and skills to solving problems in both familiar and unfamiliar contexts.

The revised mathematics syllabuses set out the learning outcomes in five strands which support continuity and progression between mathematics in the primary school and in post-primary junior cycle, and between junior cycle and senior cycle mathematics. These learning outcomes seek to engage learners, promote deeper understanding and facilitate the development and reinforcement of key skills which are seen as essential to teaching and learning across the curriculum. The key skills identified at senior cycle are: *critical and creative thinking*, *communicating*, *information processing*, *being personally effective* and *working with others*.[1] These skills are important for all learners to achieve their full potential, both during their time in school and into the future.

As well as syllabus changes and a focus on key skills, *Project Maths* advocates a changed approach to teaching and learning in mathematics which promotes understanding of the concepts involved through an inquiry-based approach that has students as active participants in developing their mathematical knowledge and skills, including their problem-solving skills. Oldham (2005) discusses issues involved in adopting the use of problem-solving and/or investigative approaches in teacher education, noting that it may be assumed that teachers are

[1] For more details on the key skills see www.ncca.ie/seniorcycle.

unlikely to implement such approaches if they have not experienced them as learners. To assist in adapting their practice to incorporate these changed approaches, an instructional model or framework can enable teachers to plan and conduct lessons in a structured manner so that effective teaching and learning takes place in the classroom.

The 5E model[2] is one example, summarised as:

- Engage
- Explore
- Explain
- Elaborate/Extend
- Evaluate

In considering classroom practice, Conway and Sloane (2006) identified a number of aspects of lesson planning that can help the teacher to promote improved mathematical understanding. This was based on a much broader consideration of a cycle of planning, teaching, reflection, evaluation and revision by groups of mathematics teachers—the Japanese lesson study model. While acknowledging that this is culturally based, they advocate that the process of lesson revision is a powerful one that can be adopted to fit the Irish school system. Sloane (2005) notes that Irish teachers can gain a lot from engaging in the process of lesson study, which 'provides an unique opportunity in mathematics instruction and in the profession of teaching in Ireland'.

The lesson study approach is effectively action research conducted by groups of teachers which is aimed at improving the quality of teaching and learning through an iterative process. The research is a collaborative process that benefits the students as well as their teachers. Callan (2006) sees such collaboration being about 'real engagement with real teaching and learning issues…a real living and dynamic culture of reflection and practical engagement by professionals'. Corcoran (2009) describes a community of practice discussing and exploring the potential of lesson study as a means of bridging the gap between planning and pupil outcomes.

The phased introduction of changed syllabuses in post-primary mathematics education is accompanied by a programme of teacher professional development that aims to bring about change in how students engage with mathematical concepts and develop their mathematical and problem-solving skills through greater emphasis on an investigative approach in which the teacher's role is less that of an 'expert' who imparts the knowledge that the students 'learn' and more that of a director and facilitator of student learning through discovery and exploration. The experience gained from engaging with the initial group of schools involved in *Project Maths* indicated that teachers needed both extensive and intensive support to meet the challenges they faced in changing their classroom practice.

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[2] Adapted from BSCS.org. The 5E model is also referred to as the e^5 model (in Victoria, for example).

The development of an instructional model or framework was undertaken as part of support structures for teachers that would focus their attention on the essential elements of good teaching and learning practice. Swan (2009) describes eight principles that should underlie all good teaching:

- Build on the knowledge learners bring to sessions
- Expose and discuss common misconceptions
- Develop effective questioning
- Use cooperative small group work
- Emphasise methods rather than answers
- Use rich collaborative tasks
- Create connections between mathematical topics
- Use technology in appropriate ways.

Le Métais (2003) recognises the teacher's need for a wider range of pedagogical strategies to enable students of different abilities to move towards becoming independent active learners. This can be challenging for teachers who have not themselves, as learners, experienced such strategies.

While the challenges and problems in mathematics education are not unique to Ireland, solutions (even successful ones) that have been developed elsewhere may not be directly transferable to the Irish context in which change in post-primary mathematics education is being implemented nationally, following its initial introduction in a small number of schools on a phased, developmental basis.

DEVELOPING THE MODEL

The framework set out in this paper grew out of the *Project Maths* experiences to date. It draws on research and developments in mathematics education elsewhere and is intended to complement the support being given to teachers of mathematics in making important changes to the way they teach the subject. It does this by structuring these changes around one key idea—*thinking about the learning and learning about the thinking.*

With this idea in mind, teachers are encouraged to take three important steps in planning and conducting their mathematics lessons:

1. Take an overview of the topic(s) involved which they share with students.
2. Ensure that they make clear connections with previous, and future, learning experiences so that they can plan a strategy to enable learners to understand the concepts involved and achieve the desired learning outcomes.
3. Engage learners in activities that promote exploration of the topic(s) and discussion of their own ideas and understanding, and that allow both teacher and learner to assess current knowledge and skills as well as continue their development.

Following these three steps can help to unlock vital parts of the learning process, so that teachers and students can **gain insights into the thinking** that will **provide evidence of learning progress** and help to **reveal misconceptions** along the way which can then be addressed. The framework is not prescriptive; rather, it is a kind of ready-reckoner for teachers to use in planning for and conducting mathematics classes.

A *CLEAR* APPROACH IN MATHEMATICS LESSONS

Connecting the learning

Leading the learning through rich tasks

Extending the learning

Assessing the learning

Reflecting on the learning

This approach complements the **Teaching and Learning Plans** which have been developed in particular syllabus topics[3]. These plans are not intended to be either exhaustive or prescriptive. Instead, they provide a structure for teachers to

- think about the learning that they intend to take place in mathematics lessons
- plan for and guide the learning experiences in a way that enables the learners to develop their knowledge and skills
- gain insights into the learners' thinking as the lesson progresses
- gather evidence that the learning outcomes are being achieved
- reflect on the lesson and refine or adapt the plan for future use.

As teachers adopt different teaching and learning approaches in their classrooms, they can develop or adapt other ideas and resources to enrich the learning experiences for their students.

The following pages elaborate the five elements involved in the ***CLEAR*** approach. These are not intended as individual points for separate consideration, but should be viewed as interwoven and interdependent threads in the canvas of mathematics on which learners can create their own picture of mathematical understanding. Examples are given to illustrate a point; these are drawn from topics and learning outcomes in the syllabus. Sometimes, explicit reference is made to key skills to draw attention to the contribution that mathematics teaching and learning makes to their development and reinforcement. While much of the emphasis is on teacher use of this approach, it can equally be adopted by learners.

[3] See www.projectmaths.ie for a range of Teaching and Learning Plans developed by the Project Maths Development Team as well as materials developed by teachers.

C **Connecting the Learning**

- Mathematics learning is linked with students' everyday lives and experience.
- Mathematical knowledge and skills are associated with key skills development.
- Prior knowledge and learning experiences are taken into account when planning learning tasks and activities.
- Classroom experiences are associated with syllabus learning outcomes.
- Individual topics are linked with other topics/syllabus strands and with other subjects or learning experiences.
- Teaching, learning and assessment are inter-connected.

At each stage of learning, it is critical that the learners can connect new knowledge with prior learning experiences, thus building their confidence that they 'can do' mathematics. This connection could be achieved in an effective manner through, for example, an open question or rich task that allows the learners to discuss and share their existing knowledge as they explore the question/task and devise possible ideas for solutions.

Through whole class or small group discussion, the development and reinforcement of the key skills can be facilitated. Care will be needed when planning the question or task so that the resulting class discussion provides the kind of learner engagement and responses which will lead to the new knowledge emerging, or that at least will provide sufficient opportunities for the teacher to link with the intended learning outcomes.

Connections within and between syllabus strands and across subjects, as well as the use of everyday contexts and experience, can help learners to appreciate that mathematics 'makes sense' and can be connected in a variety of ways to everyday life and experience. Thus, for example, an understanding of linear relationships between two variables can be developed from consideration of patterns in which a constant step-change in one (independent) variable gives rise to a constant difference in the values of the other (dependent) variable. The use of the familiar money-box problem, in which a constant amount is added to an initial or start value, allows this to be situated in a realistic context that students can relate to.

Through tabular or graphical representation of individual values for these variables, the linear pattern becomes obvious and a generalised expression can be found in which each term has a 'real' significance in the context of the original 'story'. Such a linear relationship can also emerge during the analysis of data collected in a statistical exercise or in the course of a science experiment.

In the course of this problem, the learner's skill in representing information in tabular or graphical form can be developed and reinforced, and the ground can be prepared for future learning experiences which have as their focus algebraic representation, the use of functional notation, and the solution of simultaneous linear equations.

L Leading the Learning through Rich Tasks

- Rich tasks involve active engagement and discussion which give insights into learners' thinking and expose misconceptions or misunderstanding.
- Rich tasks provide an appropriate (yet enjoyable) challenge, and enable each learner to experience success.
- Learning activities present learners with an appropriately pitched cognitive challenge; support is given in the form of question prompts that guide the learning and develop understanding rather than in the form of immediate answers.
- These tasks and activities engage all learners in solving problems and they facilitate exploration and discussion, encourage communication and collaboration, and lead to shared investigation and decision making.
- Learners devise and evaluate solution strategies rather than being provided with answers for them to memorise.
- Through suitable tasks, or extension activities, learners develop, share and refine their ideas.

Traditionally, mathematics classes involve learners being shown a typical question that focused on one or more mathematical concepts or processes and a method of solution that was generally procedural. This was followed by the learners practising the method of solution on similarly constructed questions (Lyons et al., 2003). More complex questions could then be approached using the same procedures. Student ability to complete such exercises was seen as evidence of learning, and examination questions tended to reward their skill in deciding on and applying the correct procedure accurately. Less emphasis was given to the key skills of critical and creative thinking, communicating, working with others, or learners' understanding of the underlying mathematical concept and its connectedness to a 'bigger picture' view of mathematics.

Rich tasks and questioning, on the other hand, involve learners in exploring the concepts, drawing on prior experience, intuition and imagination, and encourage the devising and refining of solution strategies. They allow learners to test ideas, explain and justify their thinking, adapt their ideas in light of emerging knowledge and so gain a deeper insight and understanding of the concepts involved and develop additional or alternative strategies for problem solving. These tasks and questions can be both challenging and enjoyable, be amenable to more than one approach for solving them, and allow all learners to experience some measure of success in working towards a solution. They provide opportunities for learners to build on prior knowledge and learning, yet also allow progression to more complex tasks and problems.

For example, in statistics, the class could be asked to find out how much time a teenager typically spends playing computer games. Such a task can be undertaken at a simple level by gathering data within the class, extended to the whole school or considered on a global scale. The task can be used to incorporate the use of ICT where, for example, the class group contributes to, or draws information from, data related to a wider population of teenagers through engagement with CensusAtSchool[4]. In each case, statistical knowledge and skills are being developed in a relevant context that will engage the learners, a variety of ideas and proposed strategies for tackling the task can be discussed, decisions made about the feasibility of these, and a chosen strategy implemented to gather or extract relevant data which can then be analysed to arrive at a solution. Such a task can involve all five of the key skills and learners can also develop the mathematical skills of displaying data, performing routine calculations and interpreting results. The task can be extended, or revisited at a later stage, to consider related statistical concepts such as random or representative sampling, bias, gender or age-based comparisons, etc.

As the class engages in this task, the teacher can assess the level of understanding that learners are bringing to the task, gain insights into their thinking—which may expose shortcomings or misconceptions that need to be corrected—and make judgements about the appropriateness of the emerging solution strategies. These insights can enable the teacher to provide additional support or 'scaffolding', probe more deeply into the original problem, or confirm the solution that has been arrived at.

The Student CD and accompanying interactive booklets which have been developed by the Project Maths Development Team[5] contain a range of activities which encourage learners to investigate and confirm their findings, thereby enriching their understanding of the different syllabus topics.

In planning tasks, teachers could ensure that they present an appropriate cognitive challenge for learners, and that this is maintained throughout the lesson. The tendency to 'do the thinking' for them would need to be resisted if learners are to develop their own skills in thinking and problem solving. Care is needed in timing teacher-initiated questions or hints which are intended to guide and prompt the learning so that they do not minimise the cognitive demand or severely restrict the thinking time needed for the learners to benefit fully from the task. Oldham and Close (2009) note that teachers will need time to experiment with a problem-solving approach if they are to engage seriously and substantially with it.

E Extending the Learning

- Extension activities can facilitate differentiation, so that all learners remain engaged, by providing alternative examples and approaches for those who may be experiencing difficulty, or presenting further challenges for those who are capable of progressing more quickly.
- Additional tasks can provide further insights into the underlying concepts and/or link with future learning
- Extension activities can also involve the transfer of knowledge and skills to new contexts and situations.

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[4] See www.censusatschool.ie for further information or examples of data submitted by participating schools.

[5] See www.projectmaths.ie for details.

Continuing the statistics example above, all learners could be kept 'on task' at an appropriate level of engagement. By careful selection of groups at each stage of the task the teacher can ensure that all learners participate in discussions and have an opportunity to contribute. In the course of the task the teacher may be able to identify opportunities for extended treatment, either within the lesson(s) or in planning for future lessons or tasks.

Depending on progress and the time available, ideas that emerge from class discussion at any stage in the task could be explored more fully. They could also prove useful points for the teacher to reflect on when reviewing the lesson. Where time is at a premium, additional constraints could be applied to focus learners on particular aspects of the problem at hand, such as restricting the scope of the statistics task above to the individual class or year group.

While, in this example, the learners are likely to gather real data themselves through their discussions and planning, they can also build an awareness of the issues involved in data collection in general and so be able to relate more knowledgeably to data which has been obtained from other sources or which is presented in media coverage of issues or events of public interest.

A Assessing the Learning

A variety of assessment approaches[6] can enable both teacher and learner to monitor and measure

- understanding of mathematical concepts
- achievement of the learning outcomes
- ability to apply knowledge and skills.

In addition, they can assist the teacher in

- evaluating the suitability of the pace of instruction
- planning the next steps in the learning
- reporting on progress and achievement.

Assessment for Learning (also referred to as *formative assessment*) is the kind of assessment associated more often with the classroom. Its purpose is to use the whole process of assessment to help learners to improve their learning, by providing feedback to the learner on the progress of his/her learning so that he/she can learn more effectively.

When the intended learning outcomes and the purposes of the learning activities are shared with learners, they can become active participants in the learning cycle. They get an insight into, not only *what* they will be learning, but *why* they are learning it, and so can be enabled to

[6] Assessment is the process of gathering, recording, interpreting, using and reporting evidence of learning in individuals, groups or systems, which relies upon a number of instruments, one of which may be a test. Educational assessment provides information about progress in learning.

make some judgements about their own progress in achieving the intended learning. By involving the learners in assessing their own and each other's work, they are motivated to achieve higher levels of attainment and develop better dispositions towards learning.

Asking questions is a natural and important part of the learning cycle. In the course of the lesson(s) a variety of questions can arise. Some of these can be questions posed by the teacher for a variety of reasons, e.g. to check for understanding, to reinforce learning, to clarify significant points in the lesson, and so on. Such questions could be closed (seeking a predetermined 'right' answer) or open (higher order questions that open up a point of discussion and allow for a variety of answers)[7], depending on the teacher's intention in posing the question. A critical aspect of effective questioning is the 'wait time' allowed before accepting answers, making further intervention or moving on to another question.

Learners can also be involved in asking questions, either of each other or of their teacher. They can spend a lot of classroom and homework time answering questions posed by the teacher, but they will benefit considerably when they are encouraged to pose questions themselves around the learning topic. What is intended here is not simply that the students ask the questions and the teacher answers them! Rather, the intention is to encourage students to adopt a broadly interrogative approach to their own learning so that they use questioning to

- explore or unlock key ideas
- identify what they already know
- establish what they need to discover
- help them in more effective note-taking.

Where it is intended to prepare for a continuation or extension of a learning activity, learners could be assigned homework which involves them designing discussion questions around the topic(s) under consideration. The quality of the questions they develop will enable the teacher to get further insights into their understanding of the topic(s).

Assessment of learning (also referred to as *summative assessment*), on the other hand, is the kind of assessment traditionally associated with examinations. Its purpose is to measure achievement, i.e. to find out what the learner has learnt and to report on that. Typically, the results of such assessment take the form of marks or grades.

Under *Project Maths* there is incremental change to the examinations, which sees greater emphasis being given to the assessment of the learners' understanding of the mathematics and their ability to apply mathematical knowledge and skills to solving problems in familiar and unfamiliar contexts. This change reflects and reinforces the changed approach to teaching and learning in mathematics. As a consequence, there is less predictability in the examination questions and the marking schemes adopt a different format and application than in the past[8].

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[7] See *Assessment for Learning (AfL)* at www.ncca.ie

[8] See information on syllabuses and assessment at www.ncca.ie/projectmaths or details of the Project Maths Trialling Report on the State Examinations website: www.examination.ie

R Reflecting on the Learning

Whether as a summary exercise or as an ongoing feature of lessons, reflecting on the learning can enable learners to

- consolidate their understanding
- monitor their own progress
- identify and correct misconceptions

and can enable teachers to

- assess the effectiveness of the lesson(s)
- adapt and improve the classroom experience.

Reflecting on the learning activity forms an intrinsic part of the learning cycle for both learner and teacher. It can be an effective means of drawing the lesson to a close or of summarising a sequence of steps or ideas before moving to another phase in the activity.

For the learner, simple reflection exercises can be used to affirm prior knowledge or to confirm new understanding, to highlight misconceptions that have been corrected and to establish what progress has been achieved. These could take the form of a checklist or a daily/weekly log in which a record is kept by the learner.

For the teacher, reflection can be ongoing throughout the lesson(s) or conducted on completion. In some cases, a simple checklist could suffice to confirm that the learning outcomes are being achieved, that the class is making progress, that the pace of instruction is appropriate, etc. In other cases, examples of learners' work can enable the teacher to gauge how effective particular learning activities have been in achieving the intended learning outcomes and to record which aspects of the activities proved more/less effective than others.

Teacher reflection can also lead to adaptation of or improvement in activities based on the experience in the classroom and the discussions/interactions/ideas that evolved as the lesson(s) progressed. By participating in group reflections, teachers can gain insights into the experience of colleagues and expand their own repertoire of teaching and learning approaches for individual topics or strands. In this regard, the lesson study approach used by Japanese teachers (Conway and Sloane, 2006) suggests that both teachers and their students stand to gain from such collaborative reflection. Where evidence from student work forms the basis for group reflection, teachers can discuss examples of activities/tasks/questions that prove more effective in the classroom or that illustrate particular points of learning, as well as considering what student answers reveal about their thinking and understanding.

The framework set out in this document is a work-in-progress; feedback is welcomed. The experiences of teachers who have adopted this approach in planning for and conducting mathematics lessons would be particularly welcome and informative. Comments and suggestions can be sent to projectmaths@ncca.ie.

REFERENCES

Callan, J. (2006) Developing Schools Enriching Learning – The SCD Experience. Maynooth: NUIM.

Conway, P.F & Sloane F.C. (2006) International Trends in Post-Primary Mathematics Education: Perspectives on Learning, Teaching and Assessment. Dublin: NCCA.

Corcoran, D. (2009). 'Mathematics education as a shared enterprise: learning by 'doing'' in Corcoran, D., Dooley, T., Close, S., and Ward, R. (Eds.) Proceedings Third National Conference on Research in Mathematics Education (pp. 34-52). Dublin: St. Patrick's College.

Le Métais, J. (2003). International developments in upper secondary education: Context, provision and issues. Dublin: National Council for Curriculum and Assessment

Lyons, M., Lynch, K., Close, S., Sheeran, E., and Boland, P. (2003). Inside Classrooms: a Study of Teaching and Learning. Dublin: Institute of Public Administration.

NCCA (2005). Review of mathematics in post-primary education: discussion paper. Dublin: National Council for Curriculum and Assessment.

Oldham, E.E. (2005) 'Solving problems with student-teachers: reactions and reflections 1995-2005' in Close, S., Dooley, T., and Corcoran, D. (Eds.) Proceedings of the First National Conference on Research in Mathematics Education (pp. 250-264). Dublin: St. Patrick's College.

Oldham, E.E. and Close, S. (2009) 'Solving problems in mathematics education: challenges for Project Maths' in in Corcoran, D., Dooley, T., Close, S., and Ward, R. (Eds.) Proceedings Third National Conference on Research in Mathematics Education (pp. 34-52). Dublin: St. Patrick's College.

Sloane, F.C. (2005) 'Japanese Teachers' Lesson Study in Mathematics Education as an exemplar of Design Research' in Close, S., Dooley, T., and Corcoran, D. (Eds.) Proceedings of the First National Conference on Research in Mathematics Education (pp. 250-264). Dublin: St. Patrick's College.

Swan, M. (2009) Standards Unit – Improving learning in mathematics: challenges and strategies. Department for Education and Skills.

LEARNING HOW TO LEARN: EXPLORING AND ENHANCING ACCESSIBILITY AND LEARNING OF ABSTRACT MATHEMATICS AT THIRD-LEVEL

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This case study concerns the learning experience of a group of students attending a core third-year undergraduate pure mathematics module (Metric Spaces). It emerged in response to a need to understand the barriers to accessing the level of abstraction required by the module. It was enabled both by the award of funding from the National Academy for the Integration of Research, Teaching and Learning (NAIRTL) and by the teaching and learning opportunities afforded by a relatively small class, namely 30 students.

This initiative explores how these students responded to the type of learning required by the module and how they were encouraged (or not) by approaches adopted in the module. This paper reports on the attitude towards, development of and facility with proof and proving in such advanced mathematics-major undergraduates.

INTRODUCTION

> Jessie: ..the problem sheet for the first time forced us to be independent and making arguments on our own rather than solving and finding matrices and eigenvalues as usual.

The module Metric Spaces is attended, principally in third year, by students from a variety of degree programmes, though there is a distinct division between those in the denominated degree programme of Financial Mathematics and Economics (FME) and those who are in a dedicated mathematics (BA, BSc or Postgraduate Diploma/Higher Diploma) programme. Typically, the division is further exacerbated by the former group dominating the class size (that year with 20 students from FME in a class of 30). The compelling evidence from the previous two years of teaching this module was of blanket disengagement from those within FME and, by dint of sheer strength of numbers, of this cohort exercising a certain stranglehold on the overall learning experience.

The subject itself has its roots in the real number system, a system that *ought* to be familiar and accessible to the students, and it takes the notion of distance between two points as central. There is tremendous scope for learning through self-discovery as many of the basic building blocks, such as deciding necessary and sufficient conditions for when a point is a limit point of a set, are or should be within reach of such students. Needless to say, establishing such facts requires the ancient practice of verification by mathematical *proof* - and mathematical proof has in my view unfortunately been contaminated by post-primary mathematical experience. Numerous publications by the National Council for Curriculum and Assessment (NCCA Discussion paper 2005, for example) charting the development of Irish mathematics education at secondary level refer to the need for, and the intention to inculcate, this skill. I contend that for many Irish students, proof has become synonymous with both memorisation and with unpleasant mathematical activity; it is hard to recover from such a firmly embedded experience.

Yet recover we must, as understanding and proving are fundamental to mathematics, and they surely must have priority within an undergraduate mathematics-major degree programme. I believe that proof and proving must be rendered essential in students' minds for the purposes of justification and verification, and also for the purposes of explanation and illumination, that is, for understanding. There is of course much literature on proof and proving, and on advanced mathematical thinking generally. Tall (1991) proposes the notion of *concept image* which plays a similar role to the *schema* introduced by Asiala, Brown, DeVries, Dubinsky, Mathews and Thomas (1997). Tall suggests that

> As a result of mentally manipulating a (mathematical) concept an individual develops an idiosyncratic personal concept image which is the product of experience and mental activity. (Tall, 1991, p. xiv)

I had taught this module for two consecutive years, adopting a full-on charm offensive in terms of seeking to energise and engage the students, especially the FME students. My efforts fell on stony ground. After two such arduous years where I encouraged classroom interaction, ran continuous assessment and urged the importance of study and reflection on a weekly basis, I felt the losing battle keenly. It was clear to me that I had to change attitudes to try to improve the mathematical experience for all concerned and conceivably enhance the learning experience. While such was the initial impetus for this pilot project, subsequent and continuing reflection suggests insight that goes beyond programme division and perceived disinterest (on the part of some students, at least) and strikes at the heart of enabling *real learning* at this level.

TEACHING TACTICS

It is pertinent that the subject area, Metric Spaces, is the students' first comprehensive encounter of an abstract analytic subject (that is, a subject that has its roots in the mathematical subject known as Analysis) which of course also has a corresponding impact on their engagement and consequent learning experience. The module is of standard format, carrying 5 ECTS and running for two (lecture) hours per week with a weekly supplemental tutorial and with continuous assessment. Students also carry five other modules in the same semester, at least two of which are within mathematics. As with most pure mathematics modules, this module is characterised by theory and proof, and so understanding and skills of reasoning are key. There is little opportunity for an automated procedural approach to flourish; rather the effort and the emphasis are in 'under the bonnet' advanced mathematical thinking.

My particular focus in teaching the module was to develop the students' competence and therefore confidence in proving results for themselves, indeed on normalising such an activity. I was less concerned with the 'big theorems' of the area but rather that students could (re)create lemmata and theorems on their own.

In an effort to mitigate a tangible groundswell of disinclination combined with the complication of a 'hard' subject area, I introduced some tactics to counteract at least some of the forces at play. In so doing, I wanted to gain some understanding of the background and expectations of the students as well as some sense of how well developed their skills of logic and reasoning were. Rather than have the students attempt to digest a substantial body of Metric Spaces knowledge in the traditional manner, I chose to assemble at least part of that body with the involvement of students. Of necessity, this required a focus on a few central concepts and definitions and a gradual building towards connections between those. I wanted students to create the connections for themselves through logical thinking.

A major resource in this endeavour was the recruitment of a graduate tutor with some expertise in the subject area, the provision of three separate weekly tutorial slots and a regimental approach to the running of those tutorials. In particular, the students were explicitly allocated to one of these groups, with a watchful eye on the composition of each group, and they were instructed to attend that group each week and only that group. Thus a tutorial group had at most *the same* 10 students in the hope that a positive group dynamic might develop. The rationale here was to promote the existence of a valuable forum, albeit in a very forced fashion, for both individual and collaborative group learning and to promote the necessity of weekly and ongoing study. A key change was to develop a team-led approach to assignments whereby, within a given tutorial group, students were assigned to teams that took ultimate responsibility for certain problems from a given problem set. Ultimately, all problems would be attempted by all students with those responsible for specific problems designated as the 'experts' who could then help others solve 'their' problems. Again teams were judiciously selected in an attempt to reduce the perceived effects of intra-programme divisions. The motivation here was again an unashamed attempt to cultivate a culture of mathematical effort and industry through collaboration. The intention was threefold: to excise an unhelpful disposition in some so as to allow us all to get on with the business of mathematical thinking; to incorporate collaboration as a useful and indeed normal mathematical practice and skill, and to stimulate engagement through peer-led learning.

In essence, the scheme was a reaction to my growing suspicion that, when the blanket of indifference is lifted, there are significant issues for a significant number of students in terms of mathematical thinking. It seemed to me that many students actually didn't know *how* to study advanced mathematics for understanding nor even realise the significant time required in the pursuit; that they didn't know therefore the exquisite buzz of enlightenment, of triumph, after the hard graft of concentration and effort. Barton (2010) highlights how *mathematicians speak like addicts about their subject, deep mathematical knowledge is a source of intense and intimate pleasure*. He suggests that the undergraduate mathematics experience might benefit from a design intended to get students similarly addicted. He writes that in order to get people addicted, classical behaviourist theory asserts that *intermittent, irregular reinforcement* is required.

> This may be what research in mathematics provides for its practitioners – expanses of frustration, lots of work, and mazes of dead-ends followed by the rare thrill of a breakthrough. But it is not how most mathematics courses are designed. Students either get little or no reinforcement at all, or get it frequently and regularly for not much effort (Barton, 2010).

It is well documented (Asiala et al., 1997) that progress in mathematical thinking is just not linear; it happens in fits and starts, with ideas and concepts beginning as difficult and awkward yet which, through regular revisiting and interrogation, gradually yield to the learner's grasp. This type of growth necessitates time and effort on an 'early and often' basis. The group initiative was an effort to inculcate such a practice. Formalising group activity in this way seemed to engender greater individual responsibility on the part of many and ultimately greater engagement. It is clear that I found myself adopting the role of *enforcer* in the above strategy. Of course, there were students for whom no such action or methods were required. These students understood how to study and think about mathematics and did not require further supports. However, the overwhelming evidence from the two preceding years alerted me to the needs of many others. What I did not know was the extent to which the perceived level of engagement was a response borne of experience, or whether the situation was compounded by individual stages of mathematical development.

The final examination, worth 70%, remained predictable in nature, with questions covering the module content and associated assignments as usual. Theoretically, rampant rote learning could result in examination success but in practice it did not as the method buckled in the face of material ridden, albeit organically, with demonstrations i.e. proofs. If stirred into action only in the run up to the examination, again the practice of too many in my experience, the fall was all the greater. With this certain knowledge, I had cajoled previous years' students into positive and timely action but to no avail. In terms of study method, the freedom of knowing and knowledge through understanding was considered no match for the burden of the old reliable rote learning with its tried and tested outcomes.

RESEARCH FOCUS AND METHODOLOGY

This paper reports on the following research question:

What does a typical third year student of mathematics understand as his/her role in the learning of advanced abstract mathematics? What constitutes *learning* in this regard?

In terms of methodology, Barrit (1986) offers the following rationale for the legitimate pursuit of qualitative inquiry, namely that it

> is not the discovery of new elements, as in natural scientific study, but rather the heightening of awareness for experience which has been forgotten and overlooked. By heightening awareness and creating dialogue, it is hoped research can lead to better understanding of the way things appear to someone else and through that insight lead to improvements in practice. (Barrit, p. 20)

Accordingly, I chose to explore this research initiative by means of a qualitative approach using audio-recorded semi-structured interviews of duration approximately one hour. I chose to collect this particular form of data in the hope of maximizing the opportunity to learn from these students by way of (interview) conversation. My primary concern was to learn about the nature of mathematical learning and growth as perceived by and experienced by these students. This could only be accomplished by a qualitative strategy, and case study design, with a grounded theoretic aspect, seemed to be the right fit.

Interview Design

Concerning this research, there is no doubt that *an interpretive orientation flows through qualitative research* (Creswell, p. 230) and that considerations of philosophy and paradigm serve to mitigate unintentional prejudice. My position as both teacher *in* the case study and researcher *of* the case study necessarily gives cause for concern. I sought to minimize this conflict by distancing myself from the data collection aspect of the process as far as was possible. Further details concerning this important issue are described below. As to the design of the interview schedule, I sought to encapsulate the central research theme concerning mathematical learning experience through a range of broad-based questions. In addition, I hoped to gain some background information in terms of influences, perception and general motivation as context. I include below a sample of the eleven interview questions most relevant to this paper:

1. Can you remember what your perception was of *learning* mathematics back in the first year of your programme?
2. Has your perception changed in your time here? How and why?
3. Do you have any sense of *becoming* a mathematician through your degree programme?
4. What do you think is required of you in order to learn abstract mathematics?
5. Leaving aside exam and assessment results, did you have any other tangible sense of improving or progressing mathematically in the course?
6. As learning activities, how did you find the assignments for Metric Spaces?

Data Collection

Conscious of my position as lecturer for the module and of the ensuing power relationship that naturally exists, I secured as a key resource for the study the recruitment of an interviewer with a strong mathematics background and with exemplary communication and people skills. I met with the interviewer a number of times to brief her on the way the module was taught, on the interview schedule and on the general purpose of the research. By way of preparation and final review, the interviewer carried out an initial mock interview with a graduate student who had studied the module previously at which I was present.

Data collection commenced early in the second semester and took the form of semi-structured individual interviews as described above. Thus from mid-February, the interviewer invited all members of the former Metric Spaces class to participate. The timing of this, although unplanned, ensured that I was removed from an obvious position of influence in that all assessment of this module had been completed by then. However, that same position as former lecturer of a module which was hardly popular meant that I was particularly reliant on students' good will and generosity to come forward. Possibly a combination of students' immersion in Semester 2's activities, together with a degree of unease or uncertainty around the notion of an interview may account for some of the difficulty experienced in persuading participants, even with the incentive of a raffle. Our efforts netted a sample of eight students from the possible 30, a sample which fortunately represented a broad spectrum of ability, experience and especially degree programme. These participants then received a letter from me describing in greater detail the nature and purpose of the study, the format of the interview and in particular emphasising the degree of anonymity and confidentiality that would protect them. It also alerted them to the intention to audio-record the interview for which their consent would be sought at the time of interview. The interviewing process was completed in April of that year. The data was subsequently transcribed by an independent third party.

Data Analysis

Through repeated immersion in the data transcripts and audio-recordings, I undertook the gradual process of data fragmentation via open and then selective coding. It was possible to do this without the aid of data analysis software due to the relatively small sample. Given my close association with the case study participants, the initial pass through the data was characterized by a certain amount of foreboding. Once I had negotiated this stage, the process became more straightforward. Note that the eight participants have been given pseudonyms in this account. Also the relevant academic year has been concealed throughout to further protect their identities.

STATEMENT AND DISCUSSION OF FINDINGS

Each participant conveyed a fascinating and unique insight into his/her experience of undergraduate mathematics. Their accounts were obviously flavoured by their backgrounds and their expectations. All eight participants had come through the same menu of mathematics modules from first year university in pursuit of a mathematics major degree. Jessie and Daragh, distinguished by European non-Irish post-primary backgrounds, were in the penultimate (third) year of an undenominated BSc programme; Chris and Pat were in their third and final year of an undenominated BA programme; and Alex, Matt, Sam and Jean were penultimate year FME students.

Recall the main motivation for this research initiative, namely to gain some insight as to how students went about learning such a subject – what they thought was required in order to learn it – and also to gauge the attitude towards and the development of proving mathematical assertions. I offer an exposition and discussion of findings under the following headings:

All Kinds of Learning

There is a problem with learning, and it starts with the word itself. Its meaning is ambiguous. The same remark applies to 'knowing'. Thus Jean, in speaking about studying school mathematics, says that she *would always know it from understanding it and not from learning it.* One infers that here *learning* is in fact 'learning off' whilst knowing (from understanding) is the highly desirable state we seek to inspire in all learners of mathematics. All six Irish students acknowledged the particular and indeed accepted role of memorization where it came to proofs in leaving certificate mathematics, a practice which persisted in first year university mathematics. Sam asserts that *memorising is just for the exam whereas, with learning, you are developing yourself in your mind* while Chris offers that *definitely with maths, there are parts you have to memorize. Just to get past the exam.* There is a general view from these six participants that memorization can and does still function well at university: *to be honest even like in 3rd year or in final year the whole learning off thing still goes on (Pat).*

Its purpose however is solely and startlingly for examination success and thus it raises serious questions about the fitness for purpose of examinations at this level, at least as they have been experienced by these students. When asked about the role of memorization, Jean says

> … if you are really stuck and you don't know what you are doing for the exam, it is handy that the theorems are there to learn off. So you can bail yourself out with knowing them.

Consider then the position the students find themselves in when faced with a module entirely made up of theorems (with the accompanying definitions, examples and proofs). The hallmark in learning such mathematics is to learn how to construct knowledge (theorems) from a bank of definitions and concepts, and to create internal connections. Even with limited interest in the subject matter itself, its power to enhance transferable skills of reasoning is significant. To develop such skills, it is necessary to be able to both understand given proofs and to exploit that understanding to derive proofs for oneself. Indeed I believe that this is what lecturers would believe and hope they are nurturing in teaching pure mathematics modules from first year onwards. But this study suggests that something is amiss.

> In a lot of courses, people tend to leave it until the last month to start learning and you might still get very good grades in the exam but after a couple of weeks you have forgotten everything completely (Matt).

Jessie coins the term 'interior learning' to describe the learning that comes from working and thinking continuously over time, the type of learning that she recognizes as meaningful. This is exactly the type of effective learning that comes with such effort. It is the type that I sought to nurture.

Expectations of and for Learning

While each student acknowledged the desirability of understanding any mathematics they encountered, there was some variation as to what that might entail or indeed to what lengths each might be prepared to go. 'Studying' can cover a spectrum from the helpful but potentially superficial reading of lecture notes and other relevant material to 'interrogating' what is read by picking apart proofs and definitions so that they become internalized. Thinking is of course essential, all of which is time consuming and certainly responsive to engagement from the outset. Late starting courted disaster. Each student confirmed how different and difficult they had found the module. *It was probably the hardest subject in maths that we had done until then (Sam)*.

I think that there is an issue of expectation here, on the part of both the lecturer and the student. As a lecturer who has gone further along the path in thinking mathematically, I am all too familiar with the effort required in conceptual understanding. I am also of the opinion that the foundation and development offered over the course of the preceding two years of pure mathematics should at least alert if not accustom students to the demands of mathematical thinking and to successful learning strategies. It is important to note that Metric Spaces is a typical module in a mathematics degree. It is not at all non-standard nor is it unreasonable to ask that students study it; hence my fascination and frustration at the paralysis that either I or it seemed to be causing. Matt's frank observation, having heard about the previous year's FME failure rate in the subject, is eye-opening:

> From the beginning … I made sure I understood everything in every lecture and after every lecture I went over it which I hadn't done with any other course. I definitely feel an understanding for Metric Spaces because of that. Other courses I can't even remember them now from last semester … but I feel Metric Spaces is still in my mind because I gave that probably nearly as much work as the rest of the courses combined last semester … With all the other courses, you just go and read the notes and you can understand it … With Metric Spaces, I found if you missed something, then everything after didn't make sense.

Constructing Knowledge: Proof and Proving

> That is really an undertaking you know, trying to do a proof. I didn't do that in 2nd year ever. We were just learning proofs (Jessie).

While abstraction is one facet of the difficulty experienced in accessing this type of mathematics, another is facility with proof and proving which has, at its core, logical thinking. The latter is a basic underlying principle in a degree in mathematics so knowing how to handle statements such as P implies Q, or P iff Q is essential and likely to be assumed at the level of third year honours students. Theorems are exactly of this form and students typically receive the theorem-proof package for their edification. It is expected that the students will seek to understand the proof but there seems to be little recognition for any further benefit to be derived from considering the proof. At such a level, the given proof seems removed from the students. It is perceived as justification of a given result, the pair forming a rigid inflexible theorem-proof or result-proof unit. Navigation from one unit to another in the manner of invoking a lemma to prove a theorem seems a difficult transition, as does seeking to connect P with Q via implication, or witnessing non-implication via counterexample. These are core skills of proving at this level and yet in reference to facility with proving:

> … even at the start of Metric Spaces, you still wouldn't have got far but after Metric spaces, you were forced to do it and shown how to do it. You were much more equipped to (Matt).

> … you learnt more [in Metric Spaces] about how to write a proof and how to think about that. I did learn how to prove things myself … (Pat).

It seems that *learning proofs* (with its connotations of memorization) and perhaps understanding proofs have been the main interaction that these students have experienced with proof. *Doing* proofs, making one's own proofs, seems quite foreign to their experience.

I learned things that I never learnt before. Metric Spaces is absolutely by far the subject where you learnt more from the point of view of making things myself, making proofs, starting up proofs, finding arguments, finding things. I didn't learn that in any other course before and it's valuable and it changed a lot for me. It changed a lot of my attitude. You went from being completely passive like in 1st year situation where you sit in a big room with hundreds of students and you are completely passive. You went from that stage to being active rather than just sitting there taking in, taking in, taking in. You would have to do things yourself. Things a few months before you would think you couldn't do these things. Mathematicians do this kind of thing. You are a student; you don't do these things. That changed a lot of my point of view. It helped me a lot. It helped me progress a lot the way we had to do simple proofs (Jessie).

This level of surprise is disconcerting, to say the least.

Thinking Mathematically

I believe that it is precisely the extent of time spent thinking as 'encouraged' by assignments that develops mathematical ability and indeed growing stamina and confidence for the next assignment. The assignments were fairly consistent in terms of level of difficulty, with the emphasis always on proof-making. The intention was for basic 'atomic' results to be derived from the students' own thinking and proving, drawing only on a limited amount of content knowledge and logical skills. I underlined the adequacy of understanding only a few key definitions and concepts together with skills of reasoning in order to push through some connection via proof. I hoped to instill some level of self-confidence by favouring quality of thinking and understanding over quantity in terms of content. In essence, students were given a controlled set of 'building blocks' [Jessie] with which to create some theory. The degree of containment and indeed security was noticed:

> …you had very few central ideas that everything came from. There wasn't anywhere too far wrong where we could go which I think helped a lot because we couldn't get it really, really wrong … You were building, building, building and you could see where you were going. I thought that was important … Very little was given to us straight up, you had to think about it first which I think is a good approach … (Pat).

Jessie offered a strong opinion on the assignments, an opinion reflecting Barton's reference to the nature of assessment:

> … the problem sheets in Metric Spaces were very hard at first sight … you start working on the first question. You think and think and think and then you get to something … I found them really good in the sense that they were the centre of the balance exactly. They weren't too easy, they weren't impossible whereas in other subjects maybe you have too easy things or really impossible stuff. Metric Spaces ones were really well calibrated …

Achievement and Affirmation

In my view, there has to be motivation, intrinsic or imposed, to continue in any struggle, and Metric Spaces is no different. While the data certainly highlighted the individual efforts given in the endeavour, there was also recognition of self-improvement. This points in the direction of self-monitoring, another concept mentioned by Barton (2010). This is an attribute associated with self-directed learning that must be nurtured; again, we are brought to the realm of expectation. Students should expect to self-regulate in terms of their own learning, yet there is a strong desire for external pressure in this regard. It is likely that previous strong conditioning in being tested by others (post-primary) has developed a dependence on and expectation of such external forces.

Those elusive 'aha' moments that critically and thankfully buoy the spirits of many flagging professional mathematicians in their research were experienced by some:

> It happened once or twice and I was so excited with myself when it happened … it was like a really good feeling to prove it yourself because you felt like you really learnt something and you really knew stuff (Pat).

However, there is a seemingly unassailable gap between how a lecturer can help in terms of a student's understanding, with the resultant and empowering sense of accomplishment and confidence, and what the student needs to do for him or herself. Jean's sentiment, expressed below, conveys at least the sense of enjoyment she received from her efforts to engage with this module. Unfortunately, the frustration is scarcely concealed: *You love doing [Metric Spaces], you just wish you understood it a bit more.*

Others speak of confidence being instilled by the lecturer; confidence is key but it is hard-won. It comes over time with perseverance, inquiry and of course achievement.

FINAL WORD

This case study has highlighted the significance of expectation and responsibility in terms of learning mathematics in the pursuit of a primary degree therein. This expectation and responsibility concerns student and lecturer alike. That 'thinking for oneself' was cited so often in the data as a feature of the learning experience in the module is disconcerting. Pursuing a degree with mathematics as a major component should have mathematical thinking at its very core. The same holds true for proof and proving. These notions are central to pure mathematics and I believe that they should be cultivated gradually from first year in university. It is startling to hear, from these students at least, that active self-composed proof and proving was beyond their realm of experience and indeed expectation. One has to wonder, and worry, about the scale of this phenomenon.

REFERENCES

Asiala, M., Brown, A., DeVries, D., Dubinsky, E., Mathews, D., & Thomas, K. (1997), A Framework for Research and Curriculum Development in Undergraduate Mathematics Education, *Research in Collegiate Mathematics Education II,* 1-32.

Barrit, L. (1986), Human sciences and the human image*, Phenomenology and Pedagogy, 4 (3),* 14 -22.

Barton, B. (2010), Inaugural Lecture *The Pleasure Principle in Undergraduate Mathematics,* University of Auckland. Transcribed in CULMS Newsletter July 2010. Available at http://www.math.auckland.ac.nz/CULMS/wp-content/uploads/2010/08/CULMS-Newsletter-No11.pdf

Creswell, J. W. (2007*), Qualitative Inquiry & Research Design, Choosing Among Five Approaches*, Sage Publications.

National Council for Curriculum and Assessment. (2005). *Review of Mathematics in Post-Primary Education*. Retrieved from http://www.ncca.ie/en/Publications/Consultative_Documents/Review_of_Mathematics_in_Post-Primary_Education.pdf

Tall, D. (Ed.), (1991), *Advanced Mathematical Thinking*, Kluwer: Dordrecht.

LESSON STUDY AND ITS IMPACT ON PARTICIPATING TEACHERS' ATTITUDES IN RELATION TO TEACHING MATHEMATICS

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This study was concerned with examining the beliefs and attitudes of four teachers, from a particular school, in relation to the teaching of mathematics. It was also interested in investigating the impact participating in one cycle of lesson study would have on these attitudes and beliefs, in particular, in relation to self-efficacy beliefs and collective teacher efficacy. The lesson study intervention was not a one-off event, since other staff groups had already engaged with the process. It is important to note that the teachers in this school were the first in-service teachers in Ireland to participate in lesson study, with a resultant pride in this situation and a possible 'Hawthorne effect'[1].

INTRODUCTION

St. Anne's National School is one of Dublin's inner city schools. It is a full stream, mixed school. At the time of the study there were 200 pupils in the school with approximately 40% of them having English as an additional language (EAL). As a designated DEIS, band one school, St. Anne's National School has targets to meet in relation to mathematics. These targets relate to the children's attitudes to mathematics, the language used in mathematics lessons, a problem solving approach being used in classes and the use of concrete materials in the teaching of mathematics. In order to meet these targets, particularly in relation to teaching using a problem solving approach, the school sought assistance from the mathematics education department in St. Patrick's College, Drumcondra. It was proposed that participating in lesson study would help the teachers develop their skills of teaching through a problem solving approach. Following a presentation on lesson study at a staff meeting the staff was interested to try it. Prior to the study reported here, two groups of teachers in St. Anne's N.S. had participated in a cycle of lesson study, one in the spring term of 2009 and one in the summer term 2009. By this time lesson study had become part of the culture of St. Anne's National School.

LESSON STUDY

Lesson study has become a widely-used and highly-refined methodology, not only in Japan, but also around the world, most notably in the US (Tall & Isoda, 2007). Lesson study consists of a collaborative effort by many teachers in a three part design: preparation *(kyozsi kenkyu)*, lesson *(kenkyu jyogyu)* and review *(jyogyu kentoukai)*. Fundamentally the purpose of Lesson Study in mathematics is to improve not only the performance of students on examinations, but to improve the nature of their mathematical thinking, including not only the ability to perform

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[1] The Hawthorne effect is a form of reactivity whereby subjects improve or modify an aspect of their behaviour being experimentally measured simply in response to the fact that they are being studied, not in response to any particular experimental manipulation.

routine tasks accurately and efficiently, but also to develop the abilities and attitudes to solve novel problems by thinking mathematically in new situations (Tall & Isoda, 2007).

Lesson study consists of cycles of instructional improvement in which teachers work together to formulate goals for student learning and long-term development. The teachers collaboratively plan a "research lesson" designed to bring to life these goals. One team member teaches the research lesson while the others in the group observe the lesson. While the lesson is in progress they collect data and make observations on student learning and development. Following the research lesson the group meets to reflect on and discuss the evidence gathered during the lesson. They use this evidence to improve the lesson and teaching of mathematics more generally. If desired, the lesson may be taught, observed and improved again in one or more additional classrooms (Lewis, 2002). Lesson study is informed by outside expertise, that is, through the participation of knowledgeable others. The knowledgeable other provides a different perspective when reacting to the lesson study work of the group, s/he provides information about subject matter content, new ideas, or reforms and s/he shares the work of other lesson study groups (Fernandez, 2005).

THE STUDY OF PARTICIPATING STAFF

In December 2009 I met with the next staff group who would participate in a cycle of lesson study for a group interview. The four teachers involved included two sixth class teachers, one fifth class teacher and the resource teacher for the three classes. There were two male and two female participants. Their teaching experience ranged from two years to ten years. Apart from Ciara, all the other teachers had taught in at least one other school before teaching in this one. Ciara, the teacher with the least teaching experience, had agreed to teach the research lesson. Individual teachers were interviewed after the research lesson. In addition, I interviewed a group of children from the participating class before and after the research lesson. When analysing the data from the interviews with the teachers in the lesson study group and the observations of their meetings a number of themes began to emerge. Among the most prominent of these were the teachers' self-efficacy beliefs and their sense of collective efficacy.

Efficacy Beliefs

Over three decades ago, Albert Bandura (1997) introduced the concept of self-efficacy perceptions or "beliefs in one's capacity to organize and execute the courses of action required to produce given attainments" (Bandura, 1997, p. 3). Since then researchers have found links between student achievement and three kinds of efficacy beliefs – (a) the self-efficacy judgements of students (Pajares, 1994, 1997), (b) teachers' beliefs in their own instructional efficacy (Tschannen-Moran, Woolfolk Hoy & Hoy, 1998), and (c) teachers' belief about the collective efficacy of their school (Goddard, Hoy, & Woolfolk Hoy, 2000).

Bandura (1986, 1997) proposes four sources of efficacy-shaping information: mastery experience, vicarious experience, social persuasion, and affective state. A mastery experience is the most powerful source of efficacy information. The perception that one's performance has been successful tends to raise efficacy beliefs, contributing to the expectation that performance will be proficient in the future. The perception that one's performance has been a failure tends to lower efficacy beliefs, contributing to the expectation that future performances will be inept. A vicarious experience is one in which the skill in question is modeled by someone else. When a model with whom the observer identifies performs well, the efficacy beliefs of the observer are most likely enhanced. When the model performs poorly, the efficacy beliefs of the observer tend to decrease. Social persuasion may entail encouraging or specific performance feedback from a supervisor or a colleague or it may involve discussions in the staff room, community, or media about the ability of teachers to influence students. Affective state refers to level of arousal, either of anxiety or excitement in relation to an incident which in turn adds to individual's perceptions of self-capability or incompetence. Just as these sources are critical for individuals, they are also important to the development of collective efficacy beliefs. According to Bandura (1997), "perceived personal and collective efficacy differ in the unit of agency, but in both forms efficacy beliefs have similar processes" (p. 478).

Bandura (1993) demonstrated that the effect of collective teacher efficacy on student attainment was stronger than the direct link between social economic status (SES) and student achievement. These findings have particular significance for DEIS schools in Ireland.

Self-efficacy Beliefs in Relation to the Teaching of Mathematics

All four teachers were very willing participants in the initial group interview. They were open and honest in their responses to my questions. There was a sense of camaraderie among the group as they laughed and joked throughout the interview. When asked to rank their enjoyment of teaching mathematics, the frustration that they feel when it comes to teaching mathematics quickly became apparent. Fiona, one of the sixth class teachers, ranked her enjoyment of teaching mathematics about 3 out of 10 because she found it so frustrating when "you're teaching something that seems so obvious and they just can't get it" (Transcript 1, p. 4). She found it "hard really" (Transcript 1, p. 5). Although Shane, the other sixth class teacher, ranked his enjoyment of teaching mathematics as 7 ½ or 8 out of 10, he reported feeling frustrated trying to teach to the wide range of abilities in his class. In an attempt to address the needs of the children in sixth class, the two sixth class teachers and the resource teacher had decided the previous month to stream the children in sixth class into three ability groups. It was decided that each teacher would teach a group for two weeks at a time with each teacher focusing on the same area of the curriculum. The teachers rotated the groups they were teaching every couple of weeks. Shane cited this arrangement as helping to improve his enjoyment of teaching mathematics as did Fiona. Shane reported that prior to this, he came away from every mathematics lesson "feeling a bit guilty. Everyday, I really did". (Transcript 1, p. 4) The other two class teachers agreed with Shane about this feeling of 'guilt'. Fiona claimed that she was ready to give up teaching before they started streaming the children. Although she laughed after stating this, her frustration was clearly evident in her tone of voice. These comments are significant in relation to the literature in relation to teacher efficacy. There was a general feeling of frustration among the teachers that they put in a lot of hard work but don't get the results they expect.

When asked what they enjoy about teaching mathematics there was a moment's silence before anyone responded. Dan cited *money* as being an area of mathematics that he enjoys teaching "because they all 'get' money. No matter where you work everyone understands money except for an odd one or two people won't get it". (Transcript 1, p. 6). Three out of the four teachers enjoyed teaching *data*. Dan felt that one needs to make one's mathematics lesson as "hands on as possible, whether it's *fractions* or anything, they'll enjoy it and then you'll enjoy it" (Transcript 1, p. 6). The strand units of *money* and *data* were two areas of the mathematics curriculum where the teachers felt that they had experienced success in their teaching and the students learning. Ciara enjoyed doing mental mathematics with her class.

When asked if there was any aspect of the mathematics curriculum that they might be apprehensive about teaching the response was immediate and definite. The teachers cited making the connection between *fractions, decimals* and *percentages* as being difficult to teach. They could also quickly list a number of difficulties they faced in their mathematics teaching. They reported poor language, reading and concentration skills, overload of the curriculum and the wide range of abilities in each class as being some of the factors that make teaching mathematics difficult. These comments indicate a need to promote positive teacher efficacy in these teachers because efficacy beliefs determine how environmental opportunities and impediments are perceived (Bandura, 2006).

In response to being asked if they felt that they had sufficient training to deliver the mathematics curriculum all four teachers agreed that you could never have enough training and they all expressed a willingness and interest to participate in continuous professional development in relation to teaching mathematics. Dan recalled having a *cuiditheoir* for mathematics visit the school in which he was working at the time. He deemed this to be "brilliant … He gave so many good ideas to hook it into your head, where it starts off and just go in and try to introduce it" (Transcript 1, p.11). This is an example of a vicarious experience whereby the skill in question was modeled by someone else. It was hoped that participation in lesson study would provide such positive vicarious experiences.

Overall, prior to engaging with lesson study the four teachers expressed a feeling of 'guilt' in relation to teaching mathematics because they felt unable to meet the needs of every child. The two sixth class teachers felt unable to deliver the mathematics curriculum without the support provided by streaming the two classes. All four teachers could list without delay the difficulties that they faced in teaching mathematics and highlighted the areas in mathematics that they felt apprehensive about teaching. However, identifying areas of the curriculum that they enjoyed teaching was more difficult for them. They each cited strand units of the curriculum where they had experienced success in their teaching as areas of the curriculum that they enjoyed teaching.

Research Lesson Teacher Profile

Ciara taught the research lesson on *Chance* to her fifth class. There were 16 pupils in her class, nine girls and seven boys. Ciara listed many difficulties that she encountered in her teaching of mathematics, including the children having poor reading skills, poor concentration skills, no self correcting skills and a wide range of abilities. Ciara stated that she had five groups of differing ability within her class of 16 children. Ciara found their "can't do it" attitude very frustrating. Ciara encouraged the children to spend some of the time working independently as she believed it was very important for their self-esteem. In general, Ciara found teaching mathematics exhausting and draining. According to Ciara teaching mathematics is "… very intense and you come out of it exhausted … it's very draining …" (Transcript 1, p. 10). She reported that she felt "uptight" (Transcript 1, p. 19) when teaching mathematics. Ciara spent a lot of time on preparing her mathematics lessons. In general, Ciara did not do whole class teaching as she did not see any benefit of it in her classroom. This teacher belief may or may not be widespread, but it contravenes the more usual approach to whole class teaching as prescribed by the curriculum. A recent study by Dooley (2010) found that deep mathematical insights could be constructed through whole class interactions, where the style of mathematics teaching allows for pupil agency and presumably teacher self-efficacy is high. Despite the many challenges Ciara faced in her teaching she described her class as a "nice bunch" who were "eager to learn and who are quite good at hands on activities" (Transcript 1, p. 1).

Preparing the Research Lesson

At the initial meeting with the knowledgeable other, Ciara explained that she had chosen the area of *chance* for the research lesson because it was an area of mathematics that she felt unsure about teaching. She felt that she lacked the relevant content knowledge to teach this strand unit effectively. The other three teachers also reported that they were unsure about teaching *chance.* Throughout that initial meeting Ciara did most of the talking. The other three teachers did not contribute much. However, with each meeting the other three teachers contributed more and more. They made suggestions for suitable activities and how the activities should be organised and the lesson delivered. They each gave their opinions on how suitable they thought the language was that they planned to use throughout the research lesson. All four teachers contributed to the discussions in relation to classroom management and the grouping of the students for the research lesson. Collective efficacy was evidently building with the group over the course of the preparation sessions.

Collaboration and Collective Teacher Efficacy Pre-'Lesson Study'

All three teachers reported that they enjoyed working and planning with others. "It's a great thing", (Transcript 1, p. 14) according to Shane. Dan reported that working collaboratively was beneficial as it allowed them to share ideas and learn from each other. Fiona felt that "it's less pressure on you as well because if you're doing something that someone else is doing then you can't go too far off the beaten track." (Transcript 1, p. 14). Shane and Dan agreed with Fiona. They also found it "reassuring" and Dan and stated that "it builds up your confidence when you find out … I'm actually doing all this stuff anyway so I must be doing something right." (Transcript 1, p. 14) Shane said that grouping the children "makes it easier on the teacher to teach them because you don't feel as bad." (Transcript 1, p. 9) Ciara found that, on the few occasions that she had in-class support, she was more relaxed. According to Goddard, Hoy and Woolfolk Hoy, (2004) schools with strong beliefs in group capability can endure pressure and crises and carry on functioning without incapacitating consequences. In addition they can learn to rise to the challenge when faced with disruptive forces. Morgan, Ludlow, Kitching, O'Leary and Clarke (2009) found that a major implication for teacher job satisfaction is the suggestion that while adverse episodes may be inevitably experienced, positive events (that occur independent of negative ones) fortify motivation and resilience.

These findings and the comments of the participating teacher would suggest that it is imperative for all schools, particularly DEIS schools, to enhance collective teacher efficacy. The deployment of Learning Support and Resource teachers may be highly useful in this regard.

Negative Aspects

The group also highlighted some of the difficulties associated with working and planning with other teachers. Due to the absence of any timetabled formal planning time, planning tends to be informal. Dan informed me that "Shane tells us what we're doing" (Transcript 1, p. 13) and Fiona stated that "I'm quite happy to be told what to do" (Transcript 1, p. 13). Although the three teachers agreed on what strand or strand unit is to be taught at any given time there was no opportunity to discuss in detail exactly what content would be covered and how it would be taught. The following extract demonstrates the difficulties that can occur when the teachers don't have time for detailed planning:

Shane: The problem solving actually was enjoyable enough but I think the problems were quite simple.

Fiona: But you see you had the top group for the problems. I had the middle group and I was bashing my head off the wall and I was really sick of problems.

Shane: Really?

Fiona: Yeah.

Dan: I had the bottom group and they couldn't read them (Transcript 1, p. 6).

Through engagement in lesson study the teachers and the knowledgeable other form a community of practice (Wenger, 1998) and are mutually engaged in a worthwhile enterprise (Corcoran, 2008). Through sustained participation in this community of practice and mutual engagement with the task of improving the teaching of mathematics it appears that any difficulties arising due to differing personalities may be overcome. Shane stated that they were unsure of whether they were allocating the correct amount of time to each strand unit, "we wouldn't be one hundred percent sure that we're allocating the right time" (Transcript 1, p. 14). When there were absent personnel or interruptions to the school day the two sixth classes cannot be streamed for mathematics. From working with one ability group at a time, Fiona then felt unable to return to teaching her whole class together, "I don't think I could handle going back to the old way" (Transcript 1, p. 11). While the streaming arrangement for teaching mathematics has improved the attitude of Fiona in relation to teaching mathematics, it may not be conducive to pupil achievement. Collaboratively negotiated practices and discussion of goals for teaching arise out of the practice of lesson study, so the practice of streaming for mathematics may well be revisited by this school staff.

Self-efficacy Post-'Lesson Study'

Each member of the lesson study group was interviewed individually a few weeks after the research lesson. Again, each teacher was very open and honest in his/her responses to my questions. Ciara, the teacher who taught the research lesson thought that the lesson went "really well" (Transcript 3, p. 1). Previously, Ciara ranked her enjoyment of teaching mathematics as 5 out of 10. She ranked her enjoyment of teaching the research lesson as 7 out of 10. She attributed the increase in her enjoyment of teaching the lesson to how organized she was. She felt that having all the relevant resources available and ready to be used contributed to the smooth running of the lesson. The other three teachers thought that they would enjoy teaching the research lesson in the future. Shane thought that the "hands on approach" (Transcript 4, p. 2), the way the students were grouped and the fun element of the lesson contributed to making the lesson go "nice and fast and that's enjoyable for them (the students) and the teacher" (Transcript 4, p. 2). Dan felt that it would be "fun" to teach the research lesson (Transcript 5, p. 1). Fiona stated that she thought she would enjoy teaching the research lesson because all the necessary resources were organised and available to be used. These comments indicate the complexity of the teacher's task in teaching mathematics, where so many variables have to be juggled. While not directly involved in teaching the research lesson, the vicarious experience of observing a successful lesson has resulted in the observing teachers feeling motivated and excited to teach the research lesson. This may have been enhanced due to the teachers themselves contributing to the planning of the lesson and making and planning the use of the resources.

Ciara reported a decrease in the feelings of 'guilt' in relation to teaching mathematics due to the way the children were grouped throughout the lesson. She considered that "there were lots of ways that people could shine in that lesson" (Transcript 3, p. 2). This realisation resonates with findings of Boaler (2008) who states that mathematics achievement is higher in classrooms where the teacher establishes that there are multiple ways to succeed and be good at mathematics. Shane didn't think that he would experience any feelings of 'guilt' were he to teach the same research lesson and that having the opportunity to participate in further lesson study cycles would help his overall feelings of 'guilt' when it comes to teaching mathematics. Dan agreed with this. However, although Fiona thought that she would enjoy teaching the research lesson, when asked did she think that participating in further lesson study cycles would reduce her feelings of 'guilt' associated with teaching mathematics she replied, "Not really. I think it's just down to being able to divide the kids up into abilities" (Transcript 6, p. 2). Fiona, despite her vicarious experience of success observing the teaching of the research lesson,still exhibited robust beliefs about how best to cope with the difficulties she faces in the teaching of mathematics.

Confidence Increased

All four teachers stated that they felt more confident to teach the strand unit of *chance* after participating in the lesson study cycle. The increase in confidence was attributed to improved content knowledge and knowledge of suitable activities and teaching methodologies. According to Dan, "any lesson like that where you've seen it done, it's so fresh in the mind, you've no problem doing it … You'd struggle to mess it up" (Transcript 5, p. 3). Three out of the four teachers felt that as time went by their confidence might diminish. The following statement by Fiona was echoed by Dan and Shane: "I think I would have felt more confident the week after we did it because I just find you start forgetting bits the more time lapses" (Transcript 6, p. 3). All four teachers cited getting ideas from others as one of the ways that they increased their mathematics pedagogical content knowledge. Ciara considered that participating in lesson study helped with "your personal development in the sense of having the opportunity to work with others which is rare" (Transcript 3, p. 7). She also thought that "it raises your confidence levels" because "if you can teach in front of a group of your peers, you know it's a big deal" (Transcript 3, p. 8). Mastery experiences are regarded as the most influential source of self-efficacy (Bong & Skaalvik, 2003; Pajares, 1997). Outcomes interpreted as successful raise self-efficacy. All four participating teachers felt that the research lesson was a success. However, participation needs to be ongoing to support prolonged feelings of self-confidence.

Knowledgeable Other

Shane acknowledged that the role of the knowledgeable other was important to the process due to her level of expertise in the area: "She's obviously a woman who is very experienced in mathematics … it's good to get other people's opinions, especially hers with her experience" (Transcript 4, p. 3). Ciara referred to the knowledgeable other as her "safety blanket" and "the queen bee of what could really work" (Transcript 3, p. 3). Persuasive communication may also raise self-efficacy. It is most effective when those who convey the efficacy information are viewed as competent and reliable (Skaalvik & Skaalvik, 2007), as the knowledgeable other was by the participating teachers. Knowledgeable others also sometimes act as cheerleaders to encourage teachers to persist in the process (Watanabe & Wang-Iverson, 2005). This single cycle of collaboration appears to have raised the self-efficacy of the participating teachers.

Collaboration and Collective Teacher Efficacy Post-'Research Lesson'

During the interviews that took place with each individual teacher after the research lesson, they all expressed how committed and motivated they felt that everyone was participating in the lesson study cycle. Prior to the research lesson the group met with the knowledgeable other and also met by themselves on two other occasions. Ciara reported that "… we went over the time. We decided to have a few more meetings ourselves and that was purely because we needed the time ourselves and we took it upon ourselves to arrange that …" (Transcript 3, p. 6). She also asserted that "if we had needed more time we would have took the time, end of story. We were, as I said, all very motivated to do it, so there was no hassle and nobody was getting fed up or in bad form. We all wanted to do it and we all wanted it to be a successful lesson" (Transcript 3, p. 6). According to Shane, "everybody was giving one hundred percent" (Transcript 4, p. 1). Each teacher conveyed satisfaction at how well they worked as a team. Shane reported that, "it was very good the way we all got together … we worked our way through it" (Transcript 4, p. 1). This commitment to the process of lesson study, as demonstrated by the four teachers, supports the findings of Goddard et al. (2004), that high expectations for action create a normative press that encourages all teachers to do what it takes to excel and discourages them from giving up when faced with difficult obstacles. Such community building was also found among lesson study practitioners by Lewis, Perry and Hurd (2009). Engagement in lesson study created high expectations among the teachers and it was the same process that enabled them to surmount any difficulties that they faced in planning and delivering the research lesson.

Learning from Others

All four teachers cited getting ideas from others in relation to teaching mathematics as one of the main benefits of participating in lesson study. Dan thought that "the process is brilliant because it's all the brains coming together and you're getting ideas off different people …" (Transcript 5, p. 1). He elaborated further, "You see how different people think, their way of looking at it. Everyone brings their strengths to the table …" (Transcript 5, p. 1). Ciara thought that "… it was really great to learn from other people and it's so rare to have the opportunity to see anyone else teaching …" (Transcript 3, p. 5). According to Shane, "… I think it's great just meeting and talking about maths in general … Get other ideas from the other teachers" (Transcript 4, p. 1). Fiona also felt that this was one of the advantages of participating in lesson study. She stated, "What I thought was good … was that we picked each others brains and we got good ideas from each other. It wasn't like it was just the same person coming up with all the ideas" (Transcript 6, p. 1). These examples of teachers' experiences of engagement with lesson study support the concept that a robust sense of group capability establishes a strong press for collective performance (Goddard et al., 2004).

One of the highlights of participating in lesson study reported by the four teachers was the opportunity to work with others. Ciara stated that "I really enjoyed it because it's rare [working with other teachers]. You never have the opportunity to do it" (Transcript 3, p. 5). The teachers felt supported by one another. According to Ciara, "They were just a great support and it was such a good environment to be working in because everyone was very open and … we were all very motivated and enthusiastic" (Transcript 3, p. 5). Dan also reported that there was a "feeling of openness" (Transcript 5, p. 6) and that "when you've got more people then you can use the resources and the ideas come and you say how we will overcome this problem and someone will come up with an idea" (Transcript 5, p. 2). Fiona thought that they "all pulled together well" (Transcript 6, p. 1). The importance of support from the wider school community was also acknowledged by Dan. He considered that, "… we're lucky here because in other schools you wouldn't have people so willing to supervise classes" (Transcript 5, p. 1).

The four participating teachers felt that they had a good working relationship and this added to the overall experience of lesson study. Shane described the whole experience of working with other teachers as

> Excellent ... I really enjoyed it, especially if you get on well with them as we did, as I say it just freshens up your ideas. It just so happened that the three of us were singing off the same sheet as in we were trying to get the job done as best we could. (Transcript 4, p. 4)

Fiona also considered having a good working relationship as important to the success of lesson study. "I thought we were a good team and that definitely helps" (Transcript 6, p. 5).

By engaging in the lesson study process, the participating teachers came together as a cohesive group to collaboratively plan and deliver the research lesson. A positive working relationship improved the teachers' awareness of one another's strengths, therefore, enhancing a strong belief in the group's capabilities. As a result of this, the collective efficacy of the group was enhanced.

Shane deemed lesson study to be of significance to the teachers involved because it gives you the opportunity to "get stuff off your chest" (Transcript 4, p. 1). He described how the experience gave them the opportunity to empathise with one another. It allowed them to:

> … just sit down and have a chat about it (teaching mathematics), what's going wrong, what's going right and even little things like you'd be frustrated at the end of a lesson and to hear another couple of teachers saying the same thing then you know that you're not on your own. I think some teachers forget that and go home thinking, it's me! (Transcript 4, p. 5)

When describing how they planned the research lesson all four teachers consistently used the pronoun "we". According to Dan "we had our [introductory mathematics] problem … we minimised the reading" (Transcript 5, p. 7). Shane described how "we came up with the idea" (Transcript 4, p. 1). At the initial meeting with Jane, Ciara did most of the talking and naturally the pronoun "I" was used most of the time. This illustrates the change in attitude in the teachers, from lesson study being about a lesson for Ciara's class to being about a group endeavour. At the post research lesson meeting the successes of the lesson were attributed to the group as were any weaknesses. This type of mastery experience is important for any organization. As a group, teachers experience successes and failures. Successes build a robust belief in the faculty's sense of collective efficacy. Ciara explained the dynamics of the group in the following statement: "We came up with the ideas and then she (knowledgeable other) questioned our ideas and then we questioned our ideas and then we came up with a group answer" (Transcript 3, p. 3). This represents a change in the groups dynamic. Prior to participating in lesson study one of the teachers described how "Shane tells us what to do" (Transcript 1, p. 4) when it came to planning in mathematics. It was unclear why Shane adopted this role. However, regardless of the reason for the group's deference to Shane when it came to planning, there was a change evident in the group dynamic after the research lesson had been taught. It changed from the group being instructed by one individual to all four teachers working together as a team and negotiating a shared approach. The experience of having to work together to overcome any difficulties may have helped to promote a strong sense of collective efficacy. According to Huber (1996), a resilient sense of collective efficacy probably requires experience in overcoming difficulties through persistent effort.

In each of the interviews with the individual teachers there was a sense of respect and admiration for one another. They acknowledged the support that they gave one another in practical terms by giving their ideas and their opinions and sourcing the required resources as well as the support that was provided by allowing them to express any concerns they had in relation to teaching mathematics. They also described the sense of group efficacy, the fact that they were all motivated to work together to produce the best lesson possible.

DISCUSSION

Analysis of the data suggests that teacher self-efficacy in relation to teaching mathematics was enhanced by engaging in this single cycle of lesson study. An increased enjoyment of teaching mathematics, reduced feelings of 'guilt' and an increased confidence to teach the strand unit of *chance* were reported. The role of the knowledgeable other was also highlighted and may have helped raise teacher self-efficacy.

Collective teacher efficacy also appears to have been enhanced by engaging in the lesson study process. One of the highlights of participating in lesson study reported by the four teachers was the opportunity to work with others. They acknowledged the support that they gave one another in practical terms by giving their ideas and their opinions and sourcing the required resources as well as the support that was provided by allowing them to express any concerns they had in relation to teaching mathematics. The data also suggests that the attitudes of the children improved in relation to problem solving and working collaboratively with others.

Teacher self-efficacy has been shown to predict student motivation and achievement; students' self-efficacy and attitudes; teachers' goals and aspirations; teachers' attitudes toward innovation and change; teachers' use of teaching strategies and their commitment to teaching. Bandura (1993) demonstrated that the effect of collective teacher efficacy on student attainment was stronger than the direct link between social economic status (SES) and student achievement. A practical implication of this reasoning may be that one should attempt to raise teachers' competencies collectively through sustained staff development programmes such as lesson study.

CONCLUSION

In 2006, Lewis, Perry and Murata, posed a challenge to researchers to investigate how lesson study improves "instruction". They suggested three possible routes: 1) Teachers' knowledge; 2) Teachers' commitment and community; and 3) Learning resources. In this study, the greatest benefit I found was the impact that participating in one cycle of lesson study had on teacher self-efficacy and collective teacher efficacy.

REFERENCES

Bandura, A. (1986). Social foundations of thought and action. A social cognitive theory. Englewood Cliffs, NJ: Prentice-Hall.

Bandura, A. (1993). Perceived self-efficacy in cognitive development and functioning. Educational Psychologist, 28(2), 117-148.

Bandura, A. (1997). Self-efficacy: The exercise of control. New York: W.H. Freeman and Company.

Bandura, A. (2006). Toward a psychology of human agency. Perspectives on Psychological Science, 1(2), 164-180.

Boaler, J. (2008). Promoting 'relational equity' and high mathematics through an innovative mixed-ability approach. Retrieved on 31 May, 2010 from http://www.unige.ch/math/EnsMath/Rome2008/WG3/Papers/Boaler.pdf

Bong, M. & Skaalvik, E. (2003). Academic self-concept and self-efficacy: How different are they really? Educational Psychology Review. 15, 1-40.

Corcoran, D. (2008). Developing mathematical knowledge for teaching: a three-tiered study of Irish pre-service teachers. PhD theis. University of Cambridge.

Dooley, T. (2010). The construction of mathematical insight by pupils in whole-class conversation. Unpublished PhD thesis, University of Cambridge.

Fernandez, C. (2005). Lesson study: A mean for elementary teachers to develop knowledge of mathematics needed for reform-minded teaching? Mathematical Thinking and Learning, 7(4), 265-289.

Goddard, R., Hoy, W. & Woolfolk Hoy, A. (2004). Collective efficacy beliefs: Theoretical developments, empirical evidence, and future directions. Educational Researcher, 33(3), 3-13.

Goddard, R., Hoy, W. & Woolfolk Hoy, H. (2000). Teachers' sense of efficacy and the organizational health of schools. The Elementary School Journal, 93, 356-372.

Huber, G. (1996). Organizational learning: The contributing processes and literatures. In M. Cohen, & L. Sproull (Eds.), Organizational learning (pp. 124-162). Thousand Oaks, CA: Sage.

Lewis, C. (2002). Lesson study: A handbook of teacher-led instructional change. Philadelphia, PA: Research for Better Schools.

Lewis, C., Perry, R. & Hurd, J. (2009). Improving mathematics instruction through lesson study: a theoretical model and North American case. Mathematics Teacher Education. 12, 285-304.

Lewis, C., Perry, R. & Murata, A. (2006). How should research contribute to instructional improvement? The case for lesson study. Educational Researcher, 356(3), 3-14.

Morgan, M., Ludlow, L., Kitching, K., O'Leary, M. & Clarke, A. (2009). What makes teachers tick? Sustaining events in new teachers' lives. British Educational Research Association, 1-18.

Pajares, F. (1994). Role of self-efficacy and self-concept beliefs in mathematical problem solving: A path analysis. Journal of Educational Psychology, 86, 193-203.

Pajares, F. (1997). Current directions in self-efficacy research. In H. Marsh, R. Craven & D. Mc Inerney (Eds.), International advances in self research (pp. 1-49). Greenwich, CT: Information Age.

Skaalvik, E. & Skaalvik, S. (2007). Dimensions of teacher self-efficacy and relations with strain factors, perceived collective teacher efficacy, and teacher burnout. Journal of Educational Psychology, 99(3), 611-625.

Tall, D. & Isoda, M. (2007). Long-term development of mathematical thinking and lesson study. Retrieved on November 9, 2009 from http://www.warwick.ac.uk/staff/David.Tall/pdfs/dot2007x-tall-isoda-lessonstudy.pdf

Tschannen-Moran, M., Woolfolk Hoy, A. & Hoy, W. K. (1998). Teacher efficacy: Its meaning and measure. Review of Educational Research, 21, 747-766.

Watanabe, T. & Wang-Iverson, P. (2005). The role of the knowledgeable others. In P. Wang-Iverson & M. Yoshida (Eds.), *Building our understanding of lesson study.* (pp. 85-92). Philadelphia, PA: Research for Better Schools.

Wenger, E. (1998). *Communities of Practice.* UK: Cambridge University Press.

THEORY TO PRACTICE: MATHEMATICS EDUCATION RESEARCH IMPACTING PRACTICE IN AN ADVANCED MATHEMATICS MODULE

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In this paper, I will describe how some of the mathematics education research at the third level (theory) has provided me with a framework with which to view my students' learning, and has impacted my teaching (practice) of an advanced mathematics module over a number of years. A result of this engagement with the literature has been the evolution of two teaching and learning tools – the Portfolio of Examples *and* In-class Exercises*. In the latter part of the paper I will describe the implementation of the* In-class Exercises *and attempt to evaluate this tool from both the lecturer's perspective and the students' perspectives.*

INTRODUCTION

In 2002, I taught a second-year Analysis module in University College Dublin (UCD) for the first time. With little experience in teaching what might be called an "advanced mathematics" course, I presented the material the only way I knew how - in a definition-theorem-proof (DTP) style. The final examination reflected what was taught: state definition X; state and prove theorem Y (as seen in lectures); prove corollary Z (either unseen or seen in problem sheets). However on grading the final papers several questions arose which caused me unease with the approach I had taken, specifically: if a student cannot state a definition correctly, then surely she should not be able to present a proof of a theorem involving that definition? What if a student correctly presents the first and last parts of a proof, but the middle part is an extract from a completely different proof - should she get any marks? Even if a proof is perfectly correct but an exact replica of the one from my notes, how do I know whether she has genuinely validated this proof for herself, or has simply memorised it? With little exposure to proof at this stage in her studies, is it reasonable to expect a student to prove an unseen corollary or result?

With an existing desire to learn about mathematics education research at the third-level, I turned to the literature with my questions. Despite being a relatively new area of research, I found much to interest me (Meehan, 2002). I have taught Analysis eight times since then and feel that my presentation of the concepts has evolved as I have engaged with the literature. In the first part of this paper, I will describe how some of the research in the area of advanced mathematical thinking (theory) has provided me with a framework with which to view my students' learning, and has impacted my teaching (practice), resulting in the evolution of two teaching and learning tools – the *Portfolio of Examples* and *In-class Exercises*. In the latter part of the paper I will describe the implementation of the *In-class Exercises* and attempt to evaluate this tool from both the lecturer's perspective and the students' perspectives. A detailed description of the *Portfolio of Examples* can be found in Curley and Meehan (2010).

THEORY TO PRACTICE

In presenting this section of the paper, I would like to make explicit how the mathematics education research that I engaged with has impacted my teaching over the last number of years. For this reason, I will present both the theory (literature review and theoretical framework) and practice together.

Theory – DTP model of mathematics

The definition-theorem-proof (DTP) model of presenting advanced mathematics at third-level has received considerable criticism in the literature. Dreyfus (1991) claims that students who have been taught using the DTP model "have been taught the products of the activity of scores of mathematicians in their final form, but they have not gained insight into the processes that have led mathematicians to create these products" (p. 28). Similarly, Burn (2002) describes this model of teaching as presenting students with the product of mathematical thought, while missing out on the genesis of this product. William Thurston (1994), who was awarded a Fields Medal in 1982, warns not only of the dangers of teaching using the DTP model, he also cautions mathematicians about presenting research results solely in this form. If, as he claims, a key role of the mathematician is to help people better understand the subject then he states that as a mathematical community "we need to pay much more attention to communicating not just our definitions, theorems, and proofs, but also our ways of thinking" (p. 168).

It should be noted however that presenting advanced mathematics using a DTP model does not mean that the lecturer puts no effort into her teaching, or indeed that the students fail to learn. Weber (2004) acknowledges that "Lecture styles that fall under the umbrella of traditional DTP instruction may vary widely and lead to drastically different learning on the part of the students" (p. 131).

Practice

On reflection, after teaching Analysis for the first time my main concern was not so much with how I had presented the material, but with how it was assessed. Suppose a final examination in advanced mathematics requires students to state some definitions, state and prove theorems discussed in class, and prove some 'small' unseen results or results from the problem sheets. If a student is awarded 50% in this examination, then what does this mean? Does she understand half the results, or does she 'half-understand' the results? Alternatively if a student receives full-marks on the examination, does this mean she understands everything or that she has a really good memory? And what exactly does it mean to say she 'understands everything' in this context?

Given that assessment drives learning, as long as I continued to examine the 'final product', I felt that students could pass my module with the aid of a good memory. I needed new assessment methods but given that what and how we assess should be rooted in what and how we teach, I needed new ways to present the material. To do this I needed a new way to think about my students' learning – I needed a theoretical framework.

Theory – the "D" in DTP

A well-known paper in the mathematics education literature at the third level is that by Tall and Vinner (1981) where they introduce the notion of concept image as "the total cognitive structure that is associated with the concept, which includes all the mental pictures and associated properties and processes. It is built up over the years through experiences of all kinds, changing as the individual meets new stimuli and matures." (p. 152). They observe that a person's concept image need not be consistent and that by evoking two "cognitive conflict factors" (p. 154) at the same time, one may become aware of the inconsistency and strive to resolve it, resulting in a more coherent, richer concept image. The concept definition is considered to be a formal description of the concept. The authors present several examples to illustrate the types of difficulties that students can encounter in Calculus and Analysis when the concept definition is not part of, or does not coincide with, their individual concept images (Tall & Vinner, 1981; Vinner, 1991).

Of course it is not surprising that many of the difficulties encountered by students in beginning proof-based courses revolve around the role of definitions in proof. Based on his study of sixteen students in a transition-to-proof course, Moore (1994) found that students experienced difficulty with proving because they could not state definitions, had inadequate concept images, were unable and/or unwilling to generate examples, and were unaware of how to use definitions to provide a proof framework. Alcock and Simpson (2002) give examples of two students attempting to prove a result in Analysis. One experiences difficulty because she has an inadequate concept image, the other because he does not appreciate the role played by formal definitions in proof production.

The "D" in DTP became my starting point. I conjectured that before students could attempt to understand relationships between concepts, or validate and produce proofs, they must have rich, coherent concept images that are consistent with, and contain, the formal mathematical definition. My question was:

> How can we encourage our students to develop rich concept images that accurately portray, and include, the definition of the concept? (Curley & Meehan, 2010, p. 49)

The literature pointed to example generation exercises as a possible solution. Dahlberg and Housman (1997) observed that on presenting students with the definition of a new concept, those who employed example generation made the most progress in understanding the concept. In addition, Moore (1994) found that the inability to generate examples was a major stumbling block for his students in attempting to prove. More explicitly, Mason and Watson (2001, 2005) advocate that students should be encouraged to construct "example spaces" in order to develop their concept images and they suggest some techniques to do this. With the help of their technique for generating "boundary examples", the *Portfolio of Examples* exercise was born.

Practice

Over the first half of the semester, the students are given example generation exercises each week to complete as homework. The examples are kept in a portfolio to be handed in towards the end of the semester. The *Portfolio of Examples* exercise constitutes 20% of the final mark.

Taking Spring Semester 2011 as an example, each student was required to generate a total of 115 examples in order to complete this exercise. Many of the example generation requests were stated in words, for example, "Give an example of a sequence that converges to one", but a significant number were also presented using mathematical notation, for example, "Give an example of a sequence (a_n) such that $\forall \varepsilon > 0, \exists N \in \mathbf{N}$ s. t. $|a_n - 1| < \varepsilon, \ \forall n \geq N$". Curley and Meehan (2010) present a more detailed description and evaluation of this exercise.

With the *Portfolio of Examples* exercise designed to encourage students to develop rich concept images that accurately portray, and include, the definition of the concept, I felt that the next step should be to encourage them to start making links and examining relationships between concept images - I turned my attention to the "T" in "DTP".

Theory – the "T" in DTP

On reflection, the first few times I taught Analysis I believe that in emphasising the "P" I may have overlooked the "T". In doing so, I think the student may have been robbed of a valuable learning opportunity. If we give a student a mathematical statement and ask her to prove it or show her a proof, how can we be sure that she ever engages with what the theorem is saying or even stops to think about whether it is true? And if this step is overlooked, then why should she care about the proof? I needed to find a way to reinstate the "T" in my lectures.

Burn (2002) makes the case for involving undergraduate mathematics students in making, refining and examining conjectures. He notes that "A student's ownership of a conjecture is the basis of proof construction" (p.25). In addition he argues that in dealing with conjectures, students are allowed to "experience and recognise the difference between plausibility and proof" (p. 23). I found this distinction useful especially given my underlying belief that even if a student never fully engages with the formal or rigorous side of Analysis, she should still be able to pass the module provided she has developed a strong conceptual or intuitive understanding of the subject. I decided that I wanted my students to make use of their concept images by examining mathematical statements with a view to determining their *plausibility*.

Practice

A sequence (a_n) is known to be monotonic increasing.

(i) Might it have an upper bound?

(ii) Might it have a lower bound?

(iii) Must it have an upper bound?

(iv) Must it have a lower bound?

Give an example to show each possibility or impossibility. (Burn, 2000, p. 31)

A sequence (a_n) is known to be unbounded.

(i) Might it contain a bounded subsequence?

(ii) Must it contain a bounded subsequence? (Burn, 2000, p. 33)

“Might/must” exercises of this form are designed to encourage the student to explore the plausibility of a mathematical statement. The student who has engaged with the *Portfolio of Examples* exercise, and has developed rich concept images that include the concept definitions, should be well-equipped to answer the questions above. For the student allergic to formal proof, these exercises play an essential part in allowing her to make progress in the subject and to demonstrate her conceptual knowledge. Therefore I believe they also have a role to play in building a student’s confidence. For these reasons, I have included exercises of this type in worksheets and final examination papers in the Analysis module.

Of course “true/false” exercises where one is asked to decide the truth or falsity of a mathematical statement and provide verification are similar in nature to “might/must” exercises, but bring the student one step closer to proof. In the spring semester 2007, I started the practice of taking time out in lectures and asking students to work on “true/false” exercises in small groups. My motivation was to use these exercises to expose and familiarise the class to some of the big results in the course, before tackling the proofs. That semester, I also decided that the in-class midterm test would consist of ten such exercises. (A more detailed description of “true/false” exercises will be presented later in the paper.)

Towards the end of the semester a small study was conducted to examine what approaches students take when determining the validity of statements (Curley & Meehan, 2009). Individual interviews were conducted with seven students from the module during which each student was given three statements from the midterm and asked to “talk out loud” as he or she decided whether each statement was true or false and attempted to provide a verification. The initial approach taken by most of the students was to either attempt to generate a counterexample, or state and attempt to use a theorem from the course.

One of the statements was a follows:

> If (a_n) and (b_n) are positive and increasing, then (a_nb_n) is increasing.

Despite all seven students correctly proclaiming the statement to be true, only one introduced the definition of increasing sequence without prompting but made no progress towards proving the result. When four of the students declared themselves to be stuck, the interviewer prompted them to consider the list of definitions. Even with this prompt, only two considered the definition of increasing sequence, and neither was able to prove the result. All indications were that it was finally time to consider how to present the “P” in DTP to my teaching.

Theory – the “P” in DTP

Mason, Burton and Stacey (1985) recommend that when conjecturing about *why* a statement is plausible, one should approach the task in three stages:

> “Convince yourself, convince a friend, convince an enemy” (p. 95).

Because convincing yourself can be quite easy, they advocate that you should next attempt to convince a friend that your argument is correct. In relation to examples, they note that although examples can convince a friend of the *plausibility* of the statement, you must attempt to justify in a step-by-step fashion, *why* your statement is true. Finally, you should consider whether your argument could convince someone who will question and attempt to pick holes in every statement. In the context of the Analysis module, a “friend” can be considered to be any other person from the class and the lecturer can play the role of “enemy”, making it clear exactly the type of argument that will convince her. By its very nature, this approach to proof is social – it involves students interacting with each other and with the lecturer.

Thurston (1994) emphasises the importance of the social aspect of mathematics:

> …the social setting is extremely important. We are inspired by other people, we seek appreciation by other people, and we like to help other people solve their mathematical problems (p. 171).

He believes that the traditional DTP model of mathematics is missing the social component that is essential for progress in the subject and considers how mathematical understanding is actually communicated within a subfield of the subject where participants share a common mathematical background. He conjectures that a significant result in the subfield which may be communicated in a one-hour seminar or a 15-20 page journal article to others from the subfield, can probably be communicated within a few minutes with face-to-face contact between two people in the subfield. As a mathematics lecturer, this highlights for me the importance of encouraging students in a class (the subfield of Introductory Analysis in this case!) to engage in mathematical dialogue with each other.

Practice

As mentioned above, in 2007, I introduced “might/must” and “true/false” exercises to encourage students to use their concept images to determine the plausibility of mathematical statements. From 2008 onwards, these exercises were more systematically included in the module. Taking Spring Semester 2011 as an example, each student was required to engage with 15 “might/must” exercises (from worksheets) and a total of 47 “true/false” exercises spread throughout worksheets (14), class test (8), final examination (5), and *In-class Exercises* (20). For the remainder of this paper, I will describe the implementation of the *In-class Exercises*, and using student feedback, provide an evaluation of “true/false” exercises. I will also describe what I have learned as a lecturer from engaging students in such exercises.

IN-CLASS EXERCISES

The Analysis module, MST20040, is offered each spring semester in UCD. It is a core module for second year BA in Mathematical Studies and BSc in Economics and Finance students, and for Higher Diploma in Mathematical Studies students. There are usually 65-70 students enrolled on the module each year. It is delivered in three lectures per week for twelve weeks, with a weekly workshop where students work on assigned worksheets in small groups. There are four assessment components: *Portfolio of Examples* (20%), *In-class Exercises* (10%), class test (20%), and final examination (50%).

During twelve of the thirty-six lectures, students are given an exercise to work on for part, or all, of the 50-minute lecture. For attempting an In-class exercise and handing it up at the end of the class, the student receives 1%, and is awarded the maximum mark of 10% for completing at least ten of the twelve exercises. Since advance notice of when an In-class exercise will occur is not given, these exercises also provide an incentive for students to attend lectures. The social aspect of the exercises is lost if the student works on them alone, and therefore I feel that attendance at lectures is essential.

The format of an *In-class Exercise* is that each student is given a handout of the exercise and asked to firstly work on it on her own. When she is content that she has considered the plausibility of the statement and perhaps outlined an initial justification of it, she then works on the exercise in a group of two or three. While the class work, I walk around and make mental notes of the approaches that students are taking. Sometimes I challenge a group if they have made an incorrect assumption or give hints if a group is stuck, but I pay special attention to not becoming an external authority. If a student asks me a question about a concept, or looks to me for validation of an approach, I will ask her to consult her friends.

I take the opportunity afforded by these exercises to stress to students the importance or writing mathematics accurately. One way of doing this is to read a piece of mathematics a student has just written and critique it. I feel that this way I can get across my point more effectively than if I had to provide written feedback on each student's exercise. I also use these exercises to address another difficulty that students experience in writing mathematics: the issue of when a result is obvious, and when it needs a proof. As these issues arise they can be brought to the attention of the whole class and discussed as a community. At the end of class, the exercises are taken up, and usually returned at the next lecture.

"True/false" exercises form the backbone of the *In-class Exercises.* For example, in Spring Semester 2011, eight of the twelve exercises contained between one and four "true/false" exercises. The first few exercises of the semester are generally quite straightforward and provide the student with the opportunity of familiarising herself with the style of the questioning. For example *In-class Exercise 4* in 2011 was as follows:

> Suppose that $(a_n)$is a sequence which is *not* bounded above. In each of the following say whether the statement is *True* or *False*, and justify your answer:
>
> (a) (a_n) contains a positive term.
>
> (b) (a_n) contains finitely many positive terms.
>
> (a) (a_n) contains infinitely many positive terms.
>
> (a) (a_n) tends to infinity.

Once the major concepts have been introduced and we start to approach the significant results of the module, the *In-class Exercises* tend to increase in difficulty. To address this, I introduce the class to the three stages of convincing described by Mason et al. (1985). For example, the following statements appear in *In-class Exercise 5* from 2011, each under the heading "Convince yourself, convince a friend":

> **Statement A**. Every bounded sequence is convergent.
>
> **Statement B.** Every convergent sequence is bounded.

The student is asked whether the statements are true or false and to explain her thinking.

My experience of students working in groups is that they can be quite uncritical of each other's work, at least in the initial stages. On first introducing these exercises, I had hoped that at the "convince a friend" stage, lively debate would ensue as students defended arguments. While this can occur, it generally does so only towards the end of the semester as students' confidence grows and they become more accustomed to group-work. To maximise the opportunity afforded by this "convincing stage", when correcting the exercises I attempt to categorise the main approaches taken. Then during the next class I present these to the class and talk about them, or else I incorporate them into another exercise.

For example, I have encountered a number of approaches taken by students in explaining why Statement B above is true. Some assume that convergent sequences are either increasing or decreasing, and hence say that they are bounded between the first term of the sequence and the limit. Others, who seem to be aware of this misconception, consider sequences which oscillate up and down while converging towards the limit. They then claim that the sequence is bounded between its first and second terms. Others conjecture that the bound is given by the maximum of the absolute value of the terms. Others introduce the definition of a sequence converging to a limit *a,* and conjecture that the sequence is bounded between something like $a-\varepsilon$ and $a+\varepsilon$. At the "convince a friend" stage these are exactly the approaches that you hope your students will critique, and therefore *In-class Exercise 6* this year consisted of five such arguments including graphs, paraphrased by me, with instructions as follows:

> Five of your friends have considered Statement B. Each friend has decided the statement is true and written down an argument to convince you that he/she is correct. Please consider each friend's argument, and give them constructive feedback on it.

On completing this exercise, some students will have already gained an understanding of an argument suitable to "convince an enemy" that Statement B is true.

Generally I tie up any loose ends and present the formal argument to the class while taking care to draw on what we have discussed in the "convince a friend" stage. I highlight the central role played by definitions in proof. I encourage students to write down what they "KNOW" and "WANT" in mathematics, that is, I encourage them to translate the problem into mathematics, and then look for the "link" between what they know and want. Finally, I take care to point out key features that make an argument suitable to "convince an enemy". I sometimes present more than one proof, depending on what valid approaches have been taken by students in the "convince a friend" stage. This provides the student with an opportunity to decide what makes a proof appealing to her, and to also see that there can be several ways to do something in mathematics.

STUDENT FEEDBACK ON TRUE/FALSE EXERCISES

At the end of Spring Semester 2011, students were invited to provide written anonymous feedback on the "true/false" exercises. The feedback form was completed by 28 students. On a Likert scale from one to five where one is "not at all difficult" and five is "very difficult", each student was asked to rate the level of difficulty involved in deciding whether a given statement is true or false. The average rating was 2.66 with eleven students giving a rating of two, and thirteen giving a rating of three. Using the same Likert scale, each was asked to rate the level of difficulty involved in providing a verification for the true/false decision. Here the average rating was 4.01 with five students giving a rating of three, seventeen a rating of four, and six a rating of five. Only five students gave the same rating to both tasks, with all others giving the verification task a higher rating than the decision task.

When asked to comment on both of their ratings, over half the class said that they found it relatively easy to decide whether a statement is true or false *and* to find a counterexample to illustrate that a statement is false. If a statement is true and requires a formal proof, they comment that is a more difficult task. (The four quotes below are all student responses taken directly from the feedback forms.)

> It is easy to see whether a statement is True or False and it is typically not difficult to find a counter example if the statement is false. The difficulty in these exercises lies in trying to prove if the statement is True.

While it is not surprising that students find determining the plausibility of a an argument easier than explaining why an argument is plausible it is still encouraging to note that most seem comfortable in deciding whether a statement is true or false and in generating examples.

In addition, students were invited to provide written feedback on three open-ended questions. The first asked them to describe how they generally approached the "true/false" exercises. The second question asked whether they found the exercises useful. Of the 28 respondents, one gave no response to this question, one responded negatively, and all others said that the exercises were useful. Eleven students commented that the exercises made you think: "They get you thinking"; "They make you think"; "It also helps us think for ourselves…"; "They allow you to really think about the questions… ". Four students commented on how they helped them to prove: "Help you to formalise your intuition"; "They allow you to […] explore various ways of going about proving". Others commented on how the exercises improved their understanding of definitions and proofs in general, and how the exercises encouraged them to keep up with the material. One student observed that the exercises had improved his ability to write mathematics:

> They made me better at explaining by writing rather than knowing something is true but not being able to show/prove it.

Finally the third question asked them to comment on whether they felt their ability to do the exercises had improved over the semester. Twenty said it had, two said it hadn't and six didn't respond. Eight commented that there is an approach to the exercises that you can learn.

> They have got me thinking more about the course, and got me into a "frame of mind", so I know what is expected from each exercise now.

A couple of students said that their confidence in their ability to prove had grown and interestingly two students said the exercises had made them more sceptical:

> I am more "sceptical" or less easily convinced that something is true/false.

LECTURER'S PERSPECTIVE

I can anticipate the following criticisms about the practicality of *In-class Exercises*: "They take up a lot of time in class"; "you must get less material covered"; and, "that is a lot of correcting". The first and second criticisms are similar and I would argue that it is not how *much* content is covered, but *how* it is covered that matters. But also, while initial progress through content is slow, as students become familiar with the concepts and the processes involved in determining plausibility and proof, the pace at which content is introduced increases. In relation to the third criticism, I estimate that correcting 50 In-class exercises takes approximately 15-20 minutes. Given that the student automatically receives 1% for handing in the exercise, there is no decision to be made about what mark to award. I look instead at each exercise to discern things like the student's general approach, any misconceptions she may have, and her use of language and notation. I tend to limit written feedback, and instead address the feedback to the whole class at the next lecture.

The single most important thing from my perspective is the opportunity that the *In-class Exercises* afford me as an educator to learn about my students' thinking. In watching them hypothesise, make mistakes, and struggle to write, and in listening to them as they propose arguments to each other and to me, I have learned a vast amount about the way they think about Analysis. I feel I have become sensitised to potential conflict factors in the subject and to the concepts and ideas that present difficulties. Consequently, I hope I am better-placed to set example generation tasks and In-class exercises that more effectively help students to develop an understanding of the subject, and examination questions that better assess this understanding.

On occasion over the past semester, I have questioned whether I was running these exercises more for my benefit and enjoyment, than that of my students. I was learning about their thinking, but were they actually learning about Analysis? Would they learn more if I stood at the board and explained the ideas carefully? Writing this paper has convinced me that they *do* have a place in the module. Also my diary entry from the penultimate lecture suggests that perhaps students do learn from engaging with them:

> I hand out *In-class Exercise* 12. True or False: If (a_n) tends to infinity and $a_n \neq 0$, then $(1/a_n)$ is bounded. […] I stop at T. She argues that since (a_n) is increasing, then $(1/a_n)$ must be bounded by "one over the first term". I explain that (a_n) is not necessarily increasing. She looks confused. I write out an example. "Oh!" she says. She is the second person to think this. A few others are trying to use the Axiom of Completeness. One who is thinking along the same lines as T is using the Monotone Convergence Theorem. Several have written down both definitions correctly from memory and are manipulating them. A number have mimicked what I do on the board and have written down "KNOW" and "WANT" with the definitions after each one. Some are trying to prove that $(1/a_n)$ is bounded using the definition, some are trying to prove that it converges and hence is bounded. G calls me over. She has proved it – well down to the last little bit. I'm delighted. B has proved it too.

CLOSING REMARKS

It is my belief that the traditional DTP model of instruction alienates all but the best students in mathematics. If we want to become less elitist and more inclusive we need to think carefully about grade descriptors for our advanced mathematics modules. We should ask ourselves what type of mathematical knowledge a student who achieves an A, B, C, or D actually has. If an advanced mathematics module only assesses a student's ability to present or produce the final product of mathematical thought, then where does this leave the student who has developed a conceptual understanding of the subject but has failed to master the formal side?

My assessment methods in Analysis have changed since 2002. I have introduced the *Portfolio of Examples* (20%), *In-class Exercises* (10%), and changed the format of the class test (20%) to consist entirely of "true/false" exercises. The fact that the continuous assessment is worth 50% reflects my belief that deep understanding cannot occur in the few days before the final examination. In the final examination, I still ask students to state definitions, but I try to never ask them to prove one of the significant results we've discussed in class. I may ask them to summarise in a few sentences the proof of a result, or I may give a proof of a result in the paper and ask them to explain why certain lines are true. I also put example generation tasks, "might/must" exercises, and "true/false" exercises on the final paper.

In many ways I have done a DIY job in teaching Analysis over the past ten years. I have read some of the literature, incorporated some ideas into my teaching, and modified and hopefully improved them as I went along. There are slicker models in the literature of how one might organise their advanced mathematics modules (Alibert & Thomas, 1991; Legrand, 2001; Larsen & Zandieh, 2008) but often the decision to adopt someone else's model or radically change your style of teaching can be too daunting to make. Each student is different, each class is different, each lecturer is different, and what might work well for one person, may not suit another person at all. In this paper I hope I have illustrated how, by engaging with the mathematics education research literature, my teaching has slowly evolved and hopefully for the better!

REFERENCES

Alcock, L.J. & Simpson, A.P. (2002). Two components in learning to reason using definitions. *Proceedings of the 2nd International Conference on Teaching of Mathematics (at the Undergraduate Level),* Hersonisoss, Greece. Published at www.math.uoc.gr/~ictm2/.

Alibert, D. & Thomas, M. (1991). Research on mathematical proof. In D. Tall (Ed.), *Advanced mathematical thinking* (pp. 215-230). Dordrecht: Kluwer Academic Publishers.

Burn, R.P. (2002). The genesis of mathematical structures. In P. Kahn and J. Kyle (Eds.), *Effective learning and teaching in mathematics and its applications* (pp. 20-33). London: Kogan Page.

Burn, R.P. (2000). *Numbers and functions: steps into Analysis.* Cambridge: Cambridge University Press.

Curley, N. & Meehan, M. (2010). Example generation exercises in an introductory analysis course. *MSOR Connections,* 10 (1), 48-51.

Curley, N. & Meehan, M. (2009). Determining the validity of mathematical statements in a third-level analysis course. In D. Corcoran, T. Dooley, S. Close & R. Ward (Eds.), *Proceedings of 3rd National Conference on Research in Mathematics Education,* 344-355.

Dahlberg, R.P. & Housman, D.L. (1997). Facilitating learning events through example generation. *Educational Studies in Mathematics,* 33, 283-299.

Dreyfus, T. (1991). Advanced mathematical thinking processes. In D. Tall (Ed.), *Advanced mathematical thinking* (pp. 25-41). Dordrecht: Kluwer Academic Publishers.

Larsen, S. & Zandieh, M. (2008). Proofs and refutations in the undergraduate mathematics classroom. *Educational Studies in Mathematics,* 67, 205-216.

Legrand, M. (2001). Scientific debate in mathematics courses. In D. Holton (Ed.), *The teaching and learning of mathematics at university level* (pp. 127-136). Dordrecht: Kluwer Academic Publishers.

Mason, J. with Burton, L. & Stacey, K. (1985). *Thinking mathematically.* London: Addison-Wesley.

Mason, J. and Watson, A. (2001). Getting students to create boundary examples. *MSOR Connections,* Vol. 1 (No. 2), 9-11.

Meehan, M. (2002). Students meeting advanced mathematics for the first time: can mathematics education research help? *Bulletin of the Irish Mathematical Society,* 49, 71-82.

Moore, R.C. (1994). Making the transition to formal proof. *Educational Studies in Mathematics,* 27, 249-266.

Tall, D. & Vinner, S. (1981). Concept image and concept definition with particular reference to limits and continuity. *Educational Studies in Mathematics,* 12 (2), 151-169.

Thurston, W.P. (1994). On proof and progress in mathematics. *Bulletin of the American Mathematical Society,* 30 (2), 161-177.

Vinner, S. (1991). The role of definitions in the teaching and learning of mathematics. In D. Tall (Ed.), *Advanced mathematical thinking* (pp. 65-81). Dordrecht: Kluwer Academic Publishers.

Watson, A. & Mason, J. (2005). *Mathematics a constructive activity – learners generating examples*. Mahwah, NJ: Lawrence Erlbaum Publishers.

Weber, K. (2004). Traditional instruction in advanced mathematics courses: a case study of one professor's lectures and proofs in an introductory real analysis course. *Journal of Mathematical Behavior,* 23, 115-133.

MATHEMATICAL MODELLING: A CLASS STUDY

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The emergence of socio-cultural approaches to learning over the last two decades has seen a greater emphasis on the development of pupils' mathematical modelling competencies and learners engaging with real-world, context-focused problems. The development of such competencies is challenging for many teachers and their pupils, particularly within the Northern Ireland context where academic selection is still an important influence on the curriculum in upper primary school. This study analyses the problem solving strategies used by pupils in a Year 7 class in Northern Ireland within a 'school mathematics' environment where the teacher traditionally acts as the source of mathematical authority and pupils are taught to carry out prescribed procedures with the focus on 'getting it right.' Although pupils were able to engage in mathematical modelling, explanations of their mathematical thinking reflect the need for a change in current practice and culture if mathematics education reform is to be successfully implemented.

INTRODUCTION

Over the last two decades, an approach to mathematics education that places a premium on real-world, context-focused problem solving has come to the forefront of international thinking on mathematics education. For hundreds of years behaviourist and cognitive theories tended to dominate the discourse on learning, with an emphasis on the learner as an isolated individual. More recently, however, a consensus has emerged that socio-cultural approaches to learning have been overlooked. Conway (2002) posits that the focus on individual capabilities fails to take account of the potentially supportive social and cultural contexts in which learning can be developed. The move towards a more socially embedded view of learning reflects a change in the relationship between school and society. Socio-cultural perspectives place a strong emphasis on learners engaging with authentic, real-world problems. In support of this approach, research by Nunes, Schleiman and Carraher (1993) demonstrated that children working as street traders in Brazil were able to perform complex calculations involving price and quantity. However, when the same calculations were reduced to arithmetic operations and presented in a formal 'school test,' their average score on the 'street test' fell from 98% to 37%. Nunes et al. (1993) concluded that while the children did learn basic arithmetic skills within more complex problem-solving activities and were able to devise their own strategies to calculate quickly and accurately, school did not utilise this out-of-school mathematical knowledge. Similarly, Desforges (1995) cites examples of pupils who failed to apply aspects of the mathematical knowledge they had learned in school to real-world problems. Brown, Collins, and Duguid (1989) challenge the assumption that conceptual knowledge can be abstracted from the situations in which it is learned and used: 'Knowledge is situated, being in part a product of the activity, context, and culture in which it is developed and used' (p. 32). This view has important implications for schooling. If the learning that takes place in school is embedded in the culture of the classroom, then children will have problems applying their knowledge outside school unless they have experience of engaging with real-world problems.

MATHEMATICAL MODELLING

Mathematising or mathematical modelling involves the application of mathematics to solve problems in the real world (Verschaffel, 2002). The origins of socio-cultural perspectives in mathematics education and their focus on real-world contexts for learning mathematics can be traced back to Realistic Mathematics Education (RME), a theoretical approach to the teaching and learning of mathematics which has been developed in The Netherlands. The Dutch reform of mathematics education began in the early 1970s and was strongly influenced by the work of Hans Freudenthal. At the heart of RME, lies Freudenthal's conception of reality as not only a context for the application of mathematical ideas but also as a source for learning mathematics (Van den Heuvel-Panhuizen, 1996, 2000). This is an important reconceptualization of the role of reality in mathematics education, as reality has traditionally served only as an area where mathematical ideas are applied. According to Freudenthal, mathematics lessons should give learners the 'guided' opportunity to 're-invent' mathematics by doing it (Van den Heuvel-Panhuizen, 1996, 2000, 2003). Instead of being receivers of mathematics as a closed structure, learners are treated as active participants in the educational process, in which they develop a range of mathematical tools and concepts themselves.

The notion of 'mathematizing' or mathematical modelling is central to RME (Van den Heuvel-Panhuizen, 1996, 2000, 2003). The Dutch researchers identified two different types of mathematization: horizontal and vertical mathematization. Horizontal mathematization involves coming up with mathematical tools to help organize and solve a real-world problem; it is the process of setting up a mathematical model. Vertical mathematization, by contrast, stands for all kinds of re-organisations and operations performed within the mathematical system itself; it is the activity of moving between different mathematical models. To put it simply, 'horizontal mathematization involves going from the world of life into the world of symbols, while vertical mathematization means moving within the world of symbols' (Van den Heuvel-Panhuizen, 2000, p. 4).

Closely related to mathematization is the 'level principle,' another important characteristic of RME (Van den Heuvel-Panhuizen, 2000, 2003). As learners progress, they pass through various levels of understanding, 'from the ability to invent informal context-related solutions, to the creation of various levels of short cuts and schematizations, to the acquisition of insight into the underlying principles and the discernment of even broader relationships' (Van den Heuvel-Panhuizen, 2000, p. 5). Teachers and educational programmes play a key role within the RME approach (Van den Heuvel-Panhuizen, 2000, 2003). Mathematics is regarded as a 'social activity' and learners are encouraged to interact and discuss their ideas and strategies with one another; this interaction stimulates reflection which enables learners to progress to higher levels of understanding (Van den Heuvel-Panhuizen, 2000).

There are many forms of representing the mathematical modelling process. According to the PISA Framework (OECD, 2003), mathematical modelling can be characterised as a cycle with five key phases: (1) Start with a problem situated in reality; (2) Organise the problem according to mathematical concepts; (3) Transform the realistic problem into a mathematical problem that faithfully represents the situation; this involves making assumptions about which features of the problem are important, generalising and formalising; (4) Solve the problem; and (5) Make sense of the mathematical solution in terms of the realistic situation. The five phases in the mathematising cycle can be represented diagrammatically (see Fig. 1).

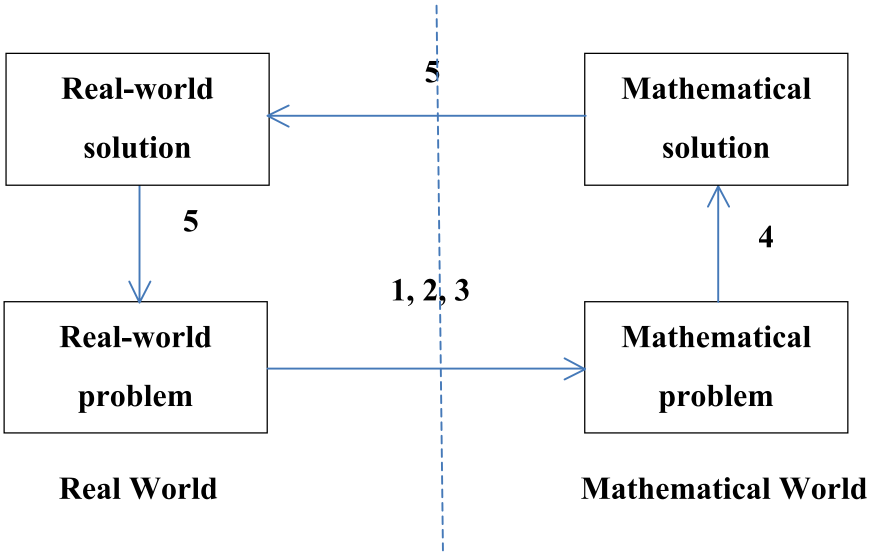


Figure 1: The mathematizing cycle (OECD, 2003, p. 38)

Challenges for Pupils

The integration of mathematical modelling into mathematics curricula has been demanded for some time in response to international comparative studies such as PISA (Maass, 2007). However, it seems that many children leave primary school without having mastered the different aptitudes required to solve mathematical application problems both efficiently and effectively (Verschaffel, De Corte, Lasure, Van Vaerenbergh, Bogaerts & Ratinckx, 1999). Verschaffel et al. (1999) suggest a number of reasons for this: First, insufficiencies in children's domain-specific knowledge and skills can interfere with their ability to solve problems. Further, when presented with novel complex problems, many pupils do not spontaneously apply useful heuristic strategies such as making a table, looking for a pattern, or drawing a sketch, etc. With respect to the metacognitive aspects of mathematical competence, there is evidence to suggest that many children fail to analyse the problem, monitor the solution process, and evaluate its outcome. Finally, it seems that even the more able pupils possess a range of beliefs and attitudes that are counterproductive. For example, they assume that a mathematics problem has only one correct answer and there is only one correct way to solve it; average pupils cannot solve unfamiliar mathematical problems by themselves; being able to solve a problem is a matter of luck; and there is a gap between the mathematics they learn in school and the mathematics they need for real life.

Verschaffel et al. (1999) posit that certain features of current practice and culture in the teaching and learning of problem solving may have a negative impact on children's abilities to solve context-based mathematical application problems. First, since word problems are commonly used to support the development of modelling competencies at primary school level, many pupils are not confronted with complex and realistic problem situations (Verschaffel et al., 1999). A typical word problem poses a quantitative question within a situation which is relatively familiar to the pupil and which can be solved by arithmetical operations performed using information within the text or otherwise inferred (Maass, 2007). As such, traditional textbook word problems are considered to be 'very simple, often artificial tasks with text' (Maass, 2011). It is therefore not surprising that pupils typically assume that a word problem is solvable, the text contains all the information needed to solve it, there is only correct answer and the final result involves precise whole numbers (Verschaffel et al., 1999). Furthermore, there is evidence to suggest that many children ignore real-life experience and engage in widespread 'suspension of sense-making' when solving real-world problems (Verschaffel, 2002). Second, typical problem-solving lessons involve the teacher explaining and demonstrating a standard problem-solving strategy followed by pupils working individually on similar routine problems without the opportunity to develop heuristic and metacognitive skills (Verschaffel et al., 1999). Third, aspects of the classroom culture such as the use of routine, mechanical problem solving strategies, mathematical misconceptions and erroneous beliefs about the nature of mathematics, often contribute to unwanted mathematical behaviour and learning outcomes (Verschaffel et al., 1999).

Challenges for Teachers

Blum (2002) argues that despite the current emphasis on mathematical modelling in mathematics education, genuine modelling activities are still rare in everyday teaching throughout Europe. A study by Maass (2011) identifies a number of possible impediments to the development of mathematical modelling in day-to-day teaching. Teachers in the study felt that there was insufficient time for modelling; they complained about the pressures to meet the demands of the curriculum (despite the fact that mathematical modelling is included in their current curriculum) and external assessment (which does not include modelling). The assessment of mathematical modelling was also considered to be problematic. Teachers in the study noted that some pupils do not like mathematical modelling, with some believing that a number of pupils are simply not capable of solving modelling tasks. However, an earlier study by Maass (2007) provides evidence that the ability to apply mathematics to real world problems can be achieved by low achieving pupils as well as more capable pupils.

THE NORTHERN IRELAND CONTEXT

One of the most distinctive aspects of the education system in Northern Ireland (NI), as compared with schools in the rest of the United Kingdom, lies in the continued use of selection at age 11 to divide pupils between grammar and secondary schools. The issue has been hotly debated in recent years. Following the publication of a major research project on the effects of the selective system (Gallagher and Smith, 2000), the Minster for Education, Martin McGuinness established the Post-Primary Review Body to consult on options for the future organisation of post-primary education. The findings of this body, published as the Burns Report, recommended the abolition of academic selection in the transfer from primary to post-primary school (Post-Primary Review Body, 2001). Alternative transfer arrangements, however, have not yet been set in place. Grammar schools, unhappy with the abolishment of the ‘eleven plus’ examination, have commissioned their own entrance examinations in English and Mathematics for those Year 7 pupils who wish to receive a post-primary grammar school education. Academic selection, with its significant influence on what is taught and how it is taught in the upper primary years, has been one of the main pressures for change to the NI primary curriculum (CCEA, 2002). It is argued that the removal of the influence of the transfer tests will enable teachers to ‘adopt a broader, more varied curriculum’ (CCEA, 2002, p. 5).

The revised curriculum for primary schools was introduced in 2007 (CCEA, 2007). At the heart of the curriculum lies an explicit emphasis on the development of pupils’ *Cross-curricular Skills* (*Communication, Using Mathematics,* and *Using Information and Communications Technology*) and their *Thinking Skills and Personal Capabilities* (*Thinking, Problem-Solving and Decision-Making; Self-management; Working with Others; Managing Information;* and *Being Creative*). The NI curriculum aims to foster these skills and capabilities by providing opportunities for children to be actively involved in their own learning across all areas of the curriculum. Assessment arrangements have been revised to support this change in emphasis: formal assessment of the *Cross-curricular Skills* for pupils at the end of Key Stage 1 and Key Stage 2 will be introduced from the 2012-13 school year for *Communication* and *Using Mathematics* with the arrangements for *Using Information and Communications Technology* coming into place from 2013-14. Although the content for *Mathematics and Numeracy* – one of the six *Areas of Learning* in the revised curriculum – has not changed considerably, there is a stronger emphasis on mathematical modelling. The curriculum stresses that pupils should apply their mathematical knowledge, skills and understanding in a variety of contexts and recommends that ideal contexts are ‘relevant real life situations that require a mathematical dimension’ (CCEA, 2007, p. 6). Not only does the NI curriculum support the application of mathematics in real life situations, greater importance is also attached to the introduction of mathematical ideas and activities through contexts that are meaningful to children.

To what extent is this reform impacting upon teaching practices within the mathematics classroom? Until the current impasse in the arrangements for transfer from primary to post-primary education is resolved, many Year 7 teachers are faced with the demands of implementing the revised curriculum along with the pressures of ensuring pupils have the best possible chance of transferring to the post-primary school of their choice. The NI curriculum places great importance on pupils solving real-world, context-focused problems but are genuine modelling activities a reality in the classroom? This research attempts to investigate the problem solving strategies of pupils in one Year 7 class on a real-world problem involving proportion.

METHODOLOGY

The research reported in this paper is part of a larger study funded by Standing Conference on Teacher Education North and South (SCoTENS). The aim of the SCoTENS study is to compare and evaluate the impact of implementing RME curriculum materials in primary schools in the North and South of Ireland. A base-line test was administered to the pupils participating in the study before the RME materials were used. Some of the questions were routine non-contextual questions; others were based on real-life contexts. This paper reports on the work of pupils in one of the schools (in the North of Ireland) on one of the questions from the base-line test. The question is based on proportion and is adapted from a task on 'caterpillars and leaves' taken from the 1996 NAEP mathematics assessment for grade 4 pupils (aged 9-10) in America. The 1996 assessment included a special study of mathematics in context and examined pupils' performance on sets of related problems based on the context of a science fair project on butterflies. The 'caterpillars and leaves' task (see Fig. 2) asks pupils to find the number of leaves needed to feed twelve caterpillars if five leaves are needed to feed two caterpillars. Pupils are invited to indicate their working out using drawings, words or numbers. This modelling task was chosen so that pupils' problem solving competencies could be analysed prior to the implementation of the RME curriculum materials. It was important to choose a problem which would not meet pupils' expectations and which would make them think about the context (Greer, Verschaffel & De Corte, 2002, cited in Maass, 2007). The problem was based on a realistic context and focused on proportional reasoning, a topic in which the pupils had not yet received any formal instruction.

The pupils in the study were in Year 7, their final year of primary education (aged 10-11). Over two-thirds of the pupils had previously finished working towards the high-stakes tests on academic ability which will determine their post-primary schools. Children's responses on the modelling task were examined in detail in order to determine the strategies that were used and to consider what kinds of errors or misconceptions were evident. The class teacher was also interviewed in order to understand her approach to teaching mathematics.

A Year 4 class has 2 caterpillars. Each day, 5 leaves are needed to feed them.

a. How many leaves would be needed each day for 12 caterpillars?

Figure 2: Caterpillars and leaves task (adapted from a 1996 NAEP mathematics assessment task)

FINDINGS

The teacher's experiences of teaching mathematics will be discussed below. Following this, the children's responses to the modelling task will be analysed.

Views from the Teacher

The class teacher has over twenty years of teaching experience and has been teaching Year 7 for approximately ten years. When asked about her perspective on the learning environment, she explained that at the beginning of the year it is "very assessment and knowledge-centred … and not centred on the learner at all." She blamed this on the fact that the children have so many tests to do, with preparation for the high-stakes tests for academic selection in the first term for Year 7 dominating the curriculum in upper primary. For this reason she claims that the children "have been pushed quite hard probably in Year 6 and moved on without maybe really grasping things and that would show up … sometimes by Year 7 you realise they don't have that understanding and then you gave to go back over things again." In general, her approach to teaching consists of teaching the basic skills first before setting children application tasks. She explains that she rarely gives pupils something to do that she hasn't "gone over" and group work is rare with many pupils having the perception that "they do maths on their own." The pressures of assessment, she believes, has resulted in a mind-set whereby they want to get everything yet and this poses many difficulties for modelling tasks which challenge children's thinking:

> And sometimes it's hard to really challenge children in a way to develop their thinking because … they want to do the work and get the right answer … but they don't really want to think about it … and they don't want to get it wrong … and they kind of give up … I think that's because they did do a test in the first term … So you really have to work hard to get them to think things through and to think of different possibilities … I think it would be really good for them but it's hard sometimes.

Test Responses

Out of the class of 31 pupils, 23 responded correctly to the problem task. One child used an appropriate method but made a calculation error. A further six pupils answered incorrectly. The different strategies that pupils used are analysed below.

Correct answer of 30 leaves – using a composite unit (see Fig. 3)

About half of the class (15 pupils out of 31) treated the given relationship of two caterpillars to five leaves as 'a composite unit' (Lamon, 1995, cited in Kenney, Lindquist & Heffernan, 2002, p. 94). To solve the problem, these pupils first used either multiplication or division to find the relationship between two caterpillars and twelve caterpillars. Since there are six times as many caterpillars, they determined that there must be six times as many leaves and proceeded to multiply the number of leaves (5) by six to obtain the answer (30). Samples of children's responses are included in Figure 3. Response C.1 gives an explanation of the strategy used whereas responses C.2-C.5 simply note the calculations used.

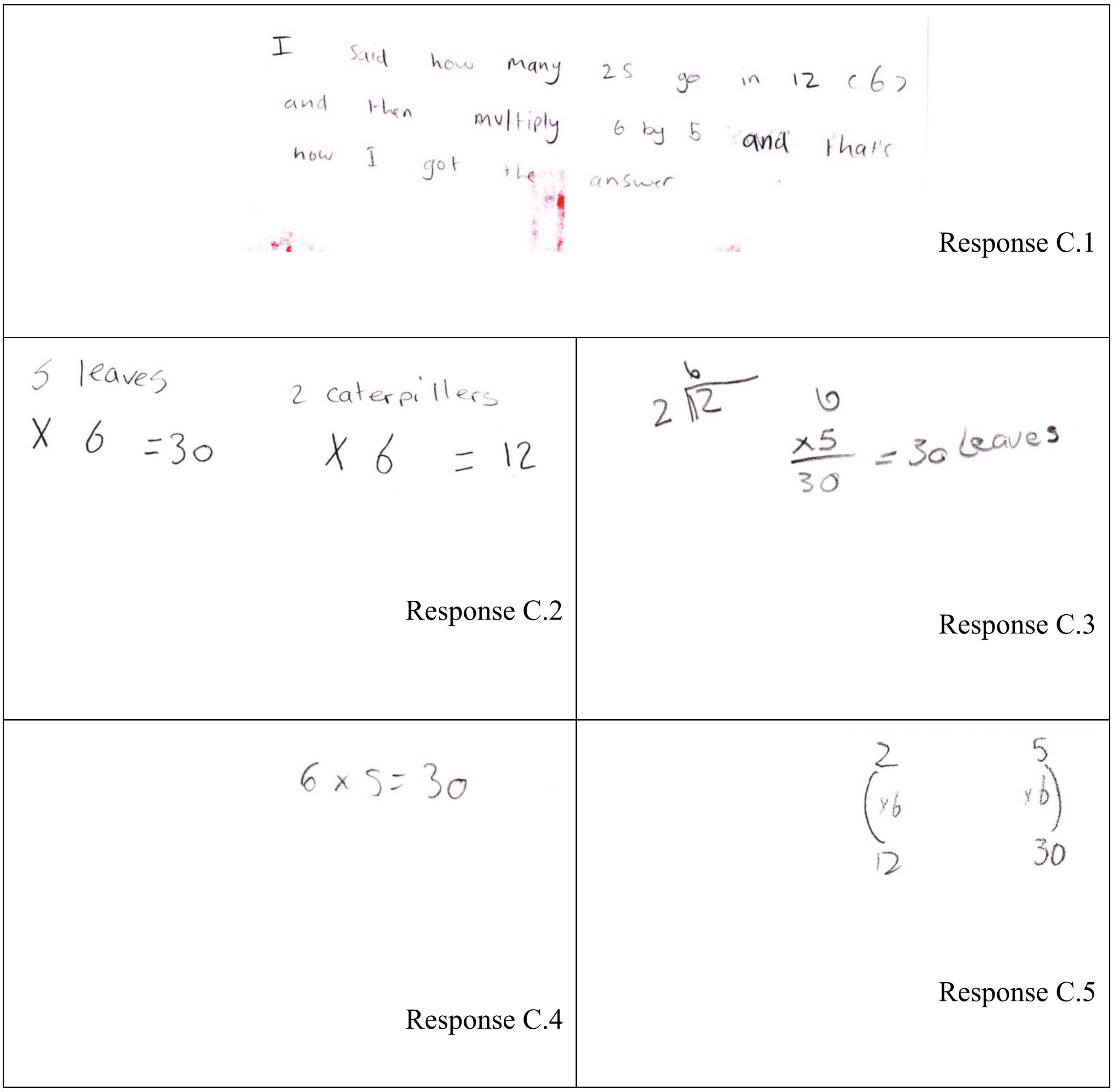


Figure 3: Correct responses – using a composite unit

Correct answer of 30 leaves – using a single unit rate (see Fig. 4)

Four pupils reasoned using a single unit rate for the number of leaves for each caterpillar. They noted that each caterpillar needs two and a half leaves each day and so they multiplied by twelve to find the number of leaves needed for twelve caterpillars. It is interesting to note that three of the four pupils transformed two and a half into decimal format and used long multiplication to find the answer (see response C.6) with one pupil making an error in her calculation. Response C.7 shows that one pupil used a picture to support her reasoning and response C.8 suggests that this pupil solved the problem mentally. The reasoning in response C.9 is unclear. The pupil begins by multiplying twelve by five, which gives the number of leaves (60) needed for twelve caterpillars if each caterpillar needs five leaves per day. Since the five leaves are shared between two caterpillars each day, it is possible that the subtraction calculation (60 – 30 = 30) demonstrates the pupil's awareness of the need to halve sixty in order to solve the problem accurately.

each caterpillar need 2 and a half leaves, so you times it by 12 and you get 30

$$\begin{array}{r} 25 \\ \times 12 \\ \hline 50 \\ 250 \\ \hline 300 \end{array}$$

Response C.6

× 12

$$\begin{array}{r} 12 \\ \times 2.5 \\ \hline 6.0 \\ 24.0 \\ \hline 30.0 \end{array}$$

Response C.7

you x 12 x 2 and the ~~half~~ add on th half

Response C.8

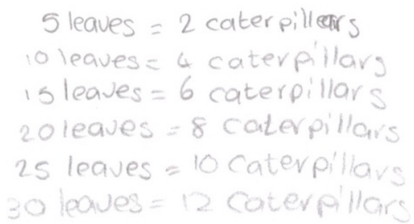
$$\begin{array}{r} 12 \\ \times .5 \\ \hline 60 \end{array} \quad \begin{array}{r} 60 \\ -\ 30 \\ \hline 30. \end{array}$$

Response C.9

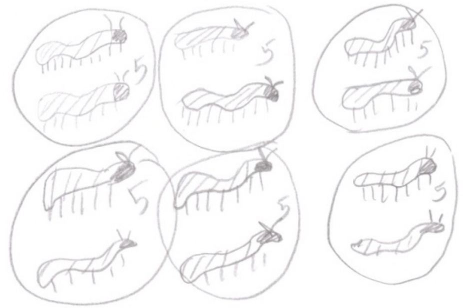
Figure 4: Correct responses – using a single unit rate

Correct answer of 30 leaves – counting up (see Fig. 5)

Four pupils solved the problem by counting up the number of leaves. Their responses provide examples of what Lamon (1999) calls the 'building up' strategy (cited in Kenney et al., 2002, p. 94). All of these pupils answered correctly but it is difficult to tell whether they used multiplication or addition to solve the problem. Three of the pupils constructed lists to keep a running total of the number of leaves needed for each pair of caterpillars until they reached twelve caterpillars, with one child noting, 'you go up in 5s for every two!' Response C.10 illustrates this reasoning. The fourth pupil used a picture to support her reasoning (see response C.11).



Response C.10

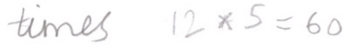


Response C.11

Figure 5: Correct responses – using counting up

Incorrect responses – misunderstanding the problem (see Fig. 6)

The most common incorrect response was 60 leaves demonstrating a misunderstanding of the proportional situation. Such responses were usually accompanied by a brief explanation in the form of 12 x 5 = 60 (see response C.12). It is possible that these pupils recognised that the problem involved a multiplicative situation, but did not have a complete understanding of the context, which required five leaves for two caterpillars, not one caterpillar.

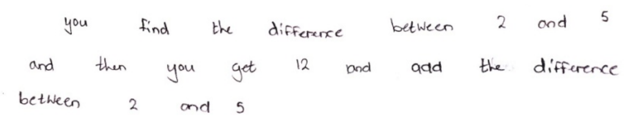


Response C.12

Figure 6: Incorrect responses – misunderstanding the problem

Incorrect responses - thinking about proportion in an additive way (see Fig. 7)

Only one pupil thought about the relationship between leaves and caterpillars in an additive way. Instead of keeping the multiplier constant, she assumed that the amount to be added must remain constant. From her response (C.13), it is likely that she reasoned that since there are three more leaves than the two caterpillars then there should be three more leaves for the twelve caterpillars, giving a result of fifteen leaves.



Response C.13

Figure 7: Incorrect responses – misunderstanding the problem

Summary

All of the pupils in the mixed-ability Year 7 class were able to engage with the modelling task, even though they previously had had no formal teaching on proportion. Although the majority of pupils used additive, multiplicative, and proportional reasoning to solve the problem, there is evidence to suggest that other pupils misunderstood the proportional situation in the problem or that they were thinking about proportion in an additive way. As Verschaffel et al. (1999) suggest, insufficiencies in the domain-specific knowledge base can have a negative impact on pupils' problem solving competencies. Furthermore, Verschaffel et al. (1999) claim that many pupils are unable to spontaneously apply useful heuristic strategies when presented with unfamiliar, complex problems. However, analysis of pupils' responses in this task reveals a range of strategies which pupils applied spontaneously, such as making a list and drawing a diagram, although it could be argued that the modelling task was not a complex one. Responses to the task demonstrate different levels of understanding. While some pupils relied on informal, context-related mathematics (such as pictorial drawings), others demonstrated a higher ability to mathematize the realistic situation and used more formal mathematics (such as multiplication and division). This is not surprising, given the range of ability within the class. Explanations of how pupils solved the problem were, in general, poorly articulated and tended to simply state the method used in the form of calculation(s) carried out, a number sentence, a list, or a diagram, for example, with little or no justification provided. This reflects the 'school mathematics tradition' (Yackel, Cobb & Wood, 1999) which their classroom follows. In this tradition, pupils typically engage in carrying out prescribed procedures or following instructions and an explanation consists of simply repeating the method used. Given that the 'teacher acts as the source of mathematical authority, the need for justification does not arise' (Yackel et al., 1999, p. 471). According to the class teacher, the pressures of high-stakes external assessment – where getting the right answer is rewarded – has had a negative impact on the development of pupils' ability to engage in productive mathematical discourse, explaining their mathematical thinking and justifying their responses.

CONCLUSION

The study demonstrates that pupils at differing levels of understanding can engage with the same problem set in a real-world context. Responses show a variety of informal and formal strategies which pupils used to solve the problem, but they also reveal limitations in the ability of pupils to articulate explanations of their working out. According to RME's level theory of learning, reflecting on the mathematizing that has occurred at one level of understanding is a condition for progressing to higher levels. In other words, the informal context-related problem solving strategies, in time, become more formal as a result of reflection. At a time when arrangements for transfer from primary to post-primary education remain uncertain, implementing the NI curriculum with its emphasis on real-world, context-focused problem solving presents a challenge to Year 7 teachers, particularly within the 'school mathematics tradition.' To stimulate progress in the development of children's mathematical modelling competencies, it is essential that pupils have the opportunity to share and discuss their problem solving strategies as this can 'function as a lever to raise their understanding' (Van den Heuvel-Panhuizen, 2000, p. 20).

ACKNOWLEDGEMENTS

The research reported in this study was made possible by a generous grant awarded by the Standing Conference on Teacher Education, North and South (SCoTENS).

REFERENCES

Blum, W. (2002). ICMI study 14: Applications and modelling in mathematics education – discussion document. *Educational Studies in Mathematics, 51* (1/2), 149-171.

Brown, S.J., Collins, A., & Duguid, P. (1989). Situated cognition and the culture of learning. *Educational Researcher, 18* (1), 32–42.

CCEA. (2002). *Curriculum review: Detailed proposals for the revised primary curriculum and its assessment arrangements.* Belfast: CCEA.

CCEA. (2007). *The Northern Ireland curriculum: Primary.* Belfast: CCEA.

Conway, P.F. (2002). Learning in communities of practice: Rethinking teaching and learning in disadvantaged contexts. *Irish Educational Studies, 21* (3), 61-91.

Desforges, C. (1995). Learning out of school. In C. Desforges (Ed.), *An introduction to teaching : Psychological perspectives* (93-112). Oxford: Blackwell.

Gallagher, T. & Smith, A. (2000). *The effects of the selective system of secondary education in Northern Ireland: Main Report.* Bangor: Department of Education for Northern Ireland.

Maass, K. (2007). Modelling tasks for low achieving students: First results of an empirical study. *Proceedings of the CERME 5, Larnaca, Cyprus,* 2120-2129.

Maass, K. (2011, March 25). How can teachers' beliefs affect their professional development? *ZDM The International Journal on Mathematics Education.* Retrieved June 3, 2011, from http://www.springerlink.com/content/x0g6j6404n378054/

Kenney, P.A., Lindquist, M.M., & Heffernan, C.L. (2002). Butterflies and caterpillars: Multiplicative and proportional reasoning in the early grades. In B. Litwiller (Ed.), *Making sense of fractions, ratios, and proportions: 2002 Yearbook* (pp. 87-99). Reston, VA:NCTM.

Nunes, T., Schliemann, A.D. & Carraher, D.W. (1993). *Street mathematics and school mathematics.* Cambridge, England: Cambridge University Press.

OECD (2003). *The PISA assessment framework: Mathematics, reading, science and problem solving knowledge and skills.* Paris: OECD.

Post-Primary Review Body. (2001). *Education for the 21st century.* Belfast: Post-Primary Review Body.

Van den Heuvel-Panhuizen, M. (1996). *Assessment and realistic mathematics education.* Utrecht: CD-β Press / Freudenthal Institute, Utrecht University.

Van den Heuvel-Panhuizen, M. (2000). *Mathematics education in the Netherlands: A guided tour. Freudenthal Institute Cd-rom for ICME9.* Utrecht: Utrecht University.

Van den Heuvel-Panhuizen, M. (2003). The didactical use of models in realistic mathematics education: An example from a longitudinal trajectory on percentage. *Educational Studies in Mathematics, 54* (1), 9-35.

Verschaffel, L. (2002, July 21-26). Taking the modelling perspective seriously at the elementary school level: Promises and pitfalls. *Proceedings of PME 26* (1)*, Norwich, England*, 64-80.

Verschaffel, L., De Corte, E., Lasure, S., Van Vaerenbergh, G., Bogaerts, H., & Ratinckx, E., (1999). Learning to solve mathematical application problems: A design experiment with fifth graders. *Mathematical Thinking and Learning, 1* (3), 195-229.

Yackel, E., Cobb, P., & Wood, T. (1999). The interactive constitution of mathematical meaning in one second grade classroom: An illustrative example. *Journal of Mathematical Behaviour, 17* (1), 469-488.

LEARNING TO TEACH MATHEMATICS BY MEANS OF REPRESENTATIONS – THANDI'S CASE

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The purpose of the study reported in this paper is to explore student teacher's choice and use of mathematical representations in a Standard 4 class in Lesotho. The participant is a full-time registered Diploma in Education Primary student teacher who is in teaching practice that takes place in second year of a three year diploma programme at the Lesotho College of Education. The teaching of mathematics is a challenging professional business for novice teachers due to their underdeveloped Pedagogical Content Knowledge (PCK). The ability to choose and use effective mathematical representations in lessons is an important component of teachers' knowledge for teaching. By mathematical representations in this study, I refer to concrete objects, images, and symbolic constructs that are used in teaching to make abstract mathematical concepts and processes accessible to learners. The findings of the lesson proceedings show that the student teacher used fake money to scaffold learners' strategies of addition and subtraction of both whole numbers and decimals. It is concluded that the choice and use of fake money as common context for Basotho learners afforded them opportunities to access mathematics.

INTRODUCTION

In this paper, I give an account of a study that was carried out in one of the primary schools in Lesotho. The purpose of the study was to explore mathematical representations that a student teacher (Thandi)[1] used in a mathematics lesson in Standard 4 (8-9 year olds) class. The lesson lasted for a period of 40 minutes and it was about the teaching and learning of addition and subtraction of money. At the time when the study was conducted, Thandi was a second-year Diploma in Education Primary (DEP) student. DEP is one of the diploma programmes offered at the Lesotho College of Education (LCE). It is a three-year programme where the second year is scheduled for Teaching Practice (TP). The lesson referred to in this paper, is one of the five lessons that I video recorded, transcribed and analysed as part of data collected for my doctoral research project. Thandi's lesson is not the best or the worst of the other four lessons. It is only chosen for this paper because my analysis of it is at an advanced stage compared to other four lessons.

BACKGROUND

The participant (Thandi) reported in this paper, is a student teacher who is training to become a primary school teacher. The current situation in Lesotho is that there is only one institution (Lesotho College of Education), which has been mandated to train students who have completed high school education to become primary school teachers. A large population of

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[1] Pseudonyms are used for all participants in this paper

primary school teachers are diploma holders and teach all subjects stipulated in the National Curriculum Development Centre (NCDC)'s documents. The three-year diploma programme for primary school trainees is designed in such a way that students spend years one and three at the college taking both content and methods courses in various learning areas. The findings presented in this paper emanate from data that were collected during second year when student teachers were on TP. The findings obtained as a result of Tier 1 data analysis in the bigger study (doctoral research project), reveal that a large percentage of candidates who enrol for the DEP programme have weak mathematical background and are generally females. However, Thandi is an exception in that she obtained a credit in Form E mathematics examination. This means she is one of the few students who pass mathematics well in Form E and opt to become primary school teachers. She practised teaching in a poorly resourced church school located about 40 kilometres away from Maseru city centre.

THEORETICAL ORIENTATIONS

The ability to represent mathematical ideas in various and flexible ways is an important component of pedagogical content knowledge (PCK). Shulman (1986, p. 9) argues that "since there are no single most powerful forms of representation, the teacher must have at hand a veritable armamentarium of alternative forms of representation". It then follows that both experienced and novice primary school teachers' PCK of mathematics can be made manifest through their ability to select and use multiple representations in lessons in order to make abstract mathematical concepts accessible to learners. Back in the 1960s a well known cognitivist Jerome Bruner in his work on the cognitive development of children proposed and identified three modes of representations:

- Enactive representation (hands on/action-based)
- Iconic representation (visuals/image-based)
- Symbolic representation (abstract and language-based)

(Bruner, 1966, p. 10)

In Bruner's point of view, modes of representation are the means through which information or knowledge are stored and encoded in a leaner's memory. Mathematics teachers in particular have to be conscious of these three modes of representations when they plan and teach mathematics especially at primary school level. In Lesotho, mathematics teachers have a tendency to focus more on symbolic representations than the other two representations perhaps that is why many learners fail mathematics in three national examinations (Standard 7, Form C, and Form E).

Enactive representations involve tasks that call for action (hands on activities) on the part of learners. In many Lesotho primary schools teachers use concrete objects such as counters, matches, sticks, and stones in mathematics lessons to assist learners to do addition, subtraction, multiplication, and division calculations. These physical objects prove to be valuable especially in the early years of schooling because learners at this stage are not yet conversant with the four basic mathematical operations namely addition, subtraction, multiplication, and division. Therefore, the models play an important role in helping learners to physically see why this sentence is true, 8 – 3 = 5.

Iconic representations are visuals that both learners and a teacher can refer to in class in order to facilitate effective learning and teaching of certain mathematical concepts. Iconic representations are resources that act as scaffolds for learners' indecisive thinking and strategies of mathematical operations. Examples of iconic representations commonly used in many primary schools include number-line, number square, number fan, number track, place value cards, multiplication square, and multiplication array. While most of these representations are commercial, student teachers have to be taught during their training ways of improvising and constructing these resources using recycled cardboard, plastic, etc. Given the economical state of Lesotho, it is logical to assume that many primary schools cannot afford commercial teaching aids. But, irrespective of financial constraints of the country, learners in all schools have an educational right to be taught mathematics well. The use of iconic representations is meant to assist learners' mathematical strategies to be strong and independent, and once this stage is reached then an iconic representation at hand can be removed.

The three modes of representations are not only hierarchical in nature but also intertwined (Rowland, Turner, Thwaites, & Huckstep, 2009). Symbolic representation being at the highest level and the most abstract representation compared to the other two (enactive and iconic). This is the most adaptable form of representation than actions and images, which have a fixed relation to that which they represent. Symbols are flexible in that they can be manipulated, ordered and classified and as such the learner is not constrained by actions or images. In the symbolic stage, knowledge is stored primarily as words, mathematical symbols, or even in other symbolic systems (Bruner, 1966).

Rowland et al. (2009) have cited an example of an empty number line to substantiate their point that the three modes of representations cannot only be used hierarchically but also in an intertwined fashion. They argue that an empty number line is an iconic representation by its nature but as learners make 'hops' or 'jumps' on it, they are using it in an enactive way. Yet the operations demonstrated by such hops and the answer reached are symbolic in nature. During their training courses, student teachers are expected to acquire and develop a web of connected representations for various mathematical concepts which they can draw on in lessons to help learners understand mathematics.

PCK AND REPRESENTATIONS

The ability of any teacher to translate his/her knowledge of the subject matter into something comprehensible to learners is an important part of that teacher's Pedagogical Content Knowledge (PCK). When working with experienced science teachers in Australia, Loughran, Mulhall, and Berry (2004) found that teachers' knowledge of their practice (teaching) is tacit. They found that although teachers find it challenging to provide reasons for teaching certain scientific concepts in particular ways, in general, teachers commonly share activities, teaching styles, and insightful thoughts of how best to teach science. Loughran et al. (2004, p. 374) take a view that researching teachers' PCK requires working at both an individual and collective level because as they put it "PCK resides in the body of science teachers as a whole while still carrying important individual diversity and idiosyncratic specialized teaching and learning practices". When exploring the notion of mathematics teachers' knowledge resource, Rowland et al. (2009, p.14) concur that within a school context, teacher's knowledge resource is "both individual – what each teacher knows, and collective – what is accessible by reference to colleagues". In class a teacher draws from his/her own content knowledge but in situations where teachers work cooperatively in school they plan lessons together and talk about methods of delivery in class. In that way each teacher gets necessary assistance from colleagues. In Tier 3 of the reported study in this paper, I worked with five student teachers as a group in workshops and with each individual participant in one's lesson to explore their understanding and use of mathematical representations.

When working with science teachers in South Africa Rollnick, Bennett, Rhemtula, Dharsey, and Ndlovu (2008) argue that the ability to choose an appropriate representation and use it effectively in lessons reflects that teacher's PCK. Shulman (1986 & 1987) too takes a view that multiple representations that teachers use in lessons are the central part of each individual teacher's PCK. It then follows that being able to choose and use representations efficiently in lessons is an important component of any teacher's PCK.

THE KNOWLEDGE QUARTET

In Thandi's lesson, there were multiple activities taking place, however I needed a theoretical tool/lens through which I could focus my eye on the teaching/learning of mathematics through representations. The 'Knowledge Quartet' proved to be a useful tool in this case. Devised by Rowland, Huckstep and Thwaites (2005), the Knowledge Quartet (KQ) is a typology that emerged from a grounded approach to data analysis of primary mathematics teaching in the UK. The KQ identifies the manner in which the student teacher's mathematical knowledge impacts on a mathematics lesson along four dimensions namely; foundation, transformation, connection, and contingency. While all four dimensions are interconnected and all are useful in looking at mathematics teaching, the 'transformation' dimension lends itself particularly well to this study in that it focuses the eye of the researcher on the uses student teachers make of representations when teaching mathematics. I also realise that 'connection' established between mathematical concepts is key in Thandi's lesson. Therefore, in this section I only discuss two dimensions that I consider to be of less relevance in this paper namely 'foundation' and 'contingency' however, useful in shedding light on the use of the whole analytical framework (Knowledge Quartet).

In the 'foundation' dimension of the KQ, the teacher's background knowledge and beliefs with regard to the meanings and descriptions on mathematical concepts and practices are manifested during teaching. A teacher's ontological position of mathematics and the rationale for teaching it at primary school level is noticeable and made manifest in mathematics lessons. According to Rowland et al. (2009) the codes for this dimension include among others: awareness of purpose, identifying errors, overt subject knowledge, use of mathematical terminology, use of textbook, reliance on mathematical procedures, and theoretical underpinning of pedagogy.

The ‘contingency’ dimension calls for a teacher to make sound decisions during the lesson about learners’ contributions. Unlike experienced teachers, student teachers lack the ability to take on and respond on the sport to unexpected learners’ contributions in class. Hume and Berry (2011) distinguish the most limiting factor as student teachers’ lack of classroom experience and experimentation. According to Rowland et al. (2009, p. 126) “there are times when the teacher is faced with an unexpected response to a question or an unexpected point within a discussion and so has to make a decision whether or not to explore the idea with the child”. A teacher has to be always alert for such moments and be ready to react appropriately to such unexpected situations during the teaching episode. It could be unfortunate if unexpected learners’ contributions can pass unnoticed in a class by the teacher because unpacking such contributions might be of special benefit to that learner or as Rowland et al. (ibid.) put it, might suggest a particularly fruitful avenue of enquiry for others. They identify the key contributory codes in this dimension of the knowledge quartet as: responding to children’s ideas; use of opportunity; and deviation from agenda. In what follows, I address the ways in which methodological issues were attended to.

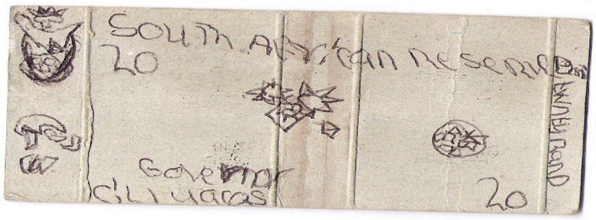
METHODOLOGICAL APPROACH

As mentioned earlier, the reported study here is part of a three-tiered longitudinal research (doctoral) project with prospective primary school teachers conducted over a period of three consecutive years (2009 – 2011). For the purposes of this paper, I choose to focus on part of Tier 3 data. This part of the study is guided by the critical question; how do student teachers on Teaching Practice use mathematical representations in lessons? The five student teachers in Tier 3 were conveniently drawn from Tier 2’s ten participants. They were a convenient sample in that I invited only those who were practising teaching of mathematics in primary schools located in Maseru district which, I could access easily and economically. Each of the five participants was observed teaching a mathematics lesson on arithmetic operations. However, this paper is about one lesson which, was taught by Thandi one of the five participants in Tier 3. There were eighty (n = 80) Standard 4 learners in Thandi’s class. The learners’ ages varied from 9 to 15 years. Below I present Thandi’s lesson synopsis:

> At the beginning of the lesson, the cooperating teacher asks learners to stand and sing a song. This looks to be a good starter for the lesson. Thandi then asks learners to take out their money. (On the previous day, Thandi had given learners home-work in which they had to construct fake money and bring it to class in the next day). Learners take out their fake moneys made from paper. Thandi mentions that money is subtracted when used to buy items in a shop. She then places pictures of different fruit on the wall in front of learners. She asks six learners to come to the front of the class and give them fake money to buy items placed on the wall. She asks each learner to say how much money they have. Learners are then asked to buy items of their choice and say whether they have change or not, and if they do, to say how much change they had. This goes on till all chosen learners have used their money. Learners are then

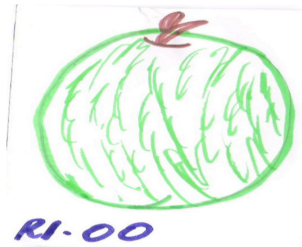
asked to go back to their seats. Thandi then distributed textbooks to learners. A pair of learners is asked to share a textbook. Learners are asked to turn to page 42 and do Exercise 4(a). The time for the lesson elapses and the lesson ends at this moment.

The picture below is a sample of the money that learners had constructed:



The teacher placed representations of fruits items on the wall and asked the six (n = 6) chosen learners to imagine being in a shop to purchase certain fruits. It is a common practice for learners in this area to buy fruits not only in supermarkets but also from the streets and at the school gates. Many people in Lesotho earn a living by selling fruits in various places including school surroundings. It is reasonable, to conclude that learners in Thandi's class are also familiar with buying fruit.

Below are some of the pictures that Thandi placed on the wall:



ANALYSIS OF THANDI'S LESSON

While the discussion of the analysis of data is organised to revolve around all four dimensions of the Knowledge Quartet (KQ), special attention is paid to transformation in this section because it focuses the eye of the researcher on representations. I also focus on connections of concepts that came about as a result of the use of fake money in Thandi's lesson.

Connection

If teachers are to make mathematics comprehensible to learners they have to make efforts to present it as a series of connected concepts, procedures, ideas, and practices. Marshall, Superfine & Canty (2010) take a view that making connections between multiple representations help students see mathematics as a web of connected ideas and NOT as a collection of arbitrary, disconnected rules and procedures. The connections can be made between concepts, operations, units, topics, and branches (e.g. Geometry and Arithmetic). The use of an iconic representation such a number square in class can help learners to recognise the connection that exists between two operations namely addition and subtraction of whole numbers. Rowland et al. (2009) argue that teachers have to bear in mind the complexity and cognitive demands of mathematical concepts and procedures in their attempt to sequence and connect mathematical content. They further identify contributory codes for this dimension (connection) as: making connections between procedures; making connections between concepts; anticipation of complexity; decisions about sequencing; and recognition of conceptual appropriateness. In Thandi's lesson, the main 'connection' that I observed is the way the use of money context managed to connect operations of addition and subtraction together. The other 'connection' was made between two topics namely whole numbers and decimal numbers and this occurred as learners were buying fruit and had to determine their change. In another incidence during the lesson, Thandi made some connection between two representations namely money and ordinary numbers:

Thandi: He has M2.00 … and from the M2.00 he has ... the banana costs M2.00

Pupils (*chorus*): Yes madam.

Thandi: From the M2.00 he has and the banana costs M2.00, is he going to get the change or not?

Pupils (*chorus*): No.

Thandi: No, because he has finished the money, ha ke re (isn't it so)? When you subtract M2.00 from M2.00 you get zero (0). Ha ke re (isn't it so)?

It seems that the connection of zero change and the number zero (0) here is critical. The connection here is between two sentences: Two Maloti take away two Maloti results in no change and 2 – 2 = 0.

Transformation

Of the four dimensions of KQ, the key dimension remains to be 'transformation' in this study that interrogates the choice and use mathematical representations in teaching. When unpacking the notion of Specialised Content Knowledge (SCK) in mathematics education, Ball, Thames and Phelps (2008) emphasize the key role representations play in making mathematics concepts understandable to learners. Ball et al. (2008, p. 393) further argue that because some representations are more powerful than others in affording learners access into mathematics concepts, teachers who have rich SCK choose and use "appropriate representations" that make content comprehensible to learners.

Every mathematics teacher has to make choices in planning and delivering a lesson. The choices involve among others selecting a key representation for the concept intended to be taught. For instance, a number line can be used for teaching addition, subtraction, multiplication, and division. But a number line is a key representation for addition and subtraction while a multiplication square or a multiplication array might prove to be key representations for performing multiplication at any level of the primary school mathematics. During lesson preparation, a teacher has to think carefully about the examples, illustrations, and analogies that he/she can use in class to make concepts, procedures, or even core vocabulary comprehensible to learners. Rowland et al. (2009) identify contributory codes in this dimension of the Knowledge Quartet as choice of representations, teacher demonstrations, and teacher's choice of examples.

In her lesson, Thandi chose to use fake money as a key representation in order to scaffold learners' skills of addition and subtraction of whole numbers and decimal numbers. When the lesson started Thandi demonstrated to learners what she meant by decomposing numbers:

Thandi: Yes, when we decompose a number we break it into pieces, ha ke re (is that so)?

Pupils (*chorus*): E-ea 'm'e (yes madam).

Thandi: If you break it up into pieces, we just take out any numbers that can add up to fifty, ha ke re (is that so)? So my own number… I can extract M20, Nka etsa (I can make) 20 + 20 + 10 = M50.00. We add up to M50.00. So which other numbers can we decompose fifty Maloti into? Which numbers can we decompose fifty Maloti into? Tefo!

Tefo: M10 + M10 + M10 + M10 + M10 = M50.00

The demonstration that Thandi made appears helpful in helping learners to comprehend the meaning of decomposing numbers. The evidence is seen by Tefo's response as shown in the excerpt above. Later in the lesson when learners were struggling to subtract decimal numbers from whole numbers, Thandi encouraged learners to use other forms of representations:

Motsamai: The banana is 1 Loti and 50 cents. We subtract M1.50.

Thandi: We subtract 1.50 Loti. It's 150 lisente (cents), from M5.00 he has, we subtract 1.50 lisente ha ke re (isn't it so)?

Pupils (*chorus*): Yes madam.

Thandi: So what is the change? What is Makoro's change? So how much is he going to get as the change for Makoro? How much is he going to get? Use your fingers, use our money … just think, think, use your fingers, your head, whatever! What do you think is going to be Makoro's change? What do you think is going to be Makoro's change, when we subtract 1.50 from the M5.00 he has? Nkele!

Nkele: Makoro's change is going to be M3.50.

Here Thandi asked learners to use fingers and their heads to think about the correct answer for the change. The excerpt suggests that the use of any of these representations (fingers or/and heads) afforded the learner (Nkele) with strategies that made it possible for her to obtain the correct answer (M3.50). It is possible that Nkele uses these representations in her daily buying to determine her change. I would like to argue that Thandi's choice of the selling and buying situation assisted learners to manage subtracting decimal numbers from whole numbers, which could have been more cognitively challenging if it was only presented symbolically as $5 – 1.5 =?$

FINDINGS AND CONCLUDING REMARKS

Analysis of the data reveals that Thandi chose to use various representations in order to support teaching and learning of mathematical operations namely addition and subtraction. Thandi had various resources at her disposal to use in this lesson; however, she carefully chose to use fake money made of paper in order to facilitate the teaching of whole numbers and decimal numbers. The home-work that was given to learners namely to construct fake money prior to this lesson, had the potential of helping learners to represent money in an enactive way (hands on activity) while at the same time helping them to think more seriously of real money used in their community. When learners encountered some subtraction difficulties, Thandi encouraged them to use the 'fingers' of their hands as representations to assist them to get to the correct answer. It is also noted in the analysis that the effective use of fake money in this lesson afforded learners opportunities to make crucial connections between subtraction of whole numbers and decimal numbers. Again, the use of fake money in the lesson encouraged discussion, and I conclude that it helped Thandi to effectively achieve her lesson objectives.

While the findings of this study cannot be generalised, I feel that this work might give some useful insights with regard to student teachers' mathematical understanding and use of representations in teaching.

REFERENCES

Ball, D. L., Thames, M. H., and Phelps, G. (2008). Content Knowledge for Teaching: What makes it special? *Journal of Teacher Education*, 59(5), pp. 389 – 407.

Bruner, J. S. (1966). *Toward a theory of instruction*. The Belknap Press of Harvard University Press.

Hume, A. and Berry, A. (2011). Constructing CoRes – a strategy for building PCK in pre-service science teacher education. *Journal of research in science education*, 41, pp. 341 – 355.

Loughran, J., Mulhall, P., and Berry, A. (2004). In search of Pedagogical Content Knowledge in science: Developing ways of articulating and documenting professional practice. *Journal of Research in Science Teaching*, 41(4), pp. 370 – 391.

Marshall, A. M., Superfine, A. C., and Canty, R. S. (2010). Star students make connections. *Teaching Children Mathematics*, 17(1), pp. 38 – 47.

Rollnick, M., Bennett, J., Rhemtula, M., Dharsey, N., and Ndlovu, T. (2008). The place of subject matter knowledge: A case study of South African teachers teaching the amount of substance and chemical equilibrium. *International Journal of Science Education*, 30(10), pp. 1365 – 1387.

Rowland, T., Huckstep, P., and Thwaites, A. (2005). Elementary teacher's mathematical knowledge: The knowledge quartet and the case of Naomi. *Journal of Mathematics Teacher Education*, 8, pp. 255 – 281.

Rowland, T., Turner, F., Thwaites, A., and Huckstep, P. (2009). *Developing Primary Mathematics Teaching*. SAGE Publishers Ltd. London EC1Y 1SP.

Shulman, L. S. (1986). Those who understand: Knowledge growth in teaching. *Educational Researcher*, 15 (2), pp. 4 – 14.

Shulman, L. S. (1987). Knowledge and Teaching: Foundations of the New Reform. *Harvard Educational Review*, 57, (1), pp. 1 – 22.

LOOK WHO'S TALKING NOW- THE ROLE OF TALK IN RELATION TO MATHEMATICAL THINKING AND PROBLEM SOLVING

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This paper gives an outline of my PhD project and some early attempts at data analysis. My research aims to explore the role of talk in relation to mathematical thinking and problem solving with a specific focus on whole class discourse. The difference between the discourse of traditional mathematics lessons and the discourse of mathematics at discipline level is discussed. The idea of a mathematical discourse community is presented and the transcript of a fourth class mathematics lesson is analysed in terms of the occurring discourse patterns. It is also explored in the context of the four central components of a discourse community, questioning, explaining mathematical thinking, source of mathematical ideas and responsibility for learning (Hufferd-Ackles, Fuson & Sherin, 2004).

My PhD project aims to explore the role of talk in relation to mathematical thinking and problem solving and in particular how best to support the learning of mathematics in whole class discourse. Traditional lessons in many mathematics classes follow a similar structure - the teacher models a strategy or procedure and this is followed by student practice (Boaler, 2009; Lyons, Lynch, Close, Sheeran & Boland, 2003). The discourse patterns of these traditional classrooms often involve repeated iterations of lower level questions and of the tripartite invitation-response-feedback/evaluation, IRE, pattern that has long been identified (Meehan, 1979). The invitation action is central to the type of mathematical thinking that students may engage in and the extent of student thinking can be limited by teacher questioning (Brodie, 2008). Repeated use of questions where there is only one right answer, combined with the evaluative teacher action, may lead to the idea that mathematics is rigid, right or wrong (Mendick, 2006) with little room for sharing emerging thinking. The centrality of the teacher's evaluative action positions her as the mathematical authority (Dooley, 2009) and can lead to the situation whereby students take little responsibility for determining whether other students' statements are mathematically correct or not (Hamm & Perry, 2003). This mode of discourse is very different to the type of informal discourse that mathematicians engage in. Discourse at discipline level includes patterns of dialogue that involve making conjectures and examining and justifying one's own mathematical thinking and the mathematical thinking of others (Lampert, 1990). This is supported by the view of mathematics put forward by Lakatos (1976), where mathematical understanding emerges when the learner follows a zig-zag path between proof and refutation. The relationship between the nature of a mathematical task and the discourse that it provokes is also worth noting and Stein, Grover and Henningsen (1996) note that that mathematical tasks presented to students will shape the opportunities for mathematical thinking.

In the reform movement in the USA, which began with the release of *Curriculum and evaluation standards for school mathematics* (NCTM, 1989), there is a big emphasis placed on meaningful classroom discourse and in particular on the creation of a classroom discourse community (Nathan & Knuth, 2003). Hufferd-Ackles, Fuson and Sherin describe a math-talk learning community as “a community in which individuals assist one another’s learning of mathematics by engaging in meaningful mathematical discourse” (2004, p. 81). Corcoran (2009) notes that this country adopted the reform agenda in primary level mathematics with the introduction of the revised curriculum. There is an emphasis put on discourse in Irish curricular and supporting literature as well as Education Research Centre, ERC, documents that have examined the impact of the revised curriculum at primary level (DES/NCCA, 1999, Surgenor, Shiel, Close & Millar, 2006). In fact, encouraging teachers to engage in discussion rich practice, Surgenor et al. recommend that

> Schools and teachers should place a stronger emphasis on teaching higher-ordermathematics skills, including Applying and Problem Solving to all pupils by implementing in a systematic way the constructivist, discussion-based approaches outlined in the Guidelines accompanying the 1999 PSCM. (ibid., p. 37)

The idea of a mathematical discourse community, where the mathematical ideas of students are made public and opportunities are created to discuss ideas while negotiating mathematical meanings, is one way to follow through on this recommendation.

To develop an understanding of what exactly is meant by a mathematical discourse community or ‘math-talk learning community’ it is worth considering the key areas of (a) questioning, (b) explaining mathematical thinking, (c) sources of mathematical ideas and (d) responsibility for learning. These areas were recognised as central by Hufferd-Ackels et al. during intensive research in real classrooms where teachers were attempting to teach in the spirit of reform. The authors identified developmental learning trajectories for both teachers and students across the four key areas. These trajectories are levelled from level 0 to level 3 and as teacher and students move through these levels, the developing classroom community moves closer to a math-talk learning community described above. The overview of the movement between the levels is given below.

Table 1: The Math-Talk learning community: Overview of shift over levels 0 - 3 (Hufferd-Ackles et al., 2004, p. 88)

Overview of Shift over levels 0 -3: The classroom community grows to support students acting in central or leading roles and shifts from a focus on answers to a focus on mathematical thinking			
Questioning	Explaining Mathematical Thinking	Source of Mathematical Ideas	Responsibility for Learning
Shift from teacher as questioner to student and teacher as questioners	Students increasingly explain and articulate their math ideas	Shift from teacher as the source of ideas to students’ ideas also influencing direction of lesson	Students increasingly take responsibility for learning and evaluation of others and self. Math sense becomes the criterion for evaluation

OUTLINE OF RESEARCH PROJECT

My research project consists of two parts. In the first part, my aim is to explore the nature of the discourse children in some Irish primary classrooms engage in during mathematics lessons on the number strand. This has involved the collection of audio-recordings of primary level mathematics lessons on number strand topics from third to sixth class. Seven recordings have been made from five different teachers. Participants were chosen on a volunteer basis after I made an initial invitation for participation to the whole staff of various schools, some of which I have a personal or professional connection with. I was not present during recording due to the constraints of my own teaching position. All recordings and invitations to participate were carried out according to the ethical guidelines laid down by St. Patrick's College. The second phase is a teaching experiment, employing a design research methodology (Cobb et al., 2003). During the school year 2010 -2011, I conducted a teaching experiment in my own classroom, the aim of which was to support mathematical thinking in whole class discourse. It was envisaged that the classroom norms around social interaction would support students in the making of mathematical conjectures. The mathematical tasks were designed according the realistic mathematics education, RME, approach and consisted of context problems that explored number concepts and linkage between these and other mathematical content domains. Audio recordings were collected in short phases across the whole school year so as to document the planned evolution of discourse within the classroom community. Analysis of this data set will be a complex process and it is hoped to adapt some of the methods outlined below that have been used on the data collected for the first part of my project. It is initial results from this phase of the research I will present in this paper.

ANALYSIS

The aim of the first part of my research was to explore the nature of the discourse children in some Irish primary classrooms engage in during mathematics lessons on the number strand. Of the seven lesson recordings I have chosen to present just one here. The sample is very small and though it can be argued that this lesson is representative of the sample (bearing many similarities in terms of both lesson and discourse structure to five out of six of the other lessons), without recourse to a much larger sample, it cannot be argued that is representative of primary mathematics lessons as a whole. However, like 95% of the children in the fourth classes studied in the 2004 National Assessment of Mathematics Achievement (NAMA) who were in classes where the teacher says the textbook is used every day, the activities of this lesson were also textbook based (Shiel, Surgenor, Close & Millar, 2006).

Lesson Synopsis

This lesson was of 44.03 minutes duration and occurred in a higher stream fourth class. The topic of the lesson was the distributive and associative properties of multiplication. It was a calm, orderly class at all times. The teacher, Liam, began by presenting the class with an array of three rows of fourteen dots on the interactive whiteboard. He stated that fourteen times three does not occur in "our tables" and it could be split up. At the suggestion of a student he split the array into two parts, seven groups of three and six groups of three and after some discussion; students solved the sums 7 x 3 and 6 x 3 separately and then mentally added the answers together to solve for the overall answer. Liam then completed a similar whole class activity for a 14 x 4 array. Then homework was corrected as whole class with students calling out the answers when prompted by the teacher. Questions were from the textbook and were missing factor questions such as 3 x ___ = 12, with all numbers within the range of the multiplication tables generally learned by heart. After this was completed, Liam asked the students to continue the exercise in the book while he checked individuals' homework. The questions were similar to the previous work but written with the product first, e.g. 15 = 5 x ___. This form of representation seemed to confuse some students. Liam also worked with two lower ability students on a one to one basis on differentiated material during this time. Afterward the set work was corrected in a whole class setting with students calling out answers at the invitation of the teacher as before.

Then Liam presented the class with the sum 3 x 4 x 5 and stated that brackets must be put in so that two of the numbers could be multiplied first. A number of these sums were worked through. In some cases, like the one above, the final answer was found, i.e., the class talked through the following solution 3 x 4 x 5 = 3 x (4 x 5) = 3 x 20 = 60. In other cases where the numbers were bigger, the final solution was not considered e.g. 5 x (6 x 5) was only brought as far as 5 x 30 and the overall solution was not discussed. Then the strategy of splitting a number into its factors in the context of a multiplication sum was discussed. For example, after some discussion 3 x 20 was broken down into 3 x 4 x 5. A number of these types of sum were discussed. Some examples had only one obvious solution e.g. 3 x 70 = 3 x 7 x 10 (within the range of tables based factors of 70 and ignoring non-tables based factoring solutions such as 70 = 2 x 35). Other examples such as 6 x 20 have more than one obvious solution, 6 x 4 x 5 or 6 x 2 x 10 but in only one case different factoring possibilities were discussed. After this the teacher directed students' attention to an exercise in their book where questions are of the form A x B = A x (C X __), where C is a factor of B. In the majority of these questions B was a multiple of ten and the "missing number" was in fact ten. The children worked on these quietly while the teacher consulted with some individuals who were struggling to understand how to complete the written exercise. Then these were corrected in a whole class setting as before.

Discourse Patterns

Wood (1994) discusses how differences in the patterns of interaction that occur in the discourse of teachers and students can result in different settings for learning. To investigate patterns of discourse within the data, I began to look at the transcripts and code teacher and student responses. For the turns of the teacher, it was necessary to break the turns into utterances according to their function, e.g., both evaluative and explanatory statements could often be found in one teacher turn. The following headings were sufficient to code all teacher utterances: explanation, invitation and evaluation. For student utterances the following codes were used: student comment, student question. The vast majority of student comments were in fact responses to Liam's questions. I confined my analysis to the sections of the discourse that were whole-class centred. These sections make up the majority of the discourse, 337 out of 385 turns. Then the data were examined to see if patterns of dialogue could be identified. There were many simple IRE patterns that were easily recognisable. For example, consider the following discussion that occurs when the class are correcting homework**:**

Excerpt from transcript	Description
Liam: ... number 3 a, three multiplied by what equals twelve?	Invitation
John: Four	Response
Liam: Very good. Sean, please, b, five multiplied by what equals thirty?	Evaluation (IRE); Invitation
Sean: Five multiplied by six	Response
Liam: Very good …	Evaluation (IRE);

(IRE) denotes a complete three part sequence. The invitation generally consisted of a known answer question and at times the evaluation was a simple repeat of the students' answer rather than statement implying praise such as "very good". Such repetitions of a student's answer usually served to finish the sequence rather than maintaining it in the public realm for further discussion (cf. Brodie, 2008). There were 54 simple IRE patterns identified. However, there were also many other segments of discourse which while not matching this structure exactly, seemed to have an underlying IRE foundation. For example, at the beginning of the lesson while considering the 14 x 4 array, Liam does a considerable amount of explaining between the questions he poses.

Excerpt from transcript	Description
Liam:…I've got fourteen rows or three sorry, three rows of fourteen dots o.k.. So we can say that that is 14 multiplied by …	Explanation Invitation
Students: Three	Response
Liam: Three, isn't that right? Now we know that we can also spilt that up because 14 multiplied by 3 is not a sum that we would know automatically through our tables. So we know that we can split that up. Can anyone give me a suggestion for what we can split that up into? -… Tom	Evaluation (EIRE); Explanation Invitation
Tom: Seven by three and seven by three	Response
Liam: We could split it up into seven by three and seven by three. So let's do that one, two three, four, five, … one, two, three, four, five, six, seven. And actually I have done this wrong. It's actually thirteen by three so we'll have to rub that out. … So this is thirteen by three so what can we split thirteen by three up into?	Evaluation (EIRE); Explanation Invitation
Aidan: Seven by three and -… six by three	Response
Liam: Seven by three and six by three. So let's do that together. We're going to put these together. One, two, three, four, five, six, … seven and we'll group them together and we'll take that down here and …	Evaluation (EIRE); Explanation

This pattern of dialogue still fits a general invitation-response-feedback pattern but is qualitatively different from the IRE pattern previously described. I have called this four part sequence EIRE, explanation-invitation-response-evaluation. In some ways this explanation action on the part of the teacher corresponds with Brodie's description of the 'insert' follow-up move a teacher may make when he or she picks up on the contribution of a learner and adds an elaboration, explanation or suggestion (2008). Brodie describes the insert action as the teacher transforming the student contribution in order to make his or her own mathematical contribution and states that it is generally considered to be more closely associated with traditional teaching rather than with reform orientated practice. Brodie also identifies other teacher follow-up moves that can be seen in other sections of dialogue which will be discussed further below.

There were occasions where the discourse pattern included at least one follow up question to the student's initial response. These have been called Extended IR before evaluation. Meehan referred to such sections of dialogue as "extended elicitation sequences" and described how in the case of known answer questions, once the initial question has been asked, the "interaction continues until the expected reply is returned" (1979, p. 287). If a student fails to answer a question successfully, the teacher may react by repeating the question, prompting or simplifying the question. For example, consider the following section of dialogue:

Excerpt from transcript	Description
Liam:… OK, this one here. Three multiplied by twenty..If I opened up that twenty into two numbers, what do you think, what could I open it up into?	Invitation
Students: Oh!!	
Liam: Who has an idea? …Yes	Incomplete response
Neil: em, three multiplied by ten	
Liam: Multiplied by?	Prompting reply
Neil: eh, multiplied by, multiplied by three. No. Three multiplied by ten multiplied by -… eh, ten	Incorrect response
Liam: No, remember we have to get two factors of the twenty, don't we. So wh-wh-what two numbers would multiply together to get twenty? What two numbers would multiply together to get twenty?	Negative evaluation; explanation; simplified elicitation
Student: Sir!	
Liam: Yeah	Response
Gavin: Five and four	Positive evaluation
Liam: Five and four, just like here.	

Liam's prompting reply at turn 227, may be considered an elicit move in Brodie's terminology. She describes the teacher aiming to get something from the learner as an elicit move and notes that such moves can function in the same way as funnelling. Wood (1994) described the difference between a funnel pattern of interaction and a focussing pattern of interaction as follows. In the funnel pattern the teacher effectively takes on the cognitively demanding aspects of the task in an attempt to guide the student to the right answer but by so doing limits the student's opportunities to engage in any meaningful mathematical thinking of his/her own. In the focussing pattern of interaction the teacher aims to focus the student's attention on the critical mathematical feature of the problem but still leaves the student with the responsibility for its solution. It may seem that this is what Liam is doing in turn 229 but focussing on the solution rather than the mathematical thinking of the student functions more like an elicit move. This narrows the range of possible contributions of the student and is different to the press move described by Kazemi and Stipek (2001) and by Brodie (2008) where the teacher may question a student with a view to creating an opportunity for conceptual thinking. The majority of the extended IR sequences in this lesson would appear to have been prompted by student errors or incomplete student answers and contained mainly known answer questions. Just one of the eleven occurrences was in response to novel student thinking and is discussed later.

Also identified was a pattern of dialogue in which Liam posed a question, a student or students responded but there was no obvious evaluation by the teacher. There were six such occurrences and they were termed IR. These cases all began with an explanation by Liam and a follow up question that was a variation of "Does that make sense?" or "Is that OK?" There were also six incidents where the audio recording did not provide any evidence of an invitation by the teacher but there was a notable class response in which the teacher also spoke. For example,

Liam: … Seven multiplied by fifty equals seven multiplied by five multiplied by …

Liam and students: Ten.

These are termed explanation-co-response in the table below. The findings are presented in the table below. The last two descriptors on the table cover non-IRE based segments of discourse and are discussed below.

Table 2: Discourse patterns by number and percentage of occurrences

Discourse Pattern	Number and percentage of occurrences
IRE	49 = 42.98%
E-I-R-E	21 = 18.42%
Extended IR before evaluation	12 = 10.53%
IR (no obvious evaluation)	8 = 7.02%
Explanation - co-response	3 = 2.63%
No discernible pattern	16 = 14.04%
Pattern broken for classroom management	5 = 4.39%
	114 = 100.01% on rounded figures above

Non - IRE based segments of discourse

It is clear that there was an underlying IRE structure in much of this lesson. However, there were also sections of dialogue where this structure was not in evidence. These were typically student dominated sections of dialogue instigated by the students themselves. Sometimes one student made a comment and another would join in. Other times a student asked a question of the teacher and other students answered instead of Liam. When each occurrence was examined in an attempt to describe the situations further, some underlying repetition was found, the results of which are shown below.

Table 3: Initial student action that breaks IRE pattern by number of occurrences

Initial student action that breaks IRE pattern	Number of occurrences
(Mis)identifies teacher/student error	2
Questions or comments on something new or different in mathematics	3
Question or comment about the nature of the task (e.g. about page number)	4
Question or comment that shows understanding or lack of (e.g. "That's easy!" or "I'm stuck")	5
Non-lesson related question	2

The following is an example of discourse involving student identification of teacher error. Liam is splitting his 14 x 4 array into two groups on the interactive whiteboard and leaves one dot behind accidentally. This section of discourse illustrates the way in which multiple students contributed to what would most likely have been a standard EIRE sequence if Liam hadn't made his mistake.

Excerpt from transcript	Description
Liam: OK, so we'll split it up. So we'll go one, two, three, four, five, six, seven and I'll group that lot together	Explanation
Finn: But Sir, I think that's eight.	Unprompted contribution (misidentification of error)
Liam: One, two-…no it's seven, that's fine. So that's one group and then this group	Explanation
Daithí: You forgot one!	Unprompted contribution (identification of error)
Students: You forgot one!	
Liam: Oh, thank you. And we'll get them again	Acknowledgement
Students: You forgot one	Unprompted contribution (identification of error)
Liam: OK, pick him up. OK, so we'll go one more time.	Acknowledgement and attempt to return to lesson
Student: Yeah!	

As illustrated in this example, one unprompted student comment often prompted other students to comment. This happened when students called out questions or comments about the work to be done or made statements about how easy or hard the tasks were. The IRE structure was broken due to student questions or comments on the mathematics three times, twice due to questions from Sam, which will be discussed below. The third time was when another student commented that rather than writing the questions with the product first e.g. 15

= 5 x ___, he had written all the "smaller numbers" first. Liam said that this was acceptable and moved on with the lesson. The non-IRE based segments of discourse were generally very short and the extract above is one of the longest examples. For the most part, students did not call out unprompted and the classroom discourse was IRE-based under the direction of the teacher. There were also five incidents where the IRE or EIRE pattern seemed to break down completely. In all of these cases this was due to classroom management issues and Liam's need to attend to one student in particular with special needs.

Discourse Community

Although it may seem unfair to analyse Irish primary mathematics lessons according to a framework developed in another educational system it is certainly worthwhile to briefly look at the four key areas of the Hufferd-Ackles et al. math-talk learning community framework to investigate the opportunities for mathematical thinking that were created in the discourse.

Questioning

The majority of Liam's questions were known answer questions where he simply asked for a solution or led students through a method. All of the written questions that students completed had only one possible answer and since much of the whole class discourse revolved around the correction of work, this may explain the high number of mathematical questions with only one answer that Liam asks. However in the introductory or explanatory phases of the lesson, Liam did ask mathematical questions with multiple possible answers. For example when considering the array 14 x 4, he says

Liam: But I can split the fourteen into two. Can you tell me two numbers that would make up fourteen?

After initially suggesting eight and eight a student eventually suggests seven and seven. Other options such as 10 x 4 and 4 x 4 are not mentioned by teacher or students and though there are multiple solution methods no others are discussed.

There are three cases where students ask mathematical questions and two are from Sam. In one of these Sam is asking if he can write the sums where the products are written first (e.g. 15 = 5 x __) "the right way round". In this case he is told to do it the way it is. His other question, as a potential source of mathematical ideas, is discussed below. The one other question came from Derek who seemed to be confused when Liam discussed the sum 6 x 20 = 6 x 4 x 5.

Explaining mathematical thinking; Source of mathematical ideas; Responsibility for learning

While it is not difficult to identify episodes within the transcript relating to the above areas, it can be harder to disentangle them from each other. The questions of teacher and students also influence these areas. There is an obvious link between the number of times the teacher asks questions probing mathematical thinking and the student act of explaining mathematical thinking. The one recorded teacher question that probes a student's mathematical thinking occurs in the following piece of dialogue. On the interactive whiteboard Liam portioned thirteen groups of three into six groups of three and seven groups of three:

Liam:	Six multiplied by three. OK so what's - and we're going to add these together then. So what's seven multiplied by three say? Sam
Sam:	There's an easier way to do it.
Liam:	What's the easier way?
Sam:	Three multiplied by three and ten multiplied by three.
Liam:	Three multiplied by three and ten multiplied by three. That's true. That's another way of working it out but we're just going to it do this, so if we, that would have been another way of splitting it up. Alright, but we've decided to split it up this time by seven multiplied by three and six multiplied by three. OK, so what's seven multiplied by three please, em, Rob?

It is evident in Liam's response that he is unsure how to handle this input. On one hand, he wishes to acknowledge the independent, flexible thinking but also wants to keep the lesson on track. In this case although he asks a follow up question to Sam's novel idea he does not probe deeper to find out why Sam thinks this method is easier. Thus we could say that, while Sam's idea was raised, it was not explored in any depth.

Later on, when Liam and the class have successfully partitioned 14 x 4 into 7 x 4 and 7 x 4, Sam asks "Teacher, could you not just do all the fours in it?" presumably asking if you could count in fours to figure it out. Liam ignores his suggestion and tells Sam that if he has a suggestion he should put his hand up. Sam's suggestion is valid and although the teacher twice refers to these sums as sums that are "not one of my tables" or ones that "we would not know automatically through our tables", one thinking strategy to solve them would be to use known facts from tables and count on. Also the visual representation of the 14 x 4 array on the whiteboard would have been a tool to aid counting in this manner. It is likely that the teacher wanted to concentrate on partitioning which seemed to be the focus of this part of the lesson. To use students' ideas productively in mathematics lessons is a highly complex task (Ball, 1993). The teacher actions carried out in these contingency moments are by their nature almost impossible to plan for (Rowland, 2007) and to maximise the opportunities for learning requires both strong mathematics content knowledge as well as pedagogical knowledge on the part of the teacher.

These episodes also illustrate that the source of mathematical ideas in the lesson is the teacher and the textbook. On the evidence of these incidents it is clear that Sam is taking responsibility for learning and responding to the mathematics presented by the teacher in his own way, attempting to figure the mathematics out by himself rather than relying on the one solution method presented by the teacher. However this attitude toward responsibility in one's own learning does not appear to be shared by rest of the class. The majority of the class passively accept the mathematics presented by the teacher, not questioning or commenting on different factoring methods for example. For the most part "Students are passive listeners; they attempt to imitate the teacher and do not take responsibility for the learning of their peers or themselves." (Hufferd-Ackles et al., 2004, p. 88)

DISCUSSION

It is the norms that emerge over time that will support the type of mathematical and communicative behaviour that is the foundation of the math-learning community. It appears from this lesson that the norms around respectful behaviour and work ethic have been clearly established. The lesson flowed smoothly with few unasked for interruptions by students. It was shaped by textbook resources and students did the work they were assigned in a calm, quiet atmosphere. By many standards, it would be considered a very good lesson. However, if one measures it against the standards suggested by the math-talk learning framework, there is little evidence to suggest that this classroom community is working at more than a level 0: Traditional teacher-directed classroom with brief answer responses from students (Hufferd - Ackles et al., 2004, p 88). It is appropriate to add the obvious caveat here. One recording is only a snapshot in time and should not imply that on a different day, perhaps with different mathematical activities, the teacher and student actions and the resulting mathematical discourse might not be of a different nature. Whether the quality of opportunities for mathematical thinking would be improved by changing the nature of the discourse remains to be seen. However, it does seem that opportunities for high level mathematical thinking were limited in this lesson and that the discourse was not truly mathematical but more a form of 'number talk' (Richards, 1991). I stated earlier that it may be unfair to analyse an Irish primary mathematics lesson based on criteria that was drawn up for a different system. But analysis using the framework drawn up by Hufferd- Ackles et al. not only illustrates differences between 'number talk' and true mathematical discourse, it may help teachers develop their practice as they aim to move from one discourse to the other.

REFERENCES

Ball, D. L. (1993). With an eye on the mathematical horizon: Dilemmas of teaching elementary school mathematics. *The Elementary School Journal*, 93(4), 373-397.

Boaler, J. (2009). *The elephant in the classroom: Helping children learn and love maths.* London: Souvenir Press.

Brodie, K., (2008). Describing teacher change: Interactions between teacher moves and learner contributions. In J.F. Matos, P. Valero & K. Yasukawa (Eds.) (2008). *Proceedings of the Fifth International Mathematics Education and Society Conference* (pp. 31 – 50)Lisbon: Centro de Investigação em Educação, Universidade de Lisboa – Department of Education, Learning and Philosophy, Aalborg University.

Cobb, P., Confrey, J., diSessa, A., Lehrer, R., & Schauble, L. (2003). Design experiments in educational research. *Educational Researcher*, 32 (1, Theme Issue: The Role of Design in Educational Research), 9-13.

Corcoran, D. (2009) Mathematics teaching as a shared enterprise: Learning by 'doing'. In D. Corcoran, T. Dooley, S. Close & R. Ward (Eds.), *Proceedings of the Third National Conference on Research in Mathematics Education* (pp. 34-52). Dublin: St. Patrick's College

DES/NCCA. (Department of Education and Science)/(National Council for Curriculum and Assessment). (1999). *Curriculum for Primary Schools –Teacher Guidelines Ma*thematics. Dublin: Stationery Office

Dooley, T. (2009). Insight in primary mathematics: Teacher ‘moves’ that facilitate the ‘stumbling of ideas across each other’. In D. Corcoran, T. Dooley, S. Close & R. Ward (Eds.), *Proceedings of the Third National Conference on Research in Mathematics Education* (pp. 102-111). Dublin: St. Patrick’s College

Hamm, J. V., & Perry, M. (2002). Learning mathematics in first-grade classrooms: On whose authority? *Journal of Educational Psychology*, 94(1), 126 - 137

Kazemi, E., & Stipek, D. (2001). Promoting conceptual thinking in four upper-elementary mathematics classrooms. *The Elementary School Journal*, 102, 59–80.

Kimberly Hufferd-Ackles, Karen C. Fuson and Miriam Gamoran Sherin (2004) Describing Levels and Components of a Math-Talk Learning Community. *Journal for Research in Mathematics Education* 35 (2), 81-116

Lakatos, I. (1976). *Proofs and refutations: The logic of mathematical discovery*. Cambridge: Cambridge University Press.

Lampert, M. (1990). When the problem is not the question and the solution is not the answer: Mathematical knowing and teaching. *American Educational Research Journal*, 27(1), 29-63.

Lyons, M., Lynch, K., Close, S., Sheeran, E., & Boland, P. (2003). *Inside classrooms: A study of teaching and learning*. Dublin: Institute of Public Administration.

Mehan, H. (1979). "What time is it, Denise?": Asking known information questions in classroom discourse. *Theory into Practice*, 18 (4, Communicating with Young Children), 285-294.

Mendick, H. (2006). *Masculinities in mathematics*. Maidenhead: Open University Press.

Nathan, M. J., Knuth, E. J., (2003). A study of whole classroom mathematical discourse and teacher change. *Cognition and Instruction*, 21(2), 175-207

Rowland, T. (2007). Developing knowledge for mathematics teaching: A theoretical loop. In S. Close, D. Corcoran & T. Dooley, (Eds.), *Proceedings of the Second National Conference on Research in Mathematics Education* (pp. 13-26). Dublin: St. Patrick’s College

Shiel, G., Surgenor, P., Close, S., & Millar, D. (2006). *The 2004 National Assessment of Mathematics*. Dublin: Educational Research Centre.

Surgenor, P. Shiel, G., Close, S., & Millar, D. (2006). *Counting on success: Mathematics achievement in Irish primary schools*. Dublin: Department of Education and Science Inspectorate.

Stein, M. K., Grover, B. W., & Henningsen, M. (1996). Building student capacity for mathematical thinking and reasoning: An analysis of mathematical tasks used in reform classrooms. *American Educational Research Journal*, 33(2), 455-488.

Wood, T., (1994). Patterns of interaction and the culture of mathematics classrooms. In S. Lerman (Ed.), *Cultural perspectives on the mathematics classroom* (149- 167). Dordrecht: Kluwer Academic Publishers.

THE NEED TO IMPROVE TEACHERS' KNOWLEDGE OF APPLICATIONS: FUTURE DIRECTIONS FOR INITIAL TEACHER TRAINING AND CONTINUOUS PROFESSIONAL DEVELOPMENT

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Research carried out in recent years has pointed to the need for mathematics teachers to develop a number of different types of knowledge in order to teach effectively. It is now accepted that knowledge of mathematics alone does not guarantee good teaching and instead it is imperative that mathematics teachers develop a number of different knowledge types including subject matter knowledge, knowledge of applications and curricular knowledge required for mathematics teaching. A critical feature of the package of knowledge designed by the author for Irish mathematics teachers (The Ladder of Knowledge) is the inclusion of knowledge of mathematical applications. The success of the new mathematics curriculum, Project Maths, relies heavily on teachers developing a good understanding of the relationship between mathematics and the real world. Its principal aims include placing a greater emphasis on the application of mathematical skills and knowledge and adopting a teaching for understanding approach. Therefore it is important that teachers first develop their own knowledge in these areas. This paper reports on a pilot study which looked at teachers' views on their own levels of knowledge in the area of mathematical applications, discusses the knowledge of applications demonstrated by second level mathematics teachers and offers recommendations for future Continuous Professional Development [CPD] initiatives and Initial Teacher Training [ITT] courses. Such changes to CPD and ITT will help overcome one of the biggest challenges facing Project Maths.

INTRODUCTION

A primary concern in Ireland among educators, policy makers and the Department of Education and Skills is the finding that many students complete their second level studies with a poor grasp of mathematics and are not prepared for the mathematics which many will face at third level or in the workplace (O'Donoghue, 2002). However, this problem is not only confined to Ireland and similar findings have been reported worldwide. For example the London Mathematical Society [LMS] (1995, p. 5) found that a large proportion of students are underprepared for third level mathematics on completion of their second level studies:

> "The problem is more serious; it is not just the case that some students are less well - prepared, but that many 'high attaining' students are seriously lacking in the fundamental notions of the subject"

This problem and the subsequent gap that developed between second and third level mathematics education forced those involved in the field of mathematics education to revise mathematics curricula (Project Maths Implementation Support Group [PMISG], 2010). New mathematics curricula, both in Ireland and abroad, have a strong focus on mathematical applications and modelling (Burkhardt, 2006). It is expected that this new focus will allow students to see the value of mathematics and enable them to be better prepared for the mathematics they will face in the workplace or at third level. However this change of focus for mathematics education means that change is also necessary in ITT and CPD initiatives in order to ensure that mathematics teachers are better equipped to teach the new curricula. This paper analyses whether or not we need to help improve teachers' levels of knowledge of mathematical applications and looks at ways in which this can be done so as to ensure Irish mathematics teachers are well prepared to deliver Project Maths effectively.

BACKGROUND TO THE STUDY

Project Maths

One of the main reasons for the authors undertaking this research was the introduction of Project Maths in Ireland in 2008. This new curriculum endeavours to teach mathematics in a way that will promote real understanding (PMISG, 2010). The NCCA (2009) continue by saying that under this new programme Leaving Certificate mathematics will strive to develop students' mathematical knowledge, skills and understanding in an effort to prepare them for the challenges they will face in the workplace. They see Project Maths as a vehicle that can promote the beauty and power of mathematics in students' everyday lives. The main objectives of Project Maths as outlined by the PMISG (2010) are to:

1. place a greater emphasis on the understanding of mathematical concepts and the application of mathematical skills and knowledge;
2. contribute to the development of higher order thinking skills, including logical reasoning and problem solving;
3. provide a solid foundation which prepares students for careers in science, technology engineering, business or humanities;
4. increase the use of context and applications so as to enable students to relate mathematics to everyday experiences.

In essence the initiative aims to strike a balance between developing an understanding of mathematical theory and developing an ability to apply mathematics to practical situations among students. While many considered the old mathematics course in Ireland to be static, formal, routine and inflexible, this new course is being viewed as a dynamic, flexible, problem focussed initiative that will change the face of mathematics education if implemented successfully (Engineers Ireland, 2010).

However, despite the positive image of Project Maths depicted thus far, it is worth noting that there is no guarantee that Project Maths will succeed. It will only work if teachers and other stakeholders in mathematics education strive to ensure that it is successfully implemented. Such reform will prove challenging for teachers. For example, this initiative will require them to extend their knowledge base and alter their teaching strategies in order to achieve the objectives outlined previously. It is essential therefore that teachers are supported in the early years and thereafter we must ensure that support and CPD opportunities are continuously available so that teachers can continue to update and develop their knowledge in the different domains necessary to teach Project Maths. The Ladder of Knowledge, a model of teacher knowledge designed by the authors, highlights the domains that mathematics teachers must continuously focus on throughout their career.

The Ladder of Knowledge

It is advisable that those involved in mathematics teacher education understand the types of knowledge a teacher must develop in order to teach effectively. Over the last two decades a number of different researchers have proposed models of teacher knowledge. For the purpose of this study, the author chose to focus on five of these models. These models were developed by Shulman (1986), Ernest (1989), Fennema & Franke (1992), Rowland (2007) and Ball et al. (2008). All these models, with the exception of that proposed by Shulman, are designed specifically for mathematics teachers. Those put forward by Rowland (2007) and Ball et al. (2008) are designed for primary school teachers while the remainder concentrate on the knowledge required by second level mathematics teachers. The knowledge domains proposed in these models are:

- Content knowledge, subject matter content knowledge and curricular knowledge (Shulman, 1986)
- Knowledge of mathematics; of other subjects; of teaching mathematics; of classroom organisation and management for mathematics; of the context of teaching mathematics; of education (Ernest, 1989)
- Mathematical knowledge, pedagogical knowledge, knowledge of mathematical representations, knowledge of learners (Fennema & Franke, 1992)
- Foundation knowledge, transformation knowledge, connection knowledge, contingency knowledge (Rowland, 2007)
- Common content knowledge, horizon knowledge, specialised content knowledge, knowledge of content and teaching, knowledge of content and students, knowledge of content and curriculum (Ball et al., 2008)

Content knowledge features in all five models. Researchers such as Ernest (1989) and Fennema & Franke (1992) explicitly stated that content knowledge was the foundation on which their model was built while other researchers also highlighted it as a core component. All the researchers who designed models previously agreed that it was essential for teachers to develop strong content knowledge and this, in turn, would allow them to develop other knowledge types which they found to be essential for teaching.

In addition to this, all the models also propose that once teachers have developed this content knowledge they must then aim to acquire pedagogical knowledge. Pedagogical knowledge is the practical knowledge that teachers require to teach mathematics. It refers to the transfer of a teacher’s subject matter knowledge into representations, explanations, analogies, illustrations, examples and demonstrations comprehensible to students. Essentially content knowledge, in tandem with pedagogical knowledge, empowers teachers to offer clear and simple explanations that students can understand as well as how to introduce topics using language and symbols that students can relate to.

Finally in each model proposed in the past, the researcher(s) identifies one knowledge domain unique to the profession of teaching. Each researcher found that it was important for a teacher's knowledge base to be different to that of a lay person or a person who majored in mathematics. As a result they all felt it necessary to include a specific knowledge domain which would differentiate the mathematics teacher from the mathematician. For example Shulman found pedagogical content knowledge to serve this purpose while Ball et al. (2008) refer to this domain as specialised content knowledge. Overall, the models proposed to date indicate that it is necessary for any model of teacher knowledge to include a knowledge domain that will distinguish the profession of mathematics teaching from other professions, especially those with high mathematical content.

In addition to these common characteristics, each model also demonstrates some unique aspects, most often the inclusion of new knowledge domains. For example, in addition to content and pedagogical knowledge, which are the core components of every model, the following knowledge domains are also included in the models chosen for analysis:

- Curricular knowledge: a knowledge of alternative teaching approaches, instruction methods, resources and texts;
- Knowledge of other subjects: a knowledge of pedagogical principles adopted in other subjects as well as the links between mathematics and other subjects;
- Knowledge of students both as individuals and members of the larger class group;
- Knowledge of representations: the knowledge required to transform their own mathematical knowledge;
- Contingency knowledge: the knowledge required to deal with the unexpected.

When all five models are analysed they yield a comprehensive list of knowledge domains necessary for mathematics teaching. Despite this in-depth list of knowledge domains, after designing their own model, Ball et al. (2008) spoke of the need for models of teacher knowledge to be continuously refined and updated in line with changes in mathematics curricula and teacher approaches.

As a result of this statement, the need for new knowledge domains to be included due to evolving mathematics curricula and with the introduction of Project Maths the authors felt it necessary to design a new model of teacher knowledge. They saw the need for a new model to cater for the needs of Irish mathematics teachers and to help these teachers overcome the obstacles with which Project Maths presents them. The model they propose to serve this purpose is the Ladder of Knowledge (Figure 1). This model is in line with those previously designed in that it demonstrates the three key characteristics previously discussed but also has unique elements that ensure it is relevant to Irish teachers in the current climate. As can be seen from Figure 1, subject matter knowledge occupies the bottom rung of the ladder due to the importance of this knowledge type. The authors note that no teacher can mount the Ladder of Knowledge without a deep understanding of content knowledge and, as with the other models, this knowledge domain acts as the cornerstone upon which this model is built. Similarly, pedagogical knowledge also features in this model and, like the models proposed in the past, the authors felt it necessary to include this knowledge domain in order to alert teachers to the fact that they need to be able to transform their mathematical content knowledge into material that students can understand and appreciate. Finally, Figure 1 highlights how competency in the knowledge domains depicted on the first two rungs make it possible for teachers to reach the third rung. Teachers must link what they know about mathematics to what they know about the effective teaching of mathematics and in doing so they will reach the third rung and develop a profound knowledge base suitable for the profession of teaching. Once they have developed this Knowledge for Effective Teaching they will have developed a type of knowledge unique to their own profession.

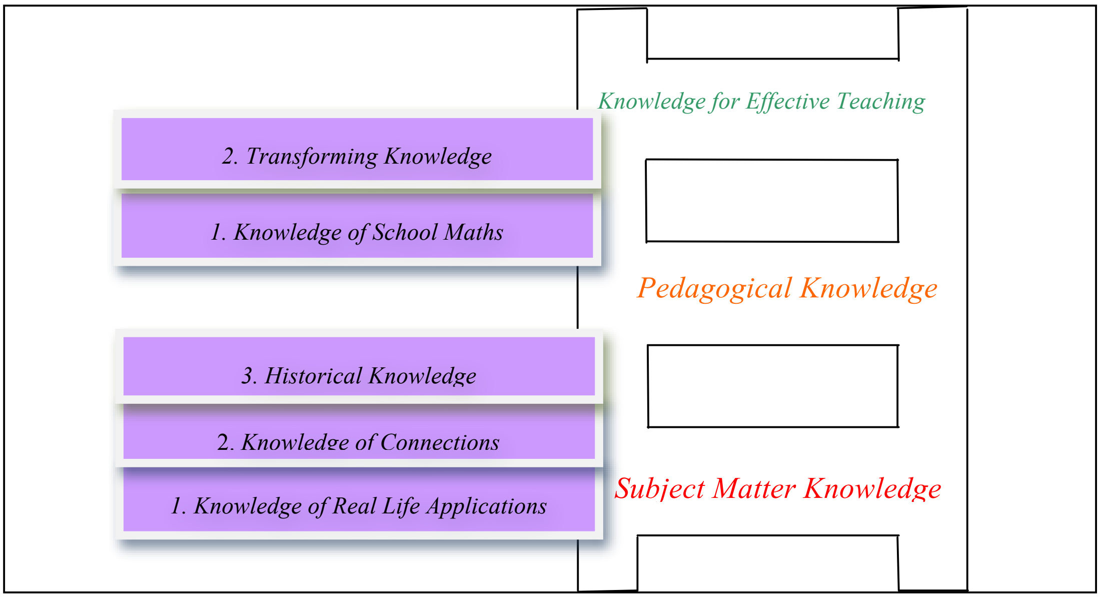

Figure 1: The Ladder of Knowledge (O'Meara, 2011)

In addition to these characteristics, which were evident in previous models, the authors also ensured that their model demonstrates new ideas. Firstly the authors adopt a novel approach by highlighting to teachers how they can progress from one rung to the next. Furthermore the authors encourage teachers to continuously revisit the rungs outlined on the ladder as well as the domains linking the rungs in order to improve and update their knowledge for mathematics teaching. The minor knowledge domains advise teachers of the different types of knowledge that they require before they can concentrate on developing types of knowledge further up the ladder. Secondly the authors include a new knowledge domain: "Knowledge of School Mathematics". Ball et al. (2008) alerted the readership to the need for teachers to understand the mathematics that is on the curriculum they are required to teach. However, they did not elaborate on or discuss this knowledge domain in detail. There is a distinct difference between the mathematics encountered by pre-service teachers at third level and the content which teachers will be required to teach at second level. Therefore it is critical for teachers to become familiar with what school mathematics entails as well as the applications and relevance of such content. The final unique aspect of this model, and possibly the most important aspect for teachers in today's society, is the inclusion of a knowledge of applications. With mathematics curricula changing worldwide a new model of teacher knowledge could ill afford to ignore this critical knowledge domain.

The Need for a Knowledge of Applications in a Model of Teacher Knowledge

In recent years researchers and international organisations have acknowledged the extensive uses of mathematics in today's society (LMS, 1995; Smith, 2004). New mathematics curricula are beginning to reflect this and, as previously discussed, the majority of these new curricula now have a strong focus on mathematical applications in the real world. Despite this, Glaister & Glaister (2000) presented evidence to suggest that a large number of teachers are unaware of the possible applications of mathematics in everyday life. Their work notes that teachers are unaware of how mathematics relates to key areas in the business and industrial communities. Furthermore Doerr (2007) found that the reason for the limited use of mathematical applications and mathematical modelling at second level to date is a lack of knowledge of applications on the part of teachers. This is a concern for those involved in mathematics education both in Ireland and internationally as, in order for teachers to deliver a curriculum which has a strong focus on mathematical applications, they must first have an extensive knowledge and understanding of such applications. As a result it is no longer viable to omit this knowledge domain from models of knowledge designed for mathematics teachers. Until teachers and teacher educators in Ireland and abroad recognise the importance of a knowledge of applications and in turn improve their proficiency in this regard they will find it difficult to effectively deliver curricula which have a very strong focus on applications and modelling.

This knowledge of applications requires teachers to develop an understanding of the uses of mathematics in the real world and an appreciation of how this subject can be used to solve numerous problems facing people outside of a school setting. If teachers were to develop this type of knowledge they would be able to highlight the numerous everyday problems which can be solved using mathematics as well as advise students of the various professions that rely heavily on mathematics (O'Meara, 2011). In essence, this knowledge domain is critical in order for teachers to deliver Project Maths effectively and to allow them to help students appreciate mathematics, enhance their interest in the subject and promote it as a discipline that they will rely heavily on once they graduate from second level education.

Due to the importance of this knowledge domain and the fact that the Ladder of Knowledge is the first model to include this knowledge type the authors felt it necessary to assess teachers' levels of knowledge of applications. The authors will first explain the methodological choices made prior to undertaking this project before discussing the results which emerged from the study.

METHODOLOGY

Having established the package of knowledge required for Irish mathematics teachers, which included a knowledge of applications, the authors then sought to determine teachers' levels of knowledge in each of the domains. This was done through qualitative and quantitative means and the findings in relation to teachers' knowledge of applications are presented in this paper.

Qualitative Data

The qualitative findings outlined in this paper were gathered through semi-structured focus groups. Focus groups were chosen for the purpose of this project as research has found that they allow participants to 'bounce' ideas off one another and as a result of this sharing of views such groups often produce data and insights that may not have been possible to attain in a one – to – one interview (Morgan, 1996). The TAP paradigm was selected for developing the questions to be used in the focus group. This paradigm, which was proposed by Foddy (1993), requires the researcher to ensure the topic for each question is properly defined (**T**opic), the questions are applicable to the respondent (**A**pplicability) and the perspective that respondents should adopt when answering the questions is specified (**P**erspective). The focus groups allowed the authors to gain an insight into the importance which teachers attribute to knowledge of applications as well as their perceived levels of competency in this domain. This qualitative data was analysed using the computer software package NVivo.

Quantitative Data

The quantitative data required for this study was collected through detailed questionnaires. Part of these questionnaires sought to assess teachers' knowledge of mathematical applications in the topic of calculus and also to determine the difference, if any, between teachers' instrumental understanding of calculus (knowledge of procedures) and their relational understanding of calculus (knowledge of applications). All data was recorded and analysed using SPSS.

Research Sample

The authors used convenience sampling to select the locations of the schools that would be contacted for this study. They selected 6 locations within a 100 km radius of their institution and another location which was convenient to them. The list of counties selected using this convenient sampling strategy were Cork, Dublin Clare, Galway, Kerry, Limerick and Tipperary. In total this meant that the sample from which individual schools could be picked consisted of 426 out of a total of 743 schools in Ireland. Once these counties had been selected, a list of all schools in these counties was drawn up and the schools to be contacted were selected through simple random sampling. 104 schools were selected using this strategy and the authors contacted these schools and asked them to participate in the study. In total 15 schools declared a willingness to participate; 4 schools (13 teachers including one out-of-field teacher) agreed to participate in the focus groups and 11 schools (22 teachers including 3 out-of-field teachers) agreed to complete the questionnaire. The sample consisted of 15 males and 19 females who ranged in age from 22 to 61 years and had 3 to 37 years teaching experience. This sample is considered small in statistical terms. The authors will now describe the qualitative and quantitative responses obtained from these teachers.

FINDINGS FROM THE PILOT STUDY

Findings from the Focus Groups

An encouraging finding to emerge from the focus groups was the widespread belief among participating teachers that it is important to develop a knowledge of applications. When teachers in this phase of the study were asked if they thought it was necessary for second level teachers to develop a knowledge of applications the majority of teachers thought that this knowledge domain was critical for the purpose of teaching. In their opinion this type of knowledge will allow them to engage students in the topic of mathematics and to arouse interest in the subject among students who are failing to see the relevance of mathematics. For example when asked if it is important for teachers to understand how mathematics is applicable to students everyday life the following were some of the responses received:

Eilis[1]: Ah extremely important, like for teachers to be able to relate it…ah if you can relate it to their, em, how they can use it in everyday life they will straight away, like it will up their interest…If we could relate it to real life and be able to say that to the students it would make learning a whole pile easier you know.

Andrew: Well yeah the practical side I would agree with, at least that gives them some reason why they're learning it…I think a lot of them have that question well why, why do we need to learn it, at least if you have a practical problem it gives them some reason why they need to learn it.

In addition to teachers seeing the importance of this knowledge domain they also admitted that this was an area that they needed help with. When teachers in this phase of the study were asked if they felt their knowledge of applications was sufficient for teaching the following are a selection of the responses:

Jason: I'd never connect differentiation [to real life] to be honest…I'd never really think of connecting differentiation to something to be honest. There's little practical side to it.

Eimear: …they do ask you oh when am I ever going to use this again…and you literally have to go [say] the only place you will see this again is on your Leaving Cert exam paper and you're not going to see it again.

Shane: As regards the applying it there to different ones [subjects] I wouldn't put anything down there. You can't connect it [calculus] to other subjects.

James: I'd like to be able to do all that stuff in actual fact [highlight the relevance of calculus and involve students in the mathematics class]…but I can't.

The responses suggest that these teachers currently feel that their knowledge of mathematical applications needs to be developed further. This makes it difficult for them to show students the value of mathematics in the workplace or in their everyday lives. Furthermore it is unlikely that teachers who are not conversant with out of school uses of mathematics will be able to deliver a curriculum which places a major focus on applications of mathematics. Responses such as that from James suggest that although teachers understand the importance of developing a knowledge of applications they need help to further their understanding in this regard.

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[1] Each teacher in this study was assigned a pseudonym in order to protect their identity and ensure confidentiality. The alias assigned to each teacher remains consistent throughout this study.

Findings from the Pilot Study

Project Maths places additional demands and challenges teachers to develop their knowledge of applications much further than heretofore. In this context, majority of teachers in this study indicated that they would avail of more opportunities to develop their knowledge of applications. 5 teachers (23%) involved in this phase of the study felt that their knowledge of applications was strong while 9 teachers (41%) felt that their knowledge in this area could be improved.

Generally speaking the focus groups confirmed that these teachers needed more support to approach applications. Such findings were also evident in this phase of the study. In section B of the questionnaire teachers were required to answer a number of calculus questions to analyse the difference between their procedural knowledge of mathematics and their knowledge of mathematical applications. The first 4 questions examined teachers' procedural knowledge, i.e., their knowledge of rules and procedures while the last 3 questions assessed their levels of relational understanding, i.e., their ability to apply such procedures to real life problems. When analysing the questionnaires the author marked each question using the same scale. Each question in Section B received one of five ratings. It was either marked fully correct, attempt made with slight error/numerical slip, attempt made but with fundamental errors/numerous numerical slips, attempt made but with no correct work to note or no attempt made. In order to compare the standard of answers the author first analysed each question individually before combining the four procedural questions and calculating the average percentage of teachers in each of the aforementioned categories. The exact same procedure was carried out when analysing the three applicable questions. The results from this analysis are highlighted in Figure 2, below.

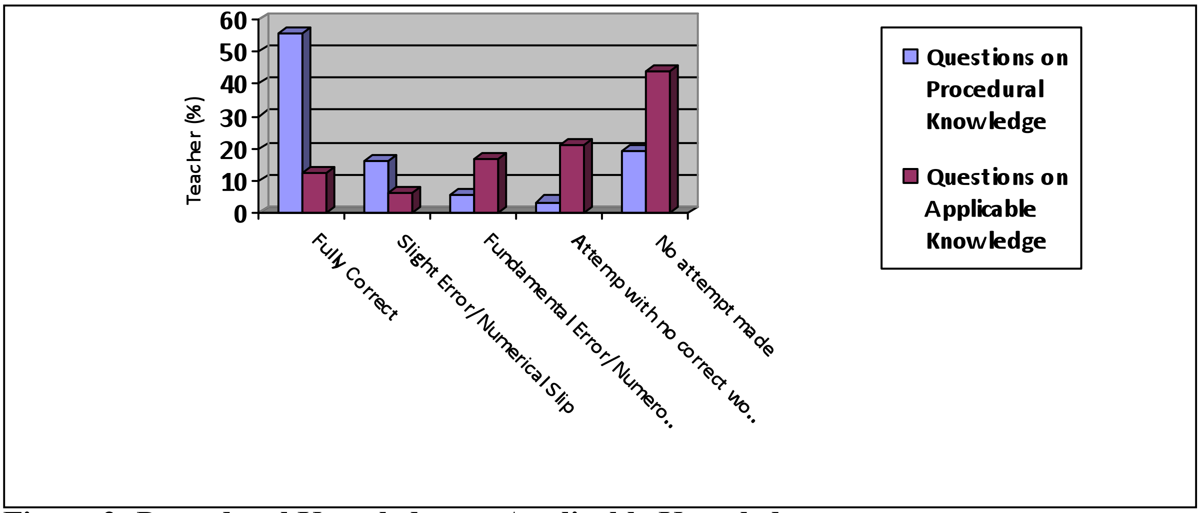

Figure 2: Procedural Knowledge vs. Applicable Knowledge

Figure 2 suggests that teachers in this phase of the study did less well with questions which assessed their relational understanding. In the main these successful mathematics teachers found it difficult to score well on this type of assessment of applications knowledge. On average 56% of participants were able to correctly answer questions using rules and procedures that appear on the Leaving Certificate syllabus with a further 16% only making a simple numerical slip. In contrast an average of 12% was able to correctly apply such procedures to everyday problems. On average, less than 4 teachers in this study were unable to attempt a question which required them to demonstrate procedural knowledge compared with over 10 teachers who could not attempt a question requiring them to apply knowledge while a further 5 teachers attempted these applicable questions but none of the work submitted was correct.

Teachers in this study also found it difficult to outline the numerous different areas where mathematics can be used in student's everyday life. When teachers in this phase of the study were asked where they would use differentiation and integration in everyday life, 10 of the 22 teachers surveyed were able to connect differentiation to 3 or more real life problems while 3 teachers were able to do likewise for integration. Such findings would again suggest that efforts need to be made to help these teachers improve in this domain. When the answers submitted by teachers were analysed further, more issues arose. When teachers outlined the areas where differentiation was applicable in everyday life the most common answers were speed and acceleration problems (16 teachers), maximum and minimum problems (5 teachers), problems involving projectiles (5 teachers) and business problems (4 teachers). The most popular links between integration and everyday life were problems relating to area (12 teachers), problems relating to volume (6 teachers) and speed and acceleration problems (4 teachers). When these applications are compared with those offered in the old Leaving Certificate textbooks there are very few discrepancies between the two. All the links provided by teachers with the exception of "business problems" appear in all textbooks available in Ireland for senior cycle mathematics. These examples provided by textbooks and in turn by teachers represent a narrow coverage of applications of calculus in the real world and they will need to be improved upon in order to teach calculus effectively in a more challenging climate.

DISCUSSION AND RECOMMENDATIONS

Overall the findings from this small scale study indicate that teachers simply do not have a good understanding of mathematical applications. They readily admit to possessing inadequate levels of knowledge in this domain and many teachers openly asked for help in this regard. Despite realising how important this knowledge domain is many teachers claimed that because applications were overlooked in the old mathematics syllabus they never got the opportunity or saw the need to develop their own knowledge of applications. However, with the change of syllabus this is no longer the case and it is now essential that teachers have a substantial knowledge of applications. In order to ensure that Project Maths becomes a successful initiative that changes mathematics education for the better steps need to be taken to help teachers improve their knowledge in this area. The authors propose the following steps in order to help teachers overcome this problem:

1. In order to facilitate trainee teachers in this regard the authors recommend a revision of teacher training courses and advocate that all such courses should include a capstone module similar to that introduced in Harvard University, Massachusetts. This capstone module should generate a knowledge of school mathematics as well as highlighting the

applicability and relevance of such content, hence enabling teachers to develop their knowledge of applications during their tertiary education.

2. Lecturers on all teacher training courses should be made aware of the Ladder of Knowledge or other models of teacher knowledge and should strive to ensure, where possible, that they share information about mathematical applications and the relevance of mathematics with their students. Only then will third level courses ensure that trainee teachers graduate with the knowledge required to teach mathematics at second level.
3. CPD needs to focus on helping teachers develop and enhance their knowledge base on a regular basis. Research has found that in order for a CPD initiative to be considered effective it must focus on improving a teacher's knowledge base (Resnick, 2005). Furthermore, Resnick found that knowledge focussed CPD can have a significantly positive effect on student learning. Therefore in order to help prepare teachers for Project Maths, CPD initiatives must focus on developing teachers' knowledge base. This study has highlighted that one of the first areas that needs to be considered for CPD is a knowledge of applications. Action needs to be undertaken in the near future to ensure that CPD which focuses on mathematical applications is made available to all mathematics teachers in Ireland.

CONCLUSION

To conclude, this study has served to highlight a challenge that currently faces Project Maths and one that may even determine its success. The findings presented in this paper suggest that teachers do realise the importance of a knowledge of mathematical applications but they currently do not believe that they have the knowledge necessary to teach a curriculum that has a strong focus on mathematical applications. However, what is encouraging is the fact that the majority of teachers in this study are willing to invest time and effort in improving their knowledge in this area and as a result there is now an onus on all other parties involved in mathematics education to help them in this regard. Implementing changes such as those advocated by the authors above will help third level institutes produce more knowledgeable and better prepared teachers while recommendations for CPD initiatives will help ensure that future programmes are worthwhile and address the necessary issues. Until such steps are taken problems relating to teachers' levels of knowledge and the standard of mathematics teaching will persist in Ireland.

REFERENCES

Ball, D., Thames, M.H. & Phelps, G. (2008) *Knowing Mathematics for Teaching: Who knows mathematics well enough to teach third grade and how can we decide.* Paper presented at the Annual Meeting of the American Educational Research Association, Montreal, 2005.

Burkhardt, H. (2006) Modelling in Mathematics Classrooms: Reflections on past developments and the future. *Zentralblatt fur Didaktik der Mathematik,*38(2), 178 – 195.

Doerr, H. (2007) What Knowledge do Teachers Need for Teaching Mathematics through Applications and Modelling? In Blum et al. (Eds.) *Modelling and Applications in Mathematics Education: The 14th ICMI Study* (pp. 69 – 78) New York: Springer.

Engineers Ireland (2010) *Report of Task Force on Education of Mathematics and Science at Second Level.* Retrieved March 10th, 2010 http://webpages.dcu.ie/~bradysa/CASTEL_report_uploads/20100211-Mathematics_and_Science_at_Second_Level.pdf.

Ernest, P. (1989) The Knowledge, beliefs and Attitudes of the Mathematics Teacher: A Model. *Journal of Education for Teaching,* 15(1), 13 – 33.

Fennema, E & Franke, F. (1992) Teachers' Knowledge and Its Impact. In D.A. Grouws (Ed.) *Handbook of Research on Mathematics Teaching and Learning* (pp. 147 – 164) New York: Macmillan Publishing Company.

Glaister, E.M. and Glaister, P. (2000) *The Role of Applications in Mathematics Teaching and the Enhancement of Mathematics Learning through Project Work*. Retrieved October 2nd, 2007 http://www.cimt.plymouth.ac.uk/journal/egappm.pdf

London Mathematical Society (2005) *Tackling the Mathematics Problem.* Southend-on-Sea: LMS.

Morgan, D.L. (1996) *Focus Groups,* in Hagan, J. & Cook, K.S. (Eds.) Annual Review of Sociology, Volume 22, California: Annual Reviews, 129 – 152.

National Council for Curriculum and Assessment (2009) *Leaving Certificate Mathematics: Draft syllabus for strands 1 – 4.* Dublin: Stationary Press.

O'Donoghue, J. (2002) *Mathematics: The transition from second level to university.* Fr. Ingram Memorial Lecture to the Irish Mathematics Teachers Association, Thurles, December, 2002.

O'Meara, N. (2011) *Improving Mathematics Teaching at Second Level through the Design of a Model of Teacher Knowledge and an Intervention Aimed at Developing Teachers' Knowledge*. Unpublished Thesis (Ph.D.), University of Limerick.

Project Maths Implementation Support Group (2010) *Report of the Project Maths Implementation Support Group.* Dublin: Stationary Press.

Resnick, B. (2005) Teaching Teachers: Professional development to improve student achievement. *Research Points: Essential Information for Education Policy,* 3(1), 1 – 4.

Rowland, T. (2007) Developing Knowledge for Mathematics Teaching: A theoretical loop. Paper presented at Second National Conference on Research in Mathematics Education in Ireland: MEI 2, Dublin, September, 2007.

Shulman, L.S. (1986) Those Who Understand: Knowledge growth in teaching. *School Psychology Review,* 16(3), 290 – 305.

Smyth, A. (2004) Making Mathematics Count. *The Report of Professor Adrian Smith's Inquiry into Post – 14 Mathematics Education.*

PROOF EVALUATION AS LEARNING ACTIVITY?

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Proof plays an important role in mathematical practice and the development of an understanding and appreciation of proof is an important objective of mathematical education. This paper suggests to consider practice of proof evaluation in the teaching of mathematics, motivated by findings from an exploratory study of the approaches taken by novice students to the validation and evaluation of mathematical proofs. The study is based on a series of tests and interviews with first year honours mathematics students at the National University of Ireland, Galway. The students were asked to evaluate and criticize several proposed proofs of mathematical statements. The participating students' engagement in these activities as well as some improvement in proof evaluation performances of individual students indicate learning effects regarding mathematical proof through practice of proof validation and proof evaluation. The considerations on proof validation and evaluation as teaching activities are discussed from a socio-cultural perspective of learning in which learning is seen as accessing and participating in the practice of an expert community.

Theoretical considerations in the first part of this paper discuss the role of proof in mathematical practice and how learning of proof appears from a socio-cultural perspective. The second part describes how the suggestion to consider practice of proof evaluation in the teaching of mathematics emerges from an exploratory study on novice students' proof evaluation performances.

THEORETICAL BACKGROUND

This section discusses the essential role of proof and proving in mathematics, relates this to students' difficulties with mathematical proof and delineates the significance of proof in the field of mathematical education. A socio-cultural approach to learning and an application of this approach to mathematical proof is explained. This approach provides a framework to consider practice of proof evaluation in the teaching of mathematics.

The Role of Proof and Proving in Mathematics.

Proof plays an essential role in mathematics. This has often been acknowledged in the literature of Mathematics and Mathematical Education. For example, the mathematician Rav (1999) emphasizes the fundamental importance of proofs in mathematics:

> "Proofs, I maintain, are the heart of mathematics."

According to Fischbein (1982) the proof of a theorem is for a mathematician:

> "the absolute guarantee of the universal validity of the theorem. He believes in that validity."

Dreyfus (1990) asserts that

> "proving is one of the central characteristics of mathematical behavior and probably the one that most clearly distinguishes mathematical behavior from behavior in other disciplines."

In mathematical practice the role of proof exceeds demonstrating the truth of a theorem and the reason why a theorem is true. Rav (1999) states that proofs do much more than just verify mathematical claims, that they are actually bearers of mathematical knowledge and also necessary to enhance that knowledge. He argues that the very act of devising a proof contributes to the development of mathematics, and sees proofs as the primary focus of mathematical interest. In his view proofs can not only yield new mathematical insights, giving them a value far beyond establishing the truth of new propositions, but can also convey new mathematical strategies and new methods for solving problems. Furthermore, proofs are important means in communities of mathematical practice. Bass (2009) emphasizes this in a special lecture on the activity of proving and accepting new proofs as certification of new knowledge and as activity of the community. He considers "proof, as a theoretical concept", and "proving, as a fundamental practice of the disciplinary community".

Acknowledging that finding new proofs is an important activity in a mathematician's work, a broader range of functions or purposes of proofs than that of establishing the truth of a statement should be recognized. Purposes of proofs in mathematical practice have been discussed within the mathematical education literature in the last four decades. Bell (1976) distinguishes three functions of proofs (verification, illumination and systematization). His distinction has been expanded by De Villiers (1999):

- *verification* (concerned with the truth of a statement)
- *explanation* (providing insight into why it is true)
- *systematization* (the organization of various results into a deductive system of axioms, major concepts and theorems)
- *discovery* (the discovery or invention of new results)
- *communication* (the transmission of mathematical knowledge)
- *intellectual challenge* (the *self-realization* derived from constructing a proof)

De Villiers' suggested model for the functions of proof has been widely accepted and applied within the mathematical education community, for example by Hanna (2000) and Weber (2002). De Villiers explicitly states that he does not claim his list of functions to be complete. Expansions have been suggested, such as an *aesthetic* function, and a function of *memorization* and *algorithmization* (De Villiers, 1999). Hanna (2000) suggests adding *construction of an empirical theory*, *exploration of the meaning of a definition or the consequences of an assumption* and *incorporation of a well-known fact into a new framework and thus viewing it from a fresh perspective*. Hanna and Barbeau (2008) refer to the first five items of De Villiers' list and claim that "it stopped short of stating that proof contains techniques and strategies useful for problem solving." They are inspired by Rav's paper (1999) in which he claims that

> "proofs rather than the statement-form of theorems are the bearers of mathematical knowledge. Theorems are in a sense just tags, labels for proofs, summaries of information, headlines of news, editorial devices. The whole arsenal of mathematical methodologies, concepts, strategies and techniques for solving problems, the establishment of interconnections between theories, the systematization of results - the entire mathematical know-how is embedded in proofs."

With Rav's considerations in mind, De Villiers' suggested list of functions of proof can be expanded by one more item:

- *mediation of problem solving competencies* (the transmission of techniques and strategies useful for problem solving).

Students' Difficulties with Mathematical Proof

Given the importance of proof and proving in the spectrum of mathematical activities, the students' understanding, appreciation and knowledge of the nature and role of proof must be considered. Proof is a difficult mathematical concept for students. Research shows that most university students have difficulties not only in construction of proofs (Schoenfeld, 1985; Senk, 1985; Martin and Harel, 1989; Hart, 1994; Moore, 1994; Bills and Tall, 1998; Harel and Sowder, 1998,; Weber, 2001), but even in knowing what constitutes a proof (Harel and Sowder, 1998; Recio and Godino, 2001). Furthermore, research demonstrates that students have difficulties to determining whether a proof is valid (Selden and Selden, 2003).

There is little or no existing research literature specifically addressing the learning of mathematical proof and proving in Ireland. Discussions with mathematics lecturers of various Irish universities indicate that Irish university students have similar difficulties with mathematical proof to those described in the studies mentioned above. Worrying tendencies in Irish second level students' abilities to construct proofs are indicated by the Chief Examiner's Report on the Higher Level Leaving Certificate Examination in Mathematics (NCCA Chief Examiner's Report, 2005) which describes a

> "noticeable decline in the capacity of candidates to engage with problems that are not of a routine and well-rehearsed type. [...] Whereas procedural competence continues to be adequate, any question that requires the candidates to display a good understanding of the concepts underlying these procedures causes unwarranted levels of difficulty."

More specifically the report notes the students' difficulties with proof by induction:

> "most [students] are completely unable to handle the substantive step" (NCCA Chief Examiner's Report, 2005).

Proof in Mathematical Education

Considering the essential role of proofs in mathematics it is not surprising that

> "there seems to be a general consensus on the fact that the development of a sense of proof constitutes an important objective of mathematical education" (Mariotti, 2006).

More focus on proof has been suggested in many syllabi in recent years. For example the guidelines from NCTM recommend that every student from pre-kindergarten through grade

12 experience proof and proving in different ways in their school mathematics (NCTM, 2000). Proof has also recently obtained a stronger status in curricula in Italy (Furinghetti & Morselli, 2010), Sweden and Estonia (Hemmi, Lepik & Viholainen, 2010).

In Ireland, while *proof* and *proving* are not explicitly mentioned, the above mentioned purposes of proof are acknowledged in some of the objectives of the newly introduced Leaving Certificate syllabus called *Project Maths.* One objective of this syllabus is to encourage and practice reasoning and justifying. Other objectives are that learners develop "relational understanding ('knowing why')" or "the ability to apply their mathematical knowledge and skill to solve problems in familiar and unfamiliar contexts" (Leaving Certificate Mathematics Syllabus for examination in 2012).

Acknowledging the importance of proof and proving in the mathematical community on the one hand and the students' difficulties with proof on the other, one aim in the field of mathematical education is to find methods to teach proof and proving which may appear less challenging to students. To explain why consideration of practice of proof evaluation in the teaching of mathematics is suggested, the following paragraph introduces a socio-cultural approach to learning and an application of this approach to mathematical proof.

A Socio-Cultural View of Learning Mathematical Proof

There has been a change in the view of learning in the field of education in the last four decades: from mostly influenced by constructivist views concerned with the development of *individual learners* (e.g. Piaget or Bruner) to more sociocultural approaches to learning in which *learning develops out of social interaction* (Vygotsky). One of these socio-cultural theories is Lave and Wenger's (1991) *theory of legitimate peripheral participation*, a situated learning theory that argues that knowledge is distributed amongst a community of practice [1] and learning appears through participation in this practice. *Artifacts* serve as mediators within a community. In regard to a community, artifacts are *products* of human activity (the practice, often work), and as well *tools* to use for activities within the community, continually being modified to increase their value and usability in the practice. Becoming knowledgeable or *learning* means increasing membership in the practice which includes the ability to use and understand its artifacts. They provide learners with opportunities to enter a community.

This theory has been widely acknowledged and used in various fields of socio-sciences and education. In mathematics, inspired by the work of Adler (1999) who considers *talk* as an artifact in mathematical practice, Hemmi (2006) considers a maths department at a Swedish university as a *community of mathematical practice*, and *proof as an artifact* within such a practice. She uses this frame to describe how students and mathematicians encounter proof at university level in Sweden. In this view learners learn to understand and use the artifact *proof* through participation in the practice.

SUGGESTIONS FOR TEACHING EMERGING FROM AN EXPLORATORY STUDY ON NOVICE STUDENTS ' PROOF EVALUATION PERFORMANCES.

Findings from a comprehensive exploratory study of the approaches taken by novice students to the validation and evaluation of mathematical proofs (Pfeiffer, 2010) suggest that inclusion of proof evaluation practice in the teaching and learning of mathematics is worthy further consideration. This section introduces the terms *proof validation* and *proof evaluation*, describes the experiments of the study and explains why evaluation of proof is considered beneficial in the teaching of mathematics in relation to the socio-cultural approaches to learning as introduced above.

Clarification of the Terms *Proof Validation* and *Proof Evaluation* and the Significance of these Activities in Mathematical Practice

With Selden and Selden (2003), the readings and considerations to determine the correctness of mathematical proofs and the mental processes associated with them are called *validations of proof*. In a mathematical community the process of accepting a proof involves more than its validation. Validation, the determination of the correctness of an argument, is a significant part of the process of accepting a proof, followed by a more extensive and open-ended process that involves a search for understanding as well as correctness, a desire for clarity and an alertness to the possibility of adaptation or extension. Seeing learning as assessing and participating in the practices of a community, I suggest to widen the context from *proof validation* to the notion of *proof evaluation*. *Proof evaluation* embraces both, determining whether a proof is correct and establishes the truth of a statement (validation) and also how good it is regarding a wider range of features (such as clarity, context, sufficiency without excess, insight, convincingness or enhancement of understanding) or purposes (as suggested in De Villier's list described above). That is, *proof evaluation* includes assessment of the significance and merits of a proposed proof.

Critical and sceptical reading of proofs are important activities in mathematical practice. Hanna (1991) quotes the view of the Russian logician Manin who emphazised that the acceptance of a proof depends significantly on a social process:

> "A proof becomes a proof after the social act of *accepting a proof* (Manin, 1997, as cited in Hanna, 1991).

Thurston's (2006) considerations also highlight the crucial role of the mathematical community in the context of proving:

> "The people who see the way to proving theorems are doing it in the context of a mathematical community; they are not doing it on their own. They depend on understanding of mathematics that they glean from other mathematicians. Once a theorem has been proven, the mathematical community depends on the social network to distribute the ideas to people who might use them further."

If proving is seen in the context of a mathematical community, the colleague mathematicians' critical and sceptical reading of proofs must be regarded as important activities in mathematical practice. Acknowledging that from a socio-cultural viewpoint it is essential for novices in a practice to become familiar with artifacts and activities in this practice and to get opportunities to participate in it, practice of the activities *proof validation* and *proof evaluation* is considered beneficial for learning of mathematics.

The Experiments

The study is based on a series of tests and interviews conducted with first year honours mathematics students at NUI Galway. In May 2008 a written exercise was carried out as one of a series of workshops. This test was attended by 37 students. The aim of the exercise was to access and analyse certain mathematical skills such as generalizing, formulating conjectures or justifying and to investigate learning progress during the first year in university. [2] One of the questions in the test was an evaluation task, which caught my attention. The undergraduates were presented with attempts of six fictional students Aoife, Barry, Cathy, Dan, Eve and Finn, to prove whether the following statement is true or false.

When you add any two even numbers, your answer is always even.

For each response, the students were asked to give a mark out of five and a line of advice. The proposed 'proofs' were chosen to provide a variety of different features to allow identification of the students' criteria for accepting or valuing a proof. One answer is an example of a correct, algebraically presented proof of the statement; one consists of a collection of examples verifying the statement; one is a correct proof of the statement written in text; one is an example of an unusual approach to justify the statement; one is wrong, but presented algebraically; another one is an example of visual reasoning. The undergraduates' responses to these proposed proofs indicate what they find essential in a good proof. In March 2009 interviews were held with eight students. Eighteen students who had attended a written exercise including an evaluation task in September 2008 as well, were invited to participate in a research project. They were chosen carefully in an effort to cover a wide spread of performances in the written experiment. Eight of these students volunteered to participate. They were all invited for interviews which took 30 to 45 minutes. Every interview was tape recorded and transcribed. The aim of the interviews was to get a deeper insight into students' opinions about valuable proofs, students' validation and evaluation processes and learning effects during the validation and evaluation processes. The students were presented with two mathematical statements and five or six proposed proofs of each statement and asked to evaluate and rank them. As in case of the written exercise, the chosen 'proofs' were of different character: for example one proof of the first statement was purely visual, one consisted only of a fairly random assortment of examples, one was completely wrong but written in "algebraic" language, one was more general than required, another was written in text. The length of this paper does not allow inclusion of all the tasks. However, to exemplify the tasks, two examples of the proposed 'proofs' are listed below. These example 'proofs' will also be used to illustrate some of the findings. The first of the two statements is similar to the statement of the written evaluation task:

***Statement I.* The squares of all even numbers are even, and the squares of all odd numbers are odd.**

***Darragh*'s answer:**

k is any whole number.
l is any whole number.
2k is an even number.
2l+1 is an odd number.
$(2k)^2 = 4k^2 = 2(2k^2)$ is even.
$(2l+1)^2 = 4l^2 + 2l + 1 = 2(2l^2 + l) + 1$ is odd.

Elaine's **answer:**

Proof: Let x be an even number and let y be an odd number.
Then x+y and x-y are both odd.
$(x+y)^2 = x^2 + 2xy + y^2$, $(x-y)^2 = x^2 - 2xy + y^2$.
So $(x+y)^2 + (x-y)^2 = 2x^2 + 2y^2 = 2(x^2 + y^2)$.
The number $2(x^2 + y^2)$ is even obviously.
Dividing by 2 we find that $x^2 + y^2$ is odd.
Then x^2 and y^2 can't be both even or both odd. So x^2 is even and y^2 is odd.

Some Findings

The written evaluations provide some understanding about the students' *criteria* when evaluating mathematical arguments. Most students recognize the difference between a demonstration by example and a mathematical proof as well as the importance of mathematical definitions, general applicability, internal correctness and convincing mathematical arguments. More demanding attributes of a proof like precise argumentation, sufficiency without excess and justification of significant assertions in a proof are important only to a relatively small proportion of the students. The students' formulaic and inflexible picture of valuable proofs must be noted. Structure and 'mathematical-looking' formalism seem more important to some students than correctness or the ideas comprising the argument.

Interpretations of the interviews provide some insight into what the participating students consider to be *features and purposes of proofs*. The evaluation task proved particularly useful to identify the purposes of proofs considered by the students. For example the excerpt of *Student F*'s evaluation of *Darragh*'s answer below indicates that *Student F* considers the purpose *verification* (De Villiers) of proof.

> *"I like that answer. - The fact that it counts for all numbers. And - ah - you can't say that it's not true."*

Student B also approves *Darragh*'s answer, however for different reasons:

> "*Better than the other ones - because you see the mathematical way, why those even and those odd."*

Student B considers the purpose *explanation* (De Villiers), to show why the statement is true, of proof.

As in the written exercise, the students considered a relatively wide range of criteria for accepting a proof. The identified evaluation criteria indicate which purposes of proofs the participating students consider relevant in their proof evaluations. In one student's evaluation the most relevant consideration is how the statement's plausibility is verified and demonstrated by the proposed answer. Consequently she favors answers consisting of examples. Generality of a proof is an important evaluation criterion to most of the students. Five students mention at some point during the interviews that they appreciate considerations and explanations about *why* the statement is true in certain proposed answers. Some of the students take into account whether the proposed proof emphasizes some mathematical content or general patterns. Some students consider sufficiency without excess in their proof evaluations. Some students appreciate a didactical value in a proof, which includes how well a reader's interest is stimulated or how well both statement and proof are being explained to the reader. The proof idea or method does not seem to play a significant role in the students' evaluations of proofs of the first statement, which is indicated by three observed phenomena. Firstly, a proposed visual approach is liked least considering the ranking of all eight students together. The intrinsic idea behind this approach seems unimportant to the students. Secondly the fact that one of the proposed arguments proves a more general statement than *Statement I*, is rarely recognized and not appreciated by the students. The third surprising fact indicating poor appreciation of proof ideas or methods is the relatively high ranking score of an irredeemably wrong approach. While some of the students noticed errors in this proof, none questioned the basic strategy.

The interviews also provide an understanding about the students' habits when evaluating a mathematical proof. For example, in some cases students just consider how the method of proof is implemented, e.g. whether it is internally correct. If the implementation is regarded sufficient by the students, they do not consider whether any purpose of the proof is met, e.g. whether the proof idea has potential to establish the truth of the statement. These habits are observed in the students' evaluations of *Elaine*'s (wrong) 'proof' (see above). Most of the students do not question *Elaine*'s method at all. On the other hand, some of the students, once satisfied with what they acknowledge as the proof idea or intended proof, do not even consider its implementation. This is indicated for example by the students' evaluations of *Darragh*'s answer. The majority of students do not detect the arithmetical error in *Darragh*'s proof.

The observed validation and evaluation habits of the participating students suggest that their proof reading performances as well as their understanding of mathematical proof may improve considerably with increasing practice and more attention to this activities.

Proof Evaluation as Learning Activity for Students?

Evidence was found in this study that students engaged in reflections about the nature of proof in their proof evaluations. In particular, comparison of different answers seemed to encourage reflections of this kind, and made some features and purposes of proofs visible to the students. Differences in performances in the two tasks also suggested possible learning effects for some of the students, caused by their reflections on their evaluations in the course of the interviews. Considering that these students were not experienced in discussing mathematical proofs and also described difficulties and tendencies to avoid engagement in proofs and proving, these findings are worthy further attention. They suggest that practice of proof evaluation may be beneficial for the learning of mathematical proof.

Proof Evaluation *in Groups*?

Data from the written exercise and interviews of the study indicate an encouragingly wide range of features, purposes and criteria that are considered by students in their validations and evaluations of mathematical proofs. This may suggest that in a collective sense, the knowledge of the first year class about the nature of mathematical proof is quite extensive. However, specific items from the wide range of criteria, features and purposes mentioned by students generally appeared in isolation; it was unusual for more than one or two of them to appear in the comments of any individual student on a proposed proof. Individual students potentially have much to learn from and teach to each other about the role and purposes of proof in mathematical practice. In particular, I suggest that interaction and discussion can contribute in an essential way to the development of the skills involved in proof validation, proof evaluation, and ultimately proof construction. Proof evaluation is an essential activity in mathematical practice that can be performed from elementary to advanced levels of mathematical learning, and seems to naturally provoke discussion. Thus the practice and discussion of proof evaluation in groups may be a particularly valuable activity both to develop a student community of mathematical practice and to enhance individual students' knowledge about proof.

These suggestions can be interpreted as consistent with the socio-cultural perspective as described above, which views the learning process as the product of a cultural and social meaning shared in a learning community.

CONCLUDING REMARKS

The importance of proof in mathematical practice on the one hand and students' difficulties and negative attitudes towards mathematical proof on the other hand give rise to more focus on proof in the teaching of mathematics. This has been acknowledged in the design of syllabi in several countries in recent years. However in new syllabi of some countries (for example in Ireland or Germany), tendencies to avoid the terms *proof* and *proving* can be noticed. Occasionally the word *proof* is replaced by terms which appear less challenging such as *justifying* or *argumentation*. Undeniably, proof is a major obstacle for students. However, the above described wide range of purposes of proof indicate the importance of proof not only for the learning of mathematics but also to gain general problem solving competencies and other transferable skills. This paper argues that the importance of proof should be emphasized in the teaching of mathematics with the aim to make features and purposes of proofs visible to learners.

Learning from a socio-cultural view as described above appears through participation in a community of practice, which includes the ability to use and understand its artifacts. Regarding proofs as artifacts in mathematical practice, I suggest to consider practice of *proof validation* and *proof evaluation* as opportunities for learners to participate in mathematical practice to enhance understanding about the artifact proof. Considering the significance of these activities in mathematical practice a focus on these habits in the teaching is considered beneficial. Proof evaluation is an essential activity in mathematical practice that can be performed from elementary to advanced levels of mathematical learning, and seems to naturally provoke discussion. Thus the practice and discussion of proof evaluation in groups may be a particularly valuable activity both to develop a student community of mathematical practice and to enhance individual students' knowledge about proof.

Findings from a study on novice students' proof evaluation performances support these considerations. A longitudinal project involving sustained practice of proof evaluations at advancing levels of mathematical complexity, and regular assessments and interviews, is suggested to enhance existing understanding of these learning effects.

NOTES

1. A *community of practice* is a group of people who share an interest, a craft, and/or a profession (Lave and Wenger, 1991).

2. Some of the questions used in the written exercises were adapted (with permission) from the *Longitudinal Proof Project* (http://www.mathsmed.co.uk/ioe-proof).

REFERENCES

Adler, J. (1999). The dilemma of transparency: seeing and seeing through through talk in the mathematics classroom. *Journal for Research in Mathematics Education* Vol. 30(1): 47-64.

Bass, H. (2009). *How Do You Know that You Know? Making Believe in Mathematics*. Distinguished University Professor Lecture, http://deepblue.lib.umich.edu/handle/2027.42/64280/Bass-2009.pdf.

Bell, A.W. (1976). A study of Pupil's Proof-Explanations in Mathematical Studies. *Educational Studies in Mathematics*, 7, 23-40.

Bills, E. and Tall, D. (1998). Operable definitions in advanced mathematics: The case of least upper bound. *Proceedings of PME*, 22(2), 104-111.

De Villiers, M. (1999). The role and function of proof in mathematics with sketchpad. *Rethinking Proof with Sketchpad*, Key Curriculum Press.

Dreyfus, T. (1990). Advanced Mathematical Thinking. In Nesher, P. and Kilpatrick, J. (Eds.), *Mathematics and Cognition: A Research Synthesis by the International Group for the Psychology of Mathematics Education* (pp. 113-134). ICMI Study Series, Cambridge University Press.

Fischbein, E. (1982). Intuition and Proof. *For the Learning of Mathematics*, 3 (2), 9-18.

Hanna, G. (1991) Mathematical Proof. In Tall, D. (Ed.) *Advanced Mathematical Thinking.* Kluwer, Dordrecht.

Hanna, G. (2000). Proof, explanation and exploration: an overview. *Educational Studies in Mathematics*, 44, 5-23.

Hanna, G. and Barbeau, E. (2008). Proofs as bearers of mathematical knowledge. *ZDM Mathematics Education,* 40, 345-353.

Harel, G. and Sowder, L. (1998). Students' Proof Schemes: Results from Explatory Studies. In Schoenfeld, A.H., Kaput, J. and Dubinsky, E. (Eds.), *Research in Collegiate Mathematics Education III*, Providence, R.I.: American Mathematical Society.

Hart, E.W. (1994). A conceptual analysis of the proof-writing performance of expert and novice students in elementary group theory. In Kaput, J.J. and Dubinsky, E. (Eds.), *Research Issues in Undergraduate Mathematics Learning: Preleminary Analysis and Results*, MAA Notes 33, Washington, D.C.

Hemmi, K. (2006). *Approaching proof in a community of mathematical practice. Doctoral Dissertation.* Department of Mathematics, Stockholm University.

Hemmi, K., Lepik, M., & Viholainen, A. (in press). Upper secondary school teachers' views of proof and proving – an explorative cross-cultural study. *Proceedings of MAVI 16*, The 16th conference on Mathematical Views, Tallinn, Estonia, June 26-29, 2010.

Küchemann, D. and Hoyles, C. (1999-2003). *The Longitudinal Proof Project. A Longitudinal Study of Mathematical Reasoning: Student Development and School Influences.* Institute of Education, University of London.

Lave, J. & Wenger, E. (1991). *Situated Learning: Legitimate peripheral participation.* Cambridge University Press, Cambridge.

Mariotti, M.A. (2006). Proof and Proving in Mathematics Education. In Gutierrez, A. and Boero, P. (Eds.), *Handbook of Research on the Psychology of Mathematics Education: Past, present and future* (pp. 173-204). Sense Publishers, Rotterdam, Netherlands.

Martin, G.W. and Harel, G. (1989). Proof frames of preservice elementary teachers. *Journal for Research in Mathematics Education*, 20, 41-51.

Moore, R.C. (1994). Making the transition to formal proof. *Educational Studies in Mathematics*, 27, 249-266.

NCCA Leaving Certificate Examination 2005: Mathematics, *Chief Examiner's Report*. Department of Education and Science. Dublin, Ireland.

National Council of Teachers of Mathematics. (2000). Principles and standards for school mathematics. Reston.

Pfeiffer, K. (2010). *Features and purposes of mathematical proofs in the view of novice students: observations from proof validation and evaluation performances.* Dissertation. National University of Ireland, Galway. http://hdl.handle.net/10379/1862.

Rav, Y. (1999). Why do we proof theorems? *Philosophia Mathematica*, 7(1), 5-41.

Recio, A.M. and Godino, J.D. (2001). Institutional and Personal Meanings of Proof. *Educational Studies in Mathematics*, 48 (1), 83-99.

Senk, S. L. (1985). How well do students write geometry proofs? *Mathematics Teacher*, 78 (6), 448-456.

Schoenfeld, A. (1985). *Mathematical Problem Solving.* Orlando, Academic Press.

Selden, A. and Selden, J. (2003). Validations of proofs written as texts: Can undergraduates tell whether an argument proves a theorem? *Journal for Research in Mathematics Education*, 34 (1), 4-36.

Thurston, W.P. (2006). On Proof and Progress in Mathematics. In Hersh, R. (Ed.) *18 Unconventional Essays on the Nature of Mathematics* (pp. 37-55). Springer.

Weber, K. (2001). Student Difficulty in Constructing Proofs: The Need for Strategic Knowledge. *Educational Studies in Mathematics*, 48, 101-119.

Weber, K. (2002). Beyond proving and explaining: Proofs that justify the use of definition and axiomatic structures and proofs that illustrate technique. *For the Learning of Mathematics*, 22 (3), 14-17.

INTEREST IN THE MATHEMATICS CLASSROOM

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In recent times there has been much focus regarding the low numbers studying Higher Level mathematics in Ireland. The Irish Government has responded through initiatives such as 'Project Maths' and the plan of awarding extra bonus points. However, while these initiatives may prove to be worthwhile, such efforts are futile unless students themselves have a desire and motivation to learn the subject (Prendergast, 2011). Hidi and Harackiewicz (2000) established that the key to influencing an individual's academic performance lies in increasing the individual's interest in the particular domain. This work is focussed upon in more detail in the author's thesis.

INTRODUCTION

Hidi and Harackiewicz (2000) established that the key to influencing an individual's academic performance lies in increasing the individual's interest in the particular domain. As Csikszentmihalyi (1990, p. 126) stated "*if intrigued by the opportunities of the domain, most students will make sure to develop the skills they need to operate within it*". Thus, the long - term key to boosting the numbers taking Higher Level mathematics in Ireland must be to increase students' interest in the subject (Prendergast, 2011). Current research figures show that there is much work to do in this respect. Statistics released by PISA (2003) show that less than half (48 per cent) of Irish students agree that they are interested in the things they learn in mathematics. This figure was slightly down on the OECD average of 53 per cent. In addition only 32 per cent of Irish students declare that they look forward to their mathematics lessons, while only 33 per cent concur that they do mathematics for the enjoyment. The same study disclosed that over two-thirds of Irish 15 year olds 'often feel bored' at school, while the OECD average for this was under 50 per cent (OECD 2003). However, despite such alarming figures, little is known about how to develop interest and utilise it to enhance students' learning. This paper examines how best to trigger, and more importantly maintain, student interest in a topic.

WHAT IS INTEREST?

Can interest be defined as a temporary fixation, attraction or appeal? Is it just an inclination or an attitude? Boekaerts and Boscolo (2002) propose that interest is conceptualised as the effect that relates one's self to the activities that provide the type of novelty, challenge, or aesthetic appeal that one desires. Hidi and Harackiewicz (2000) describe interest as an interactive relation between an individual and certain aspects of his or her environment (e.g. objects, events, ideas). It can be viewed both as a state and as an outlook of a person, and it has a cognitive as well as an affective component. Indeed, many researchers went as far as arguing that interest is a basic emotion (Silvia, 2001). Hidi (2006) considers interest to be a unique motivational variable, as well as a psychological condition that occurs during interactions between persons and their objects of interest, and is characterised by increased attention, concentration and affect.

IMPORTANCE OF INTEREST

Present-day research has demonstrated that interest has a powerful influence on academic performance (Hidi and Harackiewicz, 2000). Del Favero et al. (2007) determine that many studies have shown the energising function of interest in fostering, remembering and understanding material, and stimulating students' positive attitude towards a topic (e.g. Hidi, 1990; Mason & Boscolo, 2004; Schiefele, 1991, 1998). This view is supported by Hidi and Anderson's (1999) work who argue that interest has a profound effect on students' recollection and retrieval processes, their acquisition of knowledge, and their effort expenditure. In addition, an interested individual is more likely to develop high competency and to receive positive feedback from others (Hidi et al., 2002). Being interested may also serve as protection against the negative effects of failure (Hidi & Renninger, 2006). On top of this Alexander (1997) suggested that interest may be the key to early stages of learning, as well as to differences between expert and moderately skilled performers. When interested in a topic or domain, students are more likely to use higher-order learning, thus improving their knowledge.

TYPES OF INTEREST

There are two main types of interest, namely, situational interest and individual interest.

Situational Interest

Situational interest is environmentally triggered and involves an affective reaction and focused attention (Hidi, 2006). Boekaerts and Boscolo (2002) acknowledge that it is dependent on favourable environmental conditions, and can therefore be transient in nature. However, De Favero et al. (2007) note that it can influence learning by inducing stronger attention to learning materials and by increasing persistence in the task. Research suggests that situational interest should play an important role in learning, especially when students do not have pre-existing individual interests in academic activities, content areas, or topics (Hidi and Harackiewicz, 2000). Indeed, more specifically, the same authors suggest that the employment of situational interest could make a significant contribution to the motivation of academically unmotivated children.

Individual Interest

This is often referred to as personal interest. Boekaerts and Boscolo (2002) define individual interest as interest built on stored knowledge about a class of objects or ideas, which leads to a desire to be involved in activities related to such. It is the interest that students bring to some environment or context. Those experiencing this type of interest possess an inner drive to seek out opportunities to learn more about a specific topic. Hidi and Harackiewicz (2000) ascertained that it is a relatively enduring predisposition that develops over time and is associated with increased knowledge, value, and positive feelings. The learners' individual interests energise and motivate their thoughts and actions in a very goal-directed way (Alexander, 1997). An interested person can therefore formulate curiosity questions, and attenuate negative feelings, such as frustration and anxiety (Hidi & Renninger, 2006). Indeed, Hidi and Harackiewicz (2000) note that investigations focusing on individual interest have shown that those who are interested in particular activities or topics pay closer attention, persist for longer periods of time, learn more and enjoy their involvement to a greater degree than individuals without such interest.

Differences between Individual and Situational Interest

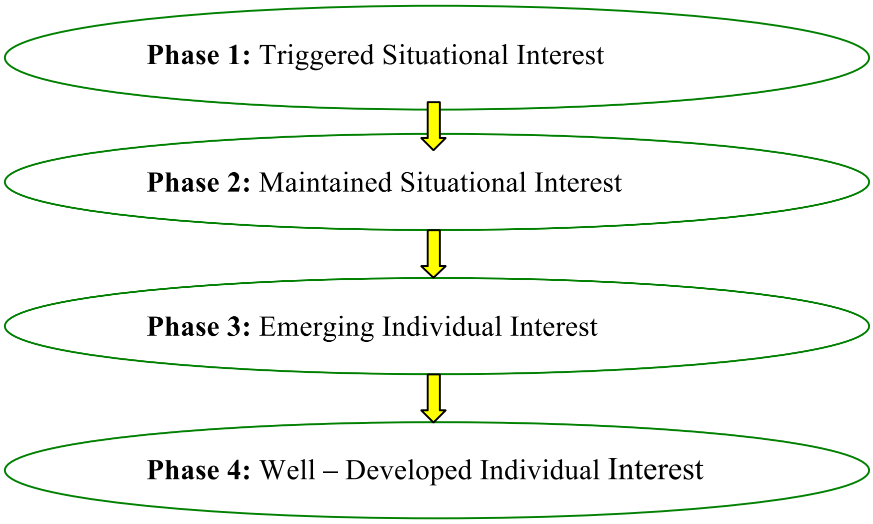
Differences between individual and situational interest are noted throughout the literature. For example Hidi and Harackiewicz (2000) determine that while individual interest develops slowly and tends to be relatively long-lasting, situational interest is triggered more suddenly by environmental factors across individuals. Research carried out by Alexander (1997) shows that situational interest is expected to play a stronger role in the early periods of learning than individual interest. Something about the topic or the context grabs the student's attention and urges them onward. However, as individuals progress toward competency in the target domain, individual interest becomes increasingly more important, with the effects of situational interest levelling off.

Nevertheless, while individual and situational interests are undoubtedly distinct, they are not completely dissimilar phenomenon. Studies carried out by researchers such as Hidi and Harackiewicz (2000) and Alexander (1997) provide evidence that both can interact and influence each other's development. For example, individual interest in a particular topic may help students persevere through boring presentations or texts about that topic, and situational interest elicited by presentations or texts may maintain motivation and performance, even when individuals have no personal interest in particular topics. In addition, situational interest can actually contribute to the development of long-lasting individual interests. For example, students who are exposed to an exciting lesson in statistics may be stimulated and pay more attention in class than they ever have before. For some students, this interest may evaporate as soon as the lesson ends. For others, the interest triggered in this situation may persist over time and may develop into individual interest in statistics.

A MODEL OF INTEREST DEVELOPMENT

The important role which both situational and individual interest can play in learning was discussed in the previous section. The activities and resources in the early stages of learning a topic must be aimed at stimulating and maintaining situational interest in students. However, over time and certainly in the latter stages of the topic individual interest must be catered for. With this in mind the authors have identified a four phase model proposed by Hidi and Renninger (2006) to support interest development in the classroom. This model builds on the work of Gottfredson (1981), Todt (1990) and Krapp (2002) in which the notion of general stages of interest development was discussed and expanded. The model identifies situational interest as providing a basis for an emerging individual interest (Hidi and Reenninger, 2006). Both situational and individual interests have been described as consisting of two stages. Situational interest involves a stage in which interest is triggered and a subsequent stage in which interest is maintained (Hidi, 2006). In individual interest, the two stages include an emerging individual interest and well-developed individual interest (Hidi and Renninger, 2006). The proposed four-phase model as denoted by Figure 1, integrates these conceptualisations. The first phase of interest development is a triggered situational interest. If sustained, this first phase evolves into the second phase, a maintained situational interest. The third phase, which is characterised by an emerging individual interest, may develop out of the second phase. The third phase of interest development can then lead to the fourth phase, a well-developed individual interest.

Differing levels of knowledge, effort and self-efficacy, have been found to typify each phase of interest (Hidi and Renninger, 2006). Hence, in a class of thirty, one is unlikely to be able to characterise all students in the same phase. The length and make-up of a given phase is influenced by individual experience and temperament and so can differ for each student (Hidi and Renninger, 2006). However, the four separate phases are considered to be sequential where students can progress from one phase to the next when interest is supported and sustained. Conversely, if not supported, students' interest can become inactive, retreat to a previous phase, or disappear altogether (Hidi and Renninger, 2006). Each phase will now be examined in more detail.



(Hidi and Renninger, 2006)

Figure 1: Interest development: A four phase model

Phase 1: Triggered Situational Interest

Triggered situational interest refers to a psychological state of interest which results from short-term changes in affective and cognitive processing (Hidi and Renninger, 2006, p. 114). It is typically externally supported and can be sparked by;

- environmental or text features such as bizarre, surprising information;
- character identification or personal relevance;
- intensity.

(Hidi and Renninger, 2006, p. 114)

Instructional practices and resources such as group work, puzzles and ICT have also been found to stimulate this type of interest (Hidi and Renninger, 2006).

Phase 2: Maintained Situational Interest

Maintained situational interest refers to a psychological state of interest that is subsequent to a triggered state, involves focused attention and persistence over an extended episode in time, and/or reoccurs and again persists (Hidi and Renninger, 2006, p. 114). This type of interest is also typically externally supported and is held and sustained through the meaningfulness of tasks and personal involvement (Hidi and Renninger, 2006). Hence the inclusion of activities in the teaching intervention such as project-based learning, cooperative group work, and one-on-one tutoring may contribute to the maintenance of situational interest.

Phase 3: Emerging Individual Interest

Emerging individual interest refers to a psychological state of interest as well as to the beginning phases of a moderately enduring predisposition to seek repeated re - engagement with particular classes of content over time (Hidi and Renninger, 2006, p. 114). Such interest is characterised by positive feelings, stored knowledge, and stored value (Hidi, 2006). Based on previous experience, students value the opportunity to re - engage with tasks and can often exceed demands in their work (De Favero et al., 2007). They are likely to be resourceful when conditions do not immediately allow a question to be answered and may be able to anticipate subsequent steps (Hidi and Renninger, 2006). This is a particularly useful trait when studying the problem solving aspect of algebra when the next step is not always obvious. Similar to the previous two phases, instructional conditions or the learning environment can enable the development of an emerging individual interest (Hidi and Renninger, 2006). However an emerging individual interest may not necessarily lead to well-developed individual interest. In contrast to Phase 1 and Phase 2, Phase 3 is typically but not exclusively self-generated (Hidi and Renninger, 2006). It does require some external support which can increase understanding and may encourage individuals to persist when confronted with difficulty.

Phase 4: Well-Developed Individual Interest

Well-developed individual interest refers to the psychological state of interest as well as to a relatively enduring predisposition to re – engage with particular classes of content over time (Hidi and Renninger, 2006, p. 114). A well-developed individual interest is characterised by positive feelings, and more stored knowledge and more stored value for particular content than for any other activity (Hidi and Renninger, 2006). This phase shares many of the same characteristics as the previous phase. For example, students are curious, resourceful and will persevere to work or address a question even in the face of frustration (Hidi and Renninger, 2006). Their interest enables them to sustain a constructive and realistic endeavour, which is again, very useful when solving real life problems with algebra. Students consider both the context and content of a problem (Hidi and Renninger, 2006). Like Phase 3, this type of interest is typically but not exclusively self-generated (Hidi and Renninger, 2006). In addition, instructional conditions or the learning environment can facilitate the development and deepening of well-developed individual interest.

HOW CAN I USE THIS IN THE MATHEMATICS CLASSROOM?

Hidi and Renninger's (2006) model provides a structure for learning in which individual student interest can be stimulated, nurtured and maintained. In a class of thirty students, there will be many differing levels of interest amongst students. The four phases proposed by Hidi and Renninger acknowledge and accommodate this. However, how can individual student's interest be progressed through the stages by mathematics teachers? Steen (1990) determined that mathematics can be made exciting for students if fresh perspectives on mathematical concepts are adopted and presented in schools. Teachers undoubtedly play a central role here. There are many recommendations offered throughout the literature. Firstly, it is important that teachers always demonstrate their own interest in the subject matter (Bergin, 1999). The next task for them is to engage their students in the topic. This can be done using certain aspects of the learning environment, such as modification of teaching materials and strategies, and how tasks are presented (Hidi and Harackiewicz, 2000). Hidi (2006) suggested other means to achieve interest such as selecting resources that trigger situational interest. These may include games, puzzles, and hands-on activities, depending on the particular topic. The textbook also has a major role to play as many lessons often revolve around the textbook. Thus it has the potential to play a major role in stimulating and promoting student interest in mathematics. According to Mikk (2000, p. 245) interest in textbooks can be increased with the inclusion of:

- Historical Data: references to discoveries and applications
- Practical Applications: concrete examples as opposed to theoretical explanations.
- Inclusion of Problems: questions aimed at reproducing, analysing and applying information.
- Humour: proverbs, riddles, jokes, etc.
- Figurative Representations: illustrations and mental images.
- Narrations about People: real life facts interlinked with historical data.

Apart from content, many researchers highlight the need for relevant, bright, attractive illustrations in textbooks to trigger initial situational interest (Mikk, 2000). These graphics not only assist with students understanding but also assist in grabbing and engaging a student's attention (Mikk, 2000). However, while resources such as games, puzzles, hands-on activities and bright illustrated textbooks definitely trigger situational interest, many of them fail to advance the student's interest through the stages of Hidi and Renninger's model. Thus the question remains, how can academically relevant interests be nurtured, utilised and indeed maintained.

A study carried out by Mitchell (1993) in the US found that the two main factors in maintaining student interest over time and developing a more individual interest were the meaningfulness of the task and student involvement. Meaningfulness refers to students' perception of topics as meaningful to their own lives and the nature and degree of their understanding. Meaningfulness appeared effective because content that is perceived as being personally meaningful to students is a direct way to empower students and hold their interest (Mitchell, 1993). Involvement refers to the degree to which students feel they are active participants in the learning process. In Mitchell's study, involvement also appeared effective because when the process of learning is experienced as absorbing, then that process is perceived as empowering to students and will therefore hold their attention (Mitchell, 1993). Basically, students are more interested when they learn by doing as opposed to sitting and listening.

Similar to empowering students through meaningfulness and involvement, Hidi and Harackiewicz (2000) found that affording students more choice, or promoting perceived autonomy can also promote individual interest. Affording more choice may be a simple undertaking such as allowing students to choose what topic to progress onto next. Perceived autonomy could involve trusting that the students have their homework done without checking each individual. Del Favero et al. (2007) also suggested that several forms of social interaction may also support the development of interest at various stages. This view was supported by Hidi and Harackiewicz (2000) who found that working in the presence of others resulted in increased interest for some individuals. This supports the case for the inclusion of group work and discussion in the classroom. Furthermore Del Favero et al. (2007) determined that problem-solving often can maintain interest through encouraging further exploration of concepts and ideas. Each of these are simple ideas and proposals which may help mathematics teachers advance their students through the four stages of Hidi and Renninger's model.

CONCLUSION

The key to influencing an individual's academic performance lies in increasing the individual's interest in the particular domain (Hidi and Harackiewicz, 2000). In this paper there are many high-quality proposals on how to enhance both situational and individual interest in mathematics. Many of these are simple ideas which can easily be adapted in any classroom. Teachers must be aware that in one particular lesson, they may encounter students who have no personal interest in the topic and also students who are very passionate about the topic (Prendergast, 2011). The four phases proposed by Hidi and Renninger acknowledge this and the teaching strategies suggested can support students' interest whether they are in the first phase or the last phase.

REFERENCES

Alexander, P.A. (1997). The Path to Competence: A Lifespan Developmental Perspective on Reading [online]. Retrieved December 2008, from www.nrconline.org/publications/ThePathToCompetence.doc.

Bergin, D.A. (1999). Influence of Classroom Interest. *Educational Psychologist*, 34, 87–98.

Boekaerts, M. and Boscolo, P. (2002). Interest in Learning, Learning to be Interested. *Learning and Instruction*, 12, 375–382.

Csikszentmihalyi, M. (1990). Literacy and intrinsic motivation. *Daedalus*, 119, 115 - 140.

Del Favero, L., Boscolo, P., Vidotto, G. and Vicentini, M. (2007). Classroom discussion and individual problem-solving in the teaching of history: Do different instructional approaches affect interest in different ways?. *Learning and Instruction*, 17, 635 – 657.

Silvia, P.J. (2001). Interest and interests: The psychology of constructive capriciousness. *Review of General Psychology*, 5, 270–290.

Hidi, S. (2006). Interest: A unique motivational variable. *Educational Research Review*, 1, 69–82.

Hidi, S., & Anderson, V. (1999). Situational interest and its impact on reading and expository Writing. In Renniger, K. A., Hidi, S., and Krapp, A. (eds.), *The role of interest in learning and development*, Hillsdale, NJ: Erlbaum.

Hidi, S., Berndorff, D., & Ainley, M. (2002). Children's argument writing, interest and self-efficacy: an intervention study. *Learning and Instruction*, 12, 429 - 446.

Hidi, S and Harackiewicz, J.M. (2000). Motivating the Academically Unmotivated: A Critical Issue for the 21st Century. *Review of Educational Research*, 70 (2) 151-179.

Hidi, S., & Renninger, A. (2006). The four-phase model of interest development. *Educational Psychologist*, 41, 111–127.

Mikk, J. (2000). *Textbook: Research and writing*. New York: Peter Lan.

Mitchell, M. (1993). Situational Interest: Its Multifaceted Structure in the Secondary School Mathematics Classroom. *Journal of Educational Psychology*. 85 (3), 424 – 436.

Prendergast, M. (2011). *Promoting student interest in mathematics: A framework for effective teaching of algebra at Junior Cycle* (Doctoral dissertation). Available from University of Limerick.

Programme for International Student Assessment (PISA) (2003). Learning for Tomorrow's World- First Results from PISA 2003, OECD [online]. Retrieved from http://www.oecd.org/dataoecd/1/60/34002216.pdf.

Steen, L.A. (1990). *On the shoulders of giants – New approaches to numeracy*. MSEB, Washington D.C.: National Academy Press.

ACTS OF KNOWING IN UNIVERSITY MATHEMATICS CURRICULA

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It has been proposed by R. Barnett and K. Coate that curricula in Higher Education can be viewed as operating in three distinct but intertwined domains that they call Knowing, Acting and Being. A key feature of their viewpoint is the position of the student as a central figure in the operation of curricula: learning is seen as manifested in decisive and purposeful acts by students and as involving a personal relationship with the knowledge field. Barnett and Coate's model of curriculum is considered here in the context of the education of mathematicians. Particular attention is focussed on the Knowing domain, what it entails for mathematics, and what sorts of curricular activities and emphases might encourage students to develop a sense of personal mathematical authority and commit acts of deliberate and purposeful mathematical knowing.

INTRODUCTION

The theme of this article is a discussion of mathematics curricula in Higher Education, in a context established by an extensive discussion of the meaning, role and possibilities of curriculum in a recent book by Ronald Barnett and Kelly Coate (2005). The meaning of the term "curriculum" does not appear to be consistent throughout educational discourse. In this article the term is intended to be used and interpreted in a broad sense, to encompass every aspect of the student's encounter with the subject during the programme of study. It includes the syllabus, the lectures, tutorials and other scheduled activities, all other learning activities, individual study, assessment and feedback, and all the resources that the student is invited to utilize.

Barnett and Coate propose that curriculum operates in three distinct but intertwined dimensions which they call *Knowing*, *Acting*, and *Being.* Superficially, the themes with which these dimensions are concerned are obvious: *knowing* is about knowledge, *acting* is about action and activity, *being* is a more nebulous notion but it is concerned with such matters as motivation, orientation and self-awareness.

The choice of the verbal forms is deliberate, emphasizing the critical role of the student's conscious and purposeful agency. Barnett and Coate argue that design of successful curricula must attend to all three domains and try to find ways of integrating them. The nature of the three domains and their interactions of course varies among disciplines, and a goal of this article is an initial exploration of what the *knowing* domain in particular constitutes in mathematics, and what this means for our practice as academic mathematicians. As a mathematician and mathematics lecturer I am obviously concerned with the design and delivery (from the academic side) of mathematics curricula. It is important to bear in mind however that the *effective* curriculum is what the student experiences, which may not be exactly what we design. Barnett and Coate make this point also, distinguishing the *curriculum-as-designed*, which is a sort of ideal theoretical object, from the *curriculum-in-*

action, which is what happens. It is the *curriculum-as-experienced* that is the potential determinant of overall quality and nature of learning. If we have the impression, or even the conviction, that learning of the quality or nature that we hope for may not be taking place, we might ask if what our students are experiencing is appropriately matched to the aspirations of our design. The *knowing-acting-being* schema might give us a means of investigating questions of this nature; at the least it gives us a framework for a discussion of curriculum.

Knowing, Acting and Being

In this section we give a brief summary of what is considered under each of the three headings by Barnett and Coate, with the intention of specializing at least part of this discussion to the case of mathematics in the next section. This is not intended to be a complete exposition of the schema, nor is it intended as an overview as the focus is on those aspects which I consider to be most relevant for mathematics. For a more comprehensive and also much more general discussion, the reader is referred to the book of Barnett and Coate.

Knowledge and Knowing

Completing a university degree programme is generally regarded as a means of accessing or acquiring knowledge of an abstruse and highly specialized nature. People go to university to learn about a particular subject and when they graduate we can reasonably expect them to have some knowledge of that subject.

What does it mean to possess *knowledge* of a subject? At the least it probably means being able to recount certain details of the body of undisputed information that is core to the subject, if such a body exists. Some academic disciplines are probably more robustly supported by a "core of information" than others, and it is probably fairly uncontroversial to posit that most of the sciences have an acknowledged body of "core" propositional knowledge that graduates could reasonably be expected to be able to declare. For example, the notion of a botany graduate who cannot say anything about the process of photosynthesis is not very sensible. However, it may also be the case that there are subjects in which no single topic is certain to be in every practitioner's conception of the "core", and it is likely that in most disciplines the boundaries of the core are debated and are far from fixed.

In any case, knowledge of facts is part of knowledge. The kind of subject knowledge that we might ideally expect a university graduate to possess goes far beyond the ability to quote facts. We would expect such a person for example to be able to apply her knowledge to investigate a new situation in the subject, to be able to adopt a position on a question and argue her case, to be able to distinguish a sound argument from a fallacious one within the context of the subject, to be able to propose a strategy for studying a problem, and so on. We might expect the graduate to have a personal mental register through which her specialist subject knowledge is accessed, organized, explored, expanded and tested. Her knowledge should be personal and while much or all of the material of it can be found in textbooks, the manner in which she knows it, can access it and and can use it is personal and places her in a unique position with respect to the discipline, meaning that her opportunities to engage with the discipline and to contribute are special.

Barnett and Coate encapsulate this idea of personal, individually constructed knowledge by emphasizing the *act of knowing* rather than the substance of knowledge. The act of knowing, they propose, "calls for will, an act of identity and a claim to ownership . . . a public act in which the individual shows herself, proclaims herself, to stand in such and such a place" (Barnett and Coate, 2005, p. 60). They further argue that university curricula have failed to attend to this "sense of immediate personal encounter" and have focussed on the material to be known rather than on the task of knowing, which is to be accomplished by persons.

> "These human - personal and social - characteristics of knowledge have in the past, we believe, all too often been overlooked by those who construct curricula. Instead, knowledge in university curricula has come to be conceived, we suggest, as consisting of a corpus – of ideas, propositions, theories, concepts - that stand outside students." (Barnett and Coate, 2005, p. 60)

As curriculum designers and deliverers, the challenge for us is to find learning and assessment activities that invite students to enter our disciplines as active practitioners rather than as observers, so that they are personally responsible for the construction, validation and revision of their own knowledge.

Skills and Acting

As Barnett and Coate point out, recent developments in Higher Education policy have led to a requirement to identify particular *skills* and *competences*, both of generic and subject-specific natures, and to emphasize them in learning outcomes and in statements of programme objectives. In the narrowest and crudest interpretation, the notion of skills is about being able to *do* things and the notion of knowledge is about being able to recite things. The sorts of tasks with which skills are concerned obviously vary widely with disciplines.

Barnett and Coate expand the "skills" discussion to a looser but more inclusive and more personal consideration that they call *acting.* Again the word is deliberately evocative of the student's agency, and again the critical role of personal engagement is emphasized:

> "In acting out the practices of a discipline, the student has to become the author of her own actions. Of course, quite often, in order to get started, those actions are modelled on the actions of others . . . Sooner or later, however, that modelling has to give way to an authentic and first-hand action that bears the student's own stamp." (Barnett and Coate, 2005, p. 62)

These authors have already identified *knowing* as a conscious, resolute and decisive *act*, and thus for them the domains of knowing and acting are unavoidably intertwined. They argue for integration of these domains in the curriculum, making the point that a representation of curriculum in terms of skills runs the risk of packaging degree programmes as toolkits of skills to be mastered. They argue that possession of skills must be accompanied by judicious consideration of their appropriate deployment in order to be truly valuable, and that this sort of consideration must involve personal engagement. This observation applies in a direct and very obvious way in professions such as medicine or teaching, but it applies equally to the task of investigating any problem in any discipline, where appropriate choice of tools and sound judgement of their capabilities, limitations and the circumstances in which they can be applied is at least as important as their competent use.

Being and Self

Having developed the discussion of *knowing* and *acting*, Barnett and Coate go on to posit that knowledge and skills, even interpreted as active "knowing" and as considered, personal and expert action, are insufficient as building blocks for contemporary curricula. They propose the inclusion of a third dimension, which they call *being*.

> "If curricula are to provide the kinds of experiences that are likely to sponsor the kinds of subjectivities called for by a world of instability, knowledge and practical dimensions are necessary claimants for our attention but they cannot be sufficient. A world of uncertainty poses challenges not just of knowing and of right action but also, and more fundamentally, on us as beings in the world. How do I understand myself? How do I orient myself? How do I stand in relation to the world?" (Barnett and Coate, 2005, p. 108)

We are not just observers of the world, we are participants in it. The bodies of knowledge that we study in universities are not inert or immutable, they have been created and assembled by human agents and they have evolved through argument and through social processes. While they are engaged by humans they are subject to expansion, reinterpretation, exploration and the possibility of discovery. While they are not engaged they are invisible. By committing acts of knowing and acts of doing within our disciplines we enter their history and we assume the potential to affect them or to use them in new ways. We become players instead of spectators. We ourselves are changed in the process, and required to renegotiate our disposition within the discipline and perhaps more widely. As we navigate our own ways through the further reaches of the domains of knowing and acting, our paths are not convergent because each is laid by personal encounters and by individual experience. We share a core of accessible knowledge and a collective toolkit of skills, but our situations are special. Each of us is uniquely positioned for action. Our actions will have repercussions and our knowledge can be used.

The schema of Coate and Barnett are not intended as a *classification* of curricular activities, nor do they suggest that the domains of *knowing*, *acting*, and *being* should be separately visible or separately experienced in the curriculum. Indeed the focus of their model is as much on the integration of the three domains in curricula as on their distinct manifestations. In mathematics, an example of an activity that might be considered as incorporating all three dimensions is the construction of proofs. Here the *knowing* domain is evident in the presentation of an argument to justify a mathematical claim; the performance of this essential activity of the particular disciplinary practice involves the *acting* domain; and the *being* domain is visible in the student's personal role as both author of mathematical proof and assessor of mathematical correctness and sufficiency.

THE CASE OF MATHEMATICS

The conceptualization of curriculum advanced by Barnett and Coate invites us to consider *knowing* as manifested in committed acts of personal conviction, steadfast and defensible claims arising from the individual's personal relationship with the knowledge field and supported by it. Such acts of knowing are not absolute decrees: they are subject to expansion, contraction, reconstruction or reversal in the light of new understanding, but such revision requires of the individual a deliberate and possibly daunting undertaking.
An *act of knowing*, for Barnett and Coate, is "a positional and personal act". An *act of knowing* is one that requires that person to declare a position, and to be prepared to justify and defend that position or thoughtfully renegotiate it if reasonable challenges show it to be untenable. To consider the meaning of an *act of knowing* in the context of Higher Education, it may be useful to also think about acts of imitation, of recollection, of memorization, of ritualized performance, and how these feature in the range of things that we routinely ask our students to do in the curriculum and in assessment.

My position in this article is to adopt the view that curricula should not only afford opportunities for students to commit personal acts of knowing, but that satisfactory participation in curricula should necessitate such acts, from the first day to the last and at advancing levels of sophistication in terms of subject knowledge as the student progresses. This position is open to question, but my purpose in this section is not to debate its merits but to ask what its implications might be for the education of mathematicians. What kind of elements of the undergraduate curriculum in Mathematics can support, encourage or demand committed knowing acts of students? What kinds of mathematical tasks demand acts of knowing? Can authentic acts of knowing be undertaken by persons who are novices in the discipline? What does it mean anyway to engage in an act of mathematical knowing? I will mention a few examples from my own practice to support a discussion of these questions, as well as looking to various works in the Mathematics and Mathematics Education literature.

What does it mean to "Know" in Mathematics?

Before attempting to specialize Barnett and Coate's interpretation of *knowing* to a particular discipline, it makes sense first to consider what it means to know within that discipline. With this in mind let us turn to the work of Hyman Bass (2009, p. 1).

> "To have a critical understanding of a discipline, it is not enough to know its products and their uses. One needs to understand as well the genesis of its knowledge, how that knowledge is warranted or certified, and how, within the norms of the field, contrary views are reconciled, if at all. Each field – physics, biology, medicine, linguistics, history, economics, law, theology, etc. – has its own versions of this."

Every academic discipline is special, and every academic discipline is special in special ways passionately articulated by its practitioners. As Bass goes on to propose, one of the important distinguishing features of mathematics is that it has a unique criterion not for the inception of new knowledge but for the acceptance of it.

> "Each field has its own norms for certifying truth. Mathematics has evolved a unique method -- deductive proof -- dating back to Greek antiquity." (Bass, 2009, p. 1)

Proof as the necessary warrant for certification of truth is an essential feature of mathematical practice. A corollary of this assertion of course is that the objects with which mathematical practice is concerned are formal ideal objects whose properties and relationships admit elaboration purely through deductive reasoning. In the arena of mathematical discourse, to *know* that a statement about mathematical objects is true means to be satisfied of the existence of a proof of that statement that will persuade any sceptic willing to engage with it. Indeed, Bass proposes that "the practice of mathematics is concerned with ways to make its practitioners convinced of the existence of proofs" (Bass, 2009, p. 3). The standard way of achieving this is to exhibit a proof, or at least an argument that could in principle be expanded unproblematically by a sufficiently knowledgeable agent to a fully detailed proof. This last hypothetical construct is vague and unsatisfying in at least two ways, but the point is that while mathematical truth (being a matter of consistency with, or consequence of, axiomatic foundations) is absolute, the mechanisms by which it is certified are socially negotiated and socially evolved exercises in careful judgement.

As Bass points out, mathematical knowledge does not come only from logical deduction, but usually has its genesis in exploration of phenomena, leading to the formulation of conjectures which may be refined as they are investigated, and whose character may progress from tentative to more assured, as the force of evidence and argument supporting them increases. A conjecture becomes a theorem when a proof of it is presented for the scrutiny of experts and is judged to be complete and correct.

Thus the role of proof in mathematical progress goes far beyond persuading obstinate pedants of mathematical facts. It also includes the discovery and communication of *reasons* that account for (possibly observable) truths. Such explanations may reveal wider generality or connections to superficially unrelated questions, or prompt further investigations. Thus a role of proof is to explore contexts in which a mathematical truth may be understood, and positions that it may occupy in organized systems of knowing. The mathematics educationalist Michael de Villiers (1999) assembles the range of functions of mathematical proof in the following widely cited (but not necessarily exhaustive) list.

- *verification* (of the truth of the statement)
- *explanation* (insight into *why* the statement is true)
- *systematization* (into a deductive system of axioms, concepts and theorems)
- *discovery* (of new results)
- *intellectual challenge* (the self-realization/fulfilment derived from constructing a proof).

The last of these items is the only one not to be directly concerned with the management of the corpus of mathematical knowledge, and it is likely that not all mathematicians would consider its inclusion in a list of *functions* of proof appropriate. The decision by de Villiers to include it is striking for its unusual but deliberate emphasis on the human experience of mathematical knowing as an essential, not incidental, feature of the subject. The state of mathematical understanding advances because practitioners are driven to search for more and more satisfying proofs, by personal fascinations that are intensely experienced. I do not mean to suggest that this distinguishes mathematics from other areas of human endeavour, although the kinds of intellectual efforts that drive progress in different areas may be qualitatively different. The transformative nature of compelling personal experiences that motivate enlightened acts of knowing, even at relatively unsophisticated stages of learning, is not always explicitly acknowledged in mathematical discourse and certainly not in mathematical curricula.

The theme of this section has been the nature of mathematical proof, both as the necessary warrant for certification of mathematical truth, and as an intellectual arena for explanation, discovery, connecting and organizing of concepts, and identification of new directions for exploration. The special position of proving at the heart of mathematical knowing is encapsulated by Yehuda Rav (1999, p. 20) in the following terms.

> " . . . proofs rather than the statement-form of theorems are the bearers of mathematical knowledge. Theorems are in a sense just tags, labels for proofs, summaries of information, headlines of news, editorial devices. The whole arsenal of mathematical methodologies, concepts, strategies and techniques for solving problems, the establishment of interconnections between theories, the systematization of results -- the entire mathematical know-how is embedded in proofs."

"Knowing" in the Curriculum

Although the discussion above was concerned with proof as the warrant for certification of mathematical truth, I do not mean to propose that every act of mathematical knowing must be an act of mathematical proving. While we know that a mathematical statement is true if we can exhibit a proof of it, knowing in mathematics is not just a matter of being able to say whether statements are true or false. Using proof to develop and interrogate our understanding is not simply a matter of constructing proofs, or of reading them. Nevertheless, mathematical understanding is created, articulated and argued through proofs. Authentic participation in mathematical practice cannot happen if the business of proving is not a significant and deliberate focus of attention, and it cannot happen without attention to the linguistic, definitional and notational precision that is necessary for elaboration through deductive reasoning.

Constructing and documenting new proofs, especially of previously unknown theorems, is one of the loftiest challenges in mathematical practice. It is difficult. We do not know at the outset how long it will take to produce a proof. We may not know if the strategy that we have in mind has the potential to prove our conjecture. We do not know how long we will wait before a useful idea flutters into our head, or how long we will take to recognize that this has happened. We do not know how many trails we will cut only to abandon them or how long we will stand at an obstacle before breaking through it or turning away in resignation and frustration. We may not know if the thing we are trying to prove is even true. We might end up proving an entirely different theorem, or radically modifying our original conjecture in order to contrive something that will yield to the methods that we can apply. Eventually, we hope that something can be salvaged from the wreckage and that we may be positioned to make and defend a claim to know.

All of these uncertainties beset the research efforts of experienced professional mathematicians, for whom strict deductive reasoning about abstract objects defined only in terms of their properties is an entirely natural form of behaviour. This natural sense is acquired through familiarity. We are habituated, through experience and through life in a peculiarly oriented community, to abstraction of essential features of the objects that we work with, to formulating definitive and precise descriptions of these features in a context-free setting, to logical inference from such descriptions as the basis for knowing, to what Barnett and Coate (2005, p. 62) call “the deep grammar of the discipline”. We understand what proof is for and how it is used in our community. Moreover, we are fluent in the language of mathematics, which overtly resembles languages of everyday life, but in which “everyday” words, potentially confusingly, have very special and absolutely precise meanings. Of course every academic discipline has its own specialized terminology, but mathematics (at least in English) abounds with appropriations of everyday words for technical usages not obviously related to their usual meanings. This is further complicated by reliance on specialized systems of manipulable symbols that do not really resemble ordinary text. In any case, an ability to both interpret and express mathematical meaning unambiguously through language is required for communication and for public acts of mathematical knowing, and abilities of this nature are generally not acquired automatically but develop through careful participation in disciplinary activities. The argument here is that constructing proofs is difficult and fraught with uncertainty, even for expert practitioners who are fluent in mathematical language and have developed unproblematic and indelible personal comprehension of the shared community conviction of what it means to prove and what it means to know. For beginning undergraduate students who do not have a deeply ingrained experiential cognizance of the meaning and nature of proof, and do not yet have ready mental access to the forms of words that express precise mathematical meaning, the task of independently constructing proofs may often be beyond the realm of feasibility.

What opportunities for acts of knowing *can* the curriculum then provide to novice students, consonant with the view of their learning as increasingly authentic (albeit peripheral) participation in real disciplinary practice? Much of the scholarly work of mathematicians involves critical engagement with proofs in order to construct or expand understanding, and critical examination of proposed proofs in order to be satisfied (or not) of their correctness. The latter occurs most obviously in the situation of refereeing new articles submitted for publication, but sceptical and inquisitive reading of proofs is a key feature of general mathematical behaviour. A mathematician embarking on a careful study a proof will investigate the argument itself, the extent of its domain of applicability, the nature of its dependence on the hypotheses and the implications of relaxing or adjusting these. John Mason and Anne Watson (2008) refer to *dimensions of possible variation* and *range of permissible change* in discussions of families of examples of mathematical objects and also of mathematical tasks assigned for students; this recalls the mathematician's habit of seeking to extend the range of situations to which the argument in a proof applies, either directly or after adaptation. In the operation of the disciplinary practice, written communication of proofs requires scepticism and an almost adversarial stance from the reader, and full disclosure by the author of the argument and the assumptions on which it rests. The reader's job is to interrogate and the author's job is to explain and to convince an inquisitive sceptic. The process, while it may not involve actual dialogue between persons, is something like a debate between the reader and author. It is described by the mathematician Paul Halmos (1985, p. 69) in advice to students as follows:

> "Don't just read it; fight it! Ask your own questions, look for your own examples, discover your own proofs. Is the hypothesis necessary? Is the converse true? What happens in the classical special case? What about the degenerate cases? Where does the proof use the hypothesis?"

For Barnett and Coate (2005, p. 60), an *act of knowing* is "a positional and personal act". It involves a deliberate and responsible taking of a stance on a question, and a preparedness to defend that stance. In order for an act of knowing to be feasible, it is necessary that a question exists upon which it is possible to have a position. If want our curricula to stimulate students to commit acts of knowing, they must not only allow but also demand that students assume and articulate personal positions on mathematical questions. A potentially valuable exercise, in my opinion, is to survey the range of tasks that we typically ask students to perform, and ask which of them and how many demand "personal and positional acts", even of a tentative nature and even at a low level. I would suggest that it is not possible to "have a position" on what the determinant of a given matrix is, but that it might be possible to "have a position" on sensible approaches to deciding whether a given matrix is invertible or not. I do not think it is possible to have a position on what the definition of a bijective function is, but it is possible to reach a position on whether a given function is bijective or not, according to the description of the function and an agreed definition of bijectivity. It is possible to have a position on whether a particular mathematical statement is true or not, and such a position might prompt an investigator to search for a proof. A mathematician however would never embark on an attempt to prove a statement without already having some conviction with regard to its likelihood of being true: why would we expect a student to do such a thing? To reject a proposed proof on the basis of an identified error in reasoning, an unwarranted assumption, a missing hypothesis or some other clearly articulated objection constitutes an act of personal position-taking and of mathematical knowing. To declare an inability to accept or reject a proposed proof, for example, on the basis of a specified point that is not fully understood, may also constitute an act of mathematical knowing. A statement of acquiescence with a proposed proof, however, may or may not involve an act of knowing: it is not possible for an observer to tell. Comparison and relative evaluation of different proposed proofs of a given statement requires the taking of positions and affords opportunities for acts of personal knowing. A detailed analysis of tasks of this nature and their potential to motivate attention to the purposes and nature of mathematical proof can be found in the recent PhD thesis of Kirsten Pfeiffer (2011).

If our curriculum presents students with complete and correct proofs to be studied (and maybe later rewritten on an examination script), we have no way of knowing the nature of the students' engagement with these proofs. The scepticism and insistent questioning that are such essential and prominent features of our disciplinary practice may be invisible to outsiders, but it is our business to make sure that they are visible to the newest members of our community. In order for this to occur it is essential that students experience situations in which they have a legitimate role as adjudicators of mathematical correctness and in which they are asked to assume and articulate and defend a position on a mathematical assertion or argument.

Schoenfeld (1994) discusses the locus of mathematical authority, both in mathematical research practice and in student communities. He proposes that the research community has collective standards for mathematical correctness, which are guaranteed and maintained through accurate and consistent community wide dissemination and quality control mechanisms. Mathematical authority resides in the community, but it also resides powerfully in persons who "having internalized the standards of correctness in their mathematical communities, apply these standards to what they know as individuals". For Schoenfeld,

> "the application of that external mathematical authority results in a very powerful personal ownership of the mathematics they can certify . . . arriving at mathematical certainty is the very personal process of applying an internalized impersonal standard."

He goes on to contrast this view of shared and individual authority with what typically holds in student communities, suggesting that

> "Most college students possess little of the sense of personal knowledge or internal authority just described. They have little idea, much less confidence, that they can serve as arbiters of mathematical correctness, either individually or collectively." (Schoenfeld, 1994, p. 62)

Obviously, we cannot expect novice students to exercise mathematical authority or mathematical authorship at the boundaries of knowledge as we aim to do ourselves within our research specializations. However, scope exists for students to use and articulate their mathematical judgement, at advancing levels of sophistication as they progress. The idea of a curriculum in which students exercise personal mathematical authority in feasible and appropriate contexts is consistent with Barnett and Coate's idea of curriculum as fostering "personal and positional" acts of knowing. If we academic mathematicians wish to facilitate or even motivate such acts, a challenge for us is to communicate our subject matter and assess our students' knowledge of it in ways that stimulate the development of their personal sense of mathematical authority, rather than inhibiting it. In order to illustrate this point in a small way, and in the very narrow context of summative assessment, here are two examination problems that I have used in recent years in a first year module in linear algebra. Each is presented in two versions.

Task 1

Version 1: What does it mean to say that a function $T{:}R^2$® R^2 is a *linear transformation?*

Version 2: Fill in the gaps in the following partial definition of a linear transformation from R^2 to R^2.

A function *T*:R^2® R^2 is a *linear transformation* if

1. *T(u+v)* = ______, for ___ *u,v Î*___.
2. *T(ku)* = ______, for ___ *u Î*___ and ____ *k Î*___ .

I have used both versions of this task in examinations, with classes of about 100 students. The first version, in practice, is an exercise in reproduction of a definition from the lecture notes or any standard source, and responses to it have included a wide range of variously garbled assortments of (some of) the relevant words. Some of these present serious dilemmas when marking: I am torn between the inclination to reward the appearance of some relevant words and the view that what has been written makes no sense whatsoever, indicates no understanding of what a linear transformation is, and deserves no marks. The second version, while it is arguably more artificial, demands that the student considers the mathematical meaning of what is written, attends to the necessity for clarity, precision and completeness in a definition as a more experienced practitioner would, and takes a position on such questions as "where does *u* live, where does *k* live, is it for *some* or for *all?*". These questions, and the question of what needs to be included in a definition, are trivial for mathematicians, but not for novice students who are still becoming accustomed to the peculiarities of our written language and notation and the meanings that these convey.

Task 2

The function *T*:R^2® R^2 is defined by *T(x,y)=(x+2y,xy+2)*, for *(x,y) Î* R^2 .

Version 1: Determine, with explanation, whether *T* is a linear transformation.

Version 2: What is wrong with the following argument claiming to show that *T* is a linear transformation?

Let $u = (1, 1)$, $v = (3,-1)$. Then $T(u) + T(v) = (3, 3) + (1,-1) = (4, 2)$.

$T(u + v) = T(4, 0) = (4, 2)$. So *T* respects addition.

Let $u = (1, 1)$. Then $T(2u) = T(2, 2) = (6, 6)$.

$2T(u)=2(3,3)=(6,6)$.

So *T* respects scalar multiplication, and *T* is a linear transformation.

The first version of this task requires a proof of a kind, but the exercise is a very standard one to which ritualized responses were difficult to distinguish from really comprehending ones. The second version requires that the student adjudicates on a matter of mathematical correctness and requires that she communicates her position on the mathematical question of whether the presented argument establishes what it claims to. The provision that the argument is not correct prompts the student to engage in a careful and sceptical reading of the text and hopefully an unravelling of the logic. Responses to the second version were individual rather than formulaic and gave me a far clearer and more detailed picture than the first of the students' understanding of linear transformations, and of their understanding of the distinction between checking a claim for a particular example and proving it globally. Most of the students identified the essential flaw in the reasoning and explained their position at least to some extent.

The shift from the first to the second version in each of these questions does not constitute a major departure. In each case both versions test the same knowledge, but in each case the second version asks the student to make (small) decisions, take positions, or exercise personal judgement, rather than repeating the words or imitating the behaviour of an external authority. Whether minor adjustments in emphasis such as these can help to raise students' awareness of their own role as "apprentice" members of the mathematical community and as negotiators of mathematical meaning may be a question worth considering.

REFERENCES

Bass, H. (2009) How do you know that you know? Making believe in mathematics. *Distinguished University Professor Lecture Series*, University of Michigan.

Barnett, R. and Coate, K. (2005) *Engaging the Curriculum in Higher Education*. Open University Press.

De Villiers, M. (1999). The role and function of proof in mathematics with Sketchpad. In *Rethinking Proof with Sketchpad*, Key Curriculum Press.

Halmos, P. (1985). *I want to be a Mathematician*. Springer-Verlag.

Mason, J. & Watson, A. (2008). Mathematics as a Constructive Activity: exploiting dimensions of possible variation. In M. Carlson & C. Rasmussen (Eds.) *Making the Connection: Research and Practice in Undergraduate Mathematics,* Washington: MAA. p189-202.

Pfeiffer, K. (2011). *Features and purposes of mathematical proofs in the view of novice students: observations from proof validation and evaluation performances*. Ph.D. Thesis, NUI Galway.

Rav, Y. (1999) Why do we prove theorems?, *Philosophiae Mathematica* 7(1), 5-41.

Schoenfeld, A. (1994). Reflections on doing and teaching mathematics. In A. Schoenfeld (ed.), *Mathematical Thinking and Problem Solving,* 53-70. Routledge.

THE LANGUAGE OF PROBLEM-SOLVING IN IMMERSION SCHOOLS - AN INVESTIGATION

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Set in the context of a growing immersion education, or gaelscoil sector in Ireland, this study looks at mathematical problem-solving in an immersion setting. The aim of the study was to examine if and how children's second language competence affects their problem-solving competence. The particular issues investigated were comprehension of problems, communicative abilities, use of mathematical terminology and the impact on problem-solving competence of any gaps in language. The method used was qualitative. A language sample was compiled from three recorded conversations of collaborative problem-solving sessions, and this sample was analysed. Findings indicated that the immersion context had little negative effects on comprehension of questions or communicative ability of the children. Mathematical terminology was not widely used, but general language was utilised to express the mathematical ideas. Gaps in language did not disadvantage children in the study tasks. Based on these findings, the conclusions drawn were that the immersion setting does not confer any particular disadvantage on mathematical problem-solving, though some concerns were raised about the difficulties experienced by some children in articulating higher-order concepts.

BACKGROUND

Irish language immersion education has been growing every year since the 1970's, and gaelscoileanna now make up almost five per cent of Primary schools in the country. It is likely that growth in this sector will continue for some time and increasing numbers of pupils will receive their education in the Irish language. Research on children's education in Gaelscoileanna thus far in Ireland has mainly concentrated on language acquisition (Harris, 1984, 2002; Harris, J., Forde, P., Archer, P., Nic Fhearaile, S., & O'Gorman, M., (2006); Harris and Murtagh, 1987, 1999; O Duibhir, 2009) or on literacy (NCCA, 2006). Research from other jurisdictions has focused on pupil attainment in mathematics in standardised tests, and indicates that immersion pupils studying in their second language (L2) are at no educational disadvantage compared to pupils who learn in their first language (L1) (Bournot-Trites and Reeder, (2006); Sierra, (2008)). While some comparative data are available at secondary level in Ireland, there has been little investigation into the experience and attainment of primary children in immersion settings in the area of mathematics. The key mathematical area of problem-solving may raise issues and challenges that could be particular to the immersion classroom, such as the comprehension of questions and the correct expression of solutions. Since the development of problem-solving skills is a priority of the Primary Mathematics Curriculum (Department of Education and Science, 1999), and Irish language education is an ever-growing area, it seemed an area ripe for investigation.

IMMERSION EDUCATION, LANGUAGE ACQUISITION AND PROBLEM-SOLVING

This study was informed by research on language acquisition, on immersion education and on mathematical communication and problem-solving. Learning through a second language is by no means an unusual experience, as large areas of the world such as India, Scandanavia and much of Africa are bilingual or multilingual and deliver their educational systems through one or more languages (Baker, 2001). In Ireland, from the time of the formation of the state in 1922, there have been various attempts to deliver some or all curriculum content through Irish (Ó hÚallacháin, 1978). In the past, misconceptions as to how the human brain dealt with multilingualism contributed to bilingual programmes being treated with some suspicion (Baker, 2001, Macnamara, 1966). Nowadays, the view of multilingualism is that, though the outward expression of languages is different, inwardly the languages are stored and processed together (Cummins, 1980). This is termed Common Underlying Proficiency (CUP), and thus content that a bilingual person learns in one language will also be available to him in the other. However, some qualifications exist. Cummins (1979) suggests that though a learner of a second language may learn language skills necessary for social interaction, or basic interpersonal communicative skills (BICS) in two years, the more abstract and higher register (cognitive academic language proficiency skills or CALPS) needed to succeed in formal education may take up to seven years to develop.

Immersion education originated in Canada in the 1970s, and since that time it has also developed in Ireland, Wales, the Basque Country and other areas where parents or the state have expressed an interest in educating their children through a minority or heritage language. Most research undertaken in Ireland has focused on language acquisition (Harris, 1984, 2002; Harris et al., 2006; Harris and Murtagh, 1987, 1999). These studies consisted of criterion-referenced tests based on the curricular objectives, with gaelscoil students showing strong achievement on these tests. Sierra's (2008) investigation into Basque-language immersion students' academic results within Spain showed that children in the primary Basque immersion setting outperformed their monolingual peers in reading comprehension, mathematics and grammar, while Basque-language secondary level children tested in Mathematics in the Programme for International Student Assessment (PISA) 2003 displayed scores above the mean. Bournot-Trites and Reeder (2006), in a study which looked at all aspects of the mathematics curriculum including problem-solving, found that the children in a total immersion environment outperformed children who received their mathematics instruction in their first language. They attributed this to positive cognitive effects of being competent in two languages as described in Cummins' Threshold Hypothesis (Cummins, 1976). These studies, however are small in number and dispersed in location.

Mathematical Problem-solving and Language

Word problems have a long history in mathematics, with examples being found on 4,000 year old Egyptian papyri, according to De Corte and Verschaffel (1991). They also identify a long history of students and teachers struggling to come to terms with these problems, terming this difficulty 'word problem depression'. The ways in which children engage with word problems, be it in an L1 or L2 situation are language dependent and are mediated through language. As identified by Durkin (1991), the transmission of meaning is one of the principal functions of language. However, in mathematics, the meanings being communicated can be complex and moreover can be conveyed by words which have other meanings in everyday language. For example, words such as ‘edge’ or ‘area’ are used in a number of contexts in ordinary language, but in the language of mathematics have very precise meanings. Children may be familiar with these words in their other, everyday meanings and may experience difficulties when greeted with the large number of new words and meanings encountered in mathematics lessons. In particular in the case of problem-solving, Durkin identifies many different demands, including syntactic, semantic, inferential, temporal and contextual knowledge. These demands arise to a greater or lesser extent in the range of problems children encounter. One way of dealing with these demands is through social collaboration, which he identifies as having potential to stimulate cognitive insights not available to children working independently.

On the other hand, Pirie (1991) finds that students’ verbalisation of their mathematical thinking can sometimes tend to lead children astray. Some features identified by her include the tendency of students to stop verbalising or to verbalise incompletely while under a high cognitive load. This might mean that the benefit to others of following the line of logic or hearing the explanation can be lost. In addition, at times students are unable to verbalise how they know what they have done. In some cases, certain cognitive activities are so automated that they are not mentioned or explained because students may not be aware they have done anything. While both these features of student discourse serve merely to provide no benefit to the other collaborators, in some cases verbalisation of mathematical thinking can lead to an undesired outcome. Pirie has observed situations where students have been found apparently to form their possibly erroneous mathematical beliefs by the force of their own explanatory words. The dynamics of classroom conversation, and how speakers can be encouraged to move from defending their own position, or from ‘going along’ with the conversation to a type of conversation where ideas are exchanged and developed in a constructive manner that respects each speaker, was the concern of Mercer, Rupert and Dawes (1999) and the characterisation of ‘exploratory talk’ and ways to foster this kind of talk led from this. It seems that the business of articulating thinking and exchanging ideas is a challenging one at the best of times, without adding the additional complication of negotiating through a second language to the mix.

Barwell, Leung, Morgan and Street (2002) recommend that mathematical language and terminology should be taught explicitly in a structured way so that ambiguities and misconceptions are eliminated. This is tempered with an awareness that broader considerations of language such as looking at the possibilities of mathematical discourse in simple, non-mathematical language, and recognising that language is interwoven with symbols, gestures and other ways of communication and making meaning. Meaning arises, they suggest, “through using words to mean things” (p. 13). However, Durkin and Shire (1991) hold that it is advisable to avoid semantically based misunderstandings by anticipating them and developing appropriate teaching strategies to help children understand the specifically mathematical meaning of the words they use. Speaking from the perspective of the multilingual classroom, Clarkson (1991) suggests that there is value in explicitly teaching mathematical vocabulary. The suggestion is made that this could be done both in the context of solving problems and in isolation.

Clarkson (ibid) has shown also that reading and comprehension of problems must be central to helping multilingual children achieve success with word problems and he identifies a role for teachers in improving the meta-linguistic awareness of children in this context. It is clear that there is neither consensus nor easy answers to the dilemma of whether or not to explicitly teach the language and vocabulary or mathematics in either L1 or L2 situations.

RESEARCH QUESTIONS

Whether the Irish language skills of 11-year old L2 learners in an immersion setting are adequate for the purposes of mathematical problem-solving formed the basis of the research question in this study. This research question was addressed by collecting and analysing a sample of language to identify if:

i) the children have sufficient language skills to understand typical problems;

ii) the communicative language of the children is of a high enough standard to facilitate meaningful and effective group work;

iii) specific mathematical terminology which may be required to deal with the problems is known to the children and can be confidently used by them to negotiate the mathematical demands of the task;

iv) gaps or errors in the children's language exist, either in terms of communicative or mathematical language, and if so, does this serve to disadvantage the children as they engage with word problems?

This analysis was adapted from Newman's framework (Zevenbergen, Dole and Wright, (2005)).

In addition, the approaches typical of second-language learners identified by Adeltula (1990) was used to characterise the children's strategies in solving the problems. This framework identifies the following approaches to solving word problems presented in the L2:

1) strategies mainly concerned with identifying numbers in the problem and performing a mathematical operation on them;

2) strategies mainly concerned with rote translation of key words to aid solving of problem;

3) strategies mainly concerned with analysing or understanding the meaning of the problem statement.

METHODS

In order to investigate the impact of L2 on mathematical problem-solving, a study was designed to provide a data sample that might be analysed to shed light on the linguistic and mathematical strategies used by children in mathematical problem-solving. This study took place among fifth class children in the school where the author was teaching at the time. This was a total immersion school in a mixed socio-economic area on the outskirts of Dublin.

Results from the standardised tests of the two previous years showed that most children in the class performed at average, higher average or above-average levels, with two children attending Learning Support for Mathematics. Attitudes to mathematics among the class were generally neutral to positive, though many children regularly expressed frustration and confusion at the language demands of routine and non-routine problems. While most standard textbooks are available as direct translations from English, supplementary and internet resources tend not to be available in Irish. Since these texts are tailored to the L1 market, at times the register of language or complexity of vocabulary can present a poor match to the requirements of an L2 learner.

Eight children, organised in three groups, undertook to solve some typical mathematics problems, some of which were standard textbook problems and some non-routine problems. The problems were chosen with a view to stimulating collaborative, problem-solving conversation across a number of mathematical areas. The mathematical problems were piloted with two groups of children who were not part of the final cohort, and some refinements were made, as well as some adjustments to the recording equipment. Each conversation lasted about thirty minutes, and both boys and girls took part. These conversations were then transcribed and analysed. The Irish language competence of the children was assessed using a comparison to O Duibhir's (2009) assessment of children in the senior classes of gaelscoileanna. Each child's contribution was analysed for each of the issues raised in Newman's framework above. The strategies used to attack each problem were characterised according to Adeltula's framework where strategy one involved simply performing operations on numbers identified, strategy two meant the children were pre-occupied with word for word translations, and strategy three involved the children attending to the whole meaning.

FINDINGS

The following is a discussion of findings of the analysis of the language sample.

Can the children understand the language of the problems?

In investigating whether or not the children comprehended the problems, each conversation was broken down into problems attempted and which of Adeltula's strategies was employed. Strategies one and two were employed by those with difficulties with language competence, while strategy three is the optimum, showing the greatest degree of higher order thinking and the least evidence of difficulty with language competence. The children's engagement with the problems will be examined below and their approaches to solving them placed in Adeltula's framework.

	Problem One	Problem Two	Problem Three	Problem Four	Problem Five
Group One	Strategy three	Strategy three		Strategy three	
Group Two		Strategy three			Strategy one
Group Three	Strategy three		Strategy three	Strategy two	

Table 1: A summary of how each group approached the problems

As the table shows, no one group attempted all problems, though each problem was attempted by at least one group. In general, the strategy most frequently used was strategy three, which involves attending to the whole meaning of the problem. The children dealt with unfamiliar words or phrases by using context, reasoning among themselves, or asking the author, all of which are reasonable and appropriate ways to deal with such a challenge. The work in conversation two on problem five, on an open-ended question involving the calculation or discovery of numerous possible alternatives, was the only one to show use of strategy one. In a long and involved conversation the children fixated on a visual aid and the real-world scenario of the problem to the extent that they failed to mathematise the problem and at length they appeared to offer various guesses, perhaps more out of weariness with the question than hope they had a reasoned solution.

The instance of strategy two, or relying on rote translation, also arose in a problem seeking all possible combinations or alternatives. Although initially focused on the translation of key words, this group did engage meaningfully with the task and made good progress. It is thus reasonable to say that the children were generally competent in understanding the problems and the challenge of operating in a second language did not present a major hindrance to meaningful comprehension.

Is the communicative language of the children of a high enough standard to facilitate meaningful and effective group work?

In assessing the children's capacity to communicate for problem-solving, each conversation was analysed to determine the relative participation of each child and to look at ways in which they exchanged ideas and dealt with the problems. The analysis was conducted on each group's conversation as a whole. Most children were active in posing and answering questions, in clarifying and building on one another's thinking, and in reasoning through the demands of each problem. Two children, one in group one, the group of two children, and one in the three-member group two contributed significantly fewer utterances to their respective conversations. Close reading of the conversation shows the child in group one was an active member, though less verbal, and made insightful suggestions that led to the correct solution of at least one problem. The child in group two was far less active generally, though his group mates were very vocal and displayed a tendency to talk over him, and each other. In follow-up interviews, each child indirectly alluded to their role in the conversation when responding to the question "Is there anything you like or dislike when working in groups?" The child from group one responded "You can just sit and listen to what the others are saying," while the response from the child in group two was "I really like working in groups because you can let the others do all the work." With the exception of these two cases, the other children communicated effectively and showed no signs in failing to function as a group member due to language constraints. Even in the case of the quieter children discussed above, it does not seem that reduced language competence prevented greater engagement. In many cases, children built on and added to the contributions of others to arrive at a solution

Is specific mathematical terminology which may be required to deal with the problems known to the children, and can it be confidently used by them to negotiate the mathematical demands of the task?

In order to determine this, each conversation was examined on a problem by problem basis for instances where they used or failed to use mathematical language when performing mathematical thinking or operations. A judgement was made as to whether their language was adequate to allow them to express themselves and to allow them to be understood by their peers.

Most of the children made mathematical statements which could be analysed to allow for some conclusions to be drawn. In general, the children did not use a particular mathematical register, instead drawing on day-to-day vocabulary to express mathematical ideas. The children were very comfortable in commentating simple mathematical operations such as addition and multiplication using the terms 'agus' (and), 'fá' (by) and 'sin' (that is-commonly used as 'equals'), and generally performed these operations routinely. In most instances, this vocabulary was adequate for the communicative requirements, particularly in the case of working out the requirements of a task, performing the mathematical operations and discussing the correct way to phrase their results to correctly solve the problem.

In the following extract co-operation is at work, having identified what needs to be done, add fractions to solve the problem together. Child A began but was overtaken by Child D who immediately identified and added the fractions, articulating as he went, while Child M neatly supplied the answer and explained a mental method for multiplying by twenty.

Child A	Aon leath, aon ceathru agus ... (One half, one quarter, and …)
Child D	Ciorcal timpeall na codáin go léir ar dtús. So, agus chuir iad go léir síos go dtí ceann amháin. OK aon ceathrú cothrom le dhá ochtú, eh, so aon ochtú dhá ochtú, agus tá leath de cothrom le ceathar ochtú. Agus cuir iad le cheile. Eh, trí agus ceathar, seacht. Agus so, tá fiche eh, so tá fiche aon ochtú ... (A circle around all the fractions first. So, put them all down to one. OK, one quarter equals two eighths, so one eighth, two eighths, and half of it is equal to four eighths. And put them together. Eh, three and four, seven. And so, twenty is, eh, so twenty is one eighth …)

Child L	B'fheidir aon cead is a seasca? (Maybe one hundred and sixty?)
Child D	Fan, má tá a haon fágtha (Wait, if one is left.)
Child L	Agus sin fiche, dó fá ocht, an codán rud, agus sin aon céad is a seasca. (And that's twenty. Two by eight, the fraction thing, that's one hundred and twenty)
Child D	Yeah tá dó fágtha...yeah fá ocht, sin é. (Yeah, two are left...yeah, by eight, that's it.)
Child L	Sé déag agus just cuir náid ag an deireadh mar sin fiche aithruithe go a dó (Sixteen and just put naught at the end because that's twenty changed to two.)

Barwell et al. (2002) have spoken strongly in favour of this way of talking about mathematics and it is clear that in many instances with these children, their general vocabulary and the ways in which they used it presented no impediments to their competence in doing mathematics. The children showed little tendency to rely on using English in areas where they may have had difficulty communicating, preferring in almost all instances to find a way to get their message across in the classroom language. There are some areas where the children struggle with language or fail to use target forms. In some instances it is the general vocabulary that is lacking. In one instance the phrase 'gach uile' meaning 'each and every' is used, where 'iad ar fad', or 'all of them' is required.

Interviewer	Seacht ochtú de na pictiúrí … (Seven eighths of the pictures)
Child G	Griangrafanna (Photographs)
Child F	Seo píosa beag de na pictiúrí agus má chuireann tú aon ochtú eile leis faigheann tú gach uile cheann. (That's a small piece of the pictures and if you put another one eight with it you get each and every one.)

Child F is very confident about what he is doing, but his language is not clear. Seven eights is not "a small piece" of a whole, though he understands that one additional eight makes the whole, here incorrectly rendered as "each and every one".

In more cases it is a failure to express mathematical ideas in mathematical terms that is the problem. There are several instances where addition is referred to as 'putting together', and indeed Child D referred to both addition and multiplication in these terms. While the children seem clear on what they mean, and no confusion is evident from the other children, a more mathematical register as stated in the mathematics curriculum (Government of Ireland, 1999) would help the development of mathematical concepts. While using a convenient phrase to describe two operations in the same terms may be nothing more than a shortcut in conversation, it may also lead to confusion in the case of the child himself or his peers, that the two operations are in fact the same thing. In another instance, when describing the process of adding fractions by finding a common denominator, Child F states that he circled all the fractions as they ‘were going to come together’. Again, while it was clear to the child what he was doing, and clear to others from the context, it is not an accurate statement of the process.

It may be said that the children generally seemed to be comfortable while dealing with the challenges of interpreting problems, attending to the mathematical operations and articulating the solutions. Difficulties sometimes arose in both general language and in mathematical expression when the children are challenged to explain their thinking and calculation procedures. It seems that it is the area of talking about mathematical processes that seemed most challenging. These conclusions must be approached with caution as relatively few of the children responded to these challenges, so the examples above are confined to the contributions of a handful of children.

Do gaps or errors in the children’s language exist, either in terms of communicative or mathematical language, and if so, does this serve to disadvantage the children as they engage with word problems?

In terms of the mathematical register used by the children, it seems there are few gaps in their language when it comes to comprehending questions, and these are dealt with either through discussion or seeking the definition of key words. They were also generally competent in performing and describing the mathematical operations needed to solve the problems. In the area of describing the thinking or calculation procedures that led to the solution of the problems, a few children dominated the conversation. These children had some problems in articulating clearly the mathematical processes, and there were also instances where general vocabulary was lacking or outside the grasp of the child at the time. It may be that this area raises the greatest concern. The mathematics curriculum (Government of Ireland, 1999) states

> Expressing mathematical ideas plays an important part in the development of mathematical concepts. One of the causes of failure in mathematics is poor comprehension of the words and phrases used. (p. 6)

The children in this study had few difficulties in making sense of the problems, performing mathematical operations and commenting on the steps that they were taking. It was only when commenting on the reasons or rationale for these steps that things became more difficult for them. It may be that the skill of expressing mathematical ideas is one which could be better developed among these children, and perhaps others. It may be that this issue requires the explicit teaching of words and phrases, to avoid this cause of failure identified in the curriculum.

DISCUSSION

It seems reasonable to say that the children had little difficulty in comprehending word problems. The children’s engagement was characterised according to the framework developed by Adeltula (1990) and in six out of eight cases the engagement with and comprehension of the problem occurred at the strategy three level, which shows higher order thinking and little difficulty with language. Research on immersion education typically shows that receptive language skills, of which reading is one, are usually acquired to native-like competence (Sierra, 2008; Baker, 2001). These findings fit into that larger trend. With regard to communicating in a group, though the children’s language displays the patterns of language error typical of immersion students, they nonetheless engaged with the problems and each other without showing evidence of serious difficulty. Though language used was largely everyday language, and though the error patterns and usage were inconsistent with a native speaker, the children were perfectly mutually comprehensible and fit for the task. Many instances were observed where the children collaborated and used each other's insight to find a solution. On this basis, the conclusion is that the standard of the children's communicative language was adequate for the purposes of effective group work.

Mathematical terminology used by the children was generally confined to descriptions of operations and common mathematical nouns, for example the word for 'fraction'. By and large, the children used general, everyday language to express mathematical ideas, for example describing addition as 'putting together'. Few examples of specifically mathematical terms were found. While some commentators (Barwell, 2003), suggest that a specific mathematical register is not required for effective discussion of mathematics, the developers of curricula do not agree. The Irish mathematics curriculum (Government of Ireland, 1999) and the British National Numeracy Strategy (DfES, 2000) are both in favour of the explicit development of mathematical vocabulary. It may also be the case, in an immersion environment, that the development of mathematical terms is seen as a language learning objective as well as a mathematical one. If it is accepted that mathematical vocabulary development is important, then the evidence of this analysis would suggest that these children need to be enabled to acquire and use more mathematical terms and a more specifically mathematical register as they engage in word problems.

Attempting to gauge whether language gaps exist is largely a subjective exercise. The analysis of the data shows that the children in this study show similar language gaps and patterns of error as other immersion children. There is little evidence of language gaps existing in the receptive skill of reading. The children's language proved competent where reading and understanding the problems was concerned. Similarly, their own discussion of the problems and their collaboration displayed few language issues, as far as communication is concerned. Problems arose in explaining their thinking and calculation procedures they followed and in providing rationales for these actions. At times, general words and phrases escaped them as they sought to express themselves, and mathematical terms were rarely used. These issues did not hinder the problem-solving per se, which was generally quite successful and collaborative. However, rationalising and explaining their thinking seems to have posed a challenge to these children. Further investigation would be needed to ascertain if this feature was a result of language skills or whether these higher order skills have been developed in the children.

REFERENCES

Adetula, L., (1990) Language Factor: Does It Affect Children's Performance on Word Problems? *Educational Studies in Mathematics,* 21, 4 351-365

Baker, C., (2001). *Foundations of bilingual education and bilingualism* (3rd ed.). Clevedon: Multilingual Matters

Barwell, R., Leung, C., Morgan, C. & Street, B., (2002) The language dimension of mathematics teaching', *Mathematics Teaching* 180 12-15

Bournot-Trites, M. And Reeder, K., (2006) Interdependence revisited: Mathematics achievement in an intensified French immersion program. *Canadian Modern Languages Review, 58* (1) 27-43

Cummins, J., (1976). 'The influence of bilingualism on cognitive growth: A synthesis of research findings and exploratory hypothesis'. *Working Papers on Bilingualism. 9* (1-43)

Cummins, J., (1979). Linguistic interdependence and the educational development of bilingual children. *Review of Educational Research 49 (2)* 222-251

Cummins, J., (1980). *The Construct of Language Proficiency in Bilingual Education.* Washington, D.C., Georgetown University Press.

De Corte, E., & Vershaffel, L., (1991). Some Factors Influencing the solution of addition and subtraction word problems. In: Durkin, K., and Shire, B., (eds.) *Language in Mathematical Education.* Milton Keynes: Open University Press.

Department for Education and Skills. (2000). *The National Numeracy Strategy: Mathematical Vocabulary*. London: Department for Education and Skills.

Durkin, K., (1991). Language in Mathematical Education: An Introduction. In: Durkin, K., and Shire, B., (eds.) *Language in Mathematical Education*. Milton Keynes: Open University Press.

Durkin, K., & Shire, B., (1991). Lexical Ambiguity in Mathematical Contexts. In: Durkin, K., and Shire, B., (eds.) *Language in Mathematical Education*. Milton Keynes: Open University Press

Government Of Ireland. (1999).*Mathematics Curriculum*. Dublin: Publications Office

Henry, A., Andrews, Á., & Ó Cainín, P. (2002). *Developing Linguistic Accuracy in Irish-Medium Primary Schools.* Bangor: Department of Education of Northern Ireland

Harris, J. (1982). Achievement in spoken Irish at the end of primary school. *Irish Journal of Education,* 16(2), 85-116.

Harris, J. (1984). *Spoken Irish in primary schools: An analysis of achievement*. Dublin: Institiuid Teangeolaiochta Eireann.

Harris, J. (2002). Research, innovation and policy change: Lessons from the ITE evaluation of the Irish programme at primary level. In J. M. Kirk & D. O Baoill (Eds.), *Language planning and education: Linguistic issues in Northern Ireland, the Republic of Ireland, and Scotland*. Belfast: Clo Ollscoil na Banriona.

Harris, J. (2007). Bilingual education and bilingualism in Ireland north and south. *The International Journal of Bilingual Education and Bilingualism, 10(4)*, 359-368.

Harris, J., & Conway, M. (2002). *Modern languages in Irish primary schools: An evaluation of the National Pilot Project*. Dublin: Institiuid Teangeolaiochta Eireann.

Harris, J., Forde, P., Archer, P., Nic Fhearaile, S., & O'Gorman, M. (2006). *Irish in primary schools: Long-term national trends in achievement.* Dublin: Department of Education and Science.

Harris, J., & Murtagh, L. (1987). Irish and English in Gaeltacht primary schools. In G. Mac Eoin, A. Ahlqvist & D. O hAodha (Eds.), *Third international conference on minority languages: Celtic papers.* Multilingual Matters 32. Clevedon: Multilingual Matters

Harris, J., & Murtagh, L. (1988). National assessment of Irish-language speaking and listening skills in primary-school children: Research issues in the evaluation of school-based heritage-language programmes. *Language Culture and Curriculum, 1(2),* 85-130.

Harris, J., & Murtagh, L. (1999). *Teaching and learning Irish in primary school: A review of research and development.* Dublin: Institiuid Teangeolaiochta Eireann.

Mac Éinrí, E., (2007) *Immersion Education: An Overview of Theory, Research and Practice.* Belfast: School of Education, QUB.

Macnamara, J.T., (1966) *Bilingualism and primary education: A study of Irish experience,* Edinburgh: University Press.

Mercer, N., Wegerif, R., Dawes, L., (1999) Children's Talk and the Development of Reasoning in the Classroom, *British Educational Research Journal* Vol. 25, No. 1. (95-111)

NCCA. (2006). *Language and literacy in Irish-medium primary schools: Descriptions of practice*. Dublin: National Council for Curriculum and Assessment.

O Duibhir, P., (2009) *The spoken Irish of sixth-class pupils in Irish immersion schools.* Unpublished Ph.D thesis, University of Dublin

Ó hÚallacháin, C., (1978). Languages of instruction in Ireland 1904-1977. *International Review of Education 14(4)* 501-510

Pirie, S., (1991). Peer Discussion in the context of mathematical problem-solving. In: Durkin, K., and Shire, B., (eds.) *Language in Mathematical Education*. Milton Keynes: Open University Press.

Sierra, J., (2008). Assessment of Bilingual Education in the Basque Country, *Language, Culture and Curriculum, 21(1),*39 – 47.

Zevenbergen, R., Dole, S., & Wright, R.J., (2005) *Teaching Mathematics in Primary Schools* Crows Nest: NSW

ON THE RELATIONSHIP BETWEEN A NOVICE TEACHER'S MATHEMATICAL KNOWLEDGE AND TEACHING ACTIONS

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The aim of this paper is to examine the relationship between mathematical knowledge for teaching (MKT) and teachers' actions when teaching mathematics, based on a qualitative case study of a novice mathematics teacher. The empirical data consist of interviews and an observed mathematics lesson during his first year of teaching mathematics. These data have been analysed by coordinating theories on teachers' goal-oriented actions, action levels and MKT. The results show that at a classroom level, the lesson provides opportunities for using the whole spectrum of MKT. However, in the different situations the teacher mainly draws on common content knowledge (CCK) and aims at maintaining control over the situation.

The main point of departure for this study is a curiosity about what kind of knowledge teachers use when undertaking different actions for teaching in the classroom. Previous research clearly points at the insufficiency of simply knowing the mathematics being taught and emphasizes the need for a specific kind of knowledge for teaching mathematics (Ball & Bass, 2000; Silverman & Thompson, 2008; Shulman, 1986; Wood, Neubrand, Seago, Agudelo-Valderrama, DeBlois & Leikin, 2009). Based on Shulman's work (1986), Ball, Thames and Phelps (2008) provide a categorisation of this specific mathematical knowledge for teaching (MKT), based on analyses of teachers' work on different levels. This categorization has been used for evaluations of prospective and in-service teachers and for examinations of the relationship between MKT and students' achievements (Hill, Rowan & Ball, 2005). Other studies have focused on how this specific kind of mathematical knowledge is developed (Rowland, 2007); on what kind of mathematical knowledge prospective teachers need in general (Hill, Sleep, Lewis & Ball, 2007) and in specific mathematical domains (Potari, Zachariades, Christou & Pitta-Pantazi, 2008); and on novice mathematics teachers' use of different kinds of MKT (Rowland, Jared & Thwaites, 2011).

In this paper I want to take a narrower approach, scrutinizing the relationship between teachers' actions and their knowledge. The way actions are related to knowledge is not obvious. A first step in elucidating this is to determine what kind of MKT teachers use for different actions. Novice teachers can make interesting contributions to our knowledge about the transition from teacher education to teaching practice. Novice teachers do not have the same experience and knowledge as their more seasoned colleagues. Still, the novice teachers' goals and intentions are to help their students learn mathematics. Thus, the analysis of novice teachers can offer valuable insight into what mathematics teachers require in the area of mathematical knowledge for teaching, and what developments need to be put in place for mathematics teachers in general.

NOVICE MATHEMATICS TEACHERS

The main aim of teacher education is to prepare prospective teachers for their forthcoming teaching practice by providing them with the knowledge for teaching that is required in the profession. The transition from teacher education to teaching in practice is a crucial step and highlights the relationship between pre-service teacher education and the practices of mathematics teachers in school. The character of the mathematical content that is addressed during teacher education seems to be significantly different from the mathematical content they are expected to teach at school.

In an ICMI study, Winsløw and his colleagues elaborate on novice mathematics teachers and their first years of teaching (2009). They recognize that students undergo a transition on an epistemological, institutional and personal level as they go from education to teaching practice. Even though all three levels are important, the epistemological level is particularly interesting for an investigation of mathematical knowledge for teaching, because it describes the shift from a more academic kind of mathematical knowledge to mathematical knowledge for teaching. It seems that a specific kind of mathematical knowledge for teaching is required, and that it differs markedly from academic mathematics. Thus, the epistemological transition from teacher education to teaching practice involves conflicting elements between the needs of schoolchildren to gain meaningful understanding of mathematics and mathematicians' preferences for formal deductive mathematical reasoning (Moreira & David, 2008). Thus, novice teachers have to create relevant connections between the academic mathematics they have learnt during teacher education and the school mathematics they will have to teach. In other words, novice teachers have to transform their knowledge of academic mathematics into both school mathematics and how it should be successfully taught.

MATHEMATICAL KNOWLEDGE FOR TEACHING

Mathematical knowledge for teaching focuses on the relationship between different kinds of knowledge that teachers need in order to teach mathematics at school. Shulman's work from 1986 can be regarded as a paradigmatic shift in that it stressed the fact that teaching requires more than academic skills and knowledge in the subject to be taught. He introduced the notions of curricular knowledge, content knowledge and pedagogical content knowledge as the three major parts of teachers' knowledge for teaching.

Shulman's categorization has been extensively referred to, further elaborated upon and developed. One of the most central and significant works in this direction is made by Ball et al. (2008), who give an account of a practice-based theory that describes mathematical knowledge for teaching, divided into six domains.

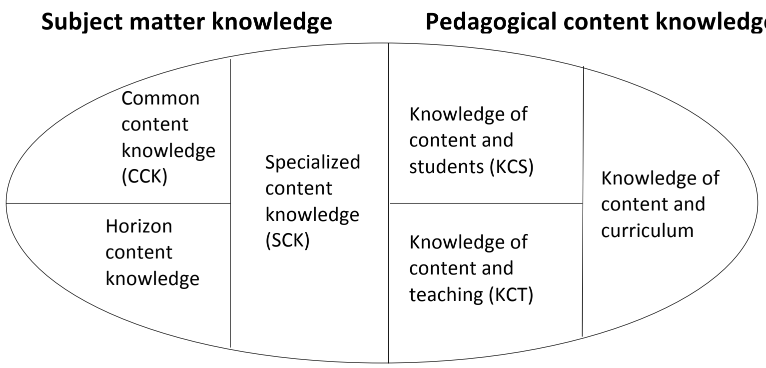

Figure 1: Domains of mathematical knowledge for teaching (Ball et al., 2008, p. 403)

Common content knowledge (CCK) refers to mathematical knowledge that is used in settings other than teaching. Briefly, teachers must know the mathematics to be taught. They must be able to solve tasks, recognize errors and use the correct words and notions. Specialized content knowledge (SCK) concerns mathematical knowledge and skills that are specifically related to teaching. "In looking for patterns in student errors or in sizing up whether a nonstandard approach would work in general, […] teachers have to do a kind of mathematics work that others do not" (Ball, 2008, p. 400). There are many different things that teachers do when teaching that require specialized content knowledge. Describing how a task may be solved may not involve SCK, but if students ask "why questions" in relation to the solution it suddenly does. Teachers probably know that some textbook exercises are more difficult than others. This may be related to CCK, but knowing why it is difficult as well as how to present it to make it easier is certainly a matter of SCK.

The domain of knowledge of content and students (KCS) concerns how mathematical knowledge is related to students' thinking and understanding of mathematics and its concepts. According to Ball and her colleagues, KCS is knowledge about the most common errors and miscalculations students are likely to make, the mathematical content with which students usually encounter difficulties, common conceptions and misconceptions, and how to facilitate and motivate students' learning of a specific content. Knowledge of content and teaching (KCT) combines mathematical knowledge with knowledge about various teaching situations. Choosing a suitable sequence of the mathematical content to be treated during a lesson; choosing suitable teaching material, tasks and other tools and aids; judging whether a student's question or comment should be further elaborated on or saved for a later occasion concern KCT.

Horizon content knowledge seems to be more tentative than the others (Ball et al., 2008). It involves the knowledge of how mathematical topics in the curriculum are related to one another. Knowing what students have done before and what they will encounter in their further studies concerns this kind of knowledge. The last category is knowledge of content and curriculum, which I interpret as an inheritance of Shulman's original categorization.

This categorization is a result of research on teachers' practices, which aimed to characterize and describe teachers' use of mathematical knowledge when teaching. As Ball and her colleagues emphasize, their theory is not about teachers' knowledge per se but rather their use of knowledge in and for teaching; the issue is mathematical knowledge for teaching, rather than teachers' knowledge. A consequence of this is that from a methodological perspective, examining teachers' mathematical knowledge for teaching actually has to be done in relation to a teaching activity or a teaching situation.

I will also draw on another of the authors' comments in the paper. They state that the categories are derived from teaching practice. By researching teachers' teaching of mathematics, they were able to identify the categories. However, a crucial issue also involves exploring teachers' use of different domains of mathematical knowledge for teaching in relation to their teaching actions, or "how different categories of knowledge come into play in the course of teaching, needs to be addressed more effectively in this work" (Ball et al., 2008, p 403). Further examining the relationship between mathematical knowledge for teaching and teaching actions is one attempt to respond to this request.

TEACHERS' ACTIONS AND DECISION MAKING

To teach mathematics is to be constantly involved in, and undertake, actions. These actions are performed in neither isolation nor randomness. Rather, they are accomplished with an overall aim to teach mathematics, and consequently, to provide opportunities for students to learn mathematics. Actions take place in a complex, interactive interplay between teacher and pupils.

Schoenfeld (2011) has developed a theoretical framework aimed at guiding the understanding of teachers' decisions and actions in the mathematics classroom. This framework builds on the assumptions that teachers are engaged in goal-oriented activities when teaching mathematics. The teacher undertakes observable actions. Because these actions are goal-orientated, goals influence actions. How an individual regards his or her environment and reacts to it are fundamentally shaped by that person's orientations. On the other hand, teachers perform actions that often have an outcome that differs from their initial intentions. In this case the teacher may reconsider the goals, which in turn will affect his or her orientation. Thus, Schoenfeld claims that there is a mutual interplay between orientations, goals and actions.

Schoenfeld uses orientations as an inclusive term to encompass beliefs, dispositions, values, tastes and preferences. It is most plausible that teachers' orientations are shaped by previous experiences of learning and teaching mathematics in specific social, cultural and contextual situations, but also from new experiences of the outcome from different actions. From a novice teacher's perspective, the first years of teaching can be regarded as a point of convergence between his or her own experiences from school mathematics as a student, from teacher education and as an in service-teacher.

According to Schoenfeld (2011), to teach mathematics is to be engaged in goal-oriented activities. He defines a goal as something that an individual wants to achieve. Teachers' different goals are influenced by their different orientations. In turn, their actions are influenced by their goals. Goals can be conscious or unconscious, and can be short-term or long-term. Teachers may have an overall goal to teach mathematics so that students can learn mathematics, but they can also have the goals of maintaining good relations with students, preparing students for assessment, and providing opportunities for students to have fun and enjoy themselves. For example, in a classroom situation with many students waiting for assistance from the teacher, guidance in small steps can be an effective way to achieve a short-term goal, namely to help as many students as possible. A more thorough explanation, which in the future could lead to the development of a deeper understanding of the mathematical content or problem, might be one way to achieve a long-term goal.

Another part of the framework is resources, the most important of which is knowledge. Schoenfeld defines knowledge as "the information that an individual has potentially available to bring to bear in order to solve problems, achieve goals or perform other such tasks" (2011, p 25). Though Schoenfeld does not make any further explicit statements about the relationship between actions, goals and orientations on the one hand and resources on the other, my interpretation is that resources are present in actions, goals and orientations. In the description of his theory Schoenfeld gives an account of various characteristics of knowledge, for example, facts or isolated pieces of knowledge, procedural and conceptual knowledge and problem-solving strategies. This categorization of mathematical knowledge is common in descriptions of the character of learners' mathematical knowledge. However, to analyse the relationship between teachers' knowledge and actions, we rather need a description of the content of the knowledge being used. Thus, if a teacher responds to a student's question, it is more relevant to know if the answer relates to the teacher's knowledge about the mathematical content, the student or both, than if the knowledge provided by the teacher is procedural or conceptual in its character. Therefore, we can coordinate Ball et al.'s categories of mathematical knowledge for teaching with Schoenfeld's theory of goal-oriented actions and MKT as the main resource of knowledge used in teaching actions.

Goals, resources and orientations have an impact on a teacher's actions, which in turn are consequences of constant decision making. Decision making involves monitoring and self-regulation. It refers to a person's own awareness of the impact of actions and the ability to adjust them in relation to the goals and resources at hand. For example, if a teacher tries to explain a mathematical issue, monitoring involves noticing how the explanation is received. Self-regulation concerns adjusting the explanation to be understandable to a learner. The role of subjective valuations serves as an explanation for why a teacher acts in a specific way. Continuing to give an explanation to a student who does not understand can be due to a teacher's subjective valuation. For example, a teacher may regard that specific explanation to be in line with mathematical rigour or correctness and therefore stick to it instead of giving a more understandable explanation that might be regarded as less mathematical.

Thus, Schoenfeld's theory of goal-oriented decision making provides a comprehensive framework to analyse teachers' actions when teaching, which involves their mathematical knowledge for teaching. However, actions for teaching can be accomplished on different levels. Concerning a lesson, teachers make decisions and actions that concern the overall arrangement of the whole learning situation (Sensevy, Schubauer-Leoni, Marcier, Ligozat & Perrot, 2005). Teachers perform actions when choosing tasks and activities, and create the overall organization of the teaching and learning situation. In the interaction with students a teacher performs actions that result in a distribution of roles and responsibilities between him- or herself and the students. With an even more fine-grained look, one can also refer to the actual development of students' knowledge and a teacher's self-monitoring process when teaching.

METHOD

In this study, I will use Schoenfeld's (2008) theory of goal-oriented decision making as a starting point for analysing teaching actions, based on the relationship between orientations, goals and actions. While a teacher's actions can be studied through observation, a teacher's goals and orientations come to the fore through interpretations of interviews. According to Sensevy et al. (2005), the actions can be sorted into three different levels, i.e. actions for: the overall organization of the teaching and learning situation; the distribution of roles and responsibilities; and the development of knowledge.

I will use Ball et al.'s categories of mathematical knowledge for teaching (2008). These categories will also be related to the transition and tensions between academic mathematics and school mathematics that were presented previously in the paper, as one aspect of mathematical knowledge for teaching. The approach of using these theories can be described as coordinating these theories "for a networked understanding of an empirical phenomenon or a piece of data" (Prediger, Bikner-Ahsbahs & Arzarello, 2008, p. 172).

To explore mathematics teachers' actions and the knowledge they use when teaching, empirical data from a mathematics lesson taught by a novice mathematics teacher will be presented and analysed in this paper. Hill et al. (2007) stress the limitations of assessing teachers' mathematical knowledge by analysing observations of a smaller number of mathematics lessons. However, for an initial study about a specific phenomenon, inductive case studies can be an appropriate way to gain an understanding of the basic phenomenon (Patton, 2002). The lesson was chosen from a number of observed lessons because it was representative of an ordinary lesson. This constitutes a case involving mathematics lessons by novice mathematics teachers, rather than this particular mathematics lesson.

The data presented in this paper constitute a small part of a more comprehensive study about novice mathematics teachers during their first years of teaching mathematics in school. Four teacher students, three female and one male, participated in this study. They were interviewed during their last semester of teacher education before graduation and were then frequently interviewed and observed during their first two years of teaching school. No specific arrangements were made because of the study, which means that the observations can be regarded as authentic. I will now give an account of a mathematics lesson, taught by the male teacher, that took place during his first semester as a newly-qualified teacher. During the lesson the teacher carried a small mp3-recorder with him. His writings on the whiteboard and the students' tasks and solutions were photographed. The audio recordings were transcribed in full and analysed with regard to the frameworks presented above (Ball et al., 2008; Schoenfeld, 2011; Sensevy et al., 2005) by identifying empirical instances of crucial theoretical concepts in the study (Stadler, 2011).

RESULTS AND ANALYSIS

The teacher is in his first year as a newly-qualified teacher in mathematics and English at an upper secondary school in Sweden. As a student teacher he had done very well in mathematics. His self-confidence is strong, concerning both his own mathematical knowledge and his ability to teach mathematics. In an interview just before he graduated, he is asked if he sometimes worries about being able to answer questions from the students.

Teacher: No, because I know so much more mathematics than they do. I can give them answers about anything. They can call me in the middle of the night, and I can give them answers. I can get something five minutes in advance and deliver during the lesson without any problem. (*Interview before graduation*)

During the lesson, the students work with repetition of powers of integers, prime number factorization and reductions and calculations with fractions. The lesson begins with the teacher writing three tasks on the whiteboard and giving the students some time to work on these tasks on their own. After a while, the teacher asks the students to explain their solutions, and then gives them some new tasks. Some of the tasks are shown in Figure 2.

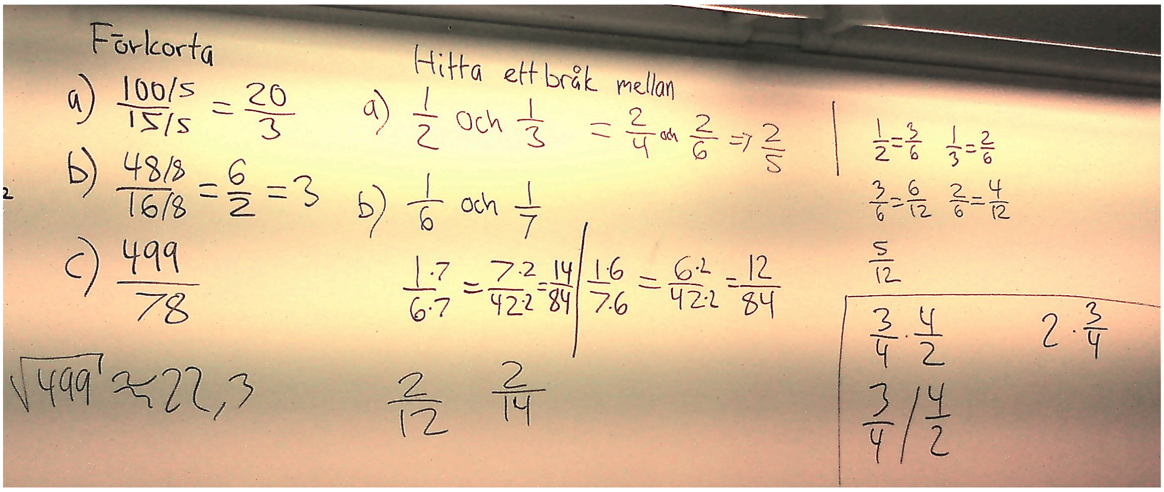

Figure 2: The tasks on the whiteboard

The teacher gives an account of his approach and the choice of exercises he presented on the whiteboard.

Teacher: I wanted the pupils to check themselves. We've worked with all these things before, but I know by experience that they forget it after some time. And also, I wanted to give them a chance before, and then talk about it as a whole class. That was my thought. (*Interview after the lesson*)

Analysis: The arrangement of the classroom setting and the setup of the lesson refer to the overall organization of the teaching and learning situation (Sensevy, 2005). On this level the teacher undertakes actions that concern, for example, the planning of the lesson and the choice of tasks. One specific goal of the lesson from the teacher's point of view is to repeat the basic arithmetical rules for calculations with fractions so that the students can remember them. The quick and superficial planning of the lesson may be due to routinized actions, whereby the lesson is worked out in a standardized way and the content and its presentation are mainly based on tasks from the textbook.

According to Schoenfeld (2011), resources like knowledge are an essential part of teachers' actions, goals and orientations. In this case, the teacher has a strong belief that a good teacher has to know a great deal of mathematics by heart and be able to deliver rapid answers. His quick preparation of lessons and his choice of tasks seem to be mainly influenced by his CCK. His responsibility when answering questions is to provide a mathematical solution for the task, while the student dimension is absent. However, in the interview after the lesson the teacher shows elements of KCS when expressing his insight into students' ways of learning and which problems they may encounter.

The teacher's planning of the lesson offers one example of the close and mutual relationship between the teacher's actions, goals and orientations. The teacher has a clear content-related goal for the lesson. At the same time, the planning and accomplishment of the lesson are done in a routinized and unreflective way, inspired by his previous experiences, values and beliefs.

One task is to find one fraction that is between *12* and *13*. When the teacher asks the students for a solution he gets an unexpected answer from Student1, a male.

Student1: You can multiply the numerator and denominator by two, and then you can take two fifths.

Teacher: Can you repeat that, please!?

Student1: You multiply two by two and one by two.

Teacher: So, you'll get two quarters? Right?

Student1: Right. And then you do the same with one third and get two sixths.

Teacher: And two fifths is in between. Yes, that's one way to do it. That was pretty smart. But what do you think most of the rest of you have done? You've done it like this: you change one half to three sixths and one third to two sixths. And then you've changed three sixths to two twelfths and two thirds to four twelfths. And between them you can take, for example, five twelfths. But your answer was good. It shows understanding.

Student2: Does the first method always work?

Teacher: Well, the question is rather if you understand it, right? It's a method that shows understanding. But sure, if you take a denominator that is in between, that'll work. Actually, I've never thought about that method myself.

Analysis: As a teaching action, classroom dialogues are an important arena for the distribution of roles and responsibilities between the teacher and the students. During the lesson, the teacher presents a couple of standardized tasks, but gets into an unexpected situation when one student suggests a solution he cannot immediately understand. In this situation the teacher provides a standardized solution to the students, which is mainly based on CCK. There could be several reasons for providing the standardized way to solve the exercise: to provide a simple or comprehensive way to solve this kind of task, or it could be due to his orientation that the teacher should deliver a solution to all exercises. However, the main goal of his actions and statements seems to be to maintain control over the situation and avoid any further distribution of initiative to the student. According to his orientation, he is the most mathematically knowledgeable person in the classroom. His response to the student's suggestion indicates that he is not prepared to risk getting into a conversation that he does not fully understand.

A closer look at what kind of MKT a teacher should need in order to act in such an unexpected situation shows that there is a need for quick and careful use of different kinds of MKT. Responding to an unexpected mathematical statement from a student not only requires an analysis of the student's thinking and reasoning, which refers to KCS; there is also an instant need to use SCK to be able to further elaborate on the question. In his answer he needs CCK to be able to elaborate on the solution from a mathematical point of view. To provide an answer that is suitable and understandable with regard to the students and the situation, he also has to draw on KCT. Thus, the situation is complex and it is not surprising that this can be a hard task for a newly qualified teacher. It is also obvious that it is the composition of the different kinds of MKT that is crucial, not the mathematical knowledge per se. It can be suggested that making quick variations between different kinds of MKT is particularly challenging to a newly qualified mathematics teacher.

The rest of the lesson, approximately 30 minutes, is spent on individual work with textbook exercises. During this time, the teacher gives 11 explanations of tasks to individual students. A majority of the questions concern arithmetic calculations with fractions, for example the rules for the addition and subtraction of fractions as well as the multiplication and division of fractions. These dialogues are characterized by the teacher's account of the calculation rules. The following dialogue serves as an example of the features of the teacher's explanations.

Student3: To multiply these (points in his notebook), I can just multiply them by each other? Is it that simple? Because in the other examples I've made the denominators equal.

Teacher: Well, you only have to do that when it's plus and minus. When it's multiplication and division, there's no need for that. In this case you can just write three times four divided by four times two, and that's it. You just have to put them on a common fraction bar, and then you're done, when it's multiplication.

After the lesson, the teacher takes the opportunity to comment on students' mathematical pre-knowledge in more general terms, but the examples seem clearly influenced by the present lesson.

Teacher: When pupils start here, they don't know things they should know from lower secondary school. Even if they multiplied and divided fractions there, they don't know how to do it when they arrive here. At least from a general point of view.

Interviewer: What are the reasons or explanations for that?

Teacher: I don't think they know what they're doing. They just copy strategies for the tests, but they lack insight into what's going on or what they're doing. I don't think they know what a fraction is. (*Interview after the lesson*)

ANALYSIS

One important scene of actions for the development of knowledge is individual dialogues between the teacher and the students. During the lesson, the students ask the teacher questions about textbook exercises they cannot manage to solve. Characteristic of the teacher's response to these questions is to give a quick instruction for how to solve the specific task at hand and sometimes provide a more general explanation of the mathematical content, for example how to add or divide fractions, use the power rules or factorize an expression. The teacher dominates these situations, without specifically considering the student's solution or exploring what he or she has done. Thus, SCK is absent in the teacher's interactions with students. The leading indicator for what he talks about with them is guided by the task rather than the student's solution at hand or the problems the student may have encountered. The teacher's explanations are guided by CCK, in this case features of fractions themselves, rather than how the students understand (or do not understand) division of fractions (KCS). Also, there are no indications of the teacher consciously varying his way of explaining things on the basis that the students are different, have different pre-conditions and pre-knowledge, or might have encountered different problems with the same task. However, by answering several similar questions, the teacher develops a local KCS with respect to the current tasks.

This way of explaining may have consequences for the actual development of students' knowledge. In terms of short-term goals, this can be a way to meet students' instant need for help with exercises. One could also regard it as an effective way to provide explanations to as many students as possible during a lesson. A clear and standardized explanation that is easy to follow can also be interpreted as an understandable explanation, which makes the students happy. The teacher is confirmed in his role as teacher, and in his orientation about what a good teacher should do. However, the dominant use of CCK and the way of elaborating with KCT as a way of maintaining control of the situation result in the students not being challenged in their ways of thinking and thus remaining at a more surface level of learning. The teacher's self-monitoring process when teaching is focused on himself rather than on the students. Because the teacher does not attempt to provide an explanation based on the students' understanding and reasoning, situations requiring SCK do not emerge.

SUMMARY AND CONCLUDING REMARKS

The aim of this paper was to examine the relationship between mathematical knowledge for teaching and teaching actions. Based on empirical data from interviews with a novice mathematics teacher and observation of a mathematics lesson, some connections have been identified between mathematical knowledge for teaching and teaching actions. Characteristic of the episode is the novice teacher's tendency to draw on CCK when his preferences are in the mathematics itself, rather than the problems and difficulties the students encounter with the mathematical content. On the level of the overall organization of the teaching and learning situation, the teacher draws on CCK when planning the lesson, even though elements of KCS can be found. Also in his interactions with students, he mainly draws on CCK with the aim of maintaining control over the situation. Actions on the level of developing the students' knowledge and the teacher's self-monitoring process still focus on CCK, with the teacher providing standardized explanations. His preference for CCK may be regarded as a typical feature of a novice teacher undergoing the transition from teacher education to teaching practice, and the transformation from academic mathematics to school mathematics and mathematical knowledge for teaching. However, the interaction with the students provides opportunities to further develop his KCS that may result in initiatives for situations involving SCK.

In this paper, Ball's categories of MKT have been used. However, there are other categorizations of mathematical knowledge for teaching, and more thorough consideration of the most suitable categorization could be further scrutinized. In this study, the results indicate a new angle of CCK as a valuable tool to allow the teacher to keep the teaching situation under control. Strategically used, this kind of knowledge can be used to silence students' questions, make an example of oneself as superior concerning content knowledge, and avoid discussions and reasoning about unfamiliar mathematical territory. This is a suggestion for supplementary evaluation of different categorizations of mathematical knowledge for teaching concerning how to use the mathematical content to retain authority in the classroom.

The results also imply the need for further elaboration on the connections and relationships between actions, goals and orientations on the one hand and the use of mathematical knowledge for teaching on the other. In the analyses, it was possible to connect actions with categories of mathematical knowledge for teaching. However, similar actions can be due to different goals and orientations. There seems to be a need for an additional theoretical or methodological framework for analysing the intentions and rationale for actions. Also, clear definitions are needed for how to operationally define an action, and how the specific analysis of goals and orientations should be accomplished. This relationship can be elaborated on from two perspectives: what the teacher actually does (which is analysed within these frameworks) and on a more theoretical level (the relationship between the frameworks themselves).

Studying the relationship between knowledge and actions also initiates a more general discussion about different theoretical perspectives on the teaching and learning of mathematics. Some readers may feel that knowledge and actions should be researched from a socio-cultural perspective, whereby Activity Theory could be regarded as a more comprehensive framework and analytic tool. However, there are crucial differences between these research approaches. Whereas Activity Theory has a clearly socio-cultural perspective, Schoenfeld adopts a more individual and cognitive approach. Therefore, some words that occur in both frameworks have significantly different meanings. The coordination of the different theoretical categorizations and frameworks in this paper has resulted in a more individual-cognitive approach to the relationship between mathematical knowledge for teaching and actions for teaching. I argue that an examination of the relationship between actions and knowledge could benefit from a more local and focused analysis that does not take into consideration the whole social perspective. Therefore, it is my intention to further elaborate on mathematical knowledge for teaching and teaching actions with the approach initiated in this paper.

REFERENCES

Ball, D. L. & Bass, H. (2000). Interweaving content and pedagogy in teaching and learning to teach: Knowing and using mathematics. In J. Boaler (ed.) *Multiple perspectives on mathematics teaching and learning*, (pp. 83-104). Westport, CT: Ablex.

Ball, D. L., Thames, M. H. & Phelps, G. (2008). Content knowledge for teaching: What makes it special? *Journal for Teacher Education*, 59 (5), 389-407.

Hill, H. C., Rowan, B. & Ball, D. L. (2005). Effects of teachers' mathematical knowledge for teaching on student achievement. *American Educational Research Journal*, 42(2), 371-406.

Hill, H. C., Sleep, L., Lewis, J.M., Ball, D.L. (2007). Assessing teachers' mathematical knowledge. What knowledge matters and what evidence counts? In F. K. Lester, Jr. (ed.) *Second handbook of research on mathematics teaching and learning: A project of the national council of teachers of mathematics*, (pp. 111-1155). Charlotte, NC: Information Age Publishing Inc.

Moreira, P.C. & David, M.M. (2008). Academic mathematics and mathematical knowledge needed in school teaching practice: some conflicting elements. *Journal of Mathematics Teacher Education*, 11(1), 23-40.

Patton, M. Q. (2002). *Qualitative research & evolution metods 3*. Thousand Oaks, California: Sage Publications Inc.

Potari, D., Zachariades, T., Christou, C. & Pitta-Pantazi, D. (2008). The relationship between teachers' mathematical knowledge and teaching in calculus. *Symposium on the Occasion of the 100th Anniversary of ICMI*. Rome, 5–8 March 2008.

Prediger, S., Bikner-Ahsbahs, A. & Arzarello, F. (2008). Networking strategies and methods for connecting theoretical approaches: First step towards a conceptual framework. *ZDM Mathematics Education*, 40, 165-178.

Rowland, T. (2007). Developing knowledge for mathematics teaching: A theoretical loop. Proceedings of Second *National Conference on Research in Mathematics Education, MEI2*. September 14th - September 15th 2007, St. Patrick's College, Dublin, Ireland.

Rowland, T., Jared, L. & Thwaites, A. (2011). Secondary mathematics teachers' content knowledge: The case of Heidi. *Proceedings of the Sevenths Congress of the European Society for Research in Mathematics Education*. January 9th - February 13th 2011, Rzeszów, Poland.

Schoenfeld, A. H. (2011). *How we think. A theory of goal-oriented decision making and its educational applications*. New York: Routledge.

Sensevy, G., Schubauer-Leoni, M.-L., Marcier, A., Ligozat, F. & Perrot, G. (2005). An attempt to model the teacher's actions in the mathematics class. *Educational Studies in Mathematics*, 29, 153-181.

Shulman, L. S. (1986). Those who understand: Knowledge growth in teaching. *Educational Researcher*, 15(2), 4-14.

Silverman, J. & Thompson, P. W. (2008). Toward a framework for the development of mathematical knowledge for teaching. *Journal of Mathematics Teacher Education*, 11, 499-511.

Stadler, E. (2011). The secondary-tertiary transition: A clash between two mathematical discourses. *Proceedings of the Sevenths Congress of the European Society for Research in Mathematics Education*. January 9th - February 13th 2011, Rzeszów, Poland.

Winsløw, C., Bergsten, C., Butlen, D., David, M., Gómez, P., Grevholm, B., Li, S., Moreira, P., Robinson, N., Sayac, N., Tatto, T., White, A. & Wood, T. (2009). First years of teaching. In R. Even, D.L. Bal (eds.), *The professional education and development of teachers of mathematics,* (pp. 93-101). New York: Springer Science+Business Media.

Wood, T., Neubrand, M., Seago, N., Agudelo-Valderrama, C., DeBlois, L. & Leikin, R. (2009). The balance of teacher knowledge: Mathematics and pedagogy. In D. Ball & R. Even (Eds.), *ICMI Study-15 Volume: The professional education and development of teachers of mathematics*, (pp. 211-226). New York: Springer Science+Business Media.

PLACE VALUE UNDERSTANDING: A CASE STUDY OF TWO FOURTH CLASS CHILDREN

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Results from international and national research show that Irish children are not performing at the level of many of their international counterparts in Mathematics (Shiel, Moran, Cosgrove & Perkins, 2010; Eivers, Close, Shiel, Millar, Clerkin, Gilleece & Kiniry, 2010). This research project aimed to look at fourth class Irish children's understanding of place value in light of international research on the topic. Task-based interviews were conducted with children in three schools to ascertain their understanding. This paper will present a case study of two of those children's understanding of place value. The results of their task-based interviews will be analysed and discussed with reference to the findings of similar research projects done around the world.

INTRODUCTION

With the current concerns about Irish students' mathematical achievements in international and national assessments, as well as the Junior and Leaving Certificate examinations, a closer look at the teaching and learning of mathematics in Ireland is crucial. Achievement of Irish children across the primary mathematics curriculum as well as the teaching and assessment of mathematics have been the focus of large-scale national research projects (Shiel & Kelly, 2001; DES, 2005a; DES, 2005b; NCCA, 2005; Shiel, Surgenor, Close & Millar, 2006; Eivers et al., 2010) Aspects of place value have been included in smaller scale projects on the teaching and learning of mathematics in Ireland (Travers, 2005; Ward, 2005; Travers, 2009), but little research has been carried out on children's understanding of the concept of place value. As number work permeates all strands of the mathematics curriculum and place value is the very basis of our base-ten number system, it is crucial that we take a closer look at the teaching and learning of place value in Irish primary schools. It makes sense to begin by finding out what children know about place value and the misconceptions they have.

THEORETICAL FRAMEWORK

Many research projects around the world have examined different aspects of place value teaching and learning. Analysis of place value development, the influence of language, mental arithmetic, teaching methodology, use of materials, curricular placement, and textbook presentations have all been researched to better understand what children know about place value. The focus of this paper is children's understanding of place value.

Levels of Understanding

Researchers around the world have tried to identify the stages or levels of understanding children have about place value. Some common themes have been identified. It is acknowledged that place value understanding develops gradually over time and that not all children go through all stages. Children start with unitary conceptual structures by making the connection between the numbers 1 to 9, the numerals and a group of objects (Resnick, 1983; Fuson, 1990), they then partition numbers into tens and ones, but have not yet developed the multiplicative concept of place value (one ten is made up of ten units) (Cobb & Wheatley, 1988). They then understand that one ten is made up of ten units and can count the tens as one group knowing that each ten is made up of ten units (Cobb & Wheatley, 1988; Fuson, Wearne, Hiebert, Murray, Human, Olivier, Carpenter & Fennema, 1997). At this stage children show no flexibility of representation and can use only canonical representation (Ross, 1990) (33 represented only as 3 tens and 3 units). As further understanding develops children can then use non-canonical representation (33 represented as 2 tens and 13 units or 1 ten and 23 units) (Ross, 1990; Miura, Okamoto, Kim, Steere, & Fayol, 1993). Ross worked on developing a "more comprehensive model of children's conceptual development in the place value domain" (1986, p. 2). Based on her finding and previous work carried out by others, Ross designed a five stage model to explain how children's place value understanding of two-digit numbers develops:

- In Stage 1, '*whole numeral',* the child sees 25 as 25 units and places no meaning on the individual digits;
- In Stage II, '*positional property',* the child sees the 5 in 25 as in the units place and the 2 in 25 as in the tens place;
- In Stage III, '*face value',* the child uses "face value' as in seeing the digit two in 25 as two of the ten blocks and the digit five as five of the unit blocks;
- Stage IV, '*construction zone'*, children represent numbers using the number of tens blocks that correspond to the tens digit and the number of unit blocks to correspond to the units digit (i.e. the child uses canonical representation). Ross classified this stage as a transitional stage;
- In Stage V, '*understanding*', the child understands that each digit (tens and units) represents part of the whole quantity. (i.e. the child uses non-canonical representation).

The results of research by Ross and others showed that many children do not develop understanding at Stage V until fourth grade**.**

Adopting a socio-constructivist approach, Jones, Thornton & Putt (1994) used the four constructs identified by previous research as essential for children's development of place value understanding (counting, partitioning, grouping, and number relationships) to develop 'a model for nurturing and assessing multidigit number sense'(1994, p. 118). They focused on four different 'levels of growth' in place value thinking in relation to these constructs to develop a framework which was then used to develop a programme for the teaching and assessment of multi-digit number sense:

- Level 1 - Pre-place value;
- Level 2- Initial place value;
- Level 3 - Extended place value 1;
- Level 4 - Extended place value 2.

As their work was with children in first grade they recommended that further research was needed to extend the framework to higher grade levels. In a follow up study, Jones, Thornton, Putt, Hill, Mogill, Rich and Van Zoest (1996) renamed Level three as 'Developing place value' and extended their framework to Level 5 which they called 'Essential place value'.

In Australia, Price (2001) used a framework he developed from previous researchers' conceptual structures for place value in his study. He proposed four conceptual structures:

- Conceptual Structure 1: Unitary construct
- Conceptual Structure 2: Tens and ones structure
- Conceptual Structure 3: Ten as a unit
- Conceptual Structure 4: Flexible representations

The Early Numeracy Research Project (Clarke, Cheeseman, Gervasoni, Gronn, Horne, McDonough, Montgomery, Roche, Sullivan, Clarke, & Rowley, 2002) developed a framework of growth points for assessing place value learning. Horne & Livy suggest that these "could more effectively be interpreted in terms of unitary, ten-structured, multi-unit and extended multi-unit concepts" (2006, p. 314).

Language and Place Value Understanding

The inconsistency of the number word system in European languages has been used to explain why Asian children outperform their Western counterparts in international mathematics assessments. Fuson and Briars (1990) contrasted the structure of the spoken system for number words ('named-value') in the English language to the written system ('positional base-ten system') for multi-digit numbers. They refer to the fact that when numbers are written the position implies each digit's value, where as the value is stated when it is spoken. They contend that this and the fact some two-digit numbers do not follow this 'named-value' structure make the concept of place value difficult for English-speaking children to grasp and that teachers need to support children in their understanding of this inconsistent system.

Following their study with children from 1st and 2nd grades using Dienes blocks to represent the 'named value system' and digit cards to represent the 'positional base-ten system' they recommended the use of Dienes blocks for teaching addition and subtraction only after children can work with number bonds to 18. They further recommended that when multi-digit work is introduced children should begin with larger numbers rather than starting with two-digit numbers because of the inconsistency of two-digit numbers. Fuson (1990a) cited the use of the abacus and the metric system of measurement as cultural advantages children from Asian countries have over English speaking children in the United States and suggested that because of this American children may need additional support for "constructing multi-unit meaning based on ten" (1990, p. 345).

Thompson (1998) investigated Fuson & Briar's recommendations. He agreed with them, but suggested that they ignored the strategy of partitioning. He recommended that when children are dealing with calculations of two-digit numbers they focus on "the 'partitionable' aspect of our system of counting words" (Thompson 1998, p. 6) and that the way that place value is taught needed to relate "more closely to the language used to express two-digit numbers in the English spoken system of number words" (1998, p. 1).

Miura, Okamoto, Kim, Chang, Steere, & Fayol (1994) compared place value understanding of Asian, European and American first grade children and found that Asian-language speakers were better able to represent numbers in different ways using Base 10 blocks, which they claimed "suggests greater flexibility of mental number manipulation" (1994, p. 411). They proposed that "numerical language characteristics" of the Asian languages may in part account for Asian students' success in mathematics achievement. Yang and Cobb (1994) found similar results when comparing children from Taiwan and the United States, but they argue that the differences are partially attributed to "deep-rooted cultural beliefs that extend beyond the classroom" (1994, p. 30).

Mental Arithmetic, Teaching Methodologies and the Use of Materials

Thompson distinguished between 'column value' as opposed to 'quantity value' in place value. He explained column value as seeing 64 as 6 tens and 4 units and quantity value as 60 and 4. Column value is used with notation boards, abacus and standard algorithms while as quantity value is used more with mental arithmetic, informal written strategies and empty number lines. Thompson argued that both are necessary and that teachers need to "ensure that connections are made between quantity value and column value" (2003, p. 188). He further contended that teachers should not be in a hurry to teach the "column value" of place value as "young children find the concept difficult" (2009, p. 5)

Thompson and Bramald (2002) investigated the relationship between children in the United Kingdom's understanding of place value and their competence at mental addition. Using adapted tasks from studies in the United States and Australia by several researchers, they compared the results of each task with the results from studies carried out by Ross (1989) and Kamii (1988) in the United States and Price (1998) in Australia. Children from the United Kingdom outperformed their American and Australian counterparts in these tasks. Thompson and Bramald attribute this partially to the emphasis on oral and mental arithmetic and the later introduction of formal written calculations in the National Numeracy Strategy, as well as the use of Gattegno tables and place value cards in classrooms in the United Kingdom. Other research studies also looked at teaching methodologies and the use of materials (Kamii and Joseph, 1988; Fuson et al., 1997; Fosnot and Dolk, 2001; Murata & Fuson, 2006; Chandler & Kamii, 2009) in developing children's understanding of place value.

Researchers around the world have looked at the teaching and learning of place value from many different perspectives. There seems to be some agreement on the type of experiences that help children construct meaning for multi-digit numbers. The importance of counting, grouping and flexibility with numbers were emphasised. Many of the researchers cited recommended using problem solving tasks, concrete materials, discussion and mental arithmetic to help children in their understanding of place value.

METHODOLOGY

The main research project aimed to investigate place value understanding of fourth class children in Ireland and to answer the following four questions:

- What do children in fourth class understand about place value?
- What aspects of the place value concept present difficulties for fourth class children?
- What misconceptions do fourth class children have about the concept of place value?
- How do fourth class Irish children compare with their international counterparts on place value tasks?

The methodology used was semi-structured task-based interviews. Children were interviewed in a private room in their own school and asked to complete a series of tasks based on international research studies to ascertain what level of understanding of place value concepts they possess. This research looked at place value understanding of whole numbers only. The subjects of this study were children in fourth class, from five classes in three schools in County Dublin. Two schools were single-sex schools and one school was a co-educational school. The sample was a non-probability sample based on convenience. This researcher conducted all interviews for consistency. Interviews were videotaped for further analysis. The video camera was placed on a tripod and focused on the child's hands and the table as it was deemed more important to see what the child was doing with the materials and hear his/her responses than to focus on the child's facial expressions. This was also decided to guarantee anonymity of the child. Teachers were asked to identify each child's achievement in Mathematics as high, middle or low in relation to other children in the class.

Interviews took place in the second half of the school year. Interviews took between 12 and 25 minutes depending on the child. The interview was moved on or terminated if the child became uneasy about a question. A variety of materials including Base-ten blocks (Dienes blocks), unifix cubes, place value board (HTU), sweets, counters and number cards were used. All numbers on number cards were presented in large font (72). The materials used were materials that are recommended by the Primary School Curriculum – Mathematics (DES, 1999) for use in teaching number in primary schools and should be available in most schools in Ireland.

This paper presents case studies of two of the participants in the study, a girl named Felicity and a boy named Stephen (Names have been changed to protect identity of children). These children were chosen because their results were surprising given their teachers' evaluation and their answers to earlier questions in the interviews, and because they show good understanding in some questions and not in others.

FINDINGS

Case Study One - Felicity

Felicity is a ten year old girl of foreign born parents living in Dublin. She attends fourth class in a co-educational school. Though English is not her first language, she has a high level of fluency in English. Her teacher identified her as one of the top in Mathematics in the class. Felicity was confident in her mathematics during the interview. At first glance Felicity showed good understanding of place value. She could read and order three digit numbers and identify hundreds, tens and units. She could use non-canonical representation, i.e. with Dienes blocks she represented the number 33 using 33 units, one ten and 23 units, two tens and 13 units and three tens and 3 units. She could represent 201 on a notation board using counters and could easily answer mental addition questions and give a clear explanation of how she worked out the answer. She showed place value understanding when asked to estimate the answer to 48 plus 39.

Felicity: Maybe somewhere around seventy.

Interviewer: OK and why do you think that?

Felicity: Mmmm, because the four and three makes a seven, so it would be somewhere around seventy or eighty.

Interviewer:	OK. So you're looking at the four and the three and that tells you it's somewhere around seventy. Do you think that it will be more than seventy or less than seventy?
Felicity:	I think it will be more.
Interviewer:	A lot more or a little bit more?
Felicity:	A lot.
Interviewer:	Why?
Felicity:	Because the ones on the end are large numbers near ten.

When asked to count 23 sweets Felicity counted correctly by ones. Then she revealed a major misconception, as shown below:

Interviewer:	Could you show me this part...the sweets that are for this part? (Circle the 2)
Felicity:	(Child moves 12 sweets one at a time, then counts the rest silently while moving them) I think it's this (points at the twelve sweets).
Interviewer:	How many are here (points at twelve sweets)?
Felicity:	(Child counts sweets) 12.
Interviewer:	12 and you think that's this part of the number (circles 2 in 23)
Felicity:	Yeah.
Interviewer:	And what about this part of the number here (circles 3 in 23)?
Felicity:	(Counts remainder again silently while moving them) I think it's three (moves three sweets away from group).
Interviewer:	So you said this part here is this part (points to twelve sweets and then to 2 in 23) and this part here is this part (points to three sweets and then to 3 in 23). What about these sweets here (points to 8 remaining sweets)?
Felicity:	Well that could be a thirteen (Moves the three sweets over to remaining eight sweets).
Interviewer:	Do you think that could be a thirteen?
Felicity:	Yes, but I need three from here (points to group of twelve sweets).
Interviewer:	OK.
Felicity:	Two.
Interviewer:	What do you think? Why do you think this here is twelve sweets (points at two in number twenty three card and then at group of twelve sweets)?
Felicity:	Because it's in tens and ten added to two is twelve.
Interviewer:	And this is the three here (points at the three sweets and the numeral three in twenty three on card)? And I wonder about these sweets here?
Felicity:	They can be the remainder.
Interviewer:	OK, but when I asked you how many altogether you said twenty three, didn't you (circles all the sweets)? So why is there a remainder here?
Felicity:	Cos twenty three is an odd number.

She again displayed this misconception when asked to count 26 unifix cubes (six lots of four cubes joined together and two loose cubes). Again she counted them correctly by ones pointing at each one as she counted. When asked to show the cubes that represented the digit two in 26 she showed twelve again with the explanation below:

Felicity:	...cos two is in the tens place, so that means that you would have to add on ten which is twelve and six is in the units place, units means just separate...

Felicity could identify that the digit three in thirty eight was worth more than the digit three in 83, explaining:

Felicity:	Because it's in the tens and tens is worth more than units.

Because of the previous misconception detected the interviewer decided to probe deeper to see if her misconception applied only to numbers in the twenties or to other two-digit numbers also.

Interviewer:	What would this three be (points to the digit three in 38)?
Felicity:	Thirteen.
Interviewer:	Thirteen? And what would this three be (points to the three in 83).
Felicity:	Just three.
Interviewer:	Just the three OK. So, thirteen that would be (points to the 3 in 38). Could you show me the thirteen with these blocks (Dienes Blocks) please?
Felicity:	Yes (shows one ten and three units). Thirteen.
Interviewer:	OK, so that's what this one here is (points at 3 in 38)?
Felicity:	Yes.

Discussion of Case Study One

Felicity has been identified as one of the strongest students in mathematics in her class and indeed she completed most tasks successfully with clear explanations of her thinking. However, some aspects of her mathematics ability were surprising. Felicity always counted by ones when asked to count or show a number while many other children counted by twos, threes, fours, tens and even eights during the study. Once she asked if she should count by fours when the cubes are grouped by fours, but resorted to counting by ones when she was told by the interviewer 'Any way you like to count them'. The only time Felicity counted by a number other than ones was when asked to count a quantity of Dienes blocks where she was able to count in tens and hundreds.

Although Felicity showed advanced understanding of place value Stage V (Ross, 1986) in some tasks she has very poor understanding of a very basic aspect of place value, which Thompson (2000) calls 'quantity value' (seeing sixty four as sixty and four) as opposed to 'column value'(seeing sixty four as six tens and four units). Felicity showed good understanding of column value in all tasks but was inconsistent when showing quantity value. While she could show thirty three as three tens and three units with the Dienes blocks, she showed the digit three in 38 as thirteen when using the same blocks in a different task. She showed the digit two in 23 and 26 as twelve in two tasks explaining that 'Because it's in ten and ten added to two is twelve' and 'cause two is in the tens place, so that means that you would have to add on ten which is twelve'.

Felicity is confident doing mental calculations and estimations. When asked the estimation question she begins by using the front-end strategy of estimation and then refines her answer when probed further. She clearly sees the 3 and 4 in question here as thirty and forty and understands that the size of the numbers in the units place will also determine the answer. While it may be expected that she would use the rounding strategy for a more accurate estimation at this level, this was not the case for most children in the study. This is consistent with the results of the Primary Curriculum Review - Phase 1 (NCCA, 2005) where when asked how often children in their class were enabled to use estimation strategies in number, approximately 60% of teachers responded less than once a week for front-end strategy and approximately 37% responded less than once a week for rounding strategy (NCCA, 2005, p. 139).

It would seem that when Felicity was questioned about her thinking and became unsure she quickly tried to make sense by resorting to her prior knowledge of numbers that was not relevant in this case – remainder and odd numbers. This would indicate that she is confident in her own ability to make sense of mathematics.

Case Study Two – Stephen

Stephen is a ten year old boy living in Dublin and attending a co-educational school. His teacher identified him as one of the weaker children in the class and he attends learning support. He was anxious throughout the interview and needed to be reassured at times, as a result the interviewer chose not to present all tasks to him. He counted by ones and used his fingers to add simple numbers. At first glance Stephen's understanding of place value appeared confused. While he could read and order three digit numbers and use non-canonical (Stage 5 (Ross, 1983)), he was confused when asked to identify place value in a three-digit number.

Interviewer: How many tens are in this number then? (356)

Stephen: Fifty.

However, surprisingly, he was successful in both digit correspondence tasks (match an individual digit in a multi-digit number to a quantity of objects) and showed flexibility with numbers.

Interviewer: Can you show me with the cubes that refer to this part here? (Circles the digit two in 26)

Stephen: Twenty. (Counts out six cubes, one group of four and two loose cubes and put them beside the digit six in 26 and pushes the rest over beside the digit two in 26)

Interviewer: OK, great. How did you do that? You didn't count twenty, you counted six. Why did you do that?

Stephen: I counted six and if you took away six, you'd be left with twenty.

He could identify that the digit three in 38 was worth more than the digit three in 83 explaining:

Stephen: Because it's thirty and this (Points to digit three in 83) is just practically three attached on to it.

Stephen could show the number 201 correctly on a notation board using counters and could identify the number 28 when shown with counters on a notation board; however he had difficulty when given four more counters and asked to do what was necessary to show 32 on the board.

Stephen:	I would just do this....if I was in a class now I would just add them together and find a way how I'm going to make thirty two with them.
Interviewer:	OK, so can you do that with them?
Stephen:	You've got twenty here (points at the two in the tens place) and the you've got (points at units place) if you took those away (points at the four extra counters) you've got twenty eight, and eight get up to twelve is four, so four add to that will make it 32, won't it.

When asked to estimate 48 plus 39 Stephen again showed flexibility with numbers.

Stephen:	77
Interviewer:	Can you tell me how you did that?
Stephen:	I did this. If we cut... say if we cut it like that (cutting motion between digits in number) I'd say ten and eight is eighteen, but it's nine and eight so you take one off, seventeen. But we add one to four (makes motion as if he's carrying one over to four) five and add three (Puts up three fingers and counts on) oh sorry 8.

This was more sophisticated than most children who answered seventy or others who reported using a standard algorithm in their head and gave an exact answer of 87.

Discussion of Case Study Two

Stephen has been identified as one of the weaker children in his class but clearly shows flexibility with numbers on occasion which needs to be encouraged. He clearly lacks confidence in his own ability and was anxious about getting the 'right' answer. Stephen counts quantities of objects by ones in most cases and still uses his fingers for basic addition facts. He has good estimation skills which need to be encouraged and developed. While he showed non-canonical understanding and was successful in both digit-correspondence tasks he had difficulty in identifying place value in a three-digit number and gave quantity value (fifty) for column value (five tens). The language of mathematics seemed to confuse Stephen at times and may have contributed to his incorrect answers.

The findings from the analysis of the digit correspondence tasks of these two children's interviews using Sharon Ross' (1986) stages of development of place value understanding do not match with the findings of her study that found most children in fourth grade (class) at Level three or four. Neither are they consistent with the findings of the pilot study (Stafford, 2007) which had similar results to Ross'. The final analysis of the interviews of all of the participants in the current study may indicate differently.

Recommendations

Both children need lots of opportunities and encouragement to count quantities of objects in twos, threes, fours, etc. While they may be able to do so, they resort to counting in ones when counting a quantity of objects. Counting and grouping are two of the constructs used by Jones, Thornton & Putt (1994) and identified as essential for children's development of place value understanding. They also need to work on further developing quantity value and column value and ensuring 'that connections are made between quantity value and column value' as recommended by Thompson (2003, 188).

Exposure to and encouragement to use the estimation strategies recommended by the Primary School Curriculum –Mathematics Teacher Guidelines (DES, 1999) would help both children develop more accurate estimations, while further developing their understanding of place value. Work with calculators may also help develop this as reported by Ward (2005).

Having opportunities to justify their thinking in group discussions with peers or to write about their thinking may also help Felicity and Stephen to self correct any misconceptions they may develop in learning new mathematical concepts.

Felicity would benefit from working one to one with a teacher or more able peer to help her to understand that the digit in the tens place means that number of groups of ten.

Stephen needs to build his confidence in his own ability. Encouragement to share his ideas with the class might help and would also expose other children to thinking about numbers in more flexible ways. He would benefit from work to consolidate his knowledge of number facts and to develop strategies for recall when he forgets them. Stephen would also gain from developing his understanding of mathematical language.

CONCLUSION

Both children present varying degrees of understanding depending on the tasks presented. Whereas both children are in the process of developing place value understanding, neither child is secure in place value understanding. Both present a case for teachers spending time to unravel children's understanding and misunderstanding of this important mathematical concept in order to help them achieve success across all strand of the Mathematics curriculum. While this is just a snapshot of two children in the research study, preliminary findings of the main study indicate that other children present similar discrepancies in their understanding of place value. By implementing assessment strategies such as conferencing, observation and discussion as recommended by the NCCA (2007) the teacher can gain a deeper insight into children's understanding of number concepts than weekly tests or standardised test can provide. Further analysis of the full data set of this study should provide interesting finding and implications for the teaching and learning of place value in Irish schools.

REFERENCES

Chandler, C., & Kamii, C. (2009). Giving change when payment is made with a dime: The difficulty of tens and ones. *Journal for Research in Mathematics Education,* 40, 97-118.

Cobb & Wheatley (1988). Children's Initial Understandings of Ten. *Focus on Learning Problems in Mathematics*, 10(3),1-28.

Clarke, D., Cheeseman, J., Gervasoni, A., Gronn, D., Horne, M., McDonough, A., Montgomery, P. Roche, A., Sullivan, P., Clarke, B., & Rowley, G. (2002). *Early Numeracy Research Project* Retrieved August 15, 2010.

Http://www.education.vic.gov.au/studentlearning/teachingresources/maths/enrp/enrplaf.htm#H3N10078

Department of Education and Science (1999). *Primary Curriculum Mathematics Teacher Guidelines.* Dublin: The Stationery Office.

Department of Education and Science (2005a) *An Evaluation of Curriculum Implementation in Primary Schools*. Dublin: The Stationery Office.

Department of Education and Science (2005b). *Literacy and Numeracy in Disadvantaged Schools: Challenges for Teachers and Learners*. Dublin: The Stationery Office.

Eivers, E., Close, S., Shiel, G., Millar, D., Clerkin, A., Gilleece., L. & Kiniry, J. (2010). *The 2009 National Assessments of Mathematics and English Reading.* Dublin: Educational Research Centre.

Fosnot, C.T. & Dolk, M. (2001) *Young Mathematicians at Work: Constructing Number Sense, Addition, and Subtraction.* New Hampshire: Heinemann.

Fuson, K. (1990). Conceptual Structures for Multiunit Numbers: Implications for Learning and Teaching Multi-digit Addition, Subtraction, and Place Value. *Cognition and Instruction*, 7(4), 343-403.

Fuson, K. (1990a). Issues in Place-Value and Multi-digit Addition and Subtraction Learning and Teaching. *Journal of Research in Mathematics Education,* 21(4), 273-280.

Fuson, K. and D. Briars (1990). Using a Base-Ten Blocks Learning/Teaching Approach for First- and Second-Grade Place Value and Multi-digit Addition and Subtraction. *Journal of Research in Mathematics Education* 21(3), 180-206.

Fuson, K., Wearne, D., Hiebert, J.C., Murray, H.G., Human, P.G., Olivier, A.I., Carpenter, T.P., & Fennema, E. (1997). Children's Conceptual Structures for Multidigit Numbers and Multidigit Addition and Subtraction. *Journal of Research in Mathematics Education* 28(2), 130-162.

Hiebert, J. & Wearne, D. (1992). Links between teaching and learning place value with understanding in first grade. *Journal of Research in Mathematics Education* 23(2), 98-122.

Horne, M. & Livy, S. (2006). Young children developing place value understandings. In J. Novotná, H. Moraová, M. Krátká, & N. Stehliková (Eds.), *Mathematics in the Centre*, Proceedings of the 30th Conference of the International Group for the Psychology of Mathematics Education, 3, 313-320. Prague, Czech: Charles University, PME.

Jones, G. A., Thornton, C. A., & Putt, I. J. (1994). A Model for Nurturing and Assessing Multidigit Number Sense among First Grade Children. *Educational Studies in Mathematics* 27(2), 117-143.

Jones, G. A., Thornton, C. A., Putt, I. J., Hill, K. M., Mogill, A. T., Rich, B. S., Van Zoest, L. A. (1996). Multidigit Number Sense: A Framework for Instruction and Assessment. *Journal of Research in Mathematics Education* 27(3), 310-336.

Kamii, Constance and Linda L. Joseph (1988). Teaching Place Value and Double Column Addition. *Arithmetic Teacher*, 48-52.

Miura, I. T., Okamoto, Y., Kim, C.C., Steere, M., & Fayol, M. (1993). First Graders' Cognitive Representation of Number and Understanding of Place Value: Cross-National Comparison-France, Japan, Korea, Sweden, and the United States. *Journal of Educational Psychology*, 85(1), 31-40.

Miura, I. T., Okamoto, Y., Kim, C.C., Chang, C.-M., Steere, M., & Fayol, M. (1994). Comparisons of Children's Cognitive Representation of Number: China, France, Japan, Korea, Sweden, and the United States. *International Journal of Behavioral Development,* 17, 401-411.

Murata & Fuson, K. (2006). Teaching as Assisting Individual Constructive Paths within an Interdependent Class Learning Zone: Japanese First Graders Learning to Add Using 10. *Journal for Research in Mathematics Education,* 37(5), 421-456.

National Council for Curriculum and Assessment (2005). *Primary Curriculum Review: Phase 1 Final Report.* Dublin: N.C.C.A.

National Council for Curriculum and Assessment (2007). *Assessment in the Primary School Curriculum: Guidelines for Schools.* Dublin: N.C.C.A.

Price, P. S. (1998). Year 3 Students' Place-Value Misconceptions: Another Look at MAB in Teaching Mathematics in New Times. *Proceedings of the 21st Annual Conference of the Mathematics Education Research Group of Australia.* Vol.2, 452-459.

Price, P. S. (2001) The Development of Year 3 Students' Place Value Understanding: Representations and Concepts, PhD thesis. Accessed September 1 2005 http://adt.library.qut.edu.au/adt-qut/public/adt-QUT20020412.102204/index.html.

Resnick, L.B. (1983). A developmental theory of number understanding. In H.P. Ginsburg, *The Development of Mathematical Thinking*. New York: Academic Press.

Ross, S. H. (1986). The Development of Children's Place-Value Numeration Concepts in Grades Two through Five. Paper presented at the annual meeting of American Research Association, San Francisco, C.A. (Eric Document Reproduction Service No. 273 482).

Ross, S. H. (1987). The Development of Children's Place Value Numeration Concepts in Grades Two through Five. Paper presented at the annual meeting of American Research Association, San Francisco, C.A. (Eric Document Reproduction Service No. 273 482).

Ross, S. H. (1989). Parts, Wholes, and Place Value: A Developmental View. *Arithmetic Teacher*, 36(6), 47-51.

Ross, S. H. (1990) Children's Acquisition of Place-Value Numeration Concepts: The Roles of Cognitive Development and Instruction. *Focus on Learning Problems in Mathematics* 12(1), 1-17.

Ross, S. H. (2002). Place Value: Problem Solving and Written Assessment. Teaching Children Mathematics, 8(7), 419-423.

Shiel, G., & Kelly, D. (2001). *The 1999 National Assessment of Mathematics Achievement*. Dublin: Educational Research Centre.

Shiel, G., Surgenor, P., Close, S., & Millar, D. (2006). *The 2004 National Assessment of Mathematics Achievement*. Dublin: Educational Research Centre.

Shiel, G., Moran, G., Cosgrove, J., & Perkins, R. (2010). *A summary of the performance of students in Ireland on the PISA 2009 test of mathematical literacy and a comparison with performance in 2003. Report to the Department of Education and Skills*. Dublin: Educational Research Centre.

Stafford, P. (2007). 'We don't need those cubes' In S. Close, D. Corcoran and T. Dooley (Eds.) *Proceedings of the Second National Conference on Research in Mathematics Education*. (pp.142-151) Dublin: St. Patrick's College.

Thompson, I. (1998). Influence of Structural Aspects of the English Counting Word System on the Teaching and Learning of Place Value. *Research in Education*, 59, 1-8.

Thompson, I. and Bramald, R. (2002). *An Investigation of the Relationship between Young Children's Understanding of Place Value and their Competence at Mental Addition (Report for the Nuffield Foundation)*. Newcastle upon Tyne: University of Newcastle upon Tyne.

Thompson, I. (2003). Place Value: the English disease? In Thompson, I. (Ed.) *Enhancing Primary Mathematics Teaching.* (pp.181-190) Berkshire: Open University Press.

Thompson, I. (2009). Place Value? *Mathematics Teaching*. 215, 4-5

Travers, J. (2005). An Exploration of Self-Reported Strategy Use for Mathematical Word Problems by Sixth Class Pupils in Irish Schools In S. Close, D. Corcoran and T. Dooley (Eds.), *Proceedings of the First National Conference on Research in Mathematics Education* (pp.153-171) Dublin: St. Patrick's College.

Travers, J. (2009). Learning support for Mathematics: Lessons from 100 lessons In D. Corcoran, T. Dooley, S. Close and R. Ward (Eds.) *Proceedings of the Third National Conference on Research in Mathematics Education* (pp.195-207) Dublin: St. Patrick's College.

Ward, R. (2005). The Role of Calculators in Teaching Whole Number and Decimal Numeration In S. Close, T. Dooley and D. Corcoran (Eds.), *Proceedings of the First National Conference on Research in Mathematics Education* (pp. 172-184) Dublin: St. Patrick's College.

Yang, M.T-L and Cobb, P. (1994). A Cross-Cultural Investigation into the Development of Place-Value Concepts of Children in Taiwan and the United States. *Educational Studies in Mathematics,* 28(1), 1-33.